

M.A. Hayat
Editor

Tumors of the Central Nervous System

Volume 7

Meningiomas and Schwannomas

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Volume 7

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System
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Nervous System

Meningiomas and Schwannomas

Edited by

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ISBN 978-94-007-2893-6 e-ISBN 978-94-007-2894-3
DOI 10.1007/978-94-007-2894-3
Springer Dordrecht Heidelberg London New York

Library of Congress Control Number: 2012933364

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Printed on acid-free paper

Springer is part of Springer Science+Business Media (www.springer.com)

Although touched by technology, surgical pathology always has been, and remains, an art. Surgical pathologists, like all artists, depict in their artwork (surgical pathology reports) their interactions with nature: emotions, observations, and knowledge are all integrated. The resulting artwork is a poor record of complex phenomena.

Richard J. Reed, MD

One Point of View

All small tumors do not always keep growing, especially small breast tumors, testicular tumors, and prostate tumors. Some small tumors may even disappear without a treatment. Indeed, because prostate tumor grows slowly, it is not unusual that a patient may die at an advanced age of some other causes, but prostate tumor is discovered in an autopsy study. In some cases of prostate tumors, the patient should be offered the option of active surveillance followed by PSA test or biopsies. Similarly, every small kidney tumor may not change or may even regress. Another example of cancer or precancer reversal is cervical cancer. Precancerous cervical cells found with Pap test, may revert to normal cells. Tumor shrinkage, regression, reversal, or stabilization is not impossible.

Another known example of cancer regression is found in pediatric neuroblastoma patients. Neuroblastoma shows one of the highest rates of spontaneous regression among malignant tumors. In addition to the well-known spontaneous regression in stage 4S disease, the high incidence of neuroblastoma remnants found during autopsy of newborns suggest that localized lesions may undergo a similar regression (Guin et al. 1969). Later studies also indicate that spontaneous regression is regularly seen in infants with localized neuroblastoma and is not limited to the first year of life (Hero et al. 2008). These and other studies justify the “wait and see” strategy, avoiding chemotherapy and radiotherapy in infants with localized neuroblastoma, unless *MYCN* gene is amplified. Infants with nonamplified *MYCN* and hyperdiploidy can be effectively treated with less intensive therapy. Infants with disseminated disease without *MYCN* have excellent survival with minimal or no treatment.

The pertinent question is: Is it always necessary to practice tumor surgery, radiotherapy, or chemotherapy? Although the conventional belief is that cancer represents an “arrow that advances unidirectionally”, it is becoming clear that for cancer to progress, they require cooperative microenvironment (niche), including immune system and hormone levels. However, it is emphasized that advanced (malignant) cancers do not show regression, and require therapy. In the light of the inadequacy of standard treatments of malignancy, clinical applications of the stem cell technology need to be expedited.

M.A. (Eric) Hayat

References

- Hero S, Simon T, Spitz R, Ernestus K, Gnekow A, Scheel-Walter H, Schwabe D, Schilling F, Benz-Bohm G, Berthold F (2008) Localized infant neuroblastomas often show spontaneous regression: results of the prospective trials NB95-S and NB 97. *J Clin Oncol* 26:1504–1510
- Guin P, Gilbert E, Jones B (1969) Incidental neuroblastoma in infants. *Am J Clin Pathol* 51:126–136

Preface

It is recognized that scientific journals and books not only provide current information but also facilitate exchange of information, resulting in rapid progress in the medical field. In this endeavor, the main role of scientific books is to present current information in more details after careful additional evaluation of the investigational results, especially those of new or relatively new therapeutic methods and their potential toxic side-effects.

Although subjects of diagnosis, drug development, therapy and its assessment, and prognosis of tumors of the central nervous system, cancer recurrence, and resistance to chemotherapy are scattered in a vast number of journals and books, there is need of combining these subjects in single volumes. An attempt will be made to accomplish this goal in the projected ten-volume series of handbooks.

In the era of cost-effectiveness, my opinion may be minority perspective, but it needs to be recognized that the potential for false-positive or false-negative interpretation on the basis of a single laboratory test in clinical pathology does exist. Interobserver or intraobserver variability in the interpretation of results in pathology is not uncommon. Interpretative differences often are related to the relative importance of the criteria being used.

Generally, no test always performs perfectly. Although there is no perfect remedy to this problem, standardized classifications with written definitions and guidelines will help. Standardization of methods to achieve objectivity is imperative in this effort. The validity of a test should be based on the careful, objective interpretation of the tomographic images, photo-micrographs, and other tests. The interpretation of the results should be explicit rather than implicit. To achieve accurate diagnosis and correct prognosis, the use of molecular criteria and targeted medicine is important. Equally important are the translation of molecular genetics into clinical practice and evidence-based therapy. Translation of medicine from the laboratory to clinical application needs to be carefully expedited. Indeed, molecular medicine has arrived.

This is the seventh volume in the series, Tumors of the Central Nervous System. As in the case of the six previously published volumes, this volume mainly contains information on the diagnosis, therapy, and prognosis of brain tumors. Various aspects of two types of brain tumors (Meningiomas and Schwannomas) are discussed. Insights on the understanding of molecular pathways involved in tumor biology are explained. For example, the role of E-cadherin gene instability, carbonic anhydrase II, urokinase plasminogen

activator, and Wnt signaling in meningiomas is discussed in detail. Such information leads to the development of effective drugs, and knowledge of involved pathways and signaling facilitates targeted therapies in cancer. Similarly, the role of molecular genetics and epigenetic mechanisms in schwannomas is explained, including the role of cyclin D1 in vestibular schwannoma.

The subtypes of meningiomas are determined using perfusion magnetic resonance imaging. The diagnosis of sporadic meningioangiomatosis using imaging technologies is explained. Diagnosis of incidentally discovered meningioma and cystic papillary meningioma is also included. Diagnosis of facial nerve schwannoma, vestibular schwannoma, and intermediate nerve schwannoma is presented. Treatments for atypical meningioma, benign meningioma, oncocytic meningioma, intracranial meningioma, and cavernous sinus meningioma are presented. Therapeutic methods such as neurosurgery, Gamma knife radiosurgery, and adjuvant radiation for this cancer are included. Large number of treatments, including radiosurgery, retrosigmoidal craniotomy, and immunotherapy, for vestibular schwannoma patients are detailed. The quality of life, after treatments of both meningioma and schwannoma patients, is discussed. Genetic and clinical features associated with recurrence in meningioma patients are explained, and the role of erythropoietin receptor in the recurrence is included.

Introduction to new technologies and their applications to tumor diagnosis, treatment, and therapy assessment are explained. For example, nanotechnology-based therapy for malignant tumors of the CNS is explained. Molecular profiling of brain tumors to select therapy in clinical trials of brain tumors is included. Several surgical treatments, including resection, and radiosurgery, are discussed. The remaining three volumes in this series will provide additional recent information on this and other aspects of CNS malignancies.

By bringing together a large number of experts (oncologists, neurosurgeons, physicians, research scientists, and pathologists) in various aspects of this medical field, it is my hope that substantial progress will be made against this terrible disease. It would be difficult for a single author to discuss effectively the complexity of diagnosis, therapy, and prognosis of any type of tumor in one volume. Another advantage of involving more than one author is to present different points of view on a specific controversial aspect of the CNS cancer. I hope these goals will be fulfilled in this and other volumes of this series. This volume was written by 104 contributors representing 12 countries. I am grateful to them for their promptness in accepting my suggestions. Their practical experience highlights their writings, which should build and further the endeavors of the reader in this important area of disease. I respect and appreciate the hard work and exceptional insight into the nature of cancer provided by these contributors. The contents of the volume are divided into two subheadings: Meningiomas and Schwannomas.

It is my hope that the current volume will join the preceding volumes of the series for assisting in the more complete understanding of globally relevant cancer syndromes. There exists a very urgent demand by the public and the scientific community to address to cancer, diagnosis, treatment, cure, and hopefully prevention. In the light of existing cancer calamity, government

funding must give priority to eradicating this deadly malignancy over military superiority.

I am thankful to Dr. Dawood Farahi and Dr. Kristie Reilly for recognizing the importance of medical research and publishing through an institution of higher education.

M.A. (Eric) Hayat

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Meningiomas

Part I

Molecular Genetics

Meningiomas: Role of Semaphorin3A Protein in Antiangiogenesis

1

Valeria Barresi

Abstract

Meningiomas represent frequent tumors mainly characterized by a benign histology and an indolent clinical course. The histological grade and extent of surgical resection are considered to be the most powerful prognosticators for these neoplasias. Nonetheless, a percentage of meningiomas do display relapses despite a total macroscopic resection, although the reasons of this phenomenon still remain to be fully clarified. Herein the prognostic value of angiogenesis and of the expression of its regulators in completely surgically resected meningiomas is discussed. In particular, the role of semaphorin3A (SEMA3A) in the regulation of angiogenesis and the correlation between its expression and the risk of recurrence of meningiomas are illustrated.

Introduction

Meningiomas are frequent arachnoidal cell neoplasms typically attached to the inner surface of the dura mater (Perry et al. 2007). They are mainly characterized by a benign histology and an indolent clinical course (Perry et al. 1999). The histological grade and the proliferation index are considered to be the most powerful histological prognosticators for the outcome of these neoplasias (Perry et al. 1999), whereas the major clinical factor in recurrence risk is the extent

of surgical resection (Simpson 1957) which is influenced by tumour site, extent of invasion and by the attachment to vital intracranial structures. A major issue relates to the recurrence of totally resected (Simpson's grade I) meningiomas and to the identification of its predicting factors. Indeed, despite total macroscopic resection, a percentage of meningiomas do develop relapses, although the reasons of this phenomenon still remain to be fully clarified.

We have recently reported that higher recurrence risk in completely resected meningiomas may be related to a high quantity of tumour neo-angiogenesis, reflected by a high microvessel density (MVD) quantified through the immunohistochemical assessment of the specific marker for neo-angiogenesis CD105 (Barresi et al. 2007). As known, neo-angiogenesis is dependent on the

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balance between angiogenic and anti-angiogenic regulators (Folkman and Shing 1992). Among the former, vascular endothelial growth factor (VEGF) plays an essential role in the positive regulation of tumor angiogenesis by promoting the migration, proliferation and tube formation of endothelial cells (Ferrara and Davis-Smyth 1997). Its up-regulation has been shown in meningiomas, suggesting that it behaves as a pro-angiogenic factor also in these tumors (Christov et al. 1999); nonetheless, the prognostic value of VEGF expression on the development of recurrences of meningiomas appears questionable (Maiuri et al. 2007).

Among the anti-angiogenic factors, attention has been recently focused on semaphorin3A (SEMA3A), which belongs to the class 3 semaphorins (SEMAs), a class of proteins secreted by vertebrates with a role in axon guidance (Luo et al. 1993). SEMA3A exerts its action through the binding to neuropilin-1 (NRP-1) (Soker et al. 1998), a protein initially identified on neuronal cells but also expressed on the endothelial cells as a trans-membrane receptor. The binding of SEMA3A to NRP-1 leads to the apoptosis and inhibition of proliferation of the endothelial cells (Guttmann-Raviv et al. 2007). Nevertheless, NRP-1 also functions as an additional receptor for VEGF and mediates its activity in the absence of its receptor VEGF-R (Soker et al. 1998). In addition, NRP-1 may also enhance the effects of the VEGF binding to VEGF-R2 (Soker et al. 2002). As a consequence, SEMA3A behaves as an anti-angiogenic factor either directly, through the binding to NRP-1, either indirectly, through the competitive inhibition of VEGF binding to NRP-1 (Miao et al. 1999; Soker et al. 2002).

Based on our finding that SEMA3A is expressed by the neoplastic cells of human meningiomas (Barresi et al. 2009), we investigated its potential anti-angiogenic activity in these tumors by analyzing its correlation with the MVD revealed in the same neoplasias (Barresi et al. 2009). As SEMA3A anti-angiogenic action is mediated by NRP-1 presence on the endothelial cells, we also tested the expression of this receptor in these meningiomas. Moreover, since the balance between VEGF and SEMA3A has been

reported as a significant prognostic factor for overall survival in patients affected by ovarian cancer (Osada et al. 2006), we evaluated the immuno-expression of VEGF and assessed the prognostic impact of VEGF/SEMA3A ratio as well as of SEMA3A expression on the recurrence-free survival of totally resected meningiomas (Barresi et al. 2009; Barresi and Tuccari 2010).

Immunohistochemistry

The analysis was carried out on a series of surgically resected meningiomas including the meningothelial, transitional, fibrous, microcystic, secretory, chordoid, atypical and anaplastic histotypes. All tumors were classified and graded according to WHO criteria (Perry et al. 2007) by two neuropathologists who independently evaluated each sample. For each case, Simpson's grade of surgical resection (Simpson 1957), the Ki-67 labelling index (LI), reflecting the tumour growth fraction as well as the follow-up data, including the patients survival and recurrences, were available. Patients were subdivided into two groups on the basis of Simpson's grade; the first included grade 1 (complete excision, including dura and bone) meningiomas, whereby the second comprised grade 2 (complete excision plus apparently reliable coagulation of dural attachments) and grade 3 cases (complete excision of the solid tumour, but with insufficient dural coagulation or bone excision). Recurrence was defined as the detection of a recurrent tumour by neuroradiological investigations in those patients with a Simpson's grade 1 meningioma.

The tumour neo-angiogenesis was quantified through the assessment of MVD performed by using the immunohistochemical detection of the specific marker for neo-angiogenesis CD105 (endoglin) (Barresi et al. 2007, 2009).

All meningiomas were fixed in 10% neutral formalin for 24 h at room temperature, embedded in paraffin at 55°C and cut into parallel consecutive 4 µm thick sections for the subsequent immunohistochemical study. Briefly, the endogenous peroxidase activity was blocked with 0.1% H₂O₂ in methanol for 20 min.; then, normal sheep serum

was applied for 30 min to prevent unspecific adherence of serum proteins. The SEMA3A, VEGF and NRP-1 antigens were unmasked by microwave oven pre-treatment in 10 mM, pH 6.0 sodium citrate buffer for 3 cycles $\times$ 5 min, whereas for CD105 epitope retrieval, specimens were pre-treated with proteinase K (S3020, DAKO Cytomation) at room temperature for 15 min. Consecutive sections were successively incubated at 4°C overnight with the primary polyclonal antibodies against SEMA3A sc-10720 (Santa Cruz Biotechnology, Santa Cruz, CA; w.d. 1:25), VEGF sc152 (Santa Cruz Biotechnology, Santa Cruz, CA; w.d. 1:100) and with the monoclonal antibodies against NRP-1 sc5307 (Santa Cruz Biotechnology, Santa Cruz, CA; w.d. 1:100) and CD105 (DAKO Corporation, Denmark, clone SN6h, w.d. 1:50). The bound primary antibodies were visualized by avidin-biotin-peroxidase detection using the Vectastain Rabbit/Mouse Elite Kit, according to the manufacturer's instructions. To reveal the immunostaining, the sections were incubated in darkness for 10 min. with 3-3' diaminobenzidine tetra hydrochloride (Sigma Chemical Co., St. Louis, MO, USA), in the amount of 100 mg in 200 ml 0.03% hydrogen peroxide in phosphate-buffered saline solution (PBS). Nuclear counterstaining was performed by Mayer's haemalum. Specificity of the binding was assessed by omitting the primary antiserum or replacing it with normal rabbit serum or phosphate buffered saline solution (PBS, pH 7.4). Moreover, normal bronchial epithelial cells and renal tubules within specimens of lung and kidney were used as positive controls for SEMA3A, VEGF, NRP-1 immuno-reactions. The syncytiotrophoblast present in specimens of human term placenta was used as a positive control for CD105 immuno-assay.

Quantification

Staining scores for SEMA3A, VEGF and NRP-1 were established semi-quantitatively. Specifically, a score for each antibody was calculated by optical analysis using the product of the percentage of positive cells and staining intensity from 1 to 3

(1, weak; 2, moderate; and 3, strong), thus ranging from 0 to 300, in comparison with expression in the positive controls. The results of immunostaining were classified as negative (–) when the score was 0–40, weakly positive (+) when the score was 41–120, and strongly positive (++) when the score was 121–300. To analyze the balance between SEMA3A and VEGF, each VEGF staining score was divided by the SEMA3A score. The resulting VEGF/SEMA3A ratio was classified as low when it was 0–3 and high when it exceeded 3.

The quantification of microvessels was performed as previously described (Barresi et al. 2007, 2008). Briefly, the three most vascularized areas detected by CD105 were initially selected (so-called hot spots) under 40 $\times$ field. Then microvessels were counted in each of these areas under a 400 $\times$ field. Single endothelial cells or cluster of endothelial cells, with or without a lumen, were considered to be individual vessels. The mean value of three $\times$ 400 field (0.30 mm²) counts was recorded as the MVD of the section. Then the MVD value was converted into the mean number of microvessels/mm² for the statistical analyses.

Statistics

Statistical analysis was performed in order to assess the correlations between SEMA3A, VEGF, NRP-1 and MVD as well as between VEGF/SEMA3A ratio, MVD or SEMA3A, VEGF and NRP-1 scores and the clinico-pathological variables of the tumours. Specifically, the Spearmann, Fisher exact, Chi-squared Mann-Whitney and Kruskal-Wallis tests were carried out. The correlations between MVD, SEMA3A, VEGF or NRP-1 expression as well as VEGF/SEMA3A ratio of the tumours on recurrence-free survival were assessed by univariate (Kaplan–Meier) and multivariate (Cox regression) methods, with the date of primary surgery as the entry data. Follow-up ranged from 5 to 120 months. The end point for the recurrence-free survival analysis was the length of survival to the detection of a recurrent tumour.

Expression and Prognostic Value of SEMA3A and VEGF/SEMA3A Ratio in Meningiomas

SEMA3A immunohistochemical expression was identified in the cytoplasm of the neoplastic cells of meningiomas (Figs. 1.1 and 1.2) (Barresi et al. 2009; Barresi and Tuccari 2010). High expression of this protein was evidenced in tumors characterized by low MVD or by the absence of newly formed vessels stained by CD105 antibody (MVD=0), independently from their histological grade. On the other

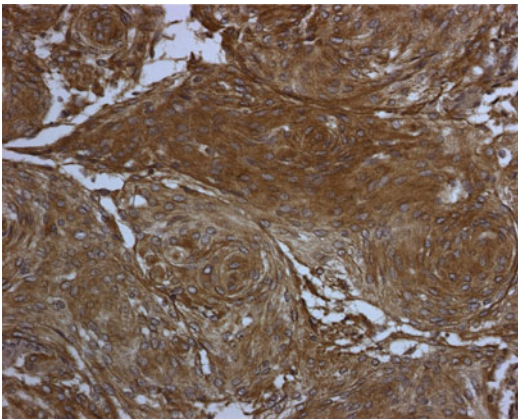


Fig. 1.1 SEMA3A immunohistochemical expression in a meningothelial meningioma (SEMA3A stain; original magnification, $\times 200$)

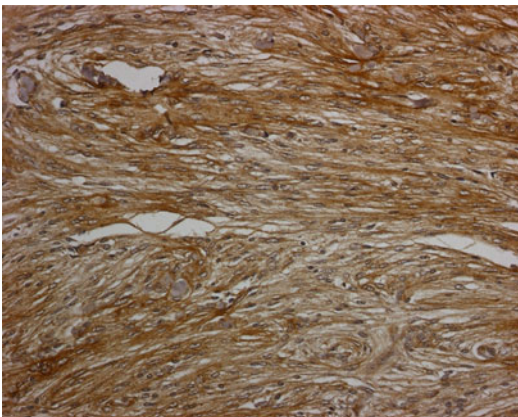


Fig. 1.2 SEMA3A immuno-expression in a transitional meningioma (SEMA3A stain; original magnification, $\times 200$)

hand, meningiomas characterized by high MVD, such as the microcystic, more frequently exhibited low or negative SEMA3A expression (Fig. 1.3). The inverse correlation between SEMA3A expression and MVD of the meningiomas was confirmed through statistical analysis ($P=0.0365$; $P=0.0118$; $r=-0.2948$; $P=0.0419$).

VEGF positive staining was found in the cytoplasm of neoplastic cells in half of the meningiomas, whereby NRP-1 positive staining was encountered in the endothelial cells of the vessels in all but two cases. VEGF expression and MVD were positively significantly correlated ($r=0.3062$; $P=0.0343$); nonetheless, some cases showing VEGF expression did not display any CD105 stained vessels (MVD=0). High SEMA3A immuno-expression was encountered in such meningiomas.

Although no significant correlations emerged between SEMA3A or VEGF expression and the histological grade or proliferation index of the meningiomas, statistical analyses revealed significant associations between high VEGF/SEMA3A ratio and high histological grade ($P=0.009$), proliferation index ($P=0.004$) as well as MVD ($P=0.001$). Univariate analyses showed that, aside from high histological grade, proliferation index and MVD, low SEMA3A expression ($P=0.0355$) and high VEGF/SEMA3A ratio ($P=0.0005$), but not VEGF

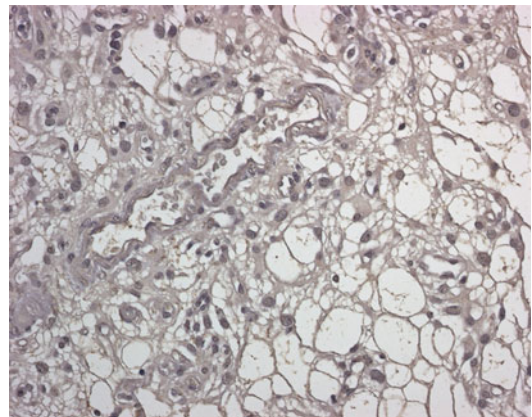


Fig. 1.3 Absence of SEMA3A immuno-staining in a microcystic meningioma (SEMA3A stain; original magnification, $\times 200$)

expression, were significant prognostic factors associated with a higher risk of recurrences in completely resected meningiomas.

Discussion

The quantity of neo-angiogenesis, reflected by the MVD, has been demonstrated as a prognostic parameter, correlated with higher biological aggressiveness, in several types of human neoplasias, including meningiomas (Saad et al. 2005; Ding et al. 2006; Barresi et al. 2007). Indeed, as long as a neoplastic mass is supplied by the only host vessels, it retains a limited volume and a low doubling time. By contrast, the so-called *angiogenic switch*, i.e. the formation of new capillaries from pre-existing ones, allows a more rapid growth and the progression of the tumour (Folkman 1995). The angiogenic switch maybe induced by the neoplastic cells through the secretion of pro-angiogenic factors and inhibited by anti-angiogenic ones. In meningiomas, neo-angiogenesis appears to be positively regulated by VEGF produced by the neoplastic cells; indeed the expression of this factor significantly correlates with the MVD of these neoplasms, as we also evidenced (Pistolesi et al. 2004; Barresi and Tuccari 2010). Nonetheless, in spite of the higher vascularity encountered in high grade tumors, VEGF expression does not significantly differ between meningiomas of benign histology and atypical or anaplastic meningiomas (Pietsch et al. 1997; Nishikawa et al. 1998; Pistolesi et al. 2004; Barresi and Tuccari 2010). Also, VEGF expression does not significantly correlates with the recurrence risk in these tumors (Maiuri et al. 2007; Barresi and Tuccari 2010).

It has been recently demonstrated that the pro-angiogenic activity of VEGF may be antagonized by the anti-angiogenic factor SEMA3A (Miao et al. 1999; Narazaki and Tosato 2006).

We demonstrated that human meningiomas express SEMA3A in their neoplastic cells (Barresi et al. 2009), similarly to other neoplasias such as gliomas (Rieger et al. 2003). Our findings suggest that in these tumors SEMA3A may exert an anti-angiogenic action (Barresi

et al. 2009), through its binding to NRP-1 which is expressed in the vessels of meningiomas (Barresi and Tuccari 2010). Consistently, we evidenced a significant inverse correlation between SEMA3A expression and MVD in meningiomas. Moreover, high SEMA3A immunohistochemical expression was encountered in the majority of cases with absent newly formed vessels (Barresi et al. 2009).

Interestingly, the amount of SEMA3A expression did not correlate with the histological grade of the meningiomas. Besides, analyzed microcystic meningiomas, which, in spite of their histological grade I, are highly vascularized tumors (Christov et al. 1999; Barresi et al. 2007), displayed negative SEMA3A and high VEGF immunohistochemical expression in association with intense neo-angiogenesis, as reflected by their high MVD (Barresi et al. 2009).

We found that some meningiomas did not present any newly formed vessels despite high VEGF expression (Barresi and Tuccari 2010); nonetheless, high SEMA3A expression was present in the same cases. We may speculate that VEGF pro-angiogenic activity is antagonized by SEMA3A presence in these cases; besides, it has been reported that the ratio of the concentration of VEGF and SEMA3A regulates the angiogenic switch in multiple myeloma (Vacca et al. 2006). Furthermore, our findings show that high VEGF/SEMA3A ratio is significantly associated with high MVD in meningiomas (Barresi and Tuccari 2010), in accordance with the hypothesis that ratio between VEGF and SEMA3A may regulate the degree of tumor angiogenesis in these neoplasias. The presence of its significant correlation with the histological grade and growth fraction suggests that VEGF/SEMA3A may also affect the growth of meningiomas. Indeed, apart from its pro-angiogenic action, VEGF has been also shown to directly influence the cell growth of several cancer cells (Masood et al. 2001). Furthermore, it has been reported that SEMA3A has a negative feedback loop that restrains VEGF signaling activity, and the abrogation of regulated SEMA3A expression is responsible for tumor cell proliferation stimulated by VEGF (Catalano et al. 2004). Therefore an altered balance between SEMA3A

and VEGF would induce tumor neo-angiogenesis, proliferation and progression.

Finally, according to our study (Barresi and Tuccari 2010), VEGF/SEMA3A ratio, but not the only VEGF expression, appears as a significant prognostic factor for the recurrence risk in those patients with completely macroscopically resected meningiomas. It has been suggested that the development of recurrences of totally removed (Simpson's grade I) meningiomas may be related to the presence of microscopic clusters of neoplastic cells left in the dura mater or in the arachnoid membrane (Kinjo et al. 1993; Kamitani et al. 2001) and we may hypothesize that this event depends upon the biological activity of these cells. Surgically not-removed neoplastic foci of these tumours may grow and give rise to recurrent tumours for their higher capability to stimulate proliferation and neo-angiogenesis in relationship to VEGF/SEMA3A balance in favour of VEGF.

In conclusion, on the basis of our results, we may hypothesize that neo-angiogenesis in meningiomas is regulated by an equilibrium in the concentrations of both VEGF and SEMA3A in the microenvironment of the tumour. Thus, neo-angiogenesis is blocked or stimulated depending on the prevalence of VEGF or SEMA3A with a high ratio between VEGF and SEMA3A as a negative predictor of recurrences in meningiomas. Our findings may open the perspective to the research of new therapeutic strategies based on VEGF inhibition and SEMA3A supplementation in order prevent recurrences of highly vascularized meningiomas.

References

- Barresi V, Tuccari G (2010) Increased ratio of vascular endothelial growth factor to semaphorin3A is a negative prognostic factor in human meningiomas. *Neuropathology* 30(5):537–546
- Barresi V, Cerasoli S, Vitarelli E, Tuccari G (2007) Density of microvessels positive for CD105 (endoglin) is related to prognosis in meningiomas. *Acta Neuropathol* 114:147–156
- Barresi V, Cerasoli S, Tuccari G (2008) Correlative evidence that tumor cell-derived caveolin-1 mediates angiogenesis in meningiomas. *Neuropathology* 28:472–478
- Barresi V, Vitarelli E, Cerasoli S (2009) Semaphorin3A immunohistochemical expression in human meningiomas: correlation with the microvessel density. *Virchows Arch* 454:563–571
- Catalano A, Caprai P, Rodilossi S, Betta P, Castellucci M, Casazza A, Tamagnone L, Procopio A (2004) Cross-talk between vascular endothelial growth factor and semaphorin-3A pathway in the regulation of normal and malignant mesothelial cell proliferation. *FASEB J* 18:358–360
- Christov C, Lechapt-Zalcman E, Adle-Biasette H, Nachev S, Gherardi RK (1999) Vascular permeability factor/vascular endothelial growth factor (VPF/VEGF) and its receptor flt-1 in microcystic meningiomas. *Acta Neuropathol* 98:414–420
- Ding S, Li C, Lin S, Liu D, Han Y, Zhang Y, Li L, Zhou L, Kumar S (2006) Comparative evaluation of microvessel density determined by CD34 or CD105 in benign and malignant gastric lesions. *Hum Pathol* 37:861–866
- Ferrara N, Davis-Smyth T (1997) The biology of vascular endothelial growth factor. *Endocr Rev* 18:4–25
- Folkman J (1995) Clinical applications of angiogenic research. *N Engl J Med* 333:1757–1763
- Folkman J, Shing Y (1992) Angiogenesis. *J Biol Chem* 267:10931–10934
- Guttmann-Raviv N, Shraga-Heled N, Varshavsky A, Guimaraes-Sternberg C, Kessler O, Neufeld G (2007) Semaphorin-3A and semaphoring-3F work together to repel endothelial cells and to inhibit their survival by induction of apoptosis. *J Biol Chem* 282:26294–26305
- Kamitani H, Masuzawa H, Kanazawa I, Kubo T (2001) Recurrence of convexity meningiomas: tumour cells in the arachnoid membrane. *Surg Neurol* 56:228–235
- Kinjo T, al-Mefty O, Kanaan I (1993) Grade zero removal of supratentorial convexity meningiomas. *Neurosurgery* 33:394–399
- Luo Y, Raible D, Raper JA (1993) Collapsin: a protein in brain that induces the collapse and paralysis of neuronal growth cones. *Cell* 75:217–227
- Maiuri F, De Caro M, Esposito F, Cappabianca P, Strazzullo V, Pettinato G, de Divitiis E (2007) Recurrences of meningiomas: predictive value of pathological features and hormonal and growth factors. *J Neurooncol* 82:63–68
- Masood R, Cai J, Zheng T, Smith DL, Hinton DR, Gill PS (2001) Vascular endothelial growth factor (VEGF) is an autocrine growth factor for VEGF receptor-positive human tumors. *Blood* 98:1904–1913
- Miao HQ, Soker S, Feiner L, Alonso JL, Raper JA, Klagsbrun M (1999) Neuropilin-1 mediates collapsin-1/semaphorin III inhibition of endothelial cell motility: functional competition of collapsin-1 and vascular endothelial growth factor-165. *J Cell Biol* 12:233–242
- Narazaki M, Tosato G (2006) Ligand-induced internalization selects use of common receptor neuropilin-1 by VEGF165 and semaphorin3A. *Blood* 107:3892–3901

- Nishikawa R, Cheng SY, Nagashima R, Huang HJ, Cavenee WK, Matsutani M (1998) Expression of vascular endothelial growth factor in human brain tumors. *Acta Neuropathol* 96:453–462
- Osada R, Horiuchi A, Kikuchi N, Ohira S, Ota M, Katsuyama Y, Konishi I (2006) Expression of semaphorins, vascular endothelial growth factor and their common receptor neuropilins and allelic loss of semaphorin locus in epithelial ovarian neoplasms: increased ratio of vascular endothelial growth factor to semaphorin is a poor prognostic factor in ovarian carcinomas. *Hum Pathol* 37:1414–1425
- Perry A, Stafford SL, Scheithauer BW, Lohse CM, Wollan PC (1999) “Malignancy” in meningiomas: a clinicopathological study of 116 patients with grading implications. *Cancer* 85:2046–2056
- Perry A, Louis DN, Scheithauer BW, Budka H, von Deimling A (2007) Meningiomas. In: Louis DN, Ohgaki H, Wiestler OD, Cavenee WK (eds) WHO classification of tumors of the central nervous system. IARCC Press, Lyon
- Pietsch T, Valter MM, Wolf HK (1997) Expression and distribution of vascular endothelial growth factor protein in human brain tumors. *Acta Neuropathol* 93:109–117
- Pistolesi S, Boldrini L, Gisfredi S, De Ieso K, Casacci T, Caniglia M, Lupi G, Leocata P, Basolo F, Pingitore R, Parenti G, Fontanini G (2004) Angiogenesis in intracranial meningiomas: immunohistochemical and molecular study. *Neuropathol Appl Neurobiol* 30:118–125
- Rieger J, Wick W, Weller M (2003) Human malignant glioma cells express semaphorins and their receptors, neuropilins and plexins. *Glia* 42:379–389
- Saad RS, El-Gohary Y, Memari E, Liu YL, Silverman JF (2005) Endoglin (CD105) and vascular endothelial growth factor as prognostic markers in esophageal adenocarcinoma. *Hum Pathol* 36:955–961
- Simpson D (1957) The recurrence of intracranial meningiomas after surgical treatment. *J Neurol Neurosurg Psychiatry* 20:22–39
- Soker S, Takashima S, Miao HQ, Neufeld G, Klagsbrun M (1998) Neuropilin-1 is expressed by endothelial and tumor cells as an isoform specific receptor for vascular endothelial growth factor. *Cell* 92:735–745
- Soker S, Miao HQ, Nomi M, Takashima S, Klagsbrun M (2002) VEGF165 mediates formation of complexes containing VEGFR-2 and neuropilin-1 that enhance VEGF165 receptor binding. *J Biol Chem* 277:357–368
- Vacca A, Scavelli C, Serini G, Di Pietro G, Citrulli T, Merchionne F, Ribatti D, Bussolino F, Guidolin D, Piaggio G, Bacigalupo A, Dammacco F (2006) Loss of inhibitory semaphorin3A (sema3A) autocrine loops in bone marrow endothelial cells of patients with multiple myeloma. *Blood* 108:1661–1667

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Abstract

Carbonic anhydrase (CA) II belongs to the family of metalloenzymes that catalyze the reversible hydration of carbon dioxide. It is widely distributed in normal organs and it is the main isoenzyme in the nervous system. In addition, CA II is expressed in several malignant tumors. Recently, CA II expression has been located in tumor vessel endothelia, suggesting a role in tumor angiogenesis. Only minority of meningiomas express CA II but expression is associated with cell proliferation and tumor malignancy grade. Thus, CA II expression may be associated with malignant progression of meningiomas.

Introduction

Meningiomas are common brain tumors accounting for up to 25% of primary brain tumors (Claus et al. 2005). Most meningiomas are benign. Atypical meningiomas (grade II) constitute about 5% of meningiomas and malignant (grade III) meningiomas 3–5%. About 90% of meningiomas occur intracranially and most of the remainder occur in spine. Incidence of meningiomas increases with age and they are more often diagnosed in women than in men, occurring at a ratio of 2:1 (Klaeboe et al. 2005). Etiology of

meningiomas is still largely unknown. There is a strong epidemiologic evidence linking ionizing radiation and meningioma (Ron et al. 1988). Surgery is still the treatment of choice for most meningiomas and in most cases it is curative. However, complete resection of meningioma is not always possible in difficult locations and despite of meningioma's benign nature, it has a tendency to recur (Marks et al. 1986). Radiation is the only adjuvant therapy for patients with malignant meningiomas, chemotherapy is seldom used. Chemotherapy is mainly applied in inoperable cases with tumor progression after radiotherapy and it has not been particularly effective even though modest activity has been shown in some subgroups of patients (Chamberlain 1996; Muhr et al. 2001). The lack of adjuvant therapy for malignant and recurring meningiomas has led to search of alternative treatment modalities. Research has mainly focused on sex

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hormone receptors. Treatment with antiestrogenic agents, as Tamoxifen, has not been effective (Schrell et al. 1990). The antiprogestosterone agent mifepristone inhibits growth of meningioma cells in vitro (Matsuda et al. 1994) but has failed to demonstrate clinical treatment effect (Grunberg et al. 2001). A large and growing body of literature has investigated carbonic anhydrases (CA) and their inhibitors. CA inhibitors already have wide clinical use as diuretics, antiepileptics and antiglaucoma drugs. Recently it has been shown that CA inhibitors also have potential as anti-tumour, anti-obesity, and anti-infective drugs (Pastorekova et al. 2004; Thiry et al. 2006).

Carbonic anhydrases are involved in numerous physiological and pathological processes. They belong to the family of zinc metalloenzymes. CAs catalyze the reversible conversion of carbon dioxide to bicarbonate and proton, and have an important role in regulating acid-base balance. The 15 human CA isoforms differ in subcellular location, tissue distribution and expression levels (reviewed in Supuran 2008). CA II is a cytosolic enzyme widely distributed in human tissues and predominantly expressed in differentiated cells.

Tissue Distribution of CA II

CA II is practically found in all human organs and is the main isoenzyme in the nervous system. In adult human brain CA II is expressed in oligodendrocytes, subset of neurons, few astrocytes, myelin, and choroid plexus epithelium. CA II is present transiently during brain development in the endothelium cells of micro vessels suggesting that it may be involved in assisting the regulation of intracellular pH in proliferating and differentiating endothelial cells (Kida et al. 2006). In addition, CA II expression has been detected in erythrocytes, eye, GI tract, bone osteoclasts, kidney, lung and testis (Supuran 2008). Interestingly, CA II expression has been recently detected in endothelium of neovessels in certain tumors (Yoshiura et al. 2005). On the other hand, deficiency of CA II is the main defect in recessive inherited osteopetrosis with renal tubular acidosis and cerebral calcification (Sly and Hu 1995).

In human cancer, CA II is found in malignant brain tumours, lung cancer cells, pancreatic and gastric carcinomas (in particular GIST) and in majority of leukaemias (Parkkila et al. 2010). In colorectal and lung cancer cells the expression of CA II decreases with progression to malignancy (Chiang et al. 2002; Kivelä et al. 2001). In contrast, astrocytomas, oligodendrogliomas and leukemias display high CA II expression (Haapasalo et al. 2006, 2007; Leppilampi et al. 2002). Apart from cancers, abnormal levels of CA II have been detected in other diseases, including glaucoma and several neurological disorders (Supuran 2008).

Function of CA II

Carbonic anhydrases catalyze the reversible reaction $\text{CO}_2 + \text{H}_2\text{O} \rightleftharpoons \text{HCO}_3^- + \text{H}^+$. Modulation of carbonic anhydrase activity provides a method to regulate the rate of HCO_3^- transport which affects the regulation of intracellular pH. The cytosolic pH must be maintained within the physiological range that is needed for biochemical processes. CA II is one of the most effective enzymes with the turnover rate of $1.3\text{--}1.9 \times 10^6/\text{s}$ (Sanyal and Maren 1981). CA II takes part in the control of bone resorption, respiration, formation of CSF, and regulates the acidity of the bile, pancreatic and gastric juice, and saliva (Supuran 2008; Swenson 1991).

Interestingly, expression of CA II was found to be upregulated by anoxic stress in CNS (Nogradi et al. 2003) possibly as a compensatory mechanism protecting CNS from pathological changes. Anoxia/hypoxia is also typical for growing tumors where it causes shift from oxidative phosphorylation to glycolysis to generate acidosis. Tumor hypoxia was also shown to be endogenous feature of meningiomas associated with an aggressive phenotype (Yoo et al. 2007). Hypoxia in tumor microenvironment triggers adaptive responses including angiogenesis, cell proliferation, and tissue invasion (Harris 2002). Yoshiura et al. (2005) investigated CA II mRNA expression in three-dimensional culture of human umbilical vein endothelial cells. Compared with the two-dimensional culture, CA II was up-regulated in three-dimensional culture, suggesting it is

needed for angiogenesis in normal endothelial cells. When the culture was exposed to hypoxic conditions, CA II was again significantly up-regulated. Thus, CA II may have a role in tumor growth and invasion through neovascularization.

Expression of CA II in Meningiomas

CA II is the most widely expressed CA isoenzyme in the central nervous system. Parkkila et al. (1995) studied the expression of CA II in several different brain tumours. Positive CA II staining was detected in astrocytomas, oligodendrogliomas and medulloblastomas. Meningiomas and neuronal tumors were negative. Later, expression of CA II in the endothelium of astrocytomas and oligodendrogliomas has been confirmed. CA expression was also significantly associated with poor prognosis in these tumors (Haapasalo et al. 2007). There is very little information concerning CA II and meningiomas. In a recent study, CA II was found to be expressed in approximately 15% of meningiomas (Korhonen et al. 2009). Expression varied among different histological subtypes of meningioma. It was more abundant in meningothelial (15.9%) and transitional (15.2%) meningiomas compared to fibrous (4.3%) subtype. Primary and recurrent meningiomas did not differ in their expression level of CA II but tumour grade was correlated with CA II expression. Expression of CA II was significantly higher in atypical and malignant meningiomas (30.6%) compared to benign meningiomas (13%). The proliferation index of CA II positive meningiomas was also found to be higher than of CA II negative meningiomas. The analysis by sex hormone receptor status revealed that androgen receptor negative meningiomas expressed CA II over twice as often as androgen receptor positive meningiomas. Progesterone or estrogen receptor status did not show any relationship to CA II expression.

CA II as Therapeutic Target Molecule

CA II is a possible target molecule for therapeutic interventions. CA II is expressed in the cytoplasm of several different tumours. It is found only in

minority of meningiomas but the finding that CA II positivity is associated with higher cell proliferation rate suggests that CA II may be involved in malignant progression of meningioma cells (Korhonen et al. 2009). Only a small percentage of meningiomas are atypical or malignant but CA II expression seems to be quite common in these tumors which also often are highly vascular. The recent discovery of CA II expression in the endothelium of neovessels of some tumour cells as melanoma, lung cancer and now also meningioma, suggests that CA II has a critical role in tumor angiogenesis (Yoshiura et al. 2005). It has been proposed that endothelial CA II expression could generate an autoantibody response which could be used in cancer immunotherapy.

CA inhibitors have been shown to inhibit growth and invasion of several different cancer types and tumor growth in vivo (Parkkila et al. 2000; Supuran et al. 2001; Ahlskog et al. 2009). The mechanism of inhibition is not yet known but may involve deterioration of tumor angiogenesis. Much research is done presently on CA inhibitors. Anticancer drug research has been mostly targeting CA IX and CA XII because they are overexpressed in several types of human tumors, whereas CA II has been a probable target for antiglaucoma drugs and anticonvulsants. Inhibition of CA II could, in theory, compromise invasion of tumor cells by prohibiting angiogenesis. Overall, research on CA inhibitors is at an early stage and wide distribution of CA II in normal tissues may limit their use in cancer therapy unless they are specifically targeted to tumor tissues or modified to inhibit only certain tumor-relevant isoenzymes.

References

- Ahlskog JK, Dumelin CE, Trüssel S, Märlind J, Neri D (2009) In vivo targeting of tumor-associated carbonic anhydrases using acetazolamide derivatives. *Bioorg Med Chem Lett* 19:4851–4856
- Chamberlain MC (1996) Adjuvant combined modality therapy for malignant meningiomas. *J Neurosurg* 84:733–736
- Chiang WL, Chu SC, Yang SS, Li MC, Lai JC, Yang SF, Chiou HL, Hsieh YS (2002) The aberrant expression of cytosolic carbonic anhydrase and its clinical significance in human non-small cell lung cancer. *Cancer Lett* 188:199–205

- Claus EB, Bondy ML, Schildkraut JM, Wiemels JL, Wrensch M, Black PM (2005) Epidemiology of intracranial meningioma. *Neurosurgery* 57:1088–1095
- Grunberg SM, Rankin C, Townsend J, Ahmadi J, Feun L, Fredericks L, Russell C, Kabbavar F, Barger GR, Stelzern KJ, Southwest Oncology Group, Burlington, VT, Southwest Oncology Group, San Antonio, TX, Eastern Cooperative Oncology Group, Brookline, MA (2001) Phase III double-blind randomized placebo-controlled study of mifepristone (RU) for the treatment of unresectable meningioma. *Proc ASCO* 20:222
- Haapasalo JA, Nordfors KM, Hilvo M, Rantala IJ, Soini Y, Parkkila AK, Pastoreková S, Pastorek J, Parkkila SM, Haapasalo HK (2006) Expression of carbonic anhydrase IX in astrocytic tumors predicts poor prognosis. *Clin Cancer Res* 12:473–477
- Haapasalo J, Nordfors KM, Järvelä S, Bragge H, Rantala I, Parkkila AK, Haapasalo H, Parkkila S (2007) Carbonic anhydrase II in the endothelium of glial tumors: a potential target for therapy. *Neuro Oncol* 3:308–313
- Harris AL (2002) Hypoxia- a key regulatory factor in tumour growth. *Nat Rev Cancer* 2:38–47
- Kida E, Palminiello S, Golabek A, Walus M, Wierzbobrowicz T, Ausma R, Albertini G, Wisniewski K (2006) Carbonic anhydrase II in the developing and adult human brain. *J Neuropathol Exp Neurol* 65:664–674
- Kivelä AJ, Saarnio J, Karttunen TJ, Kivelä J, Parkkila AK, Pastorekova S, Pastorek J, Waheed A, Sly WS, Parkkila TS, Rajaniemi H (2001) Differential expression of cytoplasm carbonic anhydrases, CA I and II, and membrane-associated isozymes, CA IX and XII, in normal mucosa of large intestine and in colorectal tumors. *Dig Dis Sci* 46:2179–2186
- Klaeboe L, Lonn S, Scheie D, Auvinen A, Christensen HC, Feychting M, Johansen C, Salminen T, Tynes T (2005) Incidence of intracranial meningiomas Denmark, Finland, Norway and Sweden 1968–1997. *Int J Cancer* 117:996–1001
- Korhonen K, Parkkila A-K, Helen P, Välimäki R, Pastorekova S, Pastorek J, Parkkila S, Haapasalo H (2009) Carbonic anhydrases in meningiomas: association of endothelial carbonic anhydrase II with aggressive tumor features. *J Neurosurg* 111:472–477
- Leppilampi M, Koistinen P, Savolainen ER (2002) The expression of carbonic anhydrase II in hematologic malignancies. *Clin Cancer Res* 8:2240–2245
- Marks SM, Whitewell HL, Lye RH (1986) Recurrence of meningiomas after operation. *Surg Neurol* 25:436–440
- Matsuda Y, Kawamoto K, Kiya K, Kurisu K, Sugiyama K, Uozumi T (1994) Antitumor effects of antiprogesterones on human meningioma cells in vitro and in vivo. *J Neurosurg* 80:527–534
- Muhr C, Gudjonsson O, Lilja A, Hartman M, Zhang ZJ, Långström B (2001) Meningioma treated with interferon-alpha, evaluated with [(11)C]-L-methionine positron emission tomography. *Clin Cancer Res* 7:2269–2276
- Nógrádi A, Domoki F, Dégi R, Borda S, Pákási M, Szabó A, Bari F (2003) Up-regulation of cerebral carbonic anhydrase by anoxic stress in piglets. *J Neurochem* 85:843–850
- Parkkila AK, Herva R, Parkkila S, Rajaniemi H (1995) Immunohistochemical demonstration of human carbonic anhydrase isoenzyme II in brain tumours. *Histochem J* 27:974–982
- Parkkila S, Rajaniemi H, Parkkila AK, Kivelä J, Waheed A, Pastorekova S, Pastorek J, Sly WS (2000) Carbonic anhydrase inhibitor suppresses invasion of renal cancer cells in vitro. *Proc Natl Acad Sci USA* 97:2220–2224
- Parkkila S, Lasota J, Fletcher JA, Ou WB, Kivelä AJ, Nuorva K, Parkkila AK, Ollikainen J, Sly WS, Waheed A, Pastorekova S, Pastorek J, Isola J, Miettinen M (2010) Carbonic anhydrase II. A novel biomarker for gastrointestinal stromal tumors. *Mod Pathol* 23(5):743–750
- Pastorekova S, Parkkila S, Pastorek J, Supuran CT (2004) Carbonic anhydrases: current state of the art, therapeutic applications and future prospects. *J Enzyme Inhib Med Chem* 19:199–229
- Ron E, Modan B, Boice JD Jr, Alfandary E, Stovall M, Chetrit A, Katz L (1988) Tumors of the brain and nervous system after radiotherapy in childhood. *N Engl J Med* 319:1033–1039
- Sanyal G, Maren TH (1981) Thermodynamics of carbonic anhydrase catalysis. A comparison between human isoenzymes B and C. *J Biol Chem* 256:608–612
- Schrell UM, Adams EF, Fahlbusch R, Greb R, Jirikowski G, Prior R, Ramalho-Ortigao FJ (1990) Hormonal dependency of meningiomas. Part I: female sex steroid receptors and their significance as specific markers for adjuvant medical therapy. *J Neurosurg* 73:743–749
- Sly WS, Hu PY (1995) Human carbonic anhydrases and carbonic anhydrase deficiencies. *Annu Rev Biochem* 64:375–401
- Supuran CT (2008) Carbonic anhydrases: novel therapeutic applications for inhibitors and activators. *Nat Rev Drug Discov* 7:168–181
- Supuran CT, Briganti F, Tilli S, Chegwiddden WR, Scozzafava A (2001) Carbonic anhydrase inhibitors: sulfonamides as antitumor agents? *Bioorg Med Chem* 9:703–714
- Swenson ER (1991) Distribution and functions of carbonic anhydrase in the gastrointestinal tract. In: Dodgson SJ, Tashian RE, Gros G, Carter ND (eds) *The carbonic anhydrases. Cellular physiology and molecular genetics*. Plenum Press, New York, pp 265–288
- Thiry A, Dogné JM, Masereel B, Supuran CT (2006) Targeting tumor-associated carbonic anhydrase IX in cancer therapy. *Trends Pharmacol Sci* 27:566–573
- Yoo H, Baia G, Smith J, McDermott M, Bollen A, Vanden BS, Lamborn K, Lal A (2007) Expression of the

- hypoxia marker carbonic anhydrase 9 is associated with anaplastic phenotypes in meningiomas. *Clin Cancer Res* 13:68–75
- Yoshiura K, Nakaoka T, Nishishita T, Sato K, Yamamoto A, Shimada S, Saida T, Kawakami Y, Takahashi TA, Fukuda H, Imajoh-Ohmi S, Oyaizu N, Yamashita N (2005) Carbonic anhydrase II is a tumor vessel endothelium-associated antigen targeted by dendritic cell therapy. *Clin Cancer Res* 11:8201–8207

Meningiomas: Role of Genetic Instabilities of the E-cadherin Gene

3

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Abstract

The numerous reports by many authors as well as our own results indicate that E-cadherin plays a role in CNS tumors – meningiomas. The suppression of E-cadherin expression is regarded as one of the main molecular events responsible for dysfunction in cell-cell adhesion. It is now apparent that downregulation of E-cadherin expression is involved in meningioma development. Our studies on meningiomas of different grades showed that 73% of meningiomas had downregulation of E-cadherin expression. Intense downregulation was noticed in tumors with grades II and III. Our analysis on the genetic level of E-cadherin changes showed loss of heterozygosity (LOH) in 32% meningiomas investigated. Also, another type of genomic instability was detected in 11% of our sample – MSI phenotype. The instability is the result of impaired cellular mismatch repair. Beta-catenin is the main effector molecule in wnt signaling and is bound to E-cadherin in adherens junctions. Beta-catenin was progressively upregulated from meningothelial to atypical, while 60% of anaplastic showed upregulation and nuclear localization of the protein. A significant association between the genetic changes of CDH1 and the nuclear localization of beta-catenin protein ($P < 0.022$) was established in our study. Better understanding of meningioma genetic profile will elucidate pathophysiological mechanisms of tumor formation, and eventually offer improved prognostic markers for meningioma.

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Introduction

Meningiomas account for approximately 25% of primary intracranial and intraspinal neoplasms originating from the meningeal coverings of the

brain and the spinal cord. Meningiomas show predominance in women, with a female to male ratio of approximately 2 : 1 for intracranial, and 10 : 1 for spinal. The incidence of meningioma is estimated to be 1 per 30,000–40,000 persons in the USA with incidence rates increasing with age (Claus et al. 2005). The risk factors for meningioma development are largely unknown, there are, however, two known risk factors – exposure to ionizing radiation and hormones. The mechanisms of their action are still unexplored.

The morphological features suggest that meningiomas are derived from arachnoidal (meningothelial) cells. The majority of meningiomas corresponds to grade I of WHO classification of CNS tumors and thus are benign, slowly growing tumors (Riemenschneider et al. 2006; Louis et al. 2007). Within the benign category there are several subtypes, including meningothelial, fibrous (fibroblastic), transitional (mixed), psammomatous and angiomatous meningiomas. Yet, these classifications are imprecise with respect to prediction of patient outcome, recurrence or response to treatment. Meningiomas associated with less favorable clinical outcome correspond to grade II (atypical) and those who will exhibit features of malignant behavior – to grade III (anaplastic). Understanding the genetic basis and molecular etiology of meningioma is essential for clinical phenotype determination as well as patient outcome.

The knowledge on genetic susceptibility for meningioma has come from studies of rare genetic syndromes. Besides many other syndromes, meningiomas are a principal feature of neurofibromatosis type 2 (NF2), a rare autosomal dominant disorder caused by germline mutation in the NF2 gene on 22q12. Loss of expression of NF2 protein product merlin (schwannomin) is consistent finding in all NF2 associated meningiomas and in about half of sporadic cases (Fuller and Perry 2005). The merlin (schwannomin) is also known as moesin-, ezrin-, radixin-like protein, being a part of the band 4.1 families of cytoskeleton-associated proteins (Fuller and Perry 2005; Ragel and Jensen 2005). The consequence of the loss of merlin protein is the decreased cell adhesion and tumorigenesis.

Merlin behaves as a tumor suppressor protein and it has been shown to interact with proteins constituting cadherin mediated adherens junctions. The majority of meningiomas suffer losses on 22q including loss of heterozygosity of the neurofibromatosis type 2 gene. Up to 60% of meningiomas carry inactivating mutations in the remaining NF2 allele. However, many meningiomas are sporadic and of unknown etiology. In approximately 60% of sporadic meningiomas, the NF2 gene is inactivated by a small mutation, and this is most frequently accompanied by loss of the second allele, usually reflected by loss of the entire chromosome 22 (Ragel and Jensen 2005; Barnholtz-Sloan and Kruchko 2007; Simon et al. 2007). The remaining percent (40%) of sporadic meningioma failed to show the aberrations of chromosome 22, nor the mutations in the NF2 gene. This discrepancy suggests that an alternative pathogenetic mechanism is responsible for the development of these tumors. The gene(s) responsible for the meningiomas with unaffected NF2 gene remains unknown. Another point in favor of additional genetic pathways existence is demonstrated in families with an increased susceptibility to meningiomas but without features of neurofibromatosis type 2. Scientists identified families with hereditary meningiomas without linkage to the NF2 locus (Barnholtz-Sloan and Kruchko 2007).

The cytogenetic alterations found in meningioma, besides abnormalities in the 22q locus, are the deletion of the short arm of chromosome 1, loss of chromosomes 6, 10, 14, 18 and 19 (Simon et al. 2007) and gains of chromosomes 1q, 9q, 12q, 15q, 17q and 20q of which many associated with tumor grade (Barnholtz-Sloan and Kruchko 2007). The possibility that other meningioma genes reside on chromosome 22 is also taken into account today, mainly because it has been demonstrated that the frequency of LOH of chromosome 22 exceeds that of NF2 gene abnormalities. Moreover, investigations on deletions of chromosome 22 have detected losses outside the genetic region of NF2 gene.

A number of investigations have studied specific genes and their role in meningioma, including genes involved in cell cycle regulation,

signaling pathways and DNA repair. Candidate genes include *LARGE*, *MNI*, *BAM22* and *INII*. *DALI* is also implicated, as well as *CDKN2A* which is associated to the anaplastic histology (Ragel and Jensen 2005; Simon et al. 2007). In addition, the changes of the well known TP53 gene showed association to meningioma as well as enhanced expression of the *sis*, *myc*, *ras* and *mos* oncogenes.

Although many molecular studies have examined genetic alterations of tumor suppressor genes and oncogenes associated with meningioma tumorigenesis and progression (Fuller and Perry 2005; Ragel and Jensen 2005; Riemenschneider et al. 2006; Simon et al. 2007), the molecular basis of etiology and pathogenesis still needs to be elucidated.

E-cadherin

E-cadherin in Normal and Tumor Cells

Novel research including our own studies demonstrates the involvement of E-cadherin in the genesis of meningiomas. The role of E-cadherin in normal cell presents this molecule as one of the most important in cell-cell adhesion. It is localized on the surfaces of cells in regions of cell-cell contact known as adherens junctions. As a member of a large family of genes coding for calcium-dependent cell adhesion molecules (CAMs), the cadherin glycoproteins are expressed by a variety of tissues, mediating adhesion through homotypic binding. Classical cadherins – E- and N-cadherins being the best characterized – play important roles in the formation of tissues during gastrulation, neurulation and organogenesis (Pećina-Šlaus 2003).

The human epithelial (E)-cadherin gene *CDH1* maps to chromosome 16q22.1. The gene that Berx et al. (1995) cloned in 1995. encompasses 16 exons and spans a region of ~100 kb. The exons range from 115 to 2,245 bp. Further analysis of the gene showed 15 introns ranging from 120 bp to 65 kb. The chromosomal location of *CDH1* on 16q22.1 was later confirmed by fluorescent in situ hybridization (FISH) analysis. The intron-exon

boundaries are highly conserved in comparison with other “classical cadherins”, and in intron 1 a 5' high-density CpG island was identified that may have a role in transcription regulation. This island covers the region from exon 1 to exon 2 of the human E-cadherin gene, while other exons lacked such features, including the biggest (exon 16 of 2,245 bp). *CDH1* encodes a 120 kDa glycoprotein with a large extra-cellular domain, a single transmembrane segment and a short cytoplasmic domain, which interacts with the actin cytoskeleton through linker molecules, alpha-, beta- and gamma-catenins. On the cytoplasmic side of the membrane, a bundle of actin filaments is linked to the E-cadherin molecules via a protein complex. Alpha-catenin and either beta- or gamma-catenins are included in this complex. Beta- and gamma-catenins share significant homology and bind to a specific domain at the E-cadherin C-terminus. Alpha-catenin links the bound beta- or gamma-catenin to the actin cytoskeleton. The C-terminal cytoplasmic domain of ~150 residues is highly conserved in sequence, and has been shown experimentally to regulate the cell-cell binding function of the extracellular domain of E-cadherin, possibly through interaction with the cytoskeleton. The juxtamembrane region of the cadherin cytoplasmic tail has been identified as a functionally active region supporting cadherin clustering and adhesive strength; one of the interacting proteins involved in clustering and cell adhesion is p120ctn. The structure of the extracellular domain of classical E-cadherin contains five tandem repeats of a 100-residue-amino-acid-motif, and the biggest part of N-terminal of these repeats contains the sites with adhesive activity. This part of the molecule also has binding sites for calcium ions situated in the pockets between the repeats. The amino acid sequences that form the Ca^{2+} binding pockets are highly conserved between different members of the cadherin family and between different species. Cell-cell adhesion is mediated through homotypic interactions of E-cadherin extracellular domains in a process of lateral dimerization. Parallel dimers are able to interdigitate with dimers from neighboring cells forming the points of adhesion.

Besides its role in normal cells, this highly conserved gene can play a major role in malignant cell transformation, and especially in tumor development and progression. The suppression of E-cadherin expression is regarded as one of the main molecular events responsible for dysfunction in cell-cell adhesion. Malignant cells are characterized in general by poor intercellular adhesion, loss of the differentiated morphology and increased cellular motility. Downregulation or a complete shutdown of E-cadherin expression, mutation of the E-cadherin gene, or other mechanisms that interfere with the integrity of the adherens junctions, are observed in carcinoma cells. In human tumors, the loss of E-cadherin-mediated cell adhesion correlates with the loss of the cellular morphology and with the acquisition of metastatic potential (Guilford 1999). Thus, a tumor invasion/suppressor role has been assigned to this gene. Additional data also support this role. Mutations in CDH1 have been described in a number of human cancers, transgenic mouse model with loss of E-cadherin expression developed invasive carcinoma from well-differentiated adenomas and finally germ-line mutations have been reported in diffuse-type stomach cancers.

E-cadherin and Wnt Signaling

It has been well documented that wnt genes and other components of wnt signaling are critical for mammalian embryogenesis (Li et al. 2006; Pečina-Šlaus 2010). Wnt signaling is essential in development because it acts as a regulator of the embryonic cell patterning, proliferation, differentiation, cell adhesion, cell survival and apoptosis. It is especially important in the development of the central nervous system because processes that include synaptic rearrangements require the expression of molecular components of the wnt pathway (Patapoutian and Reichardt 2000). The pathway regulates the normal development of the neural plate and neural tube, and later of the brain, spinal cord, and numerous sensory and motor neurons (Li et al. 2006). In addition to neural tissues, wnt pathway is also critical for sound vascular and cardiac systems development.

Furthermore, wnt signaling modulates most aspects of osteoblast physiology, including bone cell apoptosis (Pečina-Šlaus 2010). Furthermore, malfunctioning of this pathway in adult organism is responsible for tumorigenesis of many different tissues, including brain tumorigenesis (Patapoutian and Reichardt 2000; Koch et al. 2001; Howng et al. 2002).

The wnt/wingless pathway was first discovered in mouse and *Drosophila* and is one of the most interesting signal transductions, in which key components have multiple functions. In vertebrate cells, it is named after Wnt proteins, a family of highly conserved secreted signaling molecules that regulate cell-to-cell interactions during embryogenesis. Insights into the mechanisms of Wnt action have emerged from several systems: genetics in *Drosophila* and *Caenorhabditis elegans*; biochemistry in cell culture; and ectopic gene expression in *Xenopus* embryos. Many wnt genes in the mouse have been mutated, leading to very specific developmental defects. As currently understood, wnt proteins bind to receptors of the Frizzled family on the cell surface. Through several cytoplasmic relay components, the signal is transduced to beta-catenin, which then enters the nucleus to activate transcription of Wnt target genes (Pečina-Šlaus 2010). Although the main signaling molecule of the pathway is beta-catenin, APC is a critical component in the formation of multiprotein complex with axin/axin2, casein kinase I and glycogen synthase kinase 3-beta (GSK3-beta). Beta-catenin is recruited to this complex, phosphorylated, ubiquitinated and headed to the proteasome. When wnt ligand is absent, beta-catenin is being destroyed. In response to wnt signaling, or under the circumstances of mutated APC, beta-catenin is stabilized, accumulates in the cytoplasm and enters the nucleus, where it finds a partner, a member of the DNA binding protein family LEF/TCF (lymphoid enhancer factor-T cell factor). Together they activate new gene expression programs. Target genes for beta-catenin/TCF encode c-MYC and cyclin D1 proteins, explaining why constitutive activation of the wnt pathway can lead to cancer.

Besides its role in cellular architecture, E-cadherin has a role in wnt signaling too. E-cadherin is an indirect modulator of wnt signaling. Since it binds to and sequesters cytoplasmic beta-catenin it is involved, in such a fashion, in the modulation of the signal.

Changes of E-cadherin in Meningiomas

Our interest in investigating E-cadherin in meningiomas stemmed from several findings. First, classical cadherins such as E-type and N-type are involved in forming both adherens and synaptic junctions in the nervous system. Moreover, E-cadherin is transiently expressed in restricted regions of the human and mouse brain. New knowledge on wnt signaling shows that wnt proteins regulate critical developmental processes of normal brain development (Patapoutian and Reichardt 2000; Li et al. 2006). It has been shown that wnt signaling dysregulation plays an essential role in brain tumorigenesis (Koch et al. 2001; Howng et al. 2002; Nikuševa Martić et al. 2010). All these findings suggest that molecular components of wnt signaling perform important functions in CNS tissues.

Loss of Heterozygosity

We investigated loss of heterozygosity of E-cadherin (CDH1) gene in a sample of 60 meningiomas together with autologous blood tissues collected from the Department of Neurosurgery and Department of Pathology Ljudevit Jurak University Hospital “Sisters of Charity”, Zagreb, Croatia. Our methodology included magnetic resonance imaging (MRI) which detected tumor lesions in different cerebral regions. During the operative procedure the tumor was maximally reduced using a microneurosurgical technique. The patients had no family history of brain tumors and did not undergo chemotherapy or radiotherapy prior to surgery. Collection of tumor tissues included freezing in liquid nitrogen prior to transfer at -75°C . The peripheral blood samples were

collected in EDTA and processed immediately. All tumors were studied by pathologists and classified according to the WHO criteria (Louis et al. 2007). Our sample consisted of 14 meningothelial meningiomas, 10 fibrous (fibroblastic), 10 transitional (mixed), 1 psammomatous, 10 angiomatous, 10 atypical and 5 anaplastic. Thirty-four patients were female and 26 male. The age of patients varied from 35 to 84 (mean age=61.1 years). The mean age at diagnosis for females was 62.4, and for males 59.4 years. The local Ethical Committee approved our study and patients gave their informed consent. The experimental procedures included: DNA extraction, Polymerase chain reaction, Loss of heterozygosity (LOH) detection and direct sequencing.

To discover LOH of the CDH1 heterozygous samples were visualized on Spreadex EL 300 Mini gels (Elchrom Scientific, Switzerland) stained with SyberGold (Molecular Probes, Netherlands) and on 15% polyacrylamide gels stained with silver. Absence or significant decrease of one allelic band in the tumor compared with autologous blood sample was considered as LOH of CDH1 gene.

The results of our analysis showed altogether 32% of samples with LOH of the CDH1 gene (Fig. 3.1). Fibrous and angiomatous cases showed higher percent of genetic changes, 67% and 75%, respectively. When distributing total E-cadherin's changes to specific tumor types, we observed changes in 27% of meningothelial meningiomas, 67% of fibrous, 33% of transitional (mixed) and 75% of angiomatous. The oscillations of the expressed E-cadherin protein, which was primarily down-regulated, confirmed the functional consequences of the changes at the protein level. Five out of nine samples with LOH (56%) were accompanied with the downregulation of E-cadherin protein expression.

Microsatellite Instability

Certain proportion of tumors and certain proportion of meningiomas, too, demonstrate features of genomic instability. Genomic instability observed in human cancers can be divided in two

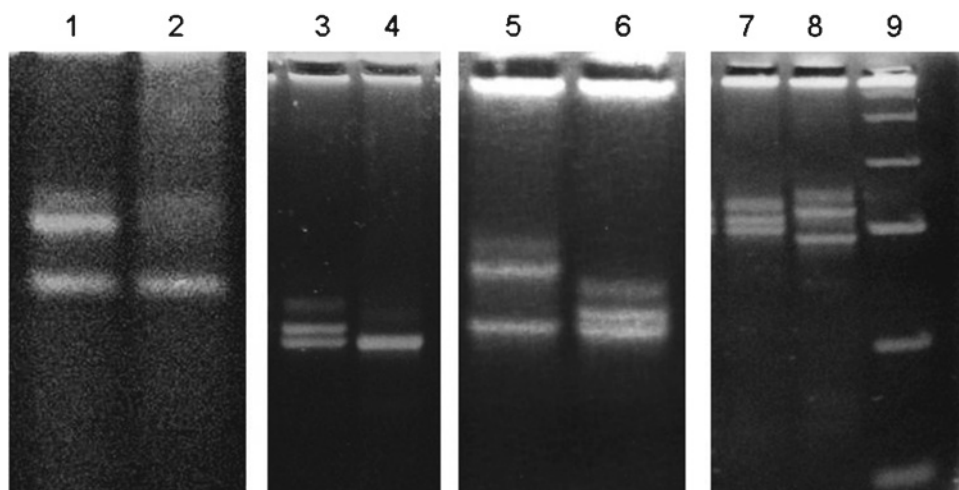


Fig. 3.1 Genomic instabilities, LOH and RER+, of the E-cadherin gene in meningioma samples on Spreadex gels (Elchrom Scientific) stained with Sybergold (Molecular Probes). Polymorphic marker D16S752 is

shown. Lanes 1, 3, 5, 7 –corresponding blood samples; lanes 2, 4 – LOHs of the E-cadherin gene; lanes 6, 8 – MSI samples; lane 9 – M3 standard

categories: microsatellite instability (MSI or MIN) and chromosomal instability (CIN). MSI is associated with the changes in the number of repetitive DNA sequences termed microsatellites. MSI involves the defects in the repair of short mismatches in the replication/repair machinery (Ionov et al. 1993) resulting from the slippage of DNA polymerase as it synthesizes segments of short repeats. CIN occurs at the chromosome level and is caused by chromosomal mis-segregation due to improper alignment of chromosomes in mitosis. The resulting daughter cells comprise aneuploidy and allelic imbalance. Besides instability at the whole chromosomal level, there are other forms of CIN, such as translocations, segmental duplications and deletions, as well as gene amplifications. Mechanisms that cause such alterations are distinct from partitioning at mitosis. Simple repeated sequences are genetically unstable, as judged by their increased mutation rate *in vivo* and *in vitro* and by their polymorphic nature in the human population (Ionov et al. 1993; Venkatesan et al. 2006).

Generally speaking the genome of tumor cells is affected by two crucial types of changes: the accumulation of somatic mutations and the acquisition of genomic instability. Although it is

usually assumed that the genetic instability is a later event in tumor progression, there are standpoints that it may be an early causative event in the formation of a specific tumor (Venkatesan et al. 2006). The usual incidence rate of spontaneous somatic mutations is much lower to the rates of genetic changes observed in tumor cells and this increased frequency is the result of genomic instability that characterizes such cells. It is now believed that the increase of frequency of genetic changes is to be at least 100 times. Body of evidence indicates that the mutation rate of normal human cells is much too low to account for the accumulation of this number of genetic alterations over a single individual's life time. We know that multiple genetic changes are required for carcinogenesis and the inadequate incidence rate observed during human life time seeks explanation. The answer to this discrepancy is given in the so called mutator hypothesis. A mutator gene is a gene whose mutation increases the level of genetic changes of the individual's genome. The mutator gene can be any gene, but usually comes from the group of genes whose products are involved in DNA repair mechanisms or genes whose products control the fidelity of DNA replication. Mutations of those genes are referred

to as mutator mutations, and cancer cells are said to exhibit a mutator phenotype with the increased rate of genomic instability.

The consequential instability observed in tumor is sometimes termed as RER+ tumor phenotype from Replication Error Positive phenotype. RER-positive tumor samples can be visualized on high resolution gels after electrophoresis techniques as samples with bands on different positions in comparison to bands of autologous normal tissue.

Interestingly, we found MSI phenotype in 11% of our investigated meningiomas. When searching for LOHs of the E-cadherin gene, the D16S752 microsatellite tetranucleotide marker revealed samples with MSI-positive phenotype, the ones that showed altered signals in the meningioma tissue. All MSI samples were reamplified and repeatedly analyzed on both Spreadex and polyacrylamide gels (Fig. 3.1). We also confirmed the (RER+) samples by direct sequencing. One meningothelial, one transitional and one anaplastic case harbored MSI.

The results obtained regarding MSI samples are indicative of involvement of replication/repair genes (hMLH1 or hMSH2, hPMS1, hPMS2), opening a potential area of interest in meningioma studies.

Changes of Protein Expression Levels

Many immunohistochemical studies have examined changes in expression of the E-cadherin gene in human malignancies. In almost all tumors

examined, the patterns of changes in the expression of this gene have been similar to that seen in colorectal cancer, i.e. loss of protein expression is positively correlated to loss of tumor differentiation, with a high grade and an advanced stage of the disorder and with poor prognosis.

We devoted a great effort to enlighten the E-cadherin protein levels in meningioma. Immunohistochemistry was performed in order to establish the levels of expression and cellular localization of E-cadherin and beta-catenin proteins. In order to determine whether E-cadherin's changes at the genetic level have consequences at the protein level in meningiomas, the expression of E-cadherin proteins was investigated. Normal levels of E-cadherin staining were scored as +++, and the protein was located along the cell membrane or inside the cytoplasm. Our analysis showed that overall 73% of meningiomas had downregulation of E-cadherin expression (Fig. 3.2). The expression of the protein according to meningioma grades was as follows. E-cadherin downregulation was observed in 50% of meningothelial; 80% of fibrous; 80% of transitional; 90% of angiomatous; 80% of atypical; and in 80% of anaplastic. Thirty-six percent of meningothelial and 40% of each fibrous, transitional and angiomatous showed intense lower expression, while in atypical and anaplastic intense downregulation or complete loss of the protein product was noted in 60% of each grade.

We also aimed to detect whether the expression and cellular localization of the main downstream wnt signaling effector molecule, beta-catenin, correlated with LOH of CDH1 gene. The features

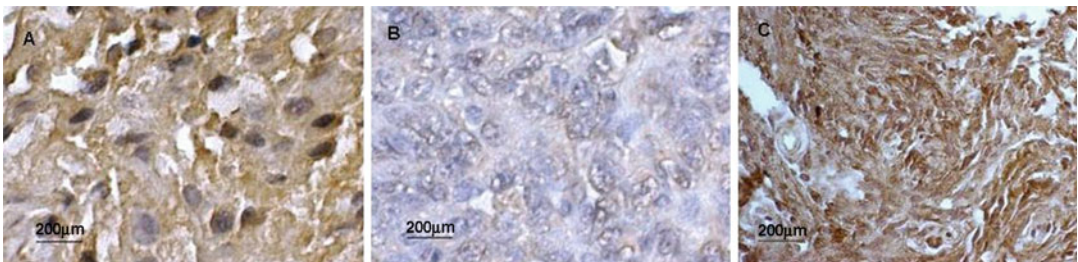


Fig. 3.2 Meningioma samples immunohistochemically stained for the expression of E-cadherin proteins. (a) Patient without LOH demonstrating strong expression of E-cadherin protein. (b) Patient with CDH1 LOH

demonstrating decreased expression of E-cadherin protein. (c) Meningioma sample immunohistochemically stained for the expression of beta-catenin proteins (nuclear localization)

that we used in statistical analysis were: PHD status, sex, age, CDH1 LOH, E-cadherin protein expression and beta-catenin protein expression and localization. The differences in the frequencies of the analyzed features were tested with the Pearson χ^2 test employing Yates correction when appropriate.

Our findings demonstrated that there is significant association between the genetic changes of CDH1 and the nuclear localization of beta-catenin protein ($\chi^2=5.25$, $df=1$, $P<0.022$). Seventy five percent of samples with genomic instabilities had nuclear localization of beta-catenin protein (Fig. 3.2). At the same time in cases where CDH1's instabilities were not detected the location of the beta-catenin protein was primarily in the membrane or was not detectable. We were not able to establish statistically relevant association between decreased expression of E-cadherin and beta-catenin transfer to the nucleus, but nevertheless noticed that 36.4% of samples with lower E-cadherin expression had beta-catenin located in the nucleus.

The relationship of beta-catenin expression to tumor grade demonstrated progressive upregulation from meningothelial (42.9%), through fibrous (50%), transitional (70%), angiomatous (80%), to atypical (90%). Upregulation of beta-catenin was noted in 60% of anaplastic cases which all had the protein localized in the nucleus.

Discussion

Our work on E-cadherin in meningiomas demonstrates that changes of E-cadherin are frequent among benign meningioma types. The observed frequency led us to conclude that gross deletions of the CDH1 gene are an important event in the mechanisms of meningioma development. Moreover, the functional consequences of the changes were confirmed at the protein level.

Besides gross deletions our analysis at the genetic level discovered MSI in 11% of meningioma sample pointing to the replication/repair machinery failure in meningioma. Other authors also reported on involvement of microsatellite

instability phenotype in meningiomas. Pykett et al. (1994) reported on 25% of meningiomas with MSI, while Sobrido et al. (2000) revealed 6.3% of MSI in the meningiomas they investigated. They also found that MSI was associated with shorter duration of clinical course.

DNA repair genes are key factors in maintaining genomic stability. Bethke et al. analyzed single nucleotide polymorphisms of the DNA repair genes in association to meningioma predisposition (Bethke et al. 2008) and found that some DNA repair gene variants are connected to higher risk of meningioma development. The similar results on DNA repair variants affecting the risk of meningioma are reported by Rajaraman et al. (2010). Detected microsatellite instability of the E-cadherin gene may have a role in meningioma development and progression indicating that mismatch repair may be targeted.

Our previous results (Pećina-Šlaus et al. 2008) on the significant association between APC genetic changes and lack of wild type protein expression or presence of mutant APC proteins in meningiomas indicate involvement of yet another component of wnt signaling (Fig. 3.3). APC changes also influenced beta-catenin expression and nuclear localization, showing beta-catenins importance in the biology of meningioma. In our paper published in 2007 (Nikuševa Martić et al. 2007) we reported on two meningiomas with heteroduplexes. In heteroduplex analysis, the conformational properties of the double stranded molecules are used to distinguish different base pairs, i.e., mutations. Annealing mutant DNA to wild-type probe gives duplexes with one or more mismatched bases (heteroduplexes). Mismatching causes the double helix to take on a conformation that retards its mobility during electrophoresis. Heteroduplex analysis of exon 3 of beta-catenin revealed two meningiomas samples with additional bands when the tumor and normal DNA samples were mixed, suggesting that those meningiomas harbor mutations of the beta-catenin gene.

It is now apparent that malignancies can in certain aspects be explained by alterations in the adhesive properties of neoplastic cells. Therefore it is also important to discuss the Epithelial-to-Mesenchymal transition (EMT) in meningiomas.

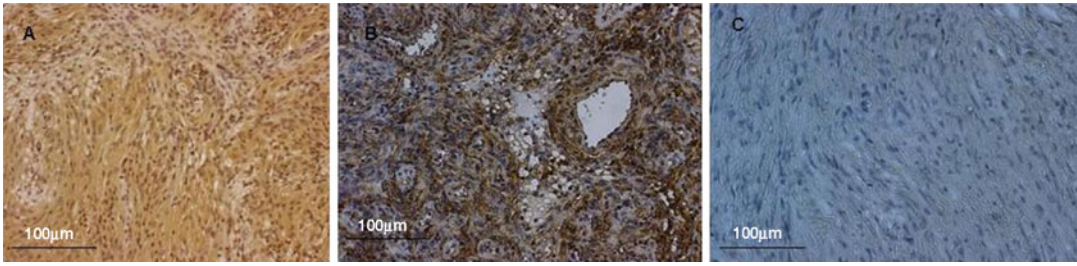


Fig. 3.3 Meningioma samples immunohistochemically stained for the expression of adenomatous polyposis coli (APC) proteins. (a) Patient demonstrating mutant APC protein (Ab-1 that recognizes the amino terminus was

used). (b) Patient positive to the expression of wild type APC protein (Ab-2 that recognizes the carboxy terminus was used). (c) Patient negative for the expression of wild type APC protein

This process is similar to developmental events but with the important difference that it is uncontrolled. Epithelial-mesenchymal conversion is an important mechanism for the initial step of metastasis and the hallmark of EMT is the loss of E-cadherin expression. The wnt pathway has a particularly tight link with EMT and it has been shown that nuclear translocation of beta-catenin can induce EMT (Wu and Zhou 2008). It is long known that meningiomas exhibit desmosomes (Schwechheimer et al. 1998), the epithelial type of cell contact, so the presence of E-cadherin in meningiomas is not unusual. Moreover, E-cadherin is considered the main cadherin type in meningiomas (Akat et al. 2008). In some meningiomas so called cadherin switch may occur, namely, E-cadherin may be replaced by N-cadherin. This phenomenon normally happens in epithelial-mesenchymal transition, but in tumors it is regarded as a sign of invasive behavior and progression. We may speculate that meningiomas, in which we identified E-cadherin losses, would later on exhibit aggressive behavior due to reduction or loss of protein product of this suppressor gene. Meningiomas need not exhibit morphological signs of malignancy, and loss of E-cadherin gene may change the situation at the tumor-brain interface and thus initiate the mechanisms of future expansion. A report by Akat et al. (2008) described a new type of adherens junction in human meningiomas and the human meningioma cell line HBL-52. This novel junction is closely related to classic adherens junctions but is

nevertheless unique and different from other junctions. The proteins it contains, among others, include E-cadherin and beta-catenin. The majority of meningiomas the authors examined were positive for E-cadherin, but the amount of expression fell in atypical or malignant cases, in which N-cadherin replaced E-cadherin.

The majority of meningiomas we analyzed were slow-growing, benign sporadic lesions with seemingly low metastatic potential. However, biologic spectrum of meningiomas is wide, heterogeneous and difficult to predict. Some of the histologically benign meningiomas recur unexpectedly even after complete resection and invade surrounding tissues. Novel revisions of meningioma classification (Riemenschneider et al. 2006; Louis et al. 2007) recommend caution on benign meningioma prognosis, proposing their proliferative activity and brain invasion as important characteristics that could indicate recurrence and should be considered in diagnostics and prognosis.

The survey by Wrobel et al. (2005) who observed that anaplastic meningiomas could be distinguished from benign by differential expression of a distinct set of genes, suggests that benign forms and those with greater likelihood of recurrence and aggressive behavior could be governed by different genetic profiles.

Our observations are in accordance with the findings of other authors investigating E-cadherin and other components of wnt signaling in meningiomas (Tohma et al. 1992; Howng et al. 2002;

Nagashima et al. 2006). For example E-cadherin's allelic losses were observed in fibrous and angiomatous meningiomas. Since the loss of E-cadherin is a well known prerequisite for tumor cell invasion, the finding suggests that those meningioma types may hide future invasive behavior. Schwechheimer et al. (1998) found that E-cadherin's expression was absent from the majority of morphologically malignant meningiomas and that the loss of its expression was correlated with tumor dedifferentiation. Utsuki et al. (2005) also reported on negative E-cadherin immunostaining in all of the fibrous meningiomas they examined. Brunner et al. (2006) found lack of E-cadherin's expression in 34% of meningiomas independent of their WHO grade as well as loss of membranous and positive nuclear immunoreactivity of beta-catenin. Panagopoulos et al. (2008) report that E-cadherin expression is present in both normal brain and arachnoid cells and that E-cadherin expression changes could not be correlated to tumor grade. Nevertheless, they observed E-cadherin expression in 59% of benign meningiomas, 67% of atypical while none of the anaplastic meningiomas expressed E-cadherin in their study. Our data on the expression according to meningioma grades demonstrate that intense downregulation of E-cadherin was noticed in tumors with grades II and III.

Wrobel et al. (2005) reported on increased expression of beta-catenin and cyclin D1 in meningiomas they examined by microarray. Our results on the relationship of beta-catenin expression to tumor grade demonstrated progressive upregulation from meningotheial to atypical. 60% of anaplastic cases showed upregulation of beta-catenin and all had the protein localized in the nucleus.

Mutation and or loss of NF2 gene were found in meningiomas of all malignancy grades which make NF2 gene a candidate gatekeeper gene of meningioma initiation, i.e. first genetic change in its pathogenesis (Riemenschneider et al. 2006). There are probably many genes other than NF2 involved in meningioma. Relevant genetic alterations of atypical and anaplastic meningiomas are

still unknown and we are unfortunately still unable to identify meningiomas that will behave aggressively. We were unable to establish correlation of NF-2 genetic changes to the results with E-cadherin (Pećina-Šlaus et al. 2010) which is consistent with the work by Brunner et al. (2006) who state that it is unlikely that loss of NF2 expression is associated with loss of the proper localization of beta-catenin and E-cadherin in meningiomas. It seems that changes of those genes are mutually exclusive.

Finally, I would like to discuss a paper by Saydam et al. (2009) which evidenced that specific microRNA, miR-200a, has a direct role in meningioma growth *via* E-cadherin and wnt/beta-catenin signaling pathway. Downregulated miR-200a in meningiomas promoted tumor growth by reducing E-cadherin and activating the wnt/beta-catenin signaling pathway. A direct correlation between the downregulation of miR-200a and the upregulation of beta-catenin was demonstrated in this study.

The numerous reports by many authors as well as our own results indicate that E-cadherin plays a role in CNS tumors – meningiomas. Changes of CDH1 gene accompanied with changes at the protein level strengthen our conclusions on its involvement in meningioma. Moreover, E-cadherin changes are in accordance to beta-catenin behavior both in the disruption of adherens junctions and wnt signaling. The association of changes to meningiomas with grades II and III suggests that it is probably connected to predisposition to progression in a subset of meningiomas. Our findings contribute to better understanding of meningioma genetic profile elucidating pathophysiological mechanisms of tumor formation, and could offer new prognostic markers for meningioma. The field is evolving very fast and relevant biomarkers will emerge in order to better diagnose and cure the meningiomas patients.

Acknowledgements This work was supported by grant 108-1081870-1905 from Ministry of Science Sports and Education, Republic of Croatia.

References

- Akat K, Bleck CK, Lee YM, Hasselman-Weiss U, Kartenbeck J (2008) Characterization of a novel type of adherens junction in meningiomas and the derived cell line HBL-52. *Cell Tissue Res* 331:401–412
- Barnholtz-Sloan JS, Kruchko C (2007) Meningiomas: causes and risk factors. *Neurosurg Focus* 23:E2
- Berx G, Staes K, van Hengel J, Molemans F, Bussemakers MJ, van Bokhoven A, van Roy F (1995) Cloning and characterization of the human invasion suppressor gene E-cadherin (CDH1). *Genomics* 26:281–289
- Bethke L, Webb E, Murray A, Schoemaker M, Johansen C, Christensen HC, Muir K, McKinney P, Hepworth S, Dimitropoulou P, Lophatananon A, Feychting M, Lönn S, Ahlbom A, Malmer B, Henriksson R, Auvinen A, Kiuru A, Salminen T, Swerdlow A, Houlston R (2008) Comprehensive analysis of the role of DNA repair gene polymorphisms on risk of glioma. *Hum Mol Genet* 17:800–805
- Brunner EC, Romeike BF, Jung M, Comtesse N, Meese E (2006) Altered expression of beta-catenin/E-cadherin in meningiomas. *Histopathology* 49:178–187
- Claus EB, Bondy ML, Wiemels JL, Wrensch M, Black PM (2005) Epidemiology of intracranial meningioma. *Neurosurgery* 57:1088–1094
- Fuller CE, Perry A (2005) Molecular diagnostics in central nervous system tumors. *Adv Anat Pathol* 12:180–194
- Guilford P (1999) E-cadherin downregulation in cancer: fuel on the fire? *Mol Med Today* 5:172–177
- Howng SL, Wu CH, Cheng TS, Sy WD, Lin PC, Wang C, Hong YR (2002) Differential expression of Wnt genes, beta-catenin and E-cadherin in human brain tumors. *Cancer Lett* 183:95–101
- Ionov Y, Peinado MA, Malkhosyan S, Shibata D, Perucho M (1993) Ubiquitous somatic mutations in simple repeated sequences reveal a new mechanism for colonic carcinogenesis. *Nature* 363:558–561
- Koch A, Waha A, Tonn JC, Sorensen N, Berthold F, Wolter M, Reifenberger J, Hartmann W, Friedl W, Reifenberger G, Wiestler OD, Pietsch T (2001) Somatic mutations of WNT/wingless signaling pathway components in primitive neuroectodermal tumors. *Int J Cancer* 93:445–449
- Li F, Chong ZZ, Maiese K (2006) Winding through the WNT pathway during cellular development and demise. *Histol Histopathol* 21:103–124
- Louis DN, Ohgaki H, Wiestler OD, Cavenee WK, Burger PC, Jouvet A, Scheithauer BW, Kleihues P (2007) The 2007 WHO classification of tumours of the central nervous system. *Acta Neuropathol* 114:97–109
- Nagashima G, Fujimoto T, Suzuki R, Asai J, Itokawa H, Noda M (2006) Dural invasion of meningioma: a histological and immunohistochemical study. *Brain Tumor Pathol* 23:13–17
- Nikuševa Martić T, Beroš V, Pečina-Šlaus N, Pečina HI, Bulić-Jakuš F (2007) Genetic changes of CDH1, APC and CTNNB1 found in human brain tumors. *Pathol Res Pract* 203:779–787
- Nikuševa Martić T, Pečina-Šlaus N, Kušec V, Kokotović T, Mušinović H, Tomas D, Zeljko M (2010) Changes of AXIN-1 and beta-catenin in neuroepithelial brain tumors. *Pathol Oncol Res* 16:75–79
- Panagopoulos AT, Lancellotti CL, Veiga JC, de Aguiar PHP, Colquhoun A (2008) Expression of cell adhesion proteins and proteins related to angiogenesis and fatty acid metabolism in benign, atypical, and anaplastic meningiomas. *J Neurooncol* 89:73–87
- Patapoutian A, Reichardt LF (2000) Roles of wnt proteins in neural development maintenance. *Curr Opin Neurobiol* 10:392–399
- Pečina-Šlaus N (2003) Tumor suppressor gene E-cadherin and its role in normal and malignant cells. *Cancer Cell Int* E3:17 [<http://www.cancerci.com/content/3/1/17>]
- Pečina-Šlaus N (2010) Wnt signal transduction pathway and apoptosis: a review. *Cancer Cell Int* E10:22 [<http://www.cancerci.com/content/10/1/22>]
- Pečina-Šlaus N, Nikuševa Martić T, Tomas D, Beroš V, Zeljko M, Čupić H (2008) Meningiomas exhibit loss of heterozygosity of the APC gene. *J Neurooncol* 87:63–70
- Pečina-Šlaus N, Nikuševa Martić T, Deak AJ, Zeljko M, Hrašćan R (2010) Genetic and protein changes of E-cadherin in meningiomas. *J Cancer Res Clin Oncol* 136:695–702
- Pykett MJ, Murphy M, Harnish PR, George DL (1994) Identification of amicrosatellite instability phenotype in meningiomas. *Cancer Res* 54:6340–6343
- Ragel BT, Jensen RL (2005) Molecular genetics of meningiomas. *Neurosurg Focus* 119:E9
- Rajaraman P, Hutchinson A, Wichner S, Black PM, Fine HA, Loeffler JS, Selker RG, Shapiro WR, Rothman N, Linet MS, Inskip PD (2010) DNA repair gene polymorphisms and risk of adult meningioma, glioma, and acoustic neuroma. *Neuro Oncol* 12:37–48
- Riemenschneider MJ, Perry A, Reifenberger G (2006) Histological classification and molecular genetics of meningiomas. *Lancet Neurol* 5:1045–1054
- Saydam O, Shen Y, Würdinger T, Senol O, Boke E, James MF, Tannous BA, Stemmer-Rachamimov AO, Yi M, Stephens RM, Fraefel C, Gusella JF, Krichevsky AM, Breakefield XO (2009) Downregulated microRNA-200a in meningiomas promotes tumor growth by reducing E-cadherin and activating the Wnt/beta-catenin signaling pathway. *Mol Cell Biol* 29:5923–5940
- Schwechheimer K, Zhou L, Birchmeier W (1998) E-cadherin in human brain tumours: loss of immunoreactivity in malignant meningiomas. *Virchows Arch* 432:163–167
- Simon M, Boström JP, Hartmann C (2007) Molecular genetics of meningiomas: from basic research to potential clinical applications. *Neurosurgery* 60:787–798
- Sobrido MJ, Pereira CR, Barros F, Forteza J, Carracedo A, Lema M (2000) Low frequency of replication errors in primary nervous system tumours. *J Neurol Neurosurg Psychiatry* 69:369–375
- Tohma Y, Yamashita T, Yamashita J (1992) Immunohistochemical localization of cell adhesion molecule

- epithelial cadherin in human arachnoid villi and meningiomas. *Cancer Res* 52:1981–1987
- Utsuki S, Oka H, Sato Y, Kawano N, Tsuchiya B, Kobayashi I, Fujii K (2005) Invasive meningioma is associated with a low expression of E-cadherin and beta-catenin. *Clin Neuropathol* 24:8–12
- Venkatesan RN, Bielas JH, Loeb LA (2006) Generation of mutator mutants during carcinogenesis. *DNA Repair (Amst)* 5:294–302
- Wrobel G, Roerig P, Kokocinski F, Neben K, Hahn M, Reifenberger G, Lichter P (2005) Microarray-based gene expression profiling of benign, atypical and anaplastic meningiomas identified novel genes associated with meningioma progression. *Int J Cancer* 114:249–256
- Wu Y, Zhou BP (2008) New insights of epithelial-mesenchymal transition in cancer metastasis. *Acta Biochim Biophys Sin* 40:643–650

Intracranial Meningiomas: Role of Exogenous Hormones

4

Sudhakar Vadivelu and Michael Schulder

Abstract

Meningiomas express multiple hormone receptors including progesterone, estrogen, androgen, growth hormone, dopamine, and somatostatin. There is a growing body of knowledge regarding the particular roles that each hormone receptor may play in modulating meningioma growth metrics. Identification of progesterone and estrogen receptor balances has shown to be of prognostic significance for meningioma growth behavior. Growth hormone may also promote meningioma growth. Somatostatin, alternatively, has been shown in robust affinity towards meningioma identification for diagnostic purposes via radionucleotide imaging. It may be used to attenuate meningioma growth, as reported in both *in vitro* and clinical reports. This strong correlation with hormonal activity suggests that patients who are treated with exogenous hormones may develop meningiomas as a result. Understanding this association may help elucidate various treatment paradigms.

At this time, unequivocal evidence of exogenous hormone involvement in the growth of meningioma remains absent. Here, we present a current understanding on the role of exogenous hormones in meningioma biology and their association with meningioma in clinical practice. Available information suggests that patients being treated with exogenous progesterone, estrogen, or growth hormone be advised that they may be at increased risk for meningioma.

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Introduction

Factors associated with the development of meningiomas include age, gender, and certain genetic combinations (Riemenschneider et al. 2006;

Claus et al. 2008). The fact that intracranial meningiomas arise in women almost two times the incidence in men (Hsu et al. 1997), along with modulation of meningioma biology during pregnancy, suggests a plausible hormonal link (Bickerstaff et al. 1958). Postmenopausal women and those with breast cancer as well have shown an increased likelihood of meningioma development (Cushing and Eisenhardt 1938; Schoenberg et al. 1975). Thus, endogenous hormonal contribution has become a focus of interest on studies attempting to understand the developmental biology of meningioma. However, another concern is the number of patients receiving exogenous hormone therapy for various organ specific disease states or for regulatory enhancement in normal organ functions, e.g., birth control or fertility enhancement. Interestingly, while current use of direct hormone replacement therapy in menopausal women is decreasing, there is increasing use of exogenous hormones for other reasons, including oral contraception (Claus et al. 2007). An obvious question is whether this use of contraceptives will lead to a greater incidence of meningioma growth in the future (Michaud et al. 2010).

In addition, an association between acromegaly and growth hormone (GH) hypersecretion has been reported (Van der Lely et al. 2001; Drake et al. 2005). Given modern society's bias in favor of greater individual height, the use of recombinant GH to bolster height in persons genetically fated to be short raises a similar concern: will iatrogenic meningioma result from such non-critical medical interventions?

Evidence of Association

Association of exogenous hormones in patients with meningiomas has been reported (Claus et al. 2007; Gruber et al. 2004; Piper et al. 1994; Shimizu et al. 2008). However, this association has not consistently appeared in all studies examining this relationship (Claus et al. 2008; Custer et al. 2006; Michaud et al. 2010; Wigertz et al. 2006). The variations in association between exogenous hormone use and meningioma may

imply that the relationship is not direct, but multifactorial. However, the general consensus from various organized task forces recognizes that a relationship exists. This has major implications on treatment risks and the advice that should be offered to patients by physicians in different specialties (Claus et al. 2007; Custer et al. 2006; Vadivelu et al. 2010).

Laboratory Evidence

Laboratory studies of resected tumors have clearly identified hormone receptor presence in meningioma, most commonly the progesterone and estrogen receptors. Expression of these receptors compared against gross classification and WHO grade identification of meningiomas demonstrate prognostic value. Pravdenkova et al. showed that meningiomas with positive progesterone expression had low proliferative indices and were karyotypically stable, versus progesterone receptor (PR) negative or estrogen receptor (ER) positive meningioma tissue. These latter groups were associated with greater proliferative indices and were more likely to have karyotype instability. The estrogen positive group, uniquely, demonstrated a greater genetic association with chromosomes 14 and 22 (Pravdenkova et al. 2006).

Receptor acceptance of various progesterone agonists has been identified in selective structural isoforms. Progesterone receptor A (PRA) at 83 kDa has been shown to bind to meningioma cell surfaces with less affinity than progesterone receptor B (PRB) at 99 kDa (Connaghan-Jones et al. 2007). This difference in binding affinities may suggest how PRB is considered to operate as the dominant receptor activator in meningiomas, versus PRA's relatively repressive activity against PRB (Fig. 4.1) (Tung et al. 1993).

Several nervous system tumors express both progesterone receptor isoforms A and B. Resected specimens evaluated for PR isoforms have shown predominance of PRB in 6 chordoma cases (Camacho-Arroyo et al. 2007), 26 (grade unspecified; Inoue et al. 2002), or 13 astrocytomas (grade III and IV; González-Agüero et al. 2001), and 16 schwannoma cases (Inoue et al. 2002).

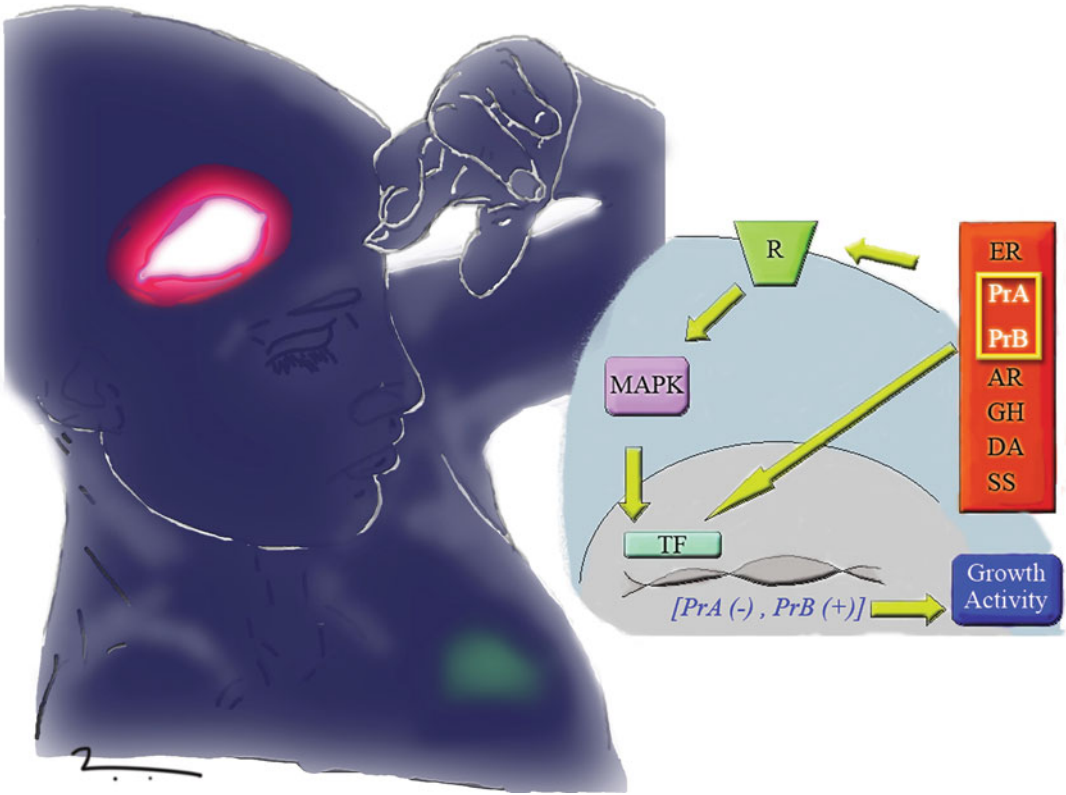


Fig. 4.1 Nongenomic and genomic actions of various known hormone receptor modulators of meningioma growth activity. Schematic illustrates exogenous hormone use for breast cancer treatment targeting the progesterone

receptor may promote meningioma growth when there is a predominance of PrB versus PrA (Reproduced with permission, Vadivelu et al. 2010)

In comparing isoform expression between PRA versus PRB, meningiomas have demonstrated mixed results. In a study of resected nervous system tumors, 35 meningiomas were examined for PR receptor level expression where no significant difference was found between PRA and PRB. Comparing meningioma PR expression versus other nervous system tumors demonstrated a significantly greater expression of PRA compared to astrocytomas and schwannomas (Inoue et al. 2002). Verheijen et al. (2001) evaluated ligand binding affinities in 61 PR positive meningiomas in an attempt to elucidate isotype predominance in meningiomas of various WHO grades. Sixty-six percent of these PR positive meningiomas demonstrated greater PRA ligand binding capacity than PRB. It is possible that PRA may be predominantly expressed in collective examination of

intracranial meningiomas while PRB predominance in a single specimen may suggest a relationship to hormonal manipulation. A distinct study classifying PRA and PRB expression predominance in patients with meningiomas who have received exogenous hormone therapy may be of use in suggesting possible therapeutic hormonal manipulation.

Case Illustration: Progesterone Agonist and Meningioma Formation

Patient 1. A 50 year old postmenopausal female complained of progressively worsening headaches. She had a history of breast carcinoma treated with mastectomy, after which she was maintained on megestrol acetate (40 mg, four

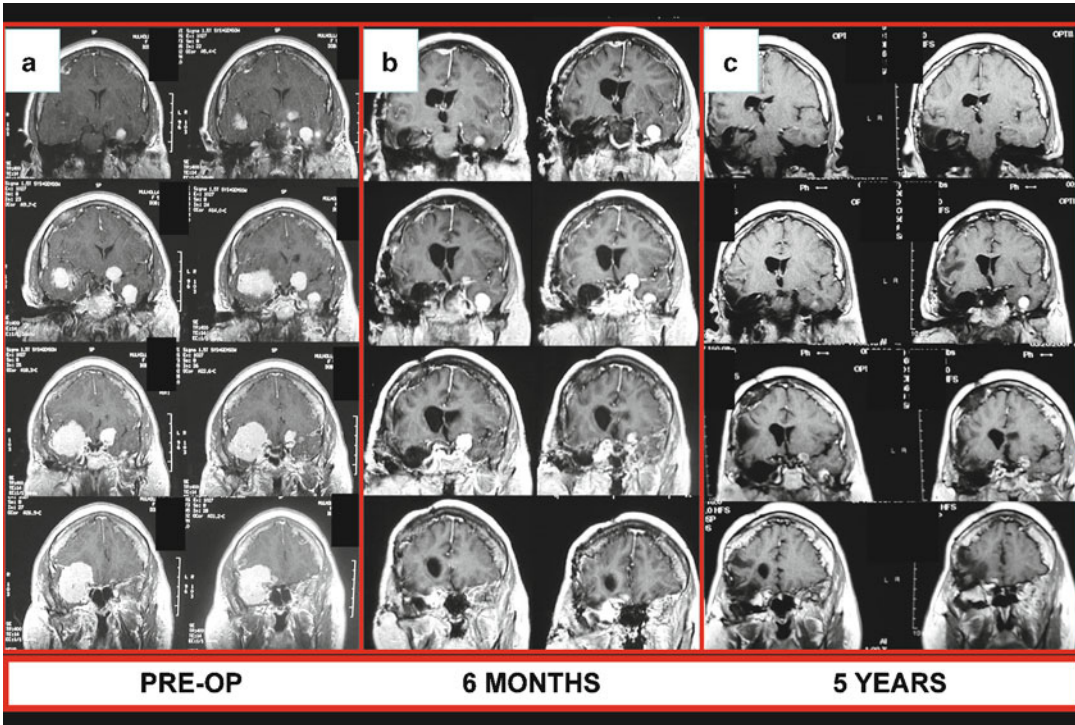


Fig. 4.2 Preoperative, 6 months postoperative, and 5 years postoperative MR T1 contrast enhanced images of patient # 1 (postmenopausal female) in coronal slices

illustrating attenuation of non-operated enhanced lesions and even complete loss of a lesion (Reproduced with permission, Vadivelu et al. 2010)

times daily) for over 7 years. Neurological examination was non-focal. Her outpatient evaluation included a contrast enhanced MRI demonstrating five intracranial meningiomas, including a large right sphenoid wing tumor that was deemed the cause of her symptoms (Fig. 4.2). Progesterone agonist therapy was discontinued. The right sphenoid wing tumor was deemed symptomatic because of its size and mass affect, and was resected a day after embolization. The patient was initially neurologically intact after surgery, but severe brain swelling on the first postoperative day led to an anterior right temporal lobectomy. She recovered well afterwards.

Histological examination of surgically removed tissue revealed features consistent with WHO grade I transitional meningioma. The tumor demonstrated high expression ($x > 25\%$) for progesterone receptor, and a lack of estrogen receptor expression. We further evaluated the

tissue for progesterone receptor isotype expression A versus B. We found high PRB expression ($x > 25\%$) versus low PRA expression.

Follow-up MR imaging demonstrated early reduction in size of multiple tumors, including the multiple tumors for which no operation was performed (Fig. 4.2). At her most recent follow up – 5 years postoperatively, and after discontinuation of megestrol acetate as the only therapeutic intervention – MRI demonstrated marked reduction in size in her multiple meningiomas and resolution of the right parasagittal lesion (Fig. 4.2).

This patient's breast cancer was being treated with megestrol acetate. Few clinical reports have demonstrated an association between progesterone agonist therapy and clinical outcomes in meningioma. In this illustrative case (Vadivelu et al. 2010), PRB predominance was observed and this may have important prognostic implications

as it has been shown to be an important survival factor in other cancers known to have a hormonal association (McGowan et al. 2007; Shabani et al. 2007). Meningiomas with higher expression of PR expression were more likely to have freedom from tumor progression, lower WHO grades, fewer karyotypic abnormalities, and lower proliferative indices (Claus et al. 2008; Pravdenkova et al. 2006); Furthermore, the high PRB expression in this patient suggests that patients whose meningiomas have similar findings may be candidates for treatment with progesterone antagonists.

Progesterone Agonist Therapy and Meningiomas

Could application of progesterone agonists be considered for direct CNS therapy to halt or promote regression of meningioma growth? Grunberg et al. (1987) concluded, initially, that progesterone receptor expression could be down-regulated in a human tumor stem cell clonogenic assay after exposure to continuous progesterone agonist therapy. However, an institutional protocol was established where they followed nine patients with intracranial meningiomas and observed them with serial imaging and clinical exams as they received up to 1 year of megestrol acetate. Unfortunately, the protocol was discontinued after progression of new or recurrent neurologic symptoms occurred in seven of their patients and the other two without relief of current symptoms (Grunberg and Weiss 1990).

Despite the notion of progesterone agonist use as a therapeutic approach to meningiomas, reports of patients with chronic use of progesterone agonist therapy for other medical disorders have demonstrated an association with the incidence of intracranial meningioma. Two cases were reported where one patient was taking an intrauterine contraceptive, levonorgestrel, for several weeks, and presented with headaches. She was found to have a sphenoid wing meningioma that did not regress in size after 6 weeks of discontinued therapy; her symptoms progressed and the tumor was resected (Piper et al. 1994). The other case was of a patient

requiring megestrol acetate for 19 years and presented with 11 intracranial meningiomas. Despite discontinuation of progesterone agonist therapy both symptoms and radiographic size of the meningioma lesion increased (Gruber et al. 2004). Thus, reported case series and single illustrations, along with some epidemiological studies, demonstrate a clear association of progesterone agonist therapy with meningioma growth. However, where along the hormonal modulation pathway intervention can yield meningioma growth *control* remains uncertain.

Regression of intracranial meningioma has been witnessed to occur. Shimizu et al. reported a male patient presenting after a motor vehicle accident with an incidental radiologic finding with features appearing similar to an intracranial meningioma. With a significant history for benign prostatic hypertrophy treated with several year use of chlormadinone acetate (progesterone agonist) the authors discontinued the progesterone agonist and switched to an alpha 1 blocker, nifedipine. Approximately, two and half years later, radiographic regression of the lesion was observed and at 4 year follow up remained stable (Shimizu et al. 2008). Our first case illustrated (Vadivelu et al. 2010) here in this chapter, describes a patient demonstrating causality between intracranial meningioma growth and long standing progesterone agonist therapy. Despite this causal relationship, it remains unclear how this *in situ* modulation could occur in our case but has not been seen other reports. It is possible that the relationship is multifactorial and has some possible underlying genetic predisposition (Nakamura et al. 2005).

Exogenous Growth Hormone

Growth hormone (GH) over the last two decades has been shown repeatedly to affect growth of neuroendocrine tumors (i.e. adrenal and pituitary tumors) and non-neuroendocrine tumors (breast, colon, prostate and meningioma as well). Laboratory evidence has suggested that the association witnessed in a few reports with acromegalic patients and incidental meningiomas

may be linked. Human meningioma are found to express GH receptor, adding IGF-1 to cultured meningioma cells augments their growth, and antagonizing the GH receptor with pegvisomant reduces tumor growth rate indices *in vitro* (McCutcheon et al. 2001). When meningioma xenografts were implanted into female athymic mice, placebo animals demonstrated 23% growth in tumor volume over an 8 week observatory period, whereas experimental mice treated with pegvisomant were shown to have a reduction in tumor volume by 32% (McCutcheon et al. 2001). However, this dramatic finding has not been observed clinically. An acromegalic patient, known to have high circulating levels of serum IGF-1, was placed in a controlled prospective trial with pegvisomant therapy. Despite her several month history of taking 10 mg/day of GH receptor antagonist therapy, the intracranial meningioma volume did not reduce and in fact was observed to continue to grow (Drake et al. 2005). Another case report did not show meningioma growth control despite normalization of serum GH and IGF-1 in a patient with acromegaly. (Curto et al. 2007).

Octreotide Imaging and Therapy for Meningioma

Similar to neuroendocrine tumors, meningiomas have high affinity somatostatin receptors. Particularly in humans, somatostatin receptor 2 (SSTR2) is commonly identified and has found an emerging role in meningioma identification, both non-invasively and through intra-operative guidance (Nathoo et al. 2007; Gay et al. 2005). This role becomes even more clear with concern over metastatic lesions versus benign meningioma, such that the combination of both ¹¹¹indium-octreotide (somatostatin) brain scintigraphy and functional MRI have provided aid in better defining a lesion to be benign or aggressive meningioma versus dural – based metastasis. Application of this diagnostic therapy may be more of use in patients with intracranial lesions who have a history of exogenous hormone therapy.

In such patients, serial imaging studies after discontinuation of hormone therapy might be deemed pragmatic. Interestingly, Nicolato (2004) published a report suggesting a greater than 10% decrease in ¹¹¹indium-octreotide brain scintigraphy index ratio of post-Gamma knife radiosurgery treatment to pre-gamma knife radiosurgery treatment performed within 1 year was a prognostic indicator of recurrent meningioma reduction at further follow – up time points suggesting that octreotide imaging may play a prognostic modality in identifying meningioma more likely to undergo delayed radiographic regression after radiosurgery.

Somatostatin analogues have been reported to have a cytostatic effect on meningioma, thus presenting an alternative therapeutic option. Human meningioma labeled for SSTR (somatostatin receptor) *in vitro* was found to have an anti-proliferative effect (Arena et al. 2004). Furthermore, a clinical study demonstrated 44% of patients with recurrent meningioma may show a 6 month progression free survival after monthly Sandostatin therapy (Chamberlain et al. 2007). Future randomized clinical studies should address this approach to meningioma management. It remains conceivable that somatostatin antagonists may play a role in the management of meningioma in patients with prior history of exogenous hormone use.

Oral Contraception

Several case control and cohort studies have resulted in equivocal conclusions regarding oral contraceptive and meningioma association. One study suggested that there was no meningioma association with use of oral contraceptives. However, the cohort included 1% of patients with “current” use of oral contraception versus never or past use (Jhawar et al. 2003). The hypothesis that exogenous hormone use may modulate meningioma growth must examine a significant proportion of patients with current use as past use may have simply missed the time point at which these exogenous hormones take effect.

The Interphone study in Finland suggested that premenopausal patients were at increased risk of developing meningioma with greater association of progesterone positivity ($x > 10\%$) (Korhonen et al. 2010). A prospective trial examining glioma and meningioma risk in a European cohort reiterated this association in that “current” users of oral contraception were at increased risk versus “never” users (Michaud et al. 2010). Future epidemiological studies should examine this potential link in greater detail, and identify the patterns of meningioma receptor expression in patients taking oral contraceptives.

Case Illustration: Exogenous GH, Oral Contraception, and Meningioma Growth

Patient 2. This 28 year old premenopausal female had presented 3 years earlier with visual disturbance and headaches that eventually resolved. Her MRI at that time demonstrated four meningiomas: midline parasellar, left anterior clinoid process, left petroclival, and the parietal superior sagittal sinus (Fig. 4.3). She now presented to our service with worsening headaches and associated nausea, vomiting, and photophobia.

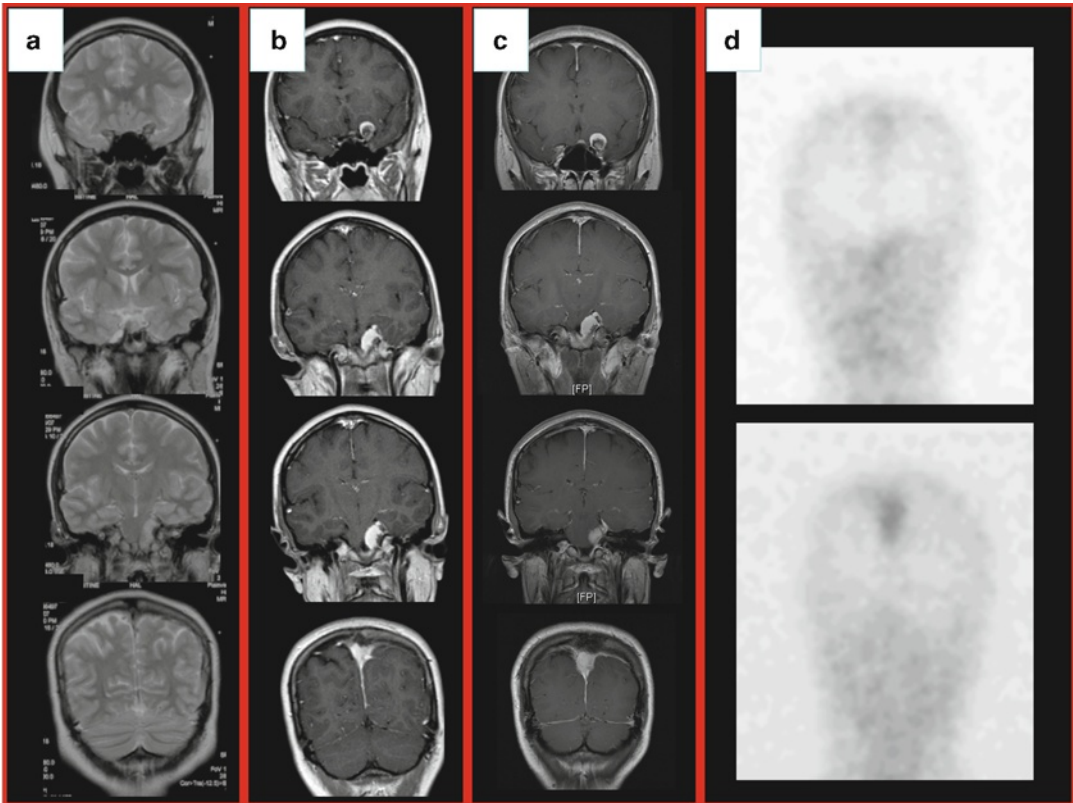


Fig. 4.3 Premenopausal female (patient # 2) with a 5 year history of oral contraception use and prior growth hormone therapy. Coronal MR demonstrating four intracranial meningiomas; midline parasellar, the left anterior clinoid process, the left petroclival region, and at the parietal sagittal sinus concomitant with a 2 year use of OCPs (column a). After three additional years of OCP use, she was referred to us at which point she was instructed to

discontinue her OCP use and obtained the following MRI demonstrating increased size in her multiple intracranial meningioma including parietal sagittal sinus venous compression (column b). Eight months follow up coronal MR (column c) is unchanged in comparison to her prior film at the time of OCP discontinuation. Octreotide scan shows uptake in the left petroclival and superior sagittal sinus lesions (column d)

As these symptoms progressed, the patient then experienced decreased left sided visual acuity and numbness in the left face and tongue. MRIs showed an increase in tumor size compared to 3 years earlier. All symptoms resolved spontaneously, and she was diagnosed with migraines.

The patient had been taking oral contraceptive pills for the previous 5 years. At the first visit, we advised her to discontinue her oral contraceptive pills. Additionally, because her parents were concerned that she would be below average height as an adult, she had received recombinant human GH between the ages of 12 and 15 years. Since stopping oral contraception, tumors have stabilized on her followup MR scans (Fig. 4.3). An octreotide nuclear scan demonstrated uptake in two of the tumors; no octreotide therapy has been prescribed as the patient is stable clinically and on imaging.

Impact on Future Clinical Encounters

Considering the association of intracranial meningioma with various exogenous progesterone agonist therapies, caution should be employed when patients are being considered for hormonal therapy. The rising use of oral contraceptives in addition to exogenous hormone therapies against various oncologic processes, such as breast or prostate cancer, lends importance to this question. To date no definitive stance on this topic exists in the neurosurgical community. At this point, several important points can be considered in the clinical setting: (1) Patients undergoing exogenous hormone therapy who present with new headaches should undergo neurologic examination for possible intracranial meningioma, (2) Patients with primary history of meningioma diagnoses should be warned of possible recurrence with use of exogenous hormone use, and (3) Patients undergoing chronic exogenous hormone therapy, specifically progesterone agonists, may have a higher risk towards developing intracranial meningioma.

Conclusion

A growing body of evidence has suggested an association between exogenous hormone use and meningioma incidence. Both *in vitro* and clinical studies have demonstrated meaningful associations between hormonal administration and tumor growth. It is suggested that progesterone agonists play a part in attenuating growth of meningioma, and yet their withdrawal also leads to tumor regression. In addition, there is tantalizing but inconclusive evidence of a connection between growth hormone and meningioma formation and enlargement. With the rising use of exogenous hormones such as oral contraceptives, progesterone agonists for oncologic control, and GH supplementation, it becomes even more imperative to elucidate alternative treatment strategies towards management of intracranial meningioma.

References

- Arena S, Barbieri F, Thellung S, Pirani P, Corsaro A, Villa V, Dadati P, Dorcaratto A, Lapertosa G, Ravetti J, Spaziante R, Schettini G, Florio T (2004) Expression of somatostatin receptor mRNA in human meningiomas and their implication in *in vitro* antiproliferative activity. *J Neurooncol* 66:155–166
- Bickerstaff ER, Small JM, Guest IA (1958) The relapsing course of certain meningiomas in relation to pregnancy and menstruation. *J Neurol Neurosurg Psychiatry* 21:89–91
- Camacho-Arroyo I, González-Agüero G, Gamboa-Domínguez A, Cerbón MA, Ondarza R (2007) Progesterone receptor isoforms expression pattern in human chordomas. *J Neurooncol* 49:1–7
- Chamberlain MC, Glantz MJ, Fadul CE (2007) Salvage therapy with long-acting somatostatin analogue. *Neurology* 69(10):969–973
- Claus EB, Black PM, Bondy ML, Calvocoressi L, Schildkraut JM, Wiemels JL, Wrensch M (2007) Exogenous hormone use and meningioma risk: what do we tell our patients? *Cancer* 110:471–476
- Claus EB, Park PJ, Carroll R, Chan J, Black PM (2008) Specific genes expressed in association with progesterone receptors in meningioma. *Cancer Res* 68:314–322
- Connaghan-Jones KD, Heneghan AF, Miura MT, Bain DL (2007) Thermodynamic analysis of progesterone receptor-promoter interactions reveals a molecular model for isoform-specific function. *Proc Natl Acad Sci USA* 13:2187–2192

- Curto L, Squadrito S, Almoto B, Longo M, Granata F, Salpietro F, Torre ML, Marini F, Trimarchi F, Cannavo S (2007) MRI finding of simultaneous coexistence of growth hormone-secreting pituitary adenoma with intracranial meningioma and carotid artery aneurysms: report of a case. *Pituitary* 10(3):299–305
- Cushing H, Eisenhardt L (1938) Meningiomas: their classification, regional behavior, life history, and surgical end results. Charles C Thomas, Springfield, p 785
- Custer B, Longstreth WT, Phillips LE, Koepsell TD, Van Belle G (2006) Hormonal exposures and the risk of intracranial meningioma in women: a population-based case-control study. *BMC Cancer* 6:152
- Drake WM, Grossman AB, Hutson RK (2005) Effect of treatment with pegvisomant on meningioma growth in vivo. *Eur J Endocrinol* 152:161–162
- Gay E, Vuillez JP, Palombi O, Brard PY, Bessou P, Passagia JG (2005) Intraoperative and postoperative gamma detection of somatostatin receptors in bone-invasive en plaque meningiomas. *Neurosurgery* 57:107–113
- González-Agüero G, Ondarza R, Gamboa-Domínguez A, Cerbón MA, Camacho-Arroyo I (2001) Progesterone receptor isoforms expression pattern in human astrocytomas. *Brain Res Bull* 56:43–48
- Gruber T, Dare AO, Balos LL, Lele S, Fenstermaker R (2004) Multiple meningiomas arising during long-term therapy with the progesterone agonist megestrol acetate. Case report. *J Neurosurg* 100:328–331
- Grunberg SM, Weiss MH (1990) Lack of efficacy of megestrol acetate in the treatment of unresectable meningioma. *J Neurooncol* 8:61–65
- Grunberg SM, Daniels AM, Muensch H, Daniels JR, Bernstein L, Kortes V, Weiss MH (1987) Correlation of meningioma hormone receptor status with hormone sensitivity in a tumor stem-cell assay. *J Neurosurg* 66:405–408
- Hsu DW, Efrid JT, Hedley-Whyte ET (1997) Progesterone and estrogen receptors in meningiomas: prognostic considerations. *J Neurosurg* 86:113–120
- Inoue T, Akahira J, Suzuki T, Darnel AD, Kaneko C, Takahashi K, Hatori M, Shirane R, Kumabe T, Kurokawa Y, Satomi S, Sasano H (2002) Progesterone production and actions in the human central nervous system and neurogenic tumors. *J Clin Endocrinol Metab* 87:5325–5331
- Jhawar BS, Fuchs CS, Colditz GA, Stampfer MJ (2003) Sex steroid hormone exposures and risk for meningioma. *J Neurosurg* 99:848–853
- Korhonen K, Raitanen J, Isola J, Haapasalo H, Salminen T, Auvinen A (2010) Exogenous sex hormone use and risk of meningioma: a population-based case control study in Finland. *Cancer Causes Control* 21(12):2149–2156
- McCutcheon IE, Flyvbjerg A, Hill H, Li J, Bennett WF, Scarlett JA, Friend KE (2001) Antitumor activity of the growth hormone receptor antagonist pegvisomant against human meningiomas in nude mice. *J Neurosurg* 94:487–492
- McGowan EM, Russell AJ, Boonyaratankornkit V, Saunders DN, Lehrbach GM, Sergio CM, Musgrove EA, Edwards DP, Sutherland RL (2007) Progestins reinstate cell cycle progression in antiestrogen-arrested breast cancer cells through the B-isoform of progesterone receptor. *Cancer Res* 67:8942–8951
- Michaud DS, Gallo V, Schlehofer B, Tjønneland A, Olsen A, Overvad K, Dahm CC, Kaaks R, Lukanova A, Boeing H, Schütze M, Trichopoulou A, Bamia C, Kyrozi A, Sacerdote C, Agnoli C, Palli D, Tumino R, Mattiello A, Bueno-de-Mesquita HB, Ros MM, Peeters PH, van Gils CH, Lund E, Bakken K, Gram IT, Barricarte A, Navarro C, Dorronsoro M, Sánchez MJ, Rodríguez L, Duell EJ, Hallmans G, Melin BS, Manjer J, Borgquist S, Khaw KT, Wareham N, Allen NE, Tsilidis KK, Romieu I, Rinaldi S, Vineis P, Riboli E (2010) Reproductive factors and exogenous hormone use in relation to risk of glioma and meningioma in a large European cohort study. *Cancer Epidemiol Biomark Prev* 19(10):2562–2569
- Nakamura Y, Shimizu T, Ohigashi Y, Itou N, Ishikawa Y (2005) Meningioma arising in Werner syndrome confirmed by mutation analysis. *J Clin Neurosci* 12:503–506
- Nathoo N, Ugokwe K, Chang AS, Li L, Ross J, Suh JH, Vogelbaum MA, Barnett GH (2007) The role of 111indium-octreotide brain scintigraphy in the diagnosis of cranial, dural-based meningiomas. *J Neurooncol* 81:167–174
- Nicolato A (2004) 111Indium-octreotide brain scintigraphy: a prognostic factor in skull base meningiomas treated with gamma knife radiosurgery. *Q J Nucl Med Mol Imaging* 48(1):26–32
- Piper JG, Follett KA, Fantin A (1994) Sphenoid wing meningioma progression after placement of a subcutaneous progesterone agonist contraceptive implant. *Neurosurgery* 34:723–725
- Pravdenkova S, Al-Mefty O, Sawyer J, Husain M (2006) Progesterone and estrogen receptors: opposing prognostic indicators in meningiomas. *J Neurosurg* 105:163–173
- Riemenschneider MJ, Perry A, Reifenberger G (2006) Histological classification and molecular genetics of meningiomas. *Lancet Neurol* 5(12):1045–1054
- Schoenberg BS, Christine BW, Whisnant JP (1975) Nervous system neoplasms and primary malignancies of other sites. The unique association between meningiomas and breast cancer. *Neurology* 25:705–712
- Shabani N, Kuhn C, Kunze S, Schulze S, Mayr D, Dian D, Ginkelmaier A, Schindelbeck C, Willgeroth F, Sommer H, Jenschke U, Frieske K, Mylonas I (2007) Prognostic significance of oestrogen receptor alpha (ERalpha) and beta (ERbeta), progesterone receptor A (PR-A) and B (PR-B) in endometrial carcinomas. *Eur J Cancer* 43:2434–2444
- Shimizu J, Matsumoto M, Yamazaki E, Yasue M (2008) Spontaneous regression of an asymptomatic meningioma associated with discontinuation of progesterone agonist administration. *Neurol Med Chir* 48:227–230

- Tung L, Mohamed MK, Hoeffler JP, Takimoto GS, Horwitz KB (1993) Antagonist-occupied human progesterone B-receptors activate transcription without binding to progesterone response elements and are dominantly inhibited by A-receptors. *Mol Endocrinol* 7:1256–1265
- Vadivelu S, Sharer L, Schulder M (2010) Regression of multiple intracranial meningiomas after cessation of long – term progesterone agonist therapy. *J Neurosurg* 112(5):920–924
- Van der Lely AJ, Hutson RK, Trainer PJ, Besser GM, Barkan AL, Katznelson L, Klibanski A, Herman-Bonert V, Melmed S, Vance ML, Freda PU, Stewart PM, Friend KE, Clemmons DR, Johannsson G, Stavrou S, Cook DM, Phillips LS, Strasburger CJ, Hackett S, Zib KA, Davis RJ, Scarlett JA, Thorner MO (2001) Long term treatment of acromegaly with pegvisomant, a growth hormone receptor antagonist. *Lancet* 358:1754–1759
- Verheijen FM, Sprong M, Jacobs HM, Donker GH, Amelink GJ, Thijssen JH, Blankenstein MA (2001) Progesterone receptor isoform expression in human meningiomas. *Eur J Cancer* 37:1488–1495
- Wigertz A, Lonn S, Mathiesen T, Ahlbom A, Hall P, Feychting M, the Swedish Interphone Study Group (2006) Risk of brain tumors associated with exposure to exogenous female sex hormones. *Am J Epidemiol* 164:629–636

Meningiomas: Clinical Needs and Molecular Insights

5

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Abstract

Meningiomas are common human brain tumors usually of benign nature. These brain tumors derive from arachnoidal cells associated with brain meninges. The etiology of incidental meningiomas is unknown. A predisposing factor is ionizing radiation. Some genetic syndromes like neurofibromatosis type 2 are associated with meningiomas. Here we highlight the clinical needs and review recent progress made in the field of genomics and proteomics. Finally, we focus on minichromosome maintenance proteins as potential novel markers for early diagnosis of meningioma.

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Introduction

Epidemiology, Etiology and Treatment

Meningiomas are common primary brain tumors accounting for approximately one third of all primary intracranial neoplasms. The incidence is 2–6 per 100,000 per year. There is a female predominance with a ratio of 2:1. Atypical and anaplastic meningiomas however are more frequent in male patients (Campbell et al. 2009). In 2.5% of patients, multiple tumors are already present at diagnosis.

Most meningiomas (90%) are categorized as benign tumors (WHO grade I), with the remaining 10% being atypical or malignant (WHO grades II and III) (Louis et al. 2007). However, the word “benign” can be misleading as they can grow

invasively, recur, affect the brain, can cause disability and even be life threatening. One of the predisposing factors for development of meningiomas is ionizing radiation (Campbell et al. 2009). Survivors of the Hiroshima bombing had a relative risk (RR) of 6.48 in comparison to a non exposed group, to develop a meningioma (Shintani et al. 1999).

Neurofibromatosis type 2 (NF2), an autosomal dominant disease, can develop multiple meningiomas (Whittle et al. 2004). In these patients a mutation in chromosome 22q leads to inactivation of suppressor-gene NF2 leading to a non-functioning proteins: merlin or schwannomin. Other genetic syndromes are associated with development of meningiomas e.g. Cowden syndrome, Gorlin syndrome and Li-Fraumeni syndrome. (Perry et al. 2004; Simon et al. 2007; Riemenschneider et al. 2006).

Surgery is the primary choice of treatment. Complete resection can be achieved in 38–80% of patients, depending on tumor localization. Despite complete resection in approximately 19% of totally removed tumors, a radiological recurrence occurs or a second primary tumor (SPT) develops within 5 years (Stafford et al. 1998). Radiotherapy is used as an additional treatment in WHO II and III meningiomas (Aghi et al. 2009). If surgery is not possible or if the tumor recurs, radiotherapy can be used as primary treatment in WHO I meningiomas (Henzel et al. 2006).

Clinical Considerations and Need for Biomarkers

Several important clinical considerations are relevant in treating meningioma patients. After removal of the primary tumor in 19% of the patients, a radiological recurrence or a second primary tumor (SPT) develops within 5 years. (Kalala et al. 2004). Moreover, in 2.5% of patients multiple tumors are already present at diagnosis. The underlying mechanism for the variability in growth rate within the group of meningiomas that are classified as WHO I is unknown. Annual growth rates can range from 1% to 21% (Firsching

et al. 1990). Rates have been described of an average of 0.24 cm per year (Olivero et al. 1995), 0.03–2.62 cm(3)/year (mean, 0.796 cm(3)/year) (Nakamura et al. 2003). Five years recurrence varies between 7% and 37% (Ayerbe et al. 1999; Kalala et al. 2004).

Since the growth rate of the benign meningiomas varies considerably, the choice of treatment is a continuing source of debate: watchful waiting and repeated scans, re-operation or radiotherapy?

No histological identification of the clinically more aggressive form of WHO I meningiomas is available (Riemenschneider et al. 2006). Because of the unpredictable growth pattern, currently, the follow-up schedule of every meningioma patient includes sequential MRI scans after surgery even if no growth is visible, overdiagnosing a substantial number of patients. If a biomarker or biological profile of the meningioma can be found, predicting the biological behavior, the psychological burden (remaining a brain tumor patient) as well as the medical financial burden for society can be diminished substantially. On the other hand, being able to identify potential aggressive biological behavior in WHO I meningiomas, can result in additional treatment, e.g. radiotherapy.

Molecular Insights

Genomics

Despite almost 40 years of cytogenetic and molecular genetic research, the NF2 tumor suppressor is still the only specific gene which has been shown to be frequently involved in the development of meningiomas. In sporadic meningiomas loss of heterozygosity of chromosome 22q is found in 47–72% (Simon et al. 2007). In only one third of the cases the NF2 gen is inactivated. Probably a different tumor suppressor gen on chromosome 22q plays a role in the tumorigenesis of sporadic meningiomas (Campbell et al. 2009). Several reports suggest that there exist additional loci contributing to the formation of meningiomas. Deletion of chromosome 1p, 3p 6q 9p, 10q and 14q in meningiomas has been described. Deletions of chromosome

1p, 9p, 10q and 14q have been associated with histopathological changes (Perry et al. 2004; Riemenschneider et al. 2006). Cytogenic alterations have been reported in meningioma, loss of 22q, losses of 1p and 14q in which a part of 1p34 might represent a candidate meningioma suppressor gene (Sulman et al. 2004). The development of microarray technology have let the cancer researches to discover genome-wide gene expression of tumor samples compared to their biological control counterparts. To explore the idea that genomic expression discovery predicts pathways and functions behind the biological phenotypes of living systems, several recent studies investigated the molecular pathophysiology of meningiomas. We have recently shown that miRNAs could also contribute to the tumorigenesis of meningiomas; down-regulated microRNA-200a in meningiomas was found to promote tumor growth by reducing E-cadherin and activating the Wnt/ β -catenin signaling pathway (Saydam et al. 2009). One study used microarrays containing 19,200 cDNAs to profile gene expression in ten meningiomas versus normal brain (Fathallah-Shaykh et al. 2003). Their results revealed that 364 genes are consistently up- or down-regulated in at least 5/10 meningiomas as compared with normal brain. Several oncogenic signaling cascade such as Wnt, MAP kinase, PI3K, and notch have been found to be upregulated in meningiomas compared to normal brain. They also found the cell cycle regulators like Cyclin D1, E2F1, BTG2, and ID1, which direct G1/S transition are up-regulated in meningiomas (Fathallah-Shaykh et al. 2003). Another microarray study has pointed out the role of several signaling pathways involved in meningiomas (Fevre-Montange et al. 2009). This microarray-based expression profiling study revealed candidate genes and pathways that may contribute to a better understanding of the recurrence of a benign meningiomas (Fevre-Montange et al. 2009). Several genes involved in cell adhesion (CD44, LOX), cell division (CKS2, BIRC5 and UBE2C), cell differentiation (Notch1) or signal transduction (ARHGAP28) were found to be upregulated, whereas tumour suppressor genes (LR1B, DRR1, PLZF, GPX3, SYNPO, TIMP3 and HOPS) and

genes involved in cell adhesion (PROS1), proliferation (SERPINF1 and PDGFD) and differentiation (AOX1) were downregulated in malignant meningiomas compared to benign tumours (Fevre-Montange et al. 2009). Recently genetic alterations associated with histologic progression of 36 paired meningioma samples were analyzed in 18 patients using 500K SNP genotyping arrays and NF2 gene sequencing (Goutagny et al. 2010). The major event associated with progression to grade III was loss of the CDKN2A/CDKN2B locus on 9p. Meningiomas displayed different patterns of alteration during progression according to their NF2 status; NF2-mutated meningiomas showed higher chromosome instability during progression than NF2-nonmutated meningiomas (Goutagny et al. 2010). In clinical translation, these studies can suggest that the genome-wide arrays could be used as biologic markers in diagnosis and/or prognosis of meningiomas.

Proteomics

Since proteins are the functional output of the genome, largely responsible for the (tumor) cell's biology and functions, and can easily be measured in clinical samples by straight-forward antibody-based detection, they may provide suitable biomarkers. In WHO II and III meningiomas, various changes in protein expressions have been found in targeted analyses of selected proteins of interest using antibodies: loss of Tumor Suppressor in Lung Cancer-1 (TCLC1) (Surace et al. 2004b), MDM2-mediated degradation and p14^{ARF} loss resulting in inactivation of p53 (Amatya et al. 2004), increased telomerase (hTERT) activity (Boldrini et al. 2003) and over-expression of the ribosomal protein S6 kinas (S6K) (Surace et al. 2004a). In addition, meningioma cell growth in vitro revealed a role in the regulation of growth of TGF- β and somatostatin (SST) (Johnson et al. 2004; Arena et al. 2004).

Global profiling of protein expression profiles, proteomics, may offer insight into the molecular mechanisms in different WHO I phenotypes as well as yield novel candidate biomarkers that provide information for both classification and

prognostication of individual tumors. Recent advances in key proteomics tools such as mass spectrometry (MS) and (bio)informatics provide tremendous opportunities for biomarker-related clinical applications (Rajcevic et al. 2009). Current MS platforms allow for the detection, quantitation and identification of hundreds to thousands of peptides and proteins in complex biological and clinical samples. Two general proteomics approaches can be discerned: (1) a high-throughput SELDI/MALDI-TOF mass spectrometry-based approach that aims at discovery of discriminating peptide peak patterns and (2) in-depth proteomics approaches that aim at identifying specific tumor markers (Jimenez et al. 2007). To obtain optimal sensitivity also in the low abundant protein expression range, the latter approach often makes use of multiple dimensions of protein and peptide fractionation prior to analysis by tandem mass spectrometry.

Recently, the high-throughput proteomics approach was employed to investigate the peptide patterns in bone invasive and noninvasive benign meningioma (Wibom et al. 2009). Interestingly, using SELDI-TOF mass spectrometry and multivariate methods, a peptide peak pattern was found that could perfectly differentiate the two groups, in tissues that were indistinguishable at the histology level (Wibom et al. 2009). Identification of the peptides underlying the signature will be important for better understanding of the mechanisms involved in meningioma invasion. MALDI-TOF mass spectrometry applied directly at tissue sections allows for peptide profiling in a spatial context. Agar et al. performed MALDI-TOF imaging mass spectrometry of recurrent and progressive meningiomas versus non progressive high grade meningiomas, high grade gliomas and non tumor brain specimens in an attempt to develop a reference database of classification of brain tumors (Agar et al. 2010). A preliminary classifier was constructed and more work is needed for maturing into a predictive algorithm.

The first study applying proteomics to the analysis of microdissected meningioma cells was reported in 2006 and employed classical in-depth proteomics based on protein fractionation in 2D gels and tandem mass spectrometry analysis of

differential spots (Okamoto et al. 2006). In this study, 24 proteins were identified that distinguished WHO I, II and III meningiomas (Okamoto et al. 2006) including cytoskeletal proteins, antioxidants, proteolysis and mitochondrial proteins. Differential protein abundancies of calponin (down in atypical and anaplastic tumors), galectin 1 (upregulated in atypical and strongly upregulated in anaplastic tumors), tropomyosin (down in anaplastic tumors) and RP/EB family member 1 (down in atypical and anaplastic tumors) were verified by western blotting and immunohistochemistry.

Recently, nanoLC-MS/MS-based proteomics has emerged as a powerful in-depth proteomics approach for profiling large sets of proteins in complex biological samples. Using this approach, we have performed a pilot study aiming to identify differentially expressed proteins between meningiomas of histological type WHO I (no clinical growth), WHO I (clinical significant growth) and WHO II (Peerdeman and Jimenez, manuscript in preparation). Proteomic analysis was performed on six samples (two in each of the above mentioned groups). Total tissue lysates were made and subjected to SDS-PAGE followed by slicing of the whole gel lane, in-gel protein digestion using trypsin and nanoLC-MS/MS on a LTQ-FTMS instrument. Proteins were identified using database searching of the IPI database with the SEQUEST search engine. The total dataset contained 1,734 different protein identified at high confidence. Importantly, unique patterns of protein expression were identified in the different groups that correlated with different biological processes. For example using gene ontology mining we found that a substantial part of the 52 proteins unique for the WHO I-no growth group were involved in the cell cycle whereas the 23 proteins uniquely present in WHO I-growth and WHO II were enriched in the gene ontologies 'response to external stimulus, defense response' 'stress response' adhesion and cell proliferation (Peerdeman and Jimenez, manuscript in preparation). These proteins will be validated using immunohistochemistry of an independent series and may provide candidate biomarkers for both classification and prognostication of individual tumors.

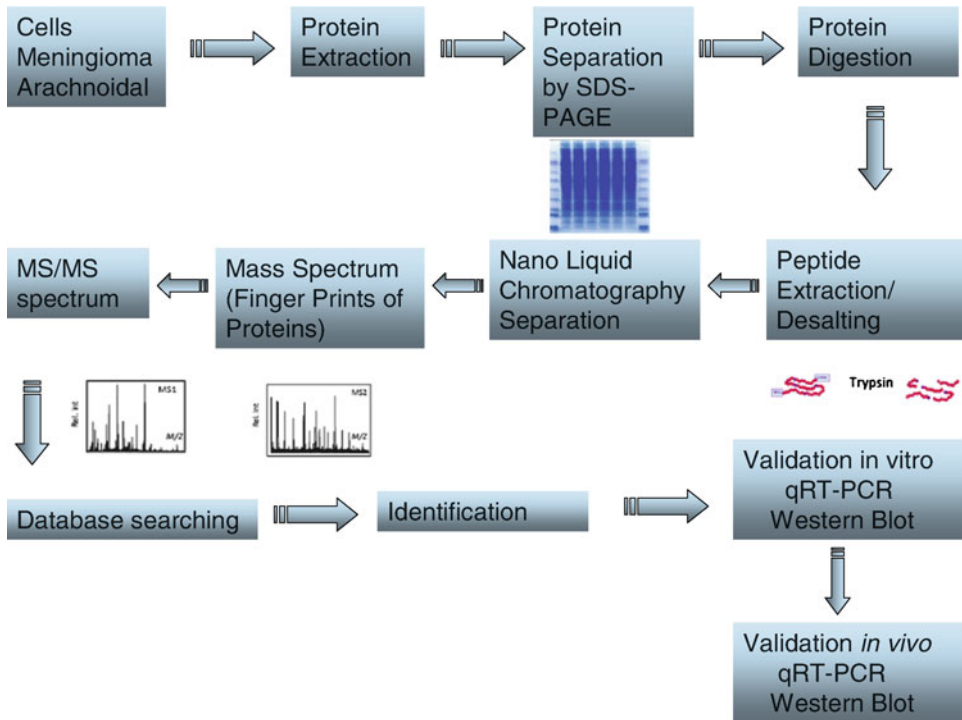


Fig. 5.1 Workflow proteomics combined with biomarker validation

Potential Role of MCMs in Meningioma Diagnosis

Still, knowledge of the biological principles underlying meningioma tumorigenesis is scarce and none of the genetic or protein aberrations has demonstrated sufficient potential to be implemented in routine diagnostics, so as of yet, no markers are available for the prediction of recurrence after resection.

Recent studies have pointed out the role(s) of minichromosome maintenance (MCM) members as diagnostic and/or prognostic markers for several malignancies including colon cancer (Giaginis et al. 2009). MCM proteins include six highly conserved DNA binding members, MCM2 through MCM7. They are considered to function as licensing components for S-phase of cell cycle.

To unravel molecular meningioma tumorigenesis and discover novel protein biomarkers for diagnostic and/or prognostic purposes, we per-

formed in-depth proteomic profiling of meningioma cells compared to normal human primary arachnoidal cells, the cell origin of meningioma tumors. A flow chart of our experimental procedures is shown in Fig. 5.1. We isolated proteins from meningioma cell line SF4433 and primary arachnoidal cells and analyzed the protein profiles by Gel-nanoLC-MS/MS, i.e. protein separation by 1D SDS-PAGE in conjunction with protein identification and quantification by shotgun nanoLC tandem mass spectrometry and spectral counting. Our approach permitted the identification and quantification of over 2,800 proteins; Statistical analysis of meningiomas revealed changes in the expression levels of 281 proteins (Saydam et al. 2009) associated with various biological functions such as DNA replication, recombination, cell cycle, and apoptosis. Network analysis exposed a network of direct relationships between the proteins involved in these cellular processes as depicted in Fig. 5.2. For 133 of these proteins, expression was

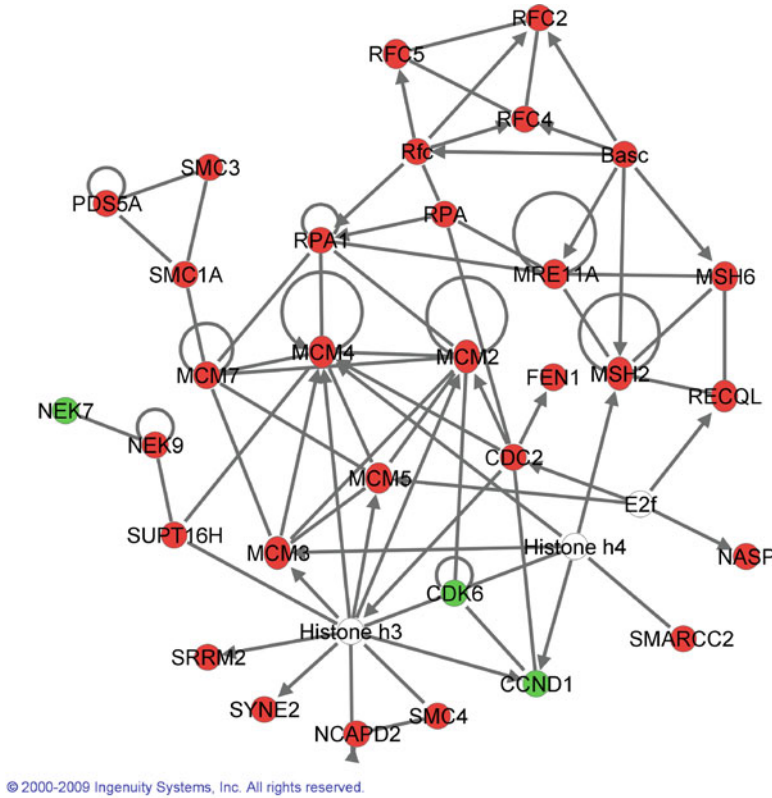


Fig. 5.2 Most significant network for on/of regulated differential proteins

exclusively detected in either arachnoidal or meningioma cells (on/off regulation). Ninety-seven proteins were found to be detected only in meningioma cells, whereas 36 proteins were solely found in arachnoidal cells (Saydam et al. 2009). Representative protein families in this network of on/off regulated proteins are the minichromosome maintenance (MCM) family, the replication factor C subunits (RFC proteins) and the structural maintenance of chromosomes (SMC) family (see Fig. 5.2). All three protein families are present in the nucleus and involved in DNA replication. Additionally several other proteins were found to be up-regulated (4–12-fold) such as replication protein A (RPA1) Epiplakin 1 (EPPK1), Flap endonuclease 1 (FEN1), BAG family molecular chaperone regulator 3 (BAG3), apoptosis inhibitor isoform 5 (API5), protein tyrosine kinase 7 (PTK7), DNA mismatch repair proteins, MSH2 and MSH6, and double-strand

break repair protein MRE11, and DNA repair protein RAD50.

Based on studies emphasizing the functional importance of the MCM family in diagnosis of several malignancies such as MCM2 and MCM5 in colon cancer, MCM2 in breast cancer, and MCM7 in prostate cancer, we decided to follow-up on the differential expression of the MCM family. To validate our proteomics discovery data, we first performed qRT-PCR reactions in meningioma tumor tissue samples WHO grade I, half of which were deleted for the *NF2* locus and half were not as determined by comparative genomic hybridization (14 samples), WHO grade II (seven samples) and WHO grade III (seven samples) compared to arachnoidal tissue controls (from fresh autopsies; three samples). WHO Grade I meningiomas are by far the most common type of meningiomas, representing an initial stage in tumor development, and as such the

normal arachnoidal tissue of origin was deemed the best control to look for changes in proteins related to tumorigenesis. A significant increase was found in *MCM2* (eight fold) and *MCM3* (five fold), *MCM4* (four fold), *MCM5* (four fold), *MCM6* (three fold), *MCM7* (five fold) expressions in meningiomas compared to arachnoidal controls (Saydam et al. 2010). It is possible that MCM family members might be also expressed in arachnoidal cells and tissue at the protein level but their expression levels were below detection limit by MS. A significant change between meningioma Grade I, II, and III samples was not observed. These data suggested that the MCM family proteins are up-regulated in meningiomas and might serve as early diagnostic markers.

Conclusions and Perspective

Despite several comprehensive genomics studies, to date, there is no biomarker in the diagnosis of meningiomas. Those studies have provided a genome-wide expression signature of meningiomas suggesting that a complicated signaling network or chromosomal alteration has been deregulated in the development or progression of this tumor. In near future, it seems possible that microarray based classification of cancer subtypes will provide more specific signature-biomarker in the diagnosis or classification of cancer. To date, only a very limited number studies have applied proteomics to the analysis of meningioma tissues of different WHO grades and growth properties, though promising preliminary results have been obtained. Recent proteomics analyses of cells in culture have identified MCMs as potential diagnostic biomarkers for meningiomas. MCM proteins were exclusively expressed in meningioma cells as compared to arachnoidal cell and MCM transcript levels were highly and significantly up-regulated in human meningioma tumor samples of all WHO stages compared to arachnoidal tissue controls. Therefore, MCM proteins may have potential use as markers for early diagnosis. We envision that in the near future with further application of mass spectrometry-based proteomics, novel protein biomarkers will

be discovered that can differentiate invasive and non-invasive grade I meningiomas.

Acknowledgements This study was supported by the Children's Tumor Foundation 2007-01-043 (O.S.) and NIH/NINDS NS24279 (OS) and by the VUmc-Cancer Center Amsterdam.

References

- Agar NY, Malcolm JG, Mohan V, Yang HW, Johnson MD, Tannenbaum A, Agar JN, Black PM (2010) Imaging of meningioma progression by matrix-assisted laser desorption ionization time-of-flight mass spectrometry. *Anal Chem* 82:2621–2625
- Aghi MK, Carter BS, Cosgrove GR, Ojemann RG, min Hanjani S, Martuza RL, Curry WT Jr, Barker FG (2009) Long-term recurrence rates of atypical meningiomas after gross total resection with or without post-operative adjuvant radiation. *Neurosurgery* 64:56–60
- Amatya VJ, Takeshima Y, Inai K (2004) Methylation of p14(ARF) gene in meningiomas and its correlation to the p53 expression and mutation. *Mod Pathol* 17:705–710
- Arena S, Barbieri F, Thellung S, Pirani P, Corsaro A, Villa V, Dadati P, Dorcaratto A, Lapertosa G, Ravetti JL, Spaziante R, Schettini G, Florio T (2004) Expression of somatostatin receptor mRNA in human meningiomas and their implication in in vitro antiproliferative activity. *J Neurooncol* 66:155–166
- Ayerbe J, Lobato RD, de la Cruz J, Alday R, Rivas JJ, Gomez PA, Cabrera A (1999) Risk factors predicting recurrence in patients operated on for intracranial meningioma. A multivariate analysis. *Acta Neurochir (Wien)* 141:921–932
- Boldrini L, Pistolesi S, Gisfredi S, Ursino S, Lupi G, Caniglia M, Pingitore R, Basolo F, Parenti G, Fontanini G (2003) Telomerase in intracranial meningiomas. *Int J Mol Med* 12:943–947
- Campbell BA, Jhamb A, Maguire JA, Toyota B, Ma R (2009) Meningiomas in 2009: controversies and future challenges. *Am J Clin Oncol* 32:73–85
- Fathallah-Shaykh HM, He B, Zhao LJ, Engelhard HH, Cerullo L, Lichter T, Byrne R, Munoz L, Von RK, Rosseau GL, Glick R, Sherman C, Farooq K (2003) Genomic expression discovery predicts pathways and opposing functions behind phenotypes. *J Biol Chem* 278:23830–23833
- Fevre-Montange M, Champier J, Durand A, Wierinckx A, Honnorat J, Guyotat J, Jouvot A (2009) Microarray gene expression profiling in meningiomas: differential expression according to grade or histopathological subtype. *Int J Oncol* 35:1395–1407
- Firsching RP, Fischer A, Peters R, Thun F, Klug N (1990) Growth rate of incidental meningiomas. *J Neurosurg* 73:545–547
- Giaginis C, Georgiadou M, Dimakopoulou K, Tsourouflis G, Gatzidou E, Kouraklis G, Theocharis S (2009)

- Clinical significance of MCM-2 and MCM-5 expression in colon cancer: association with clinicopathological parameters and tumor proliferative capacity. *Dig Dis Sci* 54:282–291
- Goutagny S, Yang HW, Zucman-Rossi J, Chan J, Dreyfuss JM, Park PJ, Black PM, Giovannini M, Carroll RS, Kalamarides M (2010) Genomic profiling reveals alternative genetic pathways of meningioma malignant progression dependent on the underlying NF2 status. *Clin Cancer Res* 16:4155–4164
- Henzel M, Gross MW, Hamm K, Surber G, Kleinert G, Failing T, Strassmann G, Engenhart-Cabillic R (2006) Stereotactic radiotherapy of meningiomas: symptomatology, acute and late toxicity. *Strahlenther Onkol* 182:382–388
- Jimenez CR, Piersma S, Pham TV (2007) High-throughput and targeted in-depth mass spectrometry-based approaches for biofluid profiling and biomarker discovery. *Biomark Med* 1:541–565
- Johnson MD, Okediji E, Woodard A (2004) Transforming growth factor-beta effects on meningioma cell proliferation and signal transduction pathways. *J Neurooncol* 66:9–16
- Kalala JP, Caemaert J, de Ridder L (2004) Primary resected meningiomas: relapses and proliferation markers. *In Vivo* 18:411–416
- Louis DN, Ohgaki H, Wiestler OD, Cavenee WK, Burger PC, Jouvet A, Scheithauer BW, Kleihues P (2007) The 2007 WHO classification of tumours of the central nervous system. *Acta Neuropathol* 114:97–109
- Nakamura M, Roser F, Michel J, Jacobs C, Samii M (2003) The natural history of incidental meningiomas. *Neurosurgery* 53:62–70
- Okamoto H, Li J, Vortmeyer AO, Jaffe H, Lee YS, Glasker S, Sohn TS, Zeng W, Ikejiri B, Proescholdt MA, Mayer C, Weil RJ, Oldfield EH, Zhuang Z (2006) Comparative proteomic profiles of meningioma subtypes. *Cancer Res* 66:10199–10204
- Olivero WC, Lister JR, Elwood PW (1995) The natural history and growth rate of asymptomatic meningiomas: a review of 60 patients. *J Neurosurg* 83:222–224
- Perry A, Gutmann DH, Reifenberger G (2004) Molecular pathogenesis of meningiomas. *J Neurooncol* 70:183–202
- Rajcevic U, Niclou SP, Jimenez CR (2009) Proteomics strategies for target identification and biomarker discovery in cancer. *Front Biosci* 14:3292–3303
- Riemenschneider MJ, Perry A, Reifenberger G (2006) Histological classification and molecular genetics of meningiomas. *Lancet Neurol* 5:1045–1054
- Saydam O, Shen Y, Wurdinger T, Senol O, Boke E, James MF, Tannous BA, Stemmer-Rachaminov AO, Yi M, Stephens RM, Fraefel C, Gusella JF, Krichevsky AM, Breakefield XO (2009) Downregulated microRNA-200a in meningiomas promotes tumor growth by reducing E-cadherin and activating the Wnt/beta-catenin signaling pathway. *Mol Cell Biol* 29:5923–5940
- Saydam O, Senol O, Schaaïj-Visser TBM, Pham TV, Piersma SR, Stemmer-Rachaminov AO, Wurdinger T, Peerdeman SM, Jimenez CR (2010) Comparative protein profiling reveals minichromosome maintenance (MCM) proteins as novel potential tumor markers for meningiomas. *J Proteome Res* 9:485–494
- Shintani T, Hayakawa N, Hoshi M, Sumida M, Kurisu K, Oki S, Kodama Y, Kajikawa H, Inai K, Kamada N (1999) High incidence of meningioma among Hiroshima atomic bomb survivors. *J Radiat Res (Tokyo)* 40:49–57
- Simon M, Bostrom JP, Hartmann C (2007) Molecular genetics of meningiomas: from basic research to potential clinical applications. *Neurosurgery* 60:787–798
- Stafford SL, Perry A, Suman VJ, Meyer FB, Scheithauer BW, Lohse CM, Shaw EG (1998) Primarily resected meningiomas: outcome and prognostic factors in 581 Mayo Clinic patients, 1978 through 1988. *Mayo Clin Proc* 73:936–942
- Sulman EP, White PS, Brodeur GM (2004) Genomic annotation of the meningioma tumor suppressor locus on chromosome 1p34. *Oncogene* 23:1014–1020
- Surace EI, Lusi E, Haipek CA, Gutmann DH (2004a) Functional significance of S6K overexpression in meningioma progression. *Ann Neurol* 56:295–298
- Surace EI, Lusi E, Murakami Y, Scheithauer BW, Perry A, Gutmann DH (2004b) Loss of tumor suppressor in lung cancer-1 (TSLC1) expression in meningioma correlates with increased malignancy grade and reduced patient survival. *J Neuropathol Exp Neurol* 63:1015–1027
- Whittle IR, Smith C, Navoo P, Collie D (2004) Meningiomas. *Lancet* 363:1535–1543
- Wibom C, Moren L, Aarhus M, Knappskog PM, Lund-Johansen M, Antti H, Bergenheim AT (2009) Proteomic profiles differ between bone invasive and noninvasive benign meningiomas of fibrous and meningothelial subtype. *J Neurooncol* 94:321–331

Meningioma: Urokinase Plasminogen Activator

6

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Abstract

Meningiomas, which arise from the meninges, are by far the most frequently occurring intracranial primary tumors in adults. Histopathological and clinical variants of meningiomas offer new challenges in diagnosis, pathology and treatment modalities. Moreover, recurrence of the more aggressive tumors demand the development of novel, targeted therapeutic strategies. Significant progress has been made in recent years in delineating the molecular mechanisms involved in meningioma formation, growth, and malignant progression. However, many questions remain unanswered. A different approach to the molecular genetics of meningiomas is to examine some aspects of carcinogenesis that are not necessarily specific to meningiomas. To that end, current literature demonstrates that the expression and cleavage of urokinase plasminogen activator (uPA) plays an important role in the tumorigenicity of brain tumors, and high endogenous levels of uPA are associated with advanced metastatic/invasive cancers. Extracellular matrix (ECM) degradation has been implicated as a key step in this process and is mediated by different classes of proteinases among which the uPA system plays a central role. In addition to its role in tumor propagation by binding to uPAR, uPA promotes proteolysis of plasminogen and activates matrix metalloproteinases (MMPs) and transduces signals affecting chemotactic, adhesive and mitogenic properties of cancer cells. Recent studies have revealed that the targeted silencing of uPA and its receptor uPAR impedes the aggressive characteristics of gliomas and meningiomas. In the present article, we review the salient features of pericellular proteases (with a focus on the uPA system), the possible signaling mechanisms involved, and the therapeutic usefulness of uPA knockdown in the management of meningiomas.

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Introduction

Central nervous system tumors are initiated when normal cells of the brain or the spinal cord grow uncontrollably and form a mass. Approximately 52,000 people are diagnosed each year with a primary brain tumor (benign or malignant) (American Cancer Society 2009). More than 100 types of primary brain tumors exist, but these tumors can be divided into two broad categories—gliomas and non-gliomas. In most cases, the cause of a brain tumor is unknown and there are no known means of preventing a brain tumor. Nevertheless, a list of risk factors including age, gender, heredity, exposure to ionizing radiation, and environmental and occupational hazards may contribute to tumorigenesis. Meningiomas are non-glioma brain tumors of which a large proportion are asymptomatic and localized. However, malignant meningiomas pose severe challenges to existing management strategies because of their invasive characteristics. Research for markers in the tumor tissue that could make meningiomas easier to diagnose and grade is ongoing. Targeted molecular therapies that can interfere with the aggressive behavior of meningiomas might also inhibit benign tumors or at least serve as adjuvants to existing strategies.

Meningioma & Grades

Meningiomas represent approximately 30% of all primary brain tumors and occur more frequently in middle-aged females than males (Campbell et al. 2009). These tumors arise from the mesoderm or meningeal cells. Progressive enlargement of the tumor leads to focal or generalized seizures and affects cranial nerves and blood vessels. In a few cases, meningioma growth can also extend into the bones of the head and face and produce perceptible changes. Most meningiomas are non-malignant tumors that grow slowly. However, some meningiomas do grow rapidly or have sudden growth spurts. The growth rate of a meningioma or how long a specific

tumor has been forming prior to diagnosis cannot be predicted. Meningiomas are graded from low to high. In general, lower grades correlate with a lower risk of recurrence or aggressive growth.

Although meningioma grading is continuously evolving, meningiomas are classified by the World Health Organization (WHO) as benign, atypical or anaplastic (Louis et al. 2000). Approximately 80–90% of meningiomas are benign with different histologic variants (Raizer 2010). While many benign meningiomas are associated with a relatively good prognosis and can be resected, growth along surgically inaccessible locations (e.g., venous sinuses and the skull base) may lead to recurrence and often cause difficulties in the management of these tumors. Tumors invading the dural sinuses, bone and vital vascular structures, such as the cavernous sinus or surrounding cranial nerves, are operable but carry a significant risk of morbidity. The recurrence rate has been reported to be between 3% and 12% within 5 years among the benign tumors (Johnson and Toms 2005; Perry et al. 2004). Approximately 5–20% of meningiomas are classified as atypical or WHO grade 2 (Raizer 2010) and have a recurrence rate of 29–40% within 5 years of surgical resection (Perry et al. 2004). Atypical tumors are histologically characterized by frequent mitoses, nuclear hypertrophy, a patternless sheet-like growth pattern, and necrotic foci. WHO grade 3 tumors comprise 1–3% of meningiomas and are characterized by poor prognosis (Raizer 2010). These tumors grow more aggressively and have recurrence rates of 50–80% (Perry et al. 2004). Despite this grading system, wide heterogeneity among benign tumors is responsible for a wide range of clinical and pathological variables in patients. To overcome this shortcoming, recent studies have used gene expression profiling and biomarker characterization to identify genes that are differentially expressed between normal brain tissues and benign and malignant meningiomas (Watson et al. 2002). However, it is still necessary to determine additional prognostic markers and identify new targets.

Therapies for Meningioma

Current therapies for meningiomas involve surgery, radiation treatment, and stereotactic radiosurgery. Meningiomas tend to have well-defined edges and do not invade surrounding tissue. As such, complete removal is sometimes possible and makes surgery the primary treatment of choice; however, the use of fractionated radiotherapy and stereotactic single-dose radiosurgery is increasing (Norden et al. 2009). The standard treatment is to remove the tumor from the portion of the lining of brain or spinal cord where it is attached along with any bone involved. Treatment options for surgically unreachable, recurrent, or malignant meningiomas remain inadequate. Considering the higher rate of recurrence in aggressive meningiomas even after surgical excision and the possibility that the recurring tumor is more aggressive than the original one, surgery should be combined with post-operative radiotherapy to improve local tumor elimination (McGregor and Sarkar 2009). Development of effective combination chemotherapy and radiation strategies has been hindered, in part, by our incomplete understanding of the signals influencing meningioma. To date, existing chemotherapy regimens have had very limited success in treating meningiomas (Alexiou et al. 2010; Johnson and Toms 2005). Data from small clinical trials and case studies suggest that most chemotherapeutic agents have minimal activity against meningiomas. However, in instances where surgery and further radiation are no longer feasible or safe, chemotherapy is utilized to destroy any tumor remaining after surgery, slow tumor growth or reduce symptoms.

During the early twentieth century, neural tissue was thought to be relatively resistant to the ill effects of ionizing radiation. As a result, patients were often exposed to relatively high doses of x-rays from diagnostic examinations and therapeutic treatment. However, radiation-induced meningiomas (RIM) resulting from childhood treatment for primary intracranial tumor or *Tinea capitis*, exposure to dental x-rays, and exposure to radioactivity have been reported.

Furthermore, meningiomas are usually not very sensitive to radiotherapy. Recent studies also indicate that radiation may increase the invasive potential of meningioma cells by stimulating the expression of proteolytic molecules, including uPA (Kargiotis et al. 2008). Radiation activates auto-phosphorylation of EGFR and enhances the activity of protein tyrosine kinases, leading to the activation of Ras proteins and downstream signaling pathways involving Akt, ERK1/2 and JNK (Valerie et al. 2007). Further, cell growth regulating pathways, such as RAF-1, MEK1/2, ERK1/2, p38 and PI3-K/Akt, are shown to be downstream responses of radiation-induced activation of the EGFR family of receptors (Valerie et al. 2007). More frequently, anaplastic meningiomas reappear after post-operative radiotherapy and cannot be controlled even by gamma knife radiosurgery. Considering that these therapeutic options are not very effective, molecular targets represent a promising therapeutic strategy for the management of atypical and anaplastic meningiomas.

A factor limiting progress in the design and development of more successful treatment for meningiomas is the lack of variety in cell lines and animal models. There is a necessity for animal models that better simulate the genetic changes in meningiomas for *in vitro* and *in vivo* studies. Many of the existing meningioma cell lines are derived from malignant meningiomas and probably contain culture-induced artifacts. However, hTERT-mediated immortalization strategies have proven to be promising in generating benign meningioma cell lines.

Extracellular Proteases in Cancer

Tumor progression depends on the tendency of the cancer cells to invade and promote angiogenesis. The entire process of tumor metastasis is a complex cascade of events comprising dislodgment of cancer cells from the primary tumor site and migration to distant sites where they form secondary tumors. Most of the deaths related to cancer result from metastatic cancer. Despite extensive, ongoing cancer research, the entire process of metastasis is still poorly understood.

Metastasis is a series of well-organized events that can be described in four stages (Bogenrieder and Herlyn 2003): (1) dislodgement of cancer cells from primary tumor mass and invasion/migration through the basement membrane and extracellular matrix (ECM) surrounding the tumor site; (2) intravasation of the detached tumor cells into the blood vessels; (3) adhesion of the metastatic tumor cells to the underlying target organ and invasion through the surrounding ECM of the target organ; and (4) proliferation and establishment at the metastatic site. The metastatic potential of any cancer depends on the ability of the cancer cells to organize and complete the entire process of events (Wong et al. 2001).

The destruction of ECM components by cancer cells is one of the major processes involved in metastasis. Dissolution of the ECM components is carried out through coordinated action of several extracellular proteases secreted either by cancer cells or by cells surrounding the tumor. These proteases not only clear the path for cancer cells to invade and migrate, but also influence the adhesion, differentiation and proliferation of cancer cells. In contrast to tight regulatory control of these proteases in the cells grown under normal physiological conditions, the cells in the tumor microenvironment tend to lose this regulatory ability. The proteases engaged in the process of tumor metastasis are primarily uPA, MMPs and cathepsins.

inactive plasminogen into active plasmin is the primary catalytic role of active uPA. Active plasmin has a broader specificity than uPA and can directly act upon various ECM components, activates other ECM-degrading enzyme precursors (e.g., uPA and MMPs), and converts them into active forms. In turn, these activated MMPs play an important role in remodeling and dissolution of the ECM (Schmitt et al. 2000). Apart from its role as an extracellular enzyme, various studies have shown the potential of uPA, mostly by its interaction with uPAR, in activating the release of growth factors, which enhance tumor growth by stimulating cell proliferation, inhibiting apoptosis, enhancing migration and promoting angiogenesis. Past *in vitro* and *in vivo* experimental studies provide evidence for the role of uPA/uPAR system in tumor progression, invasion and angiogenesis. It is not only the activation of other extracellular proteases by the uPA/uPAR system that makes them crucial for cancer progression, but also their role in mediating several intracellular signaling via their interaction with other growth factors and other cell surface receptors such as integrins. Signaling pathways activated upon binding of uPA to uPAR include the mitogen-activated protein kinase (MAPK), signal transducer and activator of transcription (Stat), Jak/Stat, and Ras/extracellular signal-regulated kinase (ERK) pathways, which lead to cell proliferation, migration, and invasion (Rao 2003).

uPA/uPAR System in Cancer

Among the extracellular proteases, the uPA system plays a vital role in the destruction of basement membrane and connective tissue surrounding the tumor. A direct correlation between uPA expression levels and tumor grade has been demonstrated in various cancers (Duffy 2004). The plasminogen activator system is comprised of uPA, its receptor uPAR, and its two serpin inhibitors, plasminogen activator inhibitor-1 (PAI-1) and plasminogen activator inhibitor-2 (PAI-2). uPA is secreted in a catalytically inactive form (pro-uPA; molecular weight of ~55 kDa), which is activated upon binding to uPAR. Conversion of

uPA in Brain Cancers

Malignant cancers of the central nervous system (CNS) invade adjacent brain structure by degrading the ECM. Since the identification of tumor cells synthesizing uPA, the body of evidence relating uPA fibrinolytic activity with the malignant biological behavior of tumors has been increasing. The uPA-mediated fibrinolytic system and its role in tumor metastasis are well documented in brain, gastric, lung, breast, prostate, and pancreatic carcinomas. Consistent with its role in the spread of cancer, uPA has been shown to be a prognostic marker in a variety of malignancies including brain tumors. Comparison

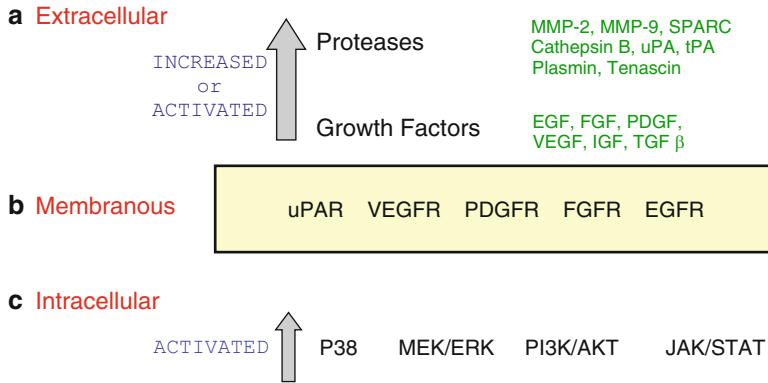
of the uPA levels in benign and malignant primary brain tumors with normal brain tissue clearly demonstrate that uPA levels are significantly elevated in glioblastoma multiforme (grade IV) as compared to normal brain tissue (Arai et al. 1998; Yamamoto et al. 1994; Zhang et al. 2000). Hsu et al. (1995) found that being over the age of 50 years and having a grade IV glioma, which showed the highest amount of uPA expression, are indicators of a shorter survival period for patients. These studies demonstrate that uPA is a biomarker for malignant brain tumors. Researchers compared the expression levels of uPA with the survival rate of cancer patients. Yamamoto et al. (1994) showed a strong immunoreactivity with uPA in tumor cells and vasculature of glioblastomas and anaplastic astrocytomas as compared to low immunoreactivity for uPA in low-grade gliomas and normal brain tissues. The location of uPA in tumor cells as well as endothelial cells, and the preferential localization of uPA near vascular proliferation and at the leading edge of the tumor are indicative of its role in invasion as well as angiogenesis (Arai et al. 1998; Yamamoto et al. 1994; Zhang et al. 2000). During the process of neovascularization, proteolytic degradation of surrounding ECM by tumor cells due to elevated expression of uPA results in remodeling of the ECM, which leads to the release of various growth factors directing the migration of endothelial cells towards the source of stimulus.

Invasiveness of Meningioma

In contrast to the extensive understanding of the invasiveness of systemic malignancies, and even brain tumors such as malignant gliomas, little is known about molecular pathogenesis or the critical molecular changes that drive meningioma tumor progression (Johnson and Toms 2005; Perry et al. 2004). The concept that all meningiomas behave as extra-axial tumors, with little impact to adjacent brain cortex, does not appear fully satisfactory. Similar to other neoplasms, meningiomas may progress from benign to ana-

plastic tumors. As with other malignancies, invasion is a hallmark feature of malignant progression in meningiomas. However, paradoxically, benign meningiomas, which typically have a broad base of attachment to the dura mater, frequently indent and occasionally invade into the brain. Because these invasive cells contribute to the failure of complete tumor resection, studies aimed at the elucidation of the molecular mechanisms that regulate the invasion process in meningiomas are highly imperative. The molecular rationalization for the ability of meningiomas to penetrate the adjacent layers is yet to be determined. Different studies demonstrate the interaction between meningiomas and adjacent brain tissue by means of angiogenic factors (Yoshioka et al. 1999), matrix metalloproteinases (Paek et al. 2002), increased proliferative activity (Aguilar et al. 2003), cortical invasion (Salpietro et al. 1994), and extensive signaling cascades (Scheme 6.1). Given the commonly invasive nature of meningiomas, it is not surprising that alterations in the regulation of ECM proteins have been found, including overexpression of matrix metalloproteinases, such as MMP-9 and MMP-2 (Siddique et al. 2003). In addition to the histomorphologic changes, the increased invasive potential of tumor cells has been used as an indicator of malignancy in meningiomas. Tumor invasiveness and the presence of peritumoral edema are the two major factors that determine the clinical management of meningioma. There are some biological, physical, and chemical factors that contribute to the peritumoral edema of meningioma, which may relate to their invading potential, interaction with adjacent cortex and producing hematoencephalic barrier breakage (Simis et al. 2008).

Invasive meningiomas overexpress MMP-2, MMP-9, SPARC, tenascin and stremelysin-3 while malignant meningiomas overexpress tissue factor pathway inhibitor-2 (TFPI-2). TFPI-2 is an ECM-associated, Kunitz-type serine protease inhibitor secreted by all vascular cells. It plays a role in tumor invasion and metastasis, presumably by plasmin-mediated matrix remodeling. Previous studies have shown that the expression of TFPI-2 is lost in high-grade tumors, including



Scheme 6.1 Meningioma invasiveness – Different factors contribute to meningioma invasiveness (a) ECM Degradation: Pericellular proteolysis facilitated by a mixture of extracellular proteases and various growth factors in

invasive meningiomas (b) Receptors: Wide range of receptors function on meningioma cells to transduce signals (c) Signaling: Components of MAP Kinase and survival pathways are constitutively active to promote proliferation

gliomas. Transfection of TFPI-2 mRNA into the human meningioma cell line IOMM-Lee inhibited tumor growth *in vitro* and *in vivo*, which suggests that TFPI-2 may have therapeutic potential for malignant meningiomas (Kondraganti et al. 2006).

It has been shown that MMP-2 and MMP-9 encoding mRNA is expressed in the majority of tumors in a consecutive series of meningiomas. Although MMP-2 expression was similar, MMP-9 expression seemed to increase from low levels in tumors that did not extend into the arachnoid membrane to higher levels in tumors that traversed the arachnoid to reach the pia mater (Nordqvist et al. 2001). MMP-9 expression was highest in anaplastic and aggressive tumors. The findings indicate that MMP-9 expression and brain invasion are relevant mechanisms that must be addressed in the treatment of aggressive and malignant meningiomas (Nordqvist et al. 2001). Even though quantification of the expression of extracellular matrix-degrading proteins revealed no strict correlation with tumor type or grade, the fact remains that meningioma cells express invasion-mediating factors, which are lacking in normal arachnoidal cells. Our previous study on meningioma cells demonstrated that RNAi treatment for MMP-9 and uPAR successfully regressed pre-established meningiomas in nude mice as well as decreased the phosphorylation status of

ERK1, ERK2, Akt, mTOR and FAK. These results suggest “cross talk” among these pathways and simultaneous upregulation of these pathways in meningioma cells (Tummalapalli et al. 2007). The tissue plasminogen activator (tPA) content of peritumoral tissue adjacent to the meningioma is significantly less whereas it remains unaffected in gliomas. Since tPA is found in the periphery of malignant tumors where there is an advancing tumor edge, this finding lends weight to the belief that plasminogen activators play a role in tumor invasiveness (Goh et al. 2005). In contrast, benign meningiomas, which are known to have well-defined capsules, have significantly less tPA in the peritumoral tissue (Goh et al. 2005).

The protein levels of cathepsins B and L were found to be significantly higher in atypical than in benign meningiomas. In contrast, their mRNA levels did not differ, indicating that the synthesis of cathepsins was accelerated at the translational level. Protein and mRNA levels of stefin B, but not cystatin C, were significantly lower in atypical than in benign meningiomas. Generally, the expression of cathepsins and inhibitors was not different between central and peripheral meningioma tissue or between histological subtypes of meningiomas. However, the exception was cathepsin L, which was significantly lower in transitional meningiomas. Using immunohistochemical

analysis and specific monoclonal antibodies, it has been shown that cathepsin B and cathepsin L levels were significantly higher in invasive types of benign meningiomas. Higher protein levels of cathepsins B and L and lower mRNA levels of stefin B can be used as potential diagnostic markers to distinguish histomorphologically benign but invasive meningiomas from histomorphologically clear benign tumors (Trinka et al. 2005).

Matrix degradation and endocytosis of cell debris are potential targets of anti-invasive therapies, particularly in atypical and anaplastic meningiomas. Other invasion inhibitors have been studied for the treatment of systemic cancers and gliomas, and some of these agents may have a therapeutic effect in meningiomas (Nakada et al. 2007; Rao 2003). One such agent is cilengitide, an inhibitor of $\alpha V\beta 3$ and $\alpha V\beta 5$ integrins, both of which are expressed in meningiomas and are potentially important for angiogenesis and invasion (Nabors et al. 2007).

Role of uPA in Meningioma

Data accumulated during the past several decades have made increasingly clear that uPA has a multifunctional role in neoplastic evolution and affects tumor angiogenesis, malignant cell proliferation, adhesion, migration, intravasation and growth at the metastatic site (Zhang et al. 2000; Duffy 2004). In agreement with its role in cancer progression and metastasis, increased expression of uPA is correlated with poor prognosis. uPA has also been demonstrated to be the most potent independent predictive factor described thus far and as a suitable target for anti-cancer therapy. Several therapeutic approaches aimed at inhibiting uPA and/or uPAR functions have been shown to possess anti-tumor effects in xenograft models. Some of these approaches include selective inhibitors of uPA activity, antagonist peptides, monoclonal antibodies and gene therapy techniques silencing uPA/uPAR expression. uPA and uPAR have been implicated in tumor cell invasion, migration, proliferation and metastasis in different cancers including glioma, meningioma, breast and prostate cancers. In addition to being a

clinically useful prognostic factor allowing estimates of the course of cancer, uPA may also serve a role in predicting response to systemic therapy.

The secretion of uPA is a constitutive property of malignant cells and is induced by some transformation-associated gene products. Alternatively, malignant cells possess altered mechanisms to control uPA enzyme activity. As stated earlier, uPA expression is correlated with malignancy in meningiomas (Nagashima et al. 2006). These results were not surprising because similar findings have been reported previously. Regardless of the tumor grade, invasive meningiomas express uPA as compared to their non-invasive counterparts. Increased uPA expression was observed in meningiomas invading not only into the brain parenchyma but also into other tissues, such as bone and dura mater. Thus, uPA may be a useful marker for assessing the invasive potential of meningiomas (Kitange et al. 2000). In another study, uPA activity and PAI antigen levels were found to be higher in benign meningiomas as compared to the normal brain tissue (Siddique et al. 2003). Although the findings suggest that uPA may be involved in the progression of meningiomas, the mechanisms of its regulation and possible functional roles remain obscured. Recently, we have shown that the simultaneous downregulation of uPA and uPAR decreases even radiation-induced migration, invasion and angiogenesis in meningioma (IOMM-Lee) cells *in vitro* and suppresses tumor growth in subcutaneous xenograft tumors *in vivo*. Our data demonstrated the importance of the uPA system in malignant meningioma proliferation (Kargiotis et al. 2008). Further, downregulation of these molecules led to pro-apoptotic signaling in different cancer cells (Kargiotis et al. 2008; Tummalapalli et al. 2007).

Meningioma growth is sustained by the dysregulated expression of steroid hormones, growth factor and their receptors, and activation of signal transduction cascades (Simon et al. 2007). Hormone receptors for estrogen, progesterone, growth hormone, somatostatin, and androgens may be expressed by meningioma cells. Progesterone receptors (PRs), which are expressed by approximately two-thirds of benign meningiomas, tend to

be lost as malignancy increases (Norden et al. 2009). Overexpression of different growth factors along with their receptors and signal transduction pathways (such as the Ras/MAPK, PI3-K/Akt, and PLC-g1-PKC) have been implicated in meningiomas but their relative significance is largely unknown (Johnson and Toms 2005; Sanson and Cornu 2000). As a result, the most important molecular targets may remain unknown.

Meningiomas express platelet-derived growth factor (PDGF) and its receptor (PDGFR), epidermal growth factor (EGF) and its receptor (EGFR), transforming growth factor- β (TGF- β , an EGFR ligand), and insulin-like growth factor II (IGF-II) and its receptor IGF binding protein 2 (IGFBP2) (Simon et al. 2007). The overexpression of ErbB 2 implicates the possible formation of their homodimers and may be a characteristic of benign tumorigenesis. Anti-ErbB 2 therapy has been utilized in the treatment of malignant tumors that overexpress ErbB 2. On the other hand, uPA reporter gene construct with expression vectors for ERK-specific protein phosphatase MKP-1 showed that a Ras/Raf-1/MEK/ERK-2/JunD pathway is induced by FGF-2 and 12-O- tetradecanoylphorbol-13-acetate, leading to the activation of the uPA gene which suggest that the growth factor mediated autocrine mechanisms may be involved in the induction of uPA and progression of meningioma (Besser et al. 1995).

Preclinical data indicate that activated MAP kinase pathways are important in meningioma pathogenesis (Johnson and Toms 2005). It has been shown that PDGF-BB and cerebrospinal fluid (which contains several growth factors) stimulate meningioma cell proliferation, in part via activation of a Ras-RAF-1-MEK-1-MAPK/ERK pathway (Johnson and Toms 2005). Signal transduction from activated receptor tyrosine kinases, such as EGFR and PDGFR, is mediated in part by the Ras/RAF/MAPK and the PI3K/Akt pathways in meningiomas. Furthermore, p38 MAP kinase inhibitor and ERK1/2 inhibitor, but not SAPK/JNK inhibitors, were shown to inhibit radiation-induced uPA expression in meningioma cells (Kargiotis et al. 2008). These results confirm that radiation-induced uPA expression is mediated through the EGFR/p38 and EGFR/

ERK-dependent MAPK pathways in meningioma cells. Blockade of uPA induction and ERK activation by the MAPK/ERK inhibitor PD98059 also demonstrates the necessity of activated MAPK pathways for the expression of uPA and the sustenance of malignant meningioma. In contrast, uPA-induced cell proliferation through ERK1/2 and p38 MAPK-mediated pathways via the activation of EGFR reveal the distinct biologic response that can be mediated by uPA and probably account for the activated MAPK pathways in meningiomas (Nicholl et al. 2005).

Meningiomas are known to have altered expression of hormones and their receptors, and these hormones are known to change the critical balance of uPA and PAI as well as uPA secretion (Murdoch et al. 1999), thereby making it possible that the hormones could influence the increased activity of uPA in invasive meningiomas as well. In meningiomas, the proportion of methylation of O⁶-methylguanine-DNA methyltransferase (MGMT) seems to be very low and unrelated to the grade of malignancy. Interestingly, the hypomethylation of uPA promoter was reported in metastatic and invasive human breast and prostate cancer cell lines, thereby tempting us to speculate on the possibility of hypomethylation of the uPA promoter in meningiomas.

Although the inhibitors and monoclonal antibodies that can interfere with the interaction of uPA with uPAR have proven successful, additional considerations have to be made because catalytically inactive uPA has also been reported to induce mitogenic effects in osteosarcoma cells by a mechanism that involved the interaction of uPA with the high affinity binding site of uPAR (Koopman et al. 1998). In addition, it has been reported that active site-inhibited uPA was able to elicit a mitogenic response in smooth muscle cells independent of high affinity binding to uPAR (Stepanova et al. 1999). The low affinity binding of uPA to the cell membrane suggests that uPA-induced signal transduction could be mediated by a hitherto unidentified membrane-associated protein. The pathway of signal transduction that is induced by uPA and the changes in cellular phenotype or response to these signals show a high degree of diversity in different cell

types, making a knockdown strategy utilizing antisense nucleotides or RNAi effective for uPA-targeted molecular therapies in the management of various cancers.

Marked differences in the E-cadherin and β -catenin expression corresponding to different stages of meningioma pathogenesis have been reported (Zhou et al. 2010). Modulation of Wnt signaling by the decrease in E-cadherin and β -catenin expression cause development of peritumoral edema, leading to the harmed cell-to-cell junctions and damaging the tumor-brain interface and the blood-brain barrier. Loss of E-cadherin-mediated cell-cell adhesion and associated expression of proteolytic enzymes, uPA in particular, characterize the transition from benign lesions to invasive, metastatic tumor—a rate-limiting step in the progression *in vivo* (Kleiner et al. 2007). Consequently, meningioma cells could infiltrate brain tissue and increase brain edema. Spread of meningioma tumor suggests specific migratory and invasive capabilities. Meningiomas are highly vascular tumors that derive their blood supply predominantly from meningeal vessels supplied by the external carotid artery with additional supply from cerebral pial vessels. Inhibitors of VEGF and VEGFR and the findings that uPA knockdown inhibits angiogenesis provide promising agents for meningioma treatment, with the potential to not only inhibit angiogenesis, but to also decrease peritumoral edema (Provias et al. 1997).

Conclusion and Future Directions

Substantial evidence suggests that studying radiation-induced meningioma, cell cycle dysregulation, telomere dynamics, and genetic instability might identify genes and proteins related to meningioma initiation and progression. Progress in identifying alternative forms of therapy in patients with recurrent meningiomas has been limited by poor understanding of the molecular pathogenesis of meningiomas and the critical molecular changes driving tumor growth. Microarray analysis on a large scale with significant sample sizes could bring about meaningful insights into tumorigenesis. Thus far,

chemotherapies have had minimal success and hormonal therapies have proven to be largely ineffective in meningioma. However, uPA gene therapy has been proven as a dependable molecular target to a partial extent in meningioma. Yet, to form a complete picture, very extensive analyses of all pericellular proteases and their activation and inhibition would be necessary. Some evidence indicates that analyzing genes and mechanisms involved in metastasis and invasion of cancers in general may yield important information relevant for the understanding of meningioma spread (and the treatment thereof). As the numbers of potential molecular targets, targeted agents, and drug combinations increase dramatically, our ability to select only the most promising ones for clinical trials becomes ever more imperative. This is particularly critical for meningiomas because the relevant patient population is small and limited resources have been devoted to the study of this tumor. There is significant experience with targeted molecular agents for systemic cancers and malignant gliomas, which can potentially be translated into effective strategies for meningiomas.

Acknowledgements We thank Shellee Abraham for manuscript preparation. We thank Diana Meister and Sushma Jasti for manuscript review.

References

- Aguiar PH, Tsanaclis AM, Tella OI Jr, Plese JP (2003) Proliferation rate of intracranial meningiomas as defined by the monoclonal antibody MIB-1: correlation with peritumoural oedema and other clinicoradiological and histological characteristics. *Neurosurg Rev* 26:221–228
- Alexiou GA, Gogou P, Markoula S, Kyritsis AP (2010) Management of meningiomas. *Clin Neurol Neurosurg* 112:177–182
- American Cancer Society (2009) Cancer facts & figures. Atlanta: American Cancer Society
- Arai Y, Kubota T, Nakagawa T, Kabuto M, Sato K, Kobayashi H (1998) Production of urokinase-type plasminogen activator (u-PA) and plasminogen activator inhibitor-1 (PAI-1) in human brain tumours. *Acta Neurochir (Wien)* 140:377–385
- Besser D, Presta M, Nagamine Y (1995) Elucidation of a signaling pathway induced by FGF-2 leading to uPA gene expression in NIH 3T3 fibroblasts. *Cell Growth Differ* 6:1009–1017

- Bogenrieder T, Herlyn M (2003) Axis of evil: molecular mechanisms of cancer metastasis. *Oncogene* 22:6524–6536
- Campbell BA, Jhamb A, Maguire JA, Toyota B, Ma R (2009) Meningiomas in 2009: controversies and future challenges. *Am J Clin Oncol* 32:73–85
- Duffy MJ (2004) The urokinase plasminogen activator system: role in malignancy. *Curr Pharm Des* 10:39–49
- Goh KY, Poon WS, Chan DT, Ip CP (2005) Tissue plasminogen activator expression in meningiomas and glioblastomas. *Clin Neurol Neurosurg* 107:296–300
- Hsu DW, Efrid JT, Hedley-Whyte ET (1995) Prognostic role of urokinase-type plasminogen activator in human gliomas. *Am J Pathol* 147:114–123
- Johnson M, Toms S (2005) Mitogenic signal transduction pathways in meningiomas: novel targets for meningioma chemotherapy? *J Neuropathol Exp Neurol* 64:1029–1036
- Kargiotis O, Chetty C, Gogineni V, Gondi CS, Pulkuri SM, Kyritsis AP, Gujrati M, Klopfenstein JD, Dinh DH, Rao JS (2008) uPA/uPAR downregulation inhibits radiation-induced migration, invasion and angiogenesis in IOMM-Lee meningioma cells and decreases tumor growth in vivo. *Int J Oncol* 33:937–947
- Kitange G, Tsunoda K, Anda T, Nakamura S, Yasunaga A, Naito S, Shibata S (2000) Immunohistochemical expression of Ets-1 transcription factor and the urokinase-type plasminogen activator is correlated with the malignant and invasive potential in meningiomas. *Cancer* 89:2292–2300
- Kleiner S, Faisal A, Nagamine Y (2007) Induction of uPA gene expression by the blockage of E-cadherin via Src- and Shc-dependent Erk signaling. *FEBS J* 274:227–240
- Kondraganti S, Gondi CS, Gujrati M, McCutcheon I, Dinh DH, Rao JS, Olivero WC (2006) Restoration of tissue factor pathway inhibitor inhibits invasion and tumor growth in vitro and in vivo in a malignant meningioma cell line. *Int J Oncol* 29:25–32
- Koopman JL, Slomp J, de Bart AC, Quax PH, Verheijen JH (1998) Mitogenic effects of urokinase on melanoma cells are independent of high affinity binding to the urokinase receptor. *J Biol Chem* 273:33267–33272
- Louis DN, Scheithauer BW, Budka H, von Deimling A, Kepes JJ (2000) Meningiomas. In: Kleihues P, Cavenee WK (eds) *Pathology and genetics of tumours of the nervous system*. IARC Press, Lyon, pp 176–184
- McGregor JM, Sarkar A (2009) Stereotactic radiosurgery and stereotactic radiotherapy in the treatment of skull base meningiomas. *Otolaryngol Clin North Am* 42:677–688
- Murdoch J, Van Kirk EA, Murdoch WJ (1999) Hormonal control of urokinase plasminogen activator secretion by sheep ovarian surface epithelial cells. *Biol Reprod* 61:1487–1491
- Nabors LB, Mikkelsen T, Rosenfeld SS, Hochberg F, Akella NS, Fisher JD, Cloud GA, Zhang Y, Carson K, Wittemer SM, Colevas AD, Grossman SA (2007) Phase I and correlative biology study of cilengitide in patients with recurrent malignant glioma. *J Clin Oncol* 25:1651–1657
- Nagashima G, Fujimoto T, Suzuki R, Asai J, Itokawa H, Noda M (2006) Dural invasion of meningioma: a histological and immunohistochemical study. *Brain Tumor Pathol* 23:13–17
- Nakada M, Nakada S, Demuth T, Tran NL, Hoelzinger DB, Berens ME (2007) Molecular targets of glioma invasion. *Cell Mol Life Sci* 64:458–478
- Nicholl SM, Roztocil E, Davies MG (2005) Urokinase-induced smooth muscle cell responses require distinct signaling pathways: a role for the epidermal growth factor receptor. *J Vasc Surg* 41:672–681
- Norden AD, Drappatz J, Wen PY (2009) Advances in meningioma therapy. *Curr Neurol Neurosci Rep* 9:231–240
- Nordqvist AC, Smurawa H, Mathiesen T (2001) Expression of matrix metalloproteinases 2 and 9 in meningiomas associated with different degrees of brain invasiveness and edema. *J Neurosurg* 95:839–844
- Paek SH, Kim CY, Kim YY, Park IA, Kim MS, Kim DG, Jung HW (2002) Correlation of clinical and biological parameters with peritumoral edema in meningioma. *J Neurooncol* 60:235–245
- Perry A, Gutmann DH, Reifenberger G (2004) Molecular pathogenesis of meningiomas. *J Neurooncol* 70:183–202
- Provias J, Claffey K, delAguila L, Lau N, Feldkamp M, Guha A (1997) Meningiomas: role of vascular endothelial growth factor/vascular permeability factor in angiogenesis and peritumoral edema. *Neurosurgery* 40:1016–1026
- Raizer J (2010) Meningiomas. *Curr Treat Options Neurol* 12:360–368
- Rao JS (2003) Molecular mechanisms of glioma invasiveness: the role of proteases. *Nat Rev Cancer* 3:489–501
- Salpietro FM, Alafaci C, Lucerna S, Iacopino DG, Todaro C, Tomasello F (1994) Peritumoral edema in meningiomas: microsurgical observations of different brain tumor interfaces related to computed tomography. *Neurosurgery* 35:638–641
- Sanson M, Cornu P (2000) Biology of meningiomas. *Acta Neurochir (Wien)* 142:493–505
- Schmitt M, Wilhelm S, Reuning U, Kruger A, Harbeck N, Lengyel E, Graeff H, Gansbacher B, Kessler H, Burgle M, Sturzebecher J, Sperl S, Magdolen V (2000) The urokinase plasminogen activation system as a novel target for tumour therapy. *Fibrinol Proteol* 14:114–132
- Siddique K, Yanamandra N, Gujrati M, Dinh D, Rao JS, Olivero W (2003) Expression of matrix metalloproteinases, their inhibitors, and urokinase plasminogen activator in human meningiomas. *Int J Oncol* 22:289–294
- Simis A, Pires de Aguiar PH, Leite CC, Santana PA Jr, Rosenberg S, Teixeira MJ (2008) Peritumoral brain

- edema in benign meningiomas: correlation with clinical, radiologic, and surgical factors and possible role on recurrence. *Surg Neurol* 70:471–477
- Simon M, Bostrom JP, Hartmann C (2007) Molecular genetics of meningiomas: from basic research to potential clinical applications. *Neurosurgery* 60:787–798
- Stepanova V, Mukhina S, Kohler E, Resink TJ, Erne P, Tkachuk VA (1999) Urokinase plasminogen activator induces human smooth muscle cell migration and proliferation via distinct receptor-dependent and proteolysis-dependent mechanisms. *Mol Cell Biol* 19:199–206
- Trinkaus M, Vranic A, Dolenc VV, Lah TT (2005) Cathepsins B and L and their inhibitors stefin B and cystatin C as markers for malignant progression of benign meningiomas. *Int J Biol Markers* 20:50–59
- Tummalapalli P, Gondi CS, Dinh DH, Gujrati M, Rao JS (2007) RNA interference-mediated targeting of urokinase plasminogen activator receptor and matrix metalloproteinase-9 gene expression in the IOMM-Lee malignant meningioma cell line inhibits tumor growth, tumor cell invasion and angiogenesis. *Int J Oncol* 31:5–17
- Valerie K, Yacoub A, Hagan MP, Curiel DT, Fisher PB, Grant S, Dent P (2007) Radiation-induced cell signaling: inside-out and outside-in. *Mol Cancer Ther* 6:789–801
- Watson MA, Gutmann DH, Peterson K, Chicoine MR, Kleinschmidt-DeMasters BK, Brown HG, Perry A (2002) Molecular characterization of human meningiomas by gene expression profiling using high-density oligonucleotide microarrays. *Am J Pathol* 161:665–672
- Wong CW, Lee A, Shientag L, Yu J, Dong Y, Kao G, Al-Mehdi AB, Bernhard EJ, Muschel RJ (2001) Apoptosis: an early event in metastatic inefficiency. *Cancer Res* 61:333–338
- Yamamoto M, Sawaya R, Mohanam S, Bindal AK, Bruner JM, Oka K, Rao VH, Tomonaga M, Nicolson GL, Rao JS (1994) Expression and localization of urokinase-type plasminogen activator in human astrocytomas in vivo. *Cancer Res* 54:3656–3661
- Yoshioka H, Hama S, Taniguchi E, Sugiyama K, Arita K, Kurisu K (1999) Peritumoral brain edema associated with meningioma: influence of vascular endothelial growth factor expression and vascular blood supply. *Cancer* 85:936–944
- Zhang X, Bu X, Zhen H, Fei Z, Wu J, Gu J, Yi S, Wang Z (2000) Expression and significance of urokinase-type plasminogen activator in human gliomas. *Chin Med J (Engl)* 113:802–804
- Zhou K, Wang G, Wang Y, Jin H, Yang S, Liu C (2010) The potential involvement of E-cadherin and beta-catenins in meningioma. *PLoS One* 5:e11231

miR-200a Regulation of the Wnt Signaling in Meningioma Tumorigenesis

7

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Abstract

A number of microRNAs (miRNAs) are dysregulated in cancer and they can exert critical roles in initiation and progression of various tumors. Meningiomas are derived from arachnoidal cells associated with brain meninges and frequently associated with loss of the neurofibromatosis 2 (*NF2*) gene. Here, we define a human meningioma-typical miRNA profile and characterize effects of one miRNA, miR-200a which is markedly downregulated in these tumors. Elevation of levels of miR-200a inhibited meningioma cell growth in culture and in a tumor model *in vivo*. This upregulation of miR-200a was associated with decreased expression of transcription factors, ZEB1 and SIP1, with consequently increased E-cadherin expression. An inverse correlation was also found between downregulation of miR-200a in meningiomas and increased expression of β -catenin and cyclin D1, with miR-200a targeting of the β -catenin mRNA and inhibiting Wnt signaling. miR-200a appears to act as a multi-functional tumor suppressor miRNA in meningiomas through effects on the E-cadherin and β -catenin signaling pathways. This reveals a previously

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unrecognized signaling cascade involved in meningioma tumor development and highlights a novel molecular interaction between miR-200a and Wnt signaling, thereby providing insights into novel therapies for meningiomas.

Introduction

Meningiomas

Meningiomas are tumors of the brain and spinal cord derived from arachnoidal cells in the meninges lining the brain, which are usually benign and account for 1–20% of all central nervous system tumors (Kleihues et al. 1993). One of the most common events associated with meningioma tumorigenesis is deletion in chromosome 22q associated with loss of the *NF2* gene (Fontaine et al. 1991; Menon et al. 1990). The *NF2* gene encodes a tumor suppressor protein, merlin, also called schwannomin, which is related to ezrin-radixin-moesin (ERM) proteins of the band 4.1 superfamily of membrane-cytoskeletal linkers (Trofatter et al. 1993). Loss/downregulation of *NF2* has also been found in malignant tumors, including mesotheliomas (Sekido 2010), gliomas (Lau et al. 2008), peripheral nerve sheath tumors (Scheithauer et al. 2009), and prostate cancer (Horiguchi et al. 2008). Several other chromosomal alterations in 1p, 3p, 6q, 10q and 14 have been also detected in meningiomas (Leone et al. 1999).

miRNA in Cancer

miRNAs are a class of small non-coding RNAs that regulate gene expression post-transcriptionally. They inhibit the expression of target mRNAs by binding to complementary sequences in untranslated regions (UTR) of mRNAs (typically the 3' UTR), and repressing translation and/or cleaving the mRNA (Ambros 2004). In humans, 1,000 different miRNAs, predicted by computational scanning in the human genome, have been identified (miRBase 10.0, [www://microrna.sanger.ac.uk](http://www.microrna.sanger.ac.uk)). Bioinformatic data suggest that they collec-

tively control about 30% of human mRNAs, with many mRNA targets per miRNA and multiple miRNAs targeting the same mRNA (Bentwich et al. 2005). miRNAs have been recently implicated as drivers in several carcinogenic processes, where they can act either as oncogenes or as tumor suppressors (Esquela-Kerscher and Slack 2006). Decreased levels of the let-7 family of tumor suppressor miRNAs are associated with increased Ras oncogene expression and reduced survival in patients with non-small cell lung cancer (NSCLC) (Johnson et al. 2005; Takamizawa et al. 2004). Among brain tumors, reduced levels of miR-128 in glioma tumors are associated with upregulation of the stem cell renewal factor, Bmi-1 (Godlewski et al. 2008). Our recent studies revealed a novel potential tumor suppressor miRNA, miR-200a, which directly targets the β -catenin and ZEP1/SIP1 mRNAs, with reduced levels of miR-200a found in meningiomas resulting in increases in Wnt signaling and expression of E-cadherin (Saydam et al. 2009).

Role of Wnt Signaling in Cancer

Wnt/ β -catenin signaling has been involved in tumorigenesis in a number of malignancies (Huang and Du 2008; Huang and He 2008; Moon et al. 2004). Under normal conditions, free β -catenin is bound by a “destruction complex” which contains adenomatous polyposis coli (APC), axin, glycogen synthase kinase 3 β (GSK3 β) and casein kinase I (CKI). This complex regulates β -catenin levels posttranslationally by phosphorylating its serine and threonine residues (Orford et al. 1997). Binding of β -catenin to axin recruits CKI to this complex, and triggers the initial phosphorylation event of β -catenin. In this way, β -catenin binds to APC and is maintained in low levels in the absence of Wnt stimulation by

proteosomal degradation after its polyubiquitination on NH₂ terminus (Orford et al. 1997; Clevers 2006; Aberle et al. 1997). Thus, the tumor suppressor protein APC acts as a negative regulator of the Wnt signaling pathway by participating in β -catenin destruction complex. Mutations in the *APC* gene lead to constitutive activation of Wnt/ β -catenin signaling as one of the main players in cancer (Huang and He 2008). Interestingly, in a recent report, miR-135a and b have been shown to target the 3' UTR of the APC transcript, and suppress its expression in colon carcinoma (Nagel et al. 2008). Thus, potentiation of β -catenin/Wnt signaling seems to be a critical step in cancer which can be regulated by miRNAs.

Methodology

Tumor and Normal Tissue Samples

Meningioma samples were obtained from discarded surgical tissues, and normal arachnoid tissue was obtained from autopsies, within 5–7 h of death. All human tissues were collected and de-identified by the Neuro-oncology Tumor Repository, snap-frozen and stored at –80°C under IRB protocols approved by Massachusetts General Hospital, Committee on Human Research.

miRNA Arrays

Total RNA was isolated from tissues with Trizol (Invitrogen, Carlsbad, CA) according to the manufacturer's instructions. RNA labeling and hybridization on a microarray was performed as previously described (Krichevsky et al. 2003; Monticelli et al. 2005). Briefly, 10 μ g total RNA from each sample were filtered through Microcon YM-100 concentrators to obtain a low molecular weight (LMW) fraction of RNA enriched in molecules under 60 bp. The LMW RNA was end-labeled with 30 μ Ci γ -33-P dATP (3,000 Ci/mmol) by T4 polynucleotide kinase (New England Biolabs, Ipswich, MA) and purified using the Qiagen Nucleotide Removal Kit (Valencia, CA). For hybridization, membranes

were first prehybridized in MicroHyb Hybridization Buffer (Invitrogen) at 37°C for 30 min followed by an overnight hybridization in the same solution containing RNA probe. Following hybridization, membranes were washed and exposed to a phosphor storage screen, scanned using a Phosphor Imager, and hybridization signals were quantified using ImageQuant software (Molecular Dynamics, Sunnyvale, CA). RNA samples were considered independent measurements for the purposes of the two-tailed, two-sample *t* test. Probe signals that showed high variance in biological replicates (more than two-fold with *p* values >0.05) were excluded from further analysis.

Data Analysis

miRNA array data was first normalized using median-centric method within BRBArrayTools (<http://linus.nci.nih.gov/BRB-ArrayTools.html>) and was further subjected to quantile normalization in Partek Genomics Suite (<http://www.partek.com/>). Differentiated miRNA lists comparing expression in control to all meningioma samples for miRNA arrays were generated using the significance analysis of microarrays (SAM) method (Tusher et al. 2001) with cutoffs for FDR <30% and *p* value <0.05. The normalized array data for the derived differentiated miRNAs list was then transformed to z-scores for data of each miRNA across samples using WPS utility program (Yi et al. 2006), which was subjected to heatmap clustering analysis and visualization using TM4 program from TIGR (<http://www.tm4.org/>).

TOPFLASH Reporter Gene Assay

To measure the TOPFLASH activity, cells were seeded at 10,000 cells per well in 96-well plates. Twenty-four hours later, TOPFLASH and pre-miR-200a and/or control precursor or inhibitor miRNAs were co-transfected into cells using lipofectamine. To monitor transfection efficiency, an expression cassette for *Renilla* luciferase (Rluc) was co-transfected. Five hours later, mWnt3a (50 ng/mL) was added. The cells were

treated for 3 days before harvest. Luciferase activity was measured using the Dual luciferase reporter 100 assay system (Promega, Madison, WI). Firefly luciferase (Fluc) activity was normalized to Rluc activity using the same kit.

Downregulated miR-200a in Meningiomas Promotes Tumor Growth by Reducing E-Cadherin and Activating the Wnt/ β -Catenin Signaling Pathway

miRNA Profiling Reveals a Set of Dysregulated miRNA in Meningiomas

In order to investigate the possible role(s) of miRNAs in meningioma tumor development and progression, we performed radioactive miRNA array screening using an array which can detect 407 known miRNAs (Krichevsky et al. 2003; Monticelli et al. 2005) and analyzed of 14 human sporadic benign meningiomas, WHO Grade I compared to three normal control arachnoid tissue samples (Saydam et al. 2009). We found 43 dysregulated miRNAs in most meningioma tumor samples (in 9–13 out of 14 tumors). Twenty of these changes in miRNA levels were confirmed by qRT-PCR, with 17 upregulated and 3 downregulated in two independent assays in a comparison of three randomly selected-meningioma samples and three arachnoidal control samples. Among several interesting dysregulated miRNAs, we focused on miR-200a, which was downregulated in 10/14 meningiomas by approximately 25-fold (Saydam et al. 2009).

miRNA-200a Triggers Apoptosis in Meningioma Cells and Inhibits Meningioma Tumor Growth *In Vivo*

Gain-of-function experiments showed that upregulation of miR-200a a growth inhibitory effect on two different benign human meningioma cell lines, SF4433 and SF4068. Five hours after transfection of meningioma cells with a precursor for

miR-200, cells were re-plated and cell counts determined 1, 2, 3, 4, and 6 days later. The growth of SF4433 and SF4068 cells was significantly reduced by about 65% and 50%, respectively, on day 6 after transfection with pre-miR-200a as compared with control cells. The inhibitory effect of pre-miR-200a on meningioma cell growth appeared to be specific for this miRNA, as transfection with a miRNA precursor for miR-575, which was also downregulated in meningiomas, did not affect the growth of SF4433 cells. The strong growth inhibitory effect of pre-miR-200a on meningioma cells in culture was then tested on tumor growth *in vivo*. Meningioma cells stably expressing Fluc, SF4433-Fluc, were transfected either with pre-miR-200a or pre-control 1, or left untransfected and 5 h later implanted subcutaneously into both flanks of nude mice (1×10^5 cells per injection site; 8 mice per group). Subsequent tumor growth was monitored by *in vivo* bioluminescence imaging over 17 days. The almost all the tumors formed by meningioma cells transfected with pre-miR-200a failed to grow, with a marked reduction in size from day 3 to 17 post-implantation, while control tumors grew slowly over this period. In control groups, *in vivo* imaging was terminated 3 weeks after implantation due to excessive size of tumors. Out of the 16 miR-200a-treated tumors only two grew over an 8 week period and they manifested comparable levels of luciferase activity to the control groups at the 3 week time of sacrifice. qRT-PCR analysis of RNA isolated from tumors at the time of sacrifice showed no difference between the levels of miR-200a in pre-miR-200a, pre-control 1, or untreated tumors, as presumably transfected precursor and derived mature miR-200a species are diluted out with tumor cell division. This finding indicates that elevated miR-200a levels in meningioma cells markedly reduce their ability to form tumors.

miR-200a Leads to E-Cadherin Upregulation and Directly Targets β -Catenin mRNA

In the light of findings by others that ZEB1 and SIP1 mRNAs are direct targets for miR-200a and

regulate E-cadherin expression (Burk et al. 2008; Gregory et al. 2008; Korpál et al. 2008; Park et al. 2008), we tested the effects of miR-200a elevation on levels of protein and mRNA for these transcription factors in meningioma cells. Overexpression of miR-200a reduced ZEB1 and SIP1 protein levels, while E-cadherin protein levels were increased. Based on studies showing the interaction between β -catenin and cadherin, as well as, the role of β -catenin in the Wnt signaling pathway, we also tested whether miR-200a levels affected β -catenin expression. Interestingly, we found that increased levels of miR-200a in meningioma cells significantly downregulated β -catenin both at the mRNA and protein levels. Thus, we hypothesized that miR-200a might inhibit meningioma tumor growth both by upregulating E-cadherin expression, and by decreasing Wnt/ β -catenin signaling. To test this hypothesis, we first examined the involvement of E-cadherin upregulation in miR-200a-mediated growth inhibition of meningioma cells. Cells were transfected with an siRNA for E-cadherin mRNA or pre-miR-200a alone, or co-transfected with both or control constructs, and 5 h later cells were replated and cell counts determined after 1, 2, 3, 4 and 5 days. siRNA reduction of E-cadherin levels alone did not significantly increase the growth of meningioma cell. When siRNA for E-cadherin mRNA and pre-miR-200a were co-transfected, growth of meningioma cells was reduced by about 50%, indicating that while miR-200a probably also regulates levels of other proteins involved in meningioma growth, elevated levels of E-cadherin are a contributing component.

miR-200a Directly Targets the β -Catenin mRNA and Inhibits β -Catenin/Wnt Signaling

To understand how increased levels of miR-200a lead to decreased levels of β -catenin, we evaluated whether the β -catenin mRNA is a direct target for miR-200a. β -catenin message is among the predicted targets of miR-200a (<http://pictar.mdc-berlin.de/data/vertebrate/target>) with one potentially conserved binding site within its 3' UTR.

In order to test whether miR-200a can bind directly to this 3' UTR site, luciferase reporter vectors encoding a fragment of wt β -catenin mRNA 3' UTR (846–892 bp; β -catenin-wt) or the same region containing mismatches in the predicted miR-200a binding site (β -catenin-mt) were transfected into HEK 293 T cells. These cells were used based on their higher transfection efficiency as compared to meningioma cells. Co-transfection of the β -catenin-wt construct and pre-miR-200a resulted in significantly decreased luciferase activity compared to transfection with pre-control 1, whereas luciferase activity was not significantly reduced in cells co-transfected with the β -catenin-mt and pre-miR-200a. These data strongly suggest that the reduced β -catenin levels in pre-miR-200a-transfected cells is a result of direct targeting of the β -catenin RNA by miR-200a.

Since Wnt/ β -catenin signaling is involved in tumorigenesis in a number of malignancies (Huang and Du 2008; Huang and He 2008; Moon et al. 2004), we hypothesized that the growth regulatory effect of miR-200a in meningioma cells might be mediated via inhibition of this signaling pathway through reduced β -catenin levels. To test this, we evaluated whether decreased growth of cells transfected with pre-miR-200a could be rescued by overexpression of β -catenin. Both SF4433 and SF4068 meningioma cell lines were transfected with pre-miR-200a and/or transduced with an adenoviral based β -catenin expression vector, Ad- β -catenin, alone or together, plus controls, and 5 h later cells were replated and cell counts determined over 5 days. Cell growth of miR-200a-transfected meningioma cells, SF4433 and SF4068 was reduced by approximately 65% and 43%, respectively, in cells transfected with pre-miR-200a, as compared to the controls, but concomitant overexpression of β -catenin increased their growth significantly (Saydam et al. 2009). This supports the hypothesis that upregulation of β -catenin can partially counteract the growth inhibitory effect of pre-miR-200a. Taken together these data suggest that decreased β -catenin levels caused by increased miR-200a contribute to miR-200a-mediated growth inhibition.

To further examine effects of miR-200a on Wnt/ β -catenin signaling, meningioma cells were

transduced with a reporter construct for the Wnt/ β -catenin pathway, Lenti-TOPFLASH (Minami et al. 2008), which contains both an expression cassette for Fluc regulated by 7X β -catenin TCF/LEF binding sites and an expression cassette for Rluc under a constitutive promoter for normalization. Cells were transfected only with the reporter construct or co-transfected with pre-control 1 or pre-miR-200a and the reporter construct, and 5 h later medium containing 50 ng/ml mouse (m) Wnt ligand (B&D System, Franklin Lakes, NJ) was added and the incubation continued for 2 days. mWnt treatment resulted in an approximately ten-fold induction in Fluc activity in cells transduced with the reporter, whereas co-transfection with pre-miR-200a reduced activity of mWnt plus the reporter by approximately 50%, as compared with pre-control 1 (Saydam et al. 2009). These data indicate that miR-200a can regulate the Wnt/ β -catenin signaling cascade, presumably by reducing levels of β -catenin. In support of this, siRNA inhibition of β -catenin expression also inhibited meningioma cell growth.

Wnt/ β -Catenin Signaling Is Dysregulated in Meningioma Tumors

The activation of Wnt signaling results in translocation of β -catenin from the plasma membrane/cytoplasm into the nucleus (Huang and Du 2008). To test whether reduced levels of miR-200a activity in arachnoidal cells (cells of meningioma origin) affect levels and cellular localization of β -catenin, we transfected these cells in culture with a miR-200a antisense inhibitor or a control oligonucleotide, or left them untransfected and performed immunofluorescence staining for β -catenin 3 days later. In controls, β -catenin showed staining consistent with localization at the plasma membrane, whereas in miR-200a-inhibitor transfected cells, β -catenin levels appeared higher and showed strong cytoplasmic and some nuclear localization (Saydam et al. 2009). These data suggest that downregulation of miR-200a in arachnoidal cells may contribute to development

of meningioma tumorigenesis by increasing levels of β -catenin in the nucleus as part of the transformed phenotype.

We also examined the expression levels of β -catenin and cyclin D1, a target gene of β -catenin, in tumor samples from meningioma patients. We performed qRT-PCR for messages in randomly selected tumor samples, including four with miR-200a downregulation and four without downregulation of miR-200a, as compared to an arachnoidal tissue sample. A significant inverse correlation was found between levels of miR-200a and those of β -catenin and cyclin D1 mRNAs in meningioma tissue samples, while decreased levels of E-cadherin mRNA were found in all meningioma samples. This suggests that in addition to miR-200a other factors are involved in the regulation of E-cadherin expression and that downregulation of E-cadherin mRNA is a common feature of meningiomas. Furthermore, immunohistochemical analysis of sections from meningioma tumors showed that β -catenin was expressed at higher levels in tumors with low miR-200a levels as compared to those in which miR-200a was not downregulated, with arachnoidal tissue showing very low levels of β -catenin. E-cadherin was found to be expressed at lower levels in these tumor samples as compared to arachnoidal tissues. Taken together these data suggest that miR-200a regulates levels of β -catenin in meningioma tumors and that downregulation of miR-200a may have an important role in meningioma growth by increasing β -catenin levels and hence facilitating Wnt signaling.

Discussion

In this review based on a previous study (Saydam et al. 2009), we define a meningioma-typical miRNA signature by miRNA expression profiling of sporadic benign meningiomas. This signature includes 20 miRNAs dysregulated in most meningioma tumor samples as compared with control arachnoidal tissues, which is the origin of these tumors. We focused on miR-200a as its expression level was the most downregulated as

compared to arachnoidal tissue of origin across 10/14 tumors by approximately 25-fold. The relevance of miR-200a downregulation to tumor growth was evaluated by gain-of-function studies in which transient overexpression of pre-miR-200a was found to inhibit growth of human meningioma cells in culture. This effect was also sufficient to curtail tumor growth *in vivo*, as meningioma cells transfected with pre-miR-200a showed regression when established as subcutaneous tumors in nude mice as compared to controls, with only 2 out of 8 eventually forming tumors over an extended period. Moreover, we found that miR-200a functions as a potential tumor suppressor by inhibiting the Wnt/ β -catenin signaling through two complementary mechanisms - direct targeting of the β -catenin mRNA leading to reduced β -catenin levels, and targeting of the mRNAs for ZEB1 and SIP1 with consequent upregulation of E-cadherin levels and sequestration of β -catenin. Dysregulation of Wnt/ β -catenin signaling and the E-cadherin cell adhesion system have been implicated as important events in the initiation and/or progression of several forms of cancer (Brunner et al. 2006; Burk et al. 2008; Jeanes et al. 2008; Peifer and Polakis 2000; Polakis 2000). The results presented here provide insight into the mechanism by which a miRNA, miR-200a, at reduced levels promotes meningioma tumor growth by elevation of β -catenin and Wnt signaling, as well as reduced levels of E-cadherin. With respect to the role of miR-200a in β -catenin signaling, we found that miR-200a directly targets the 3' UTR of β -catenin mRNA. To our knowledge, miR-200a is the first miRNA shown to be involved in the regulation of β -catenin levels. Given the wide role of Wnt/ β -catenin signaling in many cancers, including colon cancer, hepatocellular carcinoma, melanoma, and ovarian and prostate cancers (Peifer and Polakis 2000), we presume that reduction in miR-200a may contribute to increased β -catenin signaling in other forms of cancer.

To investigate the downstream effect of miR-200a levels on β -catenin responsive promoters, we overexpressed miR-200a in meningioma cells and monitored the activity of a promoter containing TCF/LEF binding sites driving luciferase

(Lenti-TOPFLASH; Minami et al. 2008) in response to the Wnt ligand, mWnt3a. Overexpression of miR-200a strongly inhibited the response of this promoter to Wnt induction, and this suppression could be overcome with an antisense inhibitor of miR-200a. Moreover, we found that β -catenin overexpression via an expression vector, Ad- β -catenin significantly overcame growth inhibition caused by elevated miR-200a in meningioma cells. Taken together, these data suggest that low levels of miR-200a in meningioma cells activate Wnt signaling by increasing levels of β -catenin mRNA and its availability to enter the nucleus and activate genes supporting growth of meningioma tumors. Furthermore, our data suggest that miR-200a can act as a key "negative regulator" of the Wnt/ β -catenin signaling pathway, and hence a potential therapeutic agent, with elevation of miR-200a levels inhibiting growth of meningioma tumors in a xenograft model. The stability of β -catenin in the cytoplasm is tightly regulated by a degradation complex containing adenomatous polyposis coli (APC), axin, casein kinase 1 (CK1) and glycogen synthase kinase 3 (GSK3) which marks β -catenin for proteolytic destruction (Jeanes et al. 2008). The activity of the Wnt/ β -catenin signaling pathway is mainly determined by the amount of the β -catenin free in the cytoplasm versus that sequestered in the plasma membrane or APC-degradation complex (Moon et al. 2002, 2004). Decreased levels of E-cadherin in meningioma cells should thus make more β -catenin available in the cytoplasm/nucleus, at the same time possibly decreasing links between E-cadherin and the cytoskeleton. This latter effect should also reduce cell-to-cell adhesion of arachnoidal cells as part of the tumorigenesis process. Disruption of cell-cell communication through miR-200a downregulation may thus promote meningioma tumor growth and progression both through decreased E-cadherin and elevated β -catenin levels.

In conclusion, our study demonstrates a key role for miR-200a downregulation in meningiomas promoting growth of these tumors. Reduced levels of miR-200a appear to contribute to tumorigenesis through two converging pathways

mediated by upregulation of three mRNA targets. Those for ZEB1 and SIP1 lead to increased levels of these transcriptional repressors which decreases levels of E-cadherin in these tumors. Upregulation of a new β -catenin mRNA target for miR-200a results in more β -catenin available to enter the nucleus and activate genes in the Wnt signaling pathway. These miR-200a-mediated tumorigenic processes are undoubtedly shared with other tumors and provide potential targets for therapeutic intervention.

Acknowledgements This study was supported by the Children's Tumor Foundation 2007-01-043 (O.S.) and NIH/NINDS NS24279 (XOB and OS). We thank Dr. Breakefield for editing of this manuscript.

References

- Aberle H, Bauer A, Stappert J, Kispert A, Kemler R (1997) Beta-catenin is a target for the ubiquitin-proteasome pathway. *EMBO J* 16:3797–3804
- Ambros V (2004) The functions of animal microRNAs. *Nature* 431:350–355
- Bentwich I, Avniel A, Karov Y, Aharonov R, Gilad S, Barad O, Barzilai A, Einat P, Einav U, Meiri E, Sharon E, Spector Y, Bentwich Z (2005) Identification of hundreds of conserved and nonconserved human microRNAs. *Nat Genet* 37:766–770
- Brunner EC, Romeike BF, Jung M, Comtesse N, Meese E (2006) Altered expression of beta-catenin/E-cadherin in meningiomas. *Histopathology* 49:178–187
- Burk U, Schubert J, Wellner U, Schmalhofer O, Vincan E, Spaderna S, Brabletz T (2008) A reciprocal repression between ZEB1 and members of the miR-200 family promotes EMT and invasion in cancer cells. *EMBO Rep* 9(6):582–589
- Clevers H (2006) Wnt/beta-catenin signaling in development and disease. *Cell* 127:469–480
- Esquela-Kerscher A, Slack FJ (2006) Oncomirs – microRNAs with a role in cancer. *Nat Rev Cancer* 6:259–269
- Fontaine B, Rouleau GA, Seizinger BR, Menon AG, Jewell AF, Martuza RL, Gusella JF (1991) Molecular genetics of neurofibromatosis 2 and related tumors (acoustic neuroma and meningioma). *Ann N Y Acad Sci* 61:338–343
- Godlewski J, Nowicki MO, Bronisz A, Williams S, Otsuki A, Nuovo G, Raychaudhury A, Newton HB, Chiocca EA, Lawler S (2008) Targeting of the Bmi-1 oncogene/stem cell renewal factor by microRNA-128 inhibits glioma proliferation and self-renewal. *Cancer Res* 68:9125–9130
- Gregory PA, Bert AG, Paterson EL, Barry SC, Tsykin A, Farshid G, Vadas MA, Khew-Goodall Y, Goodall GJ (2008) The miR-200 family and miR-205 regulate epithelial to mesenchymal transition by targeting ZEB1 and SIP1. *Nat Cell Biol* 10:593–601
- Horiguchi A, Zheng R, Shen R, Nanus DM (2008) Inactivation of the NF2 tumor suppressor protein merlin in DU145 prostate cancer cells. *Prostate* 68:975–984
- Huang D, Du X (2008) Crosstalk between tumor cells and microenvironment via Wnt pathway in colorectal cancer dissemination. *World J Gastroenterol* 14:1823–1827
- Huang H, He X (2008) Wnt/beta-catenin signaling: new (and old) players and new insights. *Curr Opin Cell Biol* 20:119–125
- Jeanes A, Gottardi CJ, Yap AS (2008) Cadherins and cancer: how does cadherin dysfunction promote tumor progression? *Oncogene* 27:6920–6929
- Johnson SM, Grosshans H, Shingara J, Byrom M, Jarvis R, Cheng A, Labourier E, Reinert KL, Brown D, Slack FJ (2005) RAS is regulated by the let-7 microRNA family. *Cell* 120:635–647
- Kleihues P, Burger PC, Scheithauer BW (1993) The new W. H. O. classification of brain tumours. *Brain Pathol* 3:255–268
- Korpai M, Lee ES, Hu G, Kang Y (2008) The miR-200 family inhibits epithelial-mesenchymal transition and cancer cell migration by direct targeting of E-cadherin transcriptional repressors ZEB1 and ZEB2. *J Biol Chem* 283:14910–14914
- Krichevsky AM, King KS, Donahue CP, Khrapko K, Kosik KS (2003) A microRNA array reveals extensive regulation of microRNAs during brain development. *RNA* 9:1274–1281
- Lau YK, Murray LB, Houshmandi SS, Xu Y, Gutmann DH, Yu Q (2008) Merlin is a potent inhibitor of glioma growth. *Cancer Res* 68:5733–5742
- Leone PE, Bello MJ, de Campos JM, Vaquero J, Sarasa JL, Pestaña A, Rey JA (1999) NF2 gene mutations and allelic status of 1p, 14q and 22q in sporadic meningiomas. *Oncogene* 18:2231–2239
- Menon AG, Gusella JF, Seizinger BR (1990) Progress toward the isolation and characterization of the genes causing neurofibromatosis. *Brain Pathol* 1:33–40
- Minami Y, Stuart SA, Ikawa T, Jiang Y, Banno A, Hunton IC, Young DJ, Naoe T, Murre C, Jamieson CH, Wang JY (2008) BCRABL-transformed GMP as myeloid leukemic stem cells. *Proc Natl Acad Sci USA* 105:17967–17972
- Monticelli S, Ansel KM, Lee DU, Rao A (2005) Regulation of gene expression in mast cells: micro-RNA expression and chromatin structural analysis of cytokine genes. *Novartis Found Symp* 271:179–187
- Moon RT, Bowerman B, Boutros M, Perrimon N (2002) The promise and perils of Wnt signaling through beta-catenin. *Science* 296:1644–1646
- Moon RT, Kohn AD, De Ferrari GV, Kaykas A (2004) WNT and beta-catenin signalling: diseases and therapies. *Nat Rev Genet* 5:691–701
- Nagel R, le Sage C, Diosdado B, van der Waal M, Oude Vrielink JA, Bolijn A, Meijer GA, Agami R (2008)

- Regulation of the adenomatous polyposis coli gene by the miR-135 family in colorectal cancer. *Cancer Res* 68:5795–5802
- Orford K, Crockett C, Jensen JP, Weissman AM, Byers SW (1997) Serine phosphorylation-regulated ubiquitination and degradation of beta-catenin. *J Biol Chem* 272:24735–24738
- Park SM, Gaur AB, Lengyel E, Peter ME (2008) The miR-200 family determines the epithelial phenotype of cancer cells by targeting the E-cadherin repressors ZEB1 and ZEB2. *Genes Dev* 22:894–907
- Peifer M, Polakis P (2000) Wnt signaling in oncogenesis and embryogenesis – a look outside the nucleus. *Science* 287:1606–1609
- Polakis P (2000) Wnt signaling and cancer. *Genes Dev* 14:1837–1851
- Saydam O, Shen Y, Wurdinger T, Senol O, Boke E, James MF, Tannous BA, Stemmer-Rachamimov AO, Yi M, Stephens RM, Fraefel C, Gusella JF, Krichevsky AM, Breakefield XO (2009) Downregulated microRNA-200a in meningiomas promotes tumor growth by reducing E-cadherin and activating the Wnt/{beta}-catenin signaling pathway. *Mol Cell Biol* 29:5923–5940
- Scheithauer BW, Erdogan S, Rodriguez FJ, Burger PC, Woodruff JM, Kros JM, Gokden M, Spinner RJ (2009) Malignant peripheral nerve sheath tumors of cranial nerves and intracranial contents: a clinicopathologic study of 17 cases. *Am J Surg Pathol* 33:325–338
- Sekido Y (2010) Genomic abnormalities and signal transduction dysregulation in malignant mesothelioma cells. *Cancer Sci* 101:1–6
- Takamizawa J, Konishi H, Yanagisawa K, Tomida S, Osada H, Endoh H, Harano T, Yatabe Y, Nagino M, Nimura Y, Mitsudomi T, Takahashi T (2004) Reduced expression of the let-7 microRNAs in human lung cancers in association with shortened postoperative survival. *Cancer Res* 64:3753–3756
- Trofatter JA, MacCollin MM, Rutter JL, Murrell JR, Duyao MP, Parry DM, Eldridge R, Kley N, Menon AG, Pulaski K (1993) A novel moesin-, ezrin-, radixin-like gene is a candidate for the neurofibromatosis 2 tumor suppressor. *Cell* 72:791–800
- Tusher VG, Tibshirani R, Chu G (2001) Significance analysis of microarrays applied to the ionizing radiation response. *Proc Natl Acad Sci USA* 98:5116–5121
- Yi M, Horton JD, Cohen JC, Hobbs HH, Stephens RM (2006) WholePathwayScope: a comprehensive pathway-based analysis tool for high-throughput data. *BMC Bioinformatics* 7:30

Part II

Diagnosis

Meningiomas: Determination of Subtypes Using Perfusion Magnetic Resonance Imaging

8

Hao Zhang, Guixiang Zhang, and Matthijs Oudkerk

Abstract

To evaluate the value of perfusion magnetic resonance imaging (MRI) in the preoperative subtyping of meningiomas by analyzing the relative cerebral blood volume (rCBV) of three benign subtypes and anaplastic meningiomas separately. Thirty seven meningiomas with peritumoral edema (15 meningothelial, 10 fibrous, 4 angiomatous and 8 anaplastic) underwent perfusion MRI by using a gradient echo echo-planar sequence. The maximal rCBV (compared with contralateral normal white matter) in both tumoral parenchyma and peritumoral edema of each tumor was measured. The mean rCBVs of each two histological subtypes were compared by using one way ANOVA and least significant difference (LSD) tests. A *P* value less than 0.05 indicated a statistically significant difference. The mean rCBV of meningothelial, fibrous, angiomatous and anaplastic meningiomas in tumoral parenchyma were 6.93 ± 3.75 , 5.61 ± 4.03 , 11.86 ± 1.93 , 5.89 ± 3.85 , respectively, and in the peritumoral edema 0.87 ± 0.62 , 1.38 ± 1.44 , 0.87 ± 0.30 , 3.28 ± 1.39 , respectively. The mean rCBV in tumoral parenchyma of angiomatous meningiomas and in the peritumoral edema of anaplastic meningiomas were statistically different ($p < 0.05$) from the other types of meningiomas. *Conclusion:* Perfusion MRI can provide useful functional information on meningiomas and help in the preoperative diagnosis of some subtypes of meningiomas.

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Introduction

Meningiomas account for about 30% of all primary intracranial tumors. Anaplastic meningiomas constitute 1–3% of all meningiomas

(Riemenschneider et al. 2006). According to their histopathological features, the aggressiveness of their growth and the risk of recurrence, the WHO (2007) has classified meningiomas into 15 subtypes (Louis et al. 2007). Although most meningiomas are benign, they have a surprisingly broad spectrum of genetic and biological behaviors (Herting et al. 2003; Okamoto et al. 2006). Previous studies (Chen et al. 1992; Ildan et al. 1999) found a correlation between the conventional MR signal intensities and the histopathological features of the tumors. If the tumor is composed primarily of fibrous or transitional elements it is significantly hypointense on T2-weighted MR images. On the other hand if the tumor is composed primarily of syncytial or angioblastic elements it is significantly hyperintense on T2-weighted MR images. Their results showed that hyperintensity on T2-weighted images associated with meningothelial, malignant, and angiomatous meningiomas, and hypointensity on T2-weighted images associated with fibroblastic and transitional meningiomas. But the majority of intracranial meningiomas exhibit a heterogeneous intensity pattern that may be related to the presence of calcification, areas of necrosis or haemorrhage, cysts, to tumor vascularity and an inherent speckling and mottling of uncertain cause, so that it may be difficult to assess and score the signal intensity of the parenchyma of the tumor (Carpeggiani et al. 1993).

Dynamic contrast-enhanced perfusion MRI studies have demonstrated that rCBV maps can be used to grade gliomas, differentiate different brain tumor types, and distinguish tumors from non-neoplastic lesions (Hakyemez et al. 2006; Kremer et al. 2002; Cha 2003). Several studies have reported that the rCBV values of meningiomas were greater than those of high-grade gliomas due to their high vascularity (Kremer et al. 2002; Cha et al. 2002). Nevertheless, the results of the differences of rCBV between typical and atypical meningiomas were inconsistent in previous studies. While Hakyemez et al. (2006) found that the rCBV of typical and atypical meningiomas were statistically different, Cha et al. (2002) and Yang et al. (2003) reported that typical and atypical meningiomas cannot be differentiated

from each other by tumoral rCBV. The typical or benign meningiomas involved in all these studies were evaluated without subtyping. Furthermore, previous studies, though rare and in small samples, have demonstrated that the vascularity of different benign meningioma subtypes such as meningothelial and angiomatous meningiomas showed different tendencies (Kimura et al. 2006). In our study the vascularity of both tumoral parenchyma and peritumoral edema of three benign meningiomas (meningothelial, fibrous, and angiomatous) and anaplastic meningiomas were calculated and compared separately, with the intention to get more useful preoperative information for clinical applications.

Materials and Methods

Patients

Thirty seven patients with newly diagnosed meningiomas with peritumoral edema (15 meningothelial, 10 fibrous, 4 angiomatous and 8 anaplastic) were enrolled in the study between January 2002 and December 2006. There were 17 male and 20 female patients, the average age was 45 years, with a range of 17–65 years. All patients underwent gross total or subtotal resection of the tumor within 30 days after MR examination. All diagnoses were pathologically confirmed.

MRI Protocol

MR examinations were performed with a 1.5T clinical system (Signa; GE Medical Systems; Milwaukee, USA). A 3-plane localizer, sagittal and axial T1-weighted SE (TR/TE=440/14 ms, matrix: 256*256; section thickness: 8 mm; field of view: 22 cm) and axial fluid attenuated inversion recovery (FLAIR, TR/TE=8,002/126 ms, TI=2,000 ms) sequences were performed before contrast injection. Hereafter, 15–20 ml contrast agent (Gd-DTPA, 0.1 mmol/kg, Omniscan, GE Medical Systems, USA) was administered intravenously with a power injector (Medrad,

Indianola, PA, USA) at a rate of 4 ml/s, followed by a 20 ml saline flush at the same rate. Five seconds after the beginning of the injection, a T2*-weighted dynamic susceptibility perfusion series was started by using a gradient-echo echo-planar sequence with the following parameters: TR/TE/excitations: 1,800/40/1; flip angle: 60°; bandwidth: 62.75; matrix: 128 *128; number of sections: 10; section thickness: 8 mm without spacing; field of view: 22 cm; acquisition time: 1 min 31 s. A series of 50 dynamic acquisitions was obtained for each section. Then post contrast T1-weighted sagittal and axial SE sequences were performed, using the same section positions and parameters as in the pre-contrast series.

Data Analysis

Image analysis was performed by using Functool software (ADW 4.0, GE Medical Systems, Milwaukee, WI, USA). A dynamic perfusion technique (Rosen et al. 1990; Weisskoff et al. 1993) with a contrast medium demonstrated cerebral hemodynamics by analyzing changes in the signal intensities after administration of contrast material were used in this study. During the first phase of passage of the contrast agent bolus, both T2* relaxation time and its equivalent T2* signal intensity decrease. The change in relaxation rate ($\Delta R2^*$) can be calculated on the basis of signal intensity by means of the following equation (Rosen et al. 1990): $\Delta R2^* = -\ln [S(t)/S_0]/\text{echo time}$, in which $S(t)$ and S_0 are the signal intensities at time t and baseline. The recirculation and T1 shortening effects caused by contrast agent leakage can be reduced by fitting a gamma-variate function to the measured $\Delta R2^*$ curve. The area under the corrected contrast agent concentration–time curve is proportional to the CBV and does not yield an absolute measurement. Therefore, it is necessary to express the measurement relative to a standard reference, usually the contralateral white matter. We refer to this as the relative CBV.

The steps to create perfusion maps had been described in detail previously by Knopp et al. (1999).

The source images are first inspected for overall image quality and motion artifacts. A single region of interest (ROI) is placed over the white matter of the contralateral unaffected centrum semiovale, and rCBV maps are generated to serve as a road map. CBV maps can then be calculated on a pixel-by-pixel basis and can be displayed as a gray-scale image. However, small but important variations in CBV are not always apparent from these maps. An alternative is to use a color overlay displayed on the raw image, in which the abnormal CBV values are often more apparent. Application of a threshold for the color overlay at a CBV the same as that of the unaffected white matter were used in this study. A color scale from dark (minimum CBV) to white (maximum CBV) is shown on the maps as reference. In lesions that are located close to major vascular structures, careful inspection of the images of different sequence can clearly indicate vessels which have to be avoided. At least 3 circular ROIs, at least 20 mm² each, were placed in the areas of highest rCBV of the solid part of the tumor parenchyma and peritumoral edema each and the maximal values were recorded. The peritumoral edema in this study was defined as high signal on FLAIR images within a 1 cm – distance from the outer enhancing tumor margin on post contrast T1-weighted images. For quantitative analysis, normal white matter within the contralateral hemisphere was used as the internal reference standard, rCBV values were calculated by dividing the maximal rCBV of a tumor by that of contralateral normal white matter.

Statistical analysis was performed with a commercially available statistical software package (SPSS, version 14.0 for Windows; SPSS, Chicago, IL, USA). The mean rCBV of each subtype were expressed as mean $\pm$ standard deviation. The differences between means of different meningiomas were first compared by using One-way ANOVA. Fisher's least significant difference (LSD) test was then performed for multiple comparisons of each two subtypes. A P value less than 0.05 indicated a statistically significant difference.

Results

All cases in our group were correctly diagnosed as meningiomas by evaluating the pre- and post contrast images of conventional MRI sequences. The conventional MR images and color rCBV maps of different subtypes are shown in Figs. 8.1, 8.2 and 8.3.

The mean rCBVs of both tumor parenchyma and peritumoral edema showed statistical differences between subtypes. The mean rCBVs in tumor parenchyma of angiomatous meningioma (11.86 ± 1.93) and in peritumoral edema of anaplastic meningioma (3.28 ± 1.39) were the highest. When comparing the mean rCBVs of each two subtypes, the mean rCBVs in tumor parenchyma of angiomatous meningioma and in

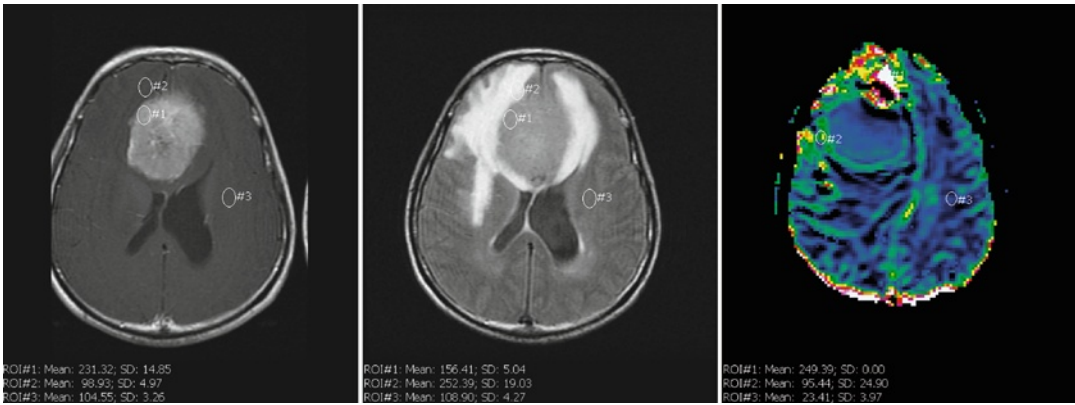


Fig. 8.1 Fibrous meningioma. FLAIR (a) and post contrast T1W (b) images show a mass with inhomogeneous parenchyma *left* of the falx cerebri accompanied by little peritumoral edema and moderate peripheral enhancement. The rCBV map (c) demonstrates the abnormal inhomogeneous

CBV of tumor parenchyma and peritumoral edema; ROIs are drawn at the maximum abnormal CBV areas of both tumor parenchyma and peritumoral edema, as well as the contralateral unaffected white matter

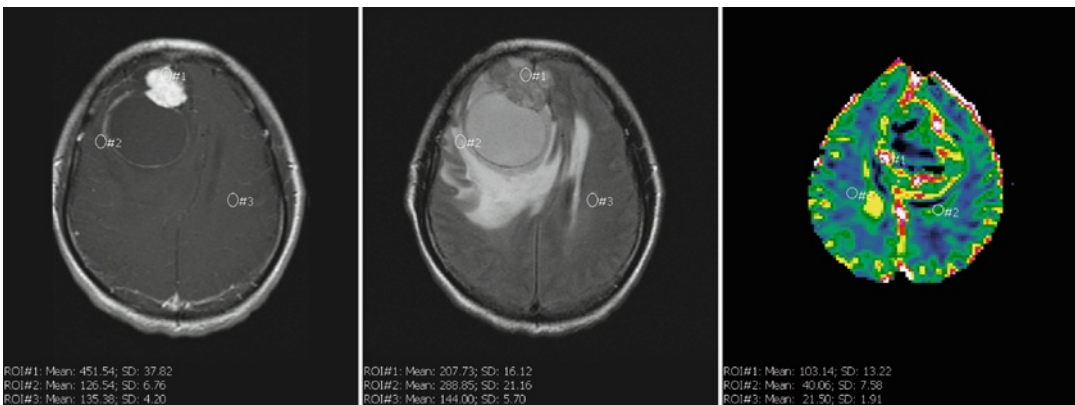


Fig. 8.2 Angiomatous meningioma. Mass right of the falx cerebri with slightly high intensity solid part and high intensity cyst accompanies by apparent peritumoral edema on FLAIR image (a). Strong homogeneous enhancement at the *solid* part of tumor is shown on post contrast T1W

(b). The rCBV map (c) demonstrates the abnormal inhomogeneous CBV of tumor parenchyma and peritumoral edema; ROIs are drawn at the maximum abnormal CBV areas of both tumor parenchyma and peritumoral edema, as well as the contralateral unaffected white matter

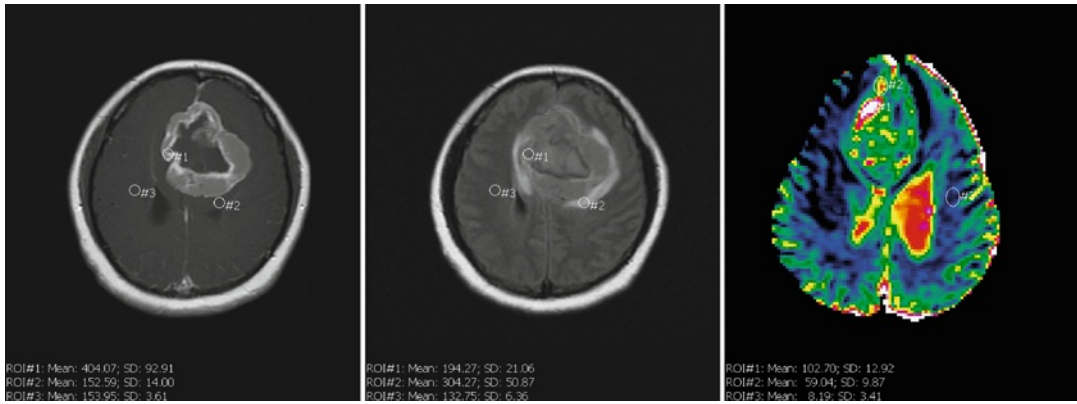


Fig. 8.3 Anaplastic meningioma. FLAIR (a) image shows a homogeneous mass on both sides of the falx cerebri accompanied by peritumoral edema. The mass is enhanced homogeneously on post contrast T1WI (b). The rCBV map (c) demonstrates the abnormal inhomogeneous

CBV of tumor parenchyma and peritumoral edema, the anterior border between tumor and peritumoral edema is blurred; ROIs are drawn at the maximum abnormal CBV areas of both tumor parenchyma and peritumoral edema, as well as the contralateral unaffected white matter

peritumoral edema of anaplastic meningioma were statistically greater ($p < 0.05$) than those of each other subtype.

Discussion

Meningiomas demonstrate some common characteristics on conventional MR images which can be used as reliable predictors for differentiating them from other brain tumors. In our study of 37 patients, all tumors were diagnosed correctly as meningiomas (without subtyping) by using conventional pre-contrast T1W, FLAIR and post-contrast T1W MR images.

Perfusion MRI technique can be used for semi-quantitative analysis of the vascularity of the brain (Villringer et al. 1988). Previous studies have indicated that the measurements of relative cerebral blood volume (rCBV) derived from dynamic susceptibility contrast (DSC) perfusion MRI data can be useful in the grading of gliomas (Knopp et al. 1999; Aronen et al. 1994). The most widely used DSC perfusion MRI technique is based on the intravascular indicator dilution theory (Zierler 1962). It has been shown that, in the absence of recirculation and contrast material

leakage, CBV is proportional to the area under the contrast agent concentration-time curve and does not yield an absolute measurement. It is, therefore, necessary to express measurements relative to a standard reference, usually contralateral white matter. Because meningiomas lack a true blood-brain barrier (BBB) and most of them have a rich blood supply, this assumption is violated. Although corrections can be applied, previous studies have noticed that CBV measurements in meningiomas are probably artificially elevated and that gradient-echo sequences are more suitable for perfusion MR of meningiomas than spin-echo echo-planar imaging techniques (Yang et al. 2003; Sugahara et al. 2001). Investigations have attempted to correct for the under- or overestimation of rCBV caused by microvascular leakage of contrast agent, but none is considered the reference standard method (Cha 2006).

The tumor rCBV of some subtypes of meningiomas has been evaluated by Kimura et al. (2006) by using continuous arterial spin labeling (CASL) and DSC perfusion MRI methods in a small group of patients and demonstrated that the rCBV was largest in the angiomatous meningiomas ($n = 2$), lowest in the fibrous-type meningioma ($n = 2$) and ranged from low to moderately

high in meningothelial cases ($n=6$). The mean rCBV of angiomatous meningiomas showed a significant difference ($P<0.05$) from each of the other two subtypes. Their results ranked in the same range as the mean rCBVs in our study which were derived from the ROIs in tumor parenchyma of the comparable subtypes of meningiomas. Yang et al. (2003) studied 15 typical meningiomas (without subtyping) and seven atypical meningiomas. The mean rCBV in tumoral parenchyma of two groups showed no statistically significant difference. In our results, the mean rCBVs of anaplastic meningioma parenchyma was statistically lower than that of angiomatous meningiomas and showed no difference with the other two subtypes. From these studies, we presumed that some common benign subtypes of meningiomas such as meningothelial and fibrous meningiomas cannot be differentiated from atypical or anaplastic meningiomas by only evaluating their parenchyma rCBV values. However, rare benign subtypes such as angiomatous meningiomas might show different perfusion characteristics.

The mean rCBV of angiomatous meningioma parenchyma was maximal compared to the subtypes we studied and this corresponded with their histological features of extremely high vascularity (Lupo et al. 2005). Despite this distinct histopathological appearance, however, atypical or anaplastic features are absent in angiomatous meningiomas (Lupo et al. 2005). Thus, the subtyping of angiomatous meningiomas does not bear consequences for tumor grading. Nevertheless, knowledge of their in vivo perfusion MRI characteristics may aid in their more correct diagnosis which may have prognostic implications.

To our knowledge, the perfusion differences in peritumoral edema of different subtypes of meningiomas have not yet been fully analyzed and established. The mean rCBVs derived from peritumoral edema of anaplastic meningiomas were statistically higher than in the other three benign subtypes in our series. This might be attributed to several factors, especially to cortical disruption and brain invasion, which are more common in anaplastic meningiomas than in grade

1 subtypes and may result in a richer peritumoral blood supply. Pathological and molecular studies of meningiomas showed that VEGF-positive cells were increased in the endothelium of intratumoral vessels and associated with microvascular proliferations in peritumoral brain tissue of anaplastic meningiomas (Yoshioka et al. 1999). Previous perfusion MR studies (Uematsu et al. 2003; Bitzer et al. 2002; Domingo et al. 1998 and Hiyama et al. 1994) demonstrated significantly decreased rCBV in peritumoral brain edema of benign meningiomas compared with normal white matter. Most meningiomas included in those studies were meningothelial or transitional ones and had not been evaluated separately by subtypes. Vasogenic edema, ischemic alterations, venous compression and possibly venous stasis have been regarded as factors affecting the regional blood volume (Bitzer et al. 2002; Domingo et al. 1998; Hiyama et al. 1994). The mean rCBVs in peritumoral edema of meningothelial and angiomatous meningiomas in these studies were similar to the values in our study. The mean rCBV of peritumoral edema in fibrous meningiomas in our study was slightly higher than that of normal white matter, a result we cannot explain. Probably, further studies which include a histopathological examination of corresponding tissues may be able to give the answer.

Conclusion

Evaluation of the rCBVs of four meningioma subtypes with DSC perfusion MRI characterizes differences between the groups better than conventional MR sequences. This may, to a certain extent, be useful for the preoperative grading of different subtypes of meningiomas. Especially the high rCBV of tumor parenchyma in angiomatous meningiomas and in peritumoral edema of anaplastic meningiomas likely can be used to differentiate them from other subtypes. Studies focused on more subtypes of meningiomas together with histopathological correlations are needed to investigate the reliability and reproducibility of this technique and to better interpret its results.

References

- Aronen HJ, Gazit IE, Louis DN, Buchbinder BR, Pardo FS, Weisskoff RM, Harsh GR, Cosgrove GR, Halpern EF, Hochberg FH (1994) Cerebral blood volume maps of gliomas: comparison with tumor grade and histologic findings. *Radiology* 191:41–51
- Bitzer M, Klose U, Geist-Barth B, Nagele T, Schick F, Morgalla M, Claussen CD, Voigt K (2002) Alterations in diffusion and perfusion in the pathogenesis of peritumoral brain edema in meningiomas. *Eur Radiol* 12:2062–2076
- Carpeggiani P, Crisi G, Trevisan C (1993) MRI of intracranial meningiomas: correlations with histology and physical consistency. *Neuroradiology* 35:532–536
- Cha S (2003) Perfusion MRI: basic principles and clinical applications. *Magn Reson Imaging Clin N Am* 11:403–413
- Cha S (2006) Update on brain tumor imaging: from anatomy to physiology. *AJNR Am J Neuroradiol* 27:475–487
- Cha S, Knopp EA, Johnson G, Wetzel SG, Litt AW, Zagzag D (2002) Intracranial mass lesions: dynamic contrast-enhanced susceptibility-weighted echo-planar perfusion MRI. *Radiology* 223:11–29
- Chen TC, Zee CS, Miller CA, Weiss MH, Tang G, Chin L, Levy ML, Apuzzo ML (1992) Magnetic resonance imaging and pathological correlates of meningiomas. *Neurosurgery* 31:1015–1022
- Domingo Z, Rowe G, Blamire AM, Cadoux-Hudson TA (1998) Role of ischaemia in the genesis of oedema surrounding meningiomas assessed using magnetic resonance imaging and spectroscopy. *Br J Neurosurg* 12:414–418
- Hakyemez B, Erdogan C, Bolca N, Yildirim N, Gokalp G, Parlak M (2006) Evaluation of different cerebral mass lesions by perfusion-weighted MRI. *J Magn Reson Imaging* 24:817–824
- Herting B, Meixensberger J, Roggendorf W, Reichmann H (2003) Metabolic patterns in meningiomas. *J Neurooncol* 65:119–123
- Hiyama H, Kubo O, Tajika Y, Tohyama T, Takakura K (1994) Meningiomas associated with peritumoral venous stasis: three types on cerebral angiogram. *Acta Neurochir* 129:31–38
- Ildan F, Tuna M, Göçer AP, Boyar B, Bağdatoğlu H, Sen O, Hacıyakupoğlu S, Burgut HR (1999) Correlation of the relationships of brain-tumor interfaces, magnetic resonance imaging, and angiographic findings to predict cleavage of meningiomas. *J Neurosurg* 91:384–390
- Kimura H, Takeuchi H, Koshimoto Y, Arishima H, Uematsu H, Kawamura Y, Kubota T, Itoh H (2006) Perfusion imaging of meningioma by using continuous arterial spin-labeling: comparison with dynamic susceptibility-weighted contrast-enhanced MR images and histopathologic features. *AJNR Am J Neuroradiol* 27:85–93
- Knopp EA, Cha S, Johnson G, Mazumdar A, Golfinos JG, Zagzag D, Miller DC, Kelly PJ, Kricheff II (1999) Glial neoplasms: dynamic contrast-enhanced T2*-weighted MRI. *Radiology* 211:791–798
- Kremer S, Grand S, Remy C, Esteve F, Lefournier V, Pasquier B, Hoffmann D, Benabid AL, Le Bas JF (2002) Cerebral blood volume mapping by MRI in the initial evaluation of brain tumors. *J Neuroradiol* 29:105–113
- Louis DN, Ohgaki H, Wiestler OD, Cavenee WK, Burger PC, Jouvet A, Scheithauer BW, Kleihues P (2007) The 2007 WHO classification of tumours of the central nervous system. *Acta Neuropathol* 114:97–109
- Lupo JM, Cha S, Chang SM, Nelson SJ (2005) Dynamic susceptibility-weighted perfusion imaging of high-grade gliomas: characterization of spatial heterogeneity. *AJNR Am J Neuroradiol* 26:1446–1454
- Okamoto H, Li J, Vortmeyer AO, Jaffe H, Lee YS, Gläsker S, Sohn TS, Zeng W, Ikejiri B, Proescholdt MA, Mayer C, Weil RJ, Oldfield EH, Zhuang Z (2006) Comparative proteomic profiles of meningioma subtypes. *Cancer Res* 66:10199–10204
- Riemenschneider MJ, Perry C, Reifenberger G (2006) Histological classification and molecular genetics of meningiomas. *Lancet Neurol* 5:1045–1054
- Rosen BR, Belliveau JW, Vevea JM, Brady TJ (1990) Perfusion imaging with NMR contrast agents. *Magn Reson Med* 14:249–265
- Sugahara T, Korogi Y, Kochi M, Ushio Y, Takahashi M (2001) Perfusion-sensitive MRI of gliomas: comparison between gradient-echo and spin-echo echo-planar imaging techniques. *AJNR Am J Neuroradiol* 22:1306–1315
- Uematsu H, Maeda M, Itoh H (2003) Peritumoral brain edema in intracranial meningiomas evaluated by dynamic perfusion-weighted MRI: a preliminary study. *Eur Radiol* 13:758–762
- Villringer A, Rosen BR, Belliveau JW, Ackerman JL, Lauffer RB, Buxton RB, Chao YS, Wedeen VJ, Brady TJ (1988) Dynamic imaging with lanthanide chelates in normal brain-contrast due to magnetic-susceptibility effects. *Magn Reson Med* 6:164–174
- Weisskoff R, Belliveau J, Kwong K, Rosen B (1993) Functional MRI of capillary hemodynamics. In: Potchen E (ed) *Magnetic resonance angiography: concepts and applications*. Mosby, St Louis, pp 473–484
- Yang S, Law M, Zagzag D, Wu HH, Cha S, Golfinos JG, Knopp EA, Johnson G (2003) Dynamic contrast-enhanced perfusion MRI measurements of endothelial permeability: differentiation between atypical and typical meningiomas. *AJNR Am J Neuroradiol* 24:1554–1559
- Yoshioka H, Hama S, Taniguchi E, Sugiyama K, Arita K, Kurisu K (1999) Peritumoral brain edema associated with meningioma: influence of vascular endothelial growth factor expression and vascular blood supply. *Cancer* 85:936–944
- Zierler KL (1962) Theoretical basis of indicator-dilution methods for measuring flow and volume. *Circ Res* 10:393–407

Intracranial Meningioma in Mice: Noninvasive Bioluminescence Imaging

9

Brian T. Ragel and Randy L. Jensen

Abstract

Meningioma research is undertaken to assess possible future treatment options for surgically or biologically incurable tumors. To date, additional surgery, radiation, and systemic therapies have been disappointing. In this chapter, we outline *in vivo* and *in vitro* meningioma research models, focusing on the intracranial meningioma mouse model that uses a novel bioluminescence imaging (BLI) system. *In vitro* and *in vivo* meningioma research utilizes cells grown from operative specimens and immortal cell lines (CH-157-MN and IOMM-Lee). *In vivo* models involve the injection of these cells into the flanks or heads of mice. The ease of monitoring tumor growth in the subcutaneous flank model is its primary advantage. The intracranial BLI mouse model places xenograft tumors in their natural location and enables sequential noninvasive measurement in a cost-effective manner. The BLI system uses the luciferase gene transfected into immortal meningioma cell lines to obtain cells that express the luciferase enzyme, which oxidizes luciferin in a reaction that releases photon energy that can be measured noninvasively. Both flank and intracranial meningioma mouse tumors exhibit microscopic, immunohistochemical, and ultrastructural features characteristic of meningiomas. Intracranial tumors exhibit characteristics of aggressive meningiomas by insinuating along arachnoid planes and invading brain. Future research with these models may hold promise for patients with refractory meningiomas.

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Introduction

Meningiomas arise from the central nervous system meninges and account for approximately 20% of all primary adult intracranial tumors (Al-Rodhan and Laws 1991). The cell of origin is a specialized meningotheial cell called an arachnoid cap cell, usually present within the

arachnoid villi but sometimes located throughout the craniospinal arachnoid space (Al-Rodhan and Laws 1991). Meningiomas are more common in women (2:1) and usually occur in older patients. Meningioma pathologic grading takes into account both the tumor subtypes known to have a higher rate of recurrence and specific histologic features that suggest a more aggressive biology (Louis et al. 2000; Burger et al. 2002). Meningiomas are graded as benign (91%), atypical (5%), and anaplastic/malignant (4%) (Mahmood et al. 1993; Gonzales 1995). Although most meningiomas can be surgically removed, their intrinsic biology and location are still the main determinants of the overall outcome of the patient (Simpson 1957; Louis et al. 2000; Burger et al. 2002). Treatment options for surgically or biologically incurable tumors, including additional surgery, radiation, and systemic therapies, have been disappointing. Meningioma researchers use *in vitro* and *in vivo* models to study possible future treatments for meningiomas (Mahmood et al. 1993; Louis et al. 2000).

***In Vitro* and *In Vivo* Meningioma Research**

Meningioma Cell Lines

The study of meningioma biology and possible treatment regimes involves *in vitro* and *in vivo* studies on immortal cell lines and cells derived from operative specimens (Ragel et al. 2008a). The two most widely used immortal cell lines in meningioma research include the CH-157-MN and IOMM-Lee cell lines. Both of these cell lines grow easily in culture and incite xenograft tumors resembling meningiomas (Ragel et al. 2008a, b). In 1995, the CH-157-MN immortal cell line was cultured from a meningioma taken from a 55-year-old woman (Tsai et al. 1995). Studies analyzing these cells grown *in vitro* show karyotypic changes typical of meningiomas (Tsai et al. 1995; Louis et al. 2000; Ragel et al. 2008a, b). Lee (1990) described his WHO grade 3 intraosseous malignant meningioma cell

line (IOMM-Lee), which was obtained from a 61-year-old man. Both of these cell lines show *in vitro* doubling times of approximately 20 h, and recent studies in our laboratory have demonstrated that they retain karyotypic, immunohistochemical, and ultrastructural features that fit their initial descriptions (e.g., vimentin positive, EMA positive, GFAP negative, and desmosomes) (Lee 1990; Tsai et al. 1995; Louis et al. 2000; Ragel and Jensen 2005; Ragel et al. 2007, 2008a, b).

In general, immortal cell lines are easy to maintain in culture and rapidly divide, making them readily available for experiments, whereas operative specimens are more difficult to maintain in culture, with longer division times, thus lengthening experiment times. The two primary concerns about immortal cell lines are that (1) genetic changes can occur over time in immortal cell lines, and (2) the aggressive nature of these cell lines may make results applicable only to aggressive meningioma subtypes, which comprise a small proportion of meningiomas. Thus, cell lines from operative specimens are also used.

Cell lines from operative specimens fall into two categories, those with simple and those with complex karyotypes. Simple karyotypes are normal or contain just one chromosome abnormality by Giemsa banding, whereas complex karyotypes have multiple chromosomal abnormalities (Ragel et al. 2008a). Meningiomas cultivated from operative specimens with simple karyotypes grew slowly in culture, with doubling times of approximately 120 h, whereas complex karyotypes had dividing times between 40 and 65 h (Ragel et al. 2008a). Unlike immortal cell lines, cell lines cultivated from operative specimens show contact inhibition, with slowed growth upon reaching Petri dish confluence. The advantage of cell lines derived from operative samples is that genetically they are closer to the actual tumor from which they were derived, making results from experimentation more applicable. In the end, cell culture experiments do not necessarily reflect *in vivo* responses, so meningioma animal models have been developed (Medhkour et al. 1989).

The Mouse Subcutaneous Flank Xenograft Meningioma Model

Meningioma animal models utilize cell lines derived from operative specimens and immortal cell lines as described above (Lee 1990; Tsai et al. 1995, 1999; Ragel et al. 2006). Historically, transplantation of human meningiomas has been described in guinea pig eyes, chorioallantoic membrane of the duck and chick, and the subcutis, subrenal capsule, flank, brain, and skull base of athymic mice (Medhkour et al. 1989; Martuza et al. 1991; McCutcheon et al. 2000). The mouse subcutaneous flank xenograft meningioma model offers the advantages of ease of tumor measurement during treatments and relatively low cost (Andersen and Jensen 1998; Jensen and Wurster 2001; Ragel et al. 2006, 2008a). Overall, our experience with the mouse xenograft flank model shows that immortal cell lines and cell lines derived from operative specimens with faster *in vitro* doubling times (i.e., complex karyotypes) consistently incite flank tumors and that these flank tumors exhibit histologic, immunohistochemical, and ultrastructural features consistent with meningiomas (e.g., vimentin positive, EMA positive, and desmosomes) (Akat et al. 2003; Ragel et al. 2008a). Experimental studies evaluating the effect of radiation, RU-486, hydroxyurea, calcium channel antagonists, pegvisomant, irinotecan, nelfinavir, imatinib, and small interfering RNA (siRNA) against urokinase plasminogen activator (uPA) have all been performed in the subcutaneous mouse meningioma model (Olson et al. 1987; Jensen and Wurster 2001; McCutcheon et al. 2001; Ragel et al. 2006; Gupta et al. 2007a, b; Kargiotis et al. 2008).

Although the xenograft flank model demonstrated microscopic, immunohistochemical, and ultrastructural features characteristic of meningiomas, it does not replicate the natural location of meningiomas and thus may lack some biological relevance. Obviously, the intracranial model is the most biologically desirable, but its use was historically limited by the inability to obtain direct serial measurements within the skull in a cost-effective manner,

making it difficult to assess treatment regimes (McCutcheon et al. 2000; Salhia et al. 2002; Kondraganti et al. 2006). To circumvent these difficulties, the intracranial mouse model using a bioluminescence imaging system was developed (Ragel et al. 2008b).

The Mouse Intracranial Xenograft Meningioma Bioluminescence Model

The mouse intracranial meningioma bioluminescence (BLI) model allows for serial measurement of tumors in a noninvasive, cost-effective manner, while replicating the macroscopic, microscopic, immunohistochemical, and ultrastructural features of aggressive meningiomas (Ragel et al. 2008b). The BLI system takes advantage of the firefly luciferase enzyme, which oxidizes luciferin in a reaction that releases photon energy that is easily measured noninvasively (Rubin et al. 2003; Szentirmai et al. 2006).

The luciferase gene is inserted into CH-157-MN and IOMM-Lee meningioma cell lines by infecting them with a modified pMMP retrovirus fused with the coding sequences for luciferase and neomycin phosphotransferase (pMMP-LucNeo) (Rubin et al. 2003). This reporter plasmid (pMMP-LucNeo) is under the control of a modified thymidine kinase promoter that constitutively expresses luciferase (Rubin et al. 2003). The neomycin phosphotransferase gene encodes for an enzyme that inhibits several aminoglycoside compounds (e.g., neomycin, kanamycin, G418). G418 is toxic to mammalian cells (e.g., CH-157-MN). Therefore, cells transfected with the pMMP-LucNeo construct are selected with G418 to make the CH-157-MN-LucNeo and IOMM-Lee-LucNeo cell lines, respectively (Rubin et al. 2003; Szentirmai et al. 2006; Ragel et al. 2008b).

CH-157-MN-LucNeo and IOMM-Lee-LucNeo can be transplanted intracranially into mice in either convexity or skull base locations with operative mortality rates of approximately 10% and tumor induction rates of greater than 90% (Ragel et al. 2008b). Depending on the number of cells initially xenografted, control animals survive

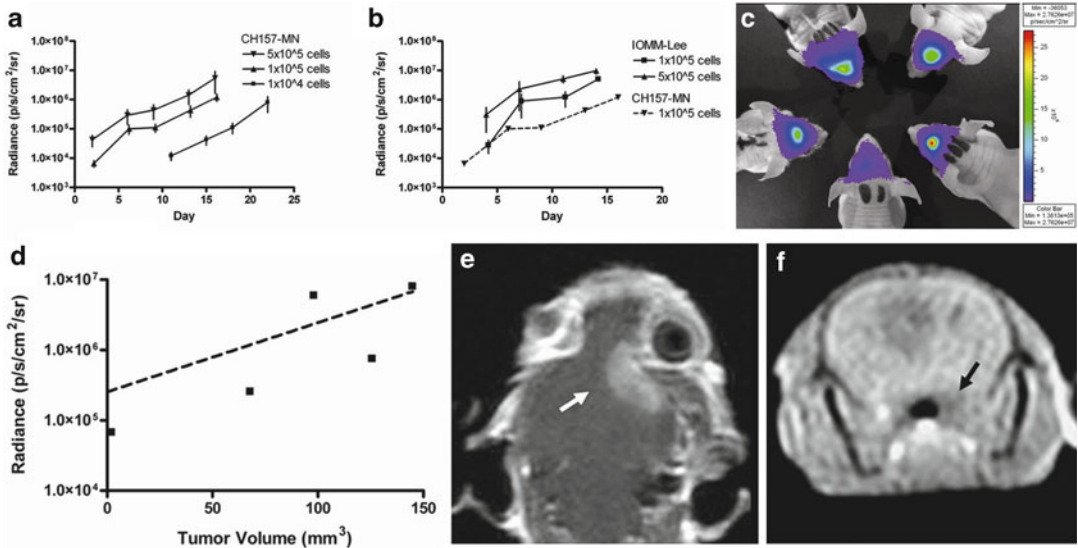


Fig. 9.1 (a–b) Logarithmic (Log 10) line graphs showing radiance values (photons/second/cm²/steradian) of intracranial meningioma xenograft mouse tumors derived from CH-157-MN-LucNeo (a) and IOMM-Lee-LucNeo (b) cell lines. Error bars represent number of mice in each group. (a) Radiance values of CH-157-MN-LucNeo skull base tumors incited by varying number of cells per mouse (1×10^4 , $n=11$; 1×10^5 , $n=10$; and, 5×10^5 , $n=4$). Note: Logarithmic graphs of tumor radiance are parallel, indicating a constant exponential growth rate, regardless of initial cell number injected. (b) Radiance values of IOMM-Lee-LucNeo skull base xenograft meningiomas (1×10^5 , $n=13$; 5×10^5 , $n=2$). The dotted line represents the CH-157-MN-LucNeo skull base tumors for comparison (1×10^5 , $n=10$). (c) Representative photo of luciferase-

expressing intracranial tumors after injection of firefly D-Luciferin. Radiance values measured with the IVIS 100 imaging system. The colored bar at right shows that higher photon emissions are red. (d) Scatter plot of CH-157-MN-LucNeo skull base meningiomas radiance (Log 10, photons/second/cm²/steradian) versus tumor volume (mm³). Each point represents one mouse. In general, higher radiance values correlated with larger tumors. Spearman correlation test, $R=0.900$. (e–f) MR imaging scans of mouse brains. (e) Axial, T1-weighted image with contrast showing vividly enhancing dural-based tumor (white arrow). (f) Coronal, T2-weighted image depicting a hypointense mass adjacent to the skull base (black arrow) (Reprinted from Ragel et al. (2008b), with permission)

between 2 and 3 weeks (Ragel et al. 2008b). Intracranial xenografted tumor cells are imaged by intraperitoneal injection of firefly D-Luciferin (Xenogen Corporation, Alameda, CA), a substrate that is oxidized by luciferase-containing cells in a reaction that releases photon energy. Photons are then assessed using the Xenogen IVIS 100 imaging system (Xenogen), by recording the bioluminescent signal emitted from xenografted tumors (Fig. 9.1a–c). Excellent correlation between BLI and MR imaging tumor volumes is noted, with a Spearman correlation test showing a strong linear correlation ($R=0.900$, $P=0.083$) (Fig. 9.1d–f) (Ragel et al. 2008b).

At autopsy, skull base meningiomas induced from CH-157-MN-LucNeo and IOMM-Lee-LucNeo showed discrete tumors that easily separated from adjacent brain but insinuated between brain and the skull base (Fig. 9.2a). Interestingly, BLI imaging shows radiance from tumor and the underlying skull base but not the brain (Fig. 9.2b).

These tumors exhibit characteristics consistent with meningiomas when evaluated histologically, immunohistochemically, and ultrastructurally (Fig. 9.3). Like meningiomas, skull base tumors derived from bioluminescent cell lines in the mouse model grow along the skull base between brain and bone, insinuating along cranial nerves

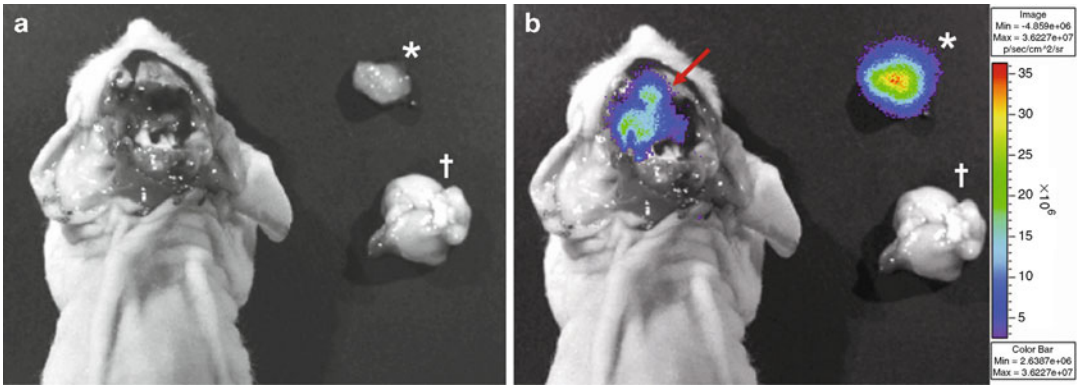


Fig. 9.2 (a–b) Representative autopsy pictures of an explanted CH-157-MN-LucNeo skull base meningioma tumor (*) and mouse brain (†). (a) Xenograft tumors grow along the skull base and are easily separated from normal

brain. (b) Bioluminescence image showing radiance readings from the tumor and skull base meninges (red arrow), but not the brain (Reprinted from Ragel et al. (2008b), with permission)

and Virchow-Robin spaces (Fig. 9.3a, c–e). Immunohistochemically, the hallmark of meningiomas is vimentin-positive staining and negative GFAP staining, with roughly 80% staining for EMA (Louis et al. 2000; Burger et al. 2002). Mouse intracranial tumors from CH-157-MN-LucNeo and IOMM-Lee-LucNeo cell lines showed 100% vimentin immunoreactivity and were negative for both EMA and GFAP (Fig. 9.3f). Immunohistochemical staining for the MIB-1 proliferative marker showed that $64\% \pm 2\%$ and $36\% \pm 3\%$ of the cells were positive for CH-157-MN-LucNeo and IOMM-Lee-LucNeo tumors, respectively (Fig. 9.3g) (Ragel et al. 2008b). These tumors also exhibit ultrastructural features of meningiomas such as desmosomes and junctional complexes (Fig. 9.3h) (Akat et al. 2003; Ragel et al. 2008b).

This novel BLI intracranial meningioma model has the advantage of enabling researchers to study these tumors in their natural location, while allowing for noninvasive serial measurements in a cost-effective manner (Szentirmai et al. 2006). Disadvantages include a stable cell transfection of a constitutively expressing bioluminescent enzyme and the possibility that transfection may alter the cell line studied (Ragel et al. 2008b). Recently, this novel BLI intracranial model has been used to study the effects of temozolomide on a malignant meningioma cell line (Baia et al. 2008).

Conclusion

Because other treatment options for surgically or biologically incurable meningiomas have been disappointing, researchers use *in vitro* and *in vivo* models to study possible future treatments. *In vitro* meningioma research uses cells grown from operative specimens and immortal cell lines (CH-157-MN and IOMM-Lee) but may not reflect *in vivo* responses. *In vivo* research involves implantation of these cell lines into animal models. Both flank and intracranial meningioma mouse tumors exhibit microscopic, immunohistochemical, and ultrastructural features characteristic of meningiomas. Flank models offer ease of tumor measurement and low cost, but are less biologically accurate. Intracranial tumors exhibit characteristics of aggressive meningiomas by insinuating along arachnoid planes and invading brain but were difficult to measure serially. The BLI system uses the luciferase gene transfected into immortal meningioma cell lines to obtain cells that express the luciferase enzyme, which oxidizes luciferin in a reaction that releases photon energy that can be measured noninvasively. Future research with these models may hold promise for patients with refractory meningiomas.

Acknowledgments We thank Kristin Kraus, M.Sc., for her excellent editorial assistance.

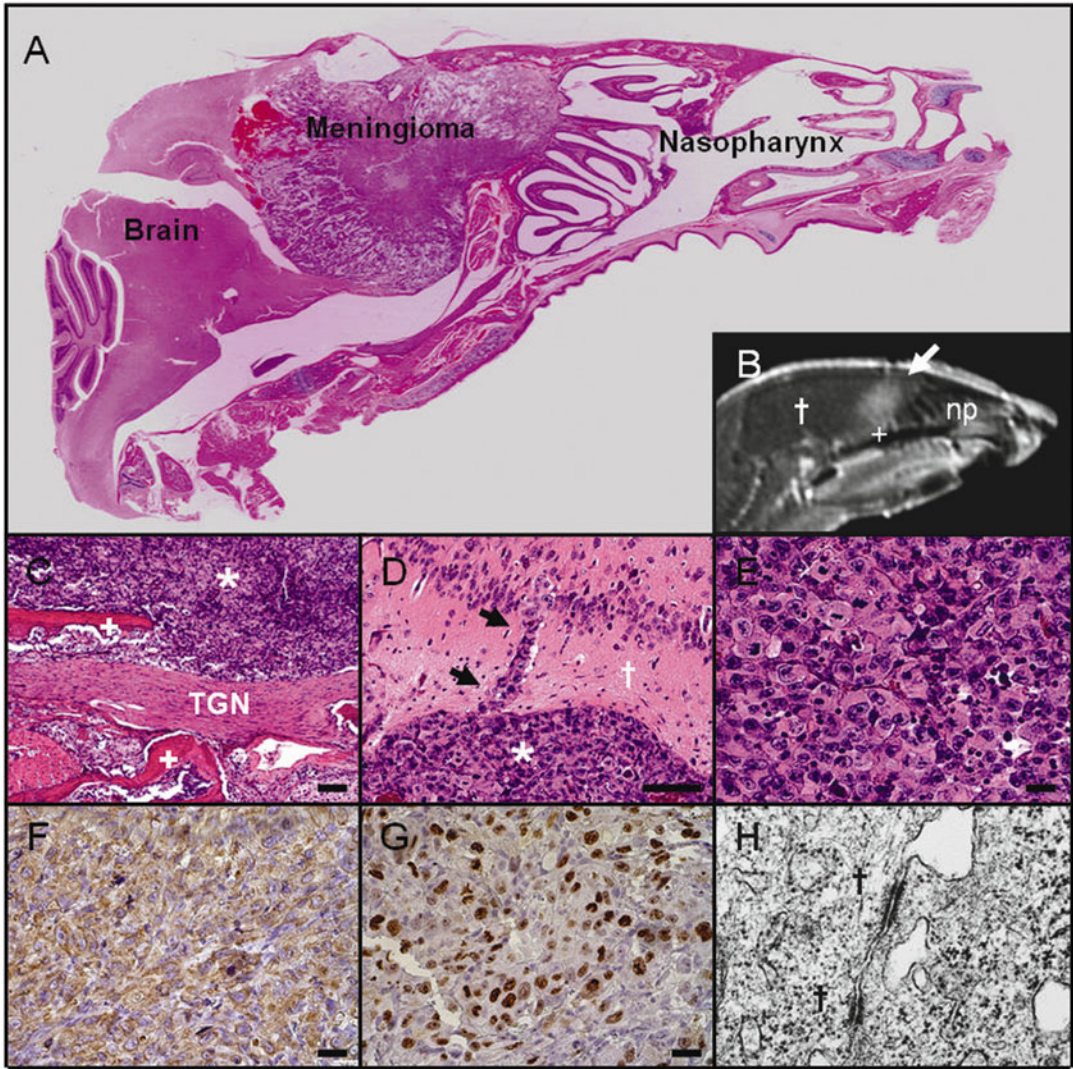


Fig. 9.3 Representative photos of CH-157-MN-LucNeo xenograft skull base meningiomas. **(a)** Parasagittal section of mouse skull showing meningioma situated between the brain and skull base. **(b)** Parasagittal, T1-weighted MR imaging with contrast showing vividly enhancing tumor (arrow) positioned between brain (†), skull base (+), and nasopharynx (np). **(c–e)** Hematoxylin and eosin stained slides. **(c)** Tumor (*) grows adjacent to the skull base (+) and trigeminal nerve (TGN) (20×, bar=0.1 mm). **(d)** Tumor (*) displaces brain (†) and insinuates along

Virchow-Robin spaces (arrows), biologic behavior consistent with aggressive meningiomas (20×, bar=100 μm). **(e)** Tumors are highly cellular with large pleomorphic nuclei (40×, bar=25 μm). **(f–g)** Immunohistochemical stained slides (40×, bar=25 μm) showing tumors to be vimentin positive (**f**, brown cytoplasmic staining), with roughly 60% of cells MIB-1 positive (**g**, brown nuclear staining). **(h)** Electron microscopy exhibiting desmosomes (†), a key ultrastructural feature of meningiomas (Reprinted from Ragel et al. (2008b), with permission)

References

- Akat K, Mennel HD, Kremer P, Gassler N, Bleck CK, Kartenbeck J (2003) Molecular characterization of desmosomes in meningiomas and arachnoidal tissue. *Acta Neuropathol (Berl)* 106:337–347

- Al-Rodhan RF, Laws ER (1991) The history of intracranial meningiomas. In: Al-Mefty O (ed) *Meningiomas*. Raven Press, Ltd, New York, pp 1–6
- Andersen C, Jensen FT (1998) Differences in blood-tumour-barrier leakage of human intracranial tumours: quantitative monitoring of vasogenic oedema and its response to

- glucocorticoid treatment. *Acta Neurochir (Wien)* 140:919–924
- Baia GS, Dinca EB, Ozawa T, Kimura ET, McDermott MW, James CD, VandenBerg SR, Lal A (2008) An orthotopic skull base model of malignant meningioma. *Brain Pathol* 18:172–179
- Burger PC, Scheithauer BW, Vogel FS (eds) (2002) *Surgical pathology of the nervous system and its coverings*. Churchill Livingstone, Edinburgh
- Gonzales MF (1995) Classification of brain tumors. In: Kaye AH, Laws ER (eds) *Brain tumors*. Churchill Livingstone, Edinburgh, p 675
- Gupta V, Samuleson CG, Su S, Chen TC (2007a) Nelfinavir potentiation of imatinib cytotoxicity in meningioma cells via survivin inhibition. *Neurosurg Focus* 23(4):E9
- Gupta V, Su YS, Samuelson CG, Liebes LF, Chamberlain MC, Hofman FM, Schonthal AH, Chen TC (2007b) Irinotecan: a potential new chemotherapeutic agent for atypical or malignant meningiomas. *J Neurosurg* 106:455–462
- Jensen RL, Wurster RD (2001) Calcium channel antagonists inhibit growth of subcutaneous xenograft meningiomas in nude mice. *Surg Neurol* 55:275–283
- Kargiotis O, Chetty C, Gogineni V, Gondi CS, Pulukuri SM, Kyritsis AP, Gujrati M, Klopfenstein JD, Dinh DH, Rao JS (2008) uPA/uPAR downregulation inhibits radiation-induced migration, invasion and angiogenesis in IOMM-Lee meningioma cells and decreases tumor growth in vivo. *Int J Oncol* 33:937–947
- Kondraganti S, Gondi CS, Gujrati M, McCutcheon I, Dinh DH, Rao JS, Olivero WC (2006) Restoration of tissue factor pathway inhibitor inhibits invasion and tumor growth in vitro and in vivo in a malignant meningioma cell line. *Int J Oncol* 29:25–32
- Lee WH (1990) Characterization of a newly established malignant meningioma cell line of the human brain: IOMM-Lee. *Neurosurgery* 27:389–395; discussion 396
- Louis DN, Scheithauer BW, Budka H, von Deimling A, Kepes JJ (eds) (2000) *Meningiomas*, World Health Organization classification of tumours: pathology and genetics: tumours of the nervous system. IARC Press, Lyon
- Mahmood A, Caccamo DV, Tomecek FJ, Malik GM (1993) Atypical and malignant meningiomas: a clinicopathological review. *Neurosurgery* 33:955–963
- Martuza RL, Malick A, Markert JM, Ruffner KL, Coen DM (1991) Experimental therapy of human glioma by means of a genetically engineered virus mutant. *Science* 252:854–856
- McCutcheon IE, Friend KE, Gerdes TM, Zhang BM, Wildrick DM, Fuller GN (2000) Intracranial injection of human meningioma cells in athymic mice: an orthotopic model for meningioma growth. *J Neurosurg* 92:306–314
- McCutcheon IE, Flyvbjerg A, Hill H, Li J, Bennett WF, Scarlett JA, Friend KE (2001) Antitumor activity of the growth hormone receptor antagonist pegvisomant against human meningiomas in nude mice. *J Neurosurg* 94:487–492
- Medhkour A, Van Roey M, Sobel RA, Fingert HJ, Lee J, Martuza RL (1989) Implantation of human meningiomas into the subrenal capsule of the nude mouse. A model for studies of tumor growth. *J Neurosurg* 71:545–550
- Olson JJ, Beck DW, Schlechte JA, Loh PM (1987) Effect of the antiprogesterone RU-38486 on meningioma implanted into nude mice. *J Neurosurg* 66:584–587
- Ragel BT, Jensen RL (2005) Molecular genetics of meningiomas. *Neurosurg Focus* 19(5):E9
- Ragel BT, Gillespie DL, Kushnir V, Polevaya N, Kelly D, Jensen RL (2006) Calcium channel antagonists augment hydroxyurea- and ru486-induced inhibition of meningioma growth in vivo and in vitro. *Neurosurgery* 59:1109–1120; discussion 1120–1121
- Ragel BT, Jensen RL, Gillespie DL, Prescott SM, Couldwell WT (2007) Celecoxib inhibits meningioma tumor growth in a mouse xenograft model. *Cancer* 109:588–597
- Ragel BT, Couldwell WT, Gillespie DL, Wendland MM, Whang K, Jensen RL (2008a) A comparison of the cell lines used in meningioma research. *Surg Neurol* 70:295–307; discussion 307
- Ragel BT, Elam IL, Gillespie DL, Flynn JR, Kelly DA, Mabey D, Feng H, Couldwell WT, Jensen RL (2008b) A novel model of intracranial meningioma in mice using luciferase-expressing meningioma cells. Laboratory investigation. *J Neurosurg* 108:304–310
- Rubin JB, Kung AL, Klein RS, Chan JA, Sun Y, Schmidt K, Kieran MW, Luster AD, Segal RA (2003) A small-molecule antagonist of CXCR4 inhibits intracranial growth of primary brain tumors. *Proc Natl Acad Sci USA* 100:13513–13518
- Salhia B, Rutka JT, Lingwood C, Nutikka A, Van Furth WR (2002) The treatment of malignant meningioma with verotoxin. *Neoplasia* 4:304–311
- Simpson D (1957) The recurrence of intracranial meningiomas after surgical treatment. *J Neurochem* 20:22–39
- Szentirmai O, Baker CH, Lin N, Szucs S, Takahashi M, Kiryu S, Kung AL, Mulligan RC, Carter BS (2006) Noninvasive bioluminescence imaging of luciferase expressing intracranial U87 xenografts: correlation with magnetic resonance imaging determined tumor volume and longitudinal use in assessing tumor growth and anti-angiogenic treatment effect. *Neurosurgery* 58:365–372
- Tsai JC, Goldman CK, Gillespie GY (1995) Vascular endothelial growth factor in human glioma cell lines: induced secretion by EGF, PDGF-BB, and bFGF. *J Neurosurg* 82:864–873
- Tsai JC, Hsiao YY, Teng LJ, Shun CT, Chen CT, Goldman CK, Kao MC (1999) Regulation of vascular endothelial growth factor secretion in human meningioma cells. *J Formos Med Assoc* 98:111–117

Incidentally Discovered Meningiomas: Growth Rates and Patterns

10

Naoya Hashimoto and Toshiki Yoshimine

Abstract

The determination of how to manage incidentally discovered meningiomas (IDMs) is becoming important, but remains controversial, mainly because the precise natural history of IDMs is not clear. We cannot predict the tumor growth or the development of symptoms. Whether all IDMs grow at the same rate over long-term follow-up has also remained unclear. This chapter discusses the natural history of IDMs and their growth rates and patterns, based on a thorough survey of the literature. Recent studies are introduced in which serial volumetric assessments of IDMs were performed over relatively long follow-up periods and patterns of growth were determined. The reports published indicate that most IDMs do not seem to grow for a certain period of time, and IDMs do not always grow exponentially, in fact exhibiting more complex patterns of growth. The authors suggest that growth patterns of IDMs might fit the Gompertzian model. This question remains unsolved, but serial monitoring of tumor volumes and regression analysis in clinical settings may reveal patterns of growth for IDMs and provide more informative data useful for determining treatment strategies.

Introduction

Meningiomas have increasingly been detected incidentally due to advances in neuroimaging, such as computed tomography (CT) and magnetic resonance imaging (MRI) (Kuratsu et al. 2000).

Neurological check-ups and neuro-radiological screening for indefinite complaints also contribute to the increased detection, especially in advanced countries (Onizuka et al. 2001). Although most incidentally discovered meningiomas (IDMs) can be cured surgically (Louis et al. 2000), the morbidity and mortality associated with surgery itself cannot be neglected, because they are asymptomatic and benign in nature (Kuratsu et al. 2000). The determination of how to manage those IDMs is becoming more

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important, but remains controversial (Nakamura et al. 2003), because we do not currently know the precise natural history of IDMs and cannot predict future potential for growth or symptomatic effects in individual cases.

Several authors have investigated the natural history of IDMs (Firsching et al. 1990; Go et al. 1998; Kuratsu et al. 2000; Nakamura et al. 2003; Olivero et al. 1995; Yano and Kuratsu 2006; Yoneoka et al. 2000). They have identified various factors predictive of tumor growth, but those have differed from study to study. In most studies, tumor growth was evaluated only by considering changes between initial and final size during follow-up. Annual growth rate (Firsching et al. 1990; Nakamura et al. 2003; Yoneoka et al. 2000) and tumor doubling time (Ttd) (Nakamura et al. 2003), used as parameters of growth rate, were also calculated only by considering changes between initial and final volumes, based on the assumption that tumors essentially grow in an exponential manner. However, whether all incidental meningiomas grow at the same rate over the course of long-term follow-up has remained unclear, as have the types of growth pattern they may exhibit throughout their natural histories (Nakasu et al. 2005).

In this chapter, issues regarding the natural history of IDMs and their growth rates and patterns are thoroughly discussed based on a survey of the literature. Recent studies (Nakasu et al. 2005; Hashiba et al. 2009) are introduced in which serial volumetric assessments were performed over relatively long follow-up periods in patients with IDMs and patterns of growth were determined for lesions by regression analysis.

Definition of IDMs

Despite the strong interests in incidental meningiomas (Firsching et al. 1990; Kuratsu et al. 2000; Nakamura et al. 2003; Niiro et al. 2000; Olivero et al. 1995; Yano and Kuratsu 2006; Yoneoka et al. 2000), the precise definition of “incidentally detected asymptomatic meningiomas” remains unclear. In this chapter, we followed the lead of other authors in judging what constitutes an IDM

(Nakamura et al. 2003; Olivero et al. 1995), defining IDMs as meningiomas harboring typical neuroimaging characteristics and discovered by neuroimaging taken for other symptoms or reasons. Importantly, the judgment that the initial symptoms are unrelated to the tumor has to be made by physicians. However, in some cases such judgment could be difficult and a report has described patients with single seizures included as asymptomatic cases (Olivero et al. 1995). As effects of meningiomas on neuro-physiological function have recently been pointed out (Tucha et al. 2003), further issues need to be clarified regarding the definition of incidental meningiomas.

Evaluation of IDM Growth

In previous studies investigating the natural history of IDMs, no standard criteria have been used for evaluation of tumor volume or radiological judgment of tumor growth (Firsching et al. 1990; Kuratsu et al. 2000; Nakamura et al. 2003; Niiro et al. 2000; Olivero et al. 1995; Yano and Kuratsu 2006; Yoneoka et al. 2000). In early studies, maximum tumor diameter was used for quantification of tumor size (Niiro et al. 2000; Olivero et al. 1995). Although this method may be useful for judging treatment response to adjuvant therapies such as radiotherapy and chemotherapy for glial tumors, commonly used methods (determination of largest diameter or the product of the two or three largest perpendicular diameters) are reportedly inaccurate for judging slight changes in the size of benign tumors such as meningiomas (Eggle et al. 1995; Hopper et al. 1996; Miller et al. 1981). Meningiomas, particularly those at the cranial base, show complex shapes and tumor growth in any direction that is not along the axis of measurement can be missed (Nakamura et al. 2003).

Volumetry appears to reflect tumor size more accurately, since this method can be used for irregularly shaped tumors and copes with changes in tumor size in any direction. It is therefore considered a more accurate method than diameter measurement (Sorensen et al. 2001) and has been used to evaluate tumor size in recent studies

(Nakamura et al. 2003; Nakasu et al. 2005; Yoneoka et al. 2000; Hashiba et al. 2009).

Given the advancement of personal and institutional circumstances for computation, volumetric analysis for each IDM should have become easier to perform. However, calculating tumor volume from MRI data for each patient remains time-consuming, as compared with visual judgment by physicians in daily clinical settings. Therefore, relatively few reports have calculated IDM size by volumetric assessment (Firsching et al. 1990; Yoneoka et al. 2000; Nakamura et al. 2003; Nakasu et al. 2005; Hashiba et al. 2009). The most recent reviews and meta-analyses of natural history and growth rates have indicated linear rates of tumor growth based on measurements of diameter, due to a lack of sufficient volumetric data (Sughrue et al. 2010).

Volumetric Accuracy

Although volumetry appears to reflect accurate tumor size, standardized methods of volumetry including acquisition of MRI data from the patient, MRI sequences and volumetric software are lacking. In most studies in which volumetric assessments have been performed (Nakamura et al. 2003; Nakasu et al. 2005; Yoneoka et al. 2000), MRI series with relatively thick slice intervals (in the range of 2.0–8.0 mm) were used without clarifying the accuracy of volumetry using MRI series with thick slice intervals. MRI data acquisition settings and volumetric software have differed from report to report. At present, determining a standard method for volumetry of IDMs seems difficult in such a complex situation of medical management around the world, but a confirmation study of volumetric accuracy and criteria for judging IDMs growth within individual institutions seems the least that should be done.

Hashiba et al. (2009) evaluated volumetric accuracy by validating potential measurement errors. In their volumetric analyses, tumor size was evaluated using Scion Image for Windows software (Scion, Frederick, MD), with the enhanced area of tumor in each slice image measured by manual tracing of tumor boundaries.

The sum of the enhanced areas was then multiplied by the slice interval of the MRI series. They noted two possible sources of errors accompanying volumetry: one caused by manual tracing of the tumor; and the other by using MRI series with thick slice intervals. A preliminary study with 10 cases showed that volumetry in an MRI series using 6.0 mm slice intervals was estimated to show acceptable accuracy for evaluating tumor volume, as compared with an MRI series using 2.0 mm slice intervals in the same patients.

Volumetric Judgment of Tumor Growth

The judgment of tumor growth should be made carefully, as modalities we have seen such as linear measurement and volumetry inevitably include measurement errors. Such errors cannot be neglected because IDMs usually show no or very small increases in size during relatively short periods of clinical observation. In 5 papers that utilized volumetry for measurement of IDMs, criteria differed from report to report (Firsching et al. 1990; Yoneoka et al. 2000; Nakamura et al. 2003; Nakasu et al. 2005; Hashiba et al. 2009). Firsching et al. (1990) did not describe the criteria for volumetric judgment of tumor growth. Yoneoka et al. (2000) reported on the growth of 37 IDMs and recruited the absolute increase in tumor volume (cm^3/year). They intentionally defined tumors showing an increase in absolute growth rate of over 1 cm^3 as showing tumor growth, without any theoretical explanations. Nakamura et al. (2003) used absolute (cm^3/year) and relative ($\%/ \text{year}$) growth rates and all 41 IDMs in their study showed increased volume in their period of observation.

In contrast, Nakasu et al. (2005) and Hashiba et al. (2009) presented apparent cut-off values for judging IDM growth, in consideration of potential measurement errors. Nakasu et al. (2005) measured volume in 31 tumors from 20 patients and repeated measurements 3 times, calculating standard deviation (SD). Mean SD was 4.1% and 2 SDs were used to define cut-offs for change, meaning that any change less than 8.2% between

the initial and final volume was considered to represent no change. Hashiba et al. (2009) also evaluated measurement error in their main study of 70 IDMs, finding that the 2 SDs as calculated from three measurements corresponded to a change of less than 15% in each tumor volume. Also in the preliminary evaluation of potential measurement errors using thick and thin slice intervals of MRI as presented in the previous paragraph, a change in tumor volume less than 15% can be thought as representing measurement error. Considering results from validation of the two types of measurement errors, the cut-off line for tumor growth was set to 15% in their study.

Most IDMs May Not Grow

Based on clinical observation and investigation on the natural history of meningiomas, we know that IDMs can be separated into tumors showing growth (growth group) and those not showing growth (no growth group) within a certain duration of follow-up (usually more than 1 year). Furthermore, even in the growth group, IDMs show very low growth rates. According to reports in which linear (diameter) measurements were utilized, Olivero et al. (1995), Go et al. (1998) and Yano and Kuratsu (2006) reported that the growth and no-growth groups comprised 22.2% and 77.8%, 11.4% and 88.6% and 37.3% and 62.7%, respectively. Mean durations of imaging follow-up were 29, 74 and over 84 months, respectively. Growth rates in the growth group from those reports were described as an average of 2.4 mm/year (Olivero et al. 1995), an annual growth rate of 12.0% (Go et al. 1998) and a mean of 1.9 mm/year (Yano and Kuratsu 2006). Likewise, the most recent meta-analysis with linear rates of tumor growth has shown that in 357 patients with IDM reported in the literature, 56% of all untreated tumors and 51% of tumors with an initial diameter at or below 2.5 cm did not demonstrate any evidence of growth on serial imaging obtained over a median follow-up period of 4.6 years (Sughrue et al. 2010).

The same results have been found in reports of volumetric analysis. Yoneoka et al. (2000)

reported that 28 of 37 patients (75.7%) showed no increase in IDM volume during the mean follow-up period of 4.2 years, although they intentionally set the cut-off value for the judgment of growth at an annual growth rate above 1 cm³. Hashiba et al. (2009) reported that in the volumetric analysis with 70 IDMs, 26 of those (37.1%) did not exhibit obvious tumor growth during a mean follow-up period of 39.3 months, after strict consideration of the judgment criteria for tumor growth.

Again, the important finding is that at least some IDMs can be categorized as showing no growth according to the literature. Whether this reflects the fact that some IDMs truly do not grow, or that slow growth is occurring but just cannot be detected with current modalities remains controversial. The possibility remains that IDMs may not show a constant growth rate throughout their development.

As to the radiological prediction of IDM growth rates, many authors have indicated that hyperintensity on T2-weighted imaging is positively correlated with tumor growth, while the presence of calcification is negatively correlated with tumor growth (Niino et al. 2000; Nakasu et al. 2005; Yano and Kuratsu 2006; Hashiba et al. 2006). In the above-mentioned study by Hashiba et al. (2009) with volumetry, presence of calcification was the only factor that significantly discriminated the non-growth and growth groups.

Rates and Patterns of IDM Growth

Considering that the majority of IDMs do not show growth, the question arises as to whether IDMs grow at the same rate. In growth analyses of cancer, cells are generally believed to show a constant doubling time, representing the period required for one cell to split into 2 cells. When cells show a constant doubling time, the growth pattern of the tumor should be exponential.

Nakasu et al. successfully introduced the concept of serial, close follow-up of imaging and regression analyses for growth pattern. They investigated patterns of growth for meningiomas, including recurrent and symptomatic cases, and

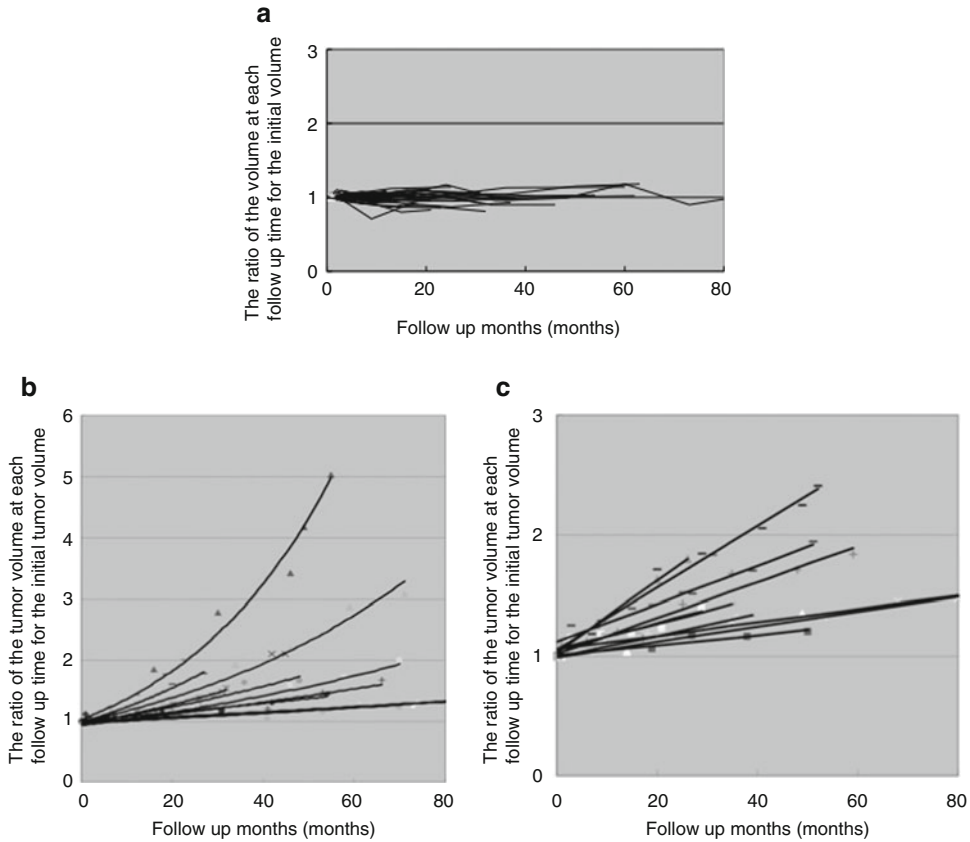


Fig. 10.1 Time courses of tumor volumes in 70 IDMs (Hashiba et al. 2009). No obvious tumor growth was seen in 26 tumors (a). Among the 44 tumors with obvious

tumor growth, regression analysis revealed that 16 tumors fit a pattern of exponential growth (b), while 15 fit a pattern of linear growth (c)

suggested three potential patterns of meningioma growth: exponential; linear; and no growth (Nakasu et al. 2005). Few studies have investigated patterns of growth among incidental meningiomas. In previous studies of the natural history of incidental meningiomas, tumor doubling time (Ttd) and annual growth rate have been used as parameters for tumor growth, calculated solely by estimating initial and final volumes during the follow-up period based on the assumption that tumors grew exponentially (Firsching et al. 1990; Nakamura et al. 2003).

Hashiba et al. (2009) subsequently performed serial volumetric assessments for 70 IDMs and reported the results of regression analyses, plotting the volume along the time axis and fitting the growth curve to either an exponential or linear

pattern in each patient. In each case in which growth fit either pattern, Ttd was estimated using a regression equation. Annual growth rate as clinically useful information in each case was also calculated as the percentage of the average annually grown volume for the initial volume. Among 70 patients with IDMs, 44 patients were included in the group with tumor growth, while 26 patients were included in the group without tumor growth. Figure 10.1a shows time courses of the tumor volumes for 26 tumors without obvious tumor growth. Of the 44 growing tumors, 16 tumors fit a pattern of exponential growth, while 15 fit a pattern of linear growth. Figure 10.1b, c show the time course of tumor volumes for the 16 tumors with exponential growth and 15 tumors with linear growth, respectively. Ttd ranged from 13 to 234 months

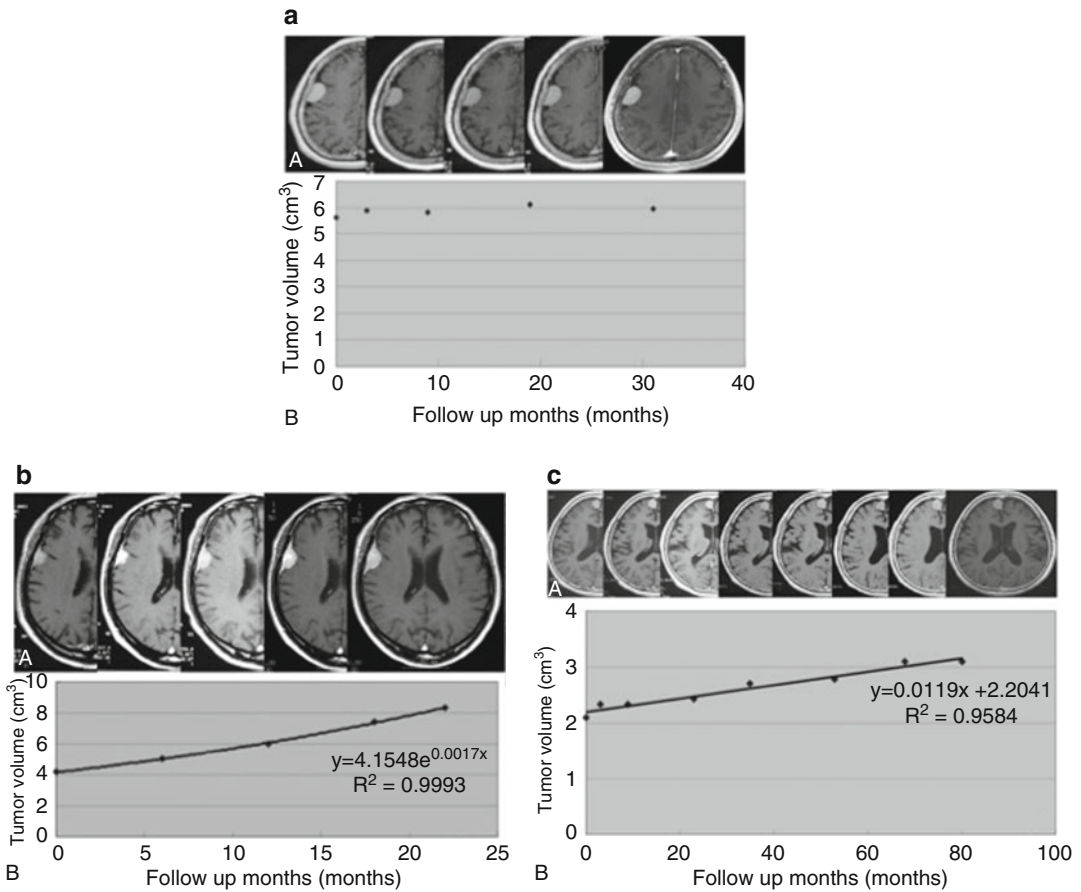


Fig. 10.2 Examples of serial changes in tumor volumes in illustrative cases without growth (a), and with exponential (b) or a linear (c) growth pattern. (a) Serial MRI images; (b) serial changes in tumor volume

(mean, 93.6 months). Annual growth rates were calculated, with a mean of 25% in the exponential pattern group and 15% in the linear pattern group, which was higher in the exponential pattern group. The remaining 13 growing tumors appeared to fit neither exponential nor linear curves, since the regression coefficients were not significant. Figure 10.2 shows examples of serial changes in tumor volumes in illustrative cases without growth (Fig. 10.2a), and with exponential (Fig. 10.2b) or linear (Fig. 10.2c) growth patterns.

The serial monitoring of tumor volumes and regression analysis by Hashiba et al. (2009) revealed that IDMs do not always grow exponentially, and can show other growth patterns. This finding suggests that Ttd or annual growth rate as

calculated by observations over a short period of time may not predict the growth potential and natural history of tumors over the lifetime of the patient. In addition, they indicated annual growth rates for both exponential and linear groups, as growth rate on a yearly basis is thought to represent useful information for clinicians. However, in tumors showing linear growth, growth rates may mathematically change over time. Tumor cell kinetics can be complex and their relationships to tumor volume have yet to be clarified, particularly in meningiomas.

In some cases, growth curves fit well to both patterns according to regression analysis, and the coefficients of determination for each regression equation were close. As meningiomas basically

show slow growth rates, growth curves may appear to fit linear patterns in the early phase of some cases, even if growth is actually exponential. Longer periods of observation will help to clarify patterns of growth.

Thirteen of the 44 tumors with growth exhibited neither exponential nor linear patterns of growth. The relatively short follow-up period may have been one reason for this, since regression coefficients calculated were not significant for some tumors in this group due to the short follow-up despite high coefficients of determination on regression analysis. Measurement error in volumetry was considered as another potential reason for this in the case of small tumors, since tumor volume is obviously influenced by measurement error.

Various studies have assumed that growth of meningiomas was either exponential or linear (Jaaskelainen et al. 1985; Nakaguchi et al. 1999; Nakamura et al. 2003). In studies like those (Nakasu et al. 2005; Hashiba et al. 2009) fitting growth curves to only two patterns of growth may raise some questions. Such assumptions have not been proven and the validity of dichotomizing tumor growth into exponential and linear growth patterns requires further assessment.

Feasibility of the Gompertzian Model of IDM Growth

In previous *in vivo* and *in vitro* studies, the Gompertzian model of growth has been widely and successfully used as a simple yet adequate descriptor of tumor growth curves (Demicheli et al. 1989; Gyllenberg and Webb 1989; Norton 1988). The growth behavior described by the Gompertzian model is exponential in the early stage, but then saturates, approaching a plateau as tumor size increases (Fig. 10.3) (Castro et al. 2003). If meningiomas grew in this fashion over the lifetime of the patient, patterns of growth would vary according to the current phase (Nakasu et al. 2005; Hashiba et al. 2009). This hypothesis may be supported from the two findings that tumors in younger patients tend to grow

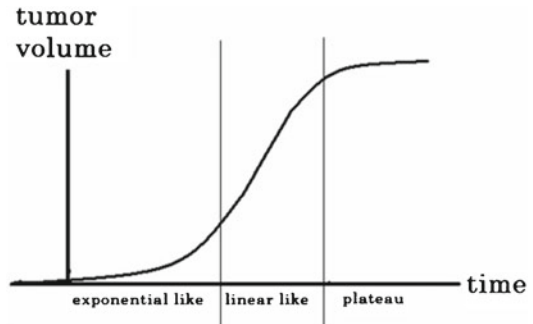


Fig. 10.3 Gompertzian model of growth

more than those in older patients (Yoneoka et al. 2000; Nakamura et al. 2003), and that smaller tumors tend to grow more than larger tumors (Olivero et al. 1995; Nakamura et al. 2003; Yano and Kuratsu 2006; Sughrue et al. 2010). Median initial tumor volume was smaller in the exponential pattern group than in the linear pattern group, which may also support this hypothesis (Hashiba et al. 2009). Unfortunately, it is not easy to test this hypothesis, since we were unable to observe tumors from their very initial appearance. Testing of this hypothesis will require follow-up of tumor volume for as long as possible.

Conclusion

Based on the recent literature, IDMs do not always grow exponentially, and in fact exhibit more complex patterns of growth. This is an unsolved question for now, but serial monitoring of tumor volumes and regression analyses in clinical settings may reveal rates and patterns of growth for incidental meningiomas and provide information useful for determining treatment strategies.

References

- Castro MA, Klamt F, Grieneisen VA, Grivicich I, Moreira JC (2003) Gompertzian growth pattern correlated with phenotypic organization of colon carcinoma, malignant glioma and non-small cell lung carcinoma cell lines. *Cell Prolif* 36:65–73
- Demicheli R, Foroni R, Ingrosso A, Pratesi G, Soranzo C, Tortoreto M (1989) An exponential-gompertzian

- description of LoVo cell tumor growth from in vivo and in vitro data. *Cancer Res* 49:6543–6546
- Eggli KD, Close P, Dillon PW, Umlauf M, Hopper KD (1995) Three-dimensional quantitation of pediatric tumor bulk. *Pediatr Radiol* 25:1–6
- Firsching RP, Fischer A, Peters R, Thun F, Klug N (1990) Growth rate of incidental meningiomas. *J Neurosurg* 73:545–547
- Go RS, Taylor BV, Kimmel DW (1998) The natural history of asymptomatic meningiomas in Olmsted county, Minnesota. *Neurology* 51:1718–1720
- Gyllenberg M, Webb GF (1989) Quiescence as an explanation of gompertzian tumor growth. *Growth Dev Aging* 53:25–33
- Hashiba T, Hashimoto N, Maruno M, Izumoto S, Suzuki T, Kagawa N, Yoshimine T (2006) Scoring radiologic characteristics to predict proliferative potential in meningiomas. *Brain Tumor Pathol* 23:49–54
- Hashiba T, Hashimoto N, Izumoto S, Suzuki T, Kagawa N, Maruno M, Kato A, Yoshimine T (2009) Serial volumetric assessment of the natural history and growth pattern of incidentally discovered meningiomas. *J Neurosurg* 110(4):675–684
- Hopper KD, Kasales CJ, Eggli KD, TenHave TR, Belman NM, Potok PS (1996) The impact of 2D versus 3D quantitation of tumor bulk determination on current methods of assessing response to treatment. *J Comput Assist Tomogr* 20:930–937
- Jaaskelainen J, Haltia M, Laasonen E, Wahlstrom T, Valtonen S (1985) The growth rate of intracranial meningiomas and its relation to histology. An analysis of 43 patients. *Surg Neurol* 24:165–172
- Kuratsu J, Kochi M, Ushio Y (2000) Incidence and clinical features of asymptomatic meningiomas. *J Neurosurg* 92:766–770
- Louis DN, Scheithauer BW, Budka H (2000) Meningiomas. In: Kleihues P (ed) WHO classification of tumors: pathology and genetics of tumors of the nervous system. IARC, Lyon, pp 176–184
- Miller AB, Hoogstraten B, Staquet M, Winkler A (1981) Reporting results of cancer treatment. *Cancer* 47:207–214
- Nakaguchi H, Fujimaki T, Matsuno A, Matsuura R, Asai A, Suzuki I (1999) Postoperative residual tumor growth of meningioma can be predicted by MIB-1 immunohistochemistry. *Cancer* 85:2249–2254
- Nakamura M, Roser F, Michel J, Jacobs C, Samii M (2003) The natural history of incidental meningiomas. *Neurosurgery* 53:62–70; discussion 62–70
- Nakasu S, Fukami T, Nakajima M, Watanabe K, Ichikawa M, Matsuda M (2005) Growth pattern changes of meningiomas: long-term analysis. *Neurosurgery* 56:946–955; discussion 946–955
- Niiri M, Yatsushiro K, Nakamura K, Kawahara Y, Kuratsu J (2000) Natural history of elderly patients with asymptomatic meningiomas. *J Neurol Neurosurg Psychiatry* 68:25–28
- Norton L (1988) A gompertzian model of human breast cancer growth. *Cancer Res* 48:7067–7071
- Olivero WC, Lister JR, Elwood PW (1995) The natural history and growth rate of asymptomatic meningiomas: a review of 60 patients. *J Neurosurg* 83:222–224
- Onizuka M, Suyama K, Shibayama A, Hiura T, Horie N, Miyazaki H (2001) A symptomatic brain tumor detected at brain check-up. *Neurol Med Chir (Tokyo)* 41:431–434; discussion 435
- Sorensen AG, Patel S, Harmath C, Bridges S, Synnott J, Sievers A (2001) Comparison of diameter and perimeter methods for tumor volume calculation. *J Clin Oncol* 19:551–557
- Sughrue ME, Rutkowski MJ, Aranda D, Barani JJ, McDermott MW, Parsa AT (2010) Treatment decision making based on the published natural history and growth rate of small meningiomas: a review and meta-analysis. *J Neurosurg*. doi:10.3171/2010.3.JNS091966
- Tucha O, Smely C, Preier M, Becker G, Paul GM, Lange KW (2003) Preoperative and postoperative cognitive functioning in patients with frontal meningiomas. *J Neurosurg* 98:21–31
- Yano S, Kuratsu J (2006) Indications for surgery in patients with asymptomatic meningiomas based on an extensive experience. *J Neurosurg* 105:538–543
- Yoneoka Y, Fujii Y, Tanaka R (2000) Growth of incidental meningiomas. *Acta Neurochir (Wien)* 142:507–511

Cystic Papillary Meningioma: Diagnosis

11

Zhi Li and Yang Li

Abstract

Meningiomas usually present as benign tumors corresponding to WHO grade I. The development of the papillary variant of meningiomas (WHO grade III) with cyst formation in the central nervous system is extremely rare. In fact, only a few cases of cystic papillary meningioma (CPM) have been described up to date. CPM is a challenging diagnostic problem and may be mistaken for several intracranial diseases with predominant cyst on neuroimaging. Usually, CPM is composed of solid and cystic part. The cyst may be intratumoral or peritumoral. The solid components of CPMs are typically composed of sheets of cells arranged in a perivascular pseudopapillary pattern along with a few well-formed papillae with a central fibrovascular core. However, the wall of cyst may be composed of islands of neoplastic cells or non-tumor components including astroglial fibres, collagen or fibroblastic tissue. There are several classification systems to identify the characteristics of intracranial cystic meningiomas, including Nauta's, Rengachary's and Zee's systems. Histologically, CPMs show characteristically immunoreactivity for vimentin, epithelial membrane antigen (EMA), but negative for glial fibrillary acidic protein (GFAP). These immunohistochemical studies may help to distinguish CPMs from other intracranial tumors with papillary architecture, such as atypical choroid plexus papilloma, papillary ependymoma and metastatic papillary adenocarcinoma. Since CPM is a rare tumor, its accurate incidence and clinical outcome are difficult to be determined. Based on the limited number of cases reported, CPM is a tumor of young patients. Brain parenchyma invasion, cranial nerve invasion and subarachnoid dissemination

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are frequently observed in this tumor. These findings suggest that CPMs may show aggressive biological behavior and a less favorable prognosis, although the studies in larger patients cohorts are needed to be conducted to confirm this results.

Introduction

Meningiomas are common tumors, accounting for 13–18% of intracranial neoplasms in adults (Kepes 1982). According to 2007 World Health Organization (WHO) classification of tumors of the central nervous system, there are 15 histological variants of meningioma, including 9 benign subtypes corresponding to WHO grade I and 6 subtypes with less favorable clinical outcomes, corresponding to WHO grade II (atypical) and III (anaplastic or malignant) (Louis et al. 2007). Papillary meningioma is a rare but distinct histological variant of meningioma and constitutes 1–2.5% in adult or 9.8% in pediatric meningiomas. In contrast to other benign variants of meningiomas, papillary meningioma has a tendency to occur in young subject. It presents a perivascular pseudopapillary pattern regarding the histological features and displays invasion of brain parenchyma, local recurrence, and distant metastasis with regard to biological behavior. It has been graded as a WHO grade III tumor. Unusually, meningiomas are associated with cysts, which occur in 1–10% of all meningiomas (Zee et al. 1995; Chan and Thompson 1984). Cystic meningioma is a challenging diagnostic problem and may be mistaken for several intracranial diseases at neuroimaging, such as pilocytic astrocytoma and ganglioglioma, all of which may show a cyst with a mural nodule. Histopathologically, it has been revealed that the meningothelial variant is the most frequent histological subtype of cystic meningiomas, whereas papillary meningiomas demonstrating cystic changes have been reported in a few cases only (Schulman et al. 1992; Lirng et al. 1995; Vaquero et al. 2002; Wakabayashi et al. 2005; Buschmann et al. 2005; Zhi et al. 2009). Herein, we demonstrate the histological characteristics of cystic

papillary meningioma. The histopathological findings and imaging features, as well as the differential diagnosis, are discussed.

Histopathological Diagnosis

Cystic papillary meningioma (CPM) is a rare tumor and its accurate incidence is difficult to be determined. Since cystic meningioma represents 1–10% of all meningiomas, the available data suggest the incidence of CPM is likely to be equal to or less than 0.01–0.025% of all meningiomas in adults and almost 0.1% in pediatric meningiomas.

Based on the limited number of cases reported, CPM is a tumor of young patients (Schulman et al. 1992; Lirng et al. 1995; Vaquero et al. 2002; Wakabayashi et al. 2005; Buschmann et al. 2005; Zhi et al. 2009). The age of the reported patients ranges between 12 and 45 years (mean 23 years). All CPMs reported to date have occurred in female, but it is difficult to determine whether or not there is a female bias due to limited cases. The localization of CPMs is varied. At presentation, the lesion of a few cases located in frontal area, cerebellopontine angle and sella turcica region. One example located in the lateral ventricle. All of reported cases showed the tumor had brain parenchyma invasion, cranial nerve invasion or subarachnoid dissemination. On magnetic resonance imaging (MRI), CPM is generally seen as solid and cystic masses (Demir et al. 2007). They may present as a multicystic lesion or a large single cystic lesion with a mural nodule. The solid portion of the lesion usually appears as enhanced intensity by contrast media. The cystic components vary from strong to low in intensity after contrast enhancement showing that the wall of cyst is composed of tumor cells or gliotic tissue (Fig. 11.1).

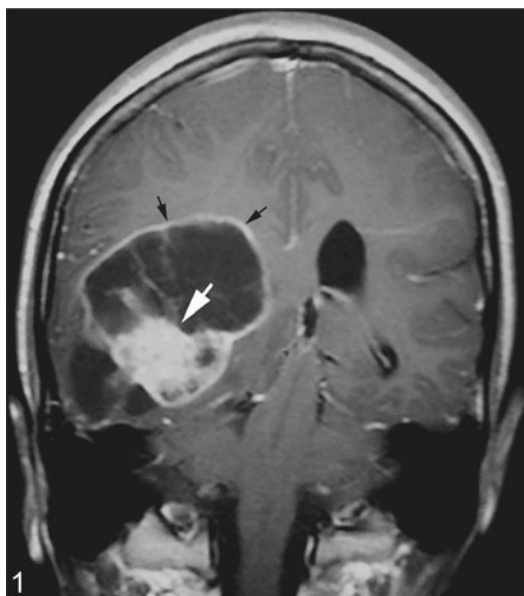


Fig. 11.1 Preoperative MRI demonstrating a lesion in the left temporal lobe containing a circumscribed cystic component and solid component. Solid component exhibits homogeneous enhancement (*white arrowed*), and the peritumoral cyst shows varied strong to low enhancement in intensity (*small black arrowed*)

Histopathology

Usually, CPM is composed of solid and cystic part. The solid components of cystic papillary meningiomas are typically composed of sheets of cells arranged in a perivascular pseudopapillary pattern along with a few well-formed papillae with a central fibrovascular core (Fig. 11.2). The papillary structures are sometimes found to intermingle with meningotheelial sheets, but whorls and psammoma bodies are usually not present in this tumor. The tumor cells may be arranged in a trabecular pattern on a collagenous stroma. The tumor cells in the papillary and trabecular area display abundant eosinophilic cytoplasm, as well as vesicular nuclei, with foci of necrosis occasionally. The wall of cyst may be composed of islands of neoplastic cells or non-tumor components including astroglial fibres, collagen or fibroblastic tissue (Fig. 11.3). In some region of the cystic components of the tumor is also found to consist of round eosinophilic cytoplasmic cells

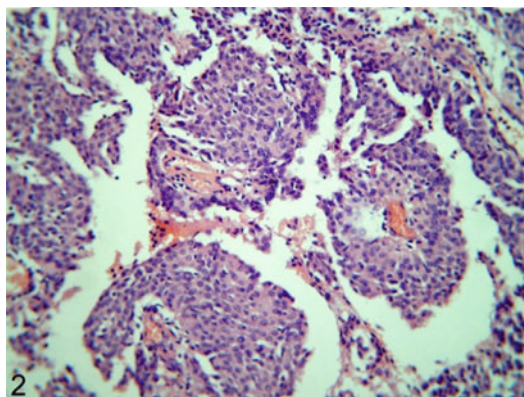


Fig. 11.2 Photomicrographs showing that the solid component of tumor is composed of sheets of cells arranged in perivascular pseudopapillary pattern with central fibrovascular core (H&E staining with original magnification of 400x)

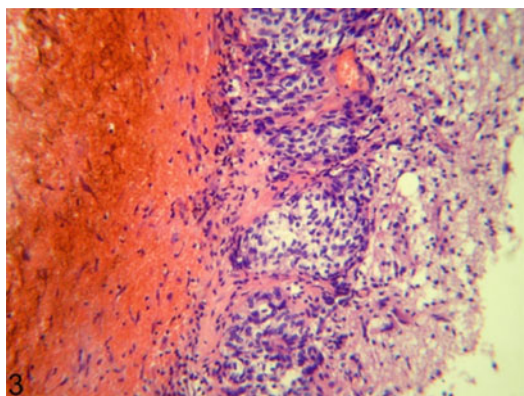


Fig. 11.3 Photomicrographs showing that the wall of cyst is composed of islands of neoplastic cells and non-tumor gliotic tissue. Focal hemorrhage is able to be observed in the cyst component (H&E staining with original magnification of 400x)

with nuclear atypia, and mitotic figures are occasionally seen. The papillary architecture may be observed in this area. It should be noted that brain parenchyma invasion and focal subarachnoid dissemination are frequently presented in this tumor.

Immunohistochemistry

Strong and diffuse positivity of Vimentin is found in all CPM. Epithelial membrane antigen (EMA)

immunoreactivity is characteristically observed in all of CPM, although focal and weak positivity for EMA is found in individual case. Immunohistochemical studies of S-100 protein have found varying positive in CPM. The vast majority of CPMs shows negative for glial fibrillary acidic protein (GFAP), but one example shows focally positive for GFAP in tumor cells. CPMs show no reactivity for neuronal or neuroendocrin markers, such as synaptophysin and chromogranin A. A variety of other molecules, including CD34, smooth muscle actin and desmin have not been shown to be expressed. The Ki-67 labeling index is 1–5% in CPMs.

Histological Differential Diagnosis

The main differential diagnoses include atypical choroid plexus papilloma, papillary ependymoma and metastatic papillary adenocarcinoma, all of which exhibit predominant papillary structures, and may sometimes cause diagnostic confusion with a papillary meningioma. However, atypical choroid plexus papilloma is usually immunoreactive for Cytokeratin and vimentin, but negative for EMA. Papillary ependymoma is GFAP positive. Metastatic papillary adenocarcinoma frequently occurs in old patients and exhibits distinct immunophenotypes for Cytokeratin, EMA positive and vimentin negative. In addition, a rare astroblastoma should also be differentiated from papillary meningioma, because astrocyte cells with a palisaded pattern and radiation towards the central blood vessels show similarities to the structures of papillary meningioma, whereas a diffuse GFAP-positive tumor cell in astroblastoma is a characteristic differentiating it from papillary meningiomas. Based on CT and MRI, pilocytic astrocytoma, ganglion cell tumors and hemangiopericytoma may also be included in the differential diagnosis because those tumors usually present a cystic lesion on imaging. However, the distinctive histological features of CPMs under the microscopy may lead to a correct diagnosis from those tumors with cyst formation.

Discussion

Cystic meningioma was first described by Penfield in 1932. It has been reported to occur with an incidence of 1–7% in adults and up to 10% in children (Zee et al. 1995; Chan and Thompson 1984). The formation of cysts in meningioma makes it mimic other neoplasms associated with more common cystic lesion, such as gliomas, hemangioblastomas, or metastatic tumors. Therefore, there are several classification systems describing the characteristics of intracranial cystic meningiomas in the literature. Nauta's system (Nauta et al. 1979) classified the cysts into four types: Type 1, an intratumoral cyst surrounded by tumor macroscopically. Type 2, a cyst still inside the tumor, but part of its rim is formed by microscopically visible tumor. Type 3, a peritumoral cyst within the adjacent brain rather than the tumor itself. Type 4, loculated CSF in the subarachnoid space between the tumor and the brain neither inside the tumor nor in the brain. Rengachary's system described the cysts as intratumoral or peritumoral (Rengachary et al. 1979). The cyst wall may be composed of islands of neoplastic cells and non-tumor components including astroglial fibres, collagen or fibroblastic tissue, but the intratumoral cysts are extremely rare. Zee's system (Zee et al. 1995) divided cystic meningiomas into three groups according to the morphology rather than histological features: type 1 includes tumors with intratumoral cystic and necrotic areas; type 2: the cyst is peritumoral, and the cyst wall enhances with contrast on MRI and is composed of tumor cells; type 3: the peritumoral cyst is in contact with the adjacent brain, and the wall is non-enhancing and composed of gliotic tissue with no tumor cells.

Although several classifications have described the morphological characteristics of cystic meningiomas, the mechanisms leading to cyst formation are still not well understood. Confluence of microcysts and intratumoral hemorrhages may be possibilities regarding the formation of intratumoral cysts (Russell et al. 1980). However,

peritumoral cysts may be the final stage of peritumoral vasogenic edema with fluid collection around the tumor, or may also result from loculation or dilatation of the subarachnoid space (Pinna et al. 1986). One reported case of CPM shows the part of tumor is located at the triangle area of lateral ventricle (Zhi et al. 2009). They suggest that the cystic meningioma may originate from the epithelium of choroid plexus, presenting in the lateral ventricle. Dilation of the lateral ventricle, induced by an extended growth pattern of tumor, might be associated with the formation of peritumoral cyst of tumor.

Histopathologically, the pervious data have revealed that the fibroblastic type is the most common form of cyst meningiomas occurring in infancy, but the meningothelial type of cyst meningiomas is predominant during adulthood. Up to date, only a few cases have demonstrated papillary type of cyst meningiomas. Papillary meningioma is a rare variant of malignant meningiomas, with a high incidence of recurrence and invasion or metastasis when compared with other benign variants of meningioma. A quarter of papillary meningiomas appear in the first two decades of life. Although surgical resection and adjuvant radiotherapy have been established as the standard treatment for papillary meningiomas, the 5-year survival rate of this variant is about 40%, which is one-half of that of other variants of meningiomas (Louis et al. 2007). The previous studies have demonstrated that about 59% (10/17) of papillary meningiomas had local recurrences after surgical resection, and approximately 70% (16/23) of patients showed histological evidence of bone and brain invasion (Ludwin et al. 1975; Radhakrishnan et al. 1993). In all of reported CPM cases, tumor cells exhibit invasive growth pattern and focal necrosis, supporting the fact that this tumor is different from other benign forms of meningiomas. Tumor cells are also observed to invade the brain parenchyma, also with involvement of the local subarachnoid space. In addition, malignant features of this tumor were also exhibited in tumor cells with high MIB-1 labeling. Based on those histological features and its biological behavior,

we suggest that this papillary meningioma with cyst formation may show aggressive biological behavior and a less favorable prognosis.

In conclusion, cystic papillary meningiomas usually exhibit a marked cyst with or without contrast enhancement on MRI and display the classical perivascular pseudopapillary pattern histologically, which has rarely been described previously. The precise mechanism of cyst formation of meningiomas remains unknown, and a long follow-up period and a larger number of case investigations are necessary to better clarify the biological characteristics and the clinical outcome of cystic papillary meningiomas.

References

- Buschmann U, Gers B, Hildebrandt G (2005) Uncommon case of a cystic papillary meningioma in an adolescent. *Childs Nerv Syst* 21:322–326
- Chan RC, Thompson GB (1984) Intracranial meningiomas in childhood. *Surg Neurol* 21:319–322
- Demir MK, Musluman M, Kilicoglu G, Hakan T, Aker FV (2007) Imaging features of unusual intracranial cystic meningiomas. *Can Assoc Radiol J* 58:109–115
- Kepes JJ (1982) Meningiomas. Biology, pathology and differential diagnosis. Masson, New York
- Lirng JF, Enterline DS, Tien RD, Fuchs H, Friedman HS, Ellington KS, MacLendon RP (1995) MRI of papillary meningiomas in children. *Pediatr Radiol* 25:S9–S13
- Louis DN, Ohgaki H, Wiestler OD, Cavenee WK (2007) WHO classification of tumor of the central nervous system, 4th edn. IARC Press, Lyon, pp 164–172
- Ludwin SK, Rubinstein LJ, Russel DS (1975) Papillary meningioma. A malignant variant of meningiomas. *Cancer* 36:1363–1373
- Nauta HJ, Tucker WS, Horsey WJ, Bilbao JM, Gonsalves C (1979) Xanthochromic cysts associated with meningiomas. *J Neurol Neurosurg Psychiatry* 42:529–535
- Pinna G, Beltramello A, Buffatti P, Signorini G, Colombari R, Brocolo A, Ore GD (1986) Cystic meningiomas, an update. *Surg Neurol* 126:441–452
- Radhakrishnan VV, Saraswathy A, Rout D (1993) Papillary meningioma: a clinicopathological study of six cases. *Indian J Cancer* 30:164–168
- Rengachary S, Batnitzky S, Kepes JJ, Morantz RA, O'Boynick P, Watanabe I (1979) Cystic lesions associated with intracranial meningiomas. *Neurosurgery* 4:107–114
- Russell EJ, George AE, Kricheff II, Budzilovich G (1980) Atypical computed tomographic features of intracranial meningioma: radiological–pathological correlation

- in a series of 131 consecutive cases. *Radiology* 135:673–682
- Schulman FY, Carpenter JL, Ribas JL, Brum DE (1992) Cystic papillary meningioma in the sella turcica of a dog. *J Am Vet Med Assoc* 200:67–69
- Vaquero J, Zurita M, Coca S (2002) Cystic papillary meningiomas. *Neuropathol Appl Neurobiol* 28:418–421
- Wakabayashi K, Suzuki N, Mori F, Kamada M, Hatanaka M (2005) Rhabdoid cystic papillary meningioma with diffuse subarachnoid dissemination. *Acta Neuropathol (Berlin)* 110:196–198
- Zee CS, Chen T, Hinton DR, Tan M, Segall HD, Apuzzo MLJ (1995) Magnetic resonance imaging of cystic meningiomas and its surgical implications. *Neurosurgery* 36:482–488
- Zhi L, Bing L, Yang L, Bo-ning L, Quan H (2009) Cystic papillary meningioma with subarachnoid dissemination: a case report and review of the literature. *Pathol Res Pract* 205:582–587

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Abstract

Meningiomas are neoplasms that arise from the leptomeningeal covering of the brain and spinal cord and represent around 20% of all central nervous system tumors. Medical writings from as early as the eighteenth century make reference to tumors arising from the meninges. The current standards for diagnosis of meningiomas are clinical and pathological findings. The World Health Organization (WHO) classifies meningiomas into three histological grades: grade I (benign), grade II (atypical), and grade III (anaplastic) in accordance with the clinical prognosis. In absolute numbers, most recurrent meningiomas correspond to histological benign tumours. The potential aggressiveness of an individual meningioma is still difficult to evaluate. Due to the potential recurrence of some meningioma regardless of the histological grade, the management of the meningioma patients must include an appropriate follow up strategy suited to the grade of surgical resection and the likelihood of recurrence in individual cases. The detection of new meningioma subgroups with clinical and biological relevance may help in these strategies. This review summarizes the most recent advances in the detection of meningioma subgroups using traditional and modern high throughput molecular techniques. Most relevant results on studies of meningioma using cytogenetics, proteomics, transcriptomics and metabolomics suggest that there are at least two molecular subgroups of histological benign meningiomas. These two subgroups exhibit very distinct genetic and biochemical properties and show different levels of metabolic aggressiveness. The assignment of individual

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meningiomas to one of these subgroups simultaneously to the traditional histopathological examination may improve the management of these tumors.

Introduction

Meningiomas are neoplasms that arise from the leptomeningeal covering of the brain and spinal cord and represent around 20% of all central nervous system tumors. Medical writings from as early as the eighteenth century make reference to tumors arising from the meninges. In 1922, Cushing first introduced the term meningioma as a simple solution to enclose the wide spectrum of meningeal neoplasms. During the first half of the twentieth century, Cushing himself and Eisenhardt established their fundamental work on these tumors, first published in 1938 (Cushing and Eisenhardt 1938). Meningiomas exhibit remarkably wide biological and histological variety. Among the potential causes for meningioma, ionizing radiation is considered a clear contribution. There is also a potential role of sex hormones, particularly progesterone receptors, although the exact link remains unproven. The association between neurofibromatosis type 2 and young onset of meningioma combined with recent discoveries related to germline mutations in the NF2 gene on chromosome 22q12 suggest some possible genetic predisposition to meningioma.

The current standards for diagnosis of meningiomas are clinical and pathological findings. The World Health Organization (WHO) classifies meningiomas into three histological grades: grade I (benign), grade II (atypical), and grade III (anaplastic) in accordance with the clinical prognosis (Louis et al. 2007). Signs of malignancy in meningioma include marked vascularity, loss of organoid structure, mitotic figures, nuclear pleomorphism, prominent nucleoli, focal necrosis, or infiltration to the adjacent brain. Atypical and anaplastic meningiomas show a high index of recurrence 5 years after complete resection and are associated with lower survival rates compared to benign meningiomas

(McLendon et al. 2006). However, in absolute numbers, most recurrent meningiomas correspond to histological benign tumours. The potential aggressiveness of an individual meningioma is still difficult to evaluate.

As a group, meningiomas are often considered benign tumors. Most meningiomas are slow growing lesions typically associated with the symptoms of progressively increasing intracranial pressure. However, they may also show concomitant hyperostosis and encasement of blood vessels. Meningioma can also grow to massive proportions compromising the likelihood of total surgical removal. As a consequence, management of the meningioma patient continues to evolve as new technologies give rise to new treatment options. Most meningiomas are slow-growing and curable by surgical resection. In this context, open surgery is still regarded as the standard treatment for large or symptomatic meningiomas. Microsurgical techniques and the introduction of computer-based localization and mapping during microsurgical resection greatly improved these treatment options. The use of stereotactic radiosurgery also provides significant benefits in the treatment of meningiomas. However, despite the significant advances in meningioma surgical treatment, there is an important subset of meningiomas that are associated to increased morbidity and mortality, often after recurrence and especially after incomplete resection. Due to the potential recurrence of some meningioma regardless of the histological grade, the management of the meningioma patients must include an appropriate follow up strategy suited to the grade of surgical resection and the likelihood of recurrence in individual cases.

This review explores the current state-of-the-art in the detection of meningioma subgroups with potential implications for therapy selection, patient management and follow up strategies.

The authors discuss the most recent advances in the fields of high throughput molecular profiling for the study of meningioma and the most relevant results associated to biological and clinical properties of these tumors.

Histopathology of Meningioma

Meningiomas appear under the microscope as remarkably diverse tumors. There are 13 recognized variants and three grade categories in the current WHO classification (McLendon et al. 2006). Most of these variants have no independent prognostic impact. The three most common variants are meningothelial, fibroblastic and transitional meningioma. However, it is common to encounter combinations of two or more patterns. Meningothelial meningiomas are characterized by lobules or whorls. Inter cellular junctions often appear fuzzy exhibiting a syncytial-like pattern. Other common features in meningothelial meningioma include clear intranuclear vacuoles, intranuclear pseudoinclusions and moderate quantities of eosinophilic cytoplasm. On the other hand, fibrous or fibroblastic meningiomas present characteristic spindled cells arranged in fascicles with interspersed collagen patterns. Finally, transitional meningiomas show typical intermediate or mixed features of meningothelial and fibroblastic meningiomas. Whorls and psammoma bodies are common in this subtype and it is considered as the most classic of all meningioma patterns. Among all meningioma variants, only four are considered intrinsically aggressive. Clear cell meningioma and chordoid meningioma exhibit WHO grade II. Papillary meningioma and rhabdoid meningioma show features of WHO grade III. Fortunately, these variants account for less than 4% of all meningiomas.

Meningioma WHO Classification and Grading

The WHO classification of meningioma aims to a better prediction of the divergent clinical characteristics of tumors, by using a histological

grading system based on statistically significant clinicopathological correlations (Perry et al. 1997; Louis et al. 2007). Most meningiomas are slowly growing benign tumors and correspond histologically to grade I according to the WHO classification. Atypical meningiomas (WHO grade II), exhibit an increased risk of recurrence compared with WHO grade I tumors. Anaplastic meningiomas are WHO grade III neoplasms and are associated with a high risk for recurrence and the least favorable prognosis. Brain invasion is considerably rare and has grading implications in meningioma. Sampling issues and imperfect histopathologic features of these tumors can make exact grading difficult. Some markers, such as vimentin, EMA, are frequently present in meningiomas but do not correlate specifically with grade.

Approximately 80% of excised meningiomas are histologically benign and they are unassociated with excess mortality when gross totally resected (McLendon et al. 2006; Wiemels et al. 2010). Those meningiomas that do not fulfill histological criteria for grades II or III and are composed predominantly of any histological pattern, other than clear cell, chordoid, papillary, or rhabdoid, are considered of grade I. Dural, bone, soft tissue, paranasal sinus, and even vascular/dural sinus invasion do not shift the diagnosis towards higher histological grades.

Atypical meningiomas account for approximately 10% of all meningioma cases. Atypical meningioma is associated with a markedly increased risk of recurrence and a small, but statistically significant increased risk of death. An estimated 40% of atypical meningiomas recur within 5 years (Perry et al. 1997). Statistically, the strongest single criterion associated with recurrence is an elevated proliferation index, defined as ≥ 4 mitoses per 10 consecutive high-powered fields (HPF), regardless of whether this finding is focal or widespread. Perry et al. suggested that Ki67, which is also a marker of proliferation, in excess of 4.2% was indicative of high tumor proliferation activity and recurrence (Perry et al. 1999). In the absence of increased mitotic activity, diagnosis of atypical meningiomas is based on the presence of either brain invasion or at least three out of five other malignancy criteria.

Although brain invasion was once considered a definite sign of malignancy, associated recurrence-free and overall survival times are similar to those of atypical meningiomas in general. Atypical meningioma also may exhibit some signs of loss of cellular differentiation combined with spontaneous necrosis which suggests superimposed hypoxia. Micronecrosis with pseudopalisading has the strongest association with recurrence.

Anaplastic meningiomas are highly aggressive rare tumors which account for only 1–2% of resected cases (Perry et al. 1999). They represent higher degrees of cell cycle deregulation and loss of differentiation with focal or diffuse findings of excessive mitotic index and/or evident anaplasia. Frankly anaplastic foci are difficult to recognize and may require immunohistochemistry, electron microscopy, or even genetic studies for confirmation. They may present either apparently *de novo* or following one or more recurrences of lower grade meningiomas. The median overall survival is less than 2 years.

Cytogenetic and Molecular Genetic Alterations

The development of cancer often follows a multi-step pathway where an increasing number of genetic aberrations accumulate due to genetic and/or chromosome instability. Specific patterns of genetic evolution typically correlate with both a more advanced stage and a more aggressive course of the disease. Meningioma pathogenesis may follow some models of similar clonal evolution based on the cytogenetic heterogeneity found through the analysis of large series of patients by conventional karyotyping. These models provide hypothetical evolution pathways for what happens during the sequential transition from normal meningeal cells to grade I, and grade II/grade III meningiomas.

Meningiomas are cytogenetically heterogeneous tumors in which chromosome gains and losses frequently occur. Cytogenetics methodologies provide an insight into these chromosome alterations which may be related to tumor initiation and progression. Tumors exhibiting complex

genetic alterations, even those with a benign histological grade, may show grossly invasive tumor growth and require closer follow up. Chromosomal aberrations can be detected by standard cytogenetic methods or by fluorescence in situ hybridization (FISH). In the last years, chromosomal aberrations in a tumour sample have been typically studied by FISH because it is a target-oriented method that allows direct observation of specific chromosomal abnormalities. Genomic alterations detectable by FISH include deletion, imbalance, polysomy, translocation, and gene amplification. FISH is more sensitive than standard cytogenetics methods because it does not require dividing cells and can be measured directly in the tumour tissue. However, cytogenetics methods provide a whole genome chromosomal profiling and may reveal chromosomal alterations in regions different to those explored by target oriented methods. The combination of both methods constitutes an exploratory approach for global chromosomal profiling with high sensitivity in specific regions. Markers detected by this exploratory approach can then be detected by more direct methodologies.

Loss of genetic material on chromosome 22 seems an essential step in meningioma pathogenesis. Monosomy of this chromosome is the most common genetic alteration in meningioma and was one of the first cytogenetic alterations described in solid tumours (Dumanski et al. 1990; Zhang et al. 1990; Lopez-Gines et al. 1995; Pfisterer et al. 2004). Changes in other chromosomes were reported in several large studies. Whether loss of 22q is the earliest initiating event in meningioma formation is controversial. Changes of other chromosomes may coincide or be associated only with progression from benign to atypical or anaplastic types. Loss of 1p and alterations in chromosome 14 are present in many atypical meningioma and are the most common chromosomal alterations in this meningioma subtype (Cerdeira-Nicolas et al. 2000; Lopez-Gines et al. 2004). Coexistence of monosomy 14 and 1p is a powerful adverse prognostic factor for early relapses (Lopez-Gines et al. 2004). Alterations in other chromosomes, like losses in 6q, 10 and 18q and gains on 1q, 9q, 12q, 15q, 17q and 20q, are

also common in atypical meningioma (Perry et al. 2004). Recent studies show that histological benign meningiomas with complex genetic alterations may be clinically aggressive (Pfisterer et al. 2008). Histological benign meningiomas with complex karyotype have higher probability of relapsing (Maillo et al. 2007). The detection of a subset of benign meningiomas with chromosomal alterations beyond monosomy 22 may have an important impact in defining aggressiveness in meningiomas. This subset of benign meningiomas shows a wide variety of chromosomal alterations all producing a complex karyotype.

Gene Expression Profiles of Meningioma

Meningiomas are common intracranial tumors, but relatively little is known about the genetic events responsible for their clinical diversity. Current data point towards meningioma initiation closely linked to the inactivation of one or more members of the highly conserved protein 4.1 superfamily (Kros et al. 2001; Robb et al. 2003). The neurofibromatosis type 2 (NF2) gene product merlin/schwannomin, the protein 4.1B (DAL-1) and the protein 4.1R belong to this family. The genetic alterations in atypical meningiomas are complex and involve losses on 1p, 6q, 10, 14q and 18q. Gains on other multiple chromosomes are also observed in atypical meningioma. Nevertheless, the relevant genes are still unknown. Anaplastic meningiomas show even more complex genetic alterations, including frequent alteration of the CDKN2A, p14ARF, and CDKN2B tumor suppressor genes at 9p21, as well as gene amplification on 17q23. Despite the current knowledge about the multiple links between gene alterations and meningioma initiation and progression, there is still a need for deeper understanding of the network of genetic alterations that take place in the pathogenesis of meningioma.

Gene expression profiling by oligonucleotide microarrays allow the screening of thousands of genes simultaneously. Several studies reported microarray analysis of meningiomas in the past with different aims, microarray platforms and

statistical approaches (Stuart et al. 2011; Aarhus et al. 2011). The result of such a variety of approaches is no shared genes deregulated in all these studies. One possible reason for these results may be the existence of different molecular and genetic subgroups of meningioma. The intrinsic variation of aggressiveness and genetic alterations within benign meningioma highly impacts the statistical results of molecular studies on meningioma. Since the year 2000, a total of 14 gene expression studies have been reported on the literature with different aims, sources of control tissue, microarray platforms, and statistical approaches. Most of these studies identified meningioma-specific genes, and genes associated with the three WHO grades and the main histological subtypes of grade I meningiomas although there is no clear overlap among studies regarding individual candidate genes. Replication of pathological and histological findings by microarray technologies has an undoubtedly added value for the study of the biology of the tumor. The results of most of these studies are limited to molecular markers of cellular features related to tumor grade and commonly observable by histopathology, like angiogenesis, apoptosis, invasion, motility, growth, proliferation and differentiation. All these are cellular processes critical for the tumor growth. However, biochemical and metabolic changes of any cell population precede morphological and cellular changes. The detection of these molecular changes may help in the identification of new molecular subgroups for better management of the tumor patient.

The application of microarray technologies for the identification of gene markers and molecular subgroups requires the use of proper quality control of the tissue material, restrictive filtering algorithms for gene selection and final validation of selected genes by real-time RT-PCR. The most recent report on gene expression profiling in meningioma explored multiple data sets and utilized restrictive criteria to identify genes differentially expressed between grade I benign and grade III anaplastic meningiomas (Stuart et al. 2011). The robustness and prognostic value of the results of this study seem very high. The study analyzed only those genes with at least a four-fold

change between grade I and grade III meningiomas. By filtering the initial set of potential biomarkers through a second and third cohorts of meningiomas and then verifying the remaining genes by quantitative polymerase chain reaction, the authors identified 9 overexpressed (TPX2, RRM2, TOP2A, PI3, BIRC5, CDC2, NUSAP1, DLG7, SOX11) and 2 underexpressed (TIMP3, KCNMA1) genes in grade III versus grade I meningiomas. As a further validation step, the authors analyzed these genes in a fourth cohort including atypical meningioma cases and found that patients with grade II meningiomas with high topoisomerase 2- α protein expressions (>5% labeling index) had shorter times to death than patients with low expressions. These results demonstrate the use of molecular phenotypic criteria for the determination of meningioma subgroups. The determination of these profiles combined with other markers of molecular aggressiveness may also aid in determining the aggressiveness of surgical resection and the necessity of combined radiation therapy.

Proteomic Profiles of Meningioma Subtypes

Proteomics is the large-scale study of proteins, particularly their structures and functions. It provides broad protein profiles which permit a direct analysis of proteins differentially or uniquely expressed by a cell or a tissue type. The proteome is the entire complement of proteins including the modifications made to a particular set of proteins, produced by an organism or system. This will vary with time and distinct requirements, or stresses, that a cell or organism undergoes. Proteomic research has been developed recently in the post-genome era. Using a combination of two-dimensional gel electrophoresis, mass spectrometry, and bioinformatics, one can now readily scan a particular cell type and establish protein expression patterns. Recent studies have shown that specific proteomic patterns and proteins can distinguish subtypes or grades of human brain tumors (Furuta et al. 2004; Iwadate et al. 2004).

There is only one proteomic study on meningiomas reported in the literature (Okamoto et al. 2006). The study conducted proteomic profiling on 10 benign meningiomas, 9 atypical meningiomas, and 5 anaplastic meningiomas. The authors reported protein expression patterns unique to meningiomas with WHO grades I, II and III and identified 24 proteins that distinguish each subtype. Fifteen proteins showed significant changes in expression level between benign and atypical meningiomas, whereas nine distinguished atypical from anaplastic meningiomas. Differential protein expression was confirmed by Western blotting and immunohistochemistry. Some of the proteins up-regulated in benign meningiomas have potential tumor suppressor activity, like Vitamin D-binding protein or caponin-1. On the other hand, several of the nine proteins that were significantly overexpressed in WHO grade 2 meningiomas have potential cancer-related functions. For example, Galectin-1 mediates adhesions between cells and with the extracellular matrix and has been shown to play a role in tumorigenesis and to enhance metastatic potential. Overall, the different proteomic profiles characterize and distinguish meningiomas of increasing grades. The proteins detected as relevant for meningioma grading enhance the understanding of the pathogenesis of meningiomas and may have implications for diagnosis, prognosis, and treatment.

Metabolic Profiles of Meningioma

Metabolomics is defined as “the quantitative measurement of the multiparametric metabolic response of living systems to pathophysiological stimuli or genetic modification” (Nicholson and Lindon 2008). High resolution NMR spectroscopy of biofluids and tissues combined with multivariate analysis methodologies, like Principal Component Analysis, represent a powerful technique for investigating the metabolome in the area of drug toxicology and disease diagnosis and prognosis (Nicholson et al. 2002). High resolution magic angle spinning NMR spectroscopy

(HR-MAS NMR) is a powerful technique for the investigation of metabolites within different intact tissues (Cheng et al. 1997; Monleon et al. 2008). The potential of HR-MAS applications to the study of biological tissues has been widely demonstrated in the investigation of different cellular alterations. This technology can supplement histopathological examination and potentially improve brain tumour diagnostics. HR-MAS spectra generate metabolic profiles that contain information on physiological and pathological status. This approach can be used to define the metabolomic phenotype of a tissue.

The metabolic characterization of tumor biopsy samples by HR-MAS NMR spectroscopy provides differential patterns for benign and atypical meningioma. The metabolic composition of the meningioma tissue is highly consistent with 'in vivo' and 'ex vivo' previous studies. Clear metabolic differences between atypical and benign meningioma exist. Metabolic differences between meningioma grades include changes in the levels of glutathione, glutamine and glutamate. Other metabolites associated with tumor malignancy that show statistically significant differences between benign and atypical meningiomas include phosphocholine and phosphoethanolamine. Alanine, which is a well-known MR spectroscopy marker for meningiomas, does not show relevant differences with tumor grade. Overall, metabolic profiling in meningioma suggests that the additional information obtained by NMR metabolomics applied to biopsies of human meningiomas may be useful for assessing tumor grade and determining optimum treatment strategies. However, and as mentioned in previous paragraphs, there is a need for determining meningioma subgroups beyond the replication of pathological and histological findings.

Recent studies suggest that there is an aggressive subset of histological benign grade I meningiomas with distinct genetic and biochemical characteristics (Pfisterer et al. 2008; Monleon et al. 2010). The measurement of a metabolic profile by HR-MAS and a cytogenetic signature by FISH shows some correlation between the genetics and biochemistry of the tumour and its clinical properties. The metabolic phenotype of

meningiomas with complex karyotype shows typical features of aggressive tumor biochemistry like increased membrane metabolites turnover and high glycolytic activity.

Cancer cells use an altered metabolism compared with that of normal differentiated adult cells in the body. This metabolic remodeling affects the utilization and/or synthesis of important metabolites including glucose, glycogen, fatty acids, amino acids, and glutamine by tumor cells. Otto Warburg was the first to describe a metabolic adaptation within solid tumors (Warburg 1956). Warburg noticed that normal tissues generate about one tenth of the total cell ATP by glycolysis in the cytoplasm and the other nine tenths by oxidative phosphorylation in mitochondria. Lower glucose levels in cells of more aggressive tumours may reflect preferential glucose consumption and a local hypoxic microenvironment. Lower levels of glucose and ascorbate together to higher levels of lactate suggest higher dependence on anaerobic breakdown of pyruvate for meningiomas with complex karyotype. Lactate accumulation also creates a microenvironment which may enhance cell proliferation. Hypoxia in the tumor microenvironment triggers a variety of genetic and adaptive responses that regulate tumor growth (Brahimi-Horn et al. 2007). Tumor hypoxia is often associated with more malignant phenotypes, resistance to therapy, and poor survival.

The creatine/phosphocreatine shuttle makes creatine another metabolite closely related to energy metabolism. Total creatine is lower in meningiomas with chromosomal instabilities. This may indicate either lower substrate availability or increased ATP usage, which suggests an environment with high energy demand. Finally, the decreased levels of fatty acids may also reflect increased β -oxidation. All these processes result in a metabolically unfavorable microenvironment with low glucose availability and local hypoxic conditions. These results suggest that meningiomas with chromosomal instabilities have this energy demanding microenvironment.

Oxidative stress is a process closely related to both cancer progression and drug resistance. Glutathione is an important cell antioxidant and,

in normal tissues, plays different roles in protection against oxidative damage (Balendiran et al. 2004), in apoptosis (Filomeni et al. 2005) and in amino acid transport (Orlowski 1976). Glutathione depletion is a common feature of apoptotic cell death. In cancer, glutathione is able to play both protective and pathogenic roles. In meningioma, levels of glutathione and closely related metabolites, like glutamate, are increased with respect to other brain tumors and healthy brain tissue (Opstad et al. 2003). A potential explanation for it is the reversible transamination reaction of alanine and α -ketoglutarate, synthesising glutamate, as an alternative energy source. This may also explain the lower levels of alanine in atypical meningioma despite higher hypoxic conditions. Another source of glutamate is glutamine by the glutaminase reaction, which also explains the lower levels found for glutamine in the same subgroup. Meningioma with chromosomal instabilities shows increased levels of glutathione and glutamate and decrease levels of alanine and glutamine, which suggest less cell differentiation, resistance to apoptosis and activation of alternative energy sources.

Phospholipids precursors, especially choline-containing compounds, are elevated in tumors (Glunde et al. 2006). Phosphocholine and choline compounds are generally considered as well established proliferation markers. In addition, high phosphocholine is also correlated with highly invasive phenotypes. Meningiomas with chromosomal instabilities have higher phosphocholine levels, which are consistent with higher proliferative rates. In addition, phosphoethanolamine, which is also considered as an intermediate in the metabolism of phospholipids, exhibits increased levels in meningioma with chromosomal instabilities. Increased phosphoethanolamine levels suggest an activation of phosphatidylethanolamine metabolism, which is involved in the modification of membrane shape in malignant tissue (Cullis et al. 1986). Additionally, free phosphoethanolamine might reflect the breakdown of phosphatidylethanolamine to diacylglycerol, and might act as a long-term second-messenger for cell proliferation.

The reviewed meningioma metabolic profile and the corresponding metabolic pathways play

critical roles in cancer metabolism. All of them suggest that meningiomas with chromosomal instabilities suffer metabolic alterations related to aggressiveness, proliferation and invasiveness. This constitutes a clear evidence of the existence of new molecular meningioma subgroups with potential clinical relevance within the group histological benign meningioma.

Concluding Remarks and Future Directions

Since the seminal work of Cushing and Eisenhardt, meningiomas have fascinated surgeons and pathologist for their wide biological, biochemical and clinical variety. The advent of high throughput techniques for molecular profiling has placed the spotlight on the early detection of new meningioma subgroups with potential clinical relevance. Different studies provided strong evidences in favour of the existence of a subset of meningiomas showing biochemical aggressiveness early on the onset of the tumor. Some reports correlate this biochemical aggressiveness with higher incidence of tumor recurrence regardless of the histological grade. The early detection of the biochemical phenotypic characteristics of meningiomas helps in detecting this biochemical aggressiveness. Undoubtedly, the use in clinical environments of the techniques and analysis described in this chapter still needs further development. At this moment, the application of these methodologies to the study of tumor tissue still requires high level instrumentation, skilled technicians in charge of the analysis and complex mathematical approaches for proper interpretation of the results. However, the future of the diagnosis and classification of meningioma based on the study of biopsies goes certainly in that direction.

Acknowledgements The authors would like to acknowledge the funding sources that helped in the development of their research, part of which is reviewed here, in the recent years. The following funding sources are gratefully acknowledged: Consellería de Sanidad (GV-AP064/07, GV-AP076/08, GV-AP014/09, GV-AP119/10 and

GV-AP023/10) and the Consellería de Educación (PROMETEO11/2011/083, FPA/2011/054, ACOMP2009-200, ACOMP2010-184 and ACOMP2011-237) de la Generalitat Valenciana, the Ministerio de Educación y Ciencia del Gobierno de España (SAF2008-00270, and Ramon y Cajal Contract 2006).

References

- Aarhus M, Lund-Johansen M, Knappskog PM (2011) Gene expression profiling of meningiomas: current status after a decade of microarray-based transcriptomic studies. *Acta Neurochir (Wien)* 153:447–456
- Balendiran GK, Dabur R, Fraser D (2004) The role of glutathione in cancer. *Cell Biochem Funct* 22:343–352
- Brahimi-Horn MC, Chiche J, Pouyssegur J (2007) Hypoxia signalling controls metabolic demand. *Curr Opin Cell Biol* 19:223–229
- Cerda-Nicolas M, Lopez-Gines C, Perez-Bacete M, Barcia-Salorio JL, Llombart-Bosch A (2000) Histopathological and cytogenetic findings in benign, atypical and anaplastic human meningiomas: a study of 60 tumors. *J Neurooncol* 47:99–108
- Cheng LL, Ma MJ, Becerra L, Ptak T, Tracey I, Lackner A, Gonzalez RG (1997) Quantitative neuropathology by high resolution magic angle spinning proton magnetic resonance spectroscopy. *Proc Natl Acad Sci USA* 94:6408–6413
- Cullis PR, Hope MJ, Tilcock CP (1986) Lipid polymorphism and the roles of lipids in membranes. *Chem Phys Lipids* 40:127–144
- Cushing H, Eisenhardt L (1938) *Meningiomas*. Hafner Publishing Co Ltd, New York
- Dumanski JP, Rouleau GA, Nordenskjold M, Collins VP (1990) Molecular genetic analysis of chromosome 22 in 81 cases of meningioma. *Cancer Res* 50:5863–5867
- Filomeni G, Aquilano K, Rotilio G, Ciriolo MR (2005) Glutathione-related systems and modulation of extracellular signal-regulated kinases are involved in the resistance of AGS adenocarcinoma gastric cells to diallyl disulfide-induced apoptosis. *Cancer Res* 65:11735–11742
- Furuta M, Weil RJ, Vortmeyer AO, Huang S, Lei J, Huang TN, Lee YS, Bhowmick DA, Lubensky IA, Oldfield EH, Zhuang Z (2004) Protein patterns and proteins that identify subtypes of glioblastoma multiforme. *Oncogene* 23:6806–6814
- Glude K, Jacobs MA, Bhujwala ZM (2006) Choline metabolism in cancer: implications for diagnosis and therapy. *Expert Rev Mol Diagn* 6:821–829
- Iwadate Y, Sakaida T, Hiwasa T, Nagai Y, Ishikura H, Takiguchi M, Yamaura A (2004) Molecular classification and survival prediction in human gliomas based on proteome analysis. *Cancer Res* 64:2496–2501
- Kros J, de Greve K, van Tilborg A, Hop W, Pieterman H, Avezaat C, Lekanne D, Deprez R, Zwarthoff E (2001) NF2 status of meningiomas is associated with tumour localization and histology. *J Pathol* 194:367–372
- Lopez-Gines C, Cerda-Nicolas M, Barcia-Salorio JL, Llombart-Bosch A (1995) Cytogenetical findings of recurrent meningiomas. A study of 10 tumors. *Cancer Genet Cytogenet* 85:113–117
- Lopez-Gines C, Cerda-Nicolas M, Gil-Benso R, Callaghan R, Collado M, Roldan P, Llombart-Bosch A (2004) Association of loss of 1p and alterations of chromosome 14 in meningioma progression. *Cancer Genet Cytogenet* 148:123–128
- Louis DN, Ohgaki H, Wiestler OD, Cavenee WK, Burger PC, Jouvet A, Scheithauer BW, Kleihues P (2007) The 2007 WHO classification of tumours of the central nervous system. *Acta Neuropathol* 114:97–109
- Maillo A, Orfao A, Espinosa AB, Sayagues JM, Merino M, Sousa P, Lara M, Tabernero MD (2007) Early recurrences in histologically benign/grade I meningiomas are associated with large tumors and coexistence of monosomy 14 and del(1p36) in the ancestral tumor cell clone. *Neuro Oncol* 9:438–446
- McLendon RE, Rosenblum MK, Bigner DD (eds) (2006) *Russell & Rubinstein's pathology of tumours of the central nervous system*. Hodder Arnold, London
- Monleon D, Morales JM, Gonzalez-Darder J, Talamantes F, Cortes O, Gil-Benso R, Lopez-Gines C, Cerda-Nicolas M, Celda B (2008) Benign and atypical meningioma metabolic signatures by high-resolution magic-angle spinning molecular profiling. *J Proteome Res* 7:2882–2888
- Monleon D, Morales JM, Gonzalez-Segura A, Gonzalez-Darder JM, Gil-Benso R, Cerda-Nicolas M, Lopez-Gines C (2010) Metabolic aggressiveness in benign meningiomas with chromosomal instabilities. *Cancer Res* 70:8426–8434
- Nicholson JK, Lindon JC (2008) Systems biology: metabolomics. *Nature* 455:1054–1056
- Nicholson JK, Connelly J, Lindon JC, Holmes E (2002) Metabonomics: a platform for studying drug toxicity and gene function. *Nat Rev Drug Discov* 1:153–161
- Okamoto H, Li J, Vortmeyer AO, Jaffe H, Lee YS, Glasker S, Sohn TS, Zeng W, Ikejiri B, Proescholdt MA, Mayer C, Weil RJ, Oldfield EH, Zhuang Z (2006) Comparative proteomic profiles of meningioma subtypes. *Cancer Res* 66:10199–10204
- Opstad KS, Provencher SW, Bell BA, Griffiths JR, Howe FA (2003) Detection of elevated glutathione in meningiomas by quantitative in vivo ¹H MRS. *Magn Reson Med* 49:632–637
- Orlowski M (1976) Possible role of glutathione in transport processes. *Adv Exp Med Biol* 69:13–28
- Perry A, Stafford SL, Scheithauer BW, Suman VJ, Lohse CM (1997) Meningioma grading: an analysis of histologic parameters. *Am J Surg Pathol* 21:1455–1465
- Perry A, Scheithauer BW, Stafford SL, Lohse CM, Wollan PC (1999) "Malignancy" in meningiomas: a clinicopathologic study of 116 patients, with grading implications. *Cancer* 85:2046–2056

- Perry A, Gutmann DH, Reifenberger G (2004) Molecular pathogenesis of meningiomas. *J Neurooncol* 70:183–202
- Pueschel J, Coons SW, Scheck AC (2004) Diagnostic and prognostic significance of genetic regional heterogeneity in meningiomas. *Neuro Oncol* 6:290–299
- Pfisterer WK, Coons SW, Aboul-Enein F, Hendricks WP, Scheck AC, Preul MC (2008) Implicating chromosomal aberrations with meningioma growth and recurrence: results from FISH and MIB-I analysis of grades I and II meningioma tissue. *J Neurooncol* 87:43–50
- Robb VA, Li W, Gascard P, Perry A, Mohandas N, Gutmann DH (2003) Identification of a third protein 4.1 tumor suppressor, protein 4.1R, in meningioma pathogenesis. *Neurobiol Dis* 13:191–202
- Stuart JE, Lusk EA, Scheck AC, Coons SW, Lal A, Perry A, Gutmann DH (2011) Identification of gene markers associated with aggressive meningioma by filtering across multiple sets of gene expression arrays. *J Neuropathol Exp Neurol* 70(1):1–12
- Warburg O (1956) On the origin of cancer cells. *Science* 123:309–314
- Wiemels J, Wrensch M, Claus EB (2010) Epidemiology and etiology of meningioma. *J Neurooncol* 99:307–314
- Zhang FR, Delattre O, Rouleau G, Couturier J, Lefrançois TG, Aurias A (1990) The neuroepithelioma breakpoint on chromosome 22 is proximal to the meningioma locus. *Genomics* 6:174–177

Prognostic Parameters in Atypical and Malignant Meningiomas

13

Joze Pizem, Andrej Cör, and Mara Popovic

Abstract

Atypical and malignant meningiomas, which account for about 20% of meningiomas, are associated with a several-fold increased risk of local recurrence, and significant morbidity and mortality compared to benign counterparts. Atypical and malignant meningiomas are, however, biologically heterogeneous, and the probability of recurrence and aggressive behaviour can be predicted by several prognostic parameters. Among the most relevant of these are brain invasion, mitotic count, Ki-67 proliferation index, extent of resection and anatomical location. Brain invasion seems to be the most powerful prognostic parameter in atypical and malignant meningiomas, although it can only be assessed in a proportion of cases, when brain tissue is identified in the surgical tumour specimen. In addition to grading meningiomas by applying current World Health Organization criteria, pathologists should provide relevant prognostic parameters in their reports, including brain invasion (present, absent, not-assessable), mitotic count and Ki-67 index. Absence of brain invasion, when brain tissue is identified in the surgical specimen, seems to bear important positive prognostic information. Since there are controversies concerning which atypical meningiomas to treat with postoperative radiation, the prognostic parameters could assist in the selection of patients for additional postoperative treatment.

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Introduction

Meningiomas are among the most common intracranial tumours and account for up to one third of all primary intracranial neoplasms (Rogers et al. 2010). Meningiomas display a

broad range of histological patterns. The current World Health Organization (WHO) classification recognizes 14 different histological subtypes, falling into three grade categories – WHO grade I (benign), WHO grade II (atypical), and WHO grade III (malignant) (Louis et al. 2007). More precise criteria for meningioma grading were first provided in the 2000 WHO classification (Kleihues and Cavenee 2000) and remained basically unchanged in the most recent 2007 WHO classification (Louis et al. 2007). Meningiomas are considered WHO grade II when they display several atypical microscopic features (≥ 4 mitoses per 10 high power fields, or at least three of the five morphological features, including sheeting architecture, hypercellularity, prominent nucleoli, foci of small cells, and necrosis), or they are of the chordoid or clear cell subtypes. Additionally, otherwise benign meningiomas with brain invasion are considered grade II according to the 2007 WHO because they have similar recurrence and mortality rates as other WHO grade II meningiomas. The diagnostic criteria for WHO grade III meningiomas are ≥ 20 mitoses per 10 high power fields, or frank anaplasia with sarcoma, carcinoma or melanoma-like features, or the papillary or rhabdoid subtypes.

In this chapter, we use the designation ‘atypical’ as synonymous with grade II and ‘malignant’ as synonymous with grade III. The discussion is pertinent to atypical and malignant (AMM) meningiomas as defined in the 2000 WHO, since there has been no relevant study of AMM using the 2007 WHO criteria. Inclusion of otherwise benign meningiomas with brain invasion among AMM is, however, unlikely to change significantly the relevance of prognostic parameters discussed below.

Because more precise meningioma grading schemes were provided relatively recently, and are only gradually and inconsistently adopted into practice, the precise incidence of atypical and malignant meningiomas is difficult to ascertain, but roughly 20% of all meningiomas are atypical and 1–2% malignant (Kane et al. 2010; Mawrin and Perry 2010; Rogers et al. 2010). Compared to benign meningiomas, AMM are relatively more often diagnosed at non-skull base locations, in male patients, and at recurrences (Kane et al. 2010). Although benign meningiomas are 2–3 times more

frequent in females, AMM are nearly equally distributed between the two sexes (Durand et al. 2009; Kane et al. 2010; Vranic et al. 2010). Among recurrent meningiomas, 50% were atypical or malignant in the study by Kane et al. (2010).

Because the diagnosis of AMM is based on sets of optional criteria (mitotic count, or certain morphological features, or histological subtype) and because there is a morphological and biological spectrum in meningiomas, the group of AMM is prognostically heterogeneous (Modha and Gutin 2005). Although limited by the relatively small number of relevant studies using current grading criteria with a substantial number of cases, and still not perfectly reproducible and objective grading criteria (Rogers et al. 2010), there is substantial evidence that the risk of aggressive behaviour of AMM can be stratified more precisely (Perry et al. 1997, 1999; Vranic et al. 2010; Yang et al. 2008). One of the most important issues in AMM is which patient should receive radiation therapy after surgery (Rogers et al. 2010). Additional prognostic parameters could assist in the selection of patients with AMM for additional postoperative radiation treatment (Modha and Gutin 2005).

We recently analysed a series of 86 primary AMM for various histopathological and clinical parameters, including brain invasion, mitotic count, various histological features used in meningioma grading, Ki-67 proliferation index, location, size, age of the patient, Simpson grade of resection, the degree of brain oedema, and cleavability (Vranic et al. 2010). Brain invasion, high mitotic count (as a continuous variable) and parasagittal-falcine location were associated with decreased recurrence free survival in multivariate analysis. When brain invasion was not included in the analysis because of the limited number of cases in which it could be assessed, high mitotic count, Ki-67 proliferation index $\geq 4\%$, the presence of macronucleoli, and parasagittal-falcine location were independent predictors of recurrence free survival. According to our results and the results of some other studies (Aghi et al. 2009; Durand et al. 2009; Perry et al. 1997, 1999; Yang et al. 2008), the most relevant prognostic parameters in AMM appear to be brain invasion, mitotic count, proliferation index, extent of surgical resection and anatomical location.

Brain Invasion

Brain invasion is defined as irregular projections of meningioma into adjacent brain tissue without an intervening layer of leptomeninges (Perry et al. 1997; Louis et al. 2007). In the past, its prognostic significance was poorly defined and controversial. It had traditionally been considered to be indicative of malignancy but, in 1999, Perry et al. (1999) showed that brain invasive meningiomas without anaplasia (i.e., otherwise benign or atypical) pursue a less aggressive clinical course than malignant (anaplastic) meningiomas. In the 2000 WHO, it was acknowledged that brain invasion increases the risk of recurrence but brain invasion was not proposed as a grading criterion (Kleihues and Cavanee 2000). This was changed in the 2007 WHO and otherwise benign meningiomas with brain invasion are now considered WHO grade II (Louis et al. 2007).

The most important limitation of brain invasion as a prognostic parameter is the fact that it is assessable only in a proportion of cases, when brain tissue is present in the surgical tumour specimen. Brain invasion has been very likely underreported (Durand et al. 2009), because of the aforementioned controversies and because identification of brain tissue in surgical specimens requires active searching and proper tissue sampling. The presence of brain tissue in surgical specimens seems to be related to a surgeon's approach, large size of the tumour, tumour grade and presence of brain oedema, while its identification is very likely related to the number of tissue blocks examined (Perry et al. 1997; Vranic et al. 2010; Yang et al. 2008). Perry et al. (1997) identified brain tissue in 15% of consecutive primary meningiomas of all grades. We identified brain tissue in 43% of primary AMM (Vranic et al. 2010). On the other hand, Yang et al. (2008) reported brain tissue in 88% of AMM, but they stated that 'adjacent brain parenchyma was intentionally removed to assess the brain invasion, when a clear arachnoid plane disappeared'. The frequency of brain invasion increases with meningioma grade, although it is difficult to estimate because the assessability of brain invasion among

different grades has not been specified in most published studies. Probably 50–70% of atypical and 80–90% of malignant meningiomas invade the brain (Perry et al. 1997; Vranic et al. 2010; Yang et al. 2008).

We showed that brain invasion was a powerful independent prognostic parameter of recurrence free survival, together with location and mitotic count (Vranic et al. 2010). Brain invasive AMM had an eight-fold increased risk of local recurrence compared to meningiomas without brain invasion but identified brain tissue in surgical specimens. Recurrence free survival of invasive malignant meningiomas was significantly worse than recurrence free survival of invasive atypical meningiomas (Fig. 13.1). Since brain invasion assessability (identification of brain tissue) was not related to recurrence free survival (Fig. 13.2), it seems that brain invasion assessability is not related to the invasion itself, i.e., the likelihood of an AMM being invasive seems not to be different (lower) when brain tissue is not identified in the tissue sample. Yang et al. (2008) reported brain invasion to be a negative prognostic parameter in their subgroup of 24 malignant but not 40 atypical meningiomas. In two seminal articles on current meningioma grading, Perry et al. (1997, 1999) analysed the prognostic significance of brain invasion in relation to other parameters. Brain invasion was a very important negative prognostic parameter (the only significant parameter among those studied) in a group of 76 primary meningiomas of all grades with brain invasion assessable. The authors also showed that brain invasive meningiomas comprise a prognostically heterogeneous group and the overall survival of patients with invasive malignant meningiomas differ significantly from those with invasive atypical meningiomas.

Brain invasion is probably one of the most important prognostic parameters in meningiomas in general as well as in AMM (Perry et al. 1997, 1999; Stafford et al. 1998; Vranic et al. 2010). Pathologists should be aware of this and should invest every effort in identifying brain tissue in surgical specimens, and report brain invasion as present, absent or not-assessable. The absence of brain invasion (when assessable) provides very

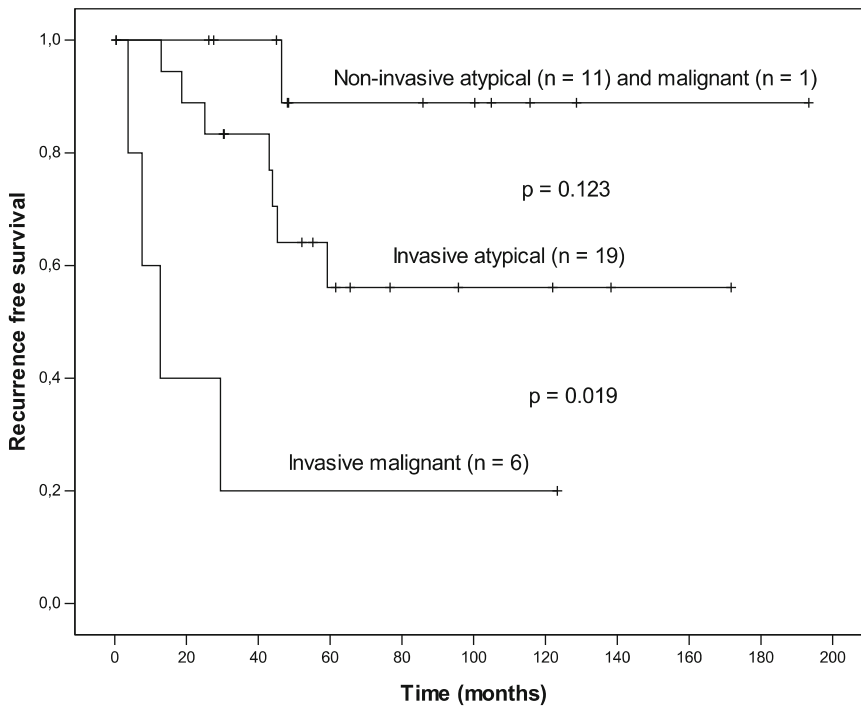


Fig. 13.1 Kaplan-Meier curves of recurrence free survival according to brain invasion (Modified from Vranic et al. 2010)

important positive prognostic information. Brain invasion cannot be predicted reliably by clinical parameters, such as peritumoural brain oedema or cleavability (Vranic et al. 2010).

Mitotic Count

Mitotic count is a powerful prognostic parameter in meningiomas and is used in meningioma grading, with cut-off ≥ 4 mitoses per 10 high power fields for atypical and ≥ 20 mitoses per 10 high power fields for malignant (Louis et al. 2007; Perry et al. 1997, 1999; Stafford et al. 1998). Based on the grading criteria, mitotic count varies greatly in AMM (Vranic et al. 2010). Perry et al. (1997) showed that meningiomas with a mitotic count of ≥ 4 are at an increased risk of recurrence compared to meningiomas that fulfil

other criteria for atypical meningiomas, indicating a potential prognostic value of mitotic count in AMM. Pasquier et al. (2008) reported high mitotic count to be adversely related to recurrence and overall survival in a series of 119 AMM, but the cut-off value was not stated. In a study of 118 atypical meningiomas with Simpson grade 1 resection by Aghi et al. (2009), mitotic count tended to be, although not significantly, associated with recurrence free survival; but the cut-off value was not provided. In our study of AMM, mitotic count as a continuous variable was an independent prognostic parameter of recurrence free survival in AMM (Vranic et al. 2010). Although there has been no other study evaluating the prognostic significance of mitotic count as a continuous variable in AMM, current evidence supports its important role as a prognostic parameter in AMM.

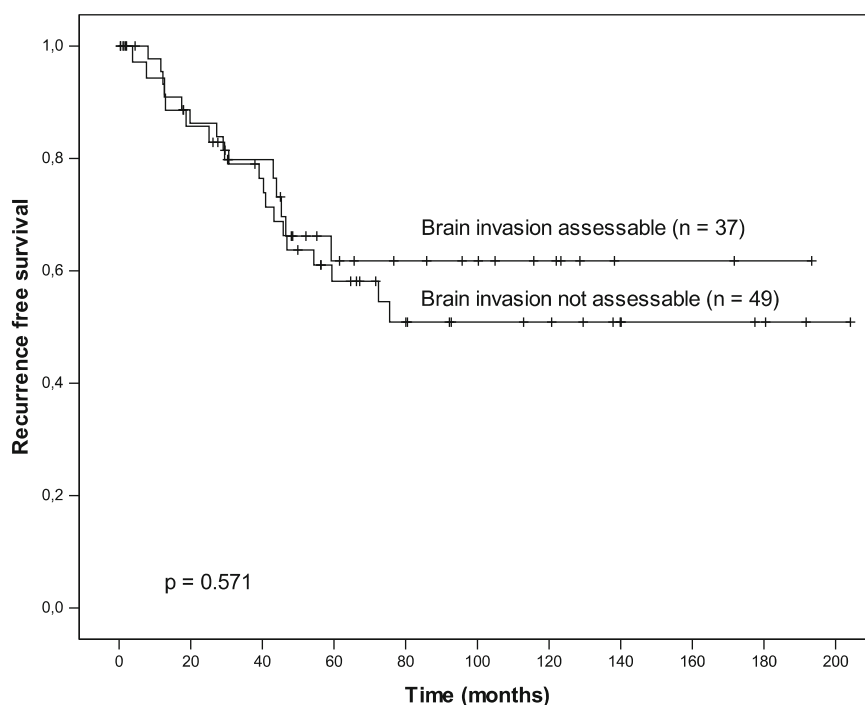


Fig. 13.2 Kaplan-Meier curves of recurrence free survival according to the presence of brain tissue in surgical specimens (brain invasion assessability) (Modified from Vranic et al. 2010)

Proliferation Index

The role of the Ki-67 proliferation index in meningiomas and its clinical relevance still remains relatively poorly defined, despite extensive research and its common use in clinical practice (Louis et al. 2007). The principal limitations seem to be lack of standardisation of immunohistochemical reaction and its quantification and difficulties in defining cut-off values. Although Ki-67 indices increase significantly from benign through atypical and malignant meningiomas, there is a considerable overlap between different grades (Korshunov et al. 2003; Perry et al. 1998). While the Ki-67 index as a continuous variable seems not to have prognostic value, there is some evidence that its categorisation into low and high values may be prognostically relevant (Aghi et al. 2009; Perry et al. 1998; Rogers et al. 2010; Yang et al. 2008). There is no consensus which cut-off

to use and cut-off values chosen in different studies were: 4%, 4.2%, 4.4%, 5% and 10% (Aghi et al. 2009; Durand et al. 2009; Ho et al. 2002; Perry et al. 1998; Rogers et al. 2010; Yang et al. 2008). Perry et al. (1998) reported a Ki-67 index of $\geq 4.2\%$ to be associated with decreased recurrence free survival in univariate but not multivariate analysis of 422 meningiomas of all grades with gross total resection; a mitotic rate of ≥ 4 per 10 high power fields but not the Ki-67 index was a significant prognostic parameter in multivariate analysis. Korshunov et al. (2003) similarly reported a Ki-67 index of $>4.4\%$ to be associated with decreased recurrence free survival in univariate analysis of 271 meningiomas of all grades and also in a subset of 208 benign meningiomas.

The prognostic significance of the Ki-67 index has been studied in four large series of AMM. Aghi et al. (2009) failed to show any prognostic significance of the Ki-67 index as a continuous

variable in a series of 108 primary atypical meningiomas with gross total resection. In a study by Yang et al. (2008), the Ki-67 index >5% tended to be adversely associated with recurrence free survival in a group of 24 malignant but not in a group of 40 atypical meningiomas. It is, however, very unusual for a malignant meningioma to have a Ki-67 index less than 5% (Korshunov et al. 2003; Rogers et al. 2010). Durand et al. (2009) reported a mean Ki-67 index of 9.9% in 81 atypical and 28.8% in 12 malignant meningiomas. A Ki-67 index with a cut-off 10% had no prognostic value in the subset of atypical meningiomas. In addition to a relatively high cut-off value, a Ki-67 index was only available for less than half of their AMM cases and its availability may have been biased. Although the study by Durand et al. (2009) was the largest study of AMM so far, the method of case selection (original diagnosis of grade II and III meningiomas followed by the exclusion of cases that did not fulfil the 2000 WHO criteria on revision) probably resulted in a study group that did not reflect the spectrum of AMM as currently defined. We used a cut-off value >4% (Vranic et al. 2010). All of the 10 malignant and 63% of the atypical meningiomas had a Ki-67 index >4%. A Ki-67 index >4% was an independent negative prognostic parameter in multivariate analysis of the entire group of 86 primary AMM, but lost its prognostic significance when brain invasion was included in the analysis. Our results showed that the Ki-67 index, when dichotomized, may be a useful prognostic parameter in AMM, especially when brain invasion cannot be assessed. The most prognostically significant cut-off value for the Ki-67 index seems to be between 4% and 5% (Perry et al. 1998; Vranic et al. 2010; Yang et al. 2008).

Extent of Surgical Resection

In various studies, the extent of surgical resection has been reported by using the Simpson grading scheme or as gross total versus subtotal resection. According to Simpson (1957), gross total resection of the tumour together with dural attachments and abnormal bone is defined as grade 1,

gross total resection with coagulation of dural attachments as grade 2, gross total resection of the tumour without additional treatment of the bone or dura grade 3, partial resection grade 4 and biopsy grade 5. Because gross total resection is variously considered Simpson grade 1, grade 1–2 or grade 1–3, and the extent of resection varies among different studies, it is difficult to interpret and compare the results (Aghi et al. 2009; Korshunov et al. 2003; Palma et al. 1997; Pasquier et al. 2008; Perry et al. 1997, 1999). There is general agreement that the extent of surgical resection of meningiomas is prognostically important (Louis et al. 2007; Modha and Gutin 2005; Rogers et al. 2010; Yang et al. 2008), although it is less clear whether the risk of recurrence increases with each individual Simpson grade and whether the prognostic value of the extent of resection varies among different meningioma grades.

In a series of benign meningiomas, Sughrue et al. (2010) reported slightly increasing recurrence rates with increasing Simpson grade but the differences were not statistically significant; 5% benign meningiomas with Simpson grade 1 resection recurred in 5 years. In a series of 108 atypical meningiomas with Simpson grade 1 resection, studied by Aghi et al. (2009), the recurrence rate at 5 years was 41%. In a heterogeneous group of 116 meningiomas (anaplastic, brain invasive and/or metastatic) reported by Perry et al. (1999), subtotal resection (probably meaning Simpson grade >3) was an independent adverse prognostic parameter with an approximately three-fold increased risk of death. Palma et al. (1997) reported Simpson grade >1 and histological grade to be independent adverse prognostic parameters in a series of 71 AMM applying their own grading system, different from the 2000 WHO or 2007 WHO. In a study by Durand et al. (2009), Simpson grade >1 was an independent prognostic parameter of overall survival in a series of 199 AMM, together with patients' age and WHO grade. In our series, AMM with Simpson grade >1 tended to recur more often but not statistically significantly (Vranic et al. 2010).

Current evidence supports the prognostic value of the extent of resection in AMM, although

the true impact of each individual Simpson grade on survival is difficult to ascertain. Of importance, a significant proportion of AMM (roughly 40–50%) with Simpson grade 1 resection will recur (Aghi et al. 2009; Durand et al. 2009; Vranic et al. 2010). The high recurrence rate in this group is probably the result of microscopic residual tumour foci in the brain parenchyma (Modha and Gutin 2005; Perry et al. 1999; Vranic et al. 2010).

Anatomical Location

Among intracranial AMM, about 30–50% are located at the convexity, 20–35% at parasagittal-falcine sites, and 20–30% at the base (Durand et al. 2009; Kane et al. 2010; Pasquier et al. 2008; Vranic et al. 2010; Yang et al. 2008). Information on the possible prognostic significance of the location in AMM and meningiomas in general is relatively sparse. In our study, recurrence free survival of parasagittal-falcine AMM was significantly shorter than AMM at other intracranial locations, in particular convexity (Vranic et al. 2010). A parasagittal-falcine location was an independent adverse prognostic parameter, with approximately four-fold increased risk of recurrence, together with brain invasion and mitotic count. The reason for the high recurrence rate of parasagittal-falcine AMM might be related to residual tumour along the superior sagittal sinus (Modha and Gutin 2005). In a study of AMM by Yang et al. (2008), location was not significantly associated with outcome, although convexity malignant meningiomas tended to be associated with longer overall survival. Similarly, Durand et al. (2009) failed to show a significant association of location with prognosis in their series of 199 AMM.

Conclusions

Because the great majority of meningiomas are benign with a relatively low recurrence rate and mortality, it is rational to identify first all those meningiomas that are at increased risk of behaving

more aggressively. This is currently best achieved by applying the WHO grading criteria for AMM (Louis et al. 2007). Using current grading criteria, AMM comprise about 20% of all meningiomas and have at least eight-fold increased risk of recurrence compared to benign meningiomas (Mawrin and Perry 2010; Perry et al. 1997). Meningioma grading seems still not to be uniformly and homogeneously applied (Rogers et al. 2010). Because several histological features have to be assessed and many histological slides examined, the grading of meningiomas is relatively tedious work. The requirement for a thorough and systematic histological analysis may be related to the gradual and inconsistent adoption of meningioma grading into clinical practice.

The group of AMM is biologically heterogeneous. The most relevant prognostic parameters in AMM are brain invasion, mitotic count, Ki-67 index and the extent of surgical resection. Considering these prognostic parameters, recurrence free survival of AMM ranges from similar to that of benign meningiomas (absence of brain invasion when assessable, low mitotic count approaching zero per ten high power fields, low proliferation index, Simpson grade 1 resection) to highly aggressive neoplasm with a recurrence rate approaching 100% and high mortality (brain invasive malignant meningiomas) (Perry et al. 1999; Vranic et al. 2010). Brain invasion seems to be the most important prognostic parameter, although it can be assessed only in a proportion of AMM, when brain tissue is identified in surgical specimens. Pathologists should actively search for brain invasion by thorough examination of surgical specimens and report brain invasion as present, absent or not-assessable. The absence of brain invasion when assessable seems to be a very important positive prognostic parameter. In addition to grading, the pathologist has the role of providing additional prognostic parameters (brain invasion, mitotic count, Ki-67 index). Because the great majority of malignant meningiomas are brain invasive and have high mitotic and Ki-67 indices, and malignant meningiomas represent a morphological and genetic continuum with atypical meningiomas, WHO grade 3 *per se* is not an independent negative prognostic parameter

in AMM when all these prognostic parameters are available (Vranic et al. 2010). Malignant meningiomas, however, represent a distinct subset of AMM with the worst prognosis.

References

- Aghi MK, Carter BS, Cosgrove GR, Ojemann RG, Amin-Hanjani S, Martuza R, Curry WT, Barker FG (2009) Long-term recurrence of atypical meningiomas after gross total resection with or without postoperative adjuvant radiation. *Neurosurgery* 64:56–60
- Durand A, Labrousse F, Jouvét A, Bauchet L, Kalamarides M, Menei P, Deruty R, Moreau JJ, Fevre-Montange M, Guyotat J (2009) WHO grade II and III meningiomas: a study of prognostic factors. *J Neurooncol* 95:367–375
- Ho DM, Hsu CY, Ting LT, Chiang H (2002) Histopathology and MIB-1 labeling index predicted recurrence of meningiomas: a proposal of diagnostic criteria for patients with atypical meningioma. *Cancer* 94:1538–1547
- Kane AJ, Sughrue ME, Rutkowski MJ, Shangari G, Fang S, McDermott MW, Berger MS, Parsa AT (2010) Anatomic location is a risk factor for atypical and malignant meningioma. *Cancer* 117:1272–1278
- Kleihues P, Cavenee WK (2000) World Health Organization classification of tumours: pathology and genetics of tumours of the central nervous system. IARC Press, Lyon
- Korshunov A, Shishikina L, Golanov A (2003) Immunohistochemical analysis of P16INK4A, P14ARF, P18INK4C, P21CIP1, P27KIP1, and P73 expression in 271 meningiomas correlation with tumor grade and clinical outcome. *Int J Cancer* 104:728–734
- Louis DN, Ohgaki H, Wiestler OD, Cavenee WK (2007) World Health Organization classification of tumours: WHO classification of tumours of the central nervous system. IARC Press, Lyon
- Mawrin C, Perry A (2010) Pathological classification and molecular genetics of meningiomas. *J Neurooncol* 99:379–391
- Modha A, Gutin PH (2005) Diagnosis and treatment of atypical and malignant meningiomas: a review. *Neurosurgery* 57:538–550
- Palma L, Celli P, Franco C, Cervoni L, Cantore G (1997) Long-term prognosis for atypical and malignant meningiomas: a study of 71 surgical cases. *J Neurosurg* 86:793–800
- Pasquier D, Bijmolt S, Veninga T, Rezvoy N, Villa S, Krengli M, Weber DC, Baumert BG, Canylmaz E, Yalman D, Szutowicz E, Tzuk-Shina T, Mirimanoff RO (2008) Atypical and malignant meningioma: outcome and prognostic factors in 119 irradiated patients. A multicenter, retrospective study of the Rare Cancer Network. *Int J Radiat Oncol Biol Phys* 71:1388–1393
- Perry A, Stafford SL, Scheithauer BW, Suman VJ, Lohse CM (1997) Meningioma grading: an analysis of histologic parameters. *Am J Surg Pathol* 21:1455–1465
- Perry A, Stafford SL, Scheithauer BW, Suman VJ, Lohse CM (1998) The prognostic significance of MIB-1, p53, and DNA flow cytometry in completely resected primary meningiomas. *Cancer* 82:2262–2269
- Perry A, Scheithauer BW, Stafford SL, Lohse CM, Wollan PC (1999) “Malignancy” in meningiomas: a clinicopathologic study of 116 patients, with grading implications. *Cancer* 85:2046–2056
- Rogers L, Gilbert M, Vogelbaum MA (2010) Intracranial meningiomas of atypical (WHO II) histology. *J Neurooncol* 99:393–405
- Simpson D (1957) The recurrence of intracranial meningiomas after surgical treatment. *J Neurol Neurosurg Psychiatry* 20:22–39
- Stafford SL, Perry A, Suman VJ, Meyer FB, Scheithauer BW, Lohse CM, Shaw EG (1998) Primarily resected meningiomas: outcome and prognostic factors in 581 Mayo Clinic patients, 1978 through 1988. *Mayo Clin Proc* 73:936–942
- Sughrue ME, Kane AJ, Shangari G, Rutkowski MJ, McDermott MW, Berger MS, Parsa AT (2010) The relevance of Simpson grade I and II resection in modern neurosurgical treatment of World Health Organization grade I meningiomas. *J Neurosurg* 113:1029–1035
- Vranic A, Popovic M, Cör A, Prestor B, Pizem J (2010) Mitotic count, brain invasion, and location are independent predictors of recurrence-free survival in primary atypical and malignant meningiomas: a study of 86 patients. *Neurosurgery* 67:1124–1132
- Yang SY, Park CK, Park SH, Kim DG, Chung YS, Jung HW (2008) Atypical and anaplastic meningiomas: prognostic implications of clinicopathological features. *J Neurol Neurosurg Psychiatry* 79:574–580

Sporadic Meningioangiomatosis: Diagnosis with Computed Tomography and Magnetic Resonance Imaging

14

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Abstract

Meningioangiomatosis (MA) is a rare, benign, hamartomatous lesion of leptomeninges and underlying cerebral cortex. Preoperative diagnosis is difficult and challenging due to its diverse clinical, pathological and imaging features. Imaging findings of this disease including computed tomography (CT) and magnetic resonance imaging (MRI) are reviewed in this chapter. CT findings include focal extensively calcified lesions, lesions with patchy calcification, and no apparent calcification. MRI findings include predominantly hypointensity on T1-weighted images and hyperintensity on T2-weighted images, and gyriform hyperintensity on FLAIR sequences. After contrast material administration, the majority of the lesions show heterogeneous enhancement. Occasionally, a non-enhancing lesion on MRI may be seen due to extensive calcification on CT. Gyriform hyperintensity on FLAIR sequence is the unique MRI feature of meningioangiomatosis, which correlates with proliferating microvessels with perivascular cuffs of spindle-cell proliferation within the cortex on pathology.

Introduction

Meningioangiomatosis is a rare benign cerebral lesion characterized by leptomeningeal calcification and meningovascular proliferation on histopathology. Although MA occurs infrequently, it

is important to establish the correct diagnosis due to its histologically benign nature. It is a surgically correctable cause of seizure. MA may or may not be associated with neurofibromatosis (NF). This chapter deals with the sporadic cases of MA. The clinical presentations of sporadic meningioangiomatosis include seizure and headache. Occasionally, it is discovered as an incidental finding. The duration of their symptoms before admission to hospital varies from 2 months to 30 years. The clinical diagnosis largely based on CT and MRI imaging findings before operation were often misdiagnosed as arteriovenous malformation (AVM), or low grade gliomas.

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Imaging Findings

The majority of these lesions are located in the cerebral cortex, including temporal lobe and frontal lobe. Other less frequent locations include thalamus, third ventricle, brain stem and cerebellar vermis. They are all well-demarcated lesions with a mean size of approximately 3 cm. The majority of these lesions are nodular in appearance and less frequently they may be irregular in contour.

In our series of seven cases (Yao et al. 2009), five demonstrated calcification on plain CT. Only one case showed slight surrounding edema and the remaining six exhibited no edema on MRI. Four of the seven cases showed mild mass effect while the other three revealed no mass effect. On plain CT scans, they appeared as thickened gyriform lesion with surrounding hypodense edema (Fig. 14.1a); nodular lesion with calcification (Fig. 14.2a); hypodense lesion with central dotted calcification (Fig. 14.3a); hypodense lesion. On MR T1-Weighted images, three were isointense to white matter (Fig. 14.3b); three showed mixed hypo-, iso- and hyperintensity (Fig. 14.1b), and one hypointensity (Fig. 14.2b); on T2-Weighted images, six showed hyperintensity (Figs. 14.1c and 14.3c) and one hypointensity (Fig. 14.2c); On FLAIR images, three demonstrated central low or mixed signal intensity surrounded with evident gyriform hyperintensity (Fig. 14.1d), three showed gyriform hyperintensity (Fig. 14.3d) and one hypointensity (Fig. 14.2d). On post-contrast MR images, three showed strong heterogeneous nodular enhancement (Fig. 14.3e), two slight peripheral band-like enhancement (Fig. 14.1e), and one no enhancement (Fig. 14.2e). One patient did not receive contrast injection.

Discussion

Meningioangiomatosis was first described by Basso and Nuzum 1915 and then was named by Worster-Drought et al. 1937. It was associated with neurofibromatosis type 2 (NF2) in nearly

14% of reported cases (Omeis et al. 2006), but the cases we reported here were sporadic and not associated with NF2. MA cases associated with neurofibromatosis are frequently discovered at autopsy since they are usually asymptomatic. Sporadic cases are often symptomatic and patients usually present with intractable seizure. In the sporadic cases, there is an equal incidence in male and female whereas in the cases associated with NF, there is a male preference of 2:1. MA associated with NF may be multifocal and is often diagnosed at autopsy due to absence of clinical symptoms. The age of onset is generally younger in the sporadic cases, focal seizures and/or headaches are the most common presenting symptoms. The mean age of our seven cases was 18.1 years old.

Meningioangiomatosis is a benign cerebral lesion characterized by leptomeningeal calcification and meningovascular proliferation. On CT scans, they present as calcified, enhancing lesions with surrounding low density edema. On MRI, they exhibit central low or mixed signal intensity on T1- and T2-weighted images and surrounding high signal edema on T2-weighted sequences. Gadolinium enhancement is seen in some of the lesions (Tien et al. 1992; Park et al. 1999; Wiebe et al. 1999). In our seven cases, five lesions exhibited calcification on plain CT scans, and the calcifications were nodular, patchy or dotted in shape. The lesions with calcification could be easily identified on plain CT scans. However, a non-calcified, isodense lesion may be missed on plain CT. On MRI, all these lesions could be easily localized in the cerebral cortex, thalamus, brain stem or in cerebellar vermis. The MRI features of these lesions on T1WI and T2WI in our series were similar to those in the literatures (Tien et al. 1992; Aizpuru et al. 1991; Gómez-Ansón et al. 1995; Meltzer et al. 1998; Kim et al. 2002b). However, on FLAIR sequence, six of our seven cases demonstrated evident gyriform hyperintensity. To our knowledge, gyriform hyperintensity on the FLAIR sequence is a characteristic feature of MA. Following administration of Gadolinium, the lesions mainly showed avid heterogeneous nodular enhancement or slight peripheral band-like enhancement. One lesion

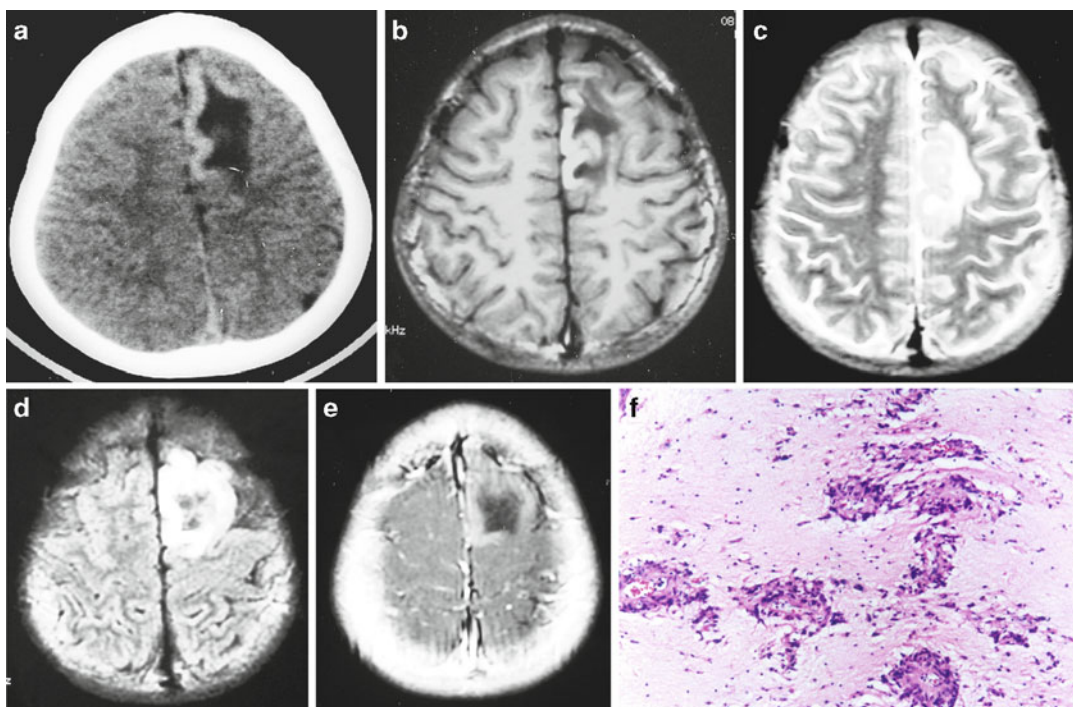


Fig. 14.1 (a) Plain CT scan shows a left frontal lobe hypodense lesion with surrounding thickened gyri-form hyperdensity. (b) MR T1WI shows the lesion with mixed hypo- and hyper- intensity, the hyperintense region has a gyri-form appearance. (c) MR T1 WI reveals a hyperintense lesion in the left frontal lobe. (d) MR FLAIR image

reveals the lesion with evident gyri-form hyperintensity. (e) Post-contrast MR T1WI shows the lesion with slight peripheral band-like enhancement. (f) Microphotography of the specimen shows perivascular spindle-cell proliferation in the white matter (HE, $\times 100$) (With permission from Yao et al. 2009)

did not show any enhancement. We find these enhancement patterns to be of limited value for the diagnosis of MA. Some authors have described edema around the lesions, but most of our cases did not show surrounding edema with the exception of one case which showed mild surrounding edema.

On pathology, characteristics of meningioangiomas are leptomeningeal calcification and meningovascular proliferation interwoven with bands of fibrous connective tissue (Rubinstein 1972). The seven cases we reported all had pathological features of abundant proliferating leptomeningeal microvessels with perivascular cuffs of spindle-cell proliferation within the cortex (Fig. 14.1f), some with calcification (Fig. 14.2e),

and some with psammoma bodies. As reported previously (Wang et al. 2006), MA might be divided into three types: predominantly cellular type, vascular type and fibrocalcifying type. Immunohistochemically, these penetrating perivascular cells were negative for glial fibrillary acidic protein and S-100 protein, and positive for vimentin staining. These findings suggest that the histogenesis of the spindle-shaped cells is most probably meningothelial (Fujimoto et al. 1993).

In correlation with pathology, the evident gyri-form hyperintense appearance of MA on FLAIR images could be explained by thickened cortex where abundant proliferating leptomeningeal microvessels and spindle cells were located. However, if the lesion was completely calcified

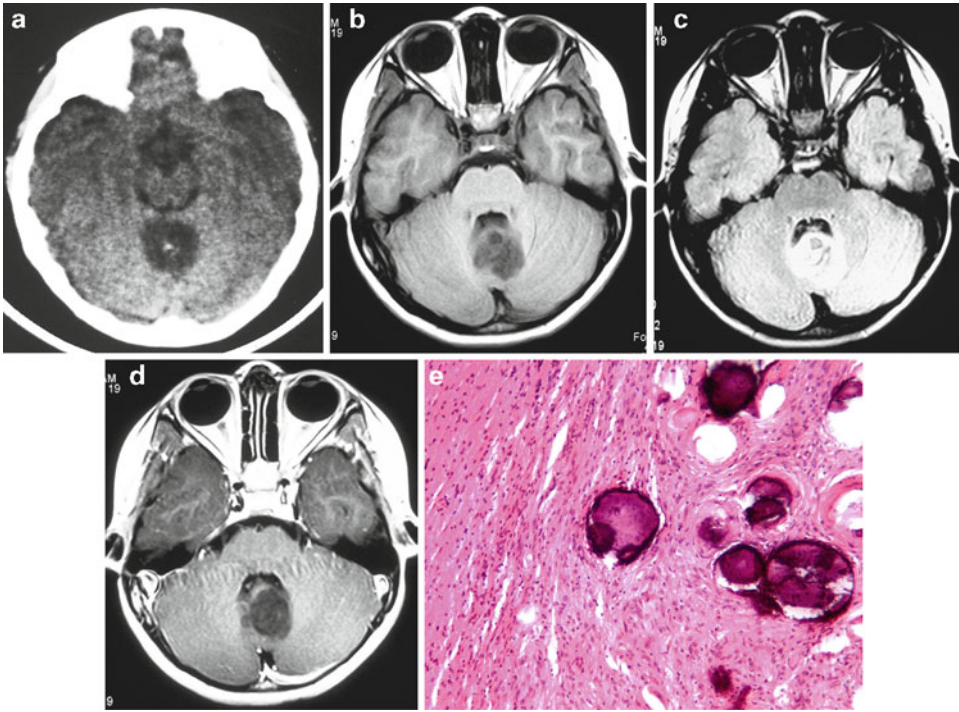


Fig. 14.2 (a) Plain CT scan shows a right temporal lobe lesion completely calcified. (b) MR T1WI demonstrates the lesion with hypointensity. (c) MR T2WI shows the lesion with hypointensity. (d) MR FLAIR image reveals the lesion

with hypointensity. (e) Microphotography of the specimen demonstrates a large number of calcification within the lesion (HE, $\times 40$) (With permission from Yao et al. 2009)

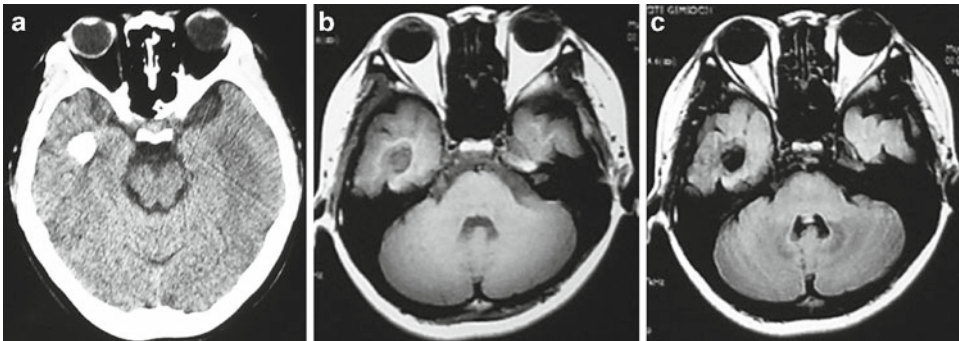


Fig. 14.3 (a) Plain CT scan shows an irregular low density lesion with calcification in the right temporal lobe. (b) MR T1WI demonstrates the lesion with predominant

isointensity. (c) MR FLAIR image reveals the lesion with diffuse gyriform hyperintensity. (With permission from Yao et al. 2009)

(fibrocalcifying type), it might show hypointensity without gyriform hyperintensity. The lesions of MA had various types of contrast enhancement pattern, varying from mild to strong heterogeneous enhancement or even no enhancement at

all. It appeared that enhancement patterns had correlation with the proliferating leptomeningeal microvessels. Although multiple lesions have occasionally been described (Park et al. 1999), none of our cases demonstrated multiple lesions.

MA associated with other tumors such as meningioma and oligodendroglioma have been reported in the literatures (López et al. 1996; Giangaspero et al. 1999; Mut et al. 2000; Kim et al. 2002a, b). In our seven cases, there were no other tumors associated with MA.

Since these lesions are a distinct clinicopathological entity and hamartomatous in nature, it is imperative to strive to make an accurate diagnosis in order to avoid unnecessary aggressive treatment (Meltzer et al. 1998; Wang et al. 2006). Differential diagnosis of MA usually include arteriovenous malformation (AVM), Sturge-Weber syndrome, low grade astrocytoma, oligodendroglioma, ganglioglioma, pleomorphic xanthoastrocytoma, desmoplastic infantile ganglioglioma, desmoplastic infantile astrocytoma, atypical or anaplastic meningioma, and granulomatous disease.

Arteriovenous malformation consists of arterial feeders, a nidus and enlarged draining veins. In general, a brain AVM extends from the pial surface of the brain to the deep white matter, and it may illustrate faint areas of hyperdensity of draining veins or denser areas of calcifications on CT which could be confused with MA. But on MRI, the most characteristic picture is that of an area with a nidus of flow void due to the flow within the vessels, which was not found in our MA cases. Sturge-Weber syndrome is also called encephalotrigeminal angiomatosis. Its pathological features are cortical calcifications and atrophy of underlying cortex, while MA is with hypertrophy of the cortex. Low grade astrocytoma can present as calcified mass on non-contrast CT and they usually exhibit hypointensity on T1WI and hyperintensity on T2WI. Gyriform hyperintensity on FLAIR seen in MA can help to differentiate the two entities. Oligodendroglioma is a slow growing glial tumor that originates from the oligodendrocyte. Coarse intratumoral calcifications are quite common within the oligodendroglioma, while calcifications in MA are more often nodular or patchy. On MRI, there are no specific features on T1WI and T2WI for differential diagnosis. However, gyriform hyperintensity on FLAIR may suggest the diagnosis of MA. Gangliogliomas are cortically based masses with

calcification seen on CT. Again, gyriform hyperintensity on MRI can suggest the diagnosis of MA. Granulomatous lesions, such as tuberculosis, can present as calcified lesions and should be included in the differential diagnosis.

In conclusion, MRI is useful for demonstrating the cortical location of MA, associated with gyriform hyperintensity on FLAIR sequence, which correlated with proliferating microvessels with perivascular cuffs of spindle-cell proliferation within the cortex on pathology. Plain CT scan is useful to illustrate the presence of calcification which could be helpful for diagnosis. The treatment of choice is surgical resection and the prognosis is usually excellent with cure of seizure in almost all cases.

References

- Aizpuru RN, Quencer RM, Norenberg M, Altman N, Smirniotopoulos J (1991) Meningioangiomatosis: clinical, radiologic, and histopathologic correlations. *Radiology* 179:819–821
- Bassoe P, Nuzum F (1915) Report of a case of central and peripheral neurofibromatosis. *J Nerv Ment Dis* 42:785–796
- Fujimoto K, Nikaidoh Y, Yuasa T, Nagata K, Ida Y, Fujioka M, Ohnishi H, Kurokawa S (1993) Meningioangiomatosis not associated with von Recklinghausen's disease—case report. *Neurol Med Chir (Tokyo)* 33:651–655
- Giangaspero F, Guiducci A, Lenz FA, Mastronardi L, Burger PC (1999) Meningioma with meningioangiomatosis: a condition mimicking invasive meningiomas in children and young adults: report of two cases and review of the literature. *Am J Surg Pathol* 23:872–875
- Gómez-Ansón B, Muñoz A, Blasco A, Madero S, Esparza J, Cordobés F, Orejón G, Mateos F (1995) Meningioangiomatosis: advanced imaging and pathological study of two cases. *Neuroradiology* 37:120–123
- Kim NR, Choe G, Shin SH, Wang KC, Cho BK, Choi KS, Chi JG (2002a) Childhood meningiomas associated with meningioangiomatosis: report of five cases and literature review. *Neuropathol Appl Neurobiol* 28:48–56
- Kim WY, Kim IO, Kim S, Cheon JE, Yeon M (2002b) Meningioangiomatosis: MR imaging and pathological correlation in two cases. *Pediatr Radiol* 32:96–98
- López JI, Ereño C, Oleage L, Areitio E (1996) Meningioangiomatosis and oligodendroglioma in a 15-year-old boy. *Arch Pathol Lab Med* 120:587–590
- Meltzer CC, Liu AY, Perrone AM, Hamilton RL (1998) Meningioangiomatosis: MR imaging with histopathologic correlation. *AJR Am J Roentgenol* 170:804–805

- Mut M, Söylemezoğlu F, Firat MM, Palaoğlu S (2000) Intraparenchymal meningioma originating from underlying meningioangiomatosis: case report and review of the literature. *J Neurosurg* 92:706–710
- Omeis I, Hillard VH, Braun A, Benzil DL, Murali R, Harter DH (2006) Meningioangiomatosis associated with neurofibromatosis: report of 2 cases in a single family and review of the literature. *Surg Neurol* 65:595–603
- Park MS, Suh DC, Choi WS, Lee SY, Kang GH (1999) Multifocal meningioangiomatosis: a report of two cases. *AJNR Am J Neuroradiol* 20:677–680
- Rubinstein LJ (1972) Atlas of tumor pathology. Second series, fascicle 6. Armed Forces Institute of Pathology, Washington, DC, pp 252–307
- Tien RD, Osumi A, Oakes JW, Madden JF, Burger PC (1992) Meningioangiomatosis: CT and MRI findings. *J Comput Assist Tomogr* 16:361–365
- Wang Y, Gao X, Yao ZW, Chen H, Zhu JJ, Wang SX, Gao MS, Zhou LF, Zhang FL (2006) Histopathological study of five cases with sporadic meningioangiomatosis. *Neuropathology* 26:249–256
- Wiebe S, Munoz DG, Smith S, Lee DH (1999) Meningioangiomatosis: a comprehensive analysis of clinical and laboratory features. *Brain* 122:709–726
- Worster-Drought C, Dickson WEC, McMenemey WH (1937) Multiple Meningeal and perineural tumors with analogous changes in the glia and ependyma. *Brain* 60:85–117
- Yao Z, Wang Y, Zee C, Feng X, Sun H (2009) Computed tomography and magnetic resonance appearance of sporadic meningioangiomatosis correlated with pathological findings. *J Comput Assist Tomogr* 33:799–804

Part III

Therapy

Atypical Meningioma: Distinguishing Features and Role of Adjuvant Radiation

15

Pankaj K. Agarwalla and Manish K. Aghi

Abstract

Meningiomas are the most common primary brain tumor in adults. Ninety percent of meningiomas are benign tumors, for which gross-total resection can achieve very low recurrence rates, with radiation therapy or radiosurgery reserved for recurrences or residual tumor that grows during serial imaging. At the other end of the spectrum, malignant meningiomas, although representing just 1% of meningiomas, have long been recognized to exhibit aggressive behavior that ultimately proves fatal regardless of extent of resection or whether adjuvant radiation and chemotherapy are utilized. Over the past two decades, there has been pathological recognition of a third type of meningioma known as “atypical,” with borderline histologic and clinical features between benign and malignant meningioma. Here, we review the histopathology, treatment, and long-term prognosis of atypical meningiomas. In particular, we review studies investigating the impact of radiation therapy and radiosurgery on the long-term recurrence rate of completely and incompletely resected atypical meningiomas.

Introduction

Meningiomas are the most common primary central nervous system tumor in adults with a reported incidence of 32% of all primary brain tumors according to the Central Brain Tumor Registry of the United States (CBTRUS) data

from 2000 to 2004 (Pearson et al. 2008). For decades, gross-total resection has been the mainstay of meningioma management, with data strongly suggesting that a complete gross-total resection can achieve very low recurrence rates of benign meningiomas. From this treatment paradigm arose important technical advances in operative approaches, particularly skull base techniques, for the resection of benign meningiomas. Unfortunately, these important surgical techniques in delicate areas often resulted in increased morbidity for patients. With improvements in radiation therapy including stereotactic

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radiotherapy, studies began to explore the possibility of near gross-total resections followed by radiation to improve surgical outcomes without significantly increasing recurrence rates in benign meningiomas. At the other end of the spectrum, malignant meningiomas have long been recognized to exhibit aggressive behavior that ultimately proves fatal regardless of extent of resection or whether adjuvant radiation and chemotherapy were utilized. Over the past two decades, there has been pathological recognition of a third type of meningioma known as “atypical,” with borderline histologic and clinical features between benign and malignant meningioma. Here, after briefly reviewing pertinent historical data regarding benign and malignant meningiomas, we will focus on atypical meningiomas with an emphasis on their histopathology, treatment, and outcomes. In particular, we will highlight advances in radiation therapy and review studies investigating the impact of radiation therapy on the long-term recurrence rate of atypical meningiomas.

Pathological Considerations

Our current classification scheme of meningiomas is the result of over a century of dedicated investigation by numerous neurosurgeons and pathologists. After Sir William Macewen described the first successful intracranial removal of a meningioma in 1881 from the left frontal convexity in a 14 year old girl, perhaps the most important subsequent contribution came from Harvey Cushing who studied the histopathology of meningioma over decades, introduced and promoted the use of the term “meningioma”, and described an important classification scheme in a volume entitled, *Meningiomas, their classification, regional behaviour, life history, and surgical end results* (al-Rodhan and Laws 1990). In this work, Cushing classified meningioma into nine types and set the stage for their surgical management. In particular, his work with W.T. Bovie and the development of diathermy was important advance in the surgical management of meningiomas. Cushing’s monograph stands as one of

the most important works on meningiomas in the twentieth century not only in terms of its classification and large series, but also as it brought to light the recurrent potential of these seemingly “benign” lesions. In fact, in their 1938 manuscript, Cushing and Eisenhardt recognized a subtype of meningiomas that they identified as malignant meningiomas (al-Rodhan and Laws 1990), which exhibited a far greater tendency to recur than benign meningiomas.

Donald Simpson would later explore the issue of recurrence in a famous series published in 1957, when he described five grades of resection with their relative rates of recurrence (al-Rodhan and Laws 1990). This landmark study demonstrated two important findings –first that despite maximal resection (Simpson Grade I), the recurrence rate of meningioma was 9% in his series and second, that histological factors can play an important role in recurrence, such as the poorer prognosis of undifferentiated vascular, reticulin-forming meningiomas (al-Rodhan and Laws 1990).

Since the 1957 report of Simpson, a number of histological classifications have been studied and have supported the notion that malignant meningiomas could be further divided into low and high malignancy subtypes (Smith et al. 2007). This distinction was first officially recognized by the World Health Organization (WHO) in 1993 when they defined atypical meningioma as a class in between benign and anaplastic meningiomas (anaplastic meningiomas are also referred to as malignant meningiomas) (Modha and Gutin 2005). In 2000, the WHO broadened the definition of atypical meningioma by adding necrosis, causing the percentage of meningiomas that were classified as atypical to increase from 18% to 23% using these criteria ($p=0.04$) (Smith et al. 2007) and meaning that atypical meningioma now includes either of the following histologic features: (1) mitotically active (four or more mitoses per high power field); or (2) three or more of the following: (i) loss of lobular architecture (sheeting), (ii) prominent nucleoli, (iii) increased cellularity, (iv) small cells with high nuclear to cytoplasmic ratios, and (v) foci of spontaneous necrosis. In 2007, the WHO

definition of atypical meningioma remained unchanged, although some pathologists had proposed broadening the definition by adding brain invasion as an alternative criterion (Commins et al. 2007). Atypical meningioma, chordoid meningioma, and clear cell meningioma currently comprise WHO grade II meningiomas, which have been shown in several reports to behave more aggressively than WHO grade I meningiomas and less aggressively than WHO grade III (also called malignant or anaplastic) meningiomas (Palma et al. 1997; Perry et al. 1997; Modha and Gutin 2005).

Given the recent classification and reclassification of atypical meningiomas, the exact incidence of atypical meningiomas is not known and likely has varying rates of diagnosis depending on institution (Smith et al. 2007). The available data suggest that 94% of meningiomas are benign, 5% atypical, and 1% malignant, which translates into an annual incidence rate for atypical and malignant meningiomas within the United States of approximately 185 atypical meningiomas and 40 malignant meningiomas (Hug et al. 2000), but a modern series using immunostaining to detect phosphorylated histone as a marker of mitoses, which allowed 17% of benign meningiomas to be reclassified as atypical (Ribalta et al. 2004), raised the interesting possibility that atypical meningiomas may be underrecognized, which, if true on a national level, could increase the number of atypical meningiomas diagnosed annually in the U.S. up to nearly 800 cases.

Early Standards of Treatment for Meningiomas

Early studies of meningiomas after the publication of Simpson's famous series but before atypical meningiomas were defined confirmed Simpson's findings that total resection of meningiomas reduces recurrence as compared to subtotal resection. These early studies also provided the first evidence for radiation as an adjuvant treatment for subtotally resected benign meningiomas and malignant meningiomas regardless of extent of resection, although these studies were

all retrospective in nature. In a study analyzing 140 patients with either benign ($n = 117$) or malignant ($n = 23$) meningiomas between 1967 and 1990 when atypical meningiomas had not been distinguished fully, Goldsmith et al. (1994) demonstrated the benefit of radiation therapy for meningiomas that underwent subtotal resection (Goldsmith et al. 1994). In particular, the 5-year progression-free survival (PFS) from meningioma for patients treated with external beam radiation therapy after 1980 was 94% (Goldsmith et al. 1994). In a longitudinal retrospective review of 262 patients with meningioma, between 1964 and 1992, Condra et al. (1997) confirmed that 15-year local control was improved with either total excision (76%) or subtotal excision followed by radiotherapy (87%) which were both significantly better than subtotal excision alone (30%) (Condra et al. 1997).

Early experience after the series by Simpson therefore confirmed that surgical resection for meningiomas could provide long-term control with low recurrence rates except when total resection was not achieved or when histological characteristics were of a malignant nature. In these situations, radiation therapy was shown to be effective in improving outcomes. The prevailing radiation at the time of these studies involved large fields of external beam radiation, which unfortunately had its own set of side effects such as visual changes, pituitary dysfunction, and subclinical cognitive changes. Because of these complications in addition to advances in radiation delivery, stereotactic radiation would remarkably change the landscape of meningioma management. The following section reviews the role of stereotactic radiation in meningiomas, both as an adjuvant to surgical resection and more recently, as an independent treatment for surgically challenging cases.

Stereotactic Radiation for Meningioma

Stereotaxis refers to the arrangement of coordinates in three-dimensional space and is a concept that has been applied to several neurosurgical

procedures, including the mapping of coordinates in space for intraoperative localization of lesions such as for tumor resection or deep electrode placement. Stereotactic radiotherapy refers to use of stereotaxis for the delivery of fractionated radiation therapy in many treatment sessions while stereotactic radiosurgery refers to the same type of radiation delivery in a single-session, i.e. “surgery with radiation” (Chin et al. 2003). There are two main types of devices used for stereotactic radiosurgery – (1) the Gamma Knife, which uses a 201-source ^{60}Co system focused onto a helmet with 4-, 8-, 14-, and 18-mm collimators, and (2) the modified linear accelerator (LINAC) systems such as the CyberKnife, which provides radiation by rotating the source around a patient’s head in arcs. Conformality, the matching of the volumetric radiation field to the target volume, is achieved by the multiple isocenters of radiation via collimation in the Gamma Knife system and by the collimation of the beam as it travels an arc in the modified LINAC systems (Chin et al. 2003).

Radiation treatment for meningiomas relies on the principles of radiobiology, most notably the concept of biologically effective dose (BED) of radiation, the computation of which allows one to compare different schemata for radiotherapy. The BED is given by the following equation: $nd(1 + d/\alpha/\beta)$, where n is the number of fractions, d is the dose per fraction, and α/β is a ratio that is specific to each tumor type and measures the survival of cells relative to radiation (Chin et al. 2003). The α/β ratio for meningiomas is relatively low which is similar to other benign lesions and even normal tissue, which equals a relatively high BED. A higher BED means that for any given single fraction of radiation, meningiomas require a higher equivalent dose of radiation to achieve a tumoricidal effect as compared to more malignant tumors. Normally, the biological difference between tumor and normal tissue as represented in the α/β ratio is exploited in fractionated schemes of radiation which reduces toxicity to normal tissue, but since meningiomas have a low α/β ratio similar to that of normal tissue, fractionation is of limited benefit (Chin et al. 2003). Although fractionated therapy has been tried in

meningiomas with good side effect profile, understanding the radiobiology of meningiomas has led single fraction stereotactic radiosurgery to become a more commonly used radiation modality in meningioma management (Chin et al. 2003).

At the University of Pittsburgh, which obtained one of the first North American Gamma Knife systems, Kondziolka et al. (1999) reviewed an early series of the first meningioma patients treated with stereotactic radiosurgery, 57% as postoperative adjuvant treatment and 43% as primary treatment for unresectable meningiomas (Kondziolka et al. 1999). The Pittsburgh series noted safety with excellent long-term control rates (Kondziolka et al. 1999). In this study, the majority of tumors (89%) that were treated with radiosurgery were skull base lesions, with the most ($n=30$) cavernous sinus meningiomas. In addition, 93% of patients treated with radiosurgery in this series required no further tumor surgery after 5–10 year follow-up. Also important is the fact that stereotactic radiosurgery in this series was being used not only for post-operative adjuvant treatment for subtotal resections, but also as a primary modality. In this series, malignant meningiomas were excluded and the maximum tumor volume that was treated was 3 cm (Kondziolka et al. 1999). Comparable levels of radiographic control have been reported using linear accelerator based stereotactic radiosurgery (Chin et al. 2003).

A critical development in the use of radiation came from multiple studies on tumors in locations where surgical approaches and resection were particularly challenging. For example, a significant number of patients with cavernous sinus meningiomas, petroclival meningiomas, parasagittal meningiomas next to critical vascular structures, and intraventricular meningiomas have been treated to good effect with various forms of radiation therapy including fractionated radiation, stereotactic radiation therapy, and stereotactic radiosurgery, typically with greater safety and comparable efficacy to surgical resection (Chin et al. 2003).

Based on these multiple studies in patients with inoperable meningiomas treated with some

form of radiation, a growing literature developed for stereotactic radiation or radiosurgery as a primary mode of treatment. In an important study of benign meningiomas treated with radiosurgery as the sole initial treatment at one institution, Pollock et al. (2003) demonstrated that radiosurgery provided equivalent tumor control to Simpson Grade I removal in patients with small to medium-sized meningiomas (<35 mm) (Pollock et al. 2003). Between 1990 and 1998, 198 patients met inclusion criteria for the study with 136 patients undergoing microsurgery and 62 undergoing radiosurgery. Of note, the group that underwent radiosurgery had a significantly higher number of skull base meningiomas. There were fewer complications in the radiosurgery group and fewer subsequent tumor treatments in the radiosurgery group, although the mean follow-up period was only 64 months. Despite its limitations, this was an important study demonstrating the benefits of radiosurgery as upfront treatment in meningiomas and confirming growing evidence of the benefit of radiosurgery as a primary means of treatment for relatively small meningiomas (Pollock et al. 2003).

One must temper such early promising results from radiation as the primary treatment modality of meningiomas by appreciating that the retrospective nature of these studies limits the data to patients with small benign tumors that are often unable to be resected due to patient preference, comorbidity, or unacceptable surgical risk. In one of the largest retrospective reviews on the treatment of meningiomas and stereotactic radiosurgery, Kondziolka et al. (2008) reviewed an 18-year interval where 972 patients with a total of 1,045 intracranial meningiomas were treated at a single institution (Kondziolka et al. 2008). The inclusion criteria in this study are an important indicator of which patients generally receive stereotactic radiosurgery. Patients were selected for radiosurgery if they had recurrent or residual tumor, if they chose treatment over observation for asymptomatic tumors, or if they chose radiosurgery over other treatment options for small symptomatic lesions. In addition, inclusion criteria

included patients who were unable to have surgery either due to unacceptable surgical risk or other medical contraindications. Exclusion criteria were larger tumors (>3.5 cm), asymptomatic tumors in elderly patients, tumors involving compression of optic nerves or chiasm, and tumors with atypical imaging features without histological diagnosis. Fifty-one percent of patients in this study did not have prior resection and out of those that did, a majority (84%) had subtotal resections. Of note, 87 tumors were considered higher grade (WHO grade II or III). Given the relatively recent timeframe of the study, long-term follow-up was limited, but the actuarial 15-year control rate was 87% for grade I meningiomas. This landmark series for stereotactic radiosurgery in patients with meningiomas as well as the series of Pollock et al. (2003) demonstrated the feasibility and effectiveness of stereotactic radiosurgery not only as an adjuvant, but also as primary treatment for benign meningiomas (Pollock et al. 2003; Kondziolka et al. 2008). These studies have defined a role for radiosurgery in the management of small to medium-sized benign meningiomas. While it is generally accepted that malignant meningiomas require radical surgical resection in addition to radiation and often chemotherapy given their rapid growth characteristics (Modha and Gutin 2005), the role of radiation in the management of atypical meningiomas, particularly after gross total resection, remains undefined, as outlined below.

Atypical Meningiomas – Impact of Extent of Resection on Recurrence

Unfortunately, atypical meningiomas can only be studied from the time the diagnosis was created in 1993 and the broadening of the diagnostic criteria in 2000 creates another confounding variable that complicates long-term studies. In fact, there is not only significant variation among institutions with regard to the diagnosis of atypical meningiomas (Simon et al. 2006; Smith et al. 2007), but within institutions, the proportion of

meningiomas diagnosed as atypical has increased over time (Pearson et al. 2008).

While a retrospective review showed that resections of atypical meningiomas are less often classified by surgeons as gross total when compared to benign meningiomas (Smith et al. 2007), a majority of atypical meningiomas are still able to undergo gross total resection as their initial treatment (Ko et al. 2007). Early studies confirmed that, as is the case with benign meningioma, gross total resection of atypical meningioma is superior to subtotal resection in improving recurrence rates. Goyal et al. (2000) studied 22 patients over a median follow-up period of 5.5 years and determined a significant difference when comparing 10-year local control rates in patients who underwent gross total resection vs. subtotal resection or unknown extent of resection (87% vs. 17%) (Goyal et al. 2000). The long-term recurrence rate of gross totally resected atypical meningiomas was found to be higher than that reported by Goyal et al. (2000) in a recent large series by Aghi et al. (2009), which reviewed the recurrence rates of atypical meningiomas in 108 patients treated at a single institution with confirmed Simpson Grade I gross total resection (Aghi et al. 2009). Of these 108 tumors, 30 (28%) recurred after an average follow-up of 39 months with 46% of patients followed to either recurrence or 5-years post-resection. The actuarial recurrence rate after 10 years was 48%, which is significantly higher than the recurrence rates noted for Simpson Grade I resections of benign meningiomas. The 30 patients that recurred after gross total resection underwent additional therapies including 22 patients with repeat craniotomy and all 30 patients receiving radiation (14 received fractionated stereotactic radiotherapy and 16 received stereotactic radiosurgery) (Aghi et al. 2009). Patients with recurrence had a 69% survival rate at 10 years. This study is one of the largest in the current literature after the 1993 establishment of a common pathological definition of atypical meningiomas to document the long-term recurrence rates of gross totally resected atypical meningiomas.

Atypical Meningioma – Impact of Postoperative Radiation on Recurrence

While radiation has been used as the primary treatment of unresectable meningiomas as described earlier in this chapter, the lack of pathologic analysis of tissue from these tumors means one has to presume them to be likely to be benign meningiomas based on the greater frequency of benign meningiomas than atypical or malignant meningiomas, meaning the response to radiation of atypical meningiomas can only be studied with certainty in the postoperative adjuvant or recurrent setting. Only 8 out of the 108 patients with gross totally resected atypical meningiomas from the study by Aghi et al. (2009) received any postoperative radiation, which suggests that, most of the time, surgeons were managing gross totally resected atypical meningiomas in the same way they manage gross totally resected benign meningiomas, without the use of post-operative radiation (Aghi et al. 2009). On the other hand, because of studies proving their tendency to grow within 2–5 years during serial imaging and studies showing the safety and efficacy of fractionated stereotactic radiation therapy or unfractionated stereotactic radiosurgery for subtotally resected or recurrent atypical meningiomas (Hug et al. 2000; Stafford et al. 2001; Milker-Zabel et al. 2005) (Table 15.1), subtotally resected atypical meningiomas receive postoperative radiation far more frequently than subtotally resected benign meningiomas, in whom growth during serial postoperative imaging is generally felt to be the indication for radiation. For example, one study surveying the practices of German neurosurgical centers, approximately 18% of atypical meningiomas that had undergone complete or near complete resection (Simpson I–III) received some form of postoperative radiotherapy, whereas 74% of atypical meningiomas received radiation after incomplete resection (Simpson IV–V) (Simon et al. 2006).

While there was initially some concern that nonfractionated radiosurgery for gross totally

Table 15.1 Series in the literature of atypical meningiomas treated with radiation

Total number of patients	Mean or median follow-up time	Mode of radiation	STR+no XRT (fraction of cases that progressed)	STR+XRT	GTR+no XRT (fraction of cases that progressed)	GTR+XRT	XRT for recurrent disease	Actual overall recurrence (actuarial recurrence rate)	Reference
46	4 years	Fractionated external beam radiotherapy	14 (8/14 = 57%)	5 (2/5 = 40%)	21 (5/21 = 23%)	4 (1/4 = 25%)	2 (1/2 = 50%)	40% at 5 years	(Condra et al. 1997)
26	31 months	LINAC-based radiosurgery	0	N/S	0	N/S	0	4/26	(Hakim et al. 1998)
15	5 years	Fractionated photon or proton beam	0	6	0	0	9	8/15 (62% at 5 years)	(Hug et al. 2000)
13	40 months	Gamma knife radiosurgery	0	N/S	0	0	N/S	32% 5 year actuarial	(Stafford et al. 2001)
18	2.3 years	Gamma knife stereotactic radiosurgery	0	N/S	0	0	N/S	17% 5 year actuarial	(Harris et al. 2003)
26	5.7 years	Fractionated stereotactic radiotherapy	0	N/S	0	0	N/S	4/26 (11% 5 year actuarial)	(Milker-Zabel et al. 2005)
21 tumors in 15 patients	35 months	Gamma knife radiosurgery	0	N/S	0	0	N/S	6/15 patients	(Huffmann et al. 2005)
25 tumors in 12 patients	44 months	LINAC –based radiosurgery	0	0	0	0	25 tumors in 12 patients	71% 5 year (below 20 Gy), 39% 5 year (20 Gy)	(Kano et al. 2007)
108	39 months	Fractionated stereotactic radiotherapy	0	0	100 (30/100 = 30%)	8 (0/8 = 0%)	0	41% 5 year actuarial overall; 30/100 (nonirradiated), 0/8 (irradiated)	(Aghi et al. 2009)

Only reports in which specific outcomes for atypical meningiomas, as defined per WHO criteria, were reported are cited

resected atypical meningiomas would be challenging and would carry potential morbidity due to the lack of an obvious target, increasing recognition of the tendency of atypical meningiomas to recur even after gross total resection has led to three reports (Table 15.1) describing the use of postoperative radiation after gross total resection of atypical meningiomas. In a large study of meningiomas, atypical meningiomas were analyzed separately and amongst all groups had a reduced 15-year local control rate of 54%, as compared to 71% for benign lesions (Condra et al. 1997). Although their sample size was small, Condra et al. (1997) demonstrated improved outcomes for patients with atypical meningiomas after total excision and adjuvant radiotherapy (Condra et al. 1997). Hakim et al. (1998) described outcomes for meningiomas treated with stereotactic radiation therapy using a LINAC-based system with 4-year survival rates for benign, atypical, and malignant meningiomas as 91%, 83.3%, and 21.5% respectively (Hakim et al. 1998). And Aghi et al. (2009) performed a retrospective review in which there was a 41% 5 year actuarial recurrence rate of gross totally resected atypical meningiomas that were not irradiated compared to no recurrences at 5 years in gross totally resected atypical meningiomas that were irradiated, but differences in sample size (100 nonirradiated tumors compared to 8 irradiated tumors) rendered the difference just shy of statistical significance (Aghi et al. 2009). The intriguing findings of these reports underscore the need for a randomized trial studying the efficacy of irradiation after gross total resection of an atypical meningioma, given the existing clinical equipoise on the subject.

Proton beam radiosurgery offers more focused dosimetry than photon-based treatments, which could potentially provide the higher equivalent dose needed to kill meningiomas. Boskos et al. (2009) studied 24 patients with non-benign meningiomas (19 with atypical, 5 with malignant histology) who received a mean total irradiation dose of 65 CGE (cobalt gray equivalents) using fractionated proton beam radiotherapy with a mean follow-up of 32 months (Boskos et al. 2009). In this series, there was no difference

between those who underwent subtotal resection ($n=18$) vs. total resection ($n=6$) in terms of overall survival or local control, but patients who underwent >60 CGE of irradiation did have a significant improvement in overall and cause-specific survival (Boskos et al. 2009). Similar results were published earlier by Hug et al. (2000) who demonstrated that >60 CGE doses using fractionated proton radiotherapy improved outcomes, but they were also able to suggest in their small sample size that proton radiation was an improvement over traditional photon-based radiation (Hug et al. 2000). Based on these early studies of proton therapy, further work on proton therapy alone for atypical and malignant meningiomas would be useful. High-dose radiation provided some benefit in another study as well (Kano et al. 2007).

As radiation therapy for meningiomas has become more prevalent and long-term follow-up has been performed, there is also a need to recognize complications that may arise, including (1) symptomatic peritumoral edema after stereotactic radiation therapy or radiosurgery, particularly after treating parasagittal meningiomas or with stereotactic radiosurgery and higher volumes/doses in another study; (2) cranial nerve deficits (specifically optic neuropathy which can happen with as little as 8 Gy to the optic nerve); (3) vascular injuries (particularly for cavernous sinus meningiomas and internal carotid artery radiation vasculopathy); and (4) cyst formation (Chin et al. 2003). Attempts to treat aggressive meningiomas with implantable radioactive seeds proved to have complications beyond those seen with stereotactic radiation or radiosurgery. Ware et al. (2004) examined the use of brachytherapy using radioactive seeds implanted at the time of surgery in patients with recurrent atypical and malignant meningiomas and found a significantly high complication rate, particularly with wound healing (Ware et al. 2004). Despite a small, but growing literature on understanding and preventing complications from stereotactic radiation therapy and stereotactic radiosurgery, for the majority of patients, this is still an extremely safe method of treatment with low complication rates. With better targeting and conformality including

expansion of proton beam therapy, complication rates should only improve.

Conclusions and Future Directions

Since their introduction as a diagnostic entity in 1993, considerable progress has already been made in our understanding of atypical meningiomas. Prognostic factors provide important clues early in the diagnosis and management of patients to help risk-stratify and to guide management. Both bone involvement and gain of 1q predict poor outcomes in patients with atypical meningiomas (Gabeau-Lacet et al. 2009a, b). Drugs such as small molecule receptor tyrosine kinase inhibitors are being studied for the treatment of atypical meningiomas as radiation sensitizers or as chemotherapy agents with tumoricidal effects independent of radiation (Modha and Gutin 2005; Milker-Zabel et al. 2008).

Despite this progress, considerable work is needed before neurosurgeons can offer their patients with atypical meningiomas the kind of evidence that is available to guide decision making for patients with benign meningiomas. To start with, either greater histologic diagnostic consensus amongst neuropathologists or definitive distinguishing molecular criteria will be needed in order to ensure that atypical meningiomas are accurately diagnosed. The most variable diagnostic criteria for atypical meningiomas is number of mitoses per high powered field, which is prone to undersampling and subjective sampling, as well as technical difficulties distinguishing mitotic figures from apoptotic cells or pyknotic nuclei (Commins et al. 2007), difficulties that can be overcome using immunohistochemical staining specific for phosphorylated histone H3, which, in one study, increased the grade of 17% of benign meningiomas to atypical (Ribalta et al. 2004). If such techniques can be standardized as diagnostic modalities, their uniform incorporation could eliminate much of the existing diagnostic variability.

In addition to this diagnostic variability, the most notable area where insight is needed into how best to manage atypical meningiomas is the

optimal management of patients after gross total resection of atypical meningiomas and risk factors predictive of recurrence. While several retrospective reports (Hug et al. 2000) have described the use of postoperative adjuvant radiation after subtotally resected atypical meningiomas have made radiation fairly standard treatment for subtotally resected atypical meningiomas, the benefit of radiation after gross total resection of an atypical meningioma has not been well defined, leading to nonuniform practice patterns. A German study reported that only 9 of 56 centers recommended radiation after gross total resection of an atypical meningioma (Simon et al. 2006) and, in a single institution study of gross totally resected meningiomas, 8 of 108 received postoperative radiation despite lacking any distinguishing factors from tumors that were not irradiated.

Clinical trials investigating radiation for gross totally resected atypical meningiomas will need to be multicenter trials due to the relatively uncommon nature of atypical meningiomas and will require centralized neuropathology review until the diagnostic variability described above is improved. Examples of such trials include Radiation Therapy Oncology Group (RTOG) trial 0539, in development at the time of this writing, and European Organization for Research and Treatment of Cancer (EORTC) randomized study 26013/26014, actively enrolling at the time of this writing, two trials that will be empowered to answer the question of whether postoperative radiation lowers the recurrence rate of atypical meningiomas after gross total resection. It is hoped that results of these trials will help guide neurosurgeons in the management of patients with atypical meningiomas.

References

- Aghi M, Carter BS, Cosgrove GR, Ojemann RG, Amin-Hanjani S, Martuza RL, Curry WT, Barker FG (2009) Long-term recurrence rates of atypical meningiomas after gross total resection with or without postoperative adjuvant radiation. *Neurosurgery* 64:56–60; discussion 60
- al-Rodhan NR, Laws ER Jr (1990) Meningioma: a historical study of the tumor and its surgical management. *Neurosurgery* 26:832–846; discussion 846–837

- Boskos C, Feuvret L, Noel G, Habrand JL, Pommier P, Alapetite C, Mammari H, Ferrand R, Boissarie G, Mazon JJ (2009) Combined proton and photon conformal radiotherapy for intracranial atypical and malignant meningioma. *Int J Radiat Oncol Biol Phys* 75:399–406
- Chin LS, Szerlip NJ, Regine WF (2003) Stereotactic radiosurgery for meningiomas. *Neurosurg Focus* 14:e6
- Commings DL, Atkinson RD, Burnett ME (2007) Review of meningioma histopathology. *Neurosurg Focus* 23:E3
- Condra KS, Buatti JM, Mendenhall WM, Friedman WA, Marcus RB Jr, Rhoton AL (1997) Benign meningiomas: primary treatment selection affects survival. *Int J Radiat Oncol Biol Phys* 39:427–436
- Gabeau-Lacet D, Aghi M, Betensky RA, Barker FG, Loeffler JS, Louis DN (2009a) Bone involvement predicts poor outcome in atypical meningioma. *J Neurosurg* 111:464–471
- Gabeau-Lacet D, Engler D, Gupta S, Scangas GA, Betensky RA, Barker FG, Loeffler JS, Louis DN, Mohapatra G (2009b) Genomic profiling of atypical meningiomas associates gain of 1q with poor clinical outcome. *J Neuropathol Exp Neurol* 68:1155–1165
- Goldsmith BJ, Wara WM, Wilson CB, Larson DA (1994) Postoperative irradiation for subtotally resected meningiomas. A retrospective analysis of 140 patients treated from 1967 to 1990. *J Neurosurg* 80:195–201
- Goyal LK, Suh JH, Mohan DS, Prayson RA, Lee J, Barnett GH (2000) Local control and overall survival in atypical meningioma: a retrospective study. *Int J Radiat Oncol Biol Phys* 46:57–61
- Hakim R, Alexander E, Loeffler JS, Shrieve DC, Wen P, Fallon MP, Stieg PE, Black PM (1998) Results of linear accelerator-based radiosurgery for intracranial meningiomas. *Neurosurgery* 42:446–453; discussion 453–444
- Harris AE, Lee JY, Omalu B, Flickinger JC, Kondziolka D, Lunsford LD (2003) The effect of radiosurgery during management of aggressive meningiomas. *Surg Neurol* 60:298–305; discussion 305
- Huffmann BC, Reinacher PC, Gilsbach JM (2005) Gamma knife surgery for atypical meningiomas. *J Neurosurg* 102(Suppl):283–286
- Hug EB, Devries A, Thornton AF, Munzenrider JE, Pardo FS, Hedley-Whyte ET, Bussiere MR, Ojemann R (2000) Management of atypical and malignant meningiomas: role of high-dose, 3D-conformal radiation therapy. *J Neurooncol* 48:151–160
- Kano H, Takahashi JA, Katsuki T, Araki N, Oya N, Hiraoka M, Hashimoto N (2007) Stereotactic radiosurgery for atypical and anaplastic meningiomas. *J Neurooncol* 84:41–47
- Ko KW, Nam DH, Kong DS, Lee JJ, Park K, Kim JH (2007) Relationship between malignant subtypes of meningioma and clinical outcome. *J Clin Neurosci* 14:747–753
- Kondziolka D, Levy EI, Niranjan A, Flickinger JC, Lunsford LD (1999) Long-term outcomes after meningioma radiosurgery: physician and patient perspectives. *J Neurosurg* 91:44–50
- Kondziolka D, Mathieu D, Lunsford LD, Martin JJ, Madhok R, Niranjan A, Flickinger JC (2008) Radiosurgery as definitive management of intracranial meningiomas. *Neurosurgery* 62:53–58; discussion 58–60
- Milker-Zabel S, Zabel A, Schulz-Ertner D, Schlegel W, Wannenmacher M, Debus J (2005) Fractionated stereotactic radiotherapy in patients with benign or atypical intracranial meningioma: long-term experience and prognostic factors. *Int J Radiat Oncol Biol Phys* 61:809–816
- Milker-Zabel S, Zabel-du Bois A, Ranai G, Trinh T, Unterberg A, Debus J, Lipson KE, Abdollahi A, Huber PE (2008) SU11657 enhances radiosensitivity of human meningioma cells. *Int J Radiat Oncol Biol Phys* 70:1213–1218
- Modha A, Gutin PH (2005) Diagnosis and treatment of atypical and anaplastic meningiomas: a review. *Neurosurgery* 57:538–550; discussion 538–550
- Palma L, Celli P, Franco C, Cervoni L, Cantore G (1997) Long-term prognosis for atypical and malignant meningiomas: a study of 71 surgical cases. *J Neurosurg* 86:793–800
- Pearson BE, Markert JM, Fisher WS, Guthrie BL, Fiveash JB, Palmer CA, Riley K (2008) Hitting a moving target: evolution of a treatment paradigm for atypical meningiomas amid changing diagnostic criteria. *Neurosurg Focus* 24:E3
- Perry A, Stafford SL, Scheithauer BW, Suman VJ, Lohse CM (1997) Meningioma grading: an analysis of histologic parameters. *Am J Surg Pathol* 21:1455–1465
- Pollock BE, Stafford SL, Utter A, Giannini C, Schreiner SA (2003) Stereotactic radiosurgery provides equivalent tumor control to Simpson Grade I resection for patients with small- to medium-size meningiomas. *Int J Radiat Oncol Biol Phys* 55:1000–1005
- Ribalta T, McCutcheon IE, Aldape KD, Bruner JM, Fuller GN (2004) The mitosis-specific antibody anti-phosphohistone-H3 (PHH3) facilitates rapid reliable grading of meningiomas according to WHO 2000 criteria. *Am J Surg Pathol* 28:1532–1536
- Simon M, Bostrom J, Koch P, Schramm J (2006) Interinstitutional variance of postoperative radiotherapy and follow up for meningiomas in Germany: impact of changes of the WHO classification. *J Neurol Neurosurg Psychiatry* 77:767–773
- Smith SJ, Boddu S, MacArthur DC (2007) Atypical meningiomas: WHO moved the goalposts? *Br J Neurosurg* 21:588–592
- Stafford SL, Pollock BE, Foote RL, Link MJ, Gorman DA, Schomberg PJ, Leavitt JA (2001) Meningioma radiosurgery: tumor control, outcomes, and complications among 190 consecutive patients. *Neurosurgery* 49:1029–1037; discussion 1037–1028
- Ware ML, Larson DA, Sneed PK, Wara WW, McDermott MW (2004) Surgical resection and permanent brachytherapy for recurrent atypical and malignant meningioma. *Neurosurgery* 54:55–63; discussion 63–54

Gamma Knife Radiosurgery for Benign Meningioma: Significance and Therapeutic Potential

16

Klaus Haselsberger and Frank Unger

Abstract

This review elucidates the value and the therapeutic spectrum of gamma knife radiosurgery (GKRS) for benign meningiomas based on results obtained at our department. Between April 1992 and May 2008 38.8% (394/1,016 patients) of patients with benign meningioma underwent primary single session GKRS. Staged GKRS was performed in 20 patients with large critically located meningiomas, 14 of whom having undergone surgery at least once. The patients' age ranged between 26 and 73 years (median 60.5). For single session GKRS median tumor volume was 6.5 cm³, median prescription dose at the tumor margin (covered by a median isodose of 45%) was 12 Gy. Median tumor volume treated by staged GKRS measured 33.3 cm³ and median treatment volume 19.0 cm³. For 41 treatments, the median prescription dose at the tumor margin was 12 Gy. Median follow-up for staged GKRS was 9.4 years.

For single treatment GKRS, the actuarial progression-free survival rate was 98.5% at 5 years and 97.2% at 10 years at a (mostly temporary) complication rate of 4.5%. In patients undergoing staged GKRS, primary tumor control was achieved in 90%. Two patients (10%) with tumor progression outlying the planning target volumes experienced tumor control after an additional radiosurgical procedure. Clinically, 45% of all patients improved within the time of follow-up, 55% remained unchanged. Radiosurgery is a neurosurgical domain that offers results comparing favorably to radical surgery for meningiomas of small to medium-size in critical locations. For larger critically located meningiomas, a combination of judicious surgery and radiosurgery yields the best functional outcome. Staged GKRS provides a substantial therapeutic improvement for large otherwise untreatable meningiomas.

Introduction

The choice of the optimum treatment for each patient harboring an intracranial meningioma constitutes a crucial issue for neurosurgeons since

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these tumors represent a common entity that accounts for ~20% of all intracranial neoplasms (Kleihues et al. 1993), incidence increasing with age (Christensen et al. 2003; Claus et al. 2005). Histologically, the vast majority are of benign character (WHO grade 1). Atypical as well as non-benign tumors referring to both histologic and cytogenetic features are found to be located more frequently at the brain surface than at the cranial base (Ketter et al. 2008). However, among so-called benign meningiomas, recent cytogenetic investigations have identified two distinct patterns of tumor growth and recurrence, respectively (Pfisterer et al. 2007) – a finding that matches with clinical experience. Due to the ready availability of specialized neuroimaging (CT, MRI) in developed countries, many tumors are discovered incidentally, specifically in elderly patients. Provided that such persons present asymptomatic, observation, including radiological and clinical controls, can be recommended.

Among treatment modalities for benign meningiomas, surgery is still mainstream since most tumors are accessible without significant surgery-related risk. Even patient age exceeding 80 years need not be an exclusive criterion for surgery, yet aged patients have to be considered on an individual basis (Sacko et al. 2007). Because of vessel encasement and/or close relationship to cranial nerves or the brainstem, there are, however, some special tumor locations (petroclival, cavernous sinus, falcatentorial) where radical resection is often extremely difficult and associated with an unacceptably high postoperative morbidity and even mortality. However, for tumor recurrence, the completeness of surgical resection is the most important parameter (followed by edema grade and brain invasion (Ketter et al. 2008; Mantle et al. 1999)). Therefore, subtotal resection without any further therapy (radiosurgery or radiotherapy) has to be regarded as insufficient (Condra et al. 1997).

Apart from the typical indication for gamma knife radiosurgery (GKRS) – small tumors (less than 3 cm in diameter) located at the skull base – there has been a shift in the treatment paradigms from radical surgery towards a combination of judicious (subtotal) surgery followed by stereotactic

radiosurgery (SRS) – “Simpson grade 4 gamma” (Mathiesen et al. 2007). For persons with larger tumors suffering from significant co-morbidity, who, consequently, are not amenable to surgery, a staged radiosurgical procedure as primary treatment constitutes a therapeutic option that has been thoroughly evaluated in our study referring to staged GKRS for large critically located meningiomas (Haselsberger et al. 2009).

This topic review elucidates the value and the therapeutic spectrum of GKRS for the treatment of benign meningioma focusing especially on large otherwise untreatable meningiomas (either recurrent/residual after surgery or primarily admitted for GKRS) undergoing a staged radiosurgical procedure.

Patients and Methods

Patient Population

Within 16 years, from April 1992 to May 2008, a total of 1,016 patients with benign intracranial meningiomas were treated at our department. During this period, a portion of 38.8% (394 patients) underwent primary single-session GKRS. Among patients harboring large tumors, 70 underwent (subtotal) surgery followed by GKRS, 28 had surgery twice and 41 had surgery twice and subsequent GKRS. In addition, 110 patients who had undergone surgery elsewhere were transferred to our department from other neurosurgical centers for GKRS of residual or recurrent tumors.

A separate group of 20 patients with large critically located meningiomas was considered for staged GKRS. A staged procedure was defined as at least two consecutive radiosurgical treatments at time intervals between 1 and 12 months. Thirteen patients were referred from other neurosurgical centers specifically for GKRS and seven were primarily admitted to our department. The majority (14/20 patients, 70%) underwent staged GKRS following surgery because of recurrent or residual tumors. Among the remaining six patients (all of whom suffering from significant co-morbidity) one had undergone tumor embolisation twice

(44 and 25 months before GKRS) which, however, had proved ineffective. Tumors with a major involvement of the cavernous sinus comprised 4/20 patients, an equal number (four patients each) was situated either at a petroclival or falcontorial location whereas huge basal meningiomas representing mainly a combination of cavernous sinus and petroclival meningiomas were diagnosed in six other cases. The only tumor located at the brain convexity was a parasagittal meningioma. Because of the extremely irregular tumor shape, this patient underwent three staged radiosurgical procedures at intervals of 4 and 5 months, respectively. Patients presented with a variety of preexisting neurological deficits, partially deteriorated after surgery. Data referring to 200 patients who were treated by GKRS at our department between April 1992 and June 1999 have been evaluated and published earlier (Kreil et al. 2005). Among those, GKRS as the only treatment was performed in 101 cases.

Radiosurgical Technique

All stereotactic procedures were performed using the Leksell gamma knife (model B until 2003, thereafter model C; Elekta Instruments AB, Stockholm, Sweden). An image-compatible stereotactic frame was applied under local anesthesia. Before 1996, CT was used for treatment planning, and thereafter MRI, using axial and coronal T1 sequences with or without contrast enhancement.

Among the series published by Kreil et al. (2005) median tumor volume was 6.5 cm^3 , median prescription dose at the tumor margin 12 Gy (range 7–25 Gy) with a median marginal dose of 45% (range 20–80%). Within this series, eight patients underwent a staged treatment (two radiosurgical procedures) because of large tumor volumes. These cases have been re-evaluated recently and included into the series dealing with staged GKRS for large critically located benign meningiomas (Haselsberger et al. 2009).

Among 20 patients undergoing staged GKRS, the diagnosis of meningioma (WHO grade 1) was confirmed by histopathological specification in 14 cases, in six other patients diagnosis was

assessed radiologically. The median total tumor volume was 33.8 cm^3 (range 13.6–79.8 cm^3). Treatment isodose and margin dose were determined corresponding to tumor size, location and proximity to critical structures. Dose selection resulted by assessing tumor volume and its relationship to the integrated logistic equation for a 3% risk of permanent brain injury modified to Flickinger (1989). Median treatment volume was 19.0 cm^3 (range 5.4–42.9 cm^3). Out of 41 treatments, the prescription dose at the tumor margin was 12 Gy for 33 treatments, 10 Gy for one treatment, 14 Gy for four treatments, 15 Gy for one treatment, and 25 Gy for two treatments (median 12 Gy to a marginal isodose of 45%). Multiple irradiation isocenters were required to cover irregular tumor shapes (median 9; range 3–22). In all cases, one or more of the 201 sources were blocked to reduce radiation exposure to the eye lens. Collimator plugging patterns were used to shape the radiation volume and to protect critical areas (brainstem, optic nerves, chiasm and optic tract). The median time interval between the staged procedures was 6 months (range 1–12 months). All patients were discharged the day after GKRS and routinely followed at intervals of 6 months, including clinical and radiological examinations. Tumor control was defined as a stable or decreased tumor volume assessed by volumetric calculations based on neuroimaging (MRI).

Follow-up data refer to the last GKRS of the staged radiosurgical treatment protocol. Statistical analysis was performed using SPSS software, V. 14.0 (SPSS Inc, Chicago, Illinois, USA). Detailed patient characteristics and individual treatment strategies have been published elsewhere (Haselsberger et al. 2009).

Results

Single Treatment Gamma Knife Radiosurgery

In the report referring to patients treated at our department between 1992 and 1999 (Kreil et al. 2005), the actuarial progression free survival rate was 98.5% at 5 years and 97.2% at 10 years.

Transient radiation induced edema was observed in two patients (1%). In 41.5% (83 patients), neurological condition improved, 54% (104 patients) remained unaltered, whereas deterioration occurred in 4.5% (9 patients), mostly temporarily (3.5%, 7 patients). Five patients (2.5%) underwent repeated microsurgical resection following GKRS.

Staged Gamma Knife Radiosurgery

Irrespective of previous treatment and tumor location, regression of tumor volume was observed in five patients (5/20, 25%) whereas 13 patients (13/20, 65%) presented with a stable tumor size. Two patients (2/20, 10%) experienced tumor regrowth after 84 and 74 months, respectively. In both patients, tumor regrowths originated from areas outlying the radiosurgical target volumes. Following an additional GKRS, tumor size in one patient has decreased in the further follow-up (37 months, April 2010), and has remained stable in the second patient, (122 months, April 2010). Currently, median follow-up is 108 months (range 18–205 months, April 2010).

Neurological evaluation showed improvements in nine patients (9/20, 45%), in terms of cranial nerve deficits, seizures, hemiparesis, etc. In nine other cases (9/20, 45%), the pre-existing neurological deficit remained stable, while two more patients (2/20, 10%), asymptomatic prior to GKRS, maintained their condition. Four patients died during follow-up, all unrelated to disease and treatment. Detailed information on neurologic outcome and radiosurgical effects on tumor volume is given elsewhere (Haselsberger et al. 2009).

Immediate adverse effects included transient nausea or vomiting in two patients and headache in five patients following frame application and removal. At the first 6 month MRI control, two patients (2/20, 10%) developed a high T2 signal intensity in the surrounding brain, suggestive of peritumoral edema which resolved completely after oral administration of steroids without evidence of any new neurological deficit. All patients were able to recover their pre-radiosurgical

functional level within one week. Of note, there were no side effects to be detected in the sole patient who underwent staged GKRS for a non-basal (parasagittal) meningioma.

Staged Gamma Knife Radiosurgery: Exemplary Cases

Case 1

A 53-year-old woman presented with reduced vision, multiple cranial nerve deficits (cranial nerves III, V, VI, VII) of the left side and (mild) hemiparesis of the right side. Within 2 years before staged GKRS she had undergone surgery three times (twice elsewhere, the third intervention taking place at our department) at time intervals of 5 and 18 months, respectively. Neurological symptoms had deteriorated after the second surgical treatment. Pre-GKRS MRI demonstrated a huge residual/recurrent skull base meningioma involving both the cavernous sinus and the petroclival region with encasement of the left carotid and basilar artery and compression of both the brainstem and the optic chiasm. Because of the unsatisfactory outcome after repeat surgery (both in terms of neurological condition and grade of resection) she was allocated to staged GKRS as an alternative treatment option. The first GKRS focused on the main tumor mass compressing the brainstem (measuring 36.7 cm³) delivering 12 Gy to the 30% marginal isodose. The second treatment 5 months thereafter covered a tumor volume of 7.5 cm³ with 15 Gy at the 50% marginal isodose. Three and 6 isocenters were used for the first and the second GKRS, respectively.

Although there was no evidence for a significant decrease in tumor volume on follow-up MRI examinations, the patient regained a normal vision with both cranial nerve deficits and hemiparesis having resolved within 2 years following the second radiosurgical treatment. 84 months after staged GKRS, a small tumor regrowth (1.9 cm³) originating from outside the previous target volumes was detected with MRI. The patient underwent an additional GKRS delivering 13 Gy to a 45% marginal isodose (9 isocenters). Tumor volume and neurological condition

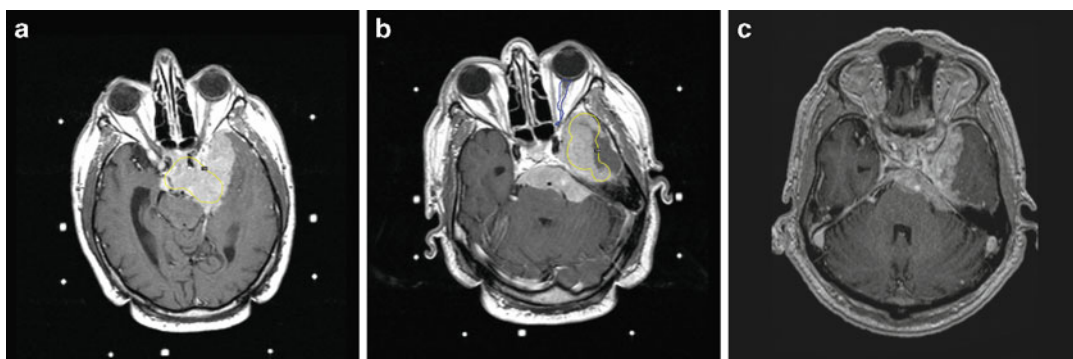


Fig. 16.1 Exemplary case 2 representing a large skull base meningioma (sphenoid wing, cavernous sinus, petroclival) – Axial contrast enhanced T1 weighted MRI used for treatment planning and follow-up. Yellow lines

indicate the 50% isodose. (a) First radiosurgical treatment, (b) Second GKRS 4 months later, (c) MRI control performed 40 months after staged GKRS documenting tumor regression

have remained stable ever since (205 months follow-up after staged GKRS and 122 months following additional GKRS; April 2010).

Case 2

A 61-year-old man presented with reduced vision, cephalgia and complex partial seizures. MRI demonstrated a large skull base meningioma (sphenoid wing, left middle fossa, clivus) with encasement of the left carotid and basilar artery. Because of a critical cardiac comorbidity, the risk of surgery was rated as high. As an alternative treatment option, the patient was allocated to GKRS. Despite tumor embolisation 44 and 25 months earlier, MRI controls revealed further tumor progression, neurological symptoms having aggravated. The first GKRS focused on the tumor mass compressing the optic chiasm and brainstem with 12 Gy to the 45% isodose. Four months later, a second GKRS was directed towards the temporal part of the tumor, delivering 12 Gy to the 40% isodose. Follow-up MRI after 51.5 months showed a considerably decreased tumor size. Neurologically, the patient was seizure-free, his vision having improved (Fig. 16.1).

Discussion

Even if histologically benign, meningiomas are capable of invading vessels, cranial nerves and brain tissue (Kotapka et al. 1994; Larson et al. 1995;

Sindou and Alaywan 1998). Therefore, in critical locations, radical surgery without disabling complications is more difficult to achieve than with most other benign tumors at the same location (Couldwell et al. 2007; De Jesús et al. 1996; Pfisterer et al. 2007). New permanent cranial nerve deficits following surgery of cavernous sinus meningiomas occurred in 18–70% (Couldwell et al. 2006), surgery related morbidity for petroclival meningiomas ranging between 7% and 40% (Little et al. 2005; Mathiesen et al. 2007).

On this basis of the above mentioned evidence, stereotactic radiosurgery (SRS), mainly GKRS, has established as an attractive alternative providing excellent long term results, particularly for “high risk” skull base meningiomas (Hasegawa et al. 2007; Kollová et al. 2007; Kondziolka et al. 2008b; Kreil et al. 2005; Lee et al. 2002). Actuarial tumor control rates at 5 years have been reported as high as 98.5% (Kreil et al. 2005), 97.9% (Kollová et al. 2007) and $96.9 \pm 3\%$ (Lee et al. 2002), actuarial tumor control rates at 10 years amounting to 97.2% (Kreil et al. 2005). With an overall morbidity rate of 7.7%, Kondziolka et al. (2008b) reported control rates between 91% and 95% after 10 years. All these data compare favorably with a Simpson grade 1 resection. With respect to the optimisation of functional outcome, a combination of (judicious) surgical resection and SRS has turned out as gold standard for the majority of larger critically

located meningiomas (Maruyama et al. 2004; Mathiesen et al. 2007; Natarajan et al. 2007).

DiBiase et al. (2004) have pointed out that proper treatment planning for GKRS should adequately cover the dural tail, a point we agree with. By applying a median prescription dose of 14 Gy to the 50% isodose line, the same authors identified a gross total volume exceeding 10 cm³ as the only significant parameter associated with a worse prognosis for radiosurgical treatment of meningiomas. By contrast, data from our own series (Haselsberger et al. 2009) equal results achieved by a Simpson grade 1 resection (recurrence rate 9%) (Simpson 1957) and are very close to those obtained by GKRS for smaller tumors (DiBiase et al. 2004; Hasegawa et al. 2007; Kollová et al. 2007; Kondziolka et al. 2008a, b; Kreil et al. 2005; Lee et al. 2002; Metellus et al. 2005). In accordance with other authors (Hasegawa et al. 2007; Kollová et al. 2007), however, we advocate a marginal dose of 12 Gy for a safe and efficient treatment, especially for larger tumors. Moreover, with the use of a staged procedure, the specific benefit of SRS, characterised by a steep radiation fall off (Iwai et al. 2001; Pendl et al. 2000), can also be maintained in the treatment of large tumors, provided that the overlap of radiation be kept to a minimum. Conversely, parasagittal and falx meningiomas with deep embedding in the cortex should be treated with special caution as there has been evidence for a higher incidence of peritumorous imaging changes after GKRS than for meningiomas located at the skull base (Chang et al. 2003).

Radiosurgery is a neurosurgical domain – unlike external beam radiation (EBRT), which is the task of radio-oncologists and radiotherapists. As there is a distinct difference between EBRT and SRS concerning both their radiobiological effects and their way of application, this very difference has prompted the American Association of Neurological Surgeons (AANS) and the Congress of Neurological Surgeons (CNS) to redefine the term “stereotactic radiosurgery” (SRS) (Barnett et al. 2007). By definition, SRS is limited to a maximum of five fractions (Barnett

et al. 2007) which is in accordance with Lars Leksell’s original description (Leksell 1951).

Nowadays, conventional radiotherapy for meningiomas has been mainly reduced to the treatment of meningeal disease and cases of atypical or malignant meningiomas. Metellus et al. (2005) have compared GKRS and conventional radiotherapy for cavernous sinus meningiomas. They concluded that both EBRT and GKRS are safe and efficient but GKRS provides a better radiological response, is far more convenient and fits into most patients’ lives much better than EBRT (Metellus et al. 2005). Moreover, in contrast with surgery, an additional GKRS can be offered readily to patients with tumor recurrence without further risks of treatment related morbidity. Cho et al. (2006) underline the value of GKRS not only because of advantages for individual patients but also because it can reduce socioeconomic loss and achieve better cost effectiveness.

Conclusion

In conclusion, at a much lower level of complications, GKRS for critically located meningiomas can and should be every bit as effective as radical surgery. We recommend small meningiomas in critical locations be treated radiosurgically rather than by open surgery. For the majority of large skull base meningiomas in delicate areas and parasagittal meningiomas, a combination of judicious surgery and radiosurgery offers the best functional outcome. Staged GKRS provides a substantial therapeutic improvement both for large recurrent or residual tumors after surgery and as primary treatment for elderly patients and persons with significant co-morbidity. Thus, GKRS constitutes an indispensable tool for a considerable portion of benign intracranial meningiomas and as such for a large number of neurosurgical patients.

Acknowledgements The authors thank Ms E. Meier for preparation of the manuscript.

References

- Barnett GH, Linskey ME, Adler JR, Cozzens JW, Friedman WA, Heilbrun MP, Lunsford LD, Schulder M, Sloan AE, The American Association of Neurological Surgeons, Congress of Neurological Surgeons Washington Committee Stereotactic Radiosurgery Task Force (2007) Stereotactic radiosurgery – an organized neurosurgery-sanctioned definition. *J Neurosurg* 106:1–5
- Chang JH, Chang JW, Choi JY, Park YG, Chung SS (2003) Complications after gamma knife radiosurgery for benign meningiomas. *J Neurol Neurosurg Psychiatry* 74:226–230
- Cho DY, Tsao M, Lee WY, Chang CS (2006) Socioeconomic costs of open surgery and gamma knife radiosurgery for benign cranial base tumors. *Neurosurgery* 58:866–873
- Christensen HC, Kosteljanetz M, Johansen C (2003) Incidence of gliomas and meningiomas in Denmark, 1943 to 1997. *Neurosurgery* 52:1327–1334
- Claus EB, Bondy ML, Schildkraut JM, Wiemels JL, Wrensch M, Black PM (2005) Epidemiology of intracranial meningioma. *Neurosurgery* 57:1088–1095
- Condra KS, Buatti JM, Mendenhall WM, Friedman WA, Marcus RB Jr, Rhoton AL (1997) Benign meningiomas: primary treatment selection affects survival. *Int J Radiat Oncol Biol Phys* 39:427–436
- Couldwell WT, Kan P, Liu JK, Apfelbaum RI (2006) Decompression of cavernous sinus meningioma for preservation and improvement of cranial nerve function. *J Neurosurg* 105:148–152
- Couldwell WT, Cole CD, Al-Mefty O (2007) Patterns of skull base meningioma progression after failed radiosurgery. *J Neurosurg* 106:30–35
- De Jesús O, Sekhar LN, Parikh HK, Wright DC, Wagner DP (1996) Long-term follow-up of patients with meningiomas involving the cavernous sinus: recurrence, progression, and quality of life. *Neurosurgery* 39:915–920
- DiBiase SJ, Kwok Y, Yovino S, Arena C, Naqvi S, Temple R, Regine WF, Amin P, Guo C, Chin LS (2004) Factors predicting local tumor control after gamma knife stereotactic radiosurgery for benign intracranial meningiomas. *Int J Radiat Oncol Biol Phys* 60:1515–1519
- Flickinger JC (1989) An integrated logistic formula for prediction of complications from radiosurgery. *Int J Radiat Oncol Biol Phys* 17:879–885
- Hasegawa T, Kida Y, Yoshimoto M, Koike J, Iizuka H, Ishii D (2007) Long-term outcomes of gamma knife surgery for cavernous sinus meningioma. *J Neurosurg* 107:745–751
- Haselsberger K, Maier T, Dominikus K, Holl H, Kurschel S, Ofner-Kopeinig P, Unger F (2009) Staged gamma knife radiosurgery for large critically located benign meningiomas: evaluation of a series comprising 20 patients. *J Neurol Neurosurg Psychiatry* 80:1172–1175
- Iwai Y, Yamanaka K, Nakajima H (2001) Two-staged gamma knife radiosurgery for the treatment of large petroclival and cavernous sinus meningiomas. *Surg Neurol* 56:308–314
- Ketter R, Rahnenführer J, Henn W, Kim YJ, Feiden W, Steudel WI, Zang ZD, Urbchat S (2008) Correspondence of tumor localization with tumor recurrence and cytogenetic progression in meningiomas. *Neurosurgery* 62:61–69
- Kleihues P, Burger PC, Scheithauer BW (1993) The new WHO classification of brain tumours. *Brain Pathol* 3:255–268
- Kollová A, Liščák R, Novotný J Jr, Vladyka V, Šimonová G, Janoušková L (2007) Gamma knife surgery for benign meningioma. *J Neurosurg* 107:325–336
- Kondziolka D, Flickinger JC, Lunsford LD (2008a) The principles of skull base radiosurgery. *Neurosurg Focus* 24:E11
- Kondziolka D, Mathieu D, Lunsford LD, Martin JJ, Madhok R, Niranjan AM, Flickinger JC (2008b) Radiosurgery as definitive management of intracranial meningiomas. *Neurosurgery* 62:53–60
- Kotapka MJ, Kalia KK, Martinez AJ, Sekhar LN (1994) Infiltration of the carotid artery by cavernous sinus meningioma. *J Neurosurg* 81:252–255
- Kreil W, Luggin J, Fuchs I, Weigl V, Eustacchio S, Papaefthymiou G (2005) Long term experience of gamma knife radiosurgery for benign skull base meningiomas. *J Neurol Neurosurg Psychiatry* 76:1425–1430
- Larson JJ, van Loveren HR, Balko G, Tew JM Jr (1995) Evidence of meningioma infiltration into cranial nerves: clinical implications for cavernous sinus meningiomas. *J Neurosurg* 83:596–599
- Lee JYK, Niranjan A, McInerney J, Kondziolka D, Flickinger JC, Lunsford LD (2002) Stereotactic radiosurgery providing long-term tumor control of cavernous sinus meningiomas. *J Neurosurg* 97:65–72
- Leksell L (1951) The stereotaxic method and radiosurgery of the brain. *Acta Chir Scand* 102:3316–3319
- Little KM, Friedman AH, Sampson JH, Wanibuchi M, Fukushima T (2005) Surgical management of petroclival meningiomas: defining resection goals based on risk of neurological morbidity and tumor recurrence rates in 137 patients. *Neurosurgery* 56:546–559
- Mantle RE, Lach B, Delgado MR, Baesa S, Bélanger G (1999) Predicting the probability of meningioma recurrence based on the quantity of peritumoral brain edema on computerized tomography scanning. *J Neurosurg* 91:375–383
- Maruyama K, Shin M, Kurita H, Kawahara N, Morita A, Kirino T (2004) Proposed treatment strategy for cavernous sinus meningiomas: a prospective study. *Neurosurgery* 55:1068–1075
- Mathiesen T, Gerlich A, Kihlström L, Svensson M, Bagger-Sjöback D (2007) Effects of using combined transpetrosal surgical approaches to treat petroclival meningiomas. *Neurosurgery* 60:982–991

- Metellus P, Regis J, Muracciole X, Fuentes S, Dufour H, Nanni I, Chinot O, Martin P, Grisoli F (2005) Evaluation of fractionated radiotherapy and gamma knife radiosurgery in cavernous sinus meningiomas: treatment surgery. *Neurosurgery* 57:873–886
- Natarajan SK, Sekhar LN, Schessel D, Morita A (2007) Petroclival meningiomas: multimodality treatment and outcomes at long-term follow-up. *Neurosurgery* 60:965–981
- Pendl G, Unger F, Papaefthymiou G, Eustacchio S (2000) Staged radiosurgical treatment for large benign cerebral lesions. *J Neurosurg* 93(Suppl 3):107–112
- Pfisterer WK, Hendricks WP, Scheck AC, Nieman RA, Birkner TH, Krampla WW, Preul MC (2007) Fluorescent in situ hybridization and ex vivo 1H magnetic resonance spectroscopic examinations of meningioma tumor tissue: is it possible to identify a clinically-aggressive subset of benign meningiomas? *Neurosurgery* 62:1048–1059
- Sacko O, Sesay M, Roux F-E, Riem T, Grenier B, Liguoro D, Loiseau H (2007) Intracranial meningioma surgery in the ninth decade of life. *Neurosurgery* 61:950–954
- Simpson D (1957) The recurrence of intracranial meningiomas after surgical treatment. *J Neurol Neurosurg Psychiatry* 20:22–39
- Sindou MP, Alaywan M (1998) Most intracranial meningiomas are not cleavable tumors: anatomic-surgical evidence and angiographic predictability. *Neurosurgery* 42:476–480

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Abstract

Oncocytic meningioma is an extremely rare tumor featuring large polygonal cells with finely granular eosinophilic cytoplasm which constitute not <75% of the total neoplastic population. On immunohistochemistry, the lesion shows diffuse and intense immunoreactivity for antimitochondrial antiserum, whose ultrastructural counterpart is represented by the presence of numerous swollen mitochondria with flattened cristae, resulting in a true oncocytic differentiation for almost all the neoplastic cells. This histopathologic pattern was introduced as an aggressive variant morphology associated with a correspondingly aggressive clinical behavior in terms of propensity toward recurrence and brain invasion, although that initial perception of aggressiveness has lately been receiving some counterweight in a series of tumors with indolent clinical course.

Our objective is to summarize the morphologic and molecular profile of oncocytic meningiomas. Accurate recognition of this variant morphology will facilitate proper diagnosis of these unusual tumors, and will enable appropriate assessment of prognosis and therapy.

Introduction

Surgery is the standard and most effective therapy for meningiomas, with gross total resection, and thus reducing the risk of both subsequent recurrence and mortality (Perry et al. 1999). Radiation is also a major treatment modality, especially

for patients for whom surgery carries high risk. Adjuvant radiotherapy has been shown to significantly reduce recurrence rates in subtotally resected or higher grade meningiomas (Aghi et al. 2009). Additionally, newer forms of delivery have become increasingly popular, especially for small tumors or postsurgical residual masses, such as stereotactic radiosurgery, fractionated stereotactic, intensity-modulated, and proton beam forms of radiotherapy.

In addition to the important prognostic variable of the extent of resection, one of the most important factors to predict tumor behavior is

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histopathologic grading. The current *WHO classification of Tumours of the Central Nervous System (CNS)* (2007) describes 16 different subtypes of meningiomas falling into three grade designations. Tumors with mixed histopathologic patterns are commonly seen. Although this broad neoplastic spectrum may lead to diagnostic difficulties, it corresponds to WHO grade I in 75–80% of cases. Gross totally resected WHO grade I meningiomas are designated “benign”, since they show no associated increased mortality rate compared with that in age and gender matched controls, but have recurrence rates of 7–25% depending on the length of follow up. Higher grade meningiomas comprise ~20% of cases and are associated with increased morbidity and mortality. Meningiomas categorized as WHO grade II (atypical) show increased mitotic activity, defined as 4 or more mitoses per 10 high-power fields (HPF, 40× objective magnification), and/or three or more of the following five features: sheet-like growth, spontaneous necrosis, increased cellularity, prominent nucleoli, and small cells with high nuclear to cytoplasmic ratio. Grade II meningiomas recur in 29–52% of the cases even with gross total resection. They also have a slightly, but statistically significantly increased risk of mortality compared with age and gender matched controls (Mawrin and Perry 2010).

Brain invasive meningiomas, even if otherwise benign in appearance, are regarded as WHO grade II based on similar recurrence and mortality rates to those of atypical meningiomas in general. Meningiomas defined as WHO grade III (anaplastic/malignant) are characterized by the presence of 20 or more mitoses per 10 HPF or the presence of frank anaplasia, defined as carcinoma, melanoma or sarcoma-like histopathology. Although rare (1–2.8% of cases), they are associated with considerable risk of death from disease, with recurrence rates of 50–94% and average survival of <2 years. Certain histopathologic subtypes of meningiomas are higher grade by definition, i.e., clear cell (WHO grade II), chordoid (WHO grade II), rhabdoid (WHO grade III), and papillary (WHO grade III), being associated with greater likelihood of recurrence and/or aggressive behavior.

The Ki-67/MIB-1 proliferation indices, even though not included in the WHO grading scheme due to lack of standardization for the assessment of labeling values given high inter-laboratory variability, increase from benign (mean 3.8%) through atypical (mean 7.2%) to anaplastic meningiomas (mean 14.7%). Thus, following the diagnosis of an atypical meningioma, a more aggressive treatment might be chosen, including radiotherapy to the tumor bed. This would similarly apply to anaplastic meningiomas, although nearly all of them will eventually recur and often exhaust both surgical and radiation therapy options (Mawrin and Perry 2010).

A histopathologic pattern which was introduced as an aggressive variant morphology associated with a correspondingly aggressive clinical course in terms of propensity toward recurrence and brain invasion is the oncocyctic meningioma (Roncaroli et al. 1997; Marucci et al. 2007). Other case series did not confirm the initial perception of aggressiveness, and reported tumors with benign clinical course (Cimmino et al. 2000; Caldarella et al. 2002; Gallina et al. 2006; Zunarelli et al. 2010).

Given the extreme rarity of oncocyctic meningiomas, with only 21 cases described in the world literature up-to-day, it was not felt that there was sufficient experience with this variant to warrant the inclusion in the current WHO scheme. Published literature and our own experience in an active academic neurosurgical pathology practice are reviewed to focus on this variant of meningioma and its clinical behavior, in order to define whether it should be graded by using the same criteria as in other variants, or should automatically be assigned a higher grade, with obvious implications in terms of patient management and treatment.

Definition

Oncocyctic meningioma is composed mostly of sheets, nests, and cords of large polygonal cells with finely granular eosinophilic cytoplasm rich in mitochondria (Figs. 17.1, 17.2, and 17.3).

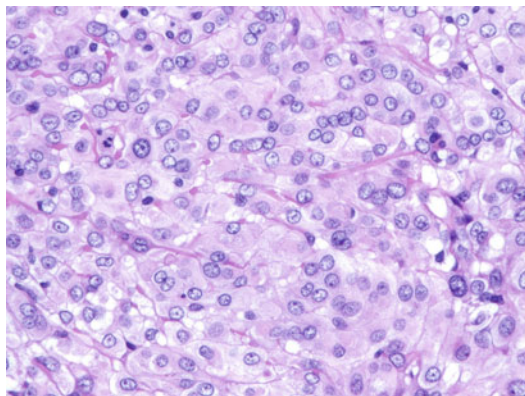


Fig. 17.1 Oncocytic meningioma is composed mostly of large polygonal cells with distinct cytoplasmic borders and finely granular eosinophilic cytoplasm. Nuclei, usually centrally located, are round to ovoid. Prominent nucleoli are seen (hematoxylin and eosin, 40×)

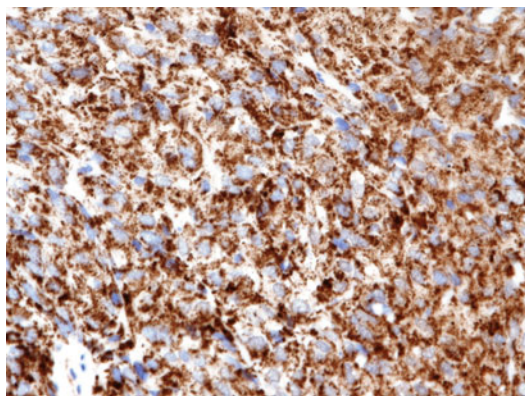


Fig. 17.2 With antimitochondrial antiserum, neoplastic cells show cytoplasm filled by strongly immunoreactive granules (antimitochondrial antiserum, 40×)

State of the Art

Cases with an Unfavorable Clinical Course

The term “oncocytic meningioma” was coined by Roncaroli et al. (1997), who retrospectively studied a series of six meningiomas with prominent oncocytic change. The tumors occurred in five female and one male patients; age ranged from 54 to 73 years (mean 62.8) at the time of surgery. Radiologically, lesions were located in the dural meninges of the left sphenoid bone

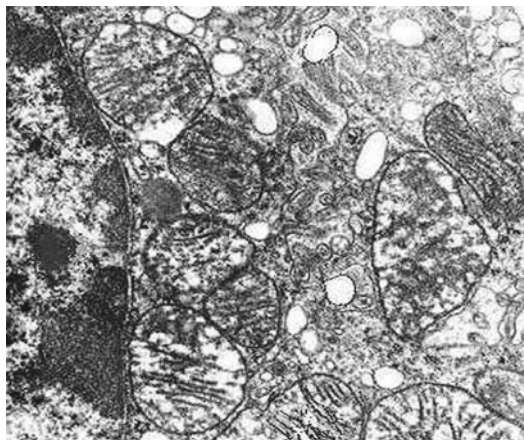


Fig. 17.3 Electron micrograph showing cytoplasmic accumulation of large mitochondria with flattened cristae in the neoplastic cells (original magnification 20000×)

(case 1), falx (case 2), left occipital region (case 3), left frontal region (case 4), right parieto-temporal region (case 5), and right sphenoid bone (case 6). Lesions showed the typical features of meningiomas as seen on magnetic resonance imaging (MRI). Therapy consisted of surgery in cases 1, 3, 4, 6 and surgery plus radiotherapy in cases 2 and 5. Information regarding past medical history of the patients, size of the tumors, and radiation treatment modalities was not available.

Histopathologically, on routine sections with hematoxylin and eosin (H&E) staining, all tumors featured sheets, nests, and cords of cells delimited by thin fibrous septa rich in vessels. Neoplastic cells were large polygonal elements with distinct cell borders, finely granular eosinophilic cytoplasm, and centrally located round to oval nuclei with prominent nucleoli, which constituted not <75% of the total neoplastic population, merging with more conventional meningioma features such as optically clear intranuclear cytoplasmic inclusions, whorls, and occasional psammoma bodies.

Mitoses ranged from 21/20 HPF in case 3 to 3/20 HPF in case 6, with a mean value of 11.3/20 HPF. Extensive necrosis was present in cases 1 and 5, whereas focal necrosis was seen in cases 2 and 3. In cases 4 and 5 brain invasion was observed. In addition, case 5 showed extensive destruction of the parietal bone, infiltration of the overlying skin, and permeation of lymphatic vessels.

On immunohistochemistry (IHC), with antimitochondrial antiserum, most neoplastic cells displayed cytoplasm filled with strongly immunoreactive granules. The authors also documented diffuse and intense reactivity for epithelial membrane antigen (EMA) in one case, and focal reactivity in the remaining five cases. Diffuse and intense reactivity for vimentin was observed in all cases. Cytokeratin (CK), S-100 protein, chromogranin, glial fibrillary acidic protein (GFAP), and anti-CD68 yielded negative reactions. Ki-67 proliferation indices ranged from 30% in cases 3 and 5 –2% in case 2.

In this manuscript, in the Table which summarises the clinical findings of this series of oncocyctic meningiomas, infiltration of the bone and overlying skin, and diffuse permeation of lymphatic vessels are presented in case 3. In the Table which also summarises the histopathologic and immunohistochemical features of oncocyctic meningiomas, the extent of EMA expression is documented as diffuse in two cases and focal in four cases. The Ki-67 proliferation index 30% is documented in case 5. Demonstration that the cytoplasmic granules corresponded to mitochondria was also found at the electron microscopy (EM) level. Most neoplastic cells featured swollen mitochondria with flattened cristae. Other more conventional features of meningioma were described, such as cytoplasmic interdigitations and desmosomes. Oncocyctic differentiation was also demonstrated by Western-blot analysis.

The follow up ranged from 7 to 82 months (mean 29.6 months). Two patients had a favorable clinical course with no evidence of disease at 8 and 82 months after resection (cases 4 and 1, respectively), whereas two patients relapsed 24 and 30 months after surgery (cases 6 and 3, respectively). In one other patient (case 2), two recurrences were observed at 18 and 27 months after surgery and radiotherapy. The sixth patient had residual tumor 7 months after surgery and radiotherapy (case 5). Given these morphologic and clinical findings, the authors pointed toward an indication of aggressive behavior of these tumors in terms of recurrence and brain invasion.

Subsequently, Marucci et al. (2007) reported the case of a 49-year-old woman with unremarkable past medical history, who presented in a confused state. Neuroimaging showed a basal-frontal tumor, measuring 5 cm in its greatest axis, which was removed through a craniotomy. On gross examination, the specimen consisted of a grayish, firm nodule measuring 5 cm in its greatest axis. One year later, the patient presented with chiasmatic syndrome and underwent a second craniotomy for the presence of residual tumor in the chiasmatic region and cavernous sinus. Due to the high operative risk, the tumor was not completely resected. Stereotactic radiosurgery was, therefore, performed as an adjuvant treatment. Four years later the tumor recurred and stereotactic radiosurgery was repeated. Eight months after the second stereotactic radiosurgery, neuroimaging showed multiple nodules in the chiasmatic region and posterior fossa. Partial surgical resection was again performed. Eighteen months after the last recurrence, the patient presented multiple intracranial nodules and a further curative approach was no longer regarded as feasible.

In all resected lesions, conventional histopathology, immunoreactivity with anti-mitochondrial antiserum and ultrastructural findings showed similar features consistent with oncocyctic meningiomas. Prominent nucleoli were sometimes observed. Mitotic index was 2/10HPF. Neither necrosis nor brain cortex invasion by tumor was documented. The only difference among various lesions was the Ki-67 proliferation index, which ranged from 2% in the first specimen to 10% in the most recent lesion. Given the dramatic progression of this tumor in spite of surgery and radiotherapy and in the absence of atypical or anaplastic features such as those described in the WHO grading scheme, the authors reported this case as an additional example of the greater aggressiveness of oncocyctic meningiomas in terms of frequent recurrences. They also suggested that radiation therapy might even worsen the course of disease. In their case, as well as in case 2 of the six cases reported by Roncaroli et al. (1997), which was treated with surgery and radiotherapy and recurred at 18 and 27 months, Marucci et al. (2007) hypothesized that oncocyctic

meningioma could be resistant to radiotherapy because it is rich in mitochondrial manganese superoxide dismutase, one of the most effective antioxidant enzymes in protecting cells against ionizing radiation damage by scavenging oxygen radicals induced by ionizing radiation. The authors also observed that radiotherapy might cause biologic changes in treated tumors, enhancing the proliferative activity and the invasive potential by activating the metalloproteinase proteolytic system.

Cases with a Favorable Clinical Course

Cimmino et al. (2000) described a case of oncocytic meningioma in a 78-year-old woman with a past medical history of bilateral mastectomy and adjuvant chemotherapy for breast cancer with positive axillary lymph-nodes. The follow up had been negative for 9 years. The patient reported frontal headache of increasing severity. Neuroradiological investigations revealed a right frontal tumor, measuring 3 cm in diameter, which was preoperatively felt to be a metastatic lesion. A craniotomy was performed. On gross examination, the specimen consisted of fragments of grayish tissue ranging from 0.5 to 0.8 cm in maximum dimension. Histopathologically, the tumor was consistent with oncocytic meningioma. Prominent nucleoli were observed. Necrosis was absent, so was brain invasion by tumor. Mitoses were extremely rare, although no mitotic count was provided. On IHC, the vast majority of the neoplastic cells were strongly immunoreactive for antimitochondrial, EMA and vimentin antisera, and faintly and focally reactive for CK. Negative immunoreactions for S-100 protein, GFAP, estrogen receptor, and progesteron receptor were documented. Ki-67 proliferation index was <1%.

Eleven months after surgery, as stated in the abstract of the manuscript, and 14 months after surgery, as stated in the text, the patient had no evidence of tumor recurrence.

Another single case of oncocytic meningioma was reported by Caldarella et al. (2002). A previously healthy 30-year-old man presented with

sudden appearance of paresthesias on his legs, arms, tongue, lips, and difficult coordination of walking. These symptoms persisted for some minutes, then the patient completely recovered. Neurological examination was negative. MRI revealed a dura-based, right parietal region tumor. The diagnosis of meningioma was suggested, and surgery was performed. On gross examination, the specimen consisted of tissue fragments with diameters ranging from 0.5 to 5 cm. The microscopic and ultrastructural findings were consistent with oncocytic meningioma. Prominent nucleoli were noted. Mitoses were 2/20HPF. Neither necrosis nor brain invasion by tumor was observed. On IHC, the authors reported diffuse immunoreactivity for EMA and vimentin antisera. Focal reactivity for CK (5% of the neoplastic cells) was present, while S-100 protein, GFAP, thyroglobulin, actin, and p53 yielded negative immunoreactions. Ki-67 proliferation index was 10%.

Eight months after surgery, the patient had no evidence of tumor recurrence.

A more sizeable series of five cases, one previously described by Caldarella et al. (2002), was reported by Gallina et al. (2006). This study included two male and three female patients; age ranged from 30 to 64 years (mean 50.2 years). On neuroimaging, all patients had dura-based lesions consistent with meningiomas. Tumors were located in the right parietal region (case 1), left sphenoid (case 2), olfactory groove (cases 3 and 4), left frontal convexity (case 5). All patients underwent gross total resection of the lesions. No additional treatments were administered.

Information regarding past medical history of patients and size of tumors was not available, apart from the case previously described by Caldarella et al. (2002). In this series, histopathologic features negatively affecting prognosis such as prominent nucleoli, necrosis and brain invasion were not observed. Mitotic figures were 2/10HPF in case 1 and 1/10HPF in case 5, absent in the three remaining cases. Ki-67 proliferation indices ranged from 1% in cases 3 and 4–10% in case 1, with a mean value of 4.4%.

The follow up period ranged from 6 to 54 months (mean 32.4 months). All patients had no evidence of tumor recurrence.

On morphologic and clinical grounds, Gallina et al. (2006) concluded that the oncocytic variant of meningioma should not be considered biologically similar to the WHO grade II and III meningiomas.

The most recent series of oncocytic meningiomas was reported by Zunarelli et al. (2010). The study included seven male and one female patients; age ranged from 26 to 59 years (mean 44.2 years) at the time of surgery. The most common presenting symptom was headache. Two patients had a clinical history of hypertension; any other significant disease was not recorded. On MRI, all patients had lesions with the characteristic imaging features of meningiomas. Tumors were located in the right optic nerve (case 1), right frontal region (cases 2, 4, 5, 6, 8), right parieto-temporal region (case 3), and right parietal region (case 7). Tumor size ranged from 1 to 7 cm in maximum dimension (mean 3.5). At surgery, all patients underwent gross total resection of the lesions. No additional treatment was administered.

As confirmed by conventional morphology, IHC, and EM, all tumors were consistent with oncocytic meningiomas. Mitotic index ranged from 1/10 HPF in cases 1, 3, 7 to 6/10 HPF in case 4, with a mean value of 1.9/10 HPF. Sheet-like growth, necrosis, increased cellularity, prominent nucleoli, small cell change, brain invasion, carcinoma-, sarcoma-, and melanoma-like features were not observed. The authors documented foci of meningeal lipidization in one case and infiltration of the dura in two other cases. They also reported mild and focal lymphoplasmacytic inflammation in one case.

On IHC, all cases were diffusely immunoreactive for antimitochondrial antiserum and focally immunoreactive for S-100 protein and EMA (5% of the neoplastic cells). Immunoreactions for GFAP and CK were negative. Ki-67 proliferation indices ranged from 2% in cases 1 and 7–20% in case 4, with a mean value of 5.75%. The authors graded the tumors according to the WHO criteria. Seven tumors corresponded to grade I (cases 1, 2, 3, 5, 6, 7, 8). Considering the mitotic index 6/10HPF and the Ki-67 proliferation index 20%, case 4 was designated grade II. No grade III tumors were documented.

The follow up ranged from 9 to 61 months (mean 39.6 months). The patients with grade I oncocytic meningiomas had no evidence of tumor recurrence. In the patient with grade II lesion, tumor relapsed at post-operative intervals of 18 and 34 months. Surgery was repeated with gross total resection of the lesions and adjacent dura, with postoperative neuroimaging confirmation. Tumors remained histopathologically unchanged at recurrence. Zunarelli et al. (2010) suggested that the aggressive clinical course of this tumor had to be related to the presence of a high mitotic count and correspondingly high Ki-67 proliferation index rather than an oncocytic morphology. Analogously, in their opinion the overall poor clinical course of the patients in the series by Roncaroli et al. (1997) might be related to the presence of morphological features predicting unfavorable behavior as defined by the WHO grading scheme for meningiomas rather than an oncocytic appearance.

Overview

In the original series by Roncaroli et al. (1997), recurrence was observed in three cases, and brain invasion by tumor was present in other two cases. This recurrence rate of 50% is definitely higher than that of other types of meningiomas, which never exceeds 20% within 20 years (Jaaskelainen 1986). Indeed, the histopathologic criteria indicating atypical meningiomas and predicting unfavorable behavior as defined by the WHO grading scheme are regular features of this series, as well as high proliferative indices.

After this study, literature on oncocytic meningioma is scant. However, the potentially aggressive behavior of this lesion as well as the propensity to invade the brain cortex and to recur were not confirmed by further anecdotal observations. Collectively, the single case reports by Cimmino et al. (2000) and Caldarella et al. (2002) describe oncocytic meningiomas with no histopathologic features correlated with higher grades/aggressive behavior with the exception of prominent nucleoli, although the follow up periods are rather short. Analogously, the study by Gallina

et al. (2006), which included a number of patients similar to that of Roncaroli et al. (1997) but with a longer follow up, do not confirm that the oncocyctic morphology in meningiomas adversely affects prognosis.

The question of whether oncocytic morphology should be predictive of recurrence and aggressive clinical course in meningiomas was raised again by Marucci et al. (2007), who reported on a case with progression after surgery and radiosurgery. A similar unfavorable clinical course was also documented in the second of the six cases reported by Roncaroli et al. (1997), which was treated by surgery and radiotherapy and recurred at 18 and 27 months. A few differences distinguish this case from the one reported in the original series. In this case the mitotic count is lower (2/10HPF versus 8/20HPF), and the Ki-67 proliferation index tends to be higher (2–10% versus 2%). Furthermore, in this case the recurrences occur later.

The most numerous series of oncocytic meningiomas and with the longest follow up so far was reported by Zunarelli et al. (2010). This series as a whole differs significantly from the original series by Roncaroli et al. (1997). The male predominance is clear cut (male: female ratio 7:1 versus 1:5), and the age of the patients is almost two decades younger (mean age 44.2 versus 62.8 years). Oncocytosis as degenerative ageing related phenomenon (Nappi et al. 1999) seems to be less likely. All tumors are right-sided, five of them in the frontal lobe, whereas in the original series no predilection for site is noted.

In the original series, histopathologic features correlated with aggressive clinical course are frequently observed, and the recurrence rate is 50%. In this series, histopathologic features predicting unfavorable clinical course are absent in all but one case, which recurred. This case accounts for 12.5% of the series. The frequency of atypical cases and recurrences is in keeping with the 20% grade II rate and with the 29–52% recurrence rates for grade II meningiomas reported by the WHO.

Differential Diagnosis

The identification of a dura-based lesion, the presence of a typical meningioma component in examples of oncocytic meningioma, plus the demonstration of cytoplasmic granular staining for anti-mitochondrial antiserum in oncocytic regions should suffice in establishing the correct diagnosis. On this backdrop, the differential diagnosis of oncocytic meningioma will have to exclude a fairly limited, but therapeutically relevant, number of alternatives. The gemistocyte-like meningioma, which shows polygonal cells with homogeneously eosinophilic cytoplasm and eccentrically located nuclei, lacks cytoplasmic granularity (Roncaroli et al. 1997). On clinical grounds, granular cell astrocytoma and granular cell schwannoma can be excluded because they are not dura-based. These tumor entities owe their cytoplasmic complexion to the accumulation of phagolysosomes rather than to mitochondria. On conventional H&E staining, this feature tends to produce a foamy rather than crystalline optical quality. On IHC, granular cell astrocytoma is immunoreactive for GFAP and CD68 (Geddes et al. 1996; Brat et al. 2002), and granular cell schwannoma is strongly and diffusely immunoreactive for S-100 protein (Carvalho et al. 1994). All these features lack in the oncocytic meningiomas, as seen at the EM and IHC levels. The distinction of oncocytic meningioma from an oncocytic variant of a pituitary adenoma is based on the immunoreactivity for neurosecretory markers synaptophysin and chromogranin and/or selected pituitary hormones by the adenoma (Fuller et al. 2007). Oncocytic meningioma will be differentiated from the recently described oncocytic ependymoma by its consistent lack of GFAP immunoreaction as well as its conspicuous interdigitating cell borders at the ultrastructural level (Vaitaj et al. 2011).

Other primary oncocytic and granular cell tumors of the neuraxis that may be encountered with reasonable, although very low, probability in the diagnostic practice include spindle cell oncocytoma of the adenohypophysis and granular

cell tumor of the neurohypophysis. As their names imply, these tumors will not be expected in extrapituitary sites.

The spindle cell morphology, hallmark of spindle cell oncocytoma of the adenohypophysis, lacks in oncocytic meningioma. On IHC, both entities could exhibit a similar immunoreactivity for EMA and vimentin, but S-100 protein expression has been documented as negative or focally positive in oncocytic meningioma. Moreover, the ultrastructural features of spindle cell oncocytoma of the adenohypophysis, lacking the features of meningotheial differentiation, argue strongly against the diagnosis of oncocytic meningioma (Fuller et al. 2007). With regard to granular cell tumor of the neurohypophysis, IHC shows variable positive immunoreactions for CD68, S-100 protein, α -1 anti-trypsin, and α -1 anti-chimotrypsin, none of which is a feature of oncocytic meningioma (Fuller and Wesseling 2007).

The oncocytic variant of choroid plexus papilloma is a very rare lesion, and the fourth ventricle has been the most common site among these cases. Immunoreactivity for S-100 protein and focal immunoreactivity for GFAP, EMA, and CK have been documented in this neoplasm (Gulwani and Roy 2008). Site of origin and IHC differentiate this neoplasm from oncoytic meningioma. Along the same line, rare instances of oncocytic paraganglioma and misplaced oncocytic adrenocortical adenoma, that were described in the CNS, tend to present neurosecretory rather than glial characteristics, as reported by Moran et al. (1997) and Schittenhelm et al. (2009).

For completeness, in the differential diagnostic checklist of primary CNS tumors with “granule-bearing” morphology, mention should be made of mitochondrial anomalies referable to paracrystalline inclusions in a case of subependymal giant cell astrocytoma (Jay et al. 1994). Granular condensations of cytoskeletal intermediate filaments may also produce refractile eosinophilic particles in the so-called gliofibrillary variant of oligodendrogliomas (Yoshida and Nakazato 2001), in non-pilocytic astrocytomas, and in ependymomas (Twiss et al. 1993). Aberrant expression of glial intermediate filaments in schwannomas may also exceptionally involve a granular pattern

(Brown and Rushing 1997). These entities tend to immunoreact for GFAP and ubiquitin. A pattern of patchy granular cell change induced by cytoplasmic accumulation of lysosomes has also been observed in a case of anaplastic medulloblastoma (Rodriguez and Scheithauer 2007).

Conversely, distinction of oncocytic meningiomas from central nervous system metastases of oncocytic carcinomas will carry a fundamentally different prognosis, and immediately impact on patient management. While any type of malignant oncocytic neoplasm may theoretically metastasize to the CNS, actual metastatic involvement has been documented for carcinomas of the thyroid (Clark et al. 2005), the parotid gland (Maiuri et al. 2003), and the pituitary (Roncaroli et al. 2003). These alternatives can be ruled out on IHC grounds, especially in view of the negative immunoreactions for CK, thyroglobulin, and synaptophysin.

Molecular Studies and Spectroscopic Analysis

The most common genetic changes found in meningiomas involve the 22q12.2 chromosomal region encompassing the neurofibromatosis 2 (NF2) gene. Allelic losses (loss of heterozygosity [LOH]) of this region are found in 40–70% of sporadic and the vast majority of NF2 associated meningiomas. Additionally, NF2 mutations are found in up to 60% of meningiomas (Mawrin and Perry 2010). The protein product of the NF2 gene is termed merlin or schwannomin. Meningioma associated NF2 mutations commonly result in a truncated, non-functional merlin protein. The frequency of NF2 mutations is roughly equal among different WHO grades, suggesting that it represents an important initiation rather than progression associated alteration; in contrast, differences in the frequency of NF2 alterations have been noted based on variant histopathology, with higher rates in non-meningothelial than in meningotheial meningiomas (Mawrin and Perry 2010). Using the real time polymerase chain reaction (RT-PCR), Buccoliero et al. (2007a) studied NF2 gene expression in 30 cases of sporadic

meningiomas, one of which was oncocytic. When NF2 gene expression of the oncocytic meningioma was compared with the average value of expression (reference value) of the entire group of meningiomas, it showed a slightly higher expression (+0.055).

Although limited to a single case, this finding is in keeping with the notion that NF2 gene expression alterations are more frequent in non-meningothelial subtypes of meningioma. Buccoliero et al. (2007b) also reviewed the cyclooxygenase-2 (COX-2) expression, as evaluated through IHC and RT-PCR, in a series of 23 sporadic meningiomas, one of which was oncocytic. COX-2 is the inducible form of the enzyme involved in the first steps of the prostaglandins and thromboxane synthesis. COX-2 up-regulation is demonstrated in tumors where it can modulate neoplastic progression, metastasis, multidrug resistance, and angiogenesis. The authors considered the possible therapeutic use of the COX-2 inhibitor nonsteroidal antiinflammatory drugs (NSAIDs) in some selected cases of meningiomas for which surgical excision is not regarded as feasible.

On IHC, the oncocytic meningioma showed a focal (defined by the authors as such when present in >25% and <50% of the neoplastic cells) and weak cytoplasmic reactivity for COX-2. At RT-PCR, when COX-2 gene expression of the oncocytic meningioma was compared with the average value of expression (reference value) of the entire group of meningiomas, it showed a lower expression (−0.43). Although limited to a single case, this result seems to rule out the possibility of utilization of NSAIDs, alone or in combination with radiotherapy, in the treatment of this subtype of meningiomas.

Tugnoli et al. (2006) reported on the magnetic resonance spectroscopy (MRS) characterisation of some histopathologic subtypes of meningiomas (3 meningothelial, 1 fibrous, 1 transitional and 1 oncocytic), which were analysed both through *in vivo* and *ex vivo* MRS experiments. The *ex vivo* investigations (high-resolution magic angle spinning [HR-MAS]), permitting an accurate description of the metabolic profile, are very helpful for the assignment of the resonances

in vivo of human tumors and for the validation of the quantification procedure of *in vivo* magnetic resonance spectra. The spectroscopic data, obtained by using one- and two dimensional experiments, showed the presence of several metabolites. Some of them were common to all meningiomas: alanine, choline-containing compounds, glutamine plus glutamate, scarce or absent creatine. At the same time, the study showed that different subtypes of meningiomas had different metabolic profiles, particularly regarding macromolecules and lipids. More specifically, the metabolic profile of oncocytic meningioma displayed an increased amount of triglycerides.

Conclusions

Meningiomas display a huge array of histopathologic patterns, associated with an equally wide range of biological features. Interestingly, all cases of oncocytic meningiomas described so far are documented in the Italian population. Whether this peculiar epidemiology reflects a difference in terms of genetic and/or environmental factors, we do not know.

As with most oncocytic tumors, these cases do not offer any explanation, local or systemic, regarding the origin of mitochondrial proliferation in tumor cells, nor do they reveal any clue as to whether oncocytic morphology necessarily associates with malignancy. Oncocytic meningioma remains enigmatic, since some of the cases described in the literature behaved like aggressive tumors, whereas some other cases did not. Greater experience is needed to determine whether or not separate grading criteria should be applied to this variant of meningioma. In the differential diagnostic checklist of CNS tumors with “finely granular eosinophilic” aspect, accurate recognition of this variant morphology will facilitate proper diagnosis of this unusual meningioma. Reporting of further cases will enable appropriate assessment of clinical behavior and optimal therapeutic approaches.

In the meantime, if a benign-appearing meningioma shows oncocytic morphology, is that

sufficient for a grade II or III designation? I feel that presently the answer to that possibility is “no”. Until such cases are studied in greater number and detail, I consider the presence of this variant morphology insufficient. In my own practice I have only applied the recommended WHO grading criteria to define oncocytic meningiomas as grade I (benign) or grade II (atypical); I have never met grade III (anaplastic) cases. In my opinion, the oncocytic morphology does not necessarily indicate clinical aggressiveness and it should not be automatically assigned a higher grade in the absence of other established histopathologic parameters of greater likelihood of recurrence and/or aggressive behavior. The diagnosis of oncocytic meningioma should not prompt adjuvant radiotherapy *per se*. As for meningiomas in general, surgery appears to be the effective mode of therapy.

Historical Annotations

After this dissertation regarding “oncocytic meningiomas”, “oncocytic morphology”, “oncocytic change”, some of the readers might be interested in a brief historical review of the nomenclature. The prefix “onco” in oncoocyte and oncology denotes tumor or volume, and the Greek word “onkoustai” means “to swell”. The term ‘oncoocyte’ was coined by Hamperl (1931) in a general and purely descriptive meaning. Thus, it is not surprising that this term has been applied to cells of different tissue and tumor types. Subsequently, Hamperl (1962) designated oncoocytes as a special type of epithelial cell having abundant cytoplasm densely filled with eosinophilic granules and rich in mitochondria at the EM level. Oncocytic transformation has been observed in various pathologic and non-pathologic conditions, possibly as a consequence of compensatory mitochondrial hyperplasia due to defects of oxidative metabolism (Nappi et al. 1999). When it occurs as part of a reactive, inflammatory or hyperplastic process, it is called oncocytosis. When it is seen as the dominant cell type in a neoplasm, the tumor is called an oncocytoma (Cotton 1990; Chang and Harawi 1992).

Although oncocytic transformation may occur in almost all tumors, it is more commonly seen in renal, thyroid, and salivary gland neoplasms, while it is rare in intracranial tumors with the exception of pituitary adenomas (Tallini et al. 1992; Roncaroli et al. 1997). A single case of suprasellar tumor showing predominant meningotheial features with intermingled nests of granular cells was reported by Friede and Yasargil (1977). Russell and Rubinstein (1989) indicated that finely granular cytoplasm was one of the possible cytological features of syncytial (meningotheial) meningiomas.

Apart from these anecdotal observations, the required fraction of such morphology in mixed tumor types to warrant the oncocytic designation is not well established, and various criteria have been proposed by which to define a neoplasm “oncocytic”. Tallini et al. (1992), addressing oncocytic tumors of the thyroid, accepted as oncocytic tumors those having not <75% of mitochondria-rich cells. All of the authors who reported on oncocytic meningiomas adopted the same criterion (not <75% of neoplastic cells with oncocytic changes) for their cases.

To my grandfather Leo

References

- Aghi MK, Carter BS, Cosgrove GR, Ojemann RG, Amin-Hanjani S, Martuza RL, Curry WT Jr, Barker FG (2009) Long-term recurrence rates of atypical meningiomas after gross total resection with or without post-operative adjuvant radiation. *Neurosurgery* 64:56–60
- Brat DJ, Scheithauer BW, Medina-Flores R, Rosenblum MK, Burger PC (2002) Infiltrative astrocytomas with granular cell features (granular cell astrocytomas). A study of histopathologic features, grading, and outcome. *Am J Surg Pathol* 26:750–757
- Brown DF, Rushing EJ (1997) Rosenthal fibers and eosinophilic granular bodies in a classic acoustic schwannoma. *Arch Pathol Lab Med* 121:1207–1209
- Buccoliero AM, Castiglione F, Degl’Innocenti DR, Gheri CF, Garbini F, Taddei A, Ammannati F, Mennonna P, Taddei GL (2007a) studied NF2 gene expression in 30 cases of sporadic meningiomas, one of which was oncocytic. *Neuropathology* 27:36–42
- Buccoliero AM, Castiglione F, Degl’Innocenti DR, Arganini L, Taddei A, Ammannati F, Mennonna P, Taddei GL (2007b) also reviewed the cyclooxygenase-2 (COX-2) expression, as evaluated thorough IHC and

- RT-PCR, in a series of 23 sporadic meningiomas, one of which was oncocytic. *Appl Immunohistochem Mol Morphol* 15:187–192
- Caldarella A, Buccoliero AM, Marini M, Taddei A, Mennonna P, Taddei GL (2002) Oncocytic meningioma: a case report. *Pathol Res Pract* 198:109–113
- Carvalho GA, Lindeke A, Tatagiba M, Ostertag H, Samii M (1994) Cranial granular-cell tumor of the trigeminal nerve. Case report. *J Neurosurg* 81:795–798
- Chang A, Harawi SJ (1992) Oncocytes, oncocytosis, and oncocytic tumors. *Pathol Annu* 27:263–304
- Cimmino A, Giangaspero F, Pennella A, Serio G, De Tomasi A, Colamaria A, Ricco R (2000) Oncocytic meningioma. Case report. *Pathologica* 92:82–85
- Clark JR, Lai P, Hall F, Borglund A, Eski S, Freeman JL (2005) Variables predicting distant metastases in thyroid cancer. *Laryngoscope* 115:661–667
- Cotton DWK (1990) Oncocytomas. *Histopathology* 16:507–509
- Friede RL, Yasargil MG (1977) Suprasellar neoplasm with a granular cell component. *J Neuropathol Exp Neurol* 26:769–782
- Fuller GN, Wesseling P (2007) Granular cell tumor of the neurohypophysis. In: Louis DN, Oghaki H, Wiestler OD, Cavenee WK (eds) WHO classification of tumours of the central nervous system, 4th edn. IARC Press, Lyon, pp 241–242
- Fuller GN, Scheithauer BW, Roncaroli F, Wesseling P (2007) Spindle cell oncocytoma of the adenohypophysis. In: Louis DN, Oghaki H, Wiestler OD, Cavenee WK (eds) WHO classification of tumours of the central nervous system, 4th edn. IARC Press, Lyon, pp 245–246
- Gallina P, Buccoliero AM, Mariotti F, Mennonna P, Di Lorenzo N (2006) Oncocytic meningiomas: cases with benign histopathologic features and a favorable clinical course. *J Neurosurg* 105:736–738
- Geddes JF, Thom M, Robinson SFD, Revesz T (1996) Granular cell change in astrocytic tumors. *Am J Surg Pathol* 20:55–63
- Gulwani H, Roy S (2008) 44-year-old man with abnormal gait. *Brain Pathol* 18:605–606
- Hamperl H (1931) Onkocyten und Geschwulste der Speicheldrüsen. *Virchows Arch [Pathol Anat]* 282:724–736
- Hamperl H (1962) Benign and malignant oncocytoma. *Cancer* 15:1019–1027
- Jaaskelainen J (1986) Seemingly complete removal of histologically benign intracranial meningioma: late recurrence rate and factors predicting recurrence in 657 patients. A multivariate analysis. *Surg Neurol* 26:461–469
- Jay V, Edwards V, Rutka JT (1994) Crystalline inclusions in a subependymal giant cell tumor in a patient with tuberous sclerosis. *Ultrastruct Pathol* 17:503–513
- Maiuri F, Esposito F, Del Basso De Caro M, Tortora F (2003) Dural cerebellopontine angle metastasis from malignant parotid oncocytoma. *J Neurooncol* 61:69–72
- Marucci G, Betts CM, Frank G, Foschini MP (2007) Oncocytic meningioma: report of a case with progression after radiosurgery. *Int J Surg Pathol* 15:77–81
- Mawrin C, Perry A (2010) Pathological classification and tumor genetics of meningiomas. *J Neurooncol* 99:379–391
- Moran CA, Rush W, Mena H (1997) Primary spinal paragangliomas: a clinicopathological and immunohistochemical study of 30 cases. *Histopathology* 31:167–173
- Nappi O, Ferrara G, Wick MR (1999) Neoplasms composed of eosinophilic polygonal cells: an overview with consideration of different cytomorphologic patterns. *Semin Diagn Pathol* 16:82–90
- Perry A, Scheithauer BW, Stafford SL, Lohse CM, Wollan PC (1999) “Malignancy” in Meningiomas. A clinicopathologic study of 116 patients, with grading implications. *Cancer* 85:2046–2055
- Perry A, Louis DN, Scheithauer BW, Budka H, von Deimling A (2007) Meningiomas. In: Louis DN, Oghaki H, Wiestler OD, Cavenee WK (eds) WHO classification of tumors of the central nervous system, 4th edn. IARC Press, Lyon, pp 164–172
- Rodriguez FJ, Scheithauer BW (2007) Anaplastic medulloblastoma with granular cell change. *Acta Neuropathol* 113:95–99
- Roncaroli F, Riccioni L, Cerati M, Capella C, Calbucci F, Trevisan C, Eusebi V (1997) Oncocytic meningioma. *Am J Surg Pathol* 21:375–382
- Roncaroli F, Nosé V, Scheithauer BW, Kovacs K, Horvath E, Young WF Jr, Lloyd RV, Bishop MC, His B, Fletcher JA (2003) Gonadotropic pituitary carcinoma: HER-2/neu expression and gene amplification. Report of two cases. *J Neurosurg* 99:402–408
- Russell DS, Rubinstein LJ (1989) Tumors of the meninges and related tissues. In: Russell DS, Rubinstein LJ (eds) Pathology of the tumors of the central nervous system, 5th edn. Edward Arnold, London, pp 449–532
- Schittenhelm J, Ebner FH, Harter P, Bornemann A (2009) Symptomatic intraspinal oncocytic adrenocortical adenoma. *Endocr Pathol* 20:73–77
- Tallini G, Carcangiu ML, Rosai J (1992) Oncocytic neoplasms of the thyroid gland. *Acta Pathol Jpn* 42:305–315
- Tugnoli V, Schenetti L, Mucci A, Parenti F, Cagnoli R, Righi V, Trinchero A, Nocetti L, Toraci C, Mavilla L, Trentini GP, Zunarelli E, Tosi MR (2006) *Ex vivo* HR-MAS MRS of human meningiomas: a comparison with *in vivo* 1H MR spectra. *Int J Mol Med* 18:859–869, 59
- Twiss JL, Anderson LJ, Horoupian DS (1993) Globular glial fibrillary acidic protein-reactive cytoplasmic inclusions in ependymoma: an immunoelectronmicroscopic study. *Acta Neuropathol* 85:658–662
- Vaitaj I, von Gunten M, Fung C, Brekenfeld C, Kappeler A, Reinert MM (2011) Oncocytic ependymoma: a new histological variant of high-grade ependymal neoplasm composed of mitochondrion-rich epithelioid cells. *Pathol Res Pract* 207:49–54
- Yoshida T, Nakazato Y (2001) Characterization of refractile eosinophilic granular cells in oligodendroglial tumors. *Acta Neuropathol* 102:11–19
- Zunarelli E, Tallarico E, Valentini A, Maiorana A (2010) Oncocytic meningioma: study of eight new cases and analysis of 13 reported cases. *Pathology* 42:587–589

Intracranial Meningiomas: Treatment and Quality of Life

18

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Abstract

Results of surgery of intracranial meningiomas are affected by location, size, histological attributes and genetic alterations of the tumors. Operative morbidity and mortality have traditionally been used to describe outcome after meningioma surgery. Yet, according to current outcomes research programs, assessment of a broad spectrum of parameters, including quality of life (QOL) indices is required to adequately reflect the results of a given treatment. An evaluation using a combination of neuropsychological tests and a structured interview was suggested to enable a comparison of both objective and subjective aspects of the complex construct QOL. Until now few information about the different aspects of QOL after meningioma surgery is available. Yet, it was shown that surgeons' assessments tend to suggest a favorable outcome, in-depth investigations of the patients' perspectives clearly showed their QOL to be adversely affected. Most of younger patients were not satisfied with life and reported a strong feeling not to be accepted in society, resulting in lowered self-confidence and general performance. It was concluded that a comprehensive evaluation after meningioma surgery is required to prevent poor long-term results after apparently successful surgery. Consideration of tumor attributes and an understanding of relevant outcome and its effect upon the patient's life are essential for a deliberate decision whether surgery is clearly indicated or another treatment should be preferred.

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Introduction

Meningiomas arise from the arachnoid cap cells and account for 13–26% of intracranial tumors (Rosenberg et al. 2009). They commonly occur as sporadic solitary benign and slow growing tumors, and the large majority of meningiomas

are associated with a relatively optimistic outcome. Generally, intracranial meningiomas present as tumors with a good chance for a total surgical removal and a curative treatment without significant mortality and morbidity. About 90% of the tumors are classified histologically as WHO grade I meningiomas. However, even benign sporadic meningiomas sometimes show relatively rapid growth and may recur after total removal (Shen et al. 2009). Grade II and III meningiomas are not well controlled because of aggressive behavior and a high recurrence risk. Atypical meningiomas WHO grade II appear in a frequency of 6–8%. WHO grade III meningiomas, comprising 2–3%, are termed anaplastic or malignant (Shen et al. 2009). In 1–8% of patients multiple meningiomas are found, with tumors presenting at different locations. Yet, it was pointed out, that according to the 2007 WHO classification grade II meningiomas now account for more than 20% of all meningiomas (Durand et al. 2009). In a retrospective multicenter study 5-year survival rates of 78% and 44% were reported for patients with grade II and grade III meningiomas, respectively. The median time interval to recurrence was 31.9 months for grade II, and 21.0 months for grade III meningiomas (Durand et al. 2009). There is still no consensus on the management of meningiomas grade II and III. While postoperative radiotherapy in grade III tumors is considered necessary it is more controversial in grade II tumors.

The benefits of radiotherapy have not been explicitly demonstrated in grade III meningiomas (Rosenberg et al. 2009). Yet, studies have often been statistically limited owing to the rarity of these tumors. Some authors even concluded that the clinical value and optimal timing of radiotherapy after surgery of meningiomas, independent of grade, still is a matter of debate (Van Nieuwenhuizen et al. 2007). Although radiotherapy is becoming a common element in the treatment, surgery remains the primary therapy in most cases. Chemotherapy has not shown significant effects on grades II and III meningiomas (Durand et al. 2009). Analyses of genetic alterations, genetic pathways of tumorigenesis and possible associations with aggressiveness and recurrence

rates may contribute to create new therapeutic agents. Furthermore there is strong evidence, that specific genetic alterations correlate with aggressiveness of growth and rate of tumor recurrence, which in some cases may be helpful for the decision about treatment (Sayagués et al. 2004).

Chromosomal Alterations

One of the most prominent chromosomal alterations described among meningiomas are whole or partial loss of 1p (22–33%) and chromosome 22. The pathways of intratumoral clonal evolution in benign meningiomas were different from those observed in meningiomas WHO grades II or III, suggesting that the tumors may progress along different genetic pathways (Sayagués et al. 2004). The combination of malignant histology, younger age, and monosomy 14 has been considered as the most relevant triad of adverse prognostic factors for recurrence of meningiomas (Espinosa et al. 2009). Losses of the chromosomes 9, 10, 14, 18, and del (1p36) were associated with a shorter recurrence-free survival (Maillo et al. 2007).

Parameters Influencing Outcome of Meningioma Surgery

The results of meningioma surgery correlate with the location of the tumors. Specific approaches have been described for skull base meningiomas (Bikmaz et al. 2007; Bambakidis et al. 2008). The significance of a total resection of these tumors is still under debate. Yet, meningiomas that occur outside of the skull base are more likely to have a higher MIB-1 labeling index and recur with a higher grade than those within the skull base, suggesting that non-skull base meningiomas may have a more aggressive biology (McGovern et al. 2009). Since most of the skull base meningiomas are slow growing grade I tumors, there is a tendency to perform a subtotal resection in cases in which severe adverse effects after radical surgery have to be suspected (Ausman 2006).

Tumor recurrence is another major clinical complication in meningiomas. The overall recurrence rate of meningiomas after 15 years ranged from 4% to 60% (Kalkanis et al. 2000). The extent of surgical resection has been considered as a prognostic factor because tumors that undergo a wider excision will be less likely to recur. Nevertheless, recurrences occur in 10–25% of cases undergoing a complete resection of the tumor (Espinosa et al. 2009). Even benign sporadic meningiomas sometimes show relatively rapid growth and may recur after total removal. Recurrence rates of these tumors after removal according to Simpson's grade I–III of 7.5% at 10 years and 9.3% at 20 years were reported (Nakasu et al. 2009).

Assessment of Outcome After Meningioma Surgery

Operative Mortality and Morbidity

Objective measures such as neurological symptoms, survival, and recurrence rates had traditionally served as the parameters by which meningioma surgery outcomes were assessed. Advanced technology concerning diagnosis, planning, and performance of meningioma surgery has resulted in a considerable decrease in severe neurological deficits. Several series with large numbers of patients with excellent outcome according to neurologic status and recurrence rate have been described (Black et al. 1998; Natarajan et al. 2007). Surgical complication rates were 7.0% and 8.8%, mortality rate was 1.8% in a group of elderly patients (older than 65 years), while no patient died in the group of younger patients (Black et al. 1998).

Yet, these objective outcomes are often influenced by patients' own perceptions of their individual experience. Often patients have strong concerns about the likelihood of returning to their previous level of functioning, both physically and emotionally (Kalkanis et al. 2000). In an editorial titled 'A revolution in skull base surgery: the quality of life matters!' James Ausman stated that it never made sense to him, that a slow-growing

skull base meningioma should be removed radically (Ausman 2006). He especially pointed out that the experience of the surgeon is a significant parameter according to results after skull base surgery. He described a sample of radically removed petrous ridge meningiomas with a complication rate as high as 89%. The annual number of skull tumors in the USA was calculated, excluding pituitary and acoustic tumors that might need surgery (Ausman and Slavin 1993). The result was that there were approximately 4,000 tumors each year – enough for each neurosurgeon in the USA to perform one procedure a year.

Quality of Life (QOL)

Although meningioma surgery offers the potential for complete physical cure, it raises concerns about postoperative neurological and psychological deficits (Taphoorn et al. 1992). It was emphasized that much of the extensive literature about outcome from complex neurosurgery focused on surgical approaches rather than outcome (Neil-Dwyer et al. 2000). While objective surgeons' assessments tend to suggest a favorable outcome, investigations of the patients' perspectives clearly shows adverse effects according to their QOL. Neurosurgical literature too often reflected the surgeon's perspective, concentrating on morbidity and mortality rather than on the perception of the patients. Specifically, taking note of features such as pain, mood, cognitive function, and the ability to cope with essential personal and family responsibilities.

Meningiomas located on the convexity, are usually accessible without significant problems. Most of them are well demarcated from the neural and vascular structures. In contrast, meningiomas arising from the skull base often are extensive and intimately associated with cranial nerves, vessels, and brainstem. Yet, a morbidity of 10% was found in patients after surgery of supratentorial convexity meningiomas (Sanai et al. 2009). The authors pointed that because of conservative recommendations for surgery of asymptomatic meningiomas and a higher frequency of radiation, these tumors are now

typically large in size. Nevertheless, following removal an uncomplicated clinical course is expected. Several studies used indicators such as analgesia or return to employment to evaluate QOL after surgery. Such approaches to QOL assessment have been criticized, because they neglect that QOL is a complex, subjective concept. Widely used outcome assessment scales show significant deficiencies and difficulties in its use, such as the failure to address activities of daily living and the ability to cope with family and social activities (Neil-Dwyer et al. 2001).

QOL has been described as a subjective, i.e. patient-rated concept, which contains more dimensions than just physical being or the ability to conclude that patients can lead an independent life. Evidence for the complex nature of QOL and the importance of patient input for its estimation has been supported by multiple studies (Kalkanis et al. 2000). An essential parameter is the effect of cognitive ability and its impact on family responsibilities, social interactions, and leisure pursuits. Various outcome assessment scales have been used. Increasing interest in parameters reflecting patients' viewpoints has led to a growing demand for standardized QOL-questionnaires. Several generic and health-specific questionnaires have been proposed to assess both functional status and the individual's perception of health. These instruments are similar in that each expresses the effects of medical care in terms that can be reported directly by a patient. Health-related QOL (HR-QOL) was defined as the individual's perception of their position in life in the context of the culture and value systems in which they live, and in relation to their expectations and concerns: a concept incorporating both functional status and the individual's perception of health (Hwang and Wang 2004). In several studies the Medical Outcome Study 36-Item Short-Form Health Survey (SF-36) was used (Ware and Sherbourne 1992).

Neuropsychological Parameters

Even subtle cognitive deficits can prevent symptom-free patients from returning to premorbid autonomy and occupation (Giovagnoli and

Boiardi 1994). It was concluded that neurocognitive parameters are significant indicators of QOL. Neuropsychological assessment using standardized tests gained more importance for evaluation of treatment benefit, efficacy and satisfaction with life (Krupp et al. 2009). Surveys included tests of cognitive functions, coping with illness, and satisfaction with life.

Structured Interviews

Also several questionnaires and tests have been extensively tested for their psychometric properties they may be insensitive to measure clinical responsiveness due to a lack of specific issues which are important to patients with specific diseases. In previous studies essential information was provided by structured interviews, offering patients the opportunity to describe their personal experience in own words. In particular, problems concerning social contact and the possibly existing desire for professional help were described in a more distinctive manner (Taphoorn et al. 1992; Krupp et al. 2009). In structured interviews topics are suggested and patients address it as they wish. These interviews provide more detailed items for exchanges with other people, perspective of life and fear of dependency, patients' impression of being a burden to caregivers and members of the family.

Outcome Including Different Parameters of Quality of Life

Only few studies have evaluated QOL after surgical treatment of meningiomas. In a retrospective study of 155 patients with a mean follow-up time of 33 months after meningioma surgery 80% reported being satisfied with their post-treatment quality of life (Kalkanis et al. 2000). Eighty-six percent of the patients reported that they could write, read, drive and return to work at their pre-morbid level of functioning, 87% described themselves as 'independent' and able to act on their own initiatives. The outcome of 82 patients of a non-selected series who

underwent meningioma resection was analyzed retrospectively using a postal survey (Mohsenipour et al. 2001). The majority of patients, about 60% in two different questionnaires, had a mild to moderate impairment of QOL. Twenty percent of the patients showed a moderate to severe impairment concerning the dimensions physical handicap, especially mobility and energy level. Most of these patients were older than 70 years and had antiepileptic medication. Analyses of different time intervals between surgery and survey did not reveal any significant effects on outcome variables. With the aim to determine outcome including QOL after surgery for petroclival meningioma 19 patients were assessed using GOS and the questionnaire SF-36 (Neil-Dwyer et al. 2000). Sixty-three percent of the patients had a good outcome, 16% a moderate, and also 16% a poor outcome, respectively, according to GOS. One patient died. Analysis of QOL using SF-36 showed that in all categories 39–72% of surviving patients were functioning below accepted norms. In view of the authors these results revealed while objective surgeons' assessment tend to suggest a favorable outcome, in-depth investigations of the patients' perspectives clearly showed their QOL to be adversely affected.

Some of the patients only achieved their pre-operative levels because of the considerable input of their caregivers and families. Few attention has been paid to the effect that the patient's disability has upon the caregivers' life and career. The QOL of patients after surgery of petroclival meningiomas was worse than that indicated in surgeon's reported results. The impact on the patient's caregiver may be profound, a burden perhaps not fully recognized by the surgeon. Fifty-six percent of the caregivers experienced a major change in life-style, 38% even had to cope with a major change with respect to their work (Lang et al. 1999).

Age of Patients

Whether there is an increased surgical risk in elderly patients remains a point of controversy. Black et al. (1998) found in their study no higher

morbidity or mortality of elderly patients, also with regards to their own control group of younger patients. In a multicenter prospective study of the National Surgical Quality Improvement Program of the USA a significant higher 30-day mortality of patients older than 70 years compared to younger subjects was found: 12.0% vs. 4.6%. After controlling for comorbidities and risk factors, perioperative mortality in elderly patients was three times that of younger patients (Patil et al. 2010).

Neurocognitive Functioning

Little is known about the effect of different treatment options on the frequency and severity of neurocognitive dysfunction of meningioma patients, a significant parameter in the evaluation of QOL. More information would improve understanding of patients complaints.

The few data available indicate that about 30% of patients after meningioma surgery experience severe long term cerebral sequelae, in most cases neurocognitive deficits (Van Nieuwenhuizen et al. 2007). It is unclear, however, whether this is due to pre-treatment brain damage by the tumor itself or due to treatment. No significant differences were found in patients who underwent surgery only, and patients who received additional radiotherapy at least 1 year before. Yet, patients of both groups had a significantly lower neurocognitive functioning than healthy controls. The most profound dysfunctions were seen in memory tasks. In another study particularly a lowered attention span and decreased executive functions were described after meningioma surgery (Tucha et al. 2003).

Different Dimensions of Quality of Life

Summarizing results presented in literature assessment of neuropsychological parameters using well-standardized psychometric instruments in combination with a disease-specific structured interview promised a comprehensive evaluation outcome after meningioma surgery.

In a retrospective study outcome of patients on an average of 15 months (standard deviation ± 3.6 months) after surgery of a supratentorial grade I meningioma in individual sessions in patients' homes was analyzed (Krupp et al. 2009). The survey included tests of cognitive performance, coping strategies, and satisfaction with life. Operative mortality of the cohort ($n=119$) was nil, operative morbidity was 13.4%, with minor deficits (GOS 5 or 4) in 10.1% and major deficits in 3.4% of cases respectively. Ninety-one patients (76%) participated in the study. A significant negative correlation between patient age and cognitive performance ($p<0.001$, goodness-of-fit-model, nonlinear regression), with a major decline beginning at the age of 55 years. Despite normal cognitive performance, 73% of younger patients (<55 years) compared with 20% of older patients ($p<0.001$) were not satisfied with life. Younger patients described their inability to accept to be hit by this severe disease as a young patient and were afraid of being considered handicapped. They reported of a strong feeling that their physical and mental abilities were not recognized, which resulted in lowered self-confidence and general performance. Sixty-two percent of these patients even felt that some persons of their social environment would not want to reach their premorbid performance level for several reasons. An unexpected effect of postoperative rehabilitation, a significant part of therapy, was found in this study. Nearly all of the younger patients tended to identify themselves with patients with major neurological deficits or emotional disturbances, finally feeling to be a member of this group of patients. Also in another study those patients who responded negatively to their current QOL tended to be younger both at the time of surgery and at interview. It was supposed that younger patients held unrealistic expectations as to their vulnerability and long-term recuperation (Kalkanis et al. 2000).

Assessing outcome of a cohort with an average patients' age of 61 years authors concluded it to be very difficult to distinguish between age- and disease-related effects in some of the measured parameters (Mohsenipour et al. 2001). Sparse information about specific coping strategies of

meningioma patients is available in literature. Depressive coping is more common among female patients and persons living as singles. Female patients also showed higher scores concerning spirituality (Krupp et al. 2009).

Conclusions

Until now few information about the different aspects of QOL after meningioma surgery is available. Studies with comprehensive assessment of outcome based on a retrospective design. Yet, summarizing results, there is convincing evidence that measurement of operative mortality and morbidity with description of neurological impairment does not provide valid evaluation of outcome after surgery. To prove the hypotheses of these surveys, prospective studies with larger cohorts and control groups are required. According to recommendation of current outcomes research programs, neuropsychological parameters as well as variables describing subjective estimation of QOL need to be integrated to comprehensively evaluate surgical procedures (Testa and Simonson 1996). According to the effort and complexity required to perform these evaluations a multicentre study should be aspired. Generally a participation of the caregiver of the patient in these studies has to be recommended. Some research outcomes groups demanded specific time consuming psychometric instruments and interviews to be performed repeatedly after certain time intervals. Yet, considering the disease and the average age of the patients, with a significant number of patients older than 70 years, the number of parameters which can be measured is clearly limited. Also the agreement to take part in such a strenuous survey probably will decline considering the statements of the patients in this study (Krupp et al. 2009).

Especially analyses of the different aspects of QOL has shown, that the frequency of deficits after meningioma surgery is significantly higher than presumed in studies measuring mortality and neurological impairment. Including patients' subjective experience of the functional consequences of disease and illness and realization

that several patients can preserve their living standards only by massive support of their caregivers reveals additional adverse effects on outcome. Yet, a major issue has to be considered: while it is argued that methods for measuring outcome should be refined, it approaches the counsel of perfection and if achieved may lead surgeons to operate on fewer patients (Neil-Dwyer et al. 2001). Also benign tumors like most of meningiomas may develop to life-threatening lesions, especially in case of growth or recurrence. Delay of surgery may lead to death of the patient. A balanced approach to high-risk, complex neurosurgery depends on the surgeon's personal experience, a critical appraisal of literature, and an understanding of relevant outcome and its effect upon the patient's life. Authors of a recent study concluded that the excellent QOL at the time of the long-term follow-up examination for patients after surgery of a petroclival meningioma warrants aggressive but judicious tumor resection, with or without radiosurgical treatment of tumor remnants (Natarajan et al. 2007). It was emphasized that by aggressive but not necessarily radical resection combined with prevention of medical complication the best results in meningioma surgery can be achieved (Black et al. 1998). Bearing in mind the possible and eventually complex deficits an experienced surgeon has to decide whether surgery is clearly indicated or another treatment should be preferred. The expertise of the surgeon is not reflected by the ability to perform the most complex approaches but in the ability to select the approach that affords both removal of the tumor and preservation of patient's quality of life (Samii and Gerganov 2008).

References

- Ausman JI (2006) A revolution in skull base surgery: the quality of life matters! *Surg Neurol* 65:635–636
- Ausman JI, Slavin KV (1993) Skull base surgery: too much, too little? *Surg Neurol* 40:437–438
- Bambakidis NC, Kakarla UK, Kim LJ, Nakaji P, Porter RW, Dasptit CP, Spetzler RF (2008) Evolution of surgical approaches in the treatment of petroclival meningiomas: a retrospective review. *Neurosurgery* 62 (Suppl 3):1182–1191
- Bikmaz K, Mrak R, Al-Mefty O (2007) Management of bone-invasive, hyperostotic sphenoid wing meningiomas. *J Neurosurg* 107:905–912
- Black P, Kathiresan S, Chung W (1998) Meningioma surgery in the elderly: a case-control study assessing morbidity and mortality. *Acta Neurochir (Wien)* 140:1013–1016
- Durand A, Labrousse F, Jouvet A, Bauchet L, Kalamaridès M, Menei P, Deruty R, Moreau JJ, Fèvre-Montange M, Guyotat J (2009) WHO grade II and III meningiomas: a study of prognostic factors. *J Neurooncol* 95:367–375
- Espinosa AB, Taberner MD, Maíllo A, Sayagués JM, Ciudad J, Merino M, Alguero MC, Lubombo AM, Sousa P, Santos-Briz A, Orfao A (2009) The cytogenetic relationship between primary and recurrent meningiomas points to the need for new treatment strategies in cases at high risk of relapse. *Clin Cancer Res* 12:772–780
- Giovagnoli AR, Boiardi A (1994) Cognitive impairment and quality of life in long-term survivors of malignant brain tumors. *Ital J Neurol Sci* 15:481–488
- Hwang JS, Wang JD (2004) Integrating health profile with survival for quality of life assessment. *Qual Life Res* 13:1–10
- Kalkanis SN, Quiñones-Hinojosa A, Buzney E, Ribaud HJ, Black PM (2000) Quality of life following surgery for intracranial meningiomas at Brigham and Women's Hospital: a study of 164 patients using a modification of the functional assessment of cancer therapy-brain questionnaire. *J Neurooncol* 48:233–241
- Krupp W, Klein C, Koschny R, Holland H, Seifert V, Meixensberger J (2009) Assessment of neuropsychological parameters and quality of life to evaluate outcome in patients with surgically treated supratentorial meningiomas. *Neurosurgery* 64:40–47
- Lang DA, Neil-Dwyer G, Garfield J (1999) Outcome after complex neurosurgery: the caregiver's burden is forgotten. *J Neurosurg* 91:359–363
- Maillo A, Orfao A, Espinosa AB, Sayagués JM, Merino M, Sousa P, Lara M, Taberner MD (2007) Early recurrences in histologically benign/grade I meningiomas are associated with large tumors and coexistence of monosomy 14 and del(1p36) in the ancestral tumor cell clone. *Neuro Oncol* 9:438–446
- McGovern SL, Aldape KD, Munsell MF, Mahajan A, Demonte F, Woo SY (2009) A comparison of World Health Organization tumor grades at recurrence in patients with non-skull base and skull base meningiomas. *J Neurosurg* 112:925–933
- Mohsenipour I, Deusch E, Gabl M, Hofer M, Twerdy K (2001) Quality of life in patients after meningioma resection. *Acta Neurochir (Wien)* 143:547–553
- Nakasu S, Fukami T, Jito J, Nozaki K (2009) Recurrence and regrowth of benign meningiomas. *Brain Tumor Pathol* 26:69–72
- Natarajan SK, Sekhar LN, Schessel D, Morita A (2007) Petroclival meningiomas: multimodality treatment and outcomes at long-term follow-up. *Neurosurgery* 60:965–979

- Neil-Dwyer G, Lang DA, Davis A (2000) Outcome from complex neurosurgery: an evidence based approach. *Acta Neurochir (Wien)* 142:367–371
- Neil-Dwyer G, Lang D, Garfield J (2001) The realities of postoperative disability and the carer's burden. *Ann R Coll Surg Engl* 83:215–218
- Patil CG, Veeravagu A, Lad S, Boakye M (2010) Craniotomy for resection of meningioma in the elderly: a multicenter, prospective analysis from the National Surgical Quality Improvement Program. *J Neurol Neurosurg Psychiatry* 81:502–505
- Rosenberg LA, Prayson RA, Lee J, Reddy C, Chao ST, Barnett GH, Vogelbaum MA, Suh JH (2009) Long-term experience with World Health Organization grade III (malignant) meningiomas at a single institution. *Int J Radiat Oncol Biol Phys* 74:427–432
- Samii M, Gerganov VM (2008) Surgery of extra-axial tumors of the cerebral base. *Neurosurgery* 62:1153–1166
- Sanai N, Sughrue ME, Shangari G, Chung K, Berger MS, McDermott MW (2009) Risk profile associated with convexity meningioma resection in the modern neurosurgical era. *J Neurosurg* 112:913–919
- Sayagués JM, Tabernero MD, Maíllo A, Espinosa A, Rasillo A, Díaz P, Ciudad J, López A, Merino M, Gonçalves JM, Santos-Briz A, Morales F, Orfao A (2004) Intratumoral patterns of clonal evolution in meningiomas as defined by multicolor interphase fluorescence in situ hybridization (FISH). *J Mol Diagn* 6:316–325
- Shen Y, Nunes F, Stemmer-Rachamimo A, James M, Mohapatra G, Plotkin S, Betensky RA, Engler DA, Roy J, Ramesh V, Gusella JF (2009) Genomic profiling distinguishes familial multiple and sporadic multiple meningiomas. *BMC Med Genomics* 2:42. doi: 10.1186/1755-8794-2-42
- Taphoorn MJ, Heimans JJ, Snoek FJ, Lindeboom J, Oosterink B, Wolbers JG, Karim AB (1992) Assessment of quality of life in patients treated for low-grade glioma: a preliminary report. *J Neurol Neurosurg Psychiatry* 55:372–376
- Testa MA, Simonson DC (1996) Assessment of quality-of-life outcomes. *N Engl J Med* 334:835–840
- Tucha O, Smely C, Preier M, Becker G, Paul G, Lange KW (2003) Preoperative and postoperative cognitive functioning in patients with frontal meningiomas. *J Neurosurg* 98:21–31
- Van Nieuwenhuizen D, Klein M, Stalpers LJ, Leenstra S, Heimans JJ, Reijneveld JC (2007) Differential effect of surgery and radiotherapy on neurocognitive functioning and health-related quality of life in WHO grade I meningioma patients. *J Neurooncol* 84:271–278
- Ware JE, Sherbourne CD (1992) The MOS 36-item short-form health survey (SF-36): conceptual framework and item selection. *Med Care* 30:473–483

Angelo Pichierri and Roberto Delfini

Abstract

Most parasellar meningiomas may eventually involve the cavernous sinus simply by following the natural anatomic pathways connecting these areas. If surgical removal is generally suggested for extracavernous masses, the optimal management of their intracavernous portion must take into account various clinical, pathological and surgical issues. Our algorithm will guide us through various options of treatment or it may suggest a multimodal approach to the patient. Surgical steps are discussed and divided in craniotomies, intracranial stages and intracavernous stages. Purely extracavernous approaches that leave the sinus unopened represent an optimal treatment for the major part of cavernous sinus meningiomas. Brainstem structures dramatically benefit from surgical decompression as do the cranial nerves. Decompressing and dissecting the tumour away from the optic pathways is crucial to lay the basis for visual function recovery and to accomplish a contingent radiotherapy or radiosurgery. Hence, the rational approach to these tumours consists of removal of the extracavernous portion of the tumour combined, in selected cases, with radiotherapy/radiosurgery to the intracavernous residuum. Aggressive attempt to achieve radical removal should be limited to patients with a complete loss of oculomotor function and to meningiomas with aggressive behaviour (histology, tendency to recur).

Introduction

The cavernous sinus is a neurovascular structure whose neurovascular relationships with the neighbouring regions are complex (Rothon 2002;

Dalgıç et al. 2010). Consequently, most parasellar meningiomas may eventually involve the cavernous sinus simply by following the natural anatomic pathways connecting these areas: examples of this behaviour are some clinoid, sphenoorbital, Meckel's cave, sphenopetroclival, tuberculum sellae or also sphenoid wing and middle fossa meningiomas (Bogaev and Sekhar 2006; Fukushima 2009; Pichierri et al. 2009; Sade and Lee 2008). In our experience, more than half of

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the patients with meningiomas involving the cavernous sinus harboured lesions larger than 5 cm in diameter (Pichierri et al. 2009). Meningiomas originating primarily within the cavernous sinus are by far less frequent (about 8%) (Heth and al-Mefty 2003; Pichierri et al. 2009).

Why is it so important to distinguish primary from secondary meningiomas? Even though the natural history of these lesions is still under investigation, many series show a slow growth pattern for intracavernous meningiomas, while meningiomas with secondary extension increase their volume at the rate of the specific site of origin (generally faster) (Bloss et al. 2010; Nakasu et al. 2009; Skeie et al. 2010; Sughrue et al. 2010). Moreover, primary cavernous sinus meningiomas are small and usually cause a varying degree of oculomotor dysfunctions, although they may be asymptomatic in a not negligible percentage of cases (about 9%) (Pichierri et al. 2009). Secondary meningiomas are often large and cause compression of the central nervous system structures. So, if surgical removal is generally suggested for the latter cases, whether or not it is worth removing a purely intracavernous benign tumour is still debatable (Lee et al. 2002; Long 2001; Pamir et al. 2005; Pichierri et al. 2009; Spiegelmann et al. 2010). Intracavernous surgery causes morbidity and may fail to eliminate the risks of recurrence even in experienced hands (Heth and al-Mefty 2003). This discussion becomes even more complex if we consider that radiation therapy series report good results in the control of lesions of limited volume (Brell et al. 2006; Kuo et al. 2004; Lee et al. 2002; Milker-Zabel et al. 2006; Skeie et al. 2010; Spiegelmann et al. 2010).

In meningiomas with secondary cavernous sinus involvement (MSCSI), portions of the tumor external to the cavernous sinus are managed excellently by cranial base surgery (Couldwell et al. 2006; Day 2000; Heth and al-Mefty 2003). As the extracavernous part of the tumour is often the cause of the worst preoperative symptoms, the current tendency is to take only this component away.

Surgical management becomes problematic when the MSCSI has already caused oculomotor dysfunctions. One option could be the

intra-operative evaluation of their consistency: in soft meningiomas, decompression of the oculomotor nerves at their entry points into the cavernous sinus can be safely performed, thus relieving the nerves without trying to dissect the tumour along with the intracavernous portions of the nerves (Couldwell et al. 2006). However, anatomical preservation often does not mean a functional preservation, since feeders and venous drainage of these nerves are critically fragile and complex (d'Avella et al. 2008; Dalgıç et al. 2010). The inferolateral trunk and the meningohypophyseal artery which feed the nerves through multiple and small collaterals may be encased in the tumour, making it extremely difficult to distinguish and preserve these functional collateral vessels within the cavernous sinus (Dalgıç et al. 2010). Arterial or venous vascular damage will cause the same amount of postoperative worsening of the nerve function as if we had directly damaged the nerves.

The best scenario would be to diagnose these tumours before they reach the cavernous structures, in order to allow a safer and more complete surgical removal. Unfortunately, this is sometimes not possible for various reasons:

1. presenting symptoms may be so subtle that patients underestimate their condition;
2. diagnoses are made later because physicians don't immediately correlate symptoms with an intracranial tumor. In the mean time, the tumor gets bigger and expands into the neighbouring compartments;
3. tumours may be asymptomatic for a long time, until they reach a critical mass and the intracranial mechanisms of space compensation exhaust.

Historical Landmarks

While in 1900 Frazier sentenced that "The cavernous sinus is not within the realm of the surgeon's knife", in 1965 Parkinson performed the first successful direct approach to the cavernous sinus for a carotido-cavernous fistula. In 1978, Trobe restated recommendations for non-surgical

treatment of intracavernous meningiomas and aneurysms. The revolution of techniques and approaches has been pioneered by Dolenc in 1980 so that intracavernous surgery was re-evaluated. In 1994, DeMonte and al-Mefty published their series of cavernous sinus meningiomas with 76% of total resections (10% of these recurred) and 13% incidence of new oculomotor deficits; DeJesus and Sekhar showed their results in 1996: 61% of total resections whose 10% recurred. O'Sullivan shared his experience in 1997: 20% of total resections (25% of these recurred) and 20% incidence of new oculomotor deficits. Various Authors (e.g. Larson, Sen, Dolenc) underlined the local aggressiveness of these meningiomas with histological studies in which they showed the infiltrative tendency of the tumour toward the nerves, the arteries and the connective planes. In 2002, Nicolato reported gamma-knife radiation therapy alone as first line treatment for small meningiomas. Abdel-Aziz found an equal rate of recurrences after total and subtotal removal in their series published in 2004. Radiosurgical effectiveness in the treatment of the remnant with lower morbidity has been further pointed out by Pamir in 2005. In 2009, Fukushima stated that radical surgery is feasible in only 10% of cavernous sinus meningiomas (medium size and soft).

Many atlases and manuscripts have been dedicated to the description of various extensive skull base approaches for the management of cavernous sinus meningiomas (Bogaev and Sekhar 2006; Dalgıç et al. 2010; Fukushima 2009). They allowed enhanced exposures and manoeuvrability inside the deeper structures while minimizing brain retraction, confirming the need to obey the old saying: "more bone, less brain". Further steps of dissection techniques described "safe" corridors to the cavernous sinus structures. They made it possible to reach the tumour without directly damaging the neurovascular structures, but some critical problems remained: the localization of the structures in this small and diseased compartment, the difficult dissection of chronically compressed and stretched nerves which adhered very tightly to the tumor and the damage of small vessels connected with intracavernous dissection.

Also, microscopic studies of cavernous sinuses affected by meningiomas showed a deep tumoral infiltration of connective intracavernous tissue, nerves and intracavernous vessels' walls in almost 50% of cases. Aggressive dissection sometimes leads to intraoperative complications (nerve transection, internal carotid artery ruptures) and late complications (delayed cranial nerve damage, postoperative internal carotid artery pseudo-aneurysms). These findings limited the concept of a truly radical excision of the tumor and the difficulties in achieving such a total removal where a limit between the tumour and the neurovascular structures is completely lacking. The possibility of following the patient by means of high quality MRI and the availability of radiation therapy led to a re-definition of the therapeutic role of surgery. The primary goal became to preserve neurological function rather than to eradicate the tumor.

Natural History

Little is known about the natural history of skull base meningiomas (Abdel-Aziz et al. 2004; Long 2001; Pichierri et al. 2009). In our own experience, we often found unpredictable behaviors: spontaneous or "radiation therapy-induced" stability of the residue, progressions even after radiation therapy, recurrences of WHO I meningiomas after a so-called "total removal". In the most recent series, tumour progressions ranged from 6% to 20% (Abdel-Aziz et al. 2004; Couldwell et al. 2006; Day 2000; Dufour et al. 2001; Heth and al-Mefty 2003; Long 2001). In our experience, malignant meningiomas (WHO III & IV) produced recurrences within weeks, but all the patients with histologically aggressive meningiomas also developed recurrences. Regarding this issue, the WHO classification seems to be deficient: in fact, histologically proven benign meningiomas show a wide range of growth rates. It is worth using the MIB-1 index as an adjunctive method for predicting the risk of recurrence (Nakasu et al. 2009). Some Authors even report an equal rate of recurrences after total and subtotal removal of benign cavernous sinus

meningiomas: if a tumor shows a high histological proliferative index, it may recur even after a so-called “total removal” (Abdel-Aziz et al. 2004). The only empirical *preoperative* way to predict the biological behavior of these meningiomas is the wait & see policy: radiological or clinical evidences of a rapid growth over time suggests an aggressive tumour. Also, extensive cancer invasion into the orbital compartment through the superior orbital fissure could be a sign associated with a higher risk of recurrence.

There is wide evidence of the effectiveness of stereotactic radiation therapy in the treatment of subtotally resected meningiomas. Nowadays, recent technical advances and the identification of safe irradiation doses allow fractionated stereotactic radiotherapy (FSRT) and radiosurgery (RS) to be used safely and efficiently, even in the cavernous sinus region, despite the closeness of the tumor to the cranial nerves and internal carotid artery. The only real drawback of the treatment seems to be the closeness of lesions to the optic pathways, the diencephalon and the brainstem (Brell et al. 2006; Dufour et al. 2001; Kuo et al. 2004; Maruyama et al. 2004; Milker-Zabel et al. 2006). In the literature regarding radiotherapy in cavernous sinus meningiomas, studies about the cognitive status of these patients is lacking. Nevertheless, many series dealing with the outcomes in other tumors state that cognitive deterioration is a concrete risk of radiation therapy (Duffner 2010; Douw et al. 2009; Welzel et al. 2008). This aspect must be clarified prior to considering the amount of radiation therapy safe for normal brain functioning. The published series' suggest that the tumour can sometimes be reduced and, in most of cases (81–88%), stabilized with an acceptable cranial nerve morbidity 5–6%) (Dufour et al. 2001; Lee et al. 2002; Nicolato et al. 2002; Pichierri et al. 2009). Risks of recurrence exist even after irradiation (Dufour et al. 2001; Pichierri et al. 2009): if a second surgery is indicated in these cases, resection of the extra-cavernous component of the remnant becomes far more challenging.

Clinical Outcome

Dissecting away the intracavernous tumour does not appreciably improve the function of the cranial nerves involved. On the other hand, pre-existing deficits often worsen or new deficits arise. Furthermore, the need to free the encased internal carotid artery often requires an extra-intracranial bypass which is burdened by additional risks. The incidence of new cranial nerve deficits or worsening of preoperative deficits, destined to be permanent, is lowered by avoiding dissection of the intracavernous portion of the tumour (Couldwell et al. 2006; Pichierri et al. 2009). Dissecting the tumour from extracavernous structures has a favourable risk-benefit ratio: the presence of an arachnoid plane between the tumour and the extracavernous segment of the nerves is a possible reason from their resistance to permanent damage during dissection. Within the cavernous sinus, the absence of a true arachnoid interface may increase the likelihood of mechanical nerve injury during dissection thus compromising the ability of the nerve to recover; surgical damage to the small vascular structures directed to the nerves is another important factor for early and delayed nerve injury (Couldwell et al. 2006). Interestingly, the trigeminal nerve actually benefits from the intracavernous dissection of the tumour, probably because of its peripheral position inside the cavernous compartment and its minor delicacy in comparison to the other intracavernous nerves. Also, decompression of the II c.n. gives immediate improvements of visual function (Couldwell et al. 2006; Sade and Lee 2008; Pichierri et al. 2009). A III c.n. paralysis is often as morbid as a moderate III c.n. paresis. However, some oculomotor improvement is possible when there is a residual function; a complete paralysis is almost always permanent. No attempt to totally remove the tumour should be made if we think we are provoking a III c.n. paralysis with our surgical manoeuvres. This is even truer if we consider that the remnant can be satisfactorily irradiated.

General Approach to Cavernous Sinus Meningiomas

Our algorithm is showed in Fig. 19.1. Primarily, we differentiate patients with objective clinical signs from those with non-impairing subjective symptoms (e.g. a mild and occasional headache). In the latter group, if the main portion of the tumour is intracavernous or if the patient is more than 70 years-old, we only suggest a 6 month clinico-radiological follow-up. FSRT may be used as a first line treatment in this subgroup of patients, if the good long-term results of radiation therapy will be confirmed. A critical discussion with the patient is crucial if he is younger and if the main portion of the tumour develops into the extracavernous space: what we know about the natural history of the disease and the Pro-Cons of the surgical intervention should be explained. The patient must know how long the enlargement of the tumour could take to cause a neurological deterioration so that he may eventually choose to maintain his good oculomotor function (Months?

Years? Half-decades?) until the tumour itself will start damaging the nerves. We think that it is not possible to talk about prevention of neurological deficits in this specific kind of meningiomas whose current surgical treatment is burdened by a 19–86% incidence of new oculomotor nerve dysfunction (Abdel-Aziz et al. 2004). Nevertheless, we address the patient to conservative surgery if he agrees.

Surgical treatment is suggested without any doubt if the patient already presents an oculomotor dysfunction or symptoms from compression of extracavernous structures (e.g. visual loss, hemiparesis, obstructive hydrocephalus, etc.). If the tumour consistency is favourable, a decompression of the nerves at their entry points into the cavernous sinus and at the superior orbital fissure can effectively improve their functionality at long term follow-up and make the compartment less rigid and more compliant to the eventual initial regrowth of the residue. An attempt at radical removal should be reserved for patients with a complete ophthalmoplegia or with a long-lasting, severe ophthalmoparesis. We previously

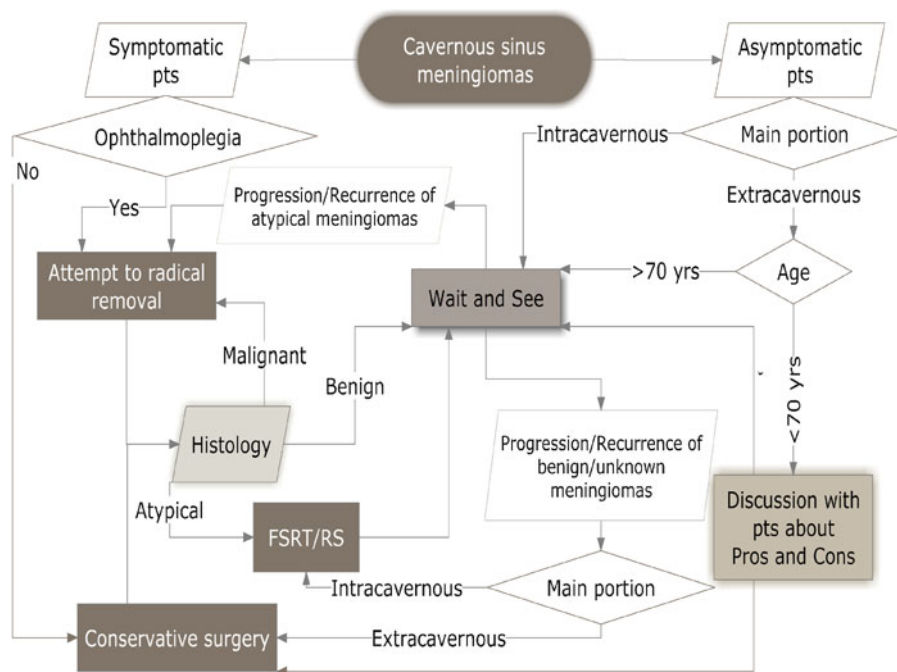


Fig. 19.1 Algorithm of treatment for patients with cavernous sinus meningiomas

discussed that total removal fails to prevent recurrences, but succeeds in extending progression-free survival. In these patients the chances of improvement are so low that the prolongation of the progression-free survival should be the target of the treatment regardless of the cost of cavernous sinus exenteration, eventually with an extra-intracranial high-flow bypass. However, management of internal carotid artery encasement in patients who don't pass the closure tests remains a delicate issue, especially in the age group. If a bypass is too risky, one option could be a tailored resection around the artery leaving a consistent layer of tumour all around its wall in order not to weaken it by dissection and to avoid intra-operative or delayed rupture of the carotid. In these cases we obviously wouldn't achieve a total resection, but if the histology of the tumour shows a low MIB-1 index the remnant could be checked periodically for volume changes using MRI; if the MIB-1 is high it could be irradiated or a bypass procedure could be planned.

Histology is crucial for all the patients who undergo surgery. A wait and see policy can be applied if a benign meningioma is confirmed (over 94% of cases) even when a partial removal has been achieved: in fact, we rely on the slow-growing nature of these lesions. Extracavernous progressions should be treated by further surgery if symptomatic. Intracavernous and asymptomatic extracavernous progressions may be treated with radiation therapy (FSRT/RS). In the rare atypical meningiomas, radiation therapy is always performed after a few weeks to delay the otherwise certain progression. If a progression occurs anyway, it represents a delicate issue: an attempt to perform radical removal is burdened by higher difficulties because the degenerative changes caused by radiation therapy upset the anatomy and lead much more easily to damage of vascular and nervous structures. We must be ready to perform a high-flow bypass if the internal carotid artery is fused with the tumour or its wall is inadvertently damaged. In patients with malignant meningiomas, exenteration of the cavernous sinus and carotid artery, eventually with a vascular bypass, should be performed during the same

hospitalization: in our series, patients with malignant meningiomas had a mean progression-free survival of 2 months and died.

Surgical Approaches to Cavernous Sinus Meningiomas (Bone)

As these meningiomas may arise from and extend to different cranial fossae, various approaches have been proposed to reach the cavernous sinus structures.

Pterional (Frontotemporal) Approach

In patients with a flat middle fossa, where the zygomatic arch does not interfere with its tangential exposure, the fronto-temporal approach is sufficient for controlling the cavernous sinus and the middle fossa floor. As for the temporal muscle, the two layers technique may improve exposure: the superficial temporal fascia is elevated from the muscle together with the galea; the muscle is cut to be reflected downward leaving a cuff on the superior temporal line. It is important to completely detach the temporal muscle from the zygomatic process of the frontal bone in order to better mobilize and transpose it inferiorly. In some cases, this two layers technique may compromise a really tangential view of the middle floor fossa if the zygoma is not removed. On the other hand, the single layer technique (with anterior transposition of the muscle) has the disadvantage of keeping all this bulky muscle above the greater sphenoid wing and anterior temporal squamous bone, thus interfering with a tangential view of the superior orbital fissure and anterior cavernous sinus region.

To take full advantage of this superficial opening, the bone work should continue deeper. Removal of the lesser sphenoid wing, drilling of the bone around the superior orbital fissure and the rotundum foramen will release and expose the superior orbital fissure and V2. These procedures are performed at various degree of completeness according to the morphology

of the lesion. The anterior clinoidectomy requires a special mention. This is a critical step for decompressing the optic nerve and gaining room for the further steps. Various techniques have been described (al-Mefty et al. 2008; Chang 2009; Noguchi et al. 2005; Takahashi et al. 2004; Yang et al. 2006). CT scans are important for choosing between an extradural or intradural removal of the ACP (Bogaev and Sekhar 2006). We prefer to detach the ACP extradurally using with the Dolenc's technique, but extradural removal can be very tricky and unsafe when the anterior clinoid process is long or when complex osseous bridges are present. The use of the bone CUSA make this step easier and safer both for the optic nerve (just medial to the anterior clinoid process), the internal carotid artery (below to the anterior clinoid process) and the III c.n. (lateral to the anterior clinoid process). The fronto-temporal dura is then opened and reflected towards the orbital side of the surgical field; the Sylvian fissure is opened widely, minimizing the tension caused by retraction of the temporal and frontal lobes. The superior orbital fissure can be safely dissected and decompressed using this approach if the tumour does not extend too far into the orbital compartment. Cutting the falciform ligament obtains early decompression of the optic nerve, completed by the unroofing of the optic canal. Clinoidocavernous meningiomas are optimally removed through this window.

Temporal Approach

It is a reduction of the previous approach, suitable for smaller lesions which extend laterally into the middle fossa and without a significant extension toward the superior orbital fissure and the optic region.

Cranio-Zygomatic Approach

Removing part of the zygomatic bone allows a really tangential view of the concave middle fossae. On the basis of preoperative MRI, we can

predict whether a zygomatic osteotomy is needed. The need to perform a zygomatic osteotomy depends on the individual conformation of the middle cranial fossa: the variable concavity of the middle fossa can be calculated on MRI. We consider the distance between the deepest point in the middle fossa and a line passing from the lowest lateral point on the temporal lobe and the cerebro-mesencephalic sulcus. Patients require zygomatic osteotomy if the distance is more than 1.5 cm (Ardeshiri and Winkler 2006; Delfini et al. 2009). The removal of the zygomatic arch is also indicated for lesions that extend from the middle cranial fossa to the infratemporal cranial fossa (Delfini et al. 2009). To be tangential to the superior orbital fissure as well, care must be paid to remove part of the zygomatic process of the frontal bone together with the zygoma. The zygomatic bone is not detached from its masseter muscle attachments, but simply retracted inferiorly together with the temporal muscle.

Cranio-Orbito-Zygomatic Approach

Removal of the posterior part of the orbital roof and the lesser wing of the sphenoid bone after a pterional approach is generally sufficient for controlling a mild orbital projection of the tumour; if the tumour extends into the orbital compartment a cranio-zygomatic approach, removal of the posterior orbital roof and lesser wing provides additional room for managing this portion. Therefore, in the majority of cases, it is advisable to flatten and thin down the orbital wall and not to perform total orbital decompression (Fukushima 2009).

Removal of the lateral and superior pillars of the orbit is only needed if the tumour overtly involves the orbital content. We perform the COZ approach in three steps: firstly the zygomatic bone is transposed inferiorly (as previously described), then a fronto-temporal craniotomy is made. Finally, the orbital osteotomy is made using an oscillating saw. The posterior parts of the superior and lateral orbital walls are removed piecemeal.

Combined Approaches

In patients in whom a consistent portion of the tumour lies in the posterior fossa, we prefer to stage surgery: middle fossa/posterior fossa approaches; bone work of complex combined approaches (e.g. posterior petrosal)/intradural approach. Only for tumours in the posterior fossa requiring a presigmoid approach, it may be worth performing a single stage posterior petrosal approach. Splitting the operation into two stages is beneficial for both the surgeon and the patient. The surgeon arrives fresh and concentrated at the most delicate part of the operation which is generally the last one, when the interface between the tumour and the neurovascular structures is reached. Although the patient has to undergo two operations, he/she will never be extubated between the two procedures which must be planned for two consecutive days, unless complications arise after the first operation. The majority of patients with sphenopetroclival meningiomas still maintain good hearing; therefore, the retrolabyrinthine combined supra-/infra-/trans-tentorial approach must be performed through the presigmoid and subtemporal dural incision (staged surgery) (Fukushima 2009).

Surgical Approaches to Cavernous Sinus Meningiomas (Intracranial Stage)

After craniotomy, both extra- and intradural approaches have been described for reaching and managing different portions of the cavernous sinus. As the major part of these meningiomas are medium or large size tumours, it is often necessary to combine intra and extradural approaches (Bogaev and Sekhar 2006; Couldwell et al. 2006; Fukushima 2009; Dalgıç et al. 2010).

Extradural Approaches

The *inferior approach* aims at the control of the petrous internal carotid artery and the inferolateral portion of the cavernous sinus. Dissection follows

an anatomical interdural route between the temporal dura and the middle fossa and, later, the Meckel's cave dura; the *anterolateral approach* exposes the superior orbital fissure and V2 and the superior and anterior portions of the cavernous sinus. The two approaches can be easily combined to gain full control of the cavernous sinus.

Intradural Approaches

The major part of these lesions need intradural approaches after various steps of extradural bone work, tailored on the basis of tumour morphology and aimed at the decompression of the optic canal and superior orbital fissure (cfr. Previous section).

Superior approach: it is the ideal extension of the anterolateral extradural approach. After the extradural bone decompression, the dura propria of the optic nerve and the rings of the internal carotid artery are opened. The dura mater is opened in a T-shaped fashion along the fronto-temporal base, and an incision is made along the axis of the Sylvian fissure to end between the optic nerve and the internal carotid artery. The fronto-temporal dura is retracted over the orbital compartment. The Sylvian fissure is widely opened. The sinus is initially approached through the clinoid (anteromedial or Dolenc) triangle.

Lateral approach: the frontotemporal dura is opened as mentioned in the superior approach. In large meningiomas with subtemporal projection, the tumour itself generally creates a good corridor, keeping the temporal lobe away from the surgical field. The lesion is debulked and the lateral wall of the cavernous sinus is exposed. The sinus is initially approached through the lateral (Parkinson) triangle. The postero-inferior triangle (just posterior to the lateral triangle), allows exposure of the sixth nerve at Dorello's canal, useful when the nerve cannot be localized in the cavernous sinus. Another way to localize the sixth nerve is by exploration of the superior orbital fissure, where it runs just superior and medial to V1. It is worth mentioning that the VI nerve is often (50% of cases) split into rootlets

(up to five); in this case localization is much more difficult. The oculomotor (supero-lateral or Fukushima) triangle, can be used to expose the medial compartment of the cavernous sinus, the meningohipophyseal trunk of the internal carotid artery and the interpeduncular fossa. In meningiomas with more complex morphology, this approach represents the ideal further step of a superior approach.

Extracavernous Decompression

As mentioned above, the optic nerve should be decompressed as soon as possible after its exposure, in order to avoid traction caused by manipulation of the tumour during surgery. The falciform ligament is cut and the dura propria of the nerve is opened (Bogaev and Sekhar 2006; Couldwell et al. 2006; Fukushima 2009; Pichierri et al. 2009). The anterior clinoid and the optic strut are eventually removed. The oculomotor nerve is decompressed at its entry point into the cavernous sinus; transversal cuts of the lateral wall of the sinus can be done along with the triangles of Parkinson (lateral) and Fukushima (superolateral) to relieve tension on the V1 and IV cranial nerves. The VI nerve would probably benefit from this manoeuvre too (Couldwell et al. 2006). The superior orbital fissure is also opened along with its dural sheath to give space to the nerves.

Intracavernous Steps

We advocate intracavernous operations when a total removal of the lesion is required, i.e. in ophthalmoplegic patients. Nevertheless, if some cleavage is maintained between the tumour and the nerves, care must be paid not to interrupt their critical vascular supply from the internal carotid artery. If the tumour infiltrates the nerves, sinus exenteration is indicated. The V1 c.n. often hides some portions of the tumour: it is important to explore the region by displacing the V1 c.n. inferiorly. If the tumour goes into Meckel's cave, the interface between the tumour and the Gasserian

ganglion must be carefully checked because this structure seems to be prone to invasion despite the lack of any neurological deficits. A strong limiting factor to a total removal is the eventual infiltration of the thin dura of the pituitary fossa with a spread of the tumour to the contralateral cavernous sinus.

The preoperative balloon occlusion test, intraoperative EEG, SSEPs, MEP and cervical carotid exposure must be feasible when an intracavernous dissection of the tumour is planned. The preoperative criteria for establishing internal carotid artery involvement are not entirely reliable as it can be present even in cases without angiographic evidence of narrowing of the internal carotid artery. Therefore, internal carotid artery encasement must be checked intraoperatively during dissection. In encasing tumours, if the patient does not tolerate internal carotid artery closure and if a bypass is therefore decided, a high-flow intra-extracranial bypass using the external/internal cervical carotid artery to the temporal M2 saphenous vein graft technique.

Conclusions

Purely extracavernous approaches that leave the sinus unopened represent an optimal treatment for the major part of cavernous sinus meningiomas because they reduce the risks of permanent postoperative morbidity without significantly higher risks of recurrence/progression. Brainstem structures dramatically benefit from surgical decompression as do the cranial nerves when freed from the tumour at their entry points into the cavernous sinus and when a superior orbital fissure decompression is performed. Decompressing and dissecting the tumour away from the optic pathways is crucial to lay the basis for visual function recovery and to accomplish a contingent radiotherapy or radiosurgery. Improvement of ophthalmoparesis and of trigeminal disease, regardless of the site of origin, is less evident and tends to stabilize. Hence, the rational approach to these tumours consists of removal of the extracavernous portion of the tumour combined, in selected cases, with radiotherapy/radiosurgery to the

intracavernous residuum. A low postoperative morbidity can be guaranteed in such patients, at the cost of living with an intracavernous remnant whose progression can be observed and eventually stabilized by radiation therapy for several years at least. Aggressive attempt to achieve radical removal – with or without extracranial bypass – should be limited to patients with a complete loss of oculomotor function and to meningiomas with aggressive behaviour (histology, tendency to recur). The risk factors for long-term recurrence are currently unpredictable. The presence of some characteristics – orbital invasion and histological aggressiveness – increases the likelihood of tumour recurrence regardless of the surgical strategy used.

References

- Abdel-Aziz KM, Froelich SC, Dagnew E, Jean W, Breneman JC, Zuccarello M, van Loveren HR, Tew JM Jr (2004) Large sphenoid wing meningiomas involving the cavernous sinus: conservative surgical strategies for better functional outcomes. *Neurosurgery* 54(6):1375–1383
- al-Mefty O, Kadri PA, Hasan DM, Isolan GR, Pravdenkova S (2008) Anterior clivectomy: surgical technique and clinical applications. *J Neurosurg* 109(5):783–793
- Ardeshiri A, Winkler PA (2006) Subtemporal approach to the tentorial incisura: normative morphometric data based on magnetic resonance imaging scans. *Neurosurgery* 58(1 Suppl):ONS22–ONS28
- Bloss HG, Proescholdt MA, Mayer C, Schreyer AG, Brawanski A (2010) Growth pattern analysis of sphenoid wing meningiomas. *Acta Neurochir* 152(1):99–103
- Bogaev CA, Sekhar LN (2006) Cavernous sinus tumors. In: Fessler RG, Sekhar LN (eds) *Atlas of neurosurgical techniques – brain*. Thieme, New York, pp 633–653
- Brell M, Villa S, Teixidor P, Lucas A, Ferran E, Marin S, Acebes JJ (2006) Fractionated stereotactic radiotherapy in the treatment of exclusive cavernous sinus meningioma: functional outcome, local control, and tolerance. *Surg Neurol* 65(1):28–33
- Chang DJ (2009) The “no-drill” technique of anterior clinoidectomy: a cranial base approach to the paraclinoid and parasellar region. *Neurosurgery* 64(3 Suppl):96–105; discussion 105–106
- Couldwell WT, Kan P, Liu JK, Apfelbaum RI (2006) Decompression of cavernous sinus meningioma for preservation and improvement of cranial nerve function. *J Neurosurg* 105:148–152
- Dalgıç A, Boyacı S, Aksoy K (2010) Anatomical study of the cavernous sinus emphasizing operative approaches. *Turk Neurosurg* 20(2):186–204
- d’Avella E, Tschabitscher M, Santoro A, Delfini R (2008) Blood supply to the intracavernous cranial nerves: comparison of the endoscopic and microsurgical perspectives. *Neurosurgery* 62(5 Suppl 2):ONS305–ONS310; discussion ONS310–ONS311
- Day JD (2000) Skull base surgical techniques for large sphenocavernous meningiomas: technical note. *Neurosurgery* 46:754–760
- Delfini R, Pichierri A, Avella E (2009) Tumors of the middle fossa. In: *Practical handbook of neurosurgery*, vol 2. Springer, Wien/New York, pp 99–114
- Douw L, Klein M, Fagel SS, van den Heuvel J, Taphoorn MJ, Aaronson NK, Postma TJ, Vandertop WP, Mooij JJ, Boerman RH, Beute GN, Sluimer JD, Slotman BJ, Reijneveld JC, Heimans JJ (2009) Cognitive and radiological effects of radiotherapy in patients with low-grade glioma: long-term follow-up. *Lancet Neurol* 8(9):810–818
- Duffner PK (2010) Risk factors for cognitive decline in children treated for brain tumors. *Eur J Paediatr Neurol* 14(2):106–115
- Dufour H, Muracciole X, Metellus P, Regis J, Chinot O, Grisoli F (2001) Long-term tumor control and functional outcome in patients with cavernous sinus meningiomas treated by radiotherapy with or without previous surgery: is there an alternative to aggressive tumor removal? *Neurosurgery* 48(2):285–294
- Fukushima T (2009) Management of cavernous sinus lesions. In: *Practical handbook of neurosurgery*, vol 2. Springer, Wien/New York, pp 129–154
- Heth JA, al-Mefty O (2003) Cavernous sinus meningiomas. *Neurosurg Focus* 14(6):1–9
- Kuo JS, Yu C, Giannotta SL, Petrovich Z, Apuzzo ML (2004) The Leksell gamma knife model U versus model C: a quantitative comparison of radiosurgical treatment parameters. *Neurosurgery* 55(1):168–172; discussion 172–173
- Lee JY, Niranjan A, McInerney J, Kondziolka D, Flickinger JC, Lunsford LD (2002) Stereotactic radiosurgery providing long-term tumor control of cavernous sinus meningiomas. *J Neurosurg* 97:65–72
- Long DM (2001) The treatment of meningiomas in the region of the cavernous sinus. *Childs Nerv Syst* 17(3):168–172
- Maruyama K, Shin M, Kurita H, Kawahara N, Morita A, Kirino T (2004) Proposed treatment strategy for cavernous sinus meningiomas: a prospective study. *Neurosurgery* 55(5):1068–1075
- Milker-Zabel S, Zabel-du Bois A, Huber P, Schlegel W, Debus J (2006) Fractionated stereotactic radiation therapy in the management of benign cavernous sinus meningiomas. *Strahlenther Onkol* 182: 635–640
- Nakasu S, Fukami T, Jito J, Nozaki K (2009) Recurrence and regrowth of benign meningiomas. *Brain Tumor Pathol* 26(2):69–72

- Nicolato A, Foroni R, Alessandrini F, Bricolo A, Gerosa M (2002) Radiosurgical treatment of cavernous sinus meningiomas: experience with 122 treated patients. *Neurosurgery* 51:1153–1159; discussion 1159–1161
- Noguchi A, Balasingam V, Shiokawa Y, McMenomey SO, Delashaw JB Jr (2005) Extradural anterior clinoidectomy. Technical note. *J Neurosurg* 102(5):945–950
- Pamir MN, Kilic T, Bayrakli F, Peker S (2005) Changing treatment strategy of cavernous sinus meningiomas: experience of a single institution. *Surg Neurol* 64(Suppl 2):S58–S66
- Pichierri A, Santoro A, Raco A, Paolini S, Cantore G, Delfini R (2009) Cavernous sinus meningiomas: retrospective analysis and proposal of a treatment algorithm. *Neurosurgery* 64:1090–1101
- Rothman AL Jr (2002) The cavernous sinus, the cavernous venous plexus and the carotid collar. *Neurosurgery* 51(Suppl 1):375–410
- Sade B, Lee JH (2008) Cavernous sinus meningiomas: conservative surgical management. In: Lee JH (ed) *Meningiomas*. Springer, Wien/New York, pp 389–392
- Skeie BS, Enger PO, Skeie GO, Thorsen F, Pedersen P (2010) Gamma knife surgery of meningiomas involving the cavernous sinus: long-term follow-up of 100 patients. *Neurosurgery* 66(4):661–668
- Spiegelmann R, Cohen ZR, Nissim O, Alezra D, Pfeffer R (2010) Cavernous sinus meningiomas: a large LINAC radiosurgery series. *J Neurooncol* 98(2):195–202
- Sughrue ME, Rutkowski MJ, Aranda D, Barani IJ, McDermott MW, Parsa AT (2010) Treatment decision making based on the published natural history and growth rate of small meningiomas. *J Neurosurg* 30:1–7
- Takahashi JA, Kawarazaki A, Hashimoto N (2004) Intradural en-bloc removal of the anterior clinoid process. *Acta Neurochir (Wien)* 146(5):505–509
- Welzel G, Fleckenstein K, Mai SK, Hermann B, Kraus-Tiefenbacher U, Wenz F (2008) Acute neurocognitive impairment during cranial radiation therapy in patients with intracranial tumors. *Strahlenther Onkol* 184(12):647–654
- Yang Y, Wang H, Shao Y, Wei Z, Zhu S, Wang J (2006) Extradural anterior clinoidectomy as an alternative approach for optic nerve decompression: anatomic study and clinical experience. *Neurosurgery* 59(4 Suppl 2):ONS253–ONS262; discussion ONS262

Part IV

Recurrence

Genetic and Clinical Features Associated with Recurrence in Atypical Meningioma

20

Darlene Gabeau and Yvonne W. Lui

Abstract

Up to 40% of meningiomas correspond to non-benign subtypes associated with two-fold increased relative risk of local failure and four-fold relative excess risk of death. Among them atypical grade II meningiomas comprise over 90% of cases and exhibit heterogeneous clinical outcomes. Several recent studies have performed detailed clinical and molecular-genetic characterization of these neoplasms in order to clarify their pathogenesis and elucidate potential predictive, prognostic and therapeutic markers that may help guide clinical management. Results of these investigations have implicated gain of genetic material on chromosome 1q with increased recurrence and/or progression of tumor and have identified associations of bone involvement with both progression and death in atypical meningioma. Important future challenges in meningioma research include the investigation of large-scale and focal genomic alterations in multi-institutional collaborations and the assessment of whether more thorough bone assessment and bone-directed treatment can improve the heterogeneous outcomes in the unpredictable atypical subtype of the disease.

Introduction

Meningiomas comprise up to 30% of intracranial neoplasms (Dziuk et al. 1998; Goyal et al. 2000; Louis et al. 2007). Although the majority of these extra-axial tumors are of the benign histologic subtype (grade I) and exhibit excellent local

control after surgical resection, approximately 10–40% correspond to atypical (grade II) and anaplastic (grade III) subtypes associated with less favorable clinical outcomes (Palma et al. 1997; Willis et al. 2005). These non-benign subtypes have been associated with two-fold increased relative risk of local failure and four-fold relative excess risk of death (Ayerbe et al. 1999; Kallio et al. 1992). Among them, atypical meningiomas (grade II) comprise up to 35% of meningioma cases (Louis et al. 2007; Perry 2004; Perry et al. 1999; Willis et al. 2005). The World Health Organization (WHO) grading system

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categorizes meningiomas based on mitotic rate and various histologic and architectural patterns. WHO grade I benign meningiomas have diverse architectural patterns and exhibit only occasional mitotic figures. WHO grade II atypical meningiomas exhibit at least four mitotic figures per ten high-power fields or have at least three other histologic features associated with higher grade, including architectural sheeting, necrosis, prominent nucleoli, hypercellularity and high nuclear-to-cytoplasmic ratio. WHO grade III malignant meningiomas exhibit frank histologic malignancy or at least 20 mitotic figures per 10 high-power fields. According to the 2000 and 2007 WHO tumor classifications, brain invasion does not necessarily imply WHO grade III meningioma; in the absence of frank anaplasia, meningiomas exhibiting brain invasion behave most like atypical meningiomas (Perry et al. 1999). Perry et al. illustrated this concept in a study of 89 patients with meningioma classified as malignant based on brain invasion. Twenty-three were morphologically benign, 61% were otherwise atypical and only 17% were frankly anaplastic. Despite brain invasion, tumors lacking anaplasia behaved like atypical meningiomas, whereas anaplastic types were fatal with median survival of 1.5 years. Based on these findings, they concluded that brain invasion constitutes a criterion for atypical meningioma, whereas frank anaplasia indicates higher grade malignancy and poor prognosis. WHO incorporated many of Perry et al.'s proposals into the 2000 and 2007 updates of meningioma grading scheme (Louis et al. 2007). This concept represented a major change in the WHO grading system and resulted in increased diagnosis of the atypical subtype (Lohmann and Brat 2000; Louis et al. 2007; Perry et al. 1999; Willis et al. 2005). For example, upon re-assigning grade based on WHO 2000 criteria versus WHO 1993 criteria, 35% of previously diagnosed grade III meningiomas were re-assigned to grade II whereas few original grade II meningiomas had to be re-assigned (Yang et al. 2008).

Atypical meningiomas are associated with a risk of recurrence of 29–40% compared to less than 20% in benign meningioma (Perry 2004). Atypical meningiomas exhibit heterogeneous

treatment response whereas outcomes for the more rare anaplastic (malignant; grade III) subtype are uniformly poor (Dziuk et al. 1998; Goyal et al. 2000; Perry 2004). A retrospective review of atypical and anaplastic meningiomas found that 50% of 108 patients responded just as well to treatment as those with benign tumors, 25% progressed slowly from atypical to anaplastic pathology and death, and 25% precipitously declined (Palma et al. 1997). Progression free survival at 5 years after definitive treatment has been reported to be 20–50% when atypical and anaplastic meningiomas are combined and 38–62% when atypical meningioma is evaluated alone (Dziuk et al. 1998; Goyal et al. 2000). In addition, atypical meningioma is associated with 57% cause specific survival at 15 years, compared to 86% for benign meningioma (Condra et al. 1997). Given the broad range of reported outcomes after radical surgery, it remains unclear which atypical meningiomas will recur and which patients will benefit from additional treatment options such as adjuvant postoperative radiation therapy. Hence, understanding the genetic and clinical features associated with clinical course of the atypical subtype is of increasing importance to help guide management, decrease inappropriate treatment, minimize treatment failure and reduce undue treatment-related toxicity.

Genetic Factors and Meningioma Recurrence

Meningiomas are among the most studied human solid tumors by karyotype analysis. Among characterized genetic alterations, loss of an entire chromosome 22 is commonly reported in meningiomas and was among the first recurring cytogenetic alterations ever described in human solid tumors (Zang 2001). The Neurofibromatosis 2 gene (NF2) localized to 22q12 is associated with 30–60% of all sporadic meningiomas (Ruttledge et al. 1994). Relatively few studies have focused specifically on the genetic characteristics of atypical meningiomas. In addition to loss on 22q, loss of 1p is a second common

chromosomal abnormality in atypical meningioma (Ozaki et al. 1999; Weber et al. 1997). Two main target regions on 1p36 and 1p32-p34 have been identified based on high-resolution microsatellite markers (Bello et al. 2000). The putative 8.2 megabase minimal region of deletion on 1p36 includes TP73, CDKN2C, RAD54L and ALPL as potential candidate genes (Murakami et al. 2003). The region of deletion on 1p33-34 has been further narrowed to 2.8 megabases using *in silico* sequence analysis and deletion mapping with microsatellite markers (Sulman et al. 2004). Other regions of chromosomal alteration reported in atypical meningioma include losses of 6q, 10, 14q, 18q and gains of 1q, 9q, 12q, 15q, 17q and 20q (Arslantas et al. 2002; Ozaki et al. 1999; Weber et al. 1997).

Array-based comparative genomic hybridization (CGH) detects DNA copy number changes and provides a global assessment of molecular events in the genome (Pinkel et al. 1998). Multiple studies using chromosomal CGH have been reported in the meningioma literature (Arslantas et al. 2002; Ozaki et al. 1999; Weber et al. 1997). However, these studies lack the improved resolution of array CGH, have not elucidated specific genes or loci associated with chromosomal changes and have not specifically studied large, carefully annotated series of atypical meningioma. Array CGH data can be integrated with underlying genome annotations, allowing identification of associations between clinical parameters, such as progression and death, and candidate tumor suppressor or oncogene loci. The potential clinical utility of array CGH-based studies is maximized with inclusion of tumor samples from patients with substantial clinical follow-up.

In order to improve the understanding of meningioma genetics and to identify potentially useful prognostic and therapeutic markers, Gabeau-Lacet et al. (2009a) profiled a large series of atypical meningiomas using high resolution oligonucleotide array CGH. This represented the first comprehensive genomic analysis of a large clinically annotated cohort of atypical meningiomas. They generated a high-resolution genomic map, permitting identification of genes targeted by frequent loss and gain of DNA sequences.

They then explored the relationship between copy number alterations, progression free survival and overall survival. The analysis revealed that 85% of atypical meningiomas exhibited loss of 22q, including the *NF2* gene. The second most frequent regions of loss were confined to the short arm of chromosome 1, particularly 1p33-p36.2 (70%) and 1p13.2 (64%). Other common regions of loss, detected in more than 50% of samples, included 14q, 10q, 8q, 7p, 21q, 19, 9q34, and 4p16. Frequent regions of gain were detected along 1q, 17q, 9q34 and 7q36. Unsupervised clustering of aCGH data revealed two groups of atypical meningioma: one group was composed of the majority of cases with 1q gain and the other group was composed of the majority of the cases with 18q loss (Fig. 20.1a, b). Further analysis of the data revealed focal regions of frequent loss on 1p36.32 (70%) and 1p13.2 (64%). Chromosomal gains on 1q were identified in 59% (28/47 cases) of the evaluated atypical meningioma cases. Correlation analysis of all frequently identified copy number alterations (>40% for losses and >25% for gains) showed a significant correlation between gain of 1q and shorter progression free survival ($p < 0.05$; Fig. 20.1c). This is the first study to correlate chromosome 1q gain with clinical endpoints in atypical meningioma. Elsewhere in the literature, gain of 1q had previously been associated with increased recurrence in neuroblastoma, Wilms' tumor and ependymoma and reduced overall survival in multiple myeloma, medulloblastoma and Ewing sarcoma.

Buckley et al. (2005) evaluated chromosome 1 copy number imbalances using a genomic tiling path method of array CGH on 82 meningiomas and revealed frequent, aberrations of chromosome 1 which varied in type (deletions and amplifications), size (monosomy and homozygous deletion) and distribution (Buckley et al. 2005). Deletion on 1p was the most frequent alteration, ranging from whole p-arm deletion (120 MB) to homozygous loss (3.5 Mb). Three distinct candidate loci on 1p and one locus on 1q exhibited tumor-specific aberrations of potential importance for tumorigenesis. Aberrations of 1q were less common and were always accompanied by full or partial loss of 1p. Loss of genetic information

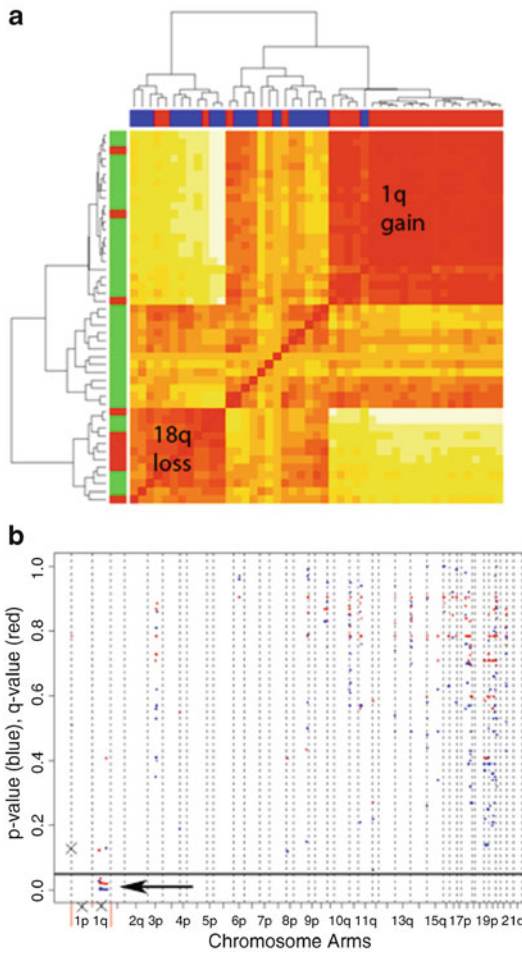


Fig. 20.1 Genomic profiling of atypical meningiomas reveals clustered patterns of chromosomal alterations and association of 1q gain with increased tumor progression. (a) A heat map based on all copy number alterations from 47 primary atypical meningiomas revealed two major genomic clusters associated with gain of 1q (indicated as red on the horizontal side bar with dendrogram) and loss of 18q (indicated as green on the vertical side bar with dendrogram). (b) Univariate marker-by-marker assessment of DNA copy number gains throughout the genome revealed association between 1q gain and shorter progression free survival ($p < 0.05$). p-values (for statistical significance) are red and q-values (for false discovery rate) are blue. The horizontal black line at 0.05 represents the threshold, below which denotes statistical significance

from 1p may represent early progression-associated changes in the transition from benign to atypical meningioma (39). The *TP73* gene, encoding a p53-related protein, is located at 1p36.2 and may represent a candidate gene warranting further

investigation for its potential role in atypical meningioma tumorigenesis. Other studies of atypical meningioma using loss of heterozygosity and/or comparative genomic hybridization analysis have reported loss on 1p in 76% (Weber et al. 1997), 50% (Maruno et al. 1998), 86% (Arslantas et al. 2002) and 40% (Simon et al. 1995) of cases.

Elsewhere in the literature, gains on 1q have been shown to occur in 30–50% of meningioma cases (Ozaki et al. 1999; Weber et al. 1997). In the Gabeau-Lacet et al. profiling study, 1q gains centered on two regions: 1q25.1 (chr1:172161698-172889501) and 1q25.3-q32.1 (chr1:182,779,475-197,636,068). The 1q25.1 interval includes two genes: *RC3H1* (roquin) and *RABGAP1L* (RAB GTPase activating protein 1-like isoform A). *RC3H1* encodes a highly conserved member of the RING type ubiquitin ligase protein family. *RABGAP1L* encodes a 298-amino acid GTPase-activating protein containing a putative phosphotyrosine binding domain. The 1q25.3-q32.1 interval includes 42 genes, some of which may have relevance to tumorigenesis: *EDEM3* (ER degradation enhancer, mannosidase alpha like), which ensures that only properly folded proteins are retained in the cell; *RNF2* (ring finger protein 2), a polycomb protein, part of the multiprotein complexes involved in transcriptional control; *PRG4* (proteoglycan 4), a large proteoglycan synthesized by chondrocytes; *TPR* (nuclear pore complex associated protein), which encodes a large coiled-coil protein that is required for the nuclear export of mRNAs and some proteins, with oncogenic fusions of the 5' end of this gene with several different kinase genes occurring in some neoplasias; *PTGS2* (prostaglandin endoperoxidase synthase 2 precursor), a key enzyme in prostaglandin biosynthesis, the expression of which is deregulated in epithelial tumors; *CDC73* (parafibromin), a member of the PAF protein complex, which associates with the RNA polymerase II subunit POLR2A and with a histone methyltransferase complex; *ASPM* (abnormal spindle)-like, microcephaly), a human ortholog of the *Drosophila* 'abnormal spindle' gene, which is essential for normal

mitotic spindle function in embryonic neuroblasts; *LHX9* (Homo sapiens LIM homeobox 9), which encodes a member of the LIM homeobox gene family of developmentally expressed transcription factors; *NEK7* (Homo sapiens NIMA (never in mitosis gene a)-related kinase 7), which shares high amino acid sequence identity with the gene product of the *Aspergillus nidulans* ‘never in mitosis A’ gene and which controls initiation of mitosis. The functional relevance of these genes in the progression of atypical meningioma is not known and will require further investigation.

There is a paucity of data evaluating the clinical significance of genomic abnormalities in atypical meningioma. Although the Gabeau-Lacet et al. series revealed an association of 1q gain with shorter progression free survival, the study remained under-powered to reveal other clinical associations of identified copy number alterations. Clarification of molecular-genetic signatures useful in atypical meningioma diagnosis and treatment stratification will require investigational efforts on even larger clinically annotated series. Given the increasing diagnosis and heterogeneous treatment outcomes of atypical meningioma, such efforts are warranted and should become a priority to improve the management of this disease.

Clinical Factors and Meningioma Recurrence

Extent of Resection

The preferred definitive treatment for meningioma involves complete excision of tumor and its dural base if tumor size and location allow this approach with acceptable morbidity. The importance of extent of resection in outcome for meningioma has been well established (Goyal et al. 2000; Palma et al. 1997; Simpson 1957). Simpson’s retrospective study graded meningiomas of various histologies based on degree of surgical excision and reported 5-year recurrence rates of 9% for gross total resection (GTR) including surrounding dura and bone (grade I resection), 16% for GTR involving dural cauterization (grade II resection),

29% for GTR with microscopic residual disease (grade III resection), 49% for extended biopsy with macroscopic residual tumor (grade IV resection), and 77% for simple decompression with or without biopsy (grade V resection). Although the series was reported prior to the advent of CT, MRI and microsurgery, its findings on the importance of extent of resection on outcome has been confirmed in multiple modern series (Goyal et al. 2000; Palma et al. 1997; Stafford et al. 1998).

With respect specifically to non-benign meningiomas, Palma et al. conducted a retrospective study of 42 cases of atypical and 29 cases of malignant meningioma, revealing that radical resection (Simpson grade I vs. II-III) and histology (atypical meningioma vs. malignant meningioma) were significantly related to progression and survival (Palma et al. 1997). Many of the “malignant” cases were classified based on the presence of brain invasion and, hence, may actually represent atypical meningiomas based on current WHO classification. Outcomes with atypical meningiomas were heterogeneous and after incomplete surgical excision patients with atypical meningiomas fared only marginally better than those with malignant meningiomas. Therefore, although the prognosis with atypical meningioma was better than with malignant meningioma, the risk of poor outcome after incomplete resection was substantial. In a subsequent retrospective study of 22 patients with atypical meningioma, Goyal et al. (2000) reported 87% local control at 10 years after GTR compared to 17% after subtotal resection (STR) (Goyal et al. 2000).

Many meningiomas cannot be totally excised without unacceptable morbidity. In particular, maximal resection of meningiomas involving the cavernous sinus, petroclival region, posterior aspect of the superior sagittal sinus, optic nerve sheath and en plaque sphenoidal tumors can result in catastrophic vascular injury and disabling cranial neuropathies (Fig. 20.2). In such cases, definitive radiation therapy or adjuvant radiation therapy after subtotal resection may be required to maximize tumor control. Radiation modalities applied for meningioma treatment include

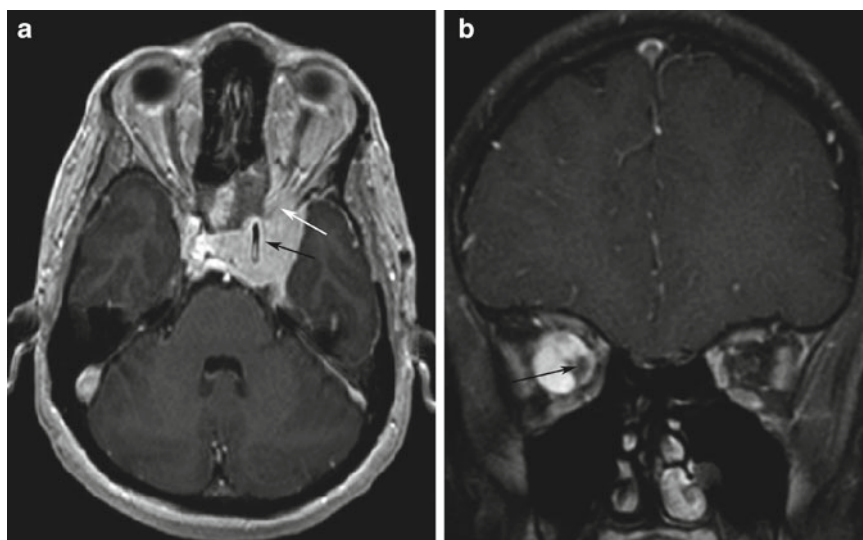


Fig. 20.2 Gadolinium-enhanced axial T1-weighted MR image in a 45 year-old female with left petroclival meningioma (**a**). Circumferential tumor around the cavernous portion of the left internal carotid artery (*black arrow*) and extension through the superior orbital fissure to involve the left orbital apex (*white arrow*). Post-contrast coronal

T1-weighted MR image in a 15 year-old female with right optic nerve sheath meningioma (**b**) shows avid enhancement of the mass surrounding the right optic nerve (*arrow*). The patient's vision was preserved. Location and involvement of adjacent critical structures makes lesions such as these not amenable to gross total resection

3D-conformal photons, stereotactic radiosurgery, fractionated stereotactic radiation therapy, proton therapy and intensity modulated radiation therapy.

Radiation therapy has historically been reserved in meningioma for cases of incomplete resection, tumor recurrence, histologic atypia or anaplasia, or inoperable disease. Among these indications, the role of radiation therapy for atypical histology remains the most nebulous. The range of reported outcomes following treatment for atypical meningioma is broad and it remains unclear which patients will benefit from immediate postoperative radiation (particularly in the setting of GTR). Furthermore, the role of radiation dose escalation on treatment outcomes is not yet established. In 2007 the European Organization for Research and Treatment of Cancer (EORTC) activated a phase II and observation trial (EORTC 22042-26042) evaluating the impact of radiotherapy on progression free survival, treatment tolerance, and post-treatment global cognitive functioning in patients with atypical and anaplastic meningiomas after various Simpson grades of resection. Such prospective

data is greatly needed since, to date, only retrospective studies have reported on these issues. For example, in a study including 46 patients with atypical pathologic features, local control at 15 years was 60% with STR and adjuvant radiation therapy versus 43% with STR alone (Condra et al. 1997). Similarly, in their study of non-benign meningiomas, Coke et al. found local failure rates of 65% after surgery alone compared to 18% after surgery plus postoperative radiation therapy. Hence, based on available retrospective results, it is generally accepted that in the setting of subtotal resection, radiation therapy significantly improves local control in atypical meningioma (Condra et al. 1997; Goyal et al. 2000).

Bone Involvement

Meningioma has long been known to induce changes in adjacent bone, including hyperostosis and direct invasion (Cushing 1922; Spiller 1929). Although some studies have implicated hyperostosis as a secondary manifestation involving

humoral mechanisms without neoplastic invasion (Heick et al. 1993), most evidence indicates that it results from direct tumor invasion into bone (Cushing 1922; Pieper et al. 1999). One study postulated that new bone growth likely results from periosteal stimulation by tumor invasion (Kim et al. 1983). Another report correlating radiographic evidence of hyperostosis with histopathology identified tumor invasion in the resected bone biopsies of 25/26 patients with radiographically identified hyperostosis (Pieper et al. 1999). In addition, histological tumor invasion of bone was identified in 9 patients who did not show evidence of hyperostosis on preoperative neuroradiological evaluation, indicating that it is difficult to delineate the extent of bone involvement definitively even with modern radiographic techniques. Additional studies involving histopathologic examination of hyperostotic bone have revealed bone infiltration by meningioma cells in 4/5 cases and 12/13 cases, supporting the direct tumor infiltration hypothesis of hyperostosis (De Jesus and Toledo 2001).

Preoperative magnetic resonance (MR) imaging is the gold standard for anatomic imaging of intracranial tumors and is routinely obtained for meningioma presurgical planning. MR provides exquisite detail and is necessary to determine the relationship of the mass to adjacent structures such as major arteries, cranial nerves, and dural sinuses. Osseous detail, however, is better appreciated using computed tomography (CT). On MR, hyperostotic bone can be difficult to see, particularly when there is involvement of the skull base. The bone may appear widened or irregular with signal hypointensity. On CT, bone changes are readily appreciated. Hyperostosis appears as an area of sclerosis with thickening of the inner table of the skull. Less characteristically, meningioma can also cause areas of bone erosion (Fig. 20.3).

More recently, intraoperative photodynamic diagnosis of osseous involvement has been reported in the literature, mostly in the form of case reports. This technique involves fluorescence-guided resection with 5-aminolevulinic acid to visualize minute areas of residual tumor along bone and guide complete surgical removal of tumor (Kajimoto et al. 2007; Morofuji et al. 2008).

Activation of 5-aminolevulinic acid elicits synthesis and accumulation of fluorescent porphyrins in epithelia and neoplastic tissues. Phase III study has shown that the intraoperative use of this 5-aminolevulinic acid fluorescence can reduce residual tumor volume and prolong progression-free survival in patients treated for malignant glioma (Stummer et al. 2006). To date, the use of 5-aminolevulinic acid for guidance of osseous resection in meningioma has not been systematically investigated.

The importance of bone involvement as a prognostic factor in meningioma is not universally accepted (Miller 1994). Furthermore, the role of bone involvement in clinical outcome specifically in atypical meningioma has not previously been reported. A study of 657 patients with seemingly complete removal of histologically benign intracranial meningioma reported overall recurrence rate of 19% at 20 years (Jaaskelainen 1986). Tumor attachment to bone was identified as a strong and independent risk factor for recurrence on multivariate analysis. Furthermore, another report noted that hyperostosis was a significant prognostic factor for excess mortality in patients who survived the first postoperative year (Kallio et al. 1992). This retrospective study of 936 meningioma cases revealed 67% relative survival rate at 15 years for patients with hyperostosis compared to 81% at 15 years for patients without hyperostosis. The impact of hyperostosis became apparent only after the seventh follow-up year. Patients with hyperostotic tumors lost 16% of their expected length of life and had a 2.1-fold relative excess risk of death, while those with no hyperostosis lost only 9% of their expected length of life. Except for hyperostosis, no attachment into adjacent structures proved significant for long-term survival. 4.7% (44 patients) of the patients were categorized as having atypical meningioma. Hyperostosis adjacent to the tumor was encountered in 20% of all patients. Because stratification by histologic grade was not reported, it remains unclear whether the effects of hyperostosis on survival differed between atypical meningioma and benign meningioma.

Gabeau-Lacet et al. (2009b) conducted a study in which 47 cases of primary atypical meningioma

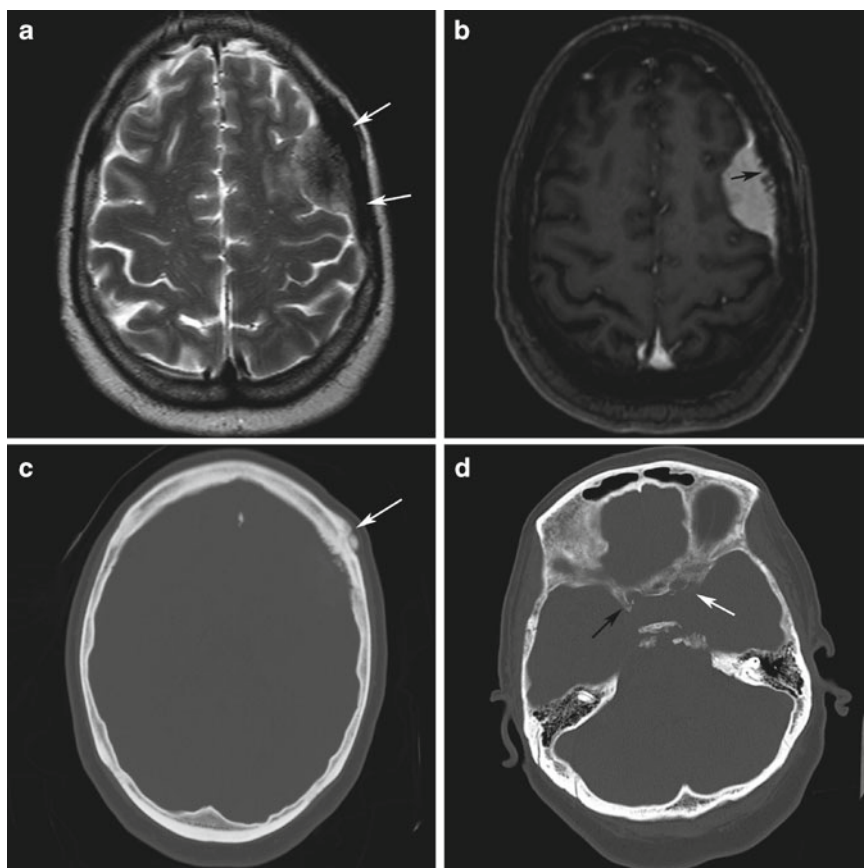


Fig. 20.3 Fifty-five year-old female with left frontal meningioma, presented with a palpable mass on her scalp. T2-weighted MR image (**a**) shows hypointense signal within the calvarium consistent with hyperostosis (*arrows*). While lesion conspicuity is high on post-contrast T1-weighted MR imaging (**b**); bone involvement is somewhat subtle. Irregularity of the inner table is seen (*arrow*).

Bone window of a CT (**c**) in the same patient shows thickening and sclerosis of the skull to better advantage (*arrow*). Bone erosion is less common, but can also be seen in meningioma. Bone window of a CT in a different patient with skull base meningioma (**d**) shows amputation of the left anterior clinoid process (*white arrow*). Note the normal anterior clinoid on the contralateral side (*black arrow*)

treated at Massachusetts General Hospital were retrospectively evaluated for clinical features associated with progression free survival and overall survival. The analysis revealed that bone involvement, including hyperostosis, was associated with increased recurrence/progression and death in atypical meningioma. The estimated median progression free survival was 56 months. Overall 3-year and 5-year progression free survival rates were 65% and 48%, respectively. The median survival time and 5- and 10-year survival rates were 158 months, 86% and 61%, respectively.

Subtotal versus gross total resection was associated with increased rate of progression ($p=0.05$) and trended toward an association with decreased survival ($p=0.09$). Bone involvement was associated with increased rate of progression ($p=0.001$) and decreased survival ($p=0.04$). Bone involvement remained significantly associated with progression after Bonferroni adjustment for multiple comparisons ($p=0.008$) and in bivariate Cox regression models. Hence, osseous involvement was associated with much worse clinical outcome in patients with atypical meningioma.

The observation that bone invasion affects clinical outcome in atypical meningioma raises a number of biological possibilities. For example, worse outcomes with bone-invasive atypical meningiomas may reflect a more aggressive tumor biology compared to those that do not invade bone. Alternatively, poor outcomes may result from failure to treat diseased bone. Indeed, pattern of failure analysis in the Gabeau-Lacet et al. cohort revealed osseous relapse in 78% of cases that initially involved bone compared to only 25% osseous relapse in cases that did not initially exhibit bone involvement. The majority of cases in the study were thought to have undergone gross total resection, i.e., the tumor and attached dura mater were apparently completely removed. However, surgical treatment of bone varied, some receiving aggressive removal and/or cranioplasty and others not. It is possible that, in cases with bone involvement, remaining tumor cells in the surrounding bone make visual estimates of complete removal misleading. Post-operative radiographic imaging may lack sensitivity to detect residual tumor cells in bone and determination of margin status in bone is not straightforward, making it difficult to assess the radical extent of bone resection. Hence, the high recurrence rates observed after surgery may reflect subclinical bony disease that would be best addressed with adjuvant radiation therapy with fields encompassing surrounding bony areas. None of the cases with bone involvement in the Gabeau-Lacet et al. study received post-operative radiation directed specifically to the remaining bone. The presence or absence of bone involvement may explain the heterogeneous treatment outcomes observed in atypical meningioma. If so, modifying radiographic assessment, pathologic evaluation, surgical management and radiotherapeutic approach to better address bony disease may be indicated.

Conclusions

Meningioma is not a uniformly benign disease. Non-benign subtypes are characterized by heterogeneous treatment outcomes in the case of atypical meningioma, and uniformly poor outcome in the

case of anaplastic meningioma. The diagnosis of atypical meningioma has increased with changes to the WHO 2000 and 2007 grading criteria. The tremendous individual variability in clinical behavior of atypical meningiomas has previously not been accounted for by known clinical and pathologic variables. Several recent studies have performed detailed clinical and molecular-genetic characterization of these neoplasms in order to clarify their pathogenesis and elucidate potential predictive, prognostic and therapeutic markers that may help guide clinical management. Results of these investigations have: (1) implicated gain of genetic material on chromosome 1q with increased recurrence and/or progression of tumor, and (2) identified associations between bone involvement and both progression and death in atypical meningioma. Careful preoperative bone assessment and aggressive bone removal and/or adjuvant radiation therapy targeting the cranium may be warranted to decrease the risk of tumor recurrence in cases involving bone. Important future challenges in atypical meningioma research include large-scale investigation of genomic alterations in multi-institutional research initiatives and the assessment of whether aggressive bone-directed treatment can improve the heterogeneous outcomes in this disease.

References

- Arslantas A, Artan S, Oner U, Durmaz R, Muslumanoglu H, Atasoy MA, Basaran N, Tel E (2002) Comparative genomic hybridization analysis of genomic alterations in benign, atypical and anaplastic meningiomas. *Acta Neurol Belg* 102(2):53–62
- Ayerbe J, Lobato RD, de la Cruz J, Alday R, Rivas JJ, Gomez PA, Cabrera A (1999) Risk factors predicting recurrence in patients operated on for intracranial meningioma. A multivariate analysis. *Acta Neurochir (Wien)* 141(9):921–932
- Bello MJ, de Campos JM, Vaquero J, Kusak ME, Sarasa JL, Rey JA (2000) High-resolution analysis of chromosome arm 1p alterations in meningioma. *Cancer Genet Cytogenet* 120(1):30–36
- Buckley PG, Jarbo C, Menzel U, Mathiesen T, Scott C, Gregory SG, Langford CF, Dumanski JP (2005) Comprehensive DNA copy number profiling of meningioma using a chromosome 1 tiling path microarray identifies novel candidate tumor suppressor loci. *Cancer Res* 65(7):2653–2661

- Condra KS, Buatti JM, Mendenhall WM, Friedman WA, Marcus RB Jr, Rhoton AL (1997) Benign meningiomas: primary treatment selection affects survival. *Int J Radiat Oncol Biol Phys* 39(2):427–436
- Cushing H (1922) The cranial hyperostoses produced by meningeal endotheliomas. *Arch Neurol Psychiatry* 8:139–154
- De Jesus O, Toledo MM (2001) Surgical management of meningioma en plaque of the sphenoid ridge. *Surg Neurol* 55(5):265–269
- Dziuk TW, Woo S, Butler EB, Thornby J, Grossman R, Dennis WS, Lu H, Carpenter LS, Chiu JK (1998) Malignant meningioma: an indication for initial aggressive surgery and adjuvant radiotherapy. *J Neurooncol* 37(2):177–188
- Gabeau-Lacet D, Gupta S, Scangas GA, Betensky RA, Barker FG 2nd, Loeffler JS, Louis DN, Mohapatra G (2009a) Genomic profiling of atypical meningiomas associates gain of 1q with poor clinical outcome. *J Neuropathol Exp Neurol* 68(10):1155–1165
- Gabeau-Lacet D, Aghi M, Betensky RA, Barker FG, Loeffler JS, Louis DN (2009b) Bone involvement predicts poor outcome in atypical meningioma. *J Neurosurg* 111(3):464–471
- Goyal LK, Suh JH, Mohan DS, Prayson RA, Lee J, Barnett GH (2000) Local control and overall survival in atypical meningioma: a retrospective study. *Int J Radiat Oncol Biol Phys* 46(1):57–61
- Heick A, Mosdal C, Jorgensen K, Klinken L (1993) Localized cranial hyperostosis of meningiomas: a result of neoplastic enzymatic activity? *Acta Neurol Scand* 87(3):243–247
- Jaaskelainen J (1986) Seemingly complete removal of histologically benign intracranial meningioma: late recurrence rate and factors predicting recurrence in 657 patients. A multivariate analysis. *Surg Neurol* 26(5):461–469
- Kajimoto Y, Kuroiwa T, Miyatake S, Ichioka T, Miyashita M, Tanaka H, Tsuji M (2007) Use of 5-aminolevulinic acid in fluorescence-guided resection of meningioma with high risk of recurrence. Case report. *J Neurosurg* 106(6):1070–1074
- Kallio M, Sankila R, Hakulinen T, Jaaskelainen J (1992) Factors affecting operative and excess long-term mortality in 935 patients with intracranial meningioma. *Neurosurgery* 31(1):2–12
- Kim KS, Rogers LF, Lee C (1983) The dural lucent line: characteristic sign of hyperostosing meningioma en plaque. *AJR Am J Roentgenol* 141(6):1217–1221
- Lohmann CM, Brat DJ (2000) A conceptual shift in the grading of meningiomas. *Adv Anat Pathol* 7(3):153–157
- Louis D, Ohgaki H, Wiestler O, Cavenee WK (2007) World Health Organization histological classification of tumours of the central nervous system. International Agency for Research on Cancer, Lyon
- Maruno M, Yoshimine T, Muhammad AK, Ninomiya H, Hayakawa T (1998) Chromosomal losses and gains in meningiomas: comparative genomic hybridization (CGH) study of the whole genome. *Neurol Res* 20(7):612–616
- Miller DC (1994) Predicting recurrence of intracranial meningiomas. A multivariate clinicopathologic model—interim report of the New York University Medical Center Meningioma Project. *Neurosurg Clin N Am* 5(2):193–200
- Morofuji Y, Matsuo T, Hayashi Y, Suyama K, Nagata I (2008) Usefulness of intraoperative photodynamic diagnosis using 5-aminolevulinic acid for meningiomas with cranial invasion: technical case report. *Neurosurgery* 62(3 Suppl 1):102–103; discussion 103–104
- Murakami M, Hashimoto N, Takahashi Y, Hosokawa Y, Inazawa J, Mineura K (2003) A consistent region of deletion on 1p36 in meningiomas: identification and relation to malignant progression. *Cancer Genet Cytogenet* 140(2):99–106
- Ozaki S, Nishizaki T, Ito H, Sasaki K (1999) Comparative genomic hybridization analysis of genetic alterations associated with malignant progression of meningioma. *J Neurooncol* 41(2):167–174
- Palma L, Celli P, Franco C, Cervoni L, Cantore G (1997) Long-term prognosis for atypical and malignant meningiomas: a study of 71 surgical cases. *J Neurosurg* 86(5):793–800
- Perry A (2004) Unmasking the secrets of meningioma: a slow but rewarding journey. *Surg Neurol* 61(2):171–173
- Perry A, Scheithauer BW, Stafford SL, Lohse CM, Wollan PC (1999) “Malignancy” in meningiomas: a clinicopathologic study of 116 patients, with grading implications. *Cancer* 85(9):2046–2056
- Pieper DR, Al-Mefty O, Hanada Y, Buechner D (1999) Hyperostosis associated with meningioma of the cranial base: secondary changes or tumor invasion. *Neurosurgery* 44(4):742–746; discussion 746–747
- Pinkel D, Segraves R, Sudar D, Clark S, Poole I, Kowbel D, Collins C, Kuo WL, Chen C, Zhai Y, Dairkee SH, Ljung BM, Gray JW, Albertson DG (1998) High resolution analysis of DNA copy number variation using comparative genomic hybridization to microarrays. *Nat Genet* 20(2):207–211
- Ruttledge MHSJ, Rangaratnam S, Phelan CM, Twist E, Merel P, Delattre O, Thomas G, Nodenskjold M, Collins VP et al (1994) Evidence for the complete inactivation of the NF2 gene in the majority of sporadic meningiomas. *Nat Genet* 6(2):180–184
- Simon M, von Deimling A, Larson JJ, Wellenreuther R, Kaskel P, Waha A, Warnick RE, Tew JM Jr, Menon AG (1995) Allelic losses on chromosomes 14, 10, and 1 in atypical and malignant meningiomas: a genetic model of meningioma progression. *Cancer Res* 55(20):4696–4701
- Simpson D (1957) The recurrence of intracranial meningiomas after surgical treatment. *J Neurol Neurosurg Psychiatry* 20(1):22–39
- Spiller W (1929) Cranial hyperostosis associated with underlying meningeal fibroblastoma. *Arch Neurol Psychiatry* 21:637–640

- Stafford SL, Perry A, Suman VJ, Meyer FB, Scheithauer BW, Lohse CM, Shaw EG (1998) Primarily resected meningiomas: outcome and prognostic factors in 581 mayo clinic patients, 1978 through 1988. *Mayo Clin Proc* 73(10):936–942
- Stummer W, Pichlmeier U, Meinel T, Wiestler OD, Zanella F, Reulen HJ (2006) Fluorescence-guided surgery with 5-aminolevulinic acid for resection of malignant glioma: a randomised controlled multicentre phase III trial. *Lancet Oncol* 7(5):392–401
- Sulman EP, White PS, Brodeur GM (2004) Genomic annotation of the meningioma tumor suppressor locus on chromosome 1p34. *Oncogene* 23(4): 1014–1020
- Weber RG, Bostrom J, Wolter M, Baudis M, Collins VP, Reifenberger G, Lichter P (1997) Analysis of genomic alterations in benign, atypical, and anaplastic meningiomas: toward a genetic model of meningioma progression. *Proc Natl Acad Sci USA* 94(26): 14719–14724
- Willis J, Smith C, Ironside JW, Erridge S, Whittle IR, Everington D (2005) The accuracy of meningioma grading: a 10-year retrospective audit. *Neuropathol Appl Neurobiol* 31(2):141–149
- Yang SY, Park CK, Park SH, Kim DG, Chung YS, Jung HW (2008) Atypical and anaplastic meningiomas: prognostic implication of clinicopathological features. *J Neurol Neurosurg Psychiatry* 79(5):574–580
- Zang KD (2001) Meningioma: a cytogenetic model of a complex benign human tumor, including data on 394 karyotyped cases. *Cytogenet Cell Genet* 93(3–4): 207–220

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Abstract

Meningiomas are mostly benign tumors that originate from the coverings of brain and spinal cord. Only a minority of cases shows progression to an anaplastic tumor (WHO grade II and III). Multiple and familial cases are rare and mostly associated with (hereditary) neurofibromatosis 2 (NF2). Meningiomas show an unexpectedly high recurrence rate. Also, completely removed low-grade tumors can recur. On cytogenetical level, meningiomas are the best-studied tumors in humans. The majority of high-grade, but only a minority of low-grade meningiomas show loss of merlin, a cytoskeleton-cytoplasm-linker protein. Merlin is the product of the NF2 gene located on chromosome 22. A second tumor suppressor gene on chromosome 22 on 22q12.3 is the gene for the tissue inhibitor of metalloproteinase 3 (*TIMP3*), which appears to be involved in meningioma progression and a high-grade meningioma phenotype. In contrast to other solid tumors, progression of meningiomas is correlated with increasing hypodiploidy, showing characteristic clonal evolutions that mostly include chromosomes 14, 18, and 19 and, more rarely, 6 and 10. Structural aberrations are rare, except for the loss of the short arm of one chromosome 1, which appears to be the decisive step for anaplastic growth. A bio-statistical approach, is proposed, using an oncogenetic tree model that estimates the most likely cytogenetic pathways of meningioma patients in terms of accumulation of chromosome changes in tumor cells. The genetic progression score (GPS) estimates the genetic status of a tumor as progression in the corresponding tumor cells along this model. High GPS values are highly correlated with early recurrence of meningiomas ($p < 10^{-4}$). This correlation holds true even when patients are stratified by WHO grade.

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Tumor location also has an impact on genetic progression. Clinical relevance of the GPS is demonstrated with respect to origin, WHO grade, and recurrence of the tumor. As a quantitative measure, the GPS allows a more precise assessment of the prognosis of meningiomas than categorical cytogenetic markers based on single chromosomal aberrations. Comparative histochemical and molecular cytogenetic studies point to the alkaline phosphatase gene (ALPL, liver-bone-kidney type) located on 1p36.1-p34 as a candidate tumor suppressor gene.

Introduction

Meningiomas are derived from the arachnoidal caps cells of the leptomeninges, the soft coverings of the brain and spinal cord. Although their matrix tissue constitutes less than 5% of the intracranial and intraspinal mass, meningiomas are estimated to constitute between 13% and 26% of the primary tumors within the CNS. Most meningiomas are sporadic, slowly growing benign tumors and correspond histologically to WHO grade I. However, certain histological subtypes and also a minority of common-type meningiomas show a more aggressive biological behaviour and are associated with an increased risk of recurrence and an unfavourable prognosis. Corresponding to their localization, meningiomas of the spinal cord tend to be significantly more benign than those of the brain basis, which in turn, are more benign than meningiomas of the brain convexity (Louis et al. 2000; Ketter et al. 2008). The current WHO classification of brain tumors (Louis et al. 2000) distinguishes three grades of meningiomas: the common type (WHO grade I), the atypical or intermediate-type (WHO grade II) and the anaplastic (WHO grade III) meningioma.

For an appropriate treatment of tumor patients, prediction of time until death or time to progression after initial treatment is an important task. Due to many clinical, topographical, radiological and surgical factors, histology is not solely decisive for the prognosis (Niedermayer et al. 1997), although mitotic activity, cellular pleomorphism with prominent nucleoli and micronecrosis, and focally raised cell density have been discussed as

indicators of a poorer prognosis (Takeuchi et al. 1997). A major challenge is the identification of genetic prognostic markers that better reflect tumor biology.

Multiple occurrence is a rare event, observed in only about 2% of cases. Frequently, multiple meningiomas occur in patients with neurofibromatosis 2 (NF2), an autosomal dominant tumor syndrome, in combination with vestibular schwannomas and ependymomas. A polyclonal origin must be supposed in these cases.

In non-NF2 cases, molecular genetic studies demonstrate a common clonal origin of multiple or recurrent meningiomas (von Deimling et al. 1999; Zhu et al. 1999). The development of familial (multiple) meningiomas within the same or different generations has so far been reported in less than 20 families. Pathologic and genetic data on these families first suggested a biologically variant type of NF2, with the development of meningiomas only, but no schwannomas. Molecular studies on a few cases indeed excluded the involvement of the NF2 gene, although a chromosome 22 was missing. Alternatively, homozygous inactivation of a different tumor suppressor gene on the same chromosome was proposed (Ruttledge et al. 1994).

The meningioma is one of the cytogenetically best-studied solid tumors. The characteristic and most frequent chromosomal aberration in meningiomas is monosomy 22 (Zang and Singer 1967), which, however seems to be not relevant for prognosis as an isolated anomaly. The progression from common-type to atypical and anaplastic meningioma is characterized by two different cytogenetic events: Firstly, further loss

of up to six other chromosomes, with mostly a typical pattern of clonal evolution, and secondly, partial or complete loss of the short arm of one chromosome 1 (Perry et al. 2004; Bello et al. 1994; Henn et al. 1995; Kolles et al. 1995; Niedermayer et al. 1997; Simon et al. 1995; Ketter et al. 2001; Zang 2001).

The identification of pathogenetic pathways in human tumors is one of the main challenges in molecular oncology. For many tumor types, genetic events defined by somatic chromosome alterations resp. gene mutations are known to accumulate over time in the course of the disease. Based on these findings, genetic changes associated with chromosome instability are believed to play an important role in both tumorigenesis and tumor progression in meningiomas. There is a vast amount of literature on linking single genetic alterations to survival (Jiang et al. 2000), but only few efforts have been made to construct more complex and comprehensive markers. Zang (2001), described a first model of clonal evolution in meningiomas based on conventional cytogenetic studies. Data of the cytogenetic findings of 394 meningiomas led to an empirical model of meningioma progression.

Cytogenetic analyses show that numerical and structural chromosome changes with pronounced hypodiploidy, or rarely hyperdiploidy (Ketter et al. 2007a), and especially deletion of the short arm of a chromosome 1, are accompanied by more aggressive biological characteristics (Perry et al. 2004; Bello et al. 1994; Henn et al. 1995; Kolles et al. 1995; Niedermayer et al. 1997; Simon et al. 1995; Ketter et al. 2001, 2007b; Zang 2001).

By using a mathematical model for estimating the most likely cytogenetic pathways in meningiomas a model for tumor recurrence could be established. In this model each tumor is represented by the genetic events that have occurred in the tumor. The most important difference of the mathematical model compared to the “hand-crafted” model (Zang 2001) is that it allows an objective assignment of the estimated time to recurrence for a single tumor based on its genetic status.

Incidence, Materials and Statistics

Patient Population

We performed a retrospective study on 661 patients (482 women and 179 men) with meningiomas operated on at the Department of Neurosurgery, University of the Saarland, between January 1973 and April 2005. The average age of the overall patient population was 57.3 years ($SD \pm 12.8$ years). The average age of the female patients was 57.6 years ($SD \pm 12.3$ years), average age of the male patients was 56.7 years ($SD \pm 14.1$ years).

Clinical Variables

The clinical variables investigated comprise patient gender and age, tumor location and histology, and the completeness of tumor extirpation.

Location: We formed the following nine groups depending on meningioma location: convexity, parasagittal region, tuberculum sellae, olfactory groove, sphenoid wing ridge, posterior cranial fossa, tentorium, ventricular, and spinal channel. The classification was carried out on the basis of the preoperative CT and NMR.

Tumor extirpation: Complete surgical extirpation of the tumor was defined as Simpson grades I and II corresponding to a macroscopically complete tumor resection with bipolar coagulation of the dura insertion.

Tumor Histology

The 661 cases investigated comprised 465 common type (WHO grade I), 156 intermediate-type (WHO atypical meningioma, grade II), and 14 anaplastic meningiomas (WHO grade III). Meningioma grade was assessed by a combined histological and morphometric approach on routinely HE and Ki-67/Feulgen stained formalin-fixed, paraffin-embedded tissue sections (Kolles et al. 1995).

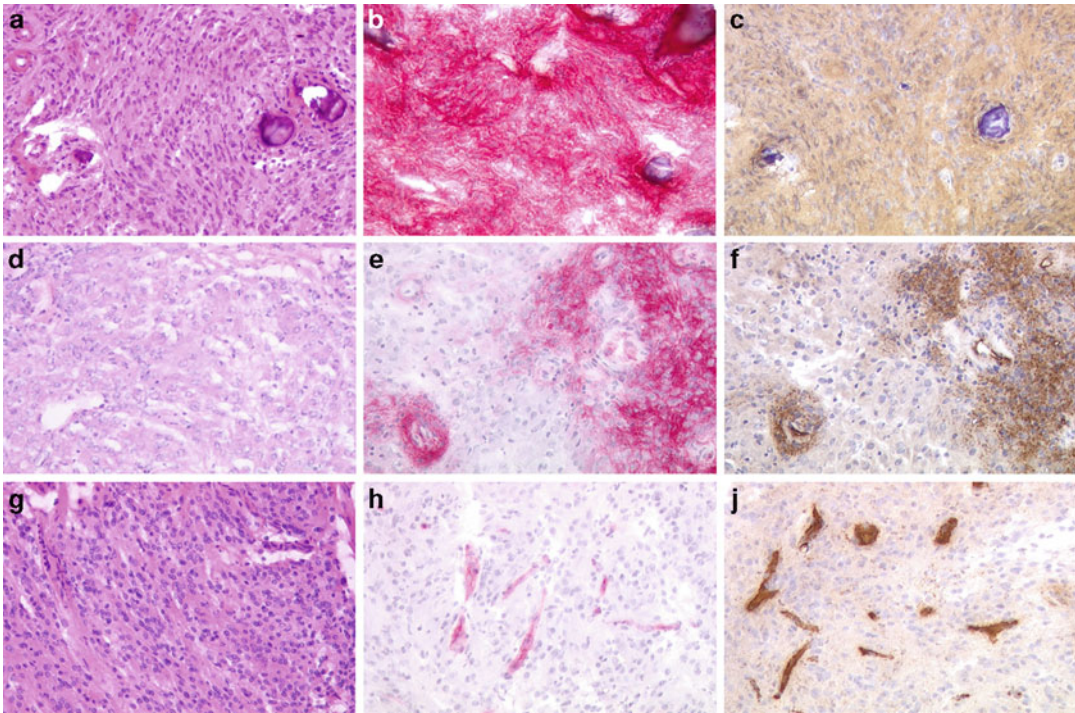


Fig. 21.1 Expression pattern of alkaline phosphatase in meningiomas of different grade (liver-bone-kidney type, ALPL, located at 1p36.1 – p34). In the rows, parallel sections of meningiomas are shown. The *upper row* (a–c) shows a WHO grade I, the *middle row* (d–f) a grade II, and the *lower row* (g–j) a grade III meningioma. The *first column* shows hematoxylin and eosin staining, the *second column* histochemical reactivity of the alkaline phosphatase, and the *third column* immunohistochemical localization of the alkaline phosphatase protein.

Histochemical and immunohistochemical staining show a very good correlation. In the grade I meningioma all tumor cells are positive; in the grade II tumor, clearly delineated areas are spared, and in the grade III meningioma only the endothelial cells of the blood vessels are positive, the tumor cells being completely negative histochemically as well as immunohistochemically. Loss of ALPL expression is highly correlated with FISH results on the terminal 1p deletion. All sections were photographed with a 20× objective (Zang 2001)

Cell Culture and Cytogenetic Preparation Follow-up

Cell cultures from 661 meningioma biopsies and chromosome preparations with GTC- banding were carried out according to standard procedures. Many tumors were characterized by multiple patterns of cytogenetic aberrations. In total, 1,068 clonal patterns were observed in the 661 tumors.

Histochemical Detection of ALPL

Histochemical findings shown in Fig. 21.1 were performed on frozen sections as described by Niedermayer et al. (1997).

The patients were investigated in the neurosurgical outpatient department of the University of the Saarland, either within routine follow-up or because of the appearance of neurological symptoms. A recurrence was evaluated as new evidence of tumor in CT or NMR after previous complete extirpation (Simpson grades 1 and 2). The Simpson grade 2 was established on the basis of the operation report and the postoperative CT or NMR investigation. Average followup time was 40.3 months (41.5 months for female patients and 37.2 months for male patients).

Statistical Analyses

Oncogenetic Tree Models

Mixtures of oncogenetic trees were used to describe the ordered accumulation of genetic aberrations during tumor progression. In a single oncogenetic tree, vertices represent genetic events and edges between vertices represent transitions between the events. Each edge is associated with the probability that the successor event will occur, given the predecessor event has already occurred. In the model genetic events are assumed to be non-reversible, thus the disease process can be fully described by the accumulation of genetic aberrations. In the mixture model, more than one tree component is estimated (Ketter et al. 2007b).

Genetic Progression Score

In oncogenetic tree models consecutive genetic aberrations are associated with corresponding conditional transition probabilities. These probabilities can be converted to average waiting times by assuming Poisson processes for the occurrence of aberrations, see Rahnenführer et al. (2005) for details. Formally, the waiting time associated to an edge of the tree with corresponding conditional probability p is given by $(1-p)/p$ multiplied with a scaling factor that is typically set to 1. The genetic progression score (GPS) of a tumor then is defined as the average waiting time of its pattern of genetic aberrations, given the underlying tree mixture model. Tumors with few aberrations that appear early in the model receive low GPS values; tumors with many late aberrations in the model are associated with high GPS values. For 221 out of the 661 patients, more than one cytogenetic pattern was detected. In these cases the GPS of the tumor was defined as the highest GPS of all clones found in the tumor.

Clinical Factors of Meningioma Prognosis

Grading of meningiomas has always been controversial. Obviously, the biological behaviour of meningiomas cannot be accounted by histological

parameters alone (Ketter et al. 2001; Zang 2001). In 1979, Zülch stated that it is not the histological grading which is most crucial for the rate of recurrence of meningiomas, but primarily the completeness of extirpation (Zülch 1979). There is agreement in the literature that radical surgical extirpation is correlated with a good prognosis (Adegbite et al. 1983; Mirimanoff et al. 1985).

It is well known that females are affected far more frequently by meningiomas than males (Zülch 1979; Adegbite et al. 1983). This observation was confirmed in our previous study (Zang 2001; Kolles et al. 1995; Ketter et al. 2001) with a ratio of 2.67:1. In particular in the 49 spinal tumors investigated, the female sex was overrepresented (87% of cases). This sex ratio was shifted when the tumors were broken down by karyotype. In the GPS group 2 ($\text{GPS} \geq 6.39$), i.e., in the tumors with a pronounced aberration of the karyotype with deletion 1p, we found a sex distribution of 1.81:1.

Cytogenetic Aspects of Meningiomas

Localization of Meningiomas

Frequency and behavior of meningiomas at different intracranial locations were first discussed in 1922 (Cushing 1922). A correlation between tumor localization and malignancy has repeatedly been observed. Meningiomas of the spinal cord tend to be significantly benign more often than those of the brain basis, which, in turn, are more often benign than meningiomas of the brain convexity (Louis et al. 2000; Ketter et al. 2008). In 1980, it was shown that tumors located at the base of the skull typically contain cells with 46 chromosomes, whereas meningiomas located at the convexity show significant numbers of chromosome aberrations (Zang 2001). In spinal tumors, almost exclusively a 22-monosomic karyotype, was detected. It was already striking at that time that meningiomas which recurred showed significantly more chromosome aberrations and a preference of the convexity. These findings are also in line with the results on a large series of meningioma patients ($N > 9,000$ cases),

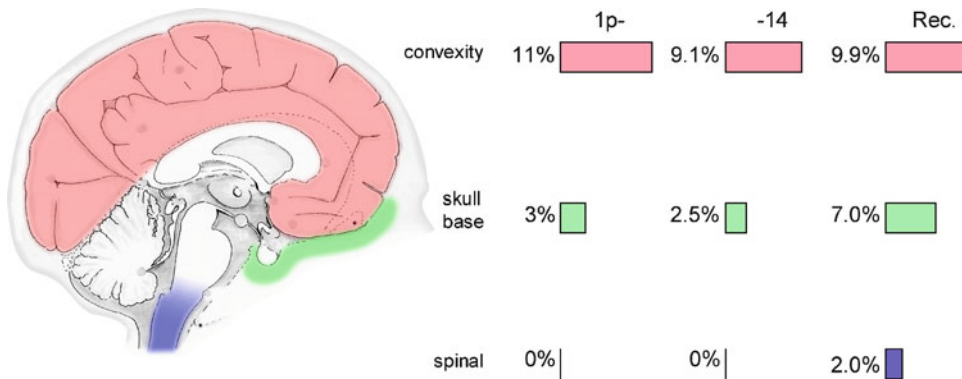


Fig. 21.2 Bar plots for chromosomal aberrations (1p-, -14) and recurrence [Rec] according to brain regions. *Red color* indicates convexity, *green color* skull base and *blue color* spinal tumors (Ketter et al. 2008)

where benign meningiomas were more frequently located at the skull than malignant meningiomas ($p < 0.02$) (Fig. 21.2) (Ketter et al. 2008).

Cytogenetic Findings and the Histomorphology

Among histopathological prognostic parameters the mitotic activity is the most important one. As the cutoffs of the mitotic activity (MI) are defined for each grade by the WHO classification of brain tumors and because the MI can be applied as the sole grading criterion, the reliable and reproducible assessment of the MI is crucial for an appropriate risk stratification. In 2007, we classified 661 meningiomas according to their genetic progression scores (GPS values) (Ketter et al. 2007b). We found a high correlation between the cytogenetic findings and the histomorphology: Higher genetic progression scores correlated highly significantly with higher histological grades ($p < 10^{-10}$, Fisher's exact test). This result is important since in the literature no clear correlation between the histological grading and the rate of recurrence in meningiomas has been reported (Perry et al. 1996, 2004; Mafillo et al. 2003). In our study, also no clear distinction between the histological grades I and II with respect to tumor recurrence could be shown (Fig. 21.1). However, WHO grade III meningiomas

show a statistically significant correlation with earlier tumor recurrence.

The Expression of the Alkaline Phosphatase ALPL in Meningiomas

More than four decades ago, Osske and Jänisch (1967) reported differences in the expression of ALPL in meningiomas. However, no clear correlation with tumor grade or subtype was found. It could be shown that there is a strong homogeneous expression of this enzyme in low-grade meningiomas, with increasing patchwork-like patterns of expression loss in anaplastic meningiomas (Zang 2001). ALPL is located on the short arm of chromosome 1 (1p34-1p36.1). In a more detailed study using interphase FISH on frozen sections of native meningioma cells (Niedermayer et al. 1997; Müller et al. 1999), the authors were able to show a complete correlation between losses in 1p and reductions in the amount and activity of alkaline phosphatase (Fig. 21.1). As the activity of alkaline phosphatase can easily be detected histochemically and is highly correlated with the presence of 1p, this prognostically relevant parameter is also available to non-cytogeneticists. The data speak in favour of a tumor suppressor gene function of ALPL; however, the mechanism of inactivation of the second allele is not yet explained.

Oncogenetic Tree Models

The basis for the oncogenetic tree mixture models are single oncogenetic trees as introduced by Desper et al. (1999) that can be used to estimate the most likely pathogenetic routes in tumors from observed subsets of genetic events. These models are of high explanatory power, but often only for a portion of the analyzed tumor samples. A subset of genetic events is only represented by this tree model if for any event in this subset all precursor events in the tree also belong to the subset. All other subsets of events are assigned likelihood zero. Von Heydebreck et al. (2004) propose to include additional hidden events in the tree and to model genetic events as leaves in the tree. This method trades feasibility of maximum-likelihood estimation of oncogenetic trees with reduced interpretability due to the introduction of hidden events.

We introduced mixture models of the single oncogenetic trees as used in Desper et al. (1999). In these mixture models, one tree component is restricted to have a star-like topology, representing independence between genetic events. Owing to the starlike component, every combination of genetic events is represented in the model. The oncogenetic tree mixture models combine interpretability of the trees of Desper et al. (1999) with an appropriate probabilistic framework.

Relevance of the Gender for Genetic Progression Models

Breaking down the oncogenetic tree by gender, there is a reversal concerning chromosomes 14 and 1p- in the male population. For the males, monosomy 14 is estimated to be an earlier event followed by the deletion of the short arm of one chromosome 1. Monosomy 14 has been found to be associated with aggressive behavior of meningiomas (Ketter et al. 2001; Perry et al. 1996; Mañillo et al. 2003; Sayaguès et al. 2004; Menon et al. 1997). However, in the literature the loss of chromosome 14 has never been correlated with the patient's gender. It is well known that females

predominate over males with a ratio of 2.67: 1 as to the overall incidence of meningiomas.

It could be shown that the deletion of the short arm of one chromosome 1 has to be regarded as a more valid cytogenetic parameter than monosomy 14 for the prediction of tumor recurrence, particularly because all anaplastic meningiomas in our series displayed a deletion of chromosome 1p (Fig. 21.3).

Predictive Value of Genetic Progression Scores for Tumor Recurrence

We introduce the GPS of a tumor as the estimated average waiting time of its observed genetic pattern in the timed oncogenetic tree. Using Cox regression analysis we demonstrate that for meningiomas the GPS has prognostic value with respect to clinical outcome and recurrence (Fig. 21.4). Previously it was shown that the information gain due to GPS holds true also for tumor samples from two other cancer types with notably different genetic backgrounds, namely glioblastoma and prostate cancer (Rahnenführer et al. 2005).

Strong correlations were found between histological measurements and the GPS. Interestingly, the histological grades I and II alone were not informative with respect to recurrence of meningioma. Figure 21.4 shows the clinical relevance of the calculated GPS classification broken down to the WHO classification. This demonstrates that the genetic aberrations, especially the deletion of 1p, are independent markers for the clinical courses of the meningiomas. Thus the GPS classification allows a prognostically significant distinction between low-risk and high-risk meningiomas at the time of primary surgery. It can be expected that a combination of both histopathological and cytogenetical description of meningiomas could result in an improved prognostic accuracy (Kolles et al. 1995; Niedermayer et al. 1997; Perry et al. 1997).

Accordingly, the deletion of the short arm of one chromosome 1 is an independent prognostic factor which correlates significantly with a raised risk of recurrence. Due to the limited follow-up

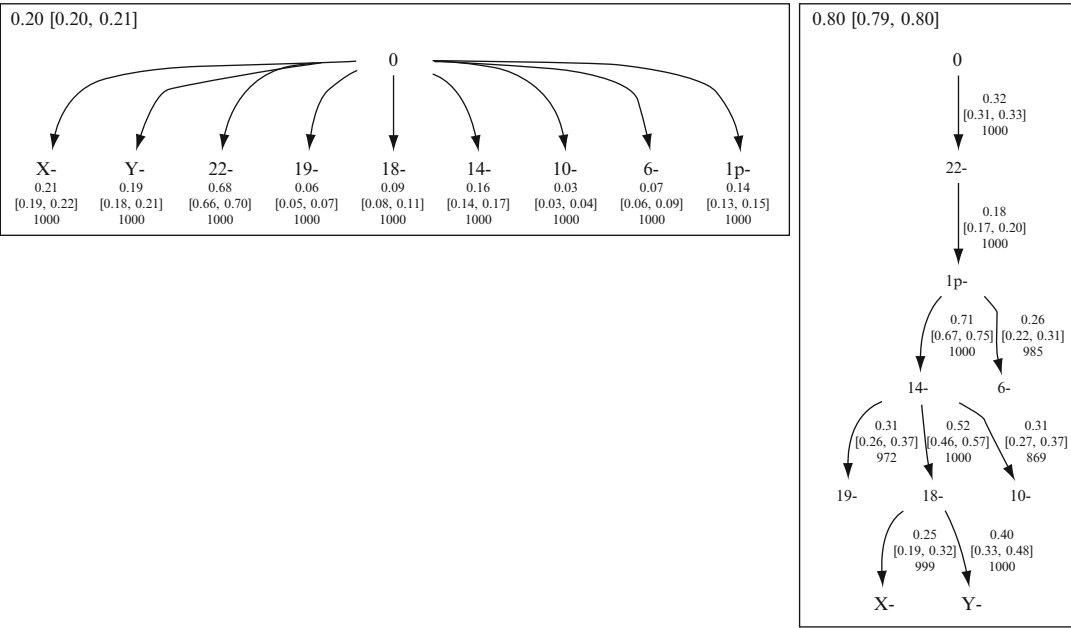


Fig. 21.3 Oncogenetic trees mixture model for the development of meningiomas. Both mixture components are labelled with their weight in the upper left corner. Trees vertices correspond to genetic loss in conventional cytogenetics. Edges are labelled with conditional probabilities

(first row) and their 95% confidence intervals (second row). The third row of the edge labels denotes the number of times the edge has appeared in 1,000 bootstrapped trees (Ketter et al. 2007b)

period, we have to expect that the recurrence rate may further increase over time.

Our results are in agreement with former cytogenetic investigations which indicated, that the deletion of the distal part of the short arm of a chromosome 1 (1p-) is associated with progression in meningiomas (Bello et al. 1994; Henn et al. 1995; Kolles et al. 1995; Müller et al. 1999; Niedermayer et al. 1997; Simon et al. 1995; Ketter et al. 2001). After initial speculation on the role of 1p deletion for tumor recurrence (Bello et al. 1994) the importance of this aberration besides monosomy 22, 14 and 10 for the development of atypical and anaplastic meningiomas (Simon et al. 1995) and for progression from typical to atypical meningioma (Zang 2001) was pointed out.

Further Parameters Involved in Meningioma Progression, Infiltration, and Recurrence

A very peculiar phenomenon in meningioma is invasion into the neighbouring anatomical structures, such as the brain tissue and the skull bone. Extracellular matrix (ECM) degradation by several proteolytic enzyme systems is a critical step in tumor infiltration. Matrix metalloproteases (MMPs) are involved in the degradation of ECM components such as collagen and proteoglycans in normal embryogenesis and remodelling, as well as in many disease processes such as tumor development, invasion and progression. Okuducu et al. (2006) found an association between the transcription factor Ets-1, which is the main

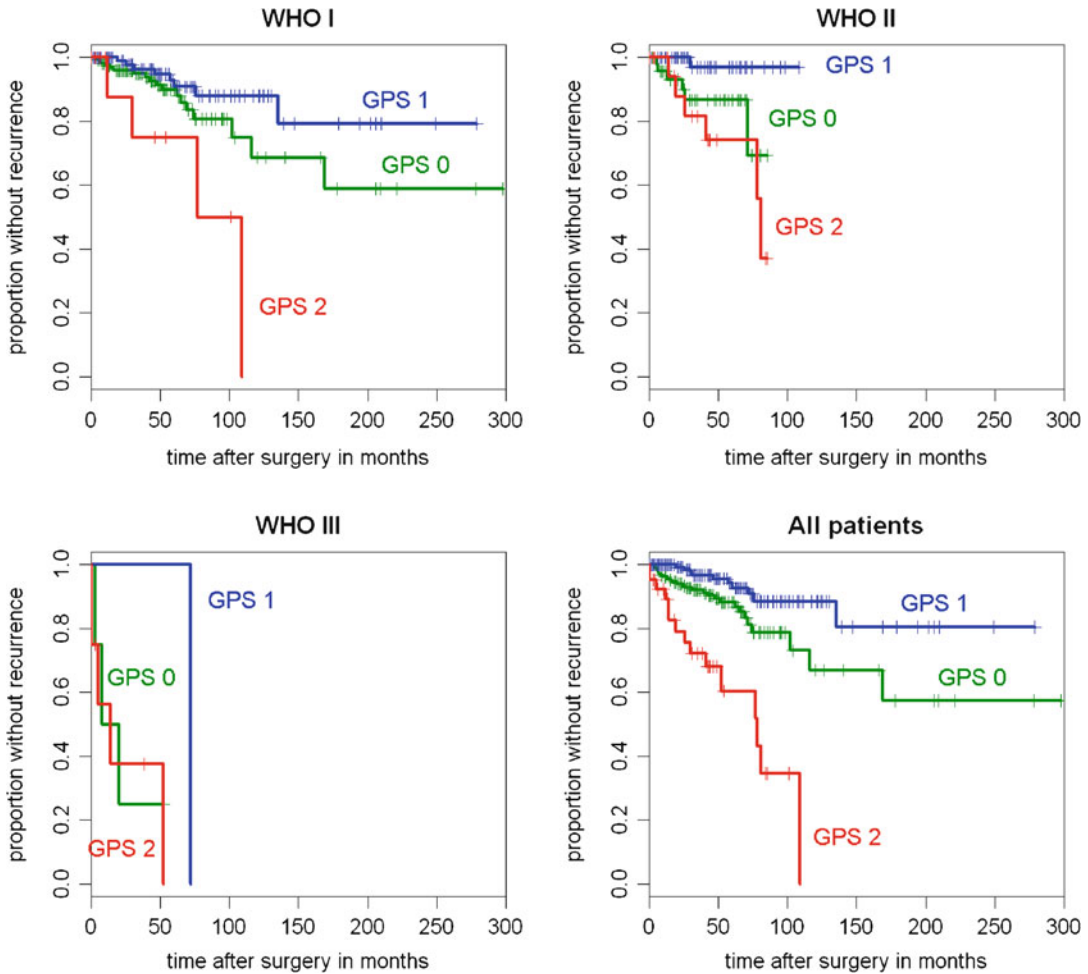


Fig. 21.4 Kaplan-Meier survival curves for time to recurrence of meningioma patients; patients are split into three subgroups according to grouping based on the genetic progression score (*green line*: $\text{GPS} \leq 1.81$, *blue line*:

$1.81 < \text{GPS} \leq 6.39$, *red line*: $6.39 < \text{GPS}$). In the first three plots only subsets of patients according to WHO grade are considered (Ketter et al. 2007b)

regulator of several MMPs, and MMP-2 and -9 expression indicating a possible role for Ets-1 in MMP regulation in meningiomas. They found that increased Ets-1 expression was associated with high WHO grades in meningiomas, offering a possible additional diagnostic and prognostic tool for the evaluation of meningiomas (Okuducu et al. 2006).

The increasing frequency of chromosome aberrations in higher-grade meningiomas might

be the result of loss of apoptotic control. Surprisingly, data on TP53 are relatively scarce and inconsistent. With usual SSCP techniques, mutations have rarely been detected. However, histochemically appears to be a significant accumulation of (inactive) p53 protein in higher-grade and recurrent meningiomas (Kamai et al. 2000).

A further gene located on chromosome 22 is the gene for the tissue inhibitor of metalloproteinase 3 (*TIMP3*) on 22q12.3. In 2010 Barski and

colleagues examined a series of 50 meningiomas, including 27 benign meningiomas (World Health Organization (WHO) grade I), 11 atypical meningiomas (WHO grade II) and 12 anaplastic meningiomas (WHO grade III). They found hypermethylation of *TIMP3* in 67% of anaplastic meningiomas, but only 22% of atypical and 17% of benign meningiomas. Therefore, *TIMP3* inactivation by methylation seems to be also involved in meningioma progression at least it is associated with a more aggressive, high-grade meningioma phenotype (Barski et al. 2010).

Especially the deletion of chromosome 1p was ascertained to be an early and crucial event in the progression in meningiomas. In order to verify the information gain due to GPS, it is of particular interest to demonstrate improved performance over established histopathological parameters. By fitting multivariate Cox regression models we also found that for meningiomas the GPS is prognostic also after adjustment for age. The GPS can thus be used to further identify subgroups. We define low-risk patients as those with histological WHO Grade I tumours that belong to GPS group 0 or 1, and high-risk patients as those who harbour tumours that are Grade II or III and / or belong to GPS group 2. High-risk patients should undergo an intensified regimen of postoperative surveillance including NMR follow-up every 6 months and glucose-PET studies to assess the biological activity of early recurrent tumour growth. Consequently, a multimodal approach to meningioma grading is most promising for identifying meningiomas with an increased tendency to recur and for planning their follow-ups.

References

- Adegbite AB, Khan MI, Paine KWE, Tan LK (1983) The recurrence of intracranial meningiomas after surgical treatment. *J Neurosurg* 58:51–56
- Barski D, Wolter M, Reifemberger G, Riemenschneider MJ (2010) Hypermethylation and transcriptional downregulation of the *TIMP3* gene is associated with allelic loss on 22q12.3 and malignancy in meningiomas. *Brain Pathol* 20(3):623–631
- Bello MJ, de Campos JM, Kusak ME, Vaquero J, Sarasa JL, Pestana A, Rey JA (1994) Allelic loss of 1p is associated with tumor progression of meningiomas. *Genes Chromosomes Cancer* 9:296–298
- Cushing H (1922) The cranial hyperostoses produced by meningeal endotheliomas. *Arch Neurol Psychiatry* 8:139–152
- Desper R, Jiang F, Kallioniemi OP, Moch H, Papadimitriou CH, Schäffer AA (1999) Inferring tree models for oncogenesis from comparative genome hybridization data. *J Comput Biol* 6:37–51
- Henn W, Cremerius U, Heide G, Lippitz B, Schröder JM, Gilsbach JM, Büll U, Zang KD (1995) Monosomy 1p is correlated with enhanced in vivo glucose metabolism in meningiomas. *Cancer Genet Cytogenet* 79:144–148
- Jiang F, Desper R, Papadimitriou CH, Schaffer AA, Kallioniemi OP, Richter J, Schraml P, Sauter G, Mihatsch MJ, Moch H (2000) Construction of evolutionary tree models for renal cell carcinoma from comparative genomic hybridization data. *Cancer Res* 60:6503–6509
- Kamai Y, Watanabe M, Nakayama T, Kanamaru K, Waga S, Shiraishi T (2000) Prognostic significance of p53 and p21 immunoreactivity and tumor mikronecrosis for recurrence of meningiomas. *J Neurooncol* 46:205–213
- Ketter R, Henn W, Niedermayer I, Steilen-Gimbel H, König J, Zang KD, Steudel WI (2001) Predictive value of progression-associated chromosomal aberrations for the prognosis of meningiomas: a retrospective study of 198 cases. *J Neurosurg* 95:601–607
- Ketter R, Kim YJ, Storck S, Rahnenführer J, Romeike BF, Steudel WI, Zang KD, Henn W (2007a) Hyperdiploidy defines a distinct cytogenetic entity of meningiomas. *J Neurooncol* 83(2):213–221
- Ketter R, Urbchat S, Henn W, Feiden W, Beerenwinkel N, Lengauer T, Steudel WI, Zang KD, Rahnenführer J (2007b) Application of oncogenetic trees mixtures as a biostatistical model of the clonal cytogenetic evolution of meningiomas. *Int J Cancer* 121:1473–1480
- Ketter R, Rahnenführer J, Henn W, Kim YJ, Feiden W, Steudel WI, Zang KD, Urbchat S (2008) Correspondence of tumor localization with tumor recurrence and cytogenetic progression in meningiomas. *Neurosurgery* 62(1):61–69
- Kolles H, Niedermayer I, Schmitt C, Henn W, Feld R, Steudel WI, Zang KD, Feiden W (1995) Triple approach for diagnosis and grading of meningiomas: histology, morphometry of Ki-67/Feulgen stainings, and cytogenetics. *Acta Neurochir (Wien)* 137:174–181
- Louis DN, Scheithauer BW, Budka H, von Deimling A, Kepes JJ (2000) Meningiomas. In: Kleihues P, Cavenee WK (eds) *Pathology and genetics. Tumors of the nervous system*. IARC, Lyon
- Maíllo A, Orfao A, Sayagués JM, Diaz P, Gomez-Moreta JA, Caballero M, Santamarta D, Santos-Briz A, Morales F, Tabernero MD (2003) New classification scheme for the prognostic stratification of meningioma on the basis of chromosome 14 abnormalities,

- patient age, and tumor histopathology. *J Clin Oncol* 21:3285–3295
- Menon GM, Rutter JL, von Sattel JP, Synder H, Murdoch C, Blumenfeld A, Martuza RL, von Deimling A, Gusella JF, Houseal TW (1997) Frequent loss of chromosome 14 in atypical and malignant meningioma: identification of a putative “tumor progression” locus. *Oncogene* 14:611–616
- Mirimanoff RO, Dosoretz DE, Linggood RM, Ojemann RG, Martuza RL (1985) Meningioma. Analysis of recurrence and progression following neurosurgical resection. *J Neurosurg* 62:18–24
- Müller P, Henn W, Niedermayer I, Ketter R, Feiden W, Steudel WI, Zang KD, Steilen-Gimbel H (1999) Deletion of chromosome 1p and loss of expression of alkaline phosphatase indicate progression of meningiomas. *Clin Cancer Res* 5:3569–3577
- Niedermayer I, Feiden W, Henn W, Steilen-Gimbel H, Steudel WI, Zang KD (1997) Loss of alkaline phosphatase activity in meningiomas: a rapid histochemical technique indicating progression-associated deletion of a putative tumor suppressor gene on the distal part of the short arm of chromosome 1. *J Neuropathol Exp Neurol* 56:879–886
- Okuducu AF, Zils U, Michaelis SA, Mawrin C, von Deimling A (2006) Increased expression of avian erythroblastosis virus E26 oncogene homolog 1 in World Health Organization grade 1 meningiomas is associated with an elevated risk of recurrence and is correlated with the expression of its target genes matrix metalloproteinase-2 and MMP-9. *Cancer* 107(6):1365–1372
- Osske G, Jänisch W (1967) Zur Histochemie der Meningeome. *Acta Neuropathol* 9:290–297
- Perry A, Jenkins RB, Dahl RJ, Moertel CA, Scheithauer BW (1996) Cytogenetic analysis of aggressive meningiomas. *Cancer* 77:2567–2573
- Perry A, Stafford SL, Scheithauer BW, Suman VJ, Lohse CM (1997) Meningioma grading: an analysis of histologic parameters. *Am J Surg Pathol* 21:1455–1465
- Perry A, Gutmann DH, Reifenberger G (2004) Molecular pathogenesis of meningiomas. *J Neurooncol* 70:183–302
- Rahnenführer R, Beerenwinkel N, Schulz WA, Hartmann C, von Deimling A, Wullich B, Lengauer T (2005) Estimating cancer survival and clinical outcome based on genetic tumor progression scores. *Bioinformatics* 21:2438–2446
- Rutledge MH, Sarrazin J, Rangartnam S, Phelan CM, Twist E, Merel P, Delattre O, Thomas G, Nordenskjöld M, Collins VP, Dumanski JP, Rouleau GA (1994) Evidence for the complete inactivation on the NF2 gene in the majority of sporadic meningiomas. *Nat Genet* 6:180–184
- Sayagués JM, Tabernero MD, Maíllo A, Espinosa A, Rasillo A, Díaz P, Ciudad J, López A, Merino M, Goncalves JM, Santos-Briz A, Morales F (2004) Intratumoral patterns of clonal evolution in meningiomas as defined by multicolor interphase fluorescence in situ hybridization (FISH). *J Mol Diagn* 6(4):316–325
- Simon M, von Deimling A, Larson JJ, Wellenreuter R, Kaskel P, Waha A, Warnick RE, Tew JR, Menon AG (1995) Allelic loss on chromosomes 14, 10 and 1 in atypical and malignant meningiomas: a genetic model of meningioma progression. *Cancer Res* 55:4696–4701
- Takeuchi H, Kubota T, Kabuto M, Kitai R, Nozaki J, Yamashita J (1997) Prediction of recurrence in histologically benign meningiomas: proliferating cell nuclear antigen and Ki-67 immunohistochemical study. *Surg Neurol* 48:501–506
- von Deimling A, Larson J, Wellenreuther R, Stangl AP, van Velthoven V, Warnick R, Tew J, Balko G, Menon AG (1999) Clonal origin of recurrent meningiomas. *Brain Pathol* 9:645–650
- von Heydebreck A, Gunawan B, Fuzesi L (2004) Maximum likelihood estimation of oncogenetic tree models. *Biostatistics* 5(4):545–556
- Zang KD (2001) Meningioma: a cytogenetic model of a complex benign human tumor, including data on 394 karyotyped cases. *Cytogenet Cell Genet* 93:207–220
- Zang KD, Singer H (1967) Chromosomal constitution of meningiomas. *Nature* 216:84–85
- Zhu JJ, Maruyama T, Jacoby LB, Hermann JG, Gusella JF, Black PM, Wu JK (1999) Clonal analysis of a case of multiple molecular genetic approaches: pathology case report. *Neurosurgery* 45:409–416
- Zülch KJ (1979) Histological typing of tumors of the central nervous system. World Health Organization, Geneva

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Abstract

The Erythropoietin – Erythropoietin receptor (Epo-EpoR) pathway is involved in tumor growth, metastasis and treatment resistance and is a potential target in tumor therapy. Expression of EpoR has been reported in various types of cancer and seems to be tissue-specific. For example, in astrocytomas EpoR decreases with grade of malignancy but is associated with patient survival. In human meningiomas EpoR expression has been confirmed by immunohistochemistry, reverse transcriptase polymerase chain reaction and Western Blot. A retrospective study showed that recurring meningiomas later had a significantly lower mean EpoR expression compared to a recurrence-free control group. No significant associations between EpoR expression and WHO grade, age, sex or brain-invasion have been observed. All three known isoforms of EpoR have been detected in meningiomas by RT-PCR. Only the truncated isoform EpoR-T is present in all meningiomas, while the full-length isoform EpoR-F and the soluble isoform EpoR-S are seen in 66% and 58% cases respectively. Lower EpoR mean levels might be a useful marker for a higher recurrence risk, but further studies are needed to verify this observation. The impact of the different EpoR isoforms on tumor growth and a possible association with peritumoral edema should be addressed in future studies.

Introduction

Meningiomas are mainly slowly growing solid tumors that account for 15–20% of primary neoplasms of the central nervous system, and have

an estimated annual incidence of about 6 per 100,000 individuals (Lantos et al. 2002). They are seen most frequently after age 40, and are more common in women than men. Although the majority of these lesions are benign, meningiomas may invade the brain and recur and in these cases are associated with increased morbidity and mortality. Depending on histological features of the tumors that affect the clinical prognosis, the World Health Organization (WHO) classifies meningiomas into three histologic

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grades, from benign grade I to anaplastic meningiomas grade III (Perry et al. 2007). Invasion of the adjacent central nervous tissue is considered a prognostic factor and such tumors are now considered WHO grade II lesions because they are more likely to recur. Brain invasion is observed in approximately 4% of the tumors examined (Perry et al. 2007) and frequencies in the paediatric age group reach values up to 17% (Perry et al. 2001). It is noteworthy that invasion has so far only been observed in the brain but not in the spinal cord. Even after total tumor resection meningiomas may recur and significantly influence the patients clinical course (Jaaskelainen 1986). The likelihood of tumor recurrence increases with WHO grade but itself does not affect tumor grading and seems to be influenced by several factors. On molecular grounds, between 40% and 60% of all meningiomas exhibit loss of heterozygosity (LOH) of chromosome 22, containing the NF2 gene. Patients suffering from neurofibromatosis type 2 may thus exhibit multiple meningeal tumors and suffer from higher recurrence rates (Perry et al. 2001). A higher risk for recurrence is observed in patients with monosomy of Chromosome 14 (Tabernero et al. 2005) indicating the presence of a hitherto unknown tumor suppressor gene at that locus. Other risk factors include history of head trauma or post therapeutic radiation treatment. Further factors that are associated with meningioma recurrence are expression status of p53 and its binding protein homologue MDM and expression of progesterone receptor (Ohkoudo et al. 1998; Strik et al. 2002). The latter might explain why meningiomas are more common in women overall, particularly in those who use postmenopausal hormone replacement therapy. Almost all WHO grade I meningiomas are treated by surgery alone when possible, in cases of WHO grade II and III meningiomas, radiotherapy and chemotherapeutic agents are also administered. Blood loss due to highly vascularised tumors during surgery as well as chemotherapy-induced anaemia might require the administration of erythropoiesis stimulating agents or blood transfusions.

The haematopoietic cytokine erythropoietin (Epo) regulates production and differentiation

of erythroid progenitor cells through its specific cell surface receptor EpoR (Jones et al. 1990). The receptor is expressed in a wide spectrum of non-haematological cells including tumors thereof (Arcasoy et al. 2003). Although EpoR expression levels in cancer cells are lower compared to erythroid cells, they are capable of mediating proliferation, invasion and angiogenesis, therefore promoting tumor growth (reviewed in Tóvári et al. 2008). Furthermore EpoR expression is associated with resistance to radiation and chemotherapy (Belenkov et al. 2004). Erythropoiesis-stimulating agents are commonly administered in cancer patients (Sinclair et al. 2007). Recently several clinical trials with recombinant human erythropoietin (rhEpo) have shown an increased mortality and tumor progression leading to study termination (Arcasoy 2008). Some groups consider administration of rhEpo helpful in improving radiosensitivity by increasing tumor oxygenation (Rades et al. 2008).

Technical Considerations for Epo-R Antibodies

There are several commercially available antibodies for EpoR. However not all of them are specific enough to mark only EpoR in immunohistochemistry or western blot. The polyclonal antibody C20 which should detect EpoR cross reacts with one of several isoforms of heat shock protein (HSP)70, which are also found in tumors (Elliott et al. 2006). The specific anti-EpoR monoclonal antibodies A82 and M20 are suitable for immunoblot assays but did not work for immunohistochemistry in our hands. The monoclonal antibodies 07-311 and M-20 detect sequence analysis confirmed EpoR peptides at 59 kDa in Western blot. Clones 07-311 and 3 8409 are suitable for immunohistochemistry. Antibodies showing a distinct lane at 66 kDa in EpoR Western blot are also likely to react with HSP70. The Epo-R studies in meningioma were performed with 20 µl/ml of monoclonal mouse anti-human EpoR clone 38409, R&D Systems, Minneapolis, USA for western blot

and immunohistochemistry (1:200 dilution). In western blot a distinct lane at 59 kDa was observed, suggesting correct detection of EpoR. Similar results in immunohistochemistry were obtained with rabbit monoclonal, Upstate, NY, USA at 1:350 dilution on the Ventana Benchmark Immunohistochemistry system with standard protocols (for details see Küster et al. 2009).

Epo-R Diagnostics – Methodology

Evaluation of Immunohistochemistry

A semiquantitative scoring for cytoplasmic EpoR staining intensity has been proven useful (scores were given as follows: 0: negative, 1: weak, 2: moderate and 3: strong expression). In most cases staining intensity is similar throughout the whole tumor tissue. In rare cases heterogeneous staining can be observed with areas of weak expression alternating with strong expression. Such cases are scored according to the predominant areas. Continuous variables should undergo Box-Cox transformation for stabilizing the variances of the raw data prior to one-way ANOVA and all pair Tukey HSD is to be used for post hoc analysis. Ordinal logistic regression can be used in order to compare different groups.

RT-PCR Primer Sequences for EpoR Isoforms

Useful primer sequences to detect full-length (EpoR-F), the truncated (EpoR-T) and the soluble (EpoR-S) isoforms are as follows: EpoR-F forward: TCATCCTGGTGCTGCTGACC, reverse: GCCTTCAAACCTCGCTCTCTGG (amplicon size: 103 bp), EpoR-T forward: TCATCCTGGTGCTGCTGACC, reverse: GGCAGTGAAGGGGACAACCA (size: 163 bp), EpoR-S forward: ACGTCTCACATCCGCTACGAG, reverse: CTGTGGAAGTGAAGCAATCA (size: 104 bp) and beta-actin: forward: AGAAAATCTGGCACCACACC, reverse: AACGGCAGAAGAGAGACCA (size: 209 bp). Conditions for all pairs are: denaturation at 95°C for 30 s, annealing at

59°C (EpoR-S 56°C) for 45 s, elongation at 72°C for 60 s using 35 (EpoR-S: 30) cycles.

Epo-R Expression in Meningiomas

Immunohistochemistry of EpoR in meningioma tumor samples typically shows a diffuse expression of varying intensity in the cytoplasm of the tumor cells independent of the meningioma subtype (Küster et al. 2009). The staining usually spares collagen deposits within the tumor as well as dura mater, lymphocytes and vessel endothelia. Delineating a distinct membranous EpoR presence is not possible, since most meningiomas exhibit a pseudosyncytial cell pattern. In some cases with strong cytoplasmic EpoR staining the expression seems to blend into the nuclei, but close examination at the focal plane of the nuclei does not confirm this observation. Typical meningioma structures such as whorls or nodules are stained in similar intensity as the surrounding tumor tissue. Around areas of necrosis a slightly stronger EpoR expression in the perinecrotic tumor cells and apoptotic bodies is observed. These observations are strikingly similar to astrocytic neoplasms in which EpoR is upregulated around areas of pseudopalisading necrosis (Mittelbronn et al. 2007) and is consistent with the finding of an EpoR-upregulation under hypoxic circumstances (Sírén et al. 2001). There is no significant difference of EpoR protein expression levels between WHO grade I, WHO grade II and WHO grade III meningiomas. Although meningiomas occur predominantly in women, no EpoR differences between male and female samples are seen. There is no association of EpoR expression and age. In brain-invasive meningiomas, which are defined as infiltration of the brain parenchyma by finger-like tumor projections without intervening leptomeninges the tumor-brain border does not exhibit any changes in the EpoR expression patterns. EpoR protein levels do not differ between brain-invasive and non-invasive meningiomas.

In primary tumors with recurrence at a later stage the EpoR expression scores are significant lower than recurrent-free tumors (radical removal

according to the Simpson classification and median follow-up of 14.5 years). There is also a negative association of EpoR expression in primary meningiomas and number of tumor recurrences. Pairwise comparison in individual cases shows no significant differences between EpoR expression scores in primary tumor and recurrence indicating consistent expression levels independent of tumor regrowth.

Western blot results are similar to the results obtained by immunohistochemistry. There is a higher mean expression levels in primary tumors compared to recurrent tumors although results are not statistically significant.

Consistent with the above, EpoR mRNA transcripts are detectable in meningioma samples. The full-length EpoR (EpoR-F) is seen in 66% of meningiomas, while the truncated form EpoR-T lacking the cytoplasmic transmembrane domain is seen in all meningiomas. In 58% of the meningioma cases the spliced EpoR-S isoform was detectable with two bands one representing the isoform without intron 4, and the splice variant including intron 4. There is no evidence that meningioma recurrence results from lack of expression of one specific isoform. Rather there seems to be a general protein reduction in tumors with known recurrences indicating that posttranscriptional modifications may account for the altered EpoR levels in these tumors. EpoR expression levels in meningiomas and other neoplasms are lower than the levels found in bone-marrow (Sinclair et al. 2007). They are nevertheless sufficient to mediate proliferative and antiapoptotic effects (Um and Lodish 2006). It has been reported that EpoR-S competes with EpoR-F for Epo binding (Sakanaka et al. 1998). Since we were able to demonstrate EpoR-S transcripts by RT-PCR, one might speculate that binding of Epo through EpoR-S might contribute to pronounced anaemia which is infrequently observed in chordoid meningioma (Couce et al. 2000). The reason why EpoR levels are reduced in recurring meningiomas is not understood, but it is possible, that a reduction of EpoR expression in recurring meningiomas could reflect an impairment of blood-brain barrier capacity or a dedifferentiation of tumour cells. Similar observations are seen with

hormonal receptors in various types of cancer. In meningiomas reduction of progesterone receptor is also associated with a higher risk of recurrence (Strik et al. 2002).

A recent study has shown that in addition to extent of surgical resection, peritumoural brain edema is associated with a higher risk for meningioma recurrence (Simis et al. 2008). Future studies should also aim to determine tumor edema for comparison with the EpoR tumor expression level.

References

- Arcasoy MO (2008) Erythropoiesis-stimulating agent use in cancer: preclinical and clinical perspectives. *Clin Cancer Res* 14:4685–4690
- Arcasoy MO, Jiang X, Haroon ZA (2003) Expression of erythropoietin receptor splice variants in human cancer. *Biochem Biophys Res Commun* 307:999–1007
- Belenkov AI, Shenouda G, Rizhevskaya E, Cournoyer D, Belzile JP, Souhami L, Devic S, Chow TY (2004) Erythropoietin induces cancer cell resistance to ionizing radiation and to cisplatin. *Mol Cancer Ther* 3:1525–1532
- Couce ME, Aker FV, Scheithauer BW (2000) Chordoid meningioma: a clinicopathologic study of 42 cases. *Am J Surg Pathol* 24:899–905
- Elliott S, Busse L, Bass MB, Lu H, Sarosi I, Sinclair AM, Spahr C, Um M, Van G, Begley CG (2006) Anti-Epo receptor antibodies do not predict Epo receptor expression. *Blood* 107:1892–1895
- Jaaskelainen J (1986) Seemingly complete removal of histologically benign intracranial meningioma: late recurrence rate and factors predicting recurrence in 657 patients. A multivariate analysis. *Surg Neurol* 25:233–242
- Jones S, D'Andrea AD, Haines LL, Wong GG (1990) Human erythropoietin receptor: cloning, expression, and biologic characterization. *Blood* 76:31–35
- Küster O, Simon P, Mittelbronn M, Tabatabai G, Hermann C, Strik H, Dietz K, Roser F, Meyermann R, Schittenhelm J (2009) Erythropoietin receptor is expressed in meningiomas and lower levels are associated with tumour recurrence. *Neuropathol Appl Neurobiol* 35:555–565
- Lantos PL, VandenBerg SR, Kleihues P (2002) Tumours of the nervous system. In: Graham DL, Lantos PL (eds) *Greenfield's neuropathology*. Arnold, London
- Mittelbronn M, Capper D, Bunz B, Dietz K, Goepfert B, Ajaaj R, Tabatabai G, Stubenvoll F, Schlaszus H, Merseburger AS, Becker R, Freudenstein D, Wick W, Weller M, Meyermann R, Simon P (2007) De novo erythropoietin receptor (EPO-R) expression in human neoplastic glial cells decreases with grade of

- malignancy but is favourable associated with patient survival. *Neuropathol Appl Neurobiol* 33:299–307
- Ohkoudo M, Sawa H, Hara M, Saruta K, Aiso T, Ohki R, Yamamoto H, Maemura E, Shiina Y, Fujii M, Saito I (1998) Expression of p53, MDM2 protein and Ki-67 antigen in recurrent meningiomas. *J Neurooncol* 38:41–49
- Perry A, Giannini C, Raghavan R, Scheithauer BW, Banerjee R, Margraf L, Bowers DC, Lytle RA, Newsham IF, Gutmann DH (2001) Aggressive phenotypic and genotypic features in pediatric and NF2-associated meningiomas: a clinicopathologic study of 53 cases. *J Neuropathol Exp Neurol* 60:994–1003
- Perry A, Louis DN, Scheithauer BW, Budka H, von Deimling A (2007) Meningiomas. In: Kleihues P, Burger PC, Scheithauer BW (eds) *World Health Organization classification of tumours. Pathology and genetics of tumours of the nervous system*. IARC Press, Lyon, pp 164–171
- Rades D, Golke H, Schild SE, Kilic E (2008) The impact of tumor expression of erythropoietin receptors and erythropoietin on clinical outcome of esophageal cancer patients treated with chemoradiation. *Int J Radiat Oncol Biol Phys* 71:152–159
- Sakanaka M, Wen TC, Matsuda S, Masuda S, Morishita E, Nagao M, Sasaki R (1998) In vivo evidence that erythropoietin protects neurons from ischemic damage. *Proc Natl Acad Sci USA* 95:4635–4640
- Simis A, Pires de Aguiar PH, Leite CC, Santana PA Jr, Rosenberg S, Teixeira MJ (2008) Peritumoral brain edema in benign meningiomas: correlation with clinical, radiologic, and surgical factors and possible role on recurrence. *Surg Neurol* 70:471–477
- Sinclair AM, Todd MD, Forsythe K, Knox SJ, Elliott S, Begley CG (2007) Expression and function of erythropoietin receptors in tumors: implications for the use of erythropoiesis-stimulating agents in cancer patients. *Cancer* 110:477–488
- Sirén AL, Knerlich F, Poser W, Gleiter CH, Brück W, Ehrenreich H (2001) Erythropoietin and erythropoietin receptor in human ischemic/hypoxic brain. *Acta Neuropathol* 101:271–276
- Strik HM, Strobel I, Pietsch-Breitfeld B, Iglesias-Rozas JR, Will B, Meyermann R (2002) The impact of progesterone receptor expression on relapse in the long-term clinical course of 93 benign meningiomas. *In Vivo* 16:265–270
- Taberner MD, Espinosa AB, Maíllo A, Sayagués JM, Alguero Mdel C, Lumbreras E, Díaz P, Gonçalves JM, Onzain I, Merino M, Morales F, Orfao A (2005) Characterization of chromosome 14 abnormalities by interphase in situ hybridization and comparative genomic hybridization in 124 meningiomas: correlation with clinical, histopathologic, and prognostic features. *Am J Clin Pathol* 123:744–751
- Tóvári J, Pirker R, Tímár J, Ostoros G, Kovács G, Döme B (2008) Erythropoietin in cancer: an update. *Curr Mol Med* 8:481–491
- Um M, Lodish HF (2006) Antiapoptotic effects of erythropoietin in differentiated neuroblastoma SH-SY5Y cells require activation of both the STAT5 and AKT signaling pathways. *J Biol Chem* 281:5648–5656

Schwannomas

Part V

Molecular Genetics

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Abstract

Recent studies suggest a role for cyclin D1 in Schwann cell differentiation. The purpose of the study was to evaluate the relationship between cyclin D1 expression and clinical and histological features of vestibular schwannoma. Twenty-one vestibular schwannoma specimens were examined. Immunohistochemistry analysis was performed with anticyclin D1. Clinical charts, audiometric data, and MRI characteristics were reviewed as well as histological features. Cyclin D1 expression was found in 52% of cases. Cyclin D1 expression was more frequent in right sided tumors ($p=0.02$), and in tumors with nuclear degenerative changes ($p<0.0001$). Cyclin D1 expression, present in nearly half of the cases, may play a role in the development of these tumors. Further studies are needed to fully understand the contributions of histopathological and immunohistochemical factors to vestibular schwannoma biological activity.

Introduction

Vestibular schwannomas (VSs) are common benign tumors originating from the Schwann-cell sheath of the eighth cranial nerve. Even though VSs are histologically benign, great variability exists in terms of clinical presentation and surgical findings. For the same tumoral size and location, the occurrence of vestibular symptoms, deteriora-

tion of hearing, and adhesion of the tumor to the facial nerve may be extremely variable. It has been proved that no correlation exists between tumor size and hearing loss (Del Río et al. 2006).

Histologically, VSs are composed of spindle-shaped Schwann cells with alternating areas of compact, elongated cells with occasional nuclear palisading (Antoni A pattern) and less cellular, loosely textured tumor areas (Antoni B). Degenerative changes such as hyperchromatic nuclei, hyalinized vessels, and xanthomatous cells have also been described in VS (Gomez-Brouchet et al. 2001; Yokoo et al. 2003). Clinical implications of these histological features remain obscure.

Unfortunately, to date it is impossible to predict the expected behavior of VS based on information

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available at diagnosis (Herwadker et al. 2005). A better understanding of the molecular mechanisms of VS development and behavior would be very useful for optimal treatment management. Most studies in the pathogenesis of VS have focused on the molecular genetic analysis of the neurofibromatosis type 2 (*NF2*) tumor suppressor gene. Inactivation of the *NF2* gene is the best known alteration in VS. Inactivation of merlin, the product of the *NF2* tumor suppressor gene, is responsible of VS development, both in sporadic and bilateral cases (Mohyuddin et al. 2002). Epigenetic changes have also been described in several genes (Bello et al. 2007), some of them with clinical implications on VS behavior (Lassaletta et al. 2006, 2007a, b). In addition to different mechanisms of gene inactivation, deregulated expression of growth regulatory genes may also play a role in the VS tumorigenesis (Lasak et al. 2002). Several studies have established *cyclin D1* as a proto-oncogene, revealing that its amplification and overexpression may contribute to uncontrolled cell growth in many human tumors (Donnellan and Chetty 1998) *Cyclin D1* elicits its function early in G1 phase, binding to and activating the cyclin dependent kinases CDK4 or CDK6 (Coqueret 2002) In a recent study, Neff et al. (2006) found expression of cyclin D3 in seven of 15 VS, whereas no cyclin D1 expression was found in any of these cases.

Objective

In this study 21 tumor samples of patients with VS were analyzed. Histological features including Antoni type and degenerative changes, as well as immunohistochemical expression of cyclin D1 were performed. The aims of the study were to determine the association of histological findings and cyclin D1 expression with clinical and radiological characteristics, in patients with VS.

Patients

The study group consisted of 21 patients who underwent surgery for removal of unilateral VS in our institution between February 2002 and January

2004. Age, gender, and tumor side were studied. Tumor size was measured in terms of the largest diameter in the axial plane of MRI. In addition, the tumor volume was also measured from MRI, using the formula for an ellipsoid, as previously described (Caye-Thomasen et al. 2005).

Equipment and Procedure

Histology

All tumor tissues obtained at surgery were fixed in 10% formalin and embedded in paraffin. Staining with hematoxylin-eosin was performed for routine microscopic diagnosis. Antoni type A regions consisted of interwoven bundles of long bipolar spindle cells, whereas Antoni B type regions showed a loose myxoid background containing more stellate tumor cells. The percentage of the different tissue type (A, B, mixed) in each tumor sample was performed independently by two pathologists. The results were grouped in three types: type A, >60% of the tumor composed of type A tissue; type B, >60% of the tumor made of type B tissue; mixed type, type A and B tissues in equal quantity, or <60% of the tumor composed of type A and B tissue. The presence of both nuclear and stromal pathologic degenerative changes was determined. Nuclear changes consisted of hyperchromatic, voluminous, irregular schwann cell nuclei. Stromal changes included hyalinized vessels, xantomatous cells, and interstitial fibrosis. Degenerative changes were graded as follows: 0 absent, $\pm$ mild, + moderate, and ++ severe.

Cyclin D1 Immunohistochemistry

The paraffin sections were dewaxed in xylene and acetone, and rehydrated. Deparaffined tissue sections were incubated overnight at 4°C with the anticyclin D1 (P2D11F11) monoclonal antibody (Novocastra, Newcastle UK) at a 1:50 dilution. Optimum primary antibody dilutions were predetermined using known positive control tissues. Multiple known positive control sections were included in each run. The tissues were incubated

in biotin-labeled goat anti-mouse serum (1:200) for 30 min, rinsed with PBS and incubated with avidin-biotin-peroxidase complex for 1 h. The signal was detected using 3, 3'-diaminobenzidine as the chromogen. Only cells with brown-colored cytoplasmic staining were considered as positive. The intensity of immunoreactivity was stratified into four categories: 0 absent, $\pm$ mild, + moderate, and ++ strong. As immunostained tissue sections often showed a heterogeneous staining pattern, the most intense immunoreactive area in each section was used to assess the intensity of immunoreactivity.

Statistical Analyses

Statistical analyses for differences between groups were performed using the Fisher exact test and the nonparametric Mann-Whitney U test. For statistical purposes, cyclin D1 expression was considered negative (0 absent) or positive ($\pm$ mild, + moderate, and ++ strong). Differences were considered significant at a level of $p < 0.05$.

Results

There were 14 women and 7 men with a mean age at surgery of 48 years (range 16–71 years). The tumor was on the left side in 13 cases (62%). Tumor size ranged from 15 to 55 mm (mean, 29 mm). Mean tumor volume at presentation was 935 mm³, ranging from 50 to 5,363 mm³.

Microscopic analysis showed 15 Antoni type A, 3 Antoni type B, and 3 Antoni mixed type tumors. Nuclear and stromal degenerative changes were found in 10 (48%) and 5 (24%) tumors respectively, with both features being present in 12 (57%) cases.

Eleven out of 21 cases (52%) showed positive staining for cyclin D1. Cyclin D1 expression was more frequent in right sided tumors (88%) than in those tumors originating in the left side (31%) ($p = 0.02$). No difference was found between cyclin D1 expression and age, gender, and tumor size. Cyclin D1 expression occurred mostly in tumors with nuclear degenerative changes ($p < 0.0001$) (Fig. 23.1), whereas no association

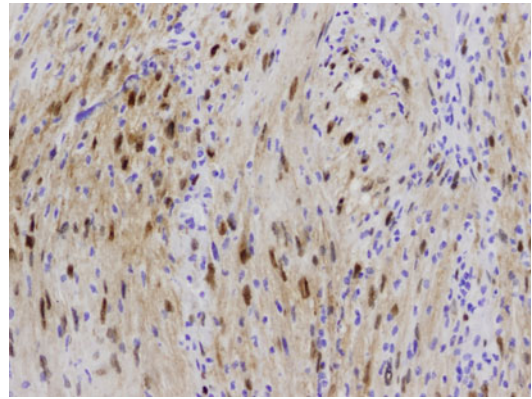


Fig. 23.1 Immunohistochemical staining of cyclin D1 in vestibular schwannoma. Nuclear staining is strong in nuclei with degenerative changes

was found between cyclin D1 expression and Antoni type or stromal degenerative changes.

Discussion

It has been shown that cyclin D1 is a key cell cycle regulatory protein for the mammalian G1-S phase transition and is involved in the regulation of proliferation and differentiation. Recent studies have established cyclin D1 as a proto-oncogene, revealing that its amplification and overexpression may contribute to uncontrolled cell growth in many human tumors, including mantle cell lymphoma, breast cancer, head and neck squamous cell carcinoma and esophageal cancer (Donellan et al. 1998). Cyclin D1 elicits its function early in G1 phase, binding to and activating the cyclin dependent kinases CDK4 or CDK6. Active CDK 4/6-cyclin D1 complexes phosphorylate the retinoblastoma target protein Rb. Upon Rb phosphorylation, the E2F transcription factors activate the expression of S-phase genes and thereby induce cell cycle progression (Coqueret et al. 2002).

The role of the cyclin D family of proteins in VS has been recently studied. In a model of Wallerian nerve degeneration, mitotic response of Schwann cells was completely inhibited in cyclin D1 knockout mice, suggesting that cyclin D1 has a specific function in Schwann cell division (Knudsen et al. 2006). Seol et al. (2005)

suggested that the loss of p27, a regulator of the cyclin D-CDK complexes during G1 to S-phase transition, may be a predictor of VS aggressiveness. In a recent study, Neff et al. (2006) found expression of cyclin D3 in seven of 15 VS, whereas no cyclin D1 expression was found in any of these cases.

Mutation of the NF2 gene product merlin can be detected in NF-2 related and sporadic schwannomas, supporting its role as a tumor suppressor gene. The significance of immunohistochemical expression of a proto-oncogene in a benign tumor such as VS remains unknown. In the study by Xiao (Xiao et al. 2005), Cyclin D1 was shown to be an essential mediator of merlin's effect on cell cycle progression, as merlin could repress cyclin D1 through inhibition of PAK. The authors suggest that how a tumor suppressor gene contributes to both malignant mesotheliomas and benign schwannomas may seem paradoxical, and that a mechanism other than tumor invasiveness must be invoked to attribute a role for NF2 inactivation in both malignant and benign tumors. Lu et al. performed in situ hybridization immunohistochemistry and western blot in tissue specimens from 54 VS patients (Lü et al. 2008). They found that merlin gradually moved from the cytoplasm to the nucleus, whereas cyclin D1 shifted from the nucleus to the cytoplasm, indicating a transfer of the cycle from G1 to S phase. They also suggest a different behavior of merlin in benign and malignant tumors.

In our study, positive expression of cyclin D1 was found in 52% of VSs. Tumors with positive cyclin D1 expression were more frequent in the right side and tumors with nuclear degenerative changes were highly associated with cyclin D1 expression. These results suggest that cyclin D1 expression plays a role in VS behavior. It appears to be a slight association between long-standing tumors and degenerative changes, as well as between nuclear degenerative changes and cyclin D1 expression. Due to the small size of our series further studies in large prospective cohorts are now required to confirm these findings, and fully establish any further clinical significance of these events.

References

- Bello MJ, Martinez-Glez V, Franco-Hernandez C, Peñalva-Granero C, de Campos JM, Isla A, Lassaletta L, Vaquero J, Rey JA (2007) DNA methylation pattern in 16 tumor-related genes in schwannomas. *Cancer Genet Cytogenet* 172:84–86
- Caye-Thomasen P, Werther K, Nalla A, Bøgg-Hansen TC, Nielsen HJ, Stangerup SE, Thomsen J (2005) VEGF and VEGF receptor-1 concentration in vestibular schwannoma homogenates correlates to tumor growth rate. *Otol Neurotol* 26:98–101
- Coqueret O (2002) Linking cyclins to transcriptional control. *Gene* 299:35–55
- Del Río L, Lassaletta L, Alfonso C, Sarriá MJ, Gavilán J (2006) Clinical tumoral size dissociation in acoustic neuroma: reality or measure distortion? *Acta Otorrinolaringol Esp* 57:345–349
- Donnellan R, Chetty R (1998) Cyclin D1 and human neoplasia. *Mol Pathol* 51:1–7
- Gomez-Bouchet A, Delisle MB, Cognard C, Bonafe A, Charlet JP, Deguine O, Frayssé B (2001) Vestibular schwannomas: correlations between magnetic resonance imaging and histopathologic appearance. *Otol Neurotol* 22:79–86
- Herwadker A, Vokurka EA, Evans DG, Ramsden RT, Jackson A (2005) Size and growth rate of sporadic vestibular schwannoma: predictive value of information available at presentation. *Otol Neurotol* 26:86–92
- Knudsen KE, Diehl JA, Haiman CA, Knudsen ES (2006) Cyclin D1: polymorphism, aberrant splicing and cancer risk. *Oncogene* 25:1620–1628
- Lasak JM, Welling DB, Akhmametyeva EM, Salloum M, Chang LS (2002) Retinoblastoma-cyclin-dependent kinase pathway deregulation in vestibular schwannomas. *Laryngoscope* 112:1555–1561
- Lassaletta L, Bello MJ, Del Río L, Alfonso C, Roda JM, Rey JA, Gavilán J (2006) DNA methylation of multiple genes in vestibular schwannoma: relationship with clinical and radiological findings. *Otol Neurotol* 27:1180–1185
- Lassaletta L, Patrón M, Del Río L, Alfonso C, Roda JM, Rey JA, Gavilán J (2007a) Cyclin D1 expression and histopathologic features in vestibular schwannomas. *Otol Neurotol* 28:939–941
- Lassaletta L, Patrón M, González T, Martínez-Glez V, Rey JA, Gavilán J (2007b) RASSF1A methylation and cyclin D1 expression in vestibular schwannomas. *Acta Neuropathol* 114:431–433
- Lü J, Zou J, Wu H, Cai L (2008) Compensative shuttling of merlin to phosphorylation on serine 518 in vestibular schwannoma. *Laryngoscope* 118:169–174
- Mohyuddin A, Neary WJ, Wallace A, Wu CL, Purcell S, Reid H, Ramsden RT, Read A, Black G, Evans DG (2002) Molecular genetic analysis of the NF2 gene in young patients with unilateral vestibular schwannomas. *J Med Genet* 39:315–322

- Neff BA, Oberstien E, Lorenz M, Chaudhury AR, Welling DB, Chang LS (2006) Cyclin D(1) and D(3) expression in vestibular schwannomas. *Laryngoscope* 116:423–426
- Seol HJ, Jung HW, Park SH, Hwang SK, Kim DG, Paek SH, Chung YS, Sub LC (2005) Aggressive vestibular schwannomas showing postoperative rapid growth – their association with decreased p27 expression. *J Neurooncol* 75:203–207
- Xiao GH, Gallagher R, Shetler J, Skele K, Altomare DA, Pestell RG, Jhanwar S, Testa JR (2005) The NF2 tumor suppressor gene product, merlin, inhibits cell proliferation and cell cycle progression by repressing cyclin D1 expression. *Mol Cell Biol* 25:2384–2394
- Yokoo H, Arai H, Isoda K, Arai M, Sasaki A, Hirato J, Nakazato Y (2003) Characterization of eosinophilic hyaline droplets in schwannoma. *Acta Neuropathol (Berl)* 105:170–176

Schwannomas: Role of Molecular Genetics and Epigenetic Mechanisms

24

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Abstract

Schwannomas are benign tumors that arise sporadically or associated to the NF2 syndrome. The characteristic alteration in schwannomas is the loss of heterozygosity of 22q (up to 50%), where *NF2* gene is located. Inactivating mutations of this gene have been found up to 66% of sporadic schwannomas and 33% of NF2-associated tumors. *NF2* nonsense and frameshift mutations, both causing truncated proteins, are associated with more severe phenotypes. Merlin, the product of *NF2* gene, belongs to the ERM family of proteins that connect the actin cytoskeleton with plasma membrane. Merlin functions as a tumor suppressor and is involved in cell-cell and cell-matrix interactions through an array of molecular pathways. Merlin over-expression can inhibit both cell proliferation and the transformation induced by oncogenes, and its down-regulation cause a lack in contact inhibition, leading to neoplastic transformation. Another oncogenic mechanism studied in schwannomas is the epigenetic hypermethylation of genic promoters. The study of epigenetic gene silencing in schwannomas has shown high frequencies of promoter hypermethylation in some well known cancer-related genes, and specifically on *NF2* gene. There are many unknown aspects in the molecular biology of schwannomas. However, current knowledge and technical advances allows us to be optimistic about future perspectives.

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Introduction

Schwannomas (or neurilemmomas) are benign tumors derived from Schwann cells. This tumor can occur as a sporadic neoplasm or associated to the Neurofibromatosis type 2 (NF2), an autosomal dominant syndrome with high penetrance caused by alterations in the *NF2* gene, in which the characteristic lesion is the bilateral vestibular

schwannoma (VS), and may appear together with meningiomas, ependimomas, spinal schwannomas and presenile crystalline opacity. Another form of neurofibromatosis, not associated with NF2 syndrome, is the schwannomatosis. This disease shows multiple lesions that are rarely of vestibular location and represents 2–5% of patients requiring surgical removal of schwannomas (MacCollin et al. 2005). The schwannomatosis is as common as the NF2 syndrome, but its familial form is an unusual presentation. Schwannomas can also be found in rare multiple neoplasms syndromes as the Carney Complex.

Vestibular schwannomas, also known as acoustic neuromas, are histologically benign tumors that originate in the vestibular branch of the VIIIth cranial nerve and occur sporadically both unilateral and bilateral. Familial bilateral presentation is the hallmark of the NF2 syndrome (Neff et al. 2006). The vestibular schwannomas account for the vast majority of tumors found in the cerebellopontine angle, the rest is composed mainly by meningiomas.

Vestibular schwannomas can be grouped into sporadic unilateral (95%), bilateral (5%)—NF2 syndrome-associated—, and cystic schwannomas (<1%). Approximately half of the cases associated with NF2 syndrome doesn't show a family history of neurofibromatosis and, therefore, they can represent *de novo* germline mutations, or cases of *NF2* gene mutations/losses in the form of mosaicism (Anderson et al. 2000). The cystic vestibular schwannomas are an aggressive group with a high degree of nuclear atypia and rapid growth, which sometimes complicates their treatment.

Unilateral hearing loss is the main clinical feature in patients affected by vestibular schwannomas. Patients with asymptomatic schwannomas associated with NF2 syndrome do not only have more severe neurologic deficit but also suffer less post-operative recovery and higher rates of recurrence (Hanemann and Evans 2006). Clinical and pathological differences exist between the bilateral vestibular schwannomas associated with NF2 syndrome and the unilateral sporadic ones. The first arise in an earlier age, are lobulated, less vascular and have a higher rates of proliferation

than those found in sporadic cases (Sobel 1993). The growth rates of sporadic tumors associated with NF2 syndrome are highly variable (Baser et al. 2002).

Some evidence suggests that the growth rate of vestibular schwannomas associated with NF2 syndrome decreases as the age of the affected patients increase; this is not dependent on the constitutional mutation status of the *NF2* gene. The frequency of these tumors is highly variable within the family and not related to the number of spinal tumors or other brain tumor types, nor the overall severity of illness (Baser et al. 2002).

Schwannoma Genetics

The only common and constant genetic alteration in schwannomas, regardless of their location, is the loss of part of chromosome 22 causing loss of heterozygosity (LOH), which can be found in up to 50% of these tumors. This loss includes the 22q12 region, which harbors the *NF2* gene, in most but not all cases (Bruder et al. 1999). In addition, inactivating mutations of this gene have been found up to 66% of sporadic schwannomas and 33% of those cases associated to NF2 syndrome. Other genetic alterations found in schwannomas include: loss of chromosome regions 1p, 7p, 13q, 15q, 16q, 19p, 20q, 21q and Xq as well as gains in the regions 9q, 11q, 17q and chromosomes 5, 7, 19 and 20 (Antinheimo et al. 2000; Warren et al. 2003).

NF2 Gene Alterations and Genotype/Phenotype Correlations

The *NF2* gene behaves as a tumor suppressor gene as shown by a number of different studies in which patients affected with NF2 syndrome inherit a mutated *NF2* allele and posteriorly acquire a somatic loss in the second allele as part of the tumorigenic process. This is the classic mechanism proposed by the two-hit hypothesis for tumor formation arising from the alteration on tumor suppressor genes. Mutational studies of the *NF2* gene combined with loss of heterozygosity

assays are able to detect changes in one of the two copies of *NF2* gene in 80–90% of the schwannomas, however, only 50–60% of the cases detect an alteration of the two copies of the gene. This may be due to the presence of different mechanisms of inactivation of *NF2* gene such as the silencing by methylation of its promoter region, detected in about 30–40% of sporadic schwannomas. This epigenetic mechanism, besides being an early event, can affect both copies of the gene. Therefore, the mutational and LOH standard approaches may not find abnormalities on these tumors.

The LOH of chromosome 22, which occurs more frequently in vestibular than spinal lesions shows a correlation with the proliferative activity of malignant schwannomas. There are also studies that highlight a genotype-phenotype correlation in *NF2* gene, from which can be inferred that nonsense mutations and frameshift changes—caused by sequence insertions or deletions leading to truncated gene products—might be associated with more severe phenotypes. Whereas, missense mutations—leading to stably expressed but non-functional products—and large *NF2* gene deletions produce milder manifestations (Patronas et al. 2001). However, those findings appear to be more valid for tumors associated with familial neurofibromatosis than for those that arise as sporadic cases.

Unlike the somatic mutations, in those of germinal origin the nonsense changes are more common than frameshift mutations, being the C>T changes the most frequent alteration found in the *NF2* gene. In older patients vestibular schwannomas shows an increased frequency of frameshift mutations, probably due to the degeneration of the DNA repair pathways in these patients (Baser 2006; Evans 2009).

Another factor that influences the severity of schwannomas associated with NF2 syndrome is the mosaicism, which can be detected *de novo* in 25–30% of patients, and whose clinical expression is milder. The offspring of patients with mosaicism who inherit the mutation will present, however, more severe clinical manifestations as they carry the defect in all their cells (Kluwe et al. 2003). Mosaicism—when in the same individual

coexist two or more populations of cells with different genotype—is produced when mutation occur after fertilization. The presence of mosaicism may explain the lower rate of mutation detection in patients with mild phenotype as this may hinder the detection in routine mutational studies made from DNA extracted from peripheral blood lymphocytes.

Merlin Protein and Tumorigenesis

The product of the *NF2* gene is the protein called merlin (moesin-ezrin-radixin-like protein), also known as schwannomin. Merlin belongs to the ERM family of structural proteins 4.1 that connect the actin cytoskeleton with the plasma membrane. All the proteins on this family share a highly conserved N-Terminal domain—the FERM domain—, an α -helical domain and a C-Terminal tail domain—that includes an actin-binding module in the ERM proteins but not in merlin—, which in turn interact with the FERM domain. Merlin function is likely dependent on its tertiary configuration as some studies have shown that *NF2* gene missense mutations are distributed throughout the FERM and C-terminal domains of the protein, disrupting merlin self-association and altering merlin localization and binding-partner interactions. Known merlin binding partners include ezrin, radixin, moesin, actin, layilin, paxillin, CD44, PKA, PAK, MYPT-1, PP1 δ , spectrin, eIF3c, PIKE, MAP, RalGDS, RhoGDI, SCHIP1 and calpain, among others. The exact function of merlin remains still unknown, but some evidence has shown its role in cell-cell and cell-matrix interactions (Scoles 2008).

Merlin shows high levels of expression during development and a significant expression in Schwann cells, cerebral meninges, lens and nerves in adults, where it localizes to the area of membrane remodeling. Over-expression of this protein can inhibit both cell proliferation, and the transformation induced by oncogenes; on the other hand, merlin functional loss cause a lack in contact inhibition, leading to neoplastic transformation (Scoles 2008).

The coding region of *NF2* gene consists of 17 exons, and merlin is transcribed by alternative splicing into two predominant isoforms and a few minor ones. Isoform I (without exon 16) seem to have the protein folding capacity to adopt the closed conformation needed to perform its tumor suppressor function. Merlin folding is regulated by phosphorylation of serine and threonine residues in response to growth conditions and cell shape. Serine 518 is phosphorylated by the p21-activated kinases (PAKs) and cAMP dependent-protein kinase A (PKA), reducing its N-term/C-term self-interaction. PAKs also weaken the ability of merlin to bind to CD44 and HRS interacting proteins, both critical for merlin growth suppression. Despite merlin lacks the actin-binding module shared by the ERM proteins, phosphorylation of merlin N-terminal domain is also involved in regulation of the association of merlin with the cortical cytoskeleton and influence actin cytoskeleton organization, separately from the role of merlin as a tumor suppressor (McClatchey and Fehon 2009; Rong et al. 2004).

In addition, merlin interacts with PIKE-L inhibiting the stimulatory activity of phosphatidylinositol 3-kinase (PI3K)/Akt signaling pathway, and in a negative feedback-loop, merlin is also phosphorylated by Akt on residues T230 and S315, abolishing merlin binding to PIKE-L and leading to ubiquitination and proteasome-mediated degradation of merlin. Moreover, Akt phosphorylation enhances the interaction between merlin N-Terminal domain and phosphatidylinositols, increasing merlin binding to CD44. These phosphorylation changes cause inactivation and subcellular localization alterations of the protein, thus causing inhibition of its proapoptotic activity and blocking its tumor suppressive activity on cell migration in Schwann cells. Dephosphorylation is mediated by MYPT-1-PP1 δ (Okada et al. 2009).

In addition to act upstream of Rac, PAK also acts as an effector of Rac1 and Cdc42 RhoGTPases, and is inhibited by merlin in a negative feedback-loop. PAK activity on this RhoGTPases regulate cellular processes like cell spreading, formation of ruffles, adhesion to the

extracellular matrix, development of polarity and contact inhibition. Studies in schwannomas that do not express merlin protein showed increased activity of Rac1 and its effector Pak1, elevated levels of Cdc42, over-expression of integrin beta, stable paxillin-containing focal adhesions, increased adhesion to extracellular matrix, and over-expression of *MAPK8* and *MAPK9*, suggesting that merlin could regulate the activation of Rac, thus conferring it an important role in cellular differentiation (Flaiz et al. 2009).

The relation between merlin and growth factors has also been subject of study in schwannomas mainly due to their therapeutic potential. Upon cell-cell contact, merlin inhibit EGFR signaling by internalizing ErbB2 receptors, thus modulating adherens junction stabilization and growth factor signaling in normal Schwann cells. The stimulation of the glial growth factor neuregulin-1 (NRG1) and its receptor ErbB2 is essential for Schwann cell differentiation, myelination, survival, and proliferation. Some studies has shown that sporadic vestibular schwannomas show an up-regulation of NRG1, ErbB2 and ErbB3, whose inhibition decreases the proliferation of vestibular schwannoma cells in vitro, and therefore could have a possible therapeutic role (Hansen et al. 2006).

On the other hand, EGF has been detected up-regulated only in vestibular schwannomas associated to NF2 syndrome. As a result, it has been inferred that EGF-induced EGFR homodimers and EGFR/ErbB2 heterodimers may be more important for signaling in the NF2-associated tumors. These molecular differences are useful to explain the elevated growth rate and invasiveness in NF2-associated tumors. It has been postulated that the observed differences in ErbB signaling between sporadic and NF2 syndrome-related schwannomas may be a consequence of the interaction between merlin haploinsufficient-Schwann normal cells and tumoral ones in NF2 syndrome patients, due to altered ErbB2 receptors internalization, thus causing a lack of the normal contact-inhibited growth regulatory signals in Schwann cells (Doherty et al. 2008).

Analysis of different schwannoma tumor series has shown that the most potent mediator of

angiogenesis and a well known cell mitogen, the *VEGF* gene, is expressed in schwannomas and appears to be directly involved in the growth rate in vestibular schwannomas, which in turn is related to a more aggressive tumoral behavior and severe clinical symptoms. *VEGF* expression levels are not related to tumor size or symptoms duration (Caye-Thomasen et al. 2005). Also, a correlation has been established between the tumor concentration of MMP9 and tumor growth rate in vestibular schwannomas (Moller et al. 2010). Vascular endothelial growth factor (VEGF) and matrix metalloproteinases (MMPs) are potent mediators of tumor progression and angiogenesis and, due to their implication on tumor growth, currently different therapies in schwannomas based on drugs against these molecules are under evaluation.

Schwannoma Epigenetics

Another phenomenon that has been studied in schwannomas, and which belongs to the epigenetic area, is the hypermethylation of genic promoters. Epigenetic alterations, those affecting gene expression in the absence of changes in DNA sequence, appear to play an important role in the tumorigenesis of many types of tumors. Epigenetic information provides patterns that can regulate and differentiate groups of cells that share the same basic genetic information. Most of the epigenetic patterns are established during differentiation and maintained from then on through cell division, allowing the cellular machinery to access only the genetic information necessary for a specific cell type at a specific given time. There are four main epigenetic mechanisms: DNA methylation, covalent histone modifications, noncovalent mechanisms—histone variants and nucleosome remodeling—, and non-coding RNAs—including microRNAs (miRNAs)—. All these changes in chromatin structure and dynamics, known as the epigenome, regulate the expression of the genome, and its alterations leads to pathology, including cancer.

Cancer cells show a global modification of their epigenetic patterns, which added to the

classical cancer genetic alterations, may favor the process of tumorigenesis, tumor growth, invasiveness and metastasis. We now know that these alterations may occur as an early event in the genesis and tumor progression. Epigenetic alterations result in global dysregulation of gene expression profiles by silencing of tumor suppressor or DNA repairing genes or by promoting tumorigenesis by activating oncogenes, therefore acting as one of the two events necessary for tumor development according to the cancer two hit hypothesis. These “epimutations” are passed on to daughter cells during mitosis, which can give tumor cells a selective advantage that results in an unrestrained growth that escape from the normal cellular control mechanisms (Ballestar and Esteller 2008; Sharma et al. 2010).

The epigenetic phenomenon that has been more widely studied in schwannomas is the gene silencing by hypermethylation of gene promoters. Cancer initiation and progression shows two major changes in DNA methylation patterns: genome-wide hypomethylation and site-specific CpG island promoter hypermethylation. The first one increases genomic instability and may cause activation of proto-oncogenes, and the second one alters the promoter region of genes inhibiting its transcriptional activation and/or favoring gene repression through interactions with histone deacetylases—another epigenetic mechanism—.

The study of epigenetic gene silencing in schwannomas has shown high frequencies of promoter hypermethylation in some well known cancer-related genes, and specifically, the cloning and functional characterization of the NF2 gene promoter region has allowed the identification of CpG sites whose methylation results in decreased activity of the promoter region of this gene (Kino et al. 2001). Posterior studies in schwannomas, both sporadic and NF2-associated have detected hypermethylation of gene promoters for *THBS1* (36%), *TP73* (19–27%), *MGMT* (20%), *NF2* (18%), *TIMP-3* (18%), *RB1/p16INK4a* (15%), *THBS1* (30%), *hMLH1* (21%), *TP53* (30%) and *MGMT* (16%) genes in large series of tumor samples (Almeida et al. 2009; Bello et al. 2007; Gonzalez-Gomez et al. 2003).

The study of promoter gene methylation status in vestibular schwannomas have also found correlations between aberrant hypermethylation of gene promoters and clinical-radiological features, methylation of *CASP8* and *RASSF1A* genes, *CASP8* hypermethylation and patient age and tumor size, hearing loss and *TP73* methylation, and an inverse correlation between *RASSF1A* hypermethylation and the clinical growth index (Lassaletta et al. 2006).

A recent study found that in about 40% of sporadic VS the alleles of the *NF2* gene that do not show losses or mutations, neither show aberrant hypermethylation of gene promoters, hence it is postulated that there must be other oncogenic mechanisms responsible for the development of this type of vestibular schwannomas (Kullar et al. 2010). Because classical genetic alterations and methylation have been discarded in these alleles, there may be other epigenetic mechanisms that have not yet been sufficiently explored in cancer as a whole and specifically in schwannomas. These changes may include epigenetic histone modifications, the regulatory action of the microRNAs or the involvement of other yet not known genes or regulatory mechanisms. Studies on this type of alternative tumorigenic mechanisms will shed more light on cancer development.

In conclusion, it seems clear that there is no single pathogenic mechanism in schwannomas. Additionally to the complexity of cellular pathways involved in this type of tumor, there are significant differences between sporadic cases and those associated with the NF2 syndrome. These differences are likely to be exploited in future molecular therapies, up to the point of individualized treatment in response to specific molecular characteristics in each case. But more close in time, current works are studying various molecular targets aimed at, in example, to limit the ability of tumor growth through inhibition of angiogenic factors like *VEGF* or acting on MAPK or PI3K pathways. Studies on therapy in schwannomas probably should also address the promising reversibility of epigenetic alterations.

As with other tumors, the molecular mechanisms underlying the pathogenesis of schwannomas have

evolved rapidly in recent years, due in part to the development of powerful techniques useful in research and clinical management. These advances allow us to be optimistic about future perspectives. However, there is still much to discover to properly understand the molecular biology of schwannomas. The search for improved quality of patient's life is, in brief, the ultimate goal that should move the engine of biomedical research.

References

- Almeida LO, Custodio AC, Pinto GR, Santos MJ, Almeida JR, Clara CA, Rey JA, Casartelli C (2009) Polymorphisms and DNA methylation of gene TP53 associated with extra-axial brain tumors. *Genet Mol Res* 8:8–18
- Anderson TD, Loevner LA, Bigelow DC, Mirza N (2000) Prevalence of unsuspected acoustic neuroma found by magnetic resonance imaging. *Otolaryngol Head Neck Surg* 122:643–646
- Antinheimo J, Sallinen SL, Sallinen P, Haapasalo H, Helin H, Horelli-Kuitunen N, Wessman M, Sainio M, Jaaskelainen J, Carpen O (2000) Genetic aberrations in sporadic and neurofibromatosis 2 (NF2)-associated schwannomas studied by comparative genomic hybridization (CGH). *Acta Neurochir (Wien)* 142:1099–1104
- Ballestar E, Esteller M (2008) Epigenetic gene regulation in cancer. *Adv Genet* 61:247–267
- Baser ME (2006) The distribution of constitutional and somatic mutations in the neurofibromatosis 2 gene. *Hum Mutat* 27:297–306
- Baser ME, Makariou EV, Parry DM (2002) Predictors of vestibular schwannoma growth in patients with neurofibromatosis Type 2. *J Neurosurg* 96:217–222
- Bello MJ, Martínez-Glez V, Franco-Hernández C, Pefla-Granero C, de Campos JM, Isla A, Lassaletta L, Vaquero J, Rey JA (2007) DNA methylation pattern in 16 tumor-related genes in schwannomas. *Cancer Genet Cytogenet* 172:84–86
- Bruder CE, Ichimura K, Tingby O, Hirakawa K, Komatsuzaki A, Tamura A, Yuasa Y, Collins VP, Dumanski JP (1999) A group of schwannomas with interstitial deletions on 22q located outside the NF2 locus shows no detectable mutations in the NF2 gene. *Hum Genet* 104:418–424
- Caye-Thomasen P, Werther K, Nalla A, Bog-Hansen TC, Nielsen HJ, Stangerup SE, Thomsen J (2005) VEGF and VEGF receptor-1 concentration in vestibular schwannoma homogenates correlates to tumor growth rate. *Otol Neurotol* 26:98–101
- Doherty JK, Ongkeko W, Crawley B, Andalibi A, Ryan AF (2008) ErbB and Nrg: potential molecular targets

- for vestibular schwannoma pharmacotherapy. *Otol Neurotol* 29:50–57
- Evans DG (2009) Neurofibromatosis type 2 (NF2): a clinical and molecular review. *Orphanet J Rare Dis* 4:16
- Flaiz C, Chernoff J, Ammoun S, Peterson JR, Hanemann CO (2009) PAK kinase regulates Rac GTPase and is a potential target in human schwannomas. *Exp Neurol* 218:137–144
- Gonzalez-Gomez P, Bello MJ, Alonso ME, Lomas J, Arjona D, Campos JM, Vaquero J, Isla A, Lassaletta L, Gutierrez M, Sarasa JL, Rey JA (2003) CpG island methylation in sporadic and neurofibromatosis type 2-associated schwannomas. *Clin Cancer Res* 9:5601–5606
- Hanemann CO, Evans DG (2006) News on the genetics, epidemiology, medical care and translational research of Schwannomas. *J Neurol* 253:1533–1541
- Hansen MR, Roehm PC, Chatterjee P, Green SH (2006) Constitutive neuregulin-1/ErbB signaling contributes to human vestibular schwannoma proliferation. *Glia* 53:593–600
- Kino T, Takeshima H, Nakao M, Nishi T, Yamamoto K, Kimura T, Saito Y, Kochi M, Kuratsu J, Saya H, Ushio Y (2001) Identification of the cis-acting region in the NF2 gene promoter as a potential target for mutation and methylation-dependent silencing in schwannoma. *Genes Cells* 6:441–454
- Kluwe L, Mautner V, Heinrich B, Dezube R, Jacoby LB, Friedrich RE, MacCollin M (2003) Molecular study of frequency of mosaicism in neurofibromatosis 2 patients with bilateral vestibular schwannomas. *J Med Genet* 40:109–114
- Kullar PJ, Pearson DM, Malley DS, Collins VP, Ichimura K (2010) CpG island hypermethylation of the NF2 gene is rare in sporadic vestibular schwannomas. *Neuropathol Appl Neurobiol* 36:505–514
- Lassaletta L, Bello MJ, Del Rio L, Alfonso C, Roda JM, Rey JA, Gavilan J (2006) DNA methylation of multiple genes in vestibular schwannoma: relationship with clinical and radiological findings. *Otol Neurotol* 27:1180–1185
- MacCollin M, Chiocca EA, Evans DG, Friedman JM, Horvitz R, Jaramillo D, Lev M, Mautner VF, Niimura M, Plotkin SR, Sang CN, Stemmer-Rachamimov A, Roach ES (2005) Diagnostic criteria for schwannomatosis. *Neurology* 64:1838–1845
- McClatchey AI, Fehon RG (2009) Merlin and the ERM proteins—regulators of receptor distribution and signaling at the cell cortex. *Trends Cell Biol* 19:198–206
- Moller MN, Werther K, Nalla A, Stangerup SE, Thomsen J, Bog-Hansen TC, Nielsen HJ, Caye-Thomasen P (2010) Angiogenesis in vestibular schwannomas: expression of extracellular matrix factors MMP-2, MMP-9, and TIMP-1. *Laryngoscope* 120:657–662
- Neff BA, Welling DB, Akhmeteyeva E, Chang LS (2006) The molecular biology of vestibular schwannomas: dissecting the pathogenic process at the molecular level. *Otol Neurotol* 27:197–208
- Okada M, Wang Y, Jang SW, Tang X, Neri LM, Ye K (2009) Akt phosphorylation of merlin enhances its binding to phosphatidylinositols and inhibits the tumor-suppressive activities of merlin. *Cancer Res* 69:4043–4051
- Patronas NJ, Courcousakis N, Bromley CM, Katzman GL, MacCollin M, Parry DM (2001) Intramedullary and spinal canal tumors in patients with neurofibromatosis 2: MR imaging findings and correlation with genotype. *Radiology* 218:434–442
- Rong R, Surace EI, Haipek CA, Gutmann DH, Ye K (2004) Serine 518 phosphorylation modulates merlin intramolecular association and binding to critical effectors important for NF2 growth suppression. *Oncogene* 23:8447–8454
- Scoles DR (2008) The merlin interacting proteins reveal multiple targets for NF2 therapy. *Biochim Biophys Acta* 1785:32–54
- Sharma S, Kelly TK, Jones PA (2010) Epigenetics in cancer. *Carcinogenesis* 31:27–36
- Sobel RA (1993) Vestibular (acoustic) schwannomas: histologic features in neurofibromatosis 2 and in unilateral cases. *J Neuropathol Exp Neurol* 52:106–113
- Warren C, James LA, Ramsden RT, Wallace A, Baser ME, Varley JM, Evans DG (2003) Identification of recurrent regions of chromosome loss and gain in vestibular schwannomas using comparative genomic hybridisation. *J Med Genet* 40:802–806

Part VI

Diagnosis

Facial Nerve Schwannoma: Diagnosis Using Magnetic Resonance Imaging

25

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Abstract

Facial nerve schwannomas are uncommon benign tumors originating along the facial nerve. These tumors often present clinically with facial nerve dysfunction. Those presenting on imaging as an enhancing cerebellopontine angle mass can be difficult to distinguish from vestibular schwannomas and meningiomas. Contrast enhancement of the geniculate ganglion on magnetic resonance imaging can help distinguish facial schwannomas from other tumor types. Proper diagnosis of cerebellopontine angle lesions is paramount as pathologic type dictates surgical approach and goals.

Introduction

Facial nerve schwannomas are an uncommon, benign tumor originating from the facial nerve (cranial nerve VII). They account for approximately 2% of all intracranial schwannomas (Nishioka et al. 2009) and occur with an increased frequency in patients with neurofibromatosis type 2 (Spilberg et al. 2009). These tumors can develop anywhere along the course of the facial nerve and generally have a slow growth rate. Given their relatively rare incidence, a delay in proper diagnosis of facial nerve schwannomas can occur (Prasai et al. 2008). Imaging characteristics such

as a computed tomographic (CT) scan finding of widening of the facial canal or magnetic resonance (MR) imaging enhancement of the geniculate ganglion can help make the diagnosis of facial schwannoma (Fabiano et al. 2010). Early diagnosis of this rare lesion is important to ensure the appropriate treatment plan is offered to the patient.

Facial Nerve Anatomy

The facial nerve contains motor and sensory divisions that have separate interfaces with the brainstem. The motor nucleus is located in the ventrolateral pontine tegmentum (Jager and Reiser 2001). Fibers leave this nucleus, loop in the internal genu of the facial nerve (around the nucleus of the abducens nerve), and exit the brainstem at the pontomedullary junction (Gilchrist 2009).

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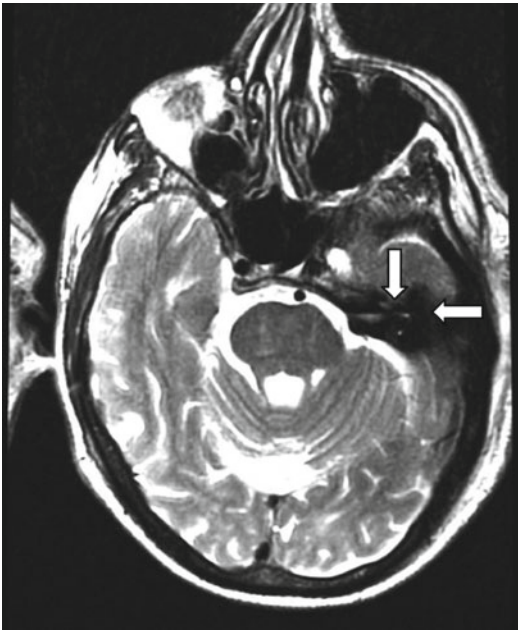


Fig. 25.1 Noncontrast axial T2-weighted MR image in a control subject. The normal course of the left facial nerve within the temporal bone can be visualized (arrows)

Exiting preganglionic parasympathetic fibers from the superior salivatory nucleus and entering sensory fibers of the solitary tract join together to form the nervus intermedius. The facial nerve roots coalesce in either the cerebellopontine cistern or the internal acoustic canal (Jager and Reiser 2001; Gilchrist 2009).

Within the temporal bone, the facial nerve travels anterolaterally from the internal acoustic canal into the facial canal, also known as the fallopian canal. The labyrinthine segment of the facial nerve then curves anterior to the arcuate eminence in the temporal bone (Fig. 25.1). This area of the temporal bone is the location of the geniculate ganglion, which contains cell bodies for specialized taste and general somatic sensory fibers. This is also the site of exit for the greater superficial petrosal nerve from the facial nerve. The facial nerve forms an angle of approximately 75° at the geniculate fossa (Jager and Reiser 2001). The tympanic segment of the facial nerve runs posteriorly, travelling inferior to the lateral semicircular canal. The mastoid segment of the

facial nerve begins at the posterior genu; this segment runs medial to the jugular bulb before the facial nerve exits the skull base at the stylomastoid foramen. The stapedius nerve arises from the mastoid segment of the facial nerve, which the chorda tympani also arises from, and the chorda tympani nerve enters the mastoid segment of the facial nerve (Jager and Reiser 2001).

The extracranial segment of the facial nerve travels through the posterior belly of the digastric muscle and enters the parotid gland. Here it divides into an upper temporofacial nerve and a lower cervicofacial nerve at the posterior border of the ramus of the mandible (Raghavan et al. 2009). These nerves then further divide in the parotid plexus to innervate the muscles of facial expression. The major terminal branches of the facial nerve are the temporal (i.e., frontal), zygomatic, buccal, marginal mandibular, and cervical nerves.

Pathophysiology

Facial schwannomas are nerve sheath tumors originating from Schwann cells of the seventh cranial nerve (Jacob et al. 2007). Schwann cells, derived embryologically from neural crest cells, are glial cells that form the myelin sheath around the facial nerve. Facial schwannomas are benign neoplasms that generally displace and compress the facial nerve and surrounding structures, rather than incorporating into them. Histopathologically, these are encapsulated, well-circumscribed lesions containing the characteristic presence of alternating cellular architecture: Antoni A and Antoni B areas. Type A tissue is highly cellular and demonstrates nuclear palisading and associated Verocay bodies (Wippold et al. 2007). Type B tissue is loosely organized with myxomatous and cystic changes and may represent degenerated Antoni A tissue (Wippold et al. 2007). Schwannomas are slow-growing tumors, and neighboring structures often accommodate this change over a period of months or years. Thus, facial schwannomas may reach considerable size prior to detection.

Presentation

The presentation of facial schwannomas is variable and depends upon the segment of the facial nerve from which the tumor arises. The most common symptom is facial weakness; however, symptoms can range from facial palsy to compressive hearing loss resulting from ossicular interference and sensorineural hearing loss due to effects on the cochlear nerve in the internal auditory canal (Marzo et al. 2009). Other presenting complaints include hemifacial spasm, taste alteration, vestibular dysfunction, and facial numbness. With the increased use of MR imaging, facial schwannomas have been detected as incidental findings. Similar patient presentations may be seen in other cases of cerebellopontine angle lesions or facial nerve disorders. Thus, it is important to keep in mind a broad differential diagnosis when evaluating patients with such complaints.

Differential Diagnoses

As facial schwannomas can occur at any point along the course of the facial nerve, they have a broad possibility of differential diagnoses that include lesions at the cerebellopontine angle, inside the temporal bone, and extracranially from the stylomastoid foramen to the parotid gland. An exhaustive list is beyond the scope of this chapter. The differential diagnosis for lesions at the cerebellopontine angle includes vestibular schwannoma, meningioma, dural-based metastasis, epidermoid cyst, lipoma, and posterior circulation aneurysm (Bonneville et al. 2001). Temporal bone lesions that may present similarly to a facial schwannoma include cholesterol granuloma, paraganglioma, endolymphatic sac tumor, sarcoidosis, chondromatous tumors, apex petrositis, idiopathic facial paralysis (“Bell’s palsy”), herpetic facial palsy, otomastoiditis, cholesteatoma, and hemangioma of the geniculate ganglion (Isaacson et al. 2005; Okada et al. 2007). Extracranial differential diagnoses for facial schwannomas include necrotizing external otitis, parotid tumors, osteoma of the mandible, dental abscess, and squamous cell carcinoma (Lemmerling et al. 2009).

CT Imaging

High-resolution CT scan is the imaging modality of choice to evaluate the temporal bone, including the internal auditory canal and facial canal. The internal auditory canal, as well as the labyrinthine, the tympanic, and mastoid segments of the facial nerve, can be identified on axial CT images (Jager and Reiser 2001). We typically obtain thin-cut (1 mm) axial images from which coronal and sagittal reconstructions are generated. Using a multislice spiral CT, acquisition is performed with a tube current of 320 milliamp-seconds (mAs) at 120 kilovolt (kV) using 1 s rotation time and a 210 mm field-of-view. The use of this sequence offers excellent visualization of the temporal bone anatomy. A normal facial nerve can be tracked as it courses through the facial canal in the temporal bone on sequential images.

Again, CT scan characteristics of a facial schwannoma will be determined by the portion of the facial nerve from which it arises. A classic facial schwannoma CT scan finding is widening of the facial canal. If the schwannoma originates from the mastoid or tympanic segment, a destruction of the ossicular chain, as well as invasion of the tympanic tegmen, and a labyrinthine invasion through the oval window is likely (Jager and Reiser 2001). Facial schwannomas uniformly enhance on a CT scan with contrast administration.

MR Imaging

MR imaging has become the standard diagnostic modality for evaluating facial nerve lesions. Of particular use when evaluating facial nerve anatomy is a high-resolution T2-weighted gradient echo sequence. This T2-weighted image should allow visualization of the course of the facial nerve from the pontine nuclei to the parotid gland. Views of the motor root of the facial nerve and nervus intermedius may be limited due to the small size of these nerves and the complexity and spatial arrangements of the temporal bone components, including the nearby mastoid air cells

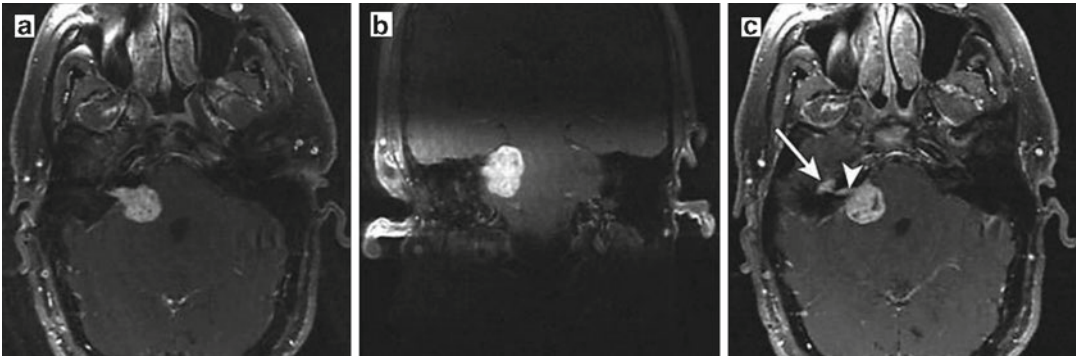


Fig. 25.2 Contrast-enhanced T1-weighted MR images. Axial (a) and coronal (b) views show an enhancing mass at the cerebellopontine angle. (c) Enhancement of the

facial nerve along the floor of the middle cranial fossa (arrowhead) and an enlarged, geniculate ganglion (arrow) are shown (With permission from Fabiano et al. 2010)

(Burmeister et al. 2009). In addition to the nerve itself, neighboring vasculature, such as the anterior inferior cerebellar artery, can be seen. The cochlea, semicircular canals, and vestibulocochlear nerve can also be visualized well on a T2-weighted image.

The hallmark of a facial schwannoma on a T2-weighted MR image is enlargement of the nerve at some point along its course. This enlargement can range from subtle to massive. Larger facial schwannomas can displace neighboring structures including the brainstem, cerebellum, and trigeminal nerve. If the tumor is large enough to obstruct cerebrospinal fluid flow, hydrocephalus can develop and be evident on the MR image. Rarely, hemorrhage within a facial schwannoma can be seen on T2 imaging with a varying intensity, depending on the age of the blood products.

The best MR imaging sequence with which to visualize a facial schwannoma is a T1-weighted image post-gadolinium diethylenetriamine penta-acetic acid (DTPA) enhancement. Facial schwannomas homogeneously enhance with contrast administration and can be seen as contrast-enhancing mass lesions along the course of the facial nerve on T1-weighted MR imaging. When the geniculate portion of the facial nerve is involved, enhancement of the geniculate ganglion can be seen (Fabiano et al. 2010) (Fig. 25.2). This can help distinguish facial nerve schwannomas from other contrast-enhancing lesions in this

location such as vestibular schwannomas and meningiomas. Following treatment with radiosurgery, a residual facial schwannoma that enhances post-contrast administration can be seen to have areas of central hypointensity. This generally occurs due to radiation effects following treatment with stereotactic radiosurgery and usually portends a good prognosis for tumor control (Kondziolka et al. 1998) (Fig. 25.3).

Other MR imaging sequences beyond T1- and T2-weighted pre-contrast and T1-weighted post-contrast images are generally used as adjunctive images and to help distinguish facial schwannomas from other lesions. Rarely, hyperintensity can be seen on T2-weighted gradient recall imaging, representing hemorrhage within the tumor. Facial schwannomas appear hypointense on diffusion-weighted MR imaging.

Treatment

The goals of treatment of facial nerve schwannomas are tumor control and preservation of facial nerve function. Options for management include observation, decompression, stripping, resection with grafting, and stereotactic radiosurgery (Marzo et al. 2009). Given the benign nature of this neoplasm, small facial schwannomas not causing a deficit can be monitored with serial (MR) imaging. Decompression of facial schwannomas, particularly at the facial canal, can

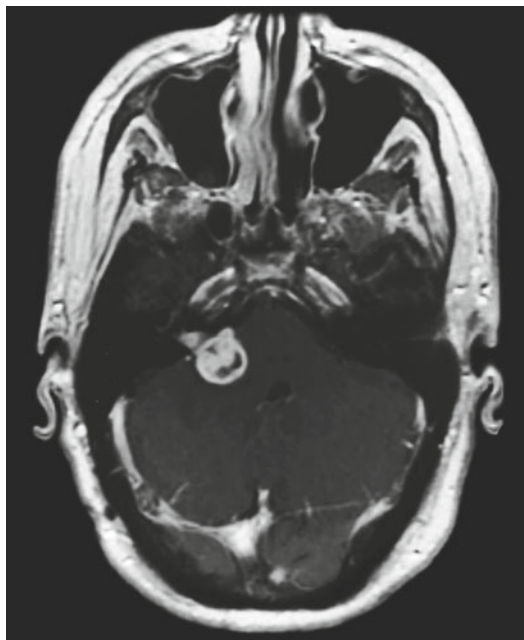


Fig. 25.3 Contrast-enhanced axial T1-weighted image of a facial schwannoma (confirmed histologically) 1 year after stereotactic radiosurgery. Areas of central hypointensity representing radiation change are seen centrally within the tumor. Such hypointense changes following stereotactic radiosurgery generally portend a good prognosis for tumor control (With permission from Fabiano et al. 2010)

be a surgical option that avoids nerve resection and grafting. Similarly, tumor stripping, without nerve transaction and grafting, can be performed. Craniotomy for tumor resection combined with grafting to reconstruct the facial nerve remains a viable option for facial nerve schwannoma management, particularly for large tumors with brainstem compression. However, many patients undergoing this procedure will have long-term facial nerve dysfunction (Sarma et al. 2002).

Stereotactic radiosurgery now provides a useful treatment option for facial schwannoma. Stereotactic radiosurgery involves treating a patient with a single fraction of high-dose radiation to a target lesion with minimal radiation to the surrounding tissues. The stereotactic coordinates are generated using a metallic frame secured to the patient's skull. Radiosurgery is in general limited to lesions less than 3 cm in size. Recent

studies have demonstrated tumor control, minimal patient morbidity, and preservation of facial nerve function with stereotactic radiosurgery (Nishioka et al. 2009; Madhok et al. 2009).

In conclusion, facial nerve schwannomas are a rare, benign neoplasm that can occur anywhere along the course of the facial nerve. Typical presentation includes an alteration in facial nerve function. MR imaging demonstrates an enhancing lesion along the course of the facial nerve and may include enhancement of the geniculate ganglion. For lesions amenable to stereotactic radiosurgery, this treatment modality provides excellent tumor control rates with limited morbidity.

Acknowledgment We thank Paul H. Dressel BFA for preparation of the images.

References

- Bonneville F, Sarrazin JL, Marsot-Dupuch K, Iffenecker C, Cordoliani YS, Doyon D, Bonneville JF (2001) Unusual lesions of the cerebellopontine angle: a segmental approach. *Radiographics* 21:419–438
- Burmeister HP, Hause F, Baltzer PAT, Schmidt P, Volk GF, Guntinas-Lichius O, Sedlacik J, Mentzel HJ, Kaiser WA (2009) Improvement of visualization of the intermediofacial nerve in the temporal bone using 3T magnetic resonance imaging: part 1: the facial nerve. *J Comput Assist Tomogr* 33:782–788
- Fabiano AJ, Plunkett RJ, Gibbons KJ (2010) Diagnosis of facial nerve schwannoma by magnetic resonance imaging enhancement of the geniculate ganglion. *Arch Neurol* 67:112–113
- Gilchrist JM (2009) Seventh cranial neuropathy. *Semin Neurol* 29:5–13
- Isaacson B, Telian SA, McKeever PE, Arts HA (2005) Hemangiomas of the geniculate ganglion. *Otol Neurotol* 26:796–802
- Jacob A, Robinson LL Jr, Bortman JS, Yu L, Dodson EE, Welling DB (2007) Nerve of origin, tumor size, hearing preservation, and facial nerve outcomes in 359 vestibular schwannoma resections at a tertiary care academic center. *Laryngoscope* 117:2087–2092
- Jager L, Reiser M (2001) CT and MR imaging of the normal and pathologic conditions of the facial nerve. *Eur J Radiol* 40:133–146
- Kondziolka D, Lunsford LD, McLaughlin MR, Flickinger JC (1998) Long-term outcomes after radiosurgery for acoustic neuromas. *N Engl J Med* 339:1426–1433
- Lemma MM, De Foer B, Verbist BM, VandeVyver V (2009) Imaging of inflammatory and infectious diseases in the temporal bone. *Neuroimaging Clin N Am* 19:321–337

- Madhok R, Kondziolka D, Flickinger JC, Lunsford LD (2009) Gamma knife radiosurgery for facial schwannomas. *Neurosurgery* 64:1102–1105
- Marzo SJ, Zender CA, Leonetti JP (2009) Facial nerve schwannoma. *Curr Opin Otolaryngol Head Neck Surg* 17:346–350
- Nishioka K, Abo D, Aoyama H, Furuta Y, Onimaru R, Onodera S, Sawamura Y, Ishikawa M, Fukuda S, Shirato H (2009) Stereotactic radiotherapy for intracranial nonacoustic schwannomas including facial nerve schwannoma. *Int J Radiat Oncol Biol Phys* 75:1415–1419
- Okada K, Ito K, Yamasoba T, Ishii M, Iwasaki S, Kaga K (2007) Benign mass lesions deep inside the temporal bone: imaging diagnosis for proper management. *Acta Otolaryngol Suppl* 127:71–77
- Prasai A, Jones SE, Cross J, Moffat DA (2008) A facial nerve schwannoma masquerading as a vestibular schwannoma. *Ear Nose Throat J* 87:E4–E6
- Raghavan P, Mukherjee S, Philips CD (2009) Imaging of the facial nerve. *Neuroimaging Clin N Am* 19:407–425
- Sarma S, Sekhar LN, Schessel DA (2002) Nonvestibular schwannomas of the brain: a 7-year experience. *Neurosurgery* 50:437–449
- Spilberg G, Marchiori E, Gasparetto EL, Cabral RF, Takayasu TC, Batista RR, Vieira IG (2009) Magnetic resonance findings of neurofibromatosis type 2: a case report. *Cases J* 2:6720
- Wippold FJ 2nd, Lubner M, Perrin RJ, Lammle M, Perry A (2007) Neuropathology for the neuroradiologist: Antoni A and Antoni B tissue patterns. *AJNR Am J Neuroradiol* 28:1633–1638

Vestibular Schwannoma: Optimizing Tumor Growth Monitoring by Volume Measurements

26

Rick van de Langenberg and Robert J. Stokroos

Abstract

Vestibular schwannomas are diagnosed more frequently nowadays, merely due to the increased practice of magnetic resonance imaging (MRI). A conservative “wait and scan policy” is more often proposed, in which symptoms are monitored and growth is followed with successive MRI. The rationale of this treatment is that symptoms may be minimal and growth is usually indolent. Intervention may therefore be avoided. Reliable judgements according to possible growth on these scan policies are essential because invasive treatment decisions are based on these observations. Currently, two dimensional measurements are used in clinical practice, which are less reliable compared to volume measurements. Therefore, volume measurements should be used in the follow up of vestibular schwannomas. Software costs and the time factor to calculate the VS volume limits the usage of volume measurements nowadays, although developments and software improvements will ease a widespread utilisation in the near future.

Introduction

Vestibular schwannomas (VS) are benign neoplasms originating from the schwann cells of the vestibular part of the eighth cranial nerve. They are located within the internal auditory canal and may extend into the cerebellopontine angle, representing 90% of the possible tumors in this area

(Kohan et al. 1997). The most common symptoms are sensorineural hearing loss, vertigo or chronic imbalance or tinnitus, and the spectrum of symptomatology may increase when a larger VS compromises more cranial nerves or other structures (Rosenberg 2000). Due to ageing of the population and the increased usage and improved techniques of MRI, incidence of VS is increasing (Stangerup et al. 2006). Several treatment strategies exist: microsurgery, stereotactic radiotherapy or a so called ‘wait and scan’ policy (Kondziolka et al. 1998; Rosenberg 2000). In this W&S policy, audiovestibular symptoms are controlled regularly and VS growth, which is

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known to be indolent, is monitored by sequential MRI (Rosenberg 2000; Stangerup et al. 2006). Since more small and asymptomatic VS are being discovered, this policy is gaining more and more popularity in VS management nowadays (Stangerup et al. 2006). A scan policy is also being adapted after stereotactic radiotherapy, in order to evaluate the therapeutic affect, or after a subtotal surgical VS resection, in which the remnant is still capable to grow (El-Kashlan et al. 2000). Because clinical symptoms do not correlate with VS growth (Walsh et al. 2000; Raut et al. 2004), sequential MRI remains the only objective method of registering tumor growth. Currently, in clinical practice, these observations are based on two dimensional (2D) measurements, although volume measurements would seem more reliable because a VS is not a flat object and grows asymmetrically in all directions, making slight volume changes not being detected with 2D measurements. In addition, a certain increase in diameter corresponds to a larger increase in volume in larger VS, because diameters are related to volume by a cubic function. Little is known regarding the increase in diameter or volume between subsequent scans which can be regarded as growth beyond measurement error, which is surprising, because important invasive treatment decisions are based on these observations.

Imaging and Measurement Methods

In the early years of VS imaging, follow up (FU) observations were performed with CT scans (Stangerup et al. 2008), which are not ideal in registering changes in soft tissues. After the introduction of magnetic resonance imaging (MRI), this imaging method became the golden standard in VS FU (Kanzaki et al. 2003; Stangerup et al. 2008). Both contrast enhanced T1- weighted images (CE T1-WI) and corresponding T2 weighted images can be used to visualise the tumor. VS Measurements can best be performed on axial CE T1-WI, making it possible to differentiate VS from surrounding tissues more easily (van de Langenberg et al. 2009). Regarding 2D measurements, several methods exist to measure

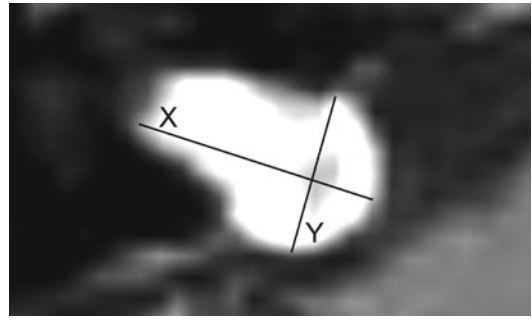


Fig. 26.1 Axial (Fig. 26.1) coupe of a *right sided* VS on contrast enhanced, T1-weighted images. Examples are given of measuring VS two dimensionally. X mediolateral diameter, Y anteroposterior diameter

VS size and growth. Most clinics measure the maximum mediolateral (ML) and/or the anteroposterior (AP) dimension, sometimes combined with the craniocaudal dimension (CC) (Fig. 26.1) (Selesnick and Johnson 1998; Fucci et al. 1999; Slattery et al. 2003; Tanaka et al. 2003; Smouha et al. 2005; Doherty and Friedman 2006). The ML dimension can be subdivided in measurements in- and excluding the portion in the internal auditory canal (IAC). Others use the guidelines for measuring VS as described by the American Academy of Otolaryngology, Head and Neck Surgery (AAO-HNS) (Walsh et al. 2000; Raut et al. 2004). This guideline advises to measure the maximum diameter of the VS in the direction parallel to the petrous ridge, the second measurement should be the maximum dimension of the VS in the orientation perpendicular to the first diameter, in which only the extracalcular portion of the VS should be measured (AAO-HNS 1995). Volume measurements are mostly performed by using (semi automatic) computer software. With the advances in computer technology, faster and more complex measurements can be performed more easily and more important, they become less time consuming. Several ways of performing volume measurements are known. Some authors estimate tumor volume by using pre-defined formulas, in which several diameters of the tumor are integrated (Charabi et al. 1995; Caye-Thomasen et al. 2006). This way of estimating tumor size is known to produce large overestimations (Shin et al. 2000), and should therefore not be used to calculate VS volume. Other authors perform measurements by

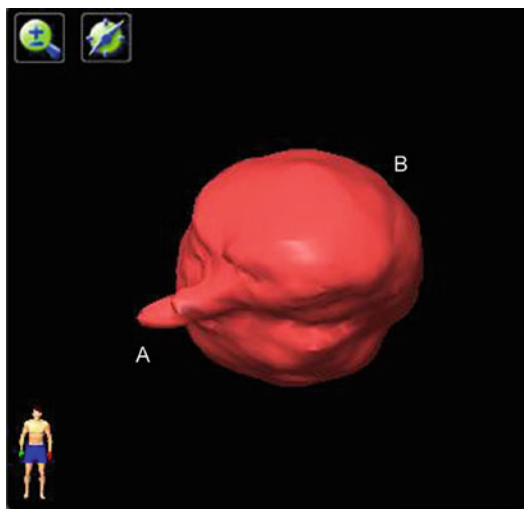


Fig. 26.2 Volumetrical representation of a vestibular schwannoma, acquired by fusing all calculated MRI slice surfaces. Clearly, the part located in the internal auditory canal (a) and in the cerebellopontine angle (b) can be seen (Image generated with iPlan® RT image version 3—Advanced Contouring Workstation (BrainLAB Oncology Solutions, Feldkirchen, Germany))

using semi-automatic computer software, able of calculating the surface of the VS on each MRI image (Fig. 26.2). These surfaces are being multiplied by the interslice distance to arrive at a tumor volume (Luppino et al. 2006; van de Langenberg et al. 2009). Some authors studied on the usage of Bayesian tissue classification, in which errors due to partial volume defects were minimised with specific software, capable of reducing the rate of error by assessing the most likely distribution of tissue proportions within one voxel (Vokurka et al. 2002). In volume measurements, MRI slice thicknesses ≤ 1 mm, are essential (Luppino et al. 2006), preferably assessed on at least a 1.5 T system in order to retrieve VS volume more precisely (van de Langenberg et al. 2009).

Measurement Error

When applying a scan policy after VS intervention or when following patients a W&S policy, it is important to generate reliable judgments according to possible growth on sequential MRI, because important decisions are based on these

observations. Both 2D and semi-automatocal volume measurements are being used in the radiological FU for VS, in which 2D measurements are still regarded as the golden standard in VS management (Luppino et al. 2006). The best measurement method would be the method that measures “real” VS size precisely. Because it is impossible to surgically remove a VS in one piece, 2D and volume measurements cannot be compared with real VS diameters and volumes. Therefore calculating reproducibility for each measurement method is the best way to gain an estimation of the agreement and reliability of each measurement method (Bland and Altman 1986). By using this information, it should be possible to generate the measurement error for each measuring modality. Three recent studies compared these measurement methods in order to find the most reliable measurement method for VS FU (Vokurka et al. 2002; Luppino et al. 2006; van de Langenberg et al. 2009). All three studies found volume measurements to be superior compared to conventional, clinically used 2D measurements in terms of the measurement error. Normally, when evaluating VS growth with 2D measurements, authors use a difference in diameter of 1 or 2 mm between two subsequent scans to speak of “significant growth” (Fucci et al. 1999; Walsh et al. 2000; Raut et al. 2004; Smouha et al. 2005; Stangerup et al. 2006), although this measure is not statistically justified and lies within measurement error (van de Langenberg et al. 2009). Therefore, one should be cautious regarding planning invasive treatment decisions based on these arbitrary growth criteria. If one should still use 2D measurements, it should best be performed on the mediolateral diameter, since this is the most reliable dimension to measure growth (van de Langenberg et al. 2009). It would be incorrect to use a 1 or 2 mm difference as criterion for growth in stage A-C VS (van de Langenberg et al. 2009). Differences in diameters between two successive MRI should exceed 2 mm before one could speak of growth beyond measurement error, and only in large, stage D VS, it is allowed to speak of growth beyond measurement error whenever a 1 mm difference is seen between two successive MRI (Table 26.1).

Table 26.1 Absolute and relative measurement errors, calculated from interobserver agreement, derived from the study of van de Langenberg et al. (2009)

VS stage	CC	CC (%)	AP	AP (%)	ML	ML (%)	Vol	Vol (%)
T1A–D (n=68)	2.98	40.3	2.23	28.3	2.12	20.9	0.11	19.7
T1 A (n=29)	2.41	52.8	1.49	28.0	1.96	27.1	0.05	28.9
T1 B (n=27)	3.23	34.5	2.53	26.5	2.33	17.8	0.06	10.4
T1 C (n=6)	3.60	32.7	3.05	28.7	2.49	14.9	0.07	8.70
T1 D (n=6)	2.96	18.8	2.90	23.7	1.08	5.00	0.26	5.70

Two independent observers measured a set of 68 vestibular schwannomas on contrast enhanced, T1 weighted magnetic resonance images. Measurement error was calculated for both two-dimensional measurements and volume measurement. VS are subdivided according to VS stage: *A* intracanalicular, *B* extending in the CPA, *C* compressing the brain stem, *D* deviating the fourth ventricle, *CC* craniocaudal axis, *AP* anteroposterior axis, *ML* mediolateral axis, *Vol* volume. Two-dimensional measurement errors in millimeters and percentages, volume measurement error in cm³ and percentages

The fact that in larger VS, measurement error is less, is not taken into account when a measurement error of for example 2 mm is pursued for all VS. A recent study of van de Langenberg et al. (2009) illustrates this. These authors measured the interobserver agreement by measuring 68 vestibular schwannomas with two independent observers and calculated the measurement error for both 2D and volume measurements. This error was subdivided into four stages, based on their morphologic appearance on MRI, ranging from small to large: *A*, purely intracanalicular, *B*, extending into the CPA, *C*, compressing the brain stem and *D*, deviating the fourth ventricle, based on a previous publication of (Hasegawa et al. 2005). Measurement error diminished when VS size increased, caused by the fact that errors due to partial volume effects in small VS will proportionally be greater. This holds true for 2D as well as volume measurements.

Measurement Error and VS Stage

When taking the different VS stages into account, measurement error regarding 2D and volume measurements were comparable in small, type *A* VS, making both imaging modalities suitable for analysis of type *A* VS. Some clinicians even suggest not to measure the small stage *A* VS anymore, because of the high measurement error (personal communication). Larger VS (stage *B*, *C* and *D*) are best measured with volume measurements on

CE-T1 WI, in order to gain the highest reliability (Table 26.1), making treatment decisions being based on the most reliable observation. When using volume measurements on CE-T1 WI, increase in volume should range from 30% to 5%, depending on VS stage, before one can speak of ‘significant’ growth (Table 26.1). Measurement error calculated by Luppino et al. (2006) was found to be 132% for diameter measurements compared to 64% for semi automatic volume measurements. The difference with the study mentioned before might be caused by the fact that a 3 mm interslice differences was used compared to the study of van de Langenberg et al., making observations less reliable. Vokurka et al. (2002) also performed volume analysis on VS. They used specific software capable of minimising the errors caused by partial volume effects. This faster method of calculating volumes was found to have a correlation of 0,99 with the “standard” manual segmentation of tumor volume calculations, indicating that this way of measuring might be a method for the near future in order to calculate volumes faster.

Considerations

Measurement errors of all techniques used and mentioned in published studies will always be influenced by the quality of the images, the software used and the experience of the radiologist performing the measurements. Published

measurement errors can therefore never simply be adopted in a certain protocol for evaluating VS, without taking the techniques of acquiring and measuring images in these studies into account. Older literature regarding 2D and volume measurements favored as 2D measurements instead of volume measurements, possibly because of unfair comparisons were made by using formula based volumes (Charabi et al. 1995), usage of CT scans in the FU (Fiirgaard et al. 1997), or utilization of slower and more time consuming software measurements. Despite the clear evidence favoring usage of volumetrics, volume measurements in the FU of VS have so far not been widely introduced in clinical practice, in which the time factor regarding volume measurements and the price of the software will have an important role. Considering the improvements seen in MRI and in specialized software, capable of performing volume measurements semi-automatically, it might be expected that measurement errors will decrease in VS measurements performed volumetrically because the human component (causing a substantial measurement error) will diminish. Anticipating on this, it should be expected that volume measurements will replace the “standard” 2D measurements in the FU within the near future.

In conclusion, more VS patients are being followed conservatively or are being followed after intervention by sequential MRI. Volume measurements are more reliable compared to 2D measurements in evaluating growth and this measure should replace 2D measurements in evaluating growth. When evaluating VS growth, one has to take VS stage into account, because measurement error strongly depends on it. The difference of 1 or 2 mm. between subsequent scans to define growth in 2D measurements lies within measurement error and should not direct clinical practice.

References

American Academy of Otolaryngology-Head and Neck Surgery Foundation, INC (1995) Committee on Hearing and Equilibrium guidelines for the evaluation

- of hearing preservation in acoustic neuroma (vestibular schwannoma). *Otolaryngol Head Neck Surg* 113:179–180
- Bland JM, Altman DG (1986) Statistical methods for assessing agreement between two methods of clinical measurement. *Lancet* 1:307–310
- Caye-Thomasen P, Hansen S, Dethloff T, Stangerup SE, Thomsen J (2006) Sublocalization and volumetric growth pattern of intracanalicular vestibular schwannomas. *Laryngoscope* 116:1131–1135
- Charabi S, Thomsen J, Mantoni M, Charabi B, Jorgensen B, Borgesen SE, Gyldensted C, Tos M (1995) Acoustic neuroma (vestibular schwannoma): growth and surgical and nonsurgical consequences of the wait-and-see policy. *Otolaryngol Head Neck Surg* 113:5–14
- Doherty JK, Friedman RA (2006) Controversies in building a management algorithm for vestibular schwannomas. *Curr Opin Otolaryngol Head Neck Surg* 14:305–313
- El-Kashlan HK, Zeitoun H, Arts HA, Hoff JT, Telian SA (2000) Recurrence of acoustic neuroma after incomplete resection. *Am J Otol* 21:389–392
- Fiirgaard B, Pedersen CB, Lundorf E (1997) The size of acoustic neuromas: CT and MRI. *Neuroradiology* 39:599–601
- Fucci MJ, Buchman CA, Brackmann DE, Berliner KI (1999) Acoustic tumor growth: implications for treatment choices. *Am J Otol* 20:495–499
- Hasegawa T, Fujitani S, Katsumata S, Kida Y, Yoshimoto M, Koike J (2005) Stereotactic radiosurgery for vestibular schwannomas: analysis of 317 patients followed more than 5 years. *Neurosurgery* 57:257–265; discussion 257–265
- Kanzaki J, Tos M, Sanna M, Moffat DA, Monsell EM, Berliner KI (2003) New and modified reporting systems from the consensus meeting on systems for reporting results in vestibular schwannoma. *Otol Neurotol* 24:642–648; discussion 648–649
- Kohan D, Downey LL, Lim J, Cohen NL, Elowitz E (1997) Uncommon lesions presenting as tumors of the internal auditory canal and cerebellopontine angle. *Am J Otol* 18:386–392
- Kondziolka D, Lunsford LD, McLaughlin MR, Flickinger JC (1998) Long-term outcomes after radiosurgery for acoustic neuromas. *N Engl J Med* 339:1426–1433
- Luppino FS, Grooters E, de Bruine FT, Zwinderman AH, van der Mey AG (2006) Volumetric measurements in vestibular schwannoma, the influence of slice thickness and patient's repositioning. *Otol Neurotol* 27:962–968
- Raut VV, Walsh RM, Bath AP, Bance ML, Guha A, Tator CH, Rutka JA (2004) Conservative management of vestibular schwannomas – second review of a prospective longitudinal study. *Clin Otolaryngol Allied Sci* 29:505–514
- Rosenberg SI (2000) Natural history of acoustic neuromas. *Laryngoscope* 110:497–508
- Selesnick SH, Johnson G (1998) Radiologic surveillance of acoustic neuromas. *Am J Otol* 19:846–849

- Shin YJ, Fraysse B, Cognard C, Gafsi I, Charlet JP, Berges C, Deguine O, Tremoulet M (2000) Effectiveness of conservative management of acoustic neuromas. *Am J Otol* 21:857–862
- Slattery WH 3rd, Fisher LM, Yoon G, Sorensen G, Lev M (2003) Magnetic resonance imaging scanner reliability for measuring changes in vestibular schwannoma size. *Otol Neurotol* 24:666–670; discussion 670–661
- Smouha EE, Yoo M, Mohr K, Davis RP (2005) Conservative management of acoustic neuroma: a meta-analysis and proposed treatment algorithm. *Laryngoscope* 115:450–454
- Stangerup SE, Caye-Thomasen P, Tos M, Thomsen J (2006) The natural history of vestibular schwannoma. *Otol Neurotol* 27:547–552
- Stangerup SE, Caye-Thomasen P, Tos M, Thomsen J (2008) Change in hearing during ‘wait and scan’ management of patients with vestibular schwannoma. *J Laryngol Otol* 122:673–681
- Tanaka Y, Hongo K, Tada T, Kobayashi S (2003) What is the best method for reporting tumor diameter in vestibular schwannoma? *Neurosurgery* 53:634–637; discussion 637–638
- van de Langenberg R, de Bondt BJ, Nelemans PJ, Baumert BG, Stokroos RJ (2009) Follow-up assessment of vestibular schwannomas: volume quantification versus two-dimensional measurements. *Neuroradiology* 51:517–524
- Vokurka EA, Herwadkar A, Thacker NA, Ramsden RT, Jackson A (2002) Using Bayesian tissue classification to improve the accuracy of vestibular schwannoma volume and growth measurement. *AJNR Am J Neuroradiol* 23:459–467
- Walsh RM, Bath AP, Bance ML, Keller A, Tator CH, Rutka JA (2000) The role of conservative management of vestibular schwannomas. *Clin Otolaryngol Allied Sci* 25:28–39

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Abstract

The intermediate nerve is rarely addressed as site of tumor origin in cerebellopontine surgery. This chapter presents typical clinical, electrophysiological and radiological signs of an intermediate nerve schwannoma. The authors describe the special surgical techniques in these rare entities with an emphasis on the intraoperative electrophysiological monitoring. In conclusion it is possible to detect the intermediate nerve as site of tumor origin during cerebellopontine surgery using selective direct nerve stimulation and multi-channel recordings. Subsequently, more intermediate nerve schwannomas may be identified as tumor origin by detailed intraoperative electrophysiological examination.

Introduction

Though facial nerve schwannomas represent 2–3% of the cerebellopontine angle tumors (King and Morrison 1990; Symore et al. 1993) the intermediate nerve is rarely addressed as site of tumor origin in the literature. So far four case reports of an intermediate nerve schwannoma of the cerebellopontine angle have been published by

Kudo et al. (1996), Scheller et al. (2008) and Sherman et al. (2002). This chapter summarizes clinical and intraoperative features that facilitate the differentiation between a facial and an intermediate nerve schwannoma.

Diagnosis

The differentiation of a facial nerve from a vestibular nerve schwannoma can be difficult before surgery, unless a pronounced facial weakness or distinct radiological signs are present (Fagan et al. 1993; Kudo et al. 1996; Lee et al. 1989). Facial nerve schwannomas typically present with hearing disturbances similar to vestibular schwannomas and facial nerve paresis which is inadequate to the tumor size (Lipkin et al. 1987;

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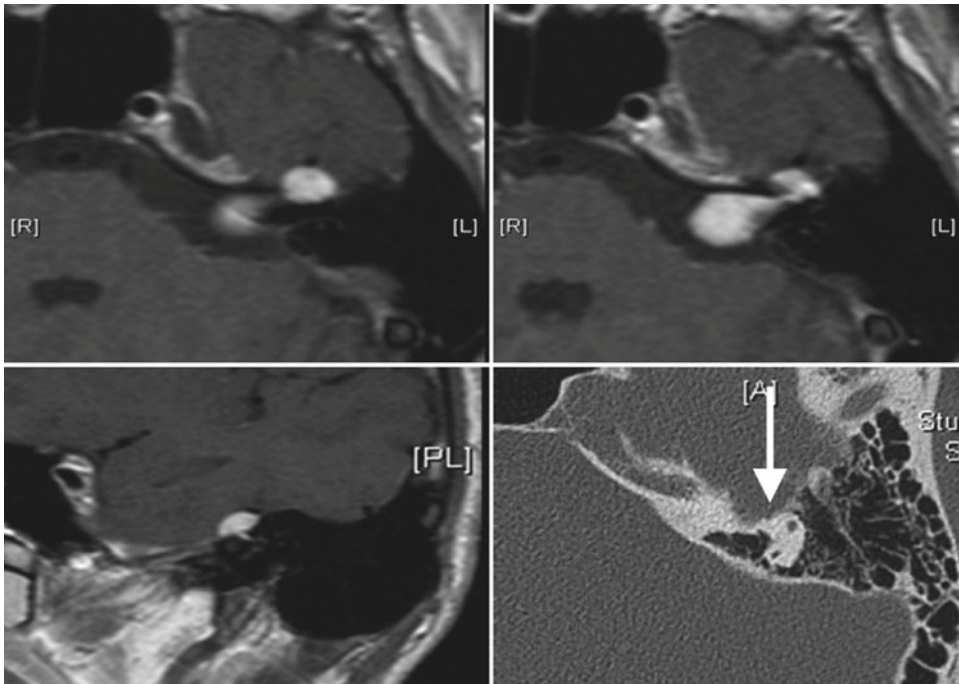


Fig. 27.1 Contrast enhanced T1-weighted MRI scans showed a lesion of the cerebellopontine angle extending into the base of the middle cranial fossa. The *upper right* MRI scan revealed an enlarged facial canal. The tumor grew along the course of the petrosus major nerve (*lower*

left MRI scan). A bone window CT scan showed a pathological enlargement of the facial nerve canal up to the external genu of the facial nerve (With kind permission of the J Neurosurg)

O'Donoghue et al. 1989; Sherman et al. 2002). Recurrent and abrupt facial nerve paresis is a highly suspicious sign of facial schwannoma (Sherman et al. 2002). Furthermore facial spasm and ear pain are characteristic of facial schwannomas (Sherman et al. 2002). An audiogram with pure tones and speech discrimination scores is necessary for hearing classification. Facial nerve function is evaluated according to the House Brackmann classification (House and Brackmann 1986).

On magnetic resonance imaging (MRI), contrast administration and T2-weighted images, facial nerve schwannomas typically extend up to the geniculate ganglion and along the course of the superficial petrosal nerve or the chorda tympani (Ashram et al. 2005; King and Morrison 1990; Scheller et al. 2008). In contrast to the diagnostic management for suspected vestibular schwannomas, a bone window CT of the petrous bone is mandatory. Pathological enlargement of the facial nerve canal on a bone

window CT scan also points to the presentation of a facial nerve schwannoma (King and Morrison 1990) (Fig. 27.1). Distal of the geniculate ganglion the tumor may erode into the middle fossa producing a smooth defect on the top of the petrous bone.

Intermediate nerve schwannomas are almost indistinguishable from facial nerve schwannomas. As reported in the literature the intermediate nerve seems to carry motor fibres selective to the orbicularis oris muscle (Ashram et al. 2005; Scheller et al. 2008; Strauss et al. 2006a). A facial nerve paresis limited of the orbicularis oris muscle and/or selective denervation potentials (e.g. fibrillation potentials) in the orbicularis oris muscle in electromyography (EMG) rather point to a tumor arising from the nervus intermedius. Deterioration of intermedius nerve function may also result in “crocodile tears”, diminished lacrimation (eye dryness), decreased taste sensation or dysgeusia and decreased

sensation in the posterior aspect of the external auditory canal (Hitzelsberger sign). Pathological findings in Schirmer tear test, nasolacrimal reflex, blink reflex test, olfacto- and gustometry may support the differential diagnosis between an intermediate and a facial nerve schwannoma. Schwannomas of the intermediate nerve seldom present with hemifacial spasm by compression of the facial nerve motor root (Kudo et al. 1996).

Therapy

In general when a patient is diagnosed as suffering from a schwannoma of the cerebellopontine angle, several treatment strategies exist: wait and scan, radiosurgery and microsurgery. The best management, especially for patients with small- to medium-sized tumors is controversial. The authors focus on the surgical treatment of this rare entity.

Surgery

In general, microsurgery is performed using a retrosigmoid approach in the supine or sitting position. For intraoperative monitoring brainstem auditory evoked potentials (BAEP) and facial nerve EMG are mandatory. The authors recommend a bipolar recording set up using at least three channels for selective monitoring of the orbicularis oris-, orbicularis oculi- and nasal muscle (Prell et al. 2007). Total intravenous anesthesia without muscle relaxants facilitates monitoring of both EMG and BAEP. The patient is given 2 g of ceftriaxone and 20 mg of dexamethasone prior to surgery. The authors strongly recommend a prophylactic neuroprotective medication consisting of nimodipine and hydroxyethylstarch from the day before surgery until the seventh postoperative day (Scheller et al. 2008). This regimen has shown to improve cranial nerve functions in several monocentric pro- and retrospective studies (Bischoff et al. 2008; Scheller et al. 2004, 2007; Strauss et al. 2006b).

Before making the incision for the retrosigmoid approach a line from theinion to the zygomatic arch signs the approximate course of the transverse sinus. A second line along the posterior border of the mastoid process represents the sigmoid sinus. The retrosigmoid skin incision is C-shaped or straight. Skin and the suboccipital muscles are infiltrated with 1% Xylocaine with epinephrine in order to reduce bleeding. Careful dissection of the soft tissue and muscle layers is more important than osteoplastic or osteoclastic craniotomy in order to reduce postoperative suboccipital headaches. Frequently the lesser occipital nerve and the occipital artery have to be transected for adequate approach. For craniotomy one or two burr holes are performed, one postero-inferior to the assumed sigmoid-transverse sinus junction, and the other accordingly at the infero-medial corner of the exposure. It may be mandatory to drill away some of the mastoid bone to expose the edge of the sigmoid sinus. It is not necessary to expose the sinus themselves. All opened air cells must be carefully closed by bone wax. After administration of mannitol a C-shaped incision of the dura mater is performed. After opening the arachnoid membrane of the cisterna magna to drain cerebrospinal fluid the tumor can usually be exposed with minimal cerebellar retraction.

The posterior tumor capsule is electrically mapped initially using 0.5 mA constant current bipolar stimulation (Neurosign 100, Inomed, Tuttlingen, Germany). In both surgical cases the authors recognized a rather large area of positive direct nerve stimulation exclusively in the orbicularis oris- and nasalis muscle on the posterior tumor surface. We have never seen activity in the orbicularis oculi muscle. The selective muscle response identifies the intermediate nerve, a phenomenon which may be also seen in conventional vestibular schwannomas, although usually the EMG activity is found on the upper tumor capsule posterior to the trigeminal nerve. With debulking of the tumor the root exit zones become visible. At this point the surgeon can see the intermediate nerve in close contact to an intact vestibulo-cochlear complex.

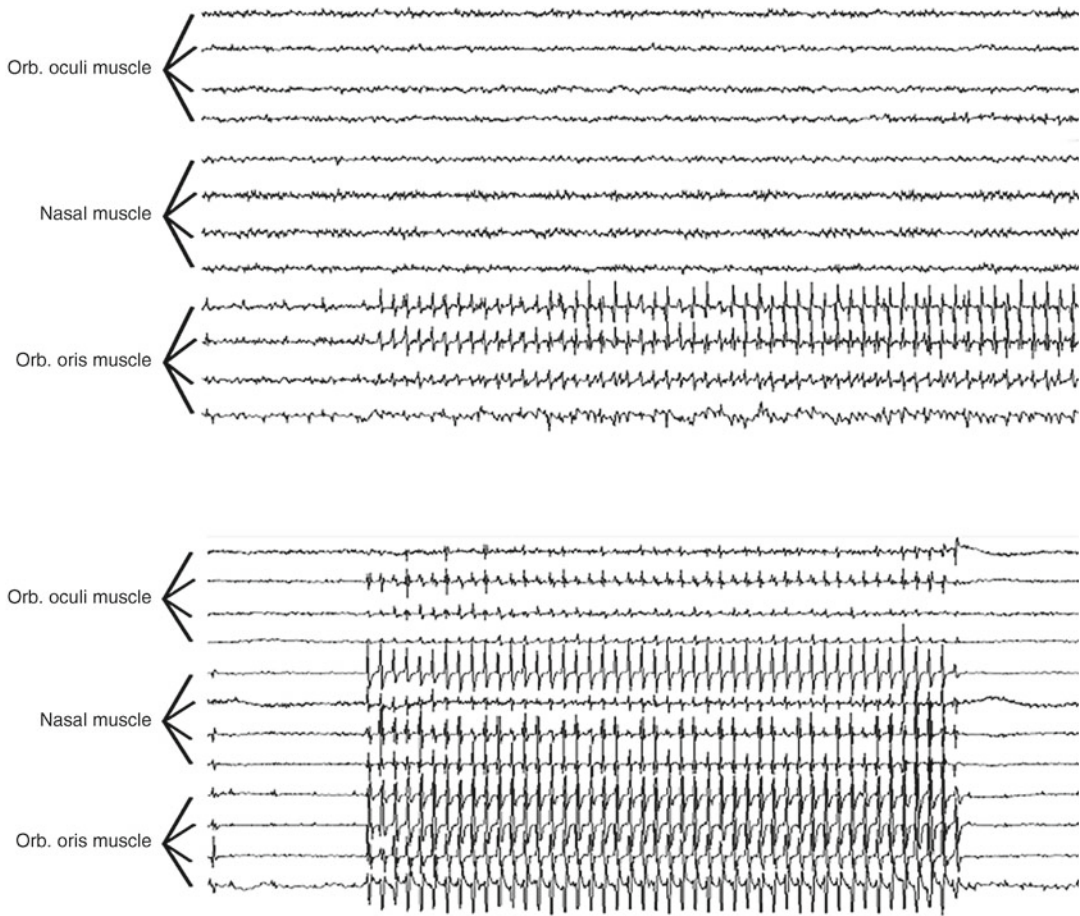


Fig. 27.2 EMG response in the orbicularis oris muscle during stimulation of the intermediate nerve (*upper potentials*). Traces (four electrodes per branch) of the

results of electrical stimulation (0.05 mA) of the facial nerve within motor response in all three branches (*lower potentials*) (With kind permission of the J Neurosurg)

The facial nerve can be localised further ventrally. Electrical stimulation usually reveals EMG responses in all three branches (orbicularis oculi-, nasalis- and orbicularis oris muscles). Electrophysiological responses by direct nerve stimulation and/or spontaneous EMG potentials during tumor resection exclusive in the orbicularis oris muscle indicate the intermediate nerve as site of tumor origin (Fig. 27.2). Under these circumstances the intermediate nerve can be clearly identified as the site of tumor origin (Fig. 27.3). The facial nerve had no contact to the tumor in the area of the root exit zone. The tumor capsule is dissected in the arachnoid plane from the brain stem and the cerebellum. The tumor can be separated from the facial nerve and the vestibulocochlear

nerve bundle in both of the authors cases was sacrificed. For further exposure of the tumor the posterior meatal lip and the petrosus bone between meatus and temporal base is drilled. Following complete drilling the dural envelope of the facial nerve is opened. Facial nerve function should be carefully monitored during intracanalicular tumor resection. Especially “A trains” are high significant indicative for a facial nerve paresis (Romstöck et al. 2000). In our first case we have drilled the facial canal up the temporal base and achieved complete tumor removal. In our last case, despite a good dissection plane between tumor and facial nerve, we temporarily lost electrical excitability and left a tumor portion within the auditory canal. Topical administered nimodipine and cessation of

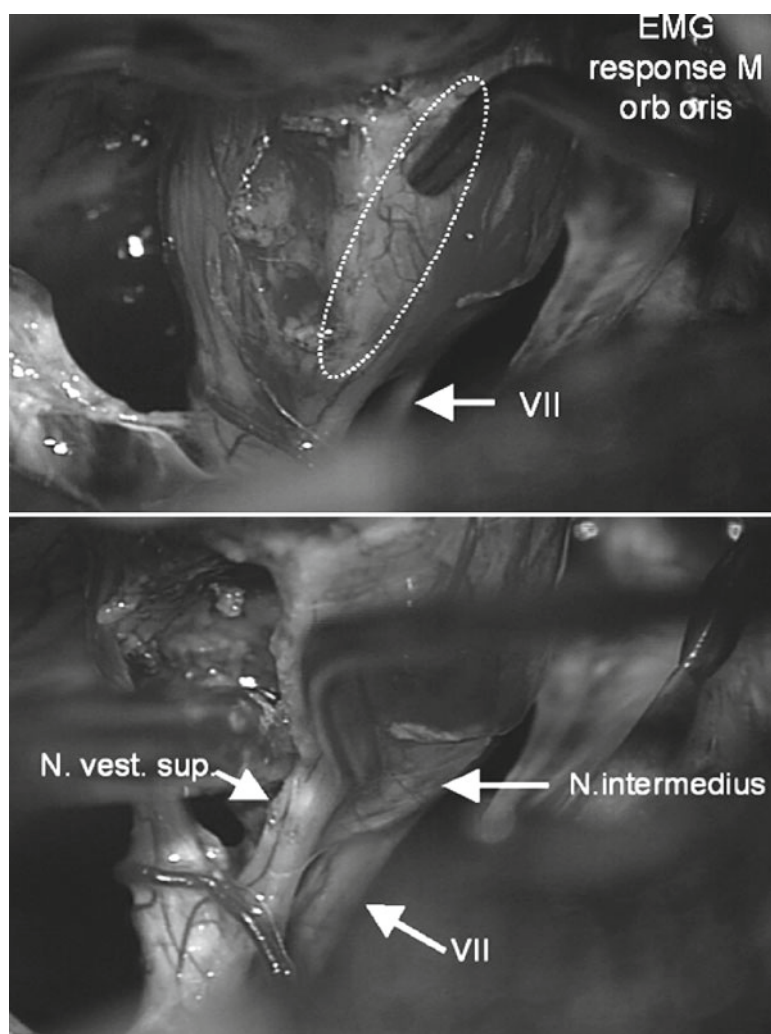


Fig. 27.3 Area of positive electrical response exclusively in the orbicularis oris muscle. Facial nerve is seen anterior and cranial (*upper photograph*). Following dissection of the superior vestibular nerve (N. vest. sup.) showing the

intermediate nerve (N. intermedius) clearly as the site of tumor origin (*lower photograph*) (With kind permission of the J Neurosurg)

surgery may be helpful in these situations, since for long term results, it is important to remove the tumor from the internal auditory canal.

Outcome

Deterioration of the facial and cochlear nerve functions are the most common postoperative complications following facial nerve schwannoma surgery.

Functional results concerning the intermediate nerve are rarely reported in the literature. Irving et al. (1995) reported an incidence of 44% for “crocodile tears”, 72% for lacrimation and 48% for taste abnormalities following vestibular surgery compared with 2% (“crocodile tears”), 4% (lacrimation) and 6% (taste abnormalities) prior to surgery. The rate of recovery was 6% for “crocodile tears”, 27% for lacrimation and 42% for taste abnormalities.

Prognosis

If total resection was achieved, postoperative MRI follow-ups after 3–6 months and then every year are recommended. In cases of incomplete resection we favour a wait and scan strategy before considering radiotherapeutic alternatives.

Discussion

In contrast to anatomical textbooks recently published morphological studies support the surgical experience that the anatomical course, the numbers of roots (1–5) and rootlets (1–5) and the arising sites of the intermediate nerve are very variable (Oh et al. 2003). The intermediate nerve can be divided into three parts during the anatomical course: a proximal segment from the brain stem to the eighth nerve, an intermediate segment between the seventh and eighth nerve and a distal segment from the seventh nerve into the internal auditory canal (Kudo et al. 1996; Rhoton et al. 1968). The intermediate nerve contains three different types of fibres: parasympathetic secretomotor fibres, taste fibres, and somatic sensory fibres (Last 1990).

At the porus acusticus internus, the intermediate nerve joins the motor root of the facial nerve to form a single structure that enters the internal acoustic canal and then passes toward the geniculate ganglion. The point at which the intermediate nerve joins the main facial nerve trunk is variable (Ashram et al. 2005). Distal of the geniculate ganglion the fibers of the intermediate nerve pass through either the greater superficial petrosal nerve or the chorda tympani nerve. The greater superficial petrosal nerve consists of preganglionic parasympathetic secretomotor fibers (that arise from the superior salivary nucleus and synapse in the pterygopalatine ganglion, the postganglionic fibers reach the lacrimal gland and the nasal and palatine mucous glands) and special sensory fibers for taste sensation from the soft palate and the anterior two-third of the tongue (the central processes of these fibers follow the intermediate nerve to the nucleus of the tractus solitarius). The parasympathetic

secretory motor fibers of the chorda tympani nerve also originate from the superior salivary nucleus and synapse in the submandibular ganglion. The postganglionic fibers innervate the submandibular and sublingual salivary glands. The somatic sensory fibers of the intermediate nerve convey sensation from the postero-superior aspect of the external auditory meatus and pass through the chorda tympani nerve.

Strauss et al. (2006a) reported of a split facial nerve course in vestibular schwannomas. They described the minor portion of the facial nerve as running along the superior vestibular nerve on the cranial dorsal tumor capsule rejoining the major portion at the level of the porus acusticus internus. This observation fits together with Rhoton's et al. description of the course of the intermediate nerve (Rhoton et al. 1968). The intermediate nerve as the origin of a schwannoma during cerebellopontine angle surgery can be determined by visual impression of the root exit zone anatomy, but even more based on electrophysiology. Preoperative electromyography may reveal denervation potentials exclusive in the orbicularis oris muscle as an indication of a damage to the supposed motoric fibers of the intermediate nerve.

As described by Ashram et al. (2005) direct intermediate nerve stimulation resulted in EMG activity only in the orbicularis oris muscle with longer latencies and smaller amplitude as compared to the facial nerve. The study of Ashram et al. (2005) and the intraoperative observations of Strauss et al. (2006a) and Scheller et al. (2008) revealed EMG activity exclusively in the orbicularis oris muscle upon stimulation of the intermediate nerve. In conclusion intraoperative direct stimulation is a feature to detect the facial and the intermediate nerve during cerebellopontine schwannoma surgery, provided at least a three channel recording set up. Furthermore these electrophysiological findings suggest that the intermediate nerve carries motor fibers for the orbicularis oris muscle. Damage to the intermediate nerve might be the explanation for the frequently postoperative recognized facial nerve weakness predominantly in the orbicularis oris muscle. In respect of the difficult and inconsistent anatomy of the intermediate nerve at least

some of the facial nerve schwannomas might actually arise from the intermediate nerve (Kudo et al. 1996; Oh et al. 2003). Several authors supposed that so called “facial schwannomas” in fact arose from the intermediate nerve because of the predilection of schwannomas for sensory nerves (Lopes et al. 1978; Shambaugh et al. 1969; Sherman et al. 2002).

Conclusion

In conclusion, it is possible to detect the intermediate nerve as site of tumor origin during cerebellopontine surgery using selective direct nerve stimulation and multi-channel recordings. Subsequently, more intermediate nerve schwannomas may be identified as tumor origin by detailed intraoperative electrophysiological examination. These electrophysiological findings suggest that the intermediate nerve carries motor fibers for the orbicularis oris muscle. Damage to the intermediate nerve might be the explanation for the frequently postoperative recognized facial nerve weakness predominantly in the orbicularis oris muscle following cerebellopontine schwannoma surgery.

References

- Ashram YA, Jackler RK, Pitts LH, Yingling CD (2005) Intraoperative electrophysiological identification of the nervus intermedius. *Otol Neurotol* 26:274–279
- Bischoff B, Romstöck J, Fahlbusch R, Buchfelder M, Strauss C (2008) Intraoperative brainstem auditory evoked potential pattern and perioperative vasoactive treatment for hearing preservation in vestibular schwannoma surgery. *J Neurol Neurosurg Psychiatry* 79:170–175
- Fagan PA, Misra SN, Doust B (1993) Facial schwannoma of the cerebellopontine angle and the internal auditory canal. *Laryngoscope* 103:442–446
- House JW, Brackmann DE (1986) Facial nerve grading system. *Otolaryngol Head Neck Surg* 93:146–147
- Irving RM, Viani L, Hardy DG, Baguley DM (1995) Nervus intermedius function after vestibular schwannoma removal: clinical features and pathophysiological mechanisms. *Laryngoscope* 105:809–813
- King TT, Morrison AW (1990) Primary facial nerve tumors within the skull. *J Neurosurg* 72:1–8
- Kudo A, Suzuki M, Kubo N, Kuroda K, Ogawa A, Iwasaki Y (1996) Schwannoma arising from the intermediate nerve and manifesting as hemifacial spasm. *J Neurosurg* 84:277–279
- Last TJ (1990) *Anatomy regional and applied*, 8th edn. Churchill Livingstone, Edinburgh, pp 640–642
- Lee KS, Britton BH, Kelly DL Jr (1989) Schwannoma of the facial nerve in the cerebellopontine angle presenting with hearing loss. *Surg Neurol* 32:231–234
- Lipkin AF, Coker NJ, Jenkins HA, Alford BR (1987) Intracranial and intratemporal facial schwannoma. *Otolaryngol Head Neck Surg* 96:71–79
- Lopes O, Castro NP, Quintanilla Ribeiro FA, Piariss PR (1978) Neurinoma of the facial nerve. *ORL J Otorhinolaryngol Relat Spec* 40:254–262
- O'Donoghue GM, Brackmann DE, House JW, Jackler RW (1989) Schwannomas of the facial nerve. *Am J Otol* 10:49–54
- Oh CS, Chung IH, Lee KS, Tanaka S (2003) Morphological study on the rootlets comprising the root of the intermediate nerve. *Anat Sci Int* 78:111–113
- Prell J, Rampp S, Romstöck J, Fahlbusch R, Strauss C (2007) Train time as a quantitative electromyographic parameter for facial nerve function in patients undergoing surgery for vestibular schwannoma. *J Neurosurg* 106:826–832
- Rhoton A, Kobayashi S, Hollinshead WH (1968) Nervus intermedius. *J Neurosurg* 29:609–618
- Romstöck J, Strauss C, Fahlbusch R (2000) Continuous electromyography monitoring of motor cranial nerves during cerebellopontine angle surgery. *J Neurosurg* 93:586–593
- Scheller C, Strauss C, Fahlbusch R, Romstöck J (2004) Delayed facial nerve paresis following acoustic neuroma resection and postoperative vasoactive treatment. *Zentralbl Neurochir* 65:103–107
- Scheller C, Richter HP, Engelhardt M, Koenig R, Antoniadis G (2007) The influence of prophylactic vasoactive treatment on cochlear and facial nerve functions following vestibular schwannoma surgery: a prospective and open-label randomized pilot-study. *Neurosurgery* 61:92–97
- Scheller C, Rächinger J, Prell J, Kornhuber M, Strauss C (2008) Schwannoma of the intermediate nerve. *J Neurosurg* 109:144–148
- Shambaugh GE, Arenberg IK, Barney PL, Valvassori GE (1969) Facial neurilemmomas: a study of four diverse cases. *Arch Otolaryngol* 90:742–755
- Sherman JD, Dagnew E, Pensak ML, van Loveren HR, Tew JM (2002) Facial nerve schwannomas. Report of 10 cases and review of the literature. *Neurosurgery* 50:450–456
- Strauss C, Prell J, Rampp S, Romstöck J (2006a) Split facial nerve course in vestibular schwannomas. *J Neurosurg* 105:698–705
- Strauss C, Romstöck J, Fahlbusch R, Rampp S, Scheller C (2006b) Preservation of facial nerve function after postoperative vasoactive treatment in vestibular schwannoma surgery. *Neurosurgery* 59:577–584
- Symore L, Cheesman AD, Kawauchi M, Bordi L (1993) Schwannomas of the facial nerve: a report of 12 cases. *Br J Neurosurg* 7:13–22

Part VII

Therapy

Keith A. Cauley

Abstract

Hydrocephalus is an uncommon and controversial complication of radiosurgery for vestibular schwannoma (VS). The controversy arises as hydrocephalus can also be associated with untreated vestibular schwannoma. Nevertheless, retrospective review of radiosurgical cases suggest that in a minority of cases of treated vestibular schwannomas, radiosurgery appears to cause or significantly accelerate subsequent hydrocephalus. This chapter discusses the evidence for this claim, and reviews the frequency of this complication, correlating variables, predisposing factors, proposed mechanisms, and patient management options.

Introduction

Both communicating and noncommunicating forms of hydrocephalus have been seen in association with vestibular schwannoma (VS). Uncommonly, hydrocephalus appears to be accelerated or worsened with radiosurgery. As the popularity of radiosurgical treatment of VS increases, larger cohort studies have demonstrated convincing statistical evidence of a causal relationship between radiosurgery for VS, and the development of hydrocephalus. The mechanisms for the development of hydrocephalus

remain speculative. The relative infrequency of the event has precluded rigorous study of risk factors for this complication of radiosurgery.

Hydrocephalus and Untreated Vestibular Schwannoma

The association between vestibular schwannoma and hydrocephalus has long been noted in the otolaryngological and neurosurgical literature (Briggs et al. 1993; Rogg et al. 2005). More recently retrospective review of larger cohorts of patients has permitted generation of prevalence statistics, and hydrocephalus has been found to be associated with vestibular schwannoma in 9–18% of cases in multiple large cohort studies (Pirouzmand et al. 2001; Tanaka et al. 2003; Rogg et al. 2005; Noren 1998).

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Mechanisms of Hydrocephalus with Untreated Vestibular Schwannoma

The causes of hydrocephalus with VS are much debated. Both obstructive and communicating hydrocephalus have been seen with VS. Case reports and small cohort studies have led to the early impression that hydrocephalus is more commonly seen in association with larger tumors (>3 cm) (Steenerson and Payne 1992). One larger cohort study which included other less common CPA masses stated a strong correlation between hydrocephalus and tumor size (Pirouzmand et al. 2001). Others have made similar observations (Atlas et al. 1996; Tanaka et al. 2003). The latter study presented raw data demonstrating very few cases of hydrocephalus associated with VS under 2 cm in size. In a more rigorous study with regard to this particular question, a strong correlation was made between obstructive hydrocephalus and tumor size, but poor correlation between tumor size and communicating hydrocephalus (Rogg et al. 2005).

An obvious mechanism for hydrocephalus is seen with large tumors which create mass effect at the cerebellar pontine angle and narrow the fourth ventricle. There is controversy even with this simple proposal, however, as it has been claimed that the slow growth of VS largely precludes a true obstructive mechanism of hydrocephalus, with the inference that many cases classified as obstructive may actually be due to another mechanism (Roche et al. 2008). To explain cases of communicating hydrocephalus, it has been proposed that shed tumor proteins alter cerebrospinal fluid characteristics with resultant obstruction of cerebrospinal fluid (CSF) resorption at the arachnoid villi (Briggs et al. 1993; Roche et al. 2008). This topic will be more fully discussed in this chapter under 'mechanisms for hydrocephalus following radiosurgery.'

Another mechanism of communicating hydrocephalus has been proposed with hydrocephalus arising as a result of tumor-associated alterations of CSF flow dynamics in the basilar cisterns (Steenerson and Payne 1992; Atlas et al. 1996), which may be related to altered CSF chemistry.

This proposal may represent a refinement of the shed tumor protein proposal, with altered CSF flow dynamics affecting CSF flow at the level of the basilar cisterns rather than at the level of resorption (Andrews et al. 2001), although it has also been interpreted to mean alteration of CSF flow due to tumor mass (Rogg et al. 2005). Currently, insufficient data exist to support or refute this claim. A recent cohort study of hydrocephalus in 28 untreated VS patients using MR imaging methods reported that ventricular obstruction accounted for <40% of cases, and communicating hydrocephalus made up the majority of cases (Rogg et al. 2005). To complicate matters further, as surgery for obstructive hydrocephalus may take years for the hydrocephalus to resolve, it has been proposed that communicating hydrocephalus due to malabsorption of CSF can coexist with obstructive hydrocephalus (Tanaka et al. 2003).

Radiosurgery for Vestibular Schwannoma

Dr. Lars Leksell is credited with the development of the gamma knife, which he first used to treat VS in 1969. The technique was initially proposed as an alternative to microsurgery (Linskey et al. 1992), and promoted as an option for patients who were considered to be poor surgical candidates. With optimization of technique and published evidence that outcomes and complication rates were at least comparable to microsurgery, radiosurgery has moved into first line therapy for VS, with only a minority of cases treated by microsurgery (Regis et al. 2008).

The initial goal of radiosurgery was to destroy the tumor. Empirical cost-benefit optimization of radiation dose has revealed that tumor growth can be arrested at lower radiation doses, with minimal side effects, and the goal of radiosurgery has shifted to tumor growth control (Abram et al. 2007). Differential dose patterns with sharp dose gradients can be obtained using multiple isocenters conforming to the shape of the tumor. Various methods of radiosurgical dosing and fractionation have been described. The details of radiosurgical

protocols is beyond the scope of this article, and the reader is referred to recent detailed review for discussion of the specifics of gamma knife methodology (Abram et al. 2007; Battista 2009). Generally, radiosurgery can be administered as a single dose (stereotactic radiosurgery) or as fractionated doses over time (stereotactic radiotherapy). An alternative stereotactic radiotherapy delivery platform has been developed using linear accelerator technology (LINAC), with success rates which are reported to be on par with gamma knife (Spiegelmann et al. 2001). There is currently insufficient data to promote one method of radiosurgery over another.

Most agree that microsurgery is the treatment of choice for large tumors, with the general consensus that complication rates of radiosurgery increase with increasing tumor size (Battista 2009). Large tumors are at risk for further expansion following radiosurgery, potentially leading to obstructive hydrocephalus (Abram et al. 2007), and are generally thought to be also increased for the development of communicating hydrocephalus (Pirouzmand et al. 2001). It is also generally accepted that patients presenting with symptomatic hydrocephalus will require either microsurgery or radiosurgery together with shunting (Noren 1998). Larger tumors also pose a technical challenge where the radiation penumbra of multiple isocenters becomes large, placing greater amount of normal tissue at risk (Abram et al. 2007) and with higher risk of facial neuropathy (Spiegelmann et al. 2001).

Radiosurgery Contributing to the Development of Hydrocephalus

Initial suspicion that hydrocephalus was caused or exacerbated by gamma knife were largely anecdotal, appearing in reports of single cases and small cohort series (Thomsen et al. 1990; Lunsford and Linskey 1992). Most subsequent reports detailing complications of radiosurgery discuss a small percentage of hydrocephalus cases, though the definition of this complication generally hinges on clinical symptoms leading to shunt placement, rather than objective imaging

findings (Andrews et al. 2001; Karpinos et al. 2002; Myrseth et al. 2005; Noren 1998; Pollock et al. 1995; Roche et al. 2008). The best argument for causality is a statistically significant increase in prevalence of hydrocephalus in radiosurgically treated cases of VS over untreated cases. With such a complication rate approaching only a few percent, statistical significance has been difficult to achieve. One group even reports statistics that 'refute' the causal connection between radiosurgery and hydrocephalus, with data from 284 cerebellopontine angle tumors 'most of which were VS' (Pirouzmand et al. 2001). Nevertheless most cohort studies report data in support of this causal claim, and the inference of a causal relationship does not appear to be going away with larger cohort studies offering greater statistical significance (Noren 1998).

Because a significant percentage of untreated vestibular schwannoma are associated with hydrocephalus, hydrocephalus after gamma knife may be seen *de novo*, or as increased over a pre-existing baseline (i.e., the hydrocephalus was already present before the use of gamma knife, but worsened following treatment, Roche et al. 2008). Obviously this latter group is more difficult to identify, as changes may be subtle or it may be argued that the hydrocephalus was increasing before treatment. One study which focused on this issue states that ~25% of cases (5 of 20) of preexisting hydrocephalus required shunting after radiosurgery. This small percentage, they conclude, indicates that 'radiosurgery does not decompensate the majority of preexisting hydrocephalus'. When it occurs, hydrocephalus has been seen to develop months after radiosurgery, with radiosurgical treatment to VP shunt time of 4–31 months with a mean time of approximately 1 year (Abram et al. 2007; Battista 2009; Roche et al. 2008).

Mechanisms of Hydrocephalus Following Radiosurgery

Several mechanisms of hydrocephalus secondary to radiosurgery have been proposed. Noncommunicating hydrocephalus can occur

either by post-treatment ‘tumor expansion’ or by swelling of tissues adjacent to the tumor, induced by the radiosurgical treatment. Both of these would logically tend to occur with larger tumors. These types of hydrocephalus have been discussed in several reports (Karpinos et al. 2002; Linskey et al. 1992; Pirouzmand et al. 2001), one series arriving at a prevalence number of 1.4% of cases (Noren 1998).

The more controversial discussion centers around communicating hydrocephalus as a complication of radiosurgery. In this proposal, hydrocephalus arises as a consequence of elevated CSF protein which alters the resorption of CSF. This is a widely accepted explanation for hydrocephalus following subarachnoid hemorrhage, and has been proposed as a mechanism for hydrocephalus in the setting of nonobstructing brain tumors (Gardner et al. 1954). In VS, it is held that high CSF protein levels are caused by “leakage of blood proteins into the CSF. The protein passes through the schwannoma, which is outside the blood brain barrier” (Noren 1998). Alternatively, tumor necrosis, which has been reported in such cases (Noren et al. 1983) may result in contamination of the CSF with tumor debris. Either of these mechanisms could potentially be magnified or hastened by radiosurgery.

In perhaps the only published study attempting to assess the properties of CSF which might contribute to hydrocephalus both before and after radiosurgery, Noren et al. (1983) reported that CSF proteins were elevated in all 7 VS patients tested. The two cases with the highest protein levels (2.72 and 5.04 g/l) demonstrated hydrocephalus. An electrophoresis pattern of CSF protein showed “a more or less pronounced barrier damage pattern” in untreated cases (without hydrocephalus). This pattern of CSF proteins apparently consisted largely of blood products. These investigators report that, in the small number of cases tested, the electrophoretic pattern of CSF did not significantly change following radiosurgery, supporting a model of further leakage of blood proteins over the introduction of new tumor necrosis proteins. These investigators also comment that their data did not support a notion of new CSF proteins arising as a consequence of

inflammation or inflammatory cells entering the CSF as a sequela of radiosurgical treatment.

These limited observations constitute much of what is known regarding the role of elevated CSF protein in the development of hydrocephalus in VS. CSF is not routinely drawn on vestibular schwannoma patients, and CSF protein values have not been reported in large cohort studies. Elevated CSF protein levels have been reported in the setting of hydrocephalus following radiosurgery of VS; however, “marked elevation of CSF proteins is characteristic of VS,” i.e., in the absence of radiosurgery (Sayk 1974). No formal study has reported a comparison of protein levels in cases with or without hydrocephalus. Though such a study may be revealing, these early investigators realized that protein level alone may not tell the whole story, and discussed (but did not show) protein electrophoresis data on the CSF of VS patients. With the general belief that hydrocephalus results from ‘sludging’ of CSF uptake mechanisms, the levels of larger or less soluble proteins may prove more important than absolute protein level. Animal studies suggest that abnormal CSF proteins must contain particulates in order to cause hydrocephalus (Davson et al. 1970).

Patient Characteristics and Factors Contributing to the Development of Hydrocephalus Following Radiosurgery

Why does hydrocephalus develop in some cases of untreated VS while not in others? Why does hydrocephalus develop following radiosurgery in some cases and not in others? As the answer to these questions would impact VS management, there have been a number of studies investigating possible correlating variables in an effort to identify risk factors for this complication.

Tumor size. As previously discussed, a correlation between VS tumor size and the incidence of accompanying hydrocephalus has long been noted. This correlation is thought to carry over to hydrocephalus as a complication of gamma knife

radiosurgery (Roche et al. 2008), occurring more commonly with tumors >2.5 cm (Battista 2009).

Patient Age. VS is typically seen in the older patient population, with a peak incidence in the sixth decade (Myrseth et al. 2007). Hydrocephalus is reported to be more common in a population that is older than the mean for VS (mean age of patients with VS, 53.6 years, mean age of patients with VS and hydrocephalus, 58.5 years (Tanaka et al. 2003)), with a correlation that is statistically significant but probably not clinically significant. The study also makes the observation that hydrocephalus following radiosurgery is more likely to occur with older patients, but the age spread of patients was wide, resulting in low statistical significance.

NF-2. VS is one manifestation of neurofibromatosis type-II, being seen in ~5% of NF-2 patients. It is generally believed that VS occurs at an earlier age in the NF-2 population, and that NF-2 should be considered in any case of VS in a young person (<30 year). VS in NF-2 are reported to be histologically distinct from sporadic VS (Battista 2009). One study of hydrocephalus following gamma knife reported a significantly higher percentage of NF-2 patients represented (13%) than were represented in the total VS population studied, implying that NF-2 constituted a risk factor for hydrocephalus (Roche et al. 2008). Though NF-2 likely predisposes to hydrocephalus, with or without radiosurgery, the numbers of reported cases is small and it has not yet been demonstrated that NF-2 represents an independent risk factor for hydrocephalus following radiosurgery.

Previous Microsurgery. The study discussed above addressed previous microsurgery as a risk factor for the development of hydrocephalus following radiosurgery, and found a significantly higher percentage of patients with hydrocephalus following radiosurgery had a history of microsurgery (20%). This also represented a small sample set of low statistical significance, which merits further study.

Radiosurgical Protocol. Though it is clear that hydrocephalus following radiosurgery must

reflect differences between tumors when the radiosurgical protocol is constant, the relative infrequency of hydrocephalus has not yet permitted comparison of rates of incidence using alternative radiosurgical protocols. Little data exist regarding the development of hydrocephalus as a function of stereotactic radiotherapy or LINAC. One study of 44 patients with VS treated by LINAC reported no cases of hydrocephalus (Spiegelmann et al. 2001). A report of 101 cases treated by fractionated delivery reports 11% of cases with hydrocephalus (Sawamura et al. 2003). Direct comparison between gamma knife and LINAC (69 patients and 56 patients) showed comparable numbers of cases of post-therapy hydrocephalus at 2.9% and 3.6% (Andrews et al. 2001), although the small cohort size limits the statistical significance of the comparison.

Management of the Post Radiosurgery VS Patient with Hydrocephalus

MRI is the appropriate imaging modality for followup of treated VS, particularly with new or worsening symptoms. Routine MR sequences are sufficient to identify hydrocephalus, with an accuracy greater than computed tomography (Gammal et al. 1987). Symptomatic post-radiosurgery hydrocephalus is generally treated with ventriculoperitoneal (VP) shunting. Tumor resection is not generally indicated. VP shunting should serve to treat both communicating and obstructive hydrocephalus. More recently endoscopic third ventriculostomy has been proposed for treatment of hydrocephalus in association with VS (Hayhurst et al. 2006). In a retrospective report of untreated VS patients with hydrocephalus, third ventriculostomy served as an alternative to VP shunt in four of five reported cases. All cases were of tumors >3 cm. Three additional cases of hydrocephalus occurred following radiosurgery. Two of the three third ventriculostomies were successful in this setting; the third tumor was exceptionally large at 6 cm and required a VP shunt. These authors point out that, in many cases, the type of hydrocephalus is not known with certainty prior to intervention, and that third

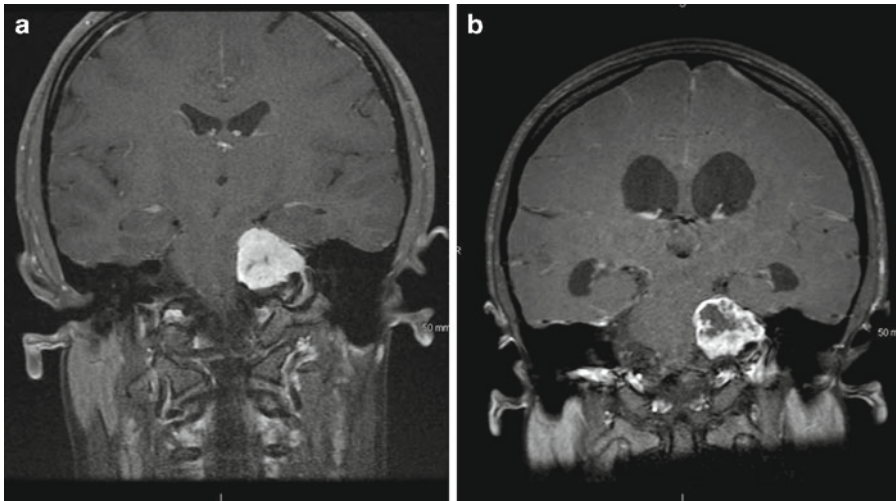


Fig. 28.1 22 year-old female with 3 cm VS. Coronal T1-weighted MR imaging postgadolinium (a) 2 weeks before gamma knife treatment and (b) 9 months after gamma knife treatment. Pretreatment images show fairly homogenous enhancement of VS, and no hydro-

cephalus. Following gamma knife, the tumor demonstrates absent central enhancement consistent with necrosis, with the concurrent development of hydrocephalus (Figure reproduced from Cauley et al. (2009), with permission)

ventriculostomy should have little impact on a communicating hydrocephalus. The authors contend, however, that this low-morbidity procedure can serve to avoid the long-term complications of VP shunting, and that the failure of the procedure resulting in VP shunting did not significantly increase morbidity over VP shunt alone. It is important to realize that the results of third ventriculostomy are less dramatic, both clinically and at imaging, than are typically seen with VP shunting. Imaging is likely to demonstrate “unimpressive improvement in the ventriculomegaly compared with patients in whom shunts are used” with subtle decrease in effacement of the subarachnoid space and cisterns at a short interval follow-up (Jallo et al. 2005). Cine MRI should be able to demonstrate flow through the floor of the third ventricle following successful ventriculostomy (Fig. 28.1).

Future Directions

With the exception of obvious ventricular obstruction, the mechanism of hydrocephalus in both treated and untreated VS remain conjectural.

Given the relatively invasive nature of lumbar puncture, it is unlikely that randomized CSF studies will ever demonstrate CSF factors to correlate with the development of hydrocephalus following radiosurgery for vestibular schwannoma. Improvements in MR imaging technique offer more detailed study of the tumor, CSF pathways, and the ventricular system. T2W FSE thin imaging (CISS or FIESTA sequences), now routinely used for evaluation of the IACs in suspected VS, could be extended to cover the skull base and ventricular system, and should offer more detailed visualization of arachnoid membranes and CSF outflow pathways, and points of potential outflow obstruction (Awaji et al. 2007). MR velocity-encoded cine phase-contrast studies can permit definitive identification of points of CSF obstruction, or serve to demonstrate that the hydrocephalus is communicating (Bradley et al. 1996; Cauley et al. 2009) (Fig. 28.2).

Tumor necrosis may contribute to elevated CSF protein and hydrocephalus. Changes in tumor enhancement properties is commonly seen following radiosurgery. Loss of central enhancement may represent tumor necrosis, and would fit with

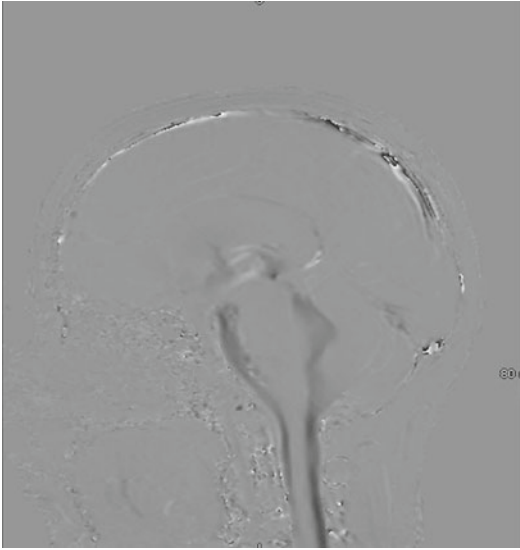


Fig. 28.2 Sagittal MR CSF cine-flow study (gradient-echo, TE, 7.9 ms, TR, 21 ms) of the same patient as in Fig. 28.1a, b. This study was performed when the patient returned with ventriculomegaly 9 months after gamma knife treatment (same time as Fig. 28.1b). CSF cine flow demonstrated free flow of CSF without evidence of intraventricular obstruction, consistent with communicating hydrocephalus (Figure reproduced from Cauley et al. (2009), with permission)

the theory of shed tumor necrosis proteins in the development of communicating hydrocephalus (Cauley et al. 2009; Linskey et al. 1992; Noren et al. 1983). Study of tumor enhancement characteristics with techniques such as perfusion imaging afford study of the time course of necrosis, possibly permitting a more quantitative assessment of tumor necrosis as a risk factor for hydrocephalus. Evaluation of tumor viability following radiosurgery may be addressed with functional imaging studies such as MR spectroscopy, single photon emission tomography, or positron emission tomography (Hasegawa et al. 2006).

Quantitative MR methods are beginning to find application to the study of VS (Sener 2003; Lin et al. 2010). Tumor apparent diffusivity coefficient (ADC) values are generally thought to correlate with cell density, and changes in fractional anisotropy and mean diffusivity may reveal a subpopulation of tumors at higher risk

for causing hydrocephalus when treated. In the future we might hope for standardization of radiosurgical doses and protocols, as well as standardization of imaging follow-up frequency (Meijer et al. 2008) and imaging protocols. Reports including details regarding patient profiles, tumor sizes, and imaging features would permit more meaningful meta-analysis studies in the future.

In conclusion, hydrocephalus can be seen in association with untreated vestibular schwannoma, and is also an uncommon and controversial complication of gamma knife radiosurgery for vestibular schwannoma, where radiosurgery is thought to worsen or accelerate hydrocephalus in some cases. More than one type of hydrocephalus can occur, although the mechanisms remain largely theoretical. Though correlations have been made between such variables as patient age and tumor size in order to determine possible risk factors, no risk factors have emerged with strong enough correlation to impact clinical management. All VS patients should be considered at risk for this complication, and clinical and imaging follow-up is recommended following radiosurgery. When hydrocephalus occurs, MR imaging and MR cine flow studies can aid in determining if the hydrocephalus is of the communicating or noncommunicating form. Though CSF analysis can offer clues as to the mechanism, the routine use of higher resolution MR imaging and quantitative MR methods are likely to offer more information on this condition in future studies. VP shunt is the conventional treatment. Third ventriculostomy has been proposed as a promising alternative to VP shunting, but is unlikely to be of value in communicating hydrocephalus.

Radiosurgery has moved into first line treatment for VS, with complication rates rivaling or surpassing those reported for microsurgery. Future studies with larger cohort sizes, greater standardization of radiosurgical protocol and imaging follow up, and improved imaging techniques should offer greater insight into the risk factors and mechanisms for this poorly understood complication of radiosurgery.

References

- Abram S, Rosenblatt P, Holcomb S (2007) Stereotactic radiation techniques in the treatment of acoustic schwannomas. *Otolaryngol Clin North Am* 40:571–588
- Andrews DW, Suarez O, Goldman HW, Downes MB, Bednarz G, Corn BW, Werner-Wasik M, Rosenstock J, Curran WJ Jr (2001) Stereotactic radiosurgery and fractionated stereotactic radiotherapy for the treatment of acoustic schwannomas: comparative observations of 125 patients treated at one institution. *Int J Radiat Oncol Biol Phys* 50:1265–1278
- Atlas MD, Perez de Tagle JR, Cook JA, Sheehy JP, Fagan PA (1996) Evolution of the management of hydrocephalus associated with acoustic neuroma. *Laryngoscope* 106:204–206
- Awaji M, Okamoto K, Nishiyama K (2007) Magnetic resonance cisternography for preoperative evaluation of arachnoid cysts. *Neuroradiology* 49:721–726
- Battista RA (2009) Gamma knife radiosurgery for vestibular schwannoma. *Otolaryngol Clin North Am* 42:635–654
- Bradley WG Jr, Scalzo D, Queralto J, Nitz WN, Atkinson DJ, Wong P (1996) Normal-pressure hydrocephalus: evaluation with cerebrospinal fluid flow measurements at MR imaging. *Radiology* 198:523–529
- Briggs RJ, Shelton C, Kwartler JA, Hitselberger W (1993) Management of hydrocephalus resulting from acoustic neuromas. *Otolaryngol Head Neck Surg* 109:1020–1024
- Cauley KA, Ratkovits B, Braff SP, Linnell G (2009) Communicating hydrocephalus after gamma knife radiosurgery for vestibular schwannoma: an MR imaging study. *AJNR Am J Neuroradiol* 30:992–994
- Davson H, Hollingsworth G, Segal MB (1970) The mechanism of drainage of the cerebrospinal fluid. *Brain* 93:665–678
- Gammal TE, Allen MB Jr, Brooks BS, Mark EK (1987) MR evaluation of hydrocephalus. *AJR Am J Roentgenol* 149:807–813
- Gardner WJ, Spittler DK, Whitten C (1954) Increased intracranial pressure caused by increased protein content in the cerebrospinal fluid; an explanation of papilledema in certain cases of small intracranial and intraspinal tumors, and in the Guillain-Barre syndrome. *N Engl J Med* 250:932–936
- Hasegawa T, Kida Y, Yoshimoto M, Koike J, Goto K (2006) Evaluation of tumor expansion after stereotactic radiosurgery in patients harboring vestibular schwannomas. *Neurosurgery* 58:1119–1128
- Hayhurst C, Javadpour M, O'Brien DF, Mallucci CL (2006) The role of endoscopic third ventriculostomy in the management of hydrocephalus associated with cerebellopontine angle tumours. *Acta Neurochir (Wien)* 148:1147–1150
- Jallo GI, Kothbauer KF, Abbott IR (2005) Endoscopic third ventriculostomy. *Neurosurg Focus* 19:1–4
- Karpinos M, Teh BS, Zeck O, Carpenter LS, Phan C, Mai WY, Lu HH, Chiu JK, Butler EB, Gormley WB, Woo SY (2002) Treatment of acoustic neuroma: stereotactic radiosurgery vs. microsurgery. *Int J Radiat Oncol Biol Phys* 54:1410–1421
- Lin YC, Wang CC, Wai YY, Wan YL, Ng SH, Chen YL, Liu HL, Wang JJ (2010) Significant temporal evolution of diffusion anisotropy for evaluating early response to radiosurgery in patients with vestibular schwannoma: findings from functional diffusion maps. *AJNR Am J Neuroradiol* 31:269–274
- Linskey ME, Lunsford LD, Flickinger JC, Kondziolka D (1992) Stereotactic radiosurgery for acoustic tumors. *Neurosurg Clin N Am* 3:191–205
- Lunsford LD, Linskey ME (1992) Stereotactic radiosurgery in the treatment of patients with acoustic tumors. *Otolaryngol Clin North Am* 25:471–491
- Meijer OW, Weijmans EJ, Knol DL, Slotman BJ, Barkhof F, Vandertop WP, Castelijns JA (2008) Tumor-volume changes after radiosurgery for vestibular schwannoma: implications for follow-up MR imaging protocol. *AJNR Am J Neuroradiol* 29:906–910
- Myrseth E, Moller P, Pedersen PH, Vassbotn FS, Wentzel-Larsen T, Lund-Johansen M (2005) Vestibular schwannomas: clinical results and quality of life after microsurgery or gamma knife radiosurgery. *Neurosurgery* 56:927–935
- Myrseth E, Pedersen PH, Moller P, Lund-Johansen M (2007) Treatment of vestibular schwannomas. Why, when, and how? *Acta Neurochir (Wien)* 149:647–660
- Noren G (1998) Long-term complications following gamma knife radiosurgery of vestibular schwannomas. *Stereotact Funct Neurosurg* 70(Suppl 1):65–73
- Noren G, Arndt J, Hindmarsh T (1983) Stereotactic radiosurgery in cases of acoustic neurinoma: further experiences. *Neurosurgery* 13:12–22
- Pirouzmand F, Tator CH, Rutka J (2001) Management of hydrocephalus associated with vestibular schwannoma and other cerebellopontine angle tumors. *Neurosurgery* 48:1246–1253
- Pollock BE, Lunsford LD, Kondziolka D, Flickinger JC, Bissonette DJ, Kelsey SF, Jannetta PJ (1995) Outcome analysis of acoustic neuroma management: a comparison of microsurgery and stereotactic radiosurgery. *Neurosurgery* 36:215–226
- Regis J, Roche PH, Thomassin JM (2008) Introduction. The evolution of the surgical management of vestibular schwannomas. *Prog Neurol Surg* 21:1–5
- Roche PH, Khalil M, Soumare O, Regis J (2008) Hydrocephalus and vestibular schwannomas: considerations about the impact of gamma knife radiosurgery. *Prog Neurol Surg* 21:200–206
- Rogg JM, Ahn SH, Tung GA, Reinert SE, Noren G (2005) Prevalence of hydrocephalus in 157 patients with vestibular schwannoma. *Neuroradiology* 47:344–351
- Sawamura Y, Shirato H, Sakamoto T, Aoyama H, Suzuki K, Onimaru R, Isu T, Fukuda S, Miyasaka K (2003) Management of vestibular schwannoma by fraction-

- ated stereotactic radiotherapy and associated cerebrospinal fluid malabsorption. *J Neurosurg* 99:685–692
- Sayk J (1974) The cerebrospinal fluid in brain tumors. In: Vinken PJ, Bruyn GW (eds) *Handbook of clinical neurology: tumors of the brain and skull*. North-Holland, Amsterdam, pp 360–417
- Sener RN (2003) Diffusion magnetic resonance imaging of solid vestibular schwannomas. *J Comput Assist Tomogr* 27:249–252
- Spiegelmann R, Lidar Z, Gofman J, Alezra D, Hadani M, Pfeffer R (2001) Linear accelerator radiosurgery for vestibular schwannoma. *J Neurosurg* 94:7–13
- Steenerson RL, Payne N (1992) Hydrocephalus in the patient with acoustic neuroma. *Otolaryngol Head Neck Surg* 107:35–39
- Tanaka Y, Kobayashi S, Hongo K, Tada T, Sato A, Takasuna H (2003) Clinical and neuroimaging characteristics of hydrocephalus associated with vestibular schwannoma. *J Neurosurg* 98: 1188–1193
- Thomsen J, Tos M, Borgeesen SE (1990) Gamma knife: hydrocephalus as a complication of stereotactic radiosurgical treatment of an acoustic neuroma. *Am J Otol* 11:330–333

Francesco Dispenza and Alessandro De Stefano

Abstract

Therapeutic strategies of vestibular schwannomas treatment are changed in the last years. The improvement of surgical equipments and skills allowed increasing the numbers of successful operations in the last 10 years. The results improved in terms of survival, quality of life, recurrences, facial nerve function, and hearing preservation. Also the age of operated patients, once a restriction, becomes progressively a relative limitation to surgical indication. However the progressive widening of MRI availability give to the Otoneurologist the possibility to follow-up the patients with a relative simple radiological examination, delaying the decision to treat surgically the neoplasm only in case of growth or symptoms occurrence. The medical society is in demand of a treatment strategy based on hard data especially on the growth pattern of these tumors. To have a complete picture of the treatment options we will discuss following the several modality to manage a solitary vestibular schwannoma; the discussion about radiotherapy-radiosurgery is treated elsewhere in this book.

Introduction

Vestibular schwannomas are benign, slow growing tumors that arise from the vestibular-cochlear nerve. This neoplasm was also called: acoustic neurinoma, acoustic neuroma or more generally

acoustic tumor. They arise from the Schwann cells of the nerves composing the acoustic bundle and are characterized by a slow growth, although not well estimable. The vestibular schwannoma could be part of a systemic disease called Neurofibromatosis type 2, in which the acoustic tumors are bilateral and more aggressive. Vestibular schwannomas represent about 6% of all intracranial tumors and accounts for about 80% of all tumors diagnosed in the cerebello-pontine angle (CPA). The incidence reported is 10–20:1,000,000 per year (Stangerup et al. 2006). Both sex are involved equally and there is not race prevalence. The incidence increased in the

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last 10 years, but it was due to the enlarged availability of MRI done for others reasons. However, the true incidence could be different and greater than the actually known, because of a lot of tumors remain silent and undiagnosed.

The typical presentation of the solitary vestibular schwannoma (VS) includes: unilateral progressive sensorineural hearing loss (SNHL) in the high frequency range, unilateral tinnitus and vertigo spells or imbalance. However abnormal clinical presentation may be present characterized by sudden sensorineural hearing loss or facial weakness. Sometime VS is completely asymptomatic and diagnosed casually. Generally the retard of diagnosis is about 5 years.

The treatment options for patients with VS consist in: total or subtotal surgical excision, radiotherapy (Gamma Knife stereotactic radiosurgery and fractionated stereotactic radiotherapy with linear accelerator) and conservative management with serial observation by magnetic resonance imaging (MRI) scanning.

Natural History and Clinical Overview

The large part of vestibular schwannomas is a benign slow growing tumor and the natural course of the disease is unreadable. The tumor may grow continuously or until a certain size, and then it can stop the growth for an incalculable time. Although slow, the growth rate is not estimable and the regulating factors are still unknown. It was estimated that the size of acoustic tumor increases in diameter at an average of 1–2 mm per year. In 30–70% a growth is demonstrable over a variable period of time, but in some patients we can observe a spontaneous tumor regression. The variable percentage of growth reported may depend at least in part on the length of the observation period (Charabi et al. 2000). In a consistent number of cases the size of tumor remain stable for a long period and after 5 years the growth seems to be arrested. However it is probable that most tumors apparently stable, will eventually exhibit a progression if observed for a decade or longer. A rapid growth could be pres-

ent in some tumor with cystic component due to accumulation of fluid into cystic space or to an improvise hemorrhage (Dispenza and De Stefano 2008), such type of tumor is quite rare. Also a malignant type of vestibular schwannoma could be rarely diagnosed, but is characterized by worse prognosis and different treatments. Progressive growth in the CPA will eventually lead to compression of the brainstem and/or the cerebellum, occlusion of the fourth ventricle, and subsequently its incarceration. This is the most dangerous eventuality, and the surgical removal becomes mandatory. The growth rate is not related with sex and age of patients, as noted by some Authors (Herwadker et al. 2005).

It has been proposed that VS may arise in the region of the transition zone between the central and peripheral myelin, also known as the Obersteiner-Redlich zone. Classically, histopathologic examination of VS reveals two morphologic patterns: Antoni A and B. Antoni type A describes a densely packed cells with small, spindle-shaped, densely staining nuclei. Antoni type B refers to a looser cellular aggregation of vacuolated, pleomorphic cells. The Antoni B seems to occur in particular in large tumors.

The typical manifestations of VS include: unilateral progressive sensori-neural hearing loss (SNHL) 68.1%, unilateral tinnitus 49.1%, and disequilibrium 50%. These symptoms can occur in various associations. Several audiological features can be present in clinical practice as unilateral sudden sensori-neural hearing loss (4–26%) and normal hearing (4–12%). Uncommon symptoms are: headaches (10.5%), otalgia, trigeminal neuralgia (9%), facial nerve palsy (6%), cerebellar dysfunction, hydrocephalus (4%) and lower cranial nerve or brainstem related symptoms. Besides some Authors described atypical presentation of VS patients with characteristic of Ménière disease or benign paroxysmal positional vertigo.

The SNHL is unilateral and slowly progressive in the history of the disease. Disturbance in word recognition over the phone is one of the first symptoms that lead the patients to the physician. The hearing may be also fluctuating mimicking a Meniere's disease or a sudden sensorineural hearing loss (SSNHL). In our personal series we found

some patients presenting with SSNHL treated elsewhere with steroid showing a transitory improvement of the hearing (unpublished data), for this reason we recommend a complete evaluation for VS in all patients presenting unilateral SSNHL.

Unilateral tinnitus is often present associated or not with hearing loss. A vestibular schwannoma should be suspected in presence of persistent unilateral tinnitus, especially if isolated and in young patients. The presence of tinnitus at initial presentation was a significant predictor of tumor growth and may be a marker of increased biological auditory nerve activity. The predictive value of tinnitus is increased within the range of small-to medium-sized tumors whose management is most controversial.

Vestibular symptoms could be present in patients with schwannoma of the VIII. The patients most often complain a progressive imbalance rather than true rotary vertigo. At some stage of the disease few patients could present vertigo spells as well as unsteadiness. Also positional vertigo were described in patients with cerebello-pontine angle tumor (De Stefano et al. 2008).

The progressive growth of the vestibular schwannoma may explicate the slow appearance of balance disturbs. The destruction of unilateral peripheral vestibular function by these slowly growing tumors is gradual and facilitates the central vestibular compensation. For these reason, in presence of big tumors, the symptoms of peripheral deficit are quite absent because of the compensation was already obtained. Usually the acoustic tumor removal doesn't produce any peripheral deficit of balance, although the bony labyrinth is completely destroyed. In case of small tumors some peripheral deficit could be detected because of the system is already functioning, but is distressed by the tumor effects.

The neurotological examination of patients with vestibular schwannoma often identifies signs and symptoms of peripheral and central vestibular dysfunction. In small tumors, nystagmus can be discovered in the horizontal plane beating to the unaffected side. A vertical nystagmus is a sign of brainstem compression by a large tumor. The examination of the gait could show a vestibular ataxia, with deviation toward the affected side.

Dizziness is also presents in case of cerebellar dysfunction due to large tumors of the CPA compressing the cerebellum. The symptoms related to cerebellar compression are difficulty in manipulating objects with hands, intention tremor and wide base gait.

Trigeminal nerve compression by large tumors is the cause of hypoesthesia of the face. The clinical examination may reveal the absence or reduction of the corneal reflex in 15% of the patients.

The facial nerve is incredibly resistant to stretch and compression by the tumor growth. Very few cases of facial weakness are encountered in the clinical practice, and usually are due to large vestibular schwannomas. Hemifacial spasms, facial weaknesses with scarce auditory symptoms are more indicative of a facial nerve tumor. However the facial function have to be assessed in preparation of the treatment; the House-Brackmann grading scale (House and Brackmann 1985) is used to quantify the function of the nerve, in particular it is helpful to monitoring the post-surgical recovery. At the time of diagnosis 98% of patients have a grade 1 facial function.

Patients with large tumor of CPA can develop papilledema and hydrocephalus (Fig. 29.1). The examination of fundus oculi should be made as integration of neurotologic examination.

The hearing have to be assessed in all patients with vestibular schwannoma. Unilateral complaints and symptoms should lead promptly to routine diagnostic pure tone and speech audiometry. Further instrumental assessments of hearing function should be done in case of hearing loss not explicable by another pathological mechanism. Asymmetric hearing loss is a suspicious condition, especially if associated with tinnitus. A decrease in speech discrimination score out of proportion to the pure tone loss is accepted as index of suspicion for acoustic tumor. Such phenomenon is known as rollover sign. It is to mention that numerous cases of vestibular schwannoma have not any hearing loss, but only an aural fullness may be present. The hearing threshold determination is mandatory to plan a correct treatment; it is generally reported as average of 500, 1,000, 2,000 and 4,000 Hz (Pure Tone

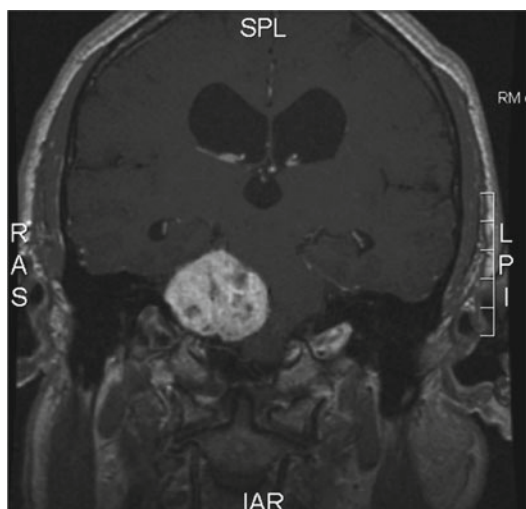


Fig. 29.1 Giant vestibular schwannoma. Coronal T1 weighted MRI scan after Gadolinium enhancement shows a giant tumor of the VIII cranial nerve that displace and compress the brainstem. The only symptoms complained by this patient were: unilateral hearing loss and sight defect. It is noticeable the hydrocephalus by the enlargement of ventricles. The tumor shows solid composition with internal areas of necrosis

Average, PTA). The definition of the hearing as serviceable is generally thought as PTA threshold better than 50 dB or speech discrimination score better than 50%, or both. However such consideration had not a large consensus and the classification of hearing status was reviewed in 2003 (Kanzaki et al. 2003). The new classification allows noting finer the degree of hearing change after surgery and makes more realistic report of results in preservation surgery. It better considers the importance of speech discrimination in the economy of the hearing with a cut off of 10% in SDS between each class (Table 29.1).

The retrocochlear evaluation is still not accepted as routine examination in case of suspected acoustic tumor, and no guidelines are present about this topic. The Auditory Brainstem Responses (ABR), although advocated as useful tool to discriminate patient to undergo radiological evaluation, is not widely accepted because of low specificity. The gold standard evaluation in case of suspected vestibular schwannomas remains the MRI with Gadolinium enhancement.

Table 29.1 Hearing classification in patients affected by vestibular schwannoma

Class	PTA (dB)	SDS (%)
A	0–20	100–80
B	21–30	79–70
C	31–40	69–60
D	41–60	59–50
E	61–80	49–40
F	81–	39–0

This classification should be adopted to report both preoperative and postoperative hearing

PTA is obtained by averaging the frequencies: 0.5, 1, 2, 4 kHz. Hearing is classified based on either PTA or SDS, which is worse. Better SDS class than PTA designates the category one class higher

PTA Pure Tone Average, SDS Speech Discrimination Score, dB decibel

The MRI is the best radiological study available for diagnosis and staging of vestibular schwannoma and others CPA tumors. It allows visualizing the inner ear structures (i.e. cochlea and vestibule), the internal auditory canal (IAC), the CPA, and the relationship between the tumors and surrounding structures (i.e.: brainstem, cerebellum, vessels, etc...). The MRI protocol includes T1-weighted images in contiguous sections through the IAC before and after contrast enhancement, and T2-weighted images obtained in section throughout the brain. On the T1 images without Gadolinium the tumor is isointense or mildly hypointense compared to the brain, after contrast enhancement the lesion appears hyperintense because of marked contrasted. On T2 images the tumor is isointense with the brain and hypointense if compared with CSF. MRI is helpful to make also some differential diagnosis preoperatively based on well-known pattern of lesions. The T2 sequences of MRI can display well the relationship between the tumor and the fundus of the IAC: the bright signal of the CSF fluid gives the size of the space between tumor and fundus (Figs. 29.1, 29.2, 29.3, and 29.4).

The radiological evaluation is mandatory to plan a treatment, whatever it will be and after this clinical step the size of the tumor must be evaluated. The size should be based on measurement on MRI or CT (if MRI is not feasible) of the

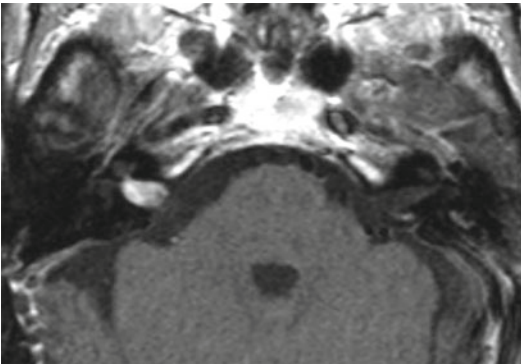


Fig. 29.2 Intracanalicular vestibular schwannoma. Axial T1 weighted MRI scan after Gadolinium enhancement showing a tumor that is content into internal auditory canal without extrameatal component (size 0 mm). The fundus in this case is completely occupied by the tumor with an initial invasion of the cochlea. In this patient every attempt to preserve hearing with surgical approach will be disastrous

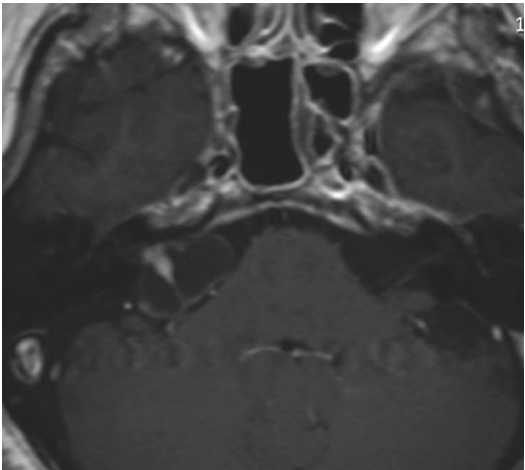


Fig. 29.4 Cystic vestibular schwannoma. Axial T1 weighted MRI scan after Gadolinium showing a tumor with prevalent cystic component. The contrast enhancement is present in the minimal solid component (*central portion*) and in the peripheral sheath

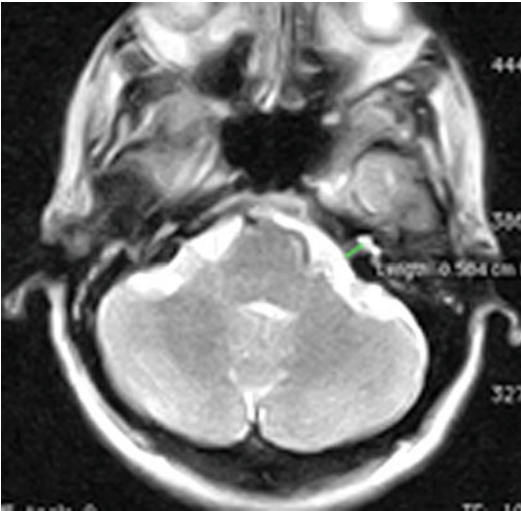


Fig. 29.3 Intracanalicular vestibular schwannoma. Axial T2 weighted MRI scan showing a tumor confined into internal auditory canal. No extrameatal component is present. The *green line* shows the right orientation to measure the tumor length inside IAC

greatest extrameatal tumor diameter (Kanzaki et al. 2003). The intrameatal portion should be clearly distinguished. If the tumor is only intrameatal (intracanalicular), without any extension out of the porus, the terms intrameatal tumor should be used (in such cases the extrameatal size is 0 mm). The intrameatal size can be measured

Table 29.2 Tumor size classification

Grade	Description	Extrameatal size (mm)
	Intrameatal tumor	0
1	Small	1–10
2	Medium	11–20
3	Moderately large	21–30
4	Large	31–40
5	Giant	>40

The size should be based on measurement of the largest extrameatal diameter on Magnetic Resonance (Computed Tomography if MRI is not practicable). It should be noted always whether the fundus is empty or if the tumor is cystic. The size of intrameatal portion can be measured in millimeters along the length of the internal auditory canal, but this measure has not to be added on the final size. Actually is the widest accepted classification that allows to report results with quite all MRI device worldwide *mm* millimeters

separately in millimetres along the length of the internal auditory canal and this portion should not be added to the final size of the tumor. This measurement system was ideated to standardize the reports of the results (Table 29.2).

In the clinical stage of the evaluation two characteristics of the tumor have to be noted: the state of the fundus of the internal auditory canal and the tumor morphology at MRI. The involvement

Table 29.3 Sanna's classification for cystic VS

Type	Cyst location/wall thickness	Subtype	Definition
A	Central/thick wall	1	Polycystic (small intratumoral cysts)
		2	Polycystic (moderate size intratumoral cysts)
		3	Monocystic (single large cyst)
B	Peripheral/thin wall	1	Anterior
		2	Medial
		3	Posterior
		4	Combined

The classification is based on overall cyst location (central or peripheral) and cyst wall thickness (thick or thin). Type A lesions are further subdivided by the cyst characteristics as polycystic or monocystic and its size. Type B lesions are further classified according to cyst orientation with respect to the internal auditory meatus

of the IAC fundus is a negative predictor factor for hearing preservation surgery, as consequence it should be noted whether the fundus is empty. The nature of the tumor may be a factor that entails challenging surgical problems. Is generally considered more difficult to remove cystic tumors than solid homogeneous tumor, for this reason cystic tumors should be considered as a separated group. Cystic vestibular schwannomas (CVS) differ from solid schwannomas by their rapid growth, frequent involvement of facial nerve, and somewhat unpredictable biologic behavior. Cystic vestibular schwannomas have been estimated to represent anywhere from 5.7% to 48% of all vestibular schwannomas. A classification (Table 29.3) of this type of VS was recently proposed (Piccirillo et al. 2009). Factors that lead to unfavorable surgical outcomes include engulfment of and adherence to neurovascular structures, hypervascular solid portions of the tumor. Theoretically, CVS could form because of one or a combination of the following mechanisms: tumor growth and subsequent central necrosis, coalescence of microcysts formed in Antoni B tissue, and/or repeated intratumoral hemorrhage. Later, gradual cyst enlargement could be attributed to osmotic gradients set up by extravasation of serum proteins from an impaired blood-tumor barrier and/or the production of mucinous material within the cyst. Recently, high levels of proteolytic enzymes (matrix metalloproteinase 2) have been identified in CVS fluid and cyst walls.

Conservative Management

The aim of all therapies should be to prevent morbidity and mortality in the safest, non-invasive fashion that can yield effective results. For this reason, the management of VS has shifted in recent decades away from microsurgical resection toward more conservative strategies. Vestibular schwannoma have all the characters of a disease manageable with the simple observation for a long period: slow growth, benign behaviour, and few symptoms in early phases. Additionally the localization of the disease requires important surgical procedures to remove the lesion that may produce more disabling effect than the tumor itself. Furthermore a considerable number of tumors seem to be quiescent and do not become clinically significant within a patient's lifetime; such tumor may never warrant treatment.

The conservative management consists of observation of the tumor without any active treatment, delaying the treatment until the lesion exhibits a dangerous modification.

Vestibular schwannomas are being diagnosed earlier and smaller because of improvement of imaging techniques and aggressive audiological screening in case of patients with tinnitus, dizziness, facial nerve weakness and unilateral hearing loss.

The two face of the medal are that delaying the treatment until the patient has an increase in symptoms is not likely in the patient's best

interest since there is greater morbidity associated with treatment of larger tumors. On the other hand, is discussable to treat a patient with a potentially risky surgical or radiation procedure when the tumor would not have grown.

The studies done on natural history of vestibular schwannomas had several points to determine: identification of predictive factors of growth, growth rate throughout the time and the impact of observation treatment option to the health of the patients.

Conservative management of VS have been reported since 1985 (Wazen et al. 1985), these authors reported 21 elderly patients monitored with serial CT scans who undergone partial tumor removal or had not been operated on. From this beginning successive reports affirmed that the majority of VS grew during the follow-up period. Others Authors reported 50 patients who benefited from wait and see policy; 78% had growth rate of <2 mm/year (Nedzelski et al. 1992). In a big observational study of the Danish Neurotologic school emerged that VS exhibit growth only in the first 5 years after diagnosis (Stangerup et al. 2006).

The results of our unpublished study (because of doing a long-term observational study) supported the current thought that VS are slow-growing tumors and in our series the growth was present in 42, 18% of patients but the growth rate is no linear and there were not present predictive growth factors yet.

It should be noted that, although serial observation may represent the conservative management, it is not without risks. Deferring definitive treatment may lead to progression of hearing impairment or to evolution from an easily respectable lesion to one that carries increased surgical risks.

The growth of VS is not related to the final symptoms but the presence of the tumor represent for the patient in follow-up a severe risk of personal status worseness, in fact audiological deterioration can occur in the absence of evidence of tumor growth, such data leads to speculate that the sensorineural hearing loss associated with VS arises not only from the pressure effect on the internal auditory artery and/or to the cochlear

nerve (Walsh et al. 2000), but perhaps also because of the deposition of tumor proteins within the inner ear (Silverstein et al. 1985).

The tumor size at initial presentation seems to be predictive of tumor growth. However the diagnosis of VS does not occur at a uniform stage in their natural history, then the tumor size at diagnosis may reflect a biological proclivity for growth, such that the detection of a large tumor at initial presentation may be indicative of past growth; to consider it as a marker for future growth potential is only a supposition.

As showed by several works, the young age is not a positive predictor of growth (Fucci et al. 1999; Herwadker et al. 2005; Stangerup et al. 2006). More aggressive treatment in younger patients based on faster tumor growth is therefore not justified by scientific observations, although other reasonable rationales for early intervention include higher cumulative lifetime risk of tumor growth and complications as well.

Conservative management presents an important concern to find the appropriate and accurate means to evaluate the tumor growth. The two main parameters to conduct a correct follow-up of the patients are: measurement of the tumor size and timing of MRI scans.

A wrong method of size measurement leads the observer to misinterpretation of the tumor behaviour and a wrong treatment may be the consequence. A two-dimensional measurement on CT, and successively on MRI, has been the method most used (Wazen et al. 1985; Nedzelski et al. 1986). In 2003 the Tokyo Consensus Meeting on System for Reporting Results in Vestibular Schwannoma accepted the measuring system, described before, with the measurement of the largest extrameatal diameter of the tumor (Kanzaki et al. 2003). Actually is still the widest adopted system because of applicable in quite all institution with all type of MRI devices (Table 29.3).

Nowadays tumor size can be calculated by estimating the mean of the supero-inferior and medio-lateral tumor extent. More recently volume measurement of the tumor was proposed as an alternative to two-dimensional measurement. Interest in volume measurement comes up from

the variable rate between diameter and volume in non-spherical VS, which is the most frequent manifestation. A mathematical formula based on medio-lateral and antero-posterior measures for obtain the volume of tumor was proposed (Charabi et al. 1995). A computer program can calculate volume of tumor also during the MRI scans. However, the results of tumor volume determination, obtained by mathematical formulas or computer program, were not similar especially for large VS (Shin et al. 2000). The volumetric determination of relative growth rate is becoming widely accepted because 2 mm growth in a 4 mm tumor is radically different from 2 mm growth in a 30 mm tumor, considering the rate of cellular proliferation. We advocate very strongly to counsel the patients to do MRI scans always in the same institute, avoiding differences in parameters and measurement that can alter the growth evaluation.

Another problem of conservative strategy is the non-observance of this management by the patients or its recede during the follow-up. Although this non-observance has not been reported in English literature we think that it represent an important difficult in this strategy. Some reasons for this non-observance may be the following: absence of traditional protocol for VS may lead patients to obtain a second opinion in other centre that applies a different policy, or patients with few symptoms may tend to wait until their hearing worseness, or elderly population with other diseases.

Today our criteria for the follow-up program are: patients with intrameatal or small – medium sized tumor (0–20 mm extrameatal) having good hearing without disabling vertigo, who refuse conservative surgical approach, patients older than 65 years, elderly patients with poor performance status, patients with only hearing ear, and patients who refuse surgery. Also in case of intralabyrinthine schwannoma, we suggest a conservative strategy in case of patients with serviceable hearing, because of whatever surgical treatment will cause anacusis in case of complete removal (Di Lella et al. 2007).

The follow-up program is developed with a yearly MRI for 5 years, followed by MRI every

other year for 4 years, and a new MRI after 5 years. After such period the observation is terminated if the tumor don't show new growth or some disabling symptoms occur. This protocol can be applied generally on all VS, but some case may require a variation in MRI scheduling.

The controversy, however, continues to surround the relative merits or otherwise of conservative management with observation. The clarification of this situation is the accurate measurement of outcomes that, over the past two decades, were clinician-assessed measurement. Today the outcome measurement is shifted to a patient-assessed outcome. One of the major methods of assessing the global impact of any intervention on patients is measurement of the healthy related quality of life.

Surgical Treatments: Indications

The modern neurotology began in early '60 when William House starts the employing of the operating microscope in his surgical approach to the neurotologic diseases. An important concept that emerged by the work of House's equip in the successive years was the multidisciplinary approach, combining the expertise of neurotologists and neurosurgeons. Such cooperation leads to the definition of the three techniques adopted by all institution worldwide: translabyrinthine, middle cranial fossa and retrosigmoid approaches to the cerebello-pontine angle.

The detailed description of the procedures is remanded to other chapters of the present book; our aim will be to address on its use in the otoneurological practice, discussing the advantage and disadvantage of the various techniques.

The translabyrinthine approach (TLA) is the most commonly used in VS removal. It allows the exposure of the CPA anterior to the sigmoid sinus, including the lateral aspect of the pons, the ventral aspect of the lateral cerebellar hemisphere and V, VI, VII, VIII cranial nerves. Also the root entry zone of the lower cranial nerves is visible at variable angulations, depending on anatomy of the jugular bulb. The translabyrinthine approach is ideal for about all lesions of the CPA in patients

without serviceable hearing and has a somewhat lower morbidity if compared to other approach that require brain retraction.

The incidence of postoperative headache is reduced because of no cerebellar retraction, completion of bony work before dura opening, scarce dissection of suboccipital musculature. The CSF leak is a rare complication following this technique. The direct and anterior prospective of view during a translabyrinthine approach allows the identification of the facial nerve at the lateral end of the internal auditory canal before the tumor dissection, and may also allow the facial nerve repair in case of accidental damage by a graft or a single anastomosis after nerve mobilization.

The main disadvantage of the translabyrinthine approach is the need to sacrifice definitively the hearing in the operated side.

The indications for this procedure generally are: nonserviceable hearing for any size of tumor, tumors larger than 2.5 cm regardless the status of the hearing. Exceptions to this rule exist in presence of particular condition (i.e.: contralateral deafness).

The middle cranial fossa approach (MFA) was adopted for small acoustic tumors with good hearing. This technique allows unroofing the IAC and the exposure of its fundus. The facial nerve, located at the lateral end of the canal where it enters the temporal bone, can be visualized and dissected from the tumor. The major indication for the middle fossa approach is removal of small VS with serviceable hearing. The middle fossa approach has the main disadvantage that is a difficult approach. Because of the superior location of the facial nerve in the canal, the dissection of the tumor cause more manipulation of the nerve than in other procedures with following facial nerve palsy. The retraction of the temporal lobe of the brain may become a problem in case of prolonged duration of the operation, causing seizures, memory loss, auditory hallucinations, and speech disturbances. Although rare, a bleeding in the posterior fossa could be difficult controlled because of limited access. Older patients do not tolerate the middle fossa approach as well as young patients due to the delicateness of the dura and retraction of the temporal lobe. Although

rare, a postoperative epidural haematoma may occur and have serious consequences if not promptly recognized.

The retrosigmoid approach (RSA) was ideated as modification of the classical suboccipital approach. In the suboccipital procedure a large bone flap is removed from the suboccipital area, with the anterior limit of the dissection being the first mastoid cell. In the retrosigmoid approach, the anterior limit of the dissection is the sigmoid sinus. Superiorly, bone is removed up to the inferior margin of the transverse sinus. The retrosigmoid approach offers a more favourable angle of view into the CPA and markedly reduced need for cerebellar retraction than the classical suboccipital technique. The retrosigmoid approach give to the surgeon a panoramic view of the posterior fossa from the tentorium to the foramen magnum. Access is provided to the cerebellar hemisphere, the lateral aspect of the pons and medulla, and from the root entry zone to the cisternal course of cranial nerves V to XI. Indications include resection of VS, meningiomas, epidermoids, cranial nerves neurectomies, vascular decompression and parenchymal lesions of the brainstem and cerebellum.

Advantages are wide access to the CPA with potential for hearing preservation. The retrosigmoid approach is capable of addressing most lesions of the CPA (Atlas et al. 1992).

Disadvantages include a higher incidence of postoperative headaches and CSF leaks (Smith et al. 1990). The rationale for the higher incidence of headache is unknown, but it is thought to be due to deterioration of posterior fossa arachnoid by bone dust dropping during intradural drilling of the posterior wall of the IAC, creating a form of chronic aseptic meningitis that is responsive to steroids. Another explication for headache is the adhesion of the posterior fossa dura to the suboccipital muscles. The higher incidence of CSF leaks is explained by the difficulty in sealing all the cells in the petrous apex, especially if it is extensively pneumatized.

The aim of the surgical treatment is dual: remove a potential life-threatening disease, and preserve the hearing and facial nerve function as possible. The decision to adopt a surgical

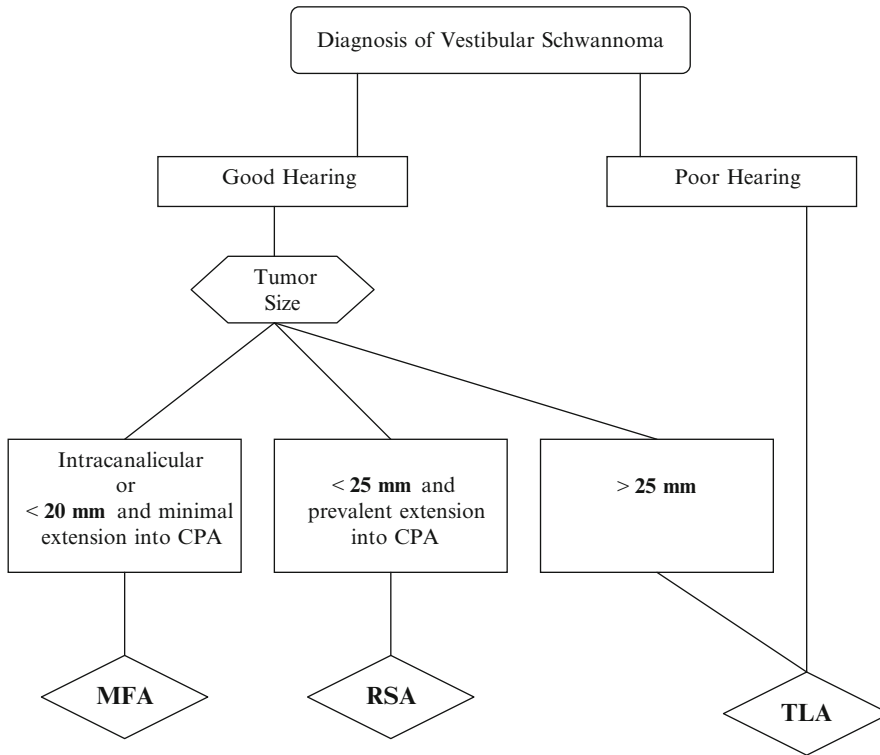


Fig. 29.5 Surgical management algorithm. Hearing status and tumor size are the two parameters that must be considered in choosing the proper surgical approach. Also the extension into CPA and the relationship between

tumor and fundus of internal auditory canal could be restriction factors (*CPA* Cerebello-Pontine Angle, *MFA* middle fossa approach, *RSA* retrosigmoid approach, *TLA* translabyrinthine approach)

approach depends upon the characteristics of the tumor, the hearing function, the performance status of the patients and the patient's needs (Fig. 29.5).

The concept of hearing preservation requires some explication, because often it is confused with the concept of useful hearing. Only the useful hearing is a measure of hearing preservation surgery, because it allows the patients to still benefit from hearing aids.

Several classification systems have been suggested for the description of the hearing loss. The aim of the classifications is always the standardization of communication between professionals, permitting also the comparison of treatment results.

As described before, the two most widely adopted systems are: that of the American Academy of Otolaryngology-Head and Neck

Surgery (AAO-HNS) (Committee on Hearing and Equilibrium 1995), and the Gardner-Robertson system (Gardner and Robertson 1988). Both are based on the pure-tone threshold average (PTA) of frequencies 0.5, 1, 2, 3 kHz, as well as the best speech discrimination score (SDS). The two systems are comparable for the first two classes (A and B for AAO-HNS and 1 and 2 for the Gardner-Robertson) and this is important because of many institutions consider patients for hearing preservation surgery only if their hearing falls within one of the first two classes. This behaviour is also known as "the 50/50 rule" (50 dB on PTA and 50% of SDS). A disadvantage of such systems is that the categories are too broad and a patient may drop in a worse class by losing as little as 5 dB on one or two frequencies; this may shift the surgical option from a hearing preservation surgery to an aggressive procedure.

Since 2003 a new classification systems are also employed for preoperative and postoperative evaluation of useful hearing, and thus an evaluation of treatment success (Kanzaki et al. 2003). It is mandatory to stress the concept that useful hearing have to allow binaural hearing, with possible localization of sound source and discrimination in noisy backgrounds. For this, the threshold difference between the two ears should be less than 15–20 dB, with a difference of less than 15% in SDS. Anyway, if the hearing obtained does not satisfy the above criteria, it could be suitable for amplification by hearing aids and thus contribute in binaural hearing, but only useful hearing is a measure of hearing preservation surgery. From there, the treatment should be personalized for each case by the judgment of the surgeon, based on experience and realistic chance of hearing preservation.

In the planning the proper treatment, the surgeons have to evaluate some factors: pre-operative hearing level, tumor size and location, and performance status of the patient.

A better pre-operative hearing level indicate minor compression or stretching of the VIII nerve, integrity of vascular supply to the nerve, and a hypothetic good result in hearing preservation with surgery. As showed in several studies patients with lower PTA and higher SDS scores have better possibility to preserve hearing function (Brackmann et al. 2000; Rohit et al. 2006). The audiological assessment with ABR may give further information about the chance to preserve hearing. The absence of a response is correlated with low possibility of hearing preservation (Slattery et al. 1997). The probability to preserve hearing is inversely proportional to the latency of waves I–V (Shelton et al. 1989).

The prognostic significance of tumor size was comprehensively investigated. As reported frequently, there is a significant correlation between tumor size and preserved hearing. However other institutions reported that the tumor size have not predictive significance (Brackmann et al. 2000). Effectively, the tumor size evaluation as predictive factor for hearing preservation has two main confounding factors: the method of measurement and the population selection. The selection of

patients is an important bias, because the subject candidates to the hearing preservation function are already a selected subpopulation of vestibular schwannoma patients. The measurement method is still an argument of controversy, especially about the inclusion or not of the intracanalicular portion in the final size of the tumor. Generally, the size considered as limit for hearing preservation surgery is 25 mm, but in our experience a tumor more than 15 mm of size have only 10% of chance to preserve a serviceable hearing.

The intracanalicular portion of the tumor plays a role in the decision making of the surgical approach. The lateral extension into the fundus of IAC is considered a prognostic factor in terms of hearing preservation. This comes up from the anatomy of the VIII nerve, which is separate in several bundles to enter the cribriform plate of the cochlea. Tumors that completely fill the IAC have a lower rate of hearing preservation than in cases in which the fundus of the canal appears uninvolved, 1–2 mm of space between tumor and fundus has been correlated with higher levels of hearing preservation (Colletti and Fiorino 2005).

In patients with unserviceable hearing the translabyrinthine approach is the standard technique adopted regardless the tumor size, because this approach give to the surgeon wide field for operation and low risks. The translabyrinthine approach is mandatory also for tumor larger than 25 mm not considering the hearing status, because of the hearing preservation is not reasonably practicable in big sized tumor. We suggest the TLA also in those patients with tumor size more than 15 mm that choose undergo to surgical removal.

The middle fossa approach is done generally in about one third of patients. This approach is used on patients with good hearing with intracanalicular tumors, or tumor with minimal extension into CPA, generally the size should be less than 20 mm.

The retrosigmoid is done in patients with good hearing and medial tumors with prevalent extension toward CPA and minimal into IAC.

The results of the surgery are quite homogeneous, with an average of hearing preservation in about 50% of patients treated with MCF or RS;

correlated with the number of treated cases, every institute may have a better results in one of the two techniques.

Cystic VS should have a different decisional approach, because of the unpredictable growth patterns and reported cases of cyst enlargement after radiosurgery, complete or subtotal microsurgical resection is often recommended. However, literature suggests that facial nerve outcomes after cystic VS resection have been less favorable in comparison to results after removal of solid VS (Piccirillo et al. 2009). Three factors need to be considered before cystic VS resection: thickness of the cyst wall, position of the cyst, and extrameatal size of the lesion, including cystic and solid components. Vestibular schwannomas with centrally located cysts and thick cyst walls or tumors with intratumoral cystic change (both Type A) tend to be less difficult to remove. Thin-walled, Type B tumors can present a more formidable challenge. Subtotal resection is frequently warranted when tumors of this nature are encountered.

It is general idea that completely excised VS do not recur and it is reasonable to consider less than complete removal despite the attendant increased risk of regrowth. The subtotal removal may be planned preoperatively under certain conditions. A planned incomplete resection may be undertaken when a more prolonged operation is judged unwise because of the patient's advanced age or performance status. Incomplete removal may be decided intraoperatively when dissection planes are poor and the surgeon think that total removal carries a high risk of damage of the facial nerve. In selected cases subtotal removal increases the odds of hearing conservation, in particular in case of contralateral ear disease. Considering the incomplete resection, it is important to distinguish whether removal has been "subtotal" or "near total" because it has substantial impact on the risk of recurrence. In subtotal excision a substantial part of tumor remains, in near total removal the residual consist of a thin capsular peel. As a general rule, the total removal of VS should be achieved whenever possible, the near total excision is most often elected when continued dissection threatens disruption of the facial nerve.

Management of Coexistent Otologic Diseases

Fortunately the existence of a Vestibular Schwannoma (VS) with a contralateral otological disease is a very rare event. However, nowadays the early diagnosis of VS caused by the advent of improved radiological imaging modalities, such as MRI, and the increased attention given to the diseases of the ear can led to the diagnosis of VS synchrony with other otological pathologies.

When this situation is present the otologist is exposed to an important management dilemma and anyone in English literature have decided a standard protocol for these patients.

The question is: which disease should be treated firstly?

Certainly a correct choice depends upon the associated disease and establishing a right protocol can be difficult. Focalizing our attention to the tumor we have three possibilities.

In the VS management the primary goal of the treatment is to ensure that there are no significant risks imposed by the VS, and then to conserve hearing and optimize the quality of life of the patient (Dispenza et al. 2008).

Non-operative management of selected patients with VS is not a new or unusual (Gardner et al. 1986). However the watchful and waiting strategy may be a correct management but, as stated before, it is not without disadvantages. The tumor may grow with consequent hearing dysfunction that may require its removal. In this case the patients loses his chance for hearing preservation surgery associated with removal of smaller VS (Naguib et al. 2005).

The risk of hearing loss is also present in the stereotactic radiosurgery as well as facial and trigeminal nerve deficit. It may block the tumor growth in the same cases but in others the tumor may continue to grow. Moreover performing the surgery after radiotherapy treatment is more demanding due the fibrosis and scar tissue.

Regarding the surgery it is obvious that the total or the partial removal of the tumor does not warrant hearing conservation or a perfect

quality of life. For these reasons no treat the tumor could be a risky choice.

All this thought become more significant if either the contralateral ear is affected by progressive hearing loss or quality of life becomes poor caused by the diseases. Then a normal protocol for the VS management could be not adequate.

In our experience we were faced with patients affected by VS and Meniere's disease and VS and contralateral otosclerosis. These pathologies are more frequent in the people and when they summarize with a VS presence can be cause of a big decisional dilemma.

Coincidence of Meniere's disease and contralateral VS represent an unusual presentation and consequently a difficult therapeutic decision-making. The main goal of treatment should be the hearing preservation and the restoration of patient normal lifestyle. It must be evaluated the true risk related to VS compared with the chances of perioperative and postoperative morbidity.

In these patients there are several problems to solve: elimination of disabling recurrent vertigo spells, management of VS and hearing preservation. In our practice we experienced five patients affected by this very rare condition.

A Meniere's disease refractory to conventional medical treatment leads to use ablative treatments: aminoglycosides (intratympanic or systemic) or surgical procedure (endolymphatic sac decompression, vestibular neurectomy, labyrinthectomy). The use of intratympanic gentamicin (IG), known since 1977 (Lange 1977), has been recommended in an attempt to minimize complications and risks involved in other procedures.

This therapeutic option was adopted by several institutions to minimize surgical risks because elderly patients and for those that refuse surgery. The total dosage of IG administrated in our patients complaint refractory MD is 81 mg (De Stefano et al. 2007).

Fluctuating hearing loss of MD usually leads to unserviceable ear in the affected side. On the other hand VS generally have a serviceable hearing or more frequently with some degree of hearing loss. This audiological deterioration is generally related to tumoral growth and seems to be caused by pressure effect on the internal

auditory artery and the cochlear nerve, but there is a significant risk of loss of serviceable hearing also in non-growing tumors. Patients with MD, even presence of moderate to profound hearing loss, and contralateral VS with serviceable hearing could be considerate having an only hearing ear (OHE) in the side of tumor (Thedinger et al. 1993).

Recent advances in surgical and diagnostic techniques for VS have markedly reduced the morbidity and mortality of patients with this type of tumor and surgical management is recommended for the majority of patients. Complications are encountered even in the hands of the most experienced surgeons. In cases of tumors smaller than 20 mm with serviceable hearing (first two class of Kanzaki Classification) the hearing preservation is achieved only in 10–30% of cases after surgical removal, this rate fell significantly in larger VS (Naguib et al. 2005). Stereotactic radiotherapy is a nonsurgical alternative in the management of VS and several studies report adequate local tumor control, but the incidence of cranial nerve injury and the risk of late complications, especially with a single large dose, remains a drawback. Hearing preservation after radiosurgical treatment was reported by several studies to be achieved in 50–70% in patients having useful hearing and unilateral tumor. This rate is not acceptable if patient present a considered OHE patient.

Based on rate of hearing preservation of non-conservative treatments as stated before and considering the small rate of tumor growth, observation result the first and the most prudent option for patients with occurrence of tumor in an OHE. Regular follow-up is made with MRI, which improved the evaluation of tumoral growth, and audiometric tests to survey the hearing function in the OHE.

During close follow-up it is possible to convert the conservative management of VS to surgical removal if there are signs of tumor growth endangering the patient's life and also if hearing in OHE deteriorate to unserviceable levels.

Because MD manifests as a peripheral auditory and vestibular disorder, those patients who develop severe to profound sensorineural hearing

loss or deafness would be excellent cochlear implantation candidates (Lustig et al. 2003). Before attempt of surgical removal of the tumor in these patients we would have strongly recommended the consideration of cochlear implantation in the deaf ear affected by MD.

Our experience showed that in these rare patients affect by MD and contralateral VS, the correct management is first the treatment of MD and second a serious follow-up of the tumor. Other treatment are very dangerous for the quality of life of the patient because hearing preservation in these patients can be very difficult also using a hearing conservative surgical techniques such as RS approach with simultaneous ABI positioning. For this ABI positioning should be considered a second line treatment.

The simultaneous presence of otosclerosis, a benign but disabling process with a VS a potentially life threatening pathology is very rare.

The first description of this association was in 1973 with a detailed account of eight patients (Clemis 1973). Except one patient described by Clemis, in which labyrinthine otosclerosis and a VS were simultaneously diagnosed, the diagnosis of otosclerosis has always antedated that of tumor. The key to tumor diagnosis in all cases presented in literature has been either the asymmetric progression of a sensorineural hearing loss (SNHL) or a sudden unilateral hearing impairment, and the differential diagnosis when such event occur should include: progressive labyrinthine otosclerosis, perilymph fistula, endolymphatic hydrops, labyrinthine vascular occlusion and cerebello-pontine angle tumor. The presence of disease causing hearing function deterioration, such as otosclerosis, may complicate the future quality of life of this patient.

We think that in these patients the future quality of life related to a sufficient hearing function is very important. For this we thought that should be convenient to treat the otosclerosis firstly, because of the surgical hearing risks is only 0, 18%, while the VS surgery presents a risk of hearing loss from 50% to 100% (Naguib et al. 2005).

Treating first the otosclerosis process allows us to warrant a good hearing in one ear at least.

Generally the SDS improves after stapedotomy because of thresholds are no longer at or beyond audiometric limits. Hearing aid use became easier because patients can comfortably obtain lower aided thresholds using less gain.

When otosclerosis and VS are diagnosed in the same patient the choice of the correct management is difficult and depends on some tumor factors, like tumor size, vertigo, and hearing loss. These patients have two pathologies, which lead to progressive hearing loss; consequently is very important decide a management. In our opinion we counsel to treat only the otosclerosis process by surgery and manage with MRI and follow-up the VS.

This procedure allows recovering one hearing function and in case of failure of stapes surgery is always possible to perform a cochlear implant in otosclerotic ear (or in VS ear if the cochlea is ossified).

The coexistence of two pathologies in the same patients such as VS and MD or otosclerosis is very rare, but it represents a big problem for a correct management.

The patient have to properly counselled regarding variability in growth rates of VS and consequent variations of management and outcome, especially for hearing. No absolute management algorithm exists and treatment must be individualized. The main difficult in decision-making is to consider each case separately for the best therapeutic choice, keeping in mind the quality of life of the patient. Our algorithm was: treatment of disabling vertigo of MD or Hearing loss of otosclerosis and conservative management of asymptomatic VS. We recommend cochlear implantation in the deaf ear before attempt surgical removal of VS

Vestibular schwannoma in an OHE is certainly a challenge. Because of no standard solution is available, it requires an open discussion with the patients and their relatives to choose a presumably optimal option in each case. Efforts should be made to increase the possible alternatives from which patients can select. It has been stated that the best option for hearing in such patients is to keep their own hearing and follow them with a wait-and-see policy (Naguib et al. 2005). However

the hearing preservation policy is a good options in these patients but its depends by size of the tumor, growth of it and location of the tumor (deep or lateral respect to the IAC) and rapid loss of hearing function are conditions that critically reduce chances for hearing preservation. If the tumor grow the otoneurosurgeon have some options to counsel to the patient: to remove the tumor, attempting hearing preservation; to refer the patient for radiosurgery; to remove the tumor and, if hearing is lost, to attempt cochlear implantation (provided the cochlear nerve has been spared); if possible an attempt to implant the other side should be done, and then to decide whether to remove or to follow-up the tumor; or to remove the tumor and, if the cochlear nerve is lost, to try an Auditory Brainstem Implant (ABI).

Cochlear implantation is the best method for hearing rehabilitation in patients with VS in the only hearing ear that loses hearing (Thedinger et al. 1993), certainly better than an ABI (Brackmann et al. 1993; Lenarz et al. 2001).

Cochlear implantation is usually a relatively easy procedure to perform, with no problem in identifying the correct placement of the prosthesis and negligible risk to the patient. The ABI, however, even in the most experienced hands, is not always easy and indeed may be impossible to insert. The operation does carry risk, albeit slight, to the stem and to the lower cranial nerves (Ramsden et al. 2005).

Cochlear implantation could be tried if hearing greatly deteriorates or is lost as a consequence of tumor growth or tumor removal. In the latter case, the cochlear nerve should be spared as well as the cochlea; also the vascular supply should be preserved to some degree, because otherwise the cochlea could be affected with time by fibrosis or ossification (Belal 2001). Cochlear implantation has been successful after VS removal through a retrosigmoid or a middle fossa approach. Cochlear implantation could be tried in the contralateral side, but it is advisable to remove the tumor afterwards to avoid the problem of follow up with a cochlear implant in place. The same problem is present when there are residual lesions.

Cochlear implantation after VS removal is not always possible; the cochlear nerve must be preserved intact and the cochlea should remain responsive to electrical stimulation. After removal of VS through a hearing preservation approach (i.e., retrosigmoid or middle cranial fossa) vascular supply to the cochlea may be compromised and the cochlea may ossify in the short-or mid-term (Belal 2001). If a CI is being considered, it should be performed near the time of the surgery for VS; promontory tests may help in confirming the utility of the cochlea but are not 100% reliable (Lambert et al. 1992). Basic studies have demonstrated that there could be as many spiral ganglion cells years after labyrinthectomy as in a successfully implanted patient; moreover, it has been proved that with a low population of spiral ganglion cells we can have a successful CI user (Chen et al. 1988). This supports the fact that we can use a CI after eliminating the posterior labyrinth, and in fact, placement of a CI after labyrinthectomy has proved to work properly (Facer et al. 2000).

The translabyrinthine approach for removal of VS is a highly destructive method and cochlear fibrosis or ossification occurs in a short time. If a CI is planned after such an approach, this should be done as close to the procedure as possible and optimally during the same operation. In view of histopathologic studies, it seemed reasonable that cochlear implantation could work after a translabyrinthine approach, provided the cochlea and the cochlear nerve could be functionally preserved.

When the CI is not actable in the VS ear or in the contralateral ear the otoneurosurgeon can use an ABI implant.

The outcomes after ABI are, however, variable, and the factors that contribute to the variability are as yet not fully understood. Accurate placement of the prosthesis is essential, and this is usually possible if there has not been excessive distortion of the brain stem from a very large tumor. It is also likely that the same pressure on the stem might cause cellular depletion of the cell in the cochlear nucleus (Ramsden et al. 2005).

Several implant teams, agree that ABI confers an awareness of environmental sounds and is an

adjuvant to lip-reading (Colletti et al. 2000; Lenarz et al. 2001). Nevertheless, there are reports of very good outcomes in a small proportion of patients, and the fact that this is possible at all means that there must be factors as yet largely unidentified that predict a good result. But, nowadays, the ABI is a valuable tool for cases in which the cochlea or the cochlear nerve could not be spared, for the moment, the results with ABI still are far from those obtained with CI in VS patient and OHE.

In fact before to try an ABI implant in an OHE patient we must keep in mind that tumor volume plays an important role in the alteration of the anatomic landmarks and the functional deterioration of the auditory pathways (Lenarz et al. 2001).

References

- Atlas MD, Harvey C, Fagan PA (1992) Hearing preservation in acoustic neuroma surgery: a continuing study. *Laryngoscope* 102:779–783
- Belal A (2001) Is cochlear implantation possible after acoustic tumor removal? *Otol Neurotol* 22(4):497–500
- Brackmann DE, Hitselberger WE, Nelson RA, Moore J, Waring MD, Portillo F, Shannon RV, Telischi FF (1993) Auditory brainstem implant: I. Issues in surgical implantation. *Otolaryngol Head Neck Surg* 108(6):624–633
- Brackmann DE, Owens RM, Friedman RA, Hitselberger WE, De la Cruz A, House JW, Nelson RA, Luxford WM, Slaterry WH III, Fayad JN (2000) Prognostic factors for hearing preservation in vestibular schwannoma surgery. *Am J Otol* 21:417–424
- Charabi S, Thomsen J, Mantoni M, Charabi B, Jorgensen B, Borgesen SE, Gyldensted C, Tos M (1995) Acoustic neuroma (vestibular schwannoma): growth and surgical and nonsurgical consequences of the wait-and-see policy. *Otolaryngol Head Neck Surg* 113(1):5–14
- Charabi S, Tos M, Thomsen J, Charabi B, Mantoni M (2000) Vestibular schwannoma growth: the continuing controversy. *Laryngoscope* 110(10 Pt 1):1720–1725
- Chen DA, Linthicum FH Jr, Rizer FM (1988) Cochlear histopathology in the labyrinthectomized ear: implications for cochlear implantation. *Laryngoscope* 98(11):1170–1172
- Clemis JD (1973) The coexistence of acoustic neuroma and otosclerosis. *Laryngoscope* 83:1953–1985
- Colletti V, Fiorino F (2005) Is the middle fossa approach the treatment of choice for intracanalicular vestibular schwannoma? *Otolaryngol Head Neck Surg* 132:459–466
- Colletti V, Fiorino FG, Carner M, Giabini N, Sacchetto L, Cumer G (2000) The retrosigmoid approach for auditory brainstem implantation. *Am J Otol* 21:826–836
- Committee on Hearing and Equilibrium (1995) Guidelines for the evaluation of hearing preservation in acoustic neuroma (vestibular schwannoma). American Academy of Otolaryngology-Head Neck Surgery Foundation, INC. *Otolaryngol Head Neck Surg* 113:179–180
- De Stefano A, Dispenza F, De Donato G, Caruso A, Taibah A, Sanna M (2007) Intratympanic gentamicin: a 1-day protocol treatment for unilateral Meniere's disease. *Am J Otolaryngol* 28(5):289–293
- De Stefano A, Dispenza F, Di Trapani G, Kulamarva G (2008) Meningioma of the cerebello-pontine angle mimicking benign paroxysmal positional vertigo. *J Otolaryngol* 37:46–48
- Di Lella F, Dispenza F, De Stefano A, Falcioni M, Sanna M (2007) Management of intralabyrinthine schwannomas. *Auris Nasus Larynx* 34(4):459–463
- Dispenza F, De Stefano A (2008) Imaging of vestibular schwannoma with prevalent cystic component: cystic vestibular schwannoma. *Otol Neurotol* 30:681–682
- Dispenza F, De Stefano A, Flanagan S, Romano G, Sanna M (2008) Decision making for solitary vestibular schwannoma and contralateral Meniere's disease. *Audiol Neurotol* 13(1):53–57
- Facer GW, Facer ML, Fowler CM, Brey RH, Peterson AM (2000) Cochlear implantation after labyrinthectomy. *Am J Otol* 21(3):336–340
- Fucci MJ, Buchman CA, Brackmann DE, Berliner KI (1999) Acoustic tumor growth: implications for treatment choices. *Am J Otol* 20:495–499
- Gardner G, Robertson JH (1988) Hearing preservation in unilateral acoustic neuroma surgery. *Ann Otol Rhinol Laryngol* 97:55–66
- Gardner G, Moretz WH Jr, Robertson JH, Clark C, Shea JJ Jr (1986) Nonsurgical management of small and intracanalicular acoustic tumors. *Otolaryngol Head Neck Surg* 94(3):328–333
- Herwadker A, Vokurka EA, Evans DG, Ramsden R, Jackson A (2005) Size and growth rate of sporadic vestibular schwannoma: predictive value of information available at presentation. *Otol Neurotol* 26:86–92
- House JW, Brackmann DE (1985) Facial nerve grading system. *Otolaryngol Head Neck Surg* 93:146–147
- Kanzaki J, Tos M, Sanna M, Moffat DA, Monsell EM, Berliner KI (2003) New and modified reporting systems from the consensus meeting on systems for reporting results in vestibular schwannoma. *Otol Neurotol* 24(4):642–648; discussion 648–9
- Lambert PR, Ruth RA, Thomas JF (1992) Promontory electrical stimulation in postoperative acoustic tumor patients. *Laryngoscope* 102(7):814–819
- Lange G (1977) Die intratympanale Behandlung des Morbus Meniere mit ototoxischen antibiotika. *Z Laryngol* 56:409–414

- Lenarz T, Moshrefi M, Matthies C, Frohne C, Lesinski-Schiedat A, Illg A, Rost U, Battmer RD, Samii M (2001) Auditory brainstem implant: part I. Auditory performance and its evolution overtime. *Otol Neurotol* 22(6):823–833
- Lustig LR, Yeagle J, Niparko JK, Minor LB (2003) Cochlear implantation in patients with bilateral Meniere's syndrome. *Otol Neurotol* 24:397–403
- Naguib MB, Cokkeser Y, Sanna M (2005) Management of acoustic neuroma in the only hearing ear. *Eur Arch Otorhinolaryngol* 262:127–130
- Nedzelski JM, Canter RJ, Kassel EE, Rowed DW, Tator CH (1986) Is no treatment good treatment in the management of acoustic neuromas in the elderly? *Laryngoscope* 96(8):825–829
- Nedzelski JM, Schessel DA, Pfeleiderer A, Kassel EE, Rowed DW (1992) Conservative management of acoustic neuromas. *Otolaryngol Clin North Am* 25(3):691–705
- Piccirillo E, Wiet MR, Flanagan S, Dispenza F, Giannuzzi A, Mancini F, Sanna M (2009) Cystic vestibular schwannoma: classification, management and facial nerve outcomes. *Otol Neurotol* 30:826–834
- Ramsden R, Khwaja S, Green K, O'Driscoll M, Mawman D (2005) Vestibular schwannoma in the only hearing ear: cochlear implant or auditory brainstem implant? *Otol Neurotol* 26:261–264
- Rohit, Piccirillo E, Jain Y, Augurio A, Sanna M (2006) Preoperative predictive factors for hearing preservation in vestibular schwannoma surgery. *Ann Otol Rhinol Laryngol* 115(1):41–46
- Shelton C, Brackmann D, House WF, Hitselberger WE (1989) Middle fossa acoustic tumor surgery: results in 106 cases. *Laryngoscope* 99:405–408
- Shin YJ, Frayssse B, Cognard C, Gafsi I, Charlet JP, Berges C, Deguine O, Tremoulet M (2000) Effectiveness of conservative management of acoustic neuromas. *Am J Otol* 21(6):857–862
- Silverstein H, McDaniel A, Norrell H, Wazen J (1985) Conservative management of acoustic neuroma in the elderly patient. *Laryngoscope* 95(7 Pt 1):766–770
- Slattery WH, Brackmann D, Hitselberger WE (1997) Middle fossa approach for hearing preservation with acoustic neuroma. *Am J Otol* 18:596–601
- Smith PG, Leonetti JP, Grubb RL (1990) Management of cerebrospinal fluid otorrhorrhea complicating the retrosigmoid approach to the cerebellopontine angle. *Am J Otol* 11:178–180
- Stangerup SE, Caye-Thomasen P, Tos M, Thomsen J (2006) The natural history of vestibular schwannoma. *Otol Neurotol* 27:547–552
- Thedinger BA, Cueva RA, Glasscock ME 3rd (1993) Treatment of an acoustic neuroma in an only-hearing ear: case reports and considerations for the future. *Laryngoscope* 103:976–980
- Walsh RM, Bath AP, Bance ML, Keller A, Rutka JA (2000) Consequences to hearing during the conservative management of vestibular schwannomas. *Laryngoscope* 110(2 Pt 1):250–255
- Wazen J, Silverstein H, Norrell H, Besse B (1985) Preoperative and postoperative growth rates in acoustic neuromas documented with CT scanning. *Otolaryngol Head Neck Surg* 93(2):151–155

Stereotactic Radiosurgery for Trigeminal Schwannoma: Tumor Control and Functional Preservation

30

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Abstract

We retrospectively reviewed records of 33 consecutive trigeminal schwannoma patients who underwent Gamma Knife® SRS. Two patients had neurofibromatosis type 2. The median patient age was 49 years (range, 15–82 years). Eleven patients had prior surgical resection of their tumors. The tumors were classified into root type (N=6), ganglion type (N=17) and dumbbell type (N=10) based on their location. The median tumor volume was 4.2 cc (range, 0.5–18.0 cc) and the median marginal dose was 15.0 Gy (range, 12–20 Gy). At an average of 6 years (range, 7.2–147.9 months), the progression free survival after SRS at 1-, 5- and 10 years, respectively was 97%, 82% and 82%. Factors associated with an improved progression free survival included female gender, smaller tumor volume, and root or ganglion type tumors. Neurological symptoms or signs improved in 11 of 33 patients (33%) and unchanged in 19 patients (58%). Three patients (9%) had symptomatic progression. Patients without prior resection were significantly more likely to have improvement in neurological symptoms or signs. Predictors of better response included female gender, smaller volume, root and ganglion type tumors, and use of SRS as the primary management.

Introduction

Trigeminal schwannomas are slow-growing, benign nerve sheath tumors that are uncommon compared to the incidence of vestibular schwannomas (de Benedittis et al. 1977; Goel et al. 2003; McCormick et al. 1988; Pollack et al. 1989; Yasui et al. 1989). They accounts for 0.2–1% of all intracranial tumors and 0.8–8% of all intracranial schwannomas (de Benedittis et al. 1977; Goel et al. 2003; Konovalov et al. 1996; McCormick et al. 1988;

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Pollack et al. 1989; Samii et al. 1995b; Taha et al. 1995; Yasui et al. 1989).

Trigeminal schwannomas are typically managed with surgical resection. Despite continued improvements in cranial base surgical techniques including endonasal surgery, tumor adherence to adjacent critical neurovascular structures makes complete removal difficult (Day and Fukushima 1998; Sarma et al. 2002; Taha et al. 1995; Yoshida and Kawase 1999). When such removal is attempted, new postoperative neurological deficits often develop in 13–86% of patients, as well as persistent high rates of cerebrospinal fluid leakage (Al-Mefty et al. 2002; Eisenberg et al. 1999; Goel et al. 2003; McCormick et al. 1988; Samii et al. 1995a, b). There is a growing literature on the value of stereotactic radiosurgery (SRS) for patients with trigeminal schwannomas (Andrews et al. 2001; Flickinger et al. 1996, 2001; Niranjana et al. 1999; Regis et al. 2002; Suh et al. 2000). Recently we retrospectively assessed tumor control, clinical response, the risk of adverse radiation effects (ARE), and variables that affected treatment outcomes (Kano et al. 2009).

Clinical Experience

Over a 16 year interval, 33 patients with trigeminal schwannomas underwent SRS using the Gamma Knife® (Elekta Instruments, Norcross, GA) at the University of Pittsburgh. There were 17 males and 16 females with a median age of 49.5 years (range, 15–82 years). Eleven patients (33%) had prior surgical resection and one patient was diagnosed by stereotactic biopsy. Twenty-two patients (67%) underwent SRS as primary management. In these patients the diagnosis was based on a combination of clinical symptoms or signs and confirmatory neuroimaging findings. All tumors extended along the course of the trigeminal nerve, had diffuse contrast enhancement identified by a magnetic resonance imaging (MRI), and had no dural tail. Six patients underwent SRS at the time of tumor recurrence identified by imaging. Tumor progression after initial management was defined as an increase in tumor volume defined by MRI.

The median duration between last surgical removal and progression was 22.1 months (range 9.4–76.5 months).

The median tumor volume was 5.0 cc (0.5–18.0 cc). A median of 6 isocenters (1–13) were used for dose planning. The median prescription dose delivered to the margins of the tumor was 15 Gy (12–22 Gy). The prescription isodose was 50% in all cases. Maximum dose varied from 24 to 40 Gy (median, 30 Gy). All patients had a minimum of 6 months follow-up (range, 7.2–147.9 months). Twenty-four patients had follow-up of 24 months or more and a mean follow-up time of 7 years.

Radiosurgery Technique

Gamma Knife radiosurgery was performed using a Model U, B, C, or 4-C Leksell Gamma Knife® (Elekta Inc, Atlanta GA). The procedure began with application of a model G Leksell stereotactic frame under conscious sedation and local scalp anesthesia except in younger children for whom general anesthesia was used. After attaching a fiducial system to the frame, all patients underwent high resolution MRI. A 3-D localizer sequence which included axial, coronal and sagittal images was performed first. The tumor was then imaged using a 3-D spoiled grass gradient recalled (SPGR) sequence images after intravenous contrast enhancement. T2 weighted MRI using Fast Spin Echo images also were acquired to assess the tumor border. The images were exported to a computer workstation via the hospital Ethernet for dose planning, using Leksell Gamma Plan® software. In all tumors the radiosurgery treatment volume conformed to the enhancing tumor volume. After SRS, the patients were instructed to have clinical and imaging assessments at 6 months intervals during the first year, then yearly for 2 years. If tumor growth was halted, additional imaging evaluations were requested at 4, 6, 8, and 12 years. If a new neurological symptom or sign developed, the patient was evaluated for tumor progression or any adverse radiation effect (ARE) and a new MRI was obtained.

Table 30.1 Effects on progression free survival in univariate analysis (N=33)

Characteristics	p value
Age (<50 vs. ≥50)	0.541
Sex (female vs. male)	0.035*
Neurofibromatosis type 2 (present vs. absent)	0.396
Tumor type (cystic vs. solid)	0.539
Tumor type (root and ganglion vs. dumbbell tumor)	0.031*
Target volume (<8.0 cc vs. ≥8.0 cc)	0.0007*
Margin dose (≥16 Gy vs. <16 Gy)	0.124
Prior surgical resection (yes vs. no)	0.702

*statistical significance

Tumor Control

Follow-up imaging studies demonstrated tumor control in 29 (88%) of 33 tumors after SRS, respectively. Six of 33 patients (18%) had root tumors, 17 patients (52%) had ganglion tumors and ten patients (30%) had dumbbell tumors. None of the six patients with root tumors showed progression after radiosurgery. One of 17 patients (6%) with ganglion tumors exhibited progression within the SRS volume. Three of ten patients (30%) with dumbbell tumors exhibited progression within the SRS volume. The root and ganglion tumors had a 1-, 5- and 10-year progression free survival of 100%, 94% and 94%, respectively. Dumbbell tumors had 1-, 5- and 10-year progression free survival of 90%, 56% and 56%, respectively. The dumbbell tumors were associated with lower rates of control ($p=0.031$). A target volume of <8.0 cc was significantly associated with better progression free survival ($p=0.0007$). Other variables including patient age, neurofibromatosis type 2, cystic tumor, margin dose, and prior surgical resection were not associated progression free survival (Table 30.1). The median tumor volume of dumbbell type tumors was 8.7 cc and of the root and ganglion type tumors was 2.9 cc ($p=0.015$). This may indicate that dumbbell type tumors had worse PFS, due to larger volume than other type tumors. One of ten dumbbell type tumors (10%) had prior surgical removal, while two of six root type tumors (33%) and eight of 17 ganglion type tumors (47%) had prior surgical removal. We suspect that root and

ganglion type tumors are recognized at an earlier stage, which facilitates earlier and more successful intervention with SRS.

Functional Preservation

In our series, all patients had neurological symptoms related to their trigeminal schwannoma. After SRS, 11 of 33 patients (33%) had clinical improvement in symptoms or signs. Symptoms improved at a median of 9.5 months (range, 2.7–88.5 months) after SRS. One of 11 patients (9%) who had prior surgical removal exhibited improvement in neurological symptoms and signs, while 10 of 22 patients (46%) who had no prior surgical removal exhibited improvement in neurological symptoms and signs. Patients who had no prior surgical removal experienced significantly better improvement in neurological symptoms and signs ($p=0.037$). Eight of 11 improved patients (73%) had reduction in their tumor volume. We noted improvement in oculomotor nerve function in one of two patients and in trochlear nerve function in one of two patients. Seven (30%) of 23 patients with trigeminal neuropathy improved, as did two (25%) of eight patients with facial pain. Five of nine patients with abducens nerve palsies improved. Hearing did not improve in three patients with preoperative hearing loss. Nineteen patients (58%) had no change in their pre-SRS neurological deficits. Three patients (9%) had progression of their neurological deficits. Two patients developed new trigeminal deficits (facial sensory loss and pain) and both had tumor progression (Table 30.2). Ten of 22 patients (46%) who underwent primary SRS (no prior surgery) exhibited improvement of neurological symptoms. One of 11 patients (9%) who had prior surgical removal had improvement in neurological symptoms.

Radiosurgical Morbidity

Two patients (6.1%) developed suspected adverse radiation effects (ARE). Two root type trigeminal

Table 30.2 Cranial nerve clinical response (N=33)

Before SRS		After SRS			
Resection (N = 11)/no resection (N = 22)		Improved	Unchanged	Worse	New symptom
Cranial nerve	Resection/ no resection	Resection/ no resection	Resection/ no resection	Resection/ no resection	Resection/no resection
Oculomotor	1/1	1/0	0/1	0/0	0/0
Trochlear	0/2	0/1	0/1	0/0	0/0
Trigeminal: neuropathy	7/16	0/7	6/7	1/2	0/1
: pain	2/6	0/2	2/3	0/1	0/1
Abducence	2/7	1/4	1/1	0/2	0/0
Facial	0/1	0/0	0/1	0/0	0/0
Acoustic	1/2	0/0	1/2	0/0	0/0

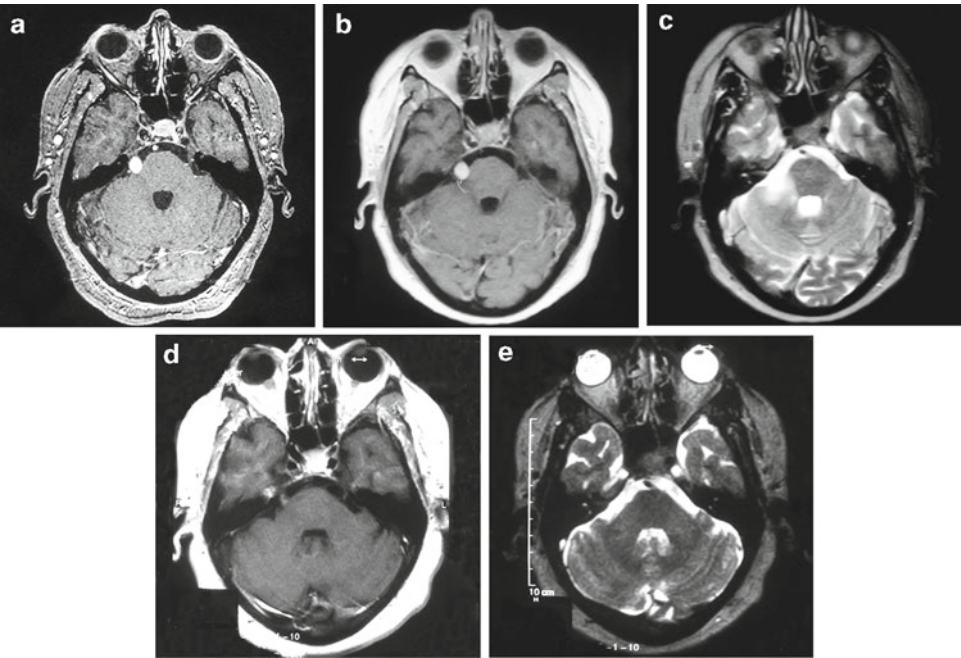


Fig. 30.1 T1-weighted contrast-enhanced axial MR images of a 57-year-old female show a root type trigeminal schwannoma in the right cerebello-pontine angle at the time of SRS (a). Axial T1-weighted contrast-enhanced MR images obtained 2 years after SRS show no residual enhancing tumor, while Axial T2-weighted MR images obtained 2 years after SRS show increased peritumoral T2

signal changes (b, c). This patient was successfully managed with oral corticosteroids. Axial T1-weighted contrast-enhanced MR images obtained 8 years after SRS show regression of the enhancing tumor (d). Axial T2-weighted MR images obtained 8 years after SRS show complete resolution of peritumoral T2 signal changes (e)

schwannoma patients (who received respectively 13 and 15 Gy at the tumor margin) showed increased peritumoral T2 signal changes. Neither

patient had new symptoms and both patients were successfully managed with oral corticosteroids (Figs. 30.1, 30.2, and 30.3).

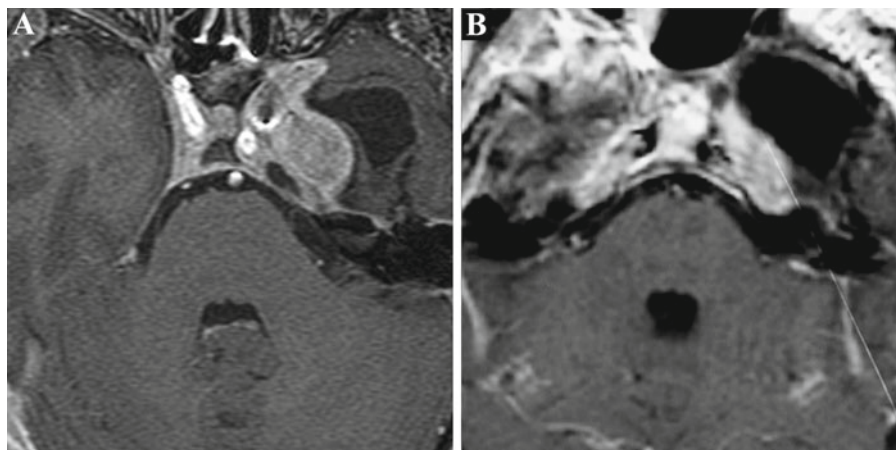


Fig. 30.2 T1-weighted contrast-enhanced axial MR images of a 34-year-old male at the time of SRS show a left-sided ganglion type trigeminal schwannoma (a).

Axial T1-weighted contrast-enhanced MR images obtained 30 months after SRS revealed complete resolution of the tumor (b)

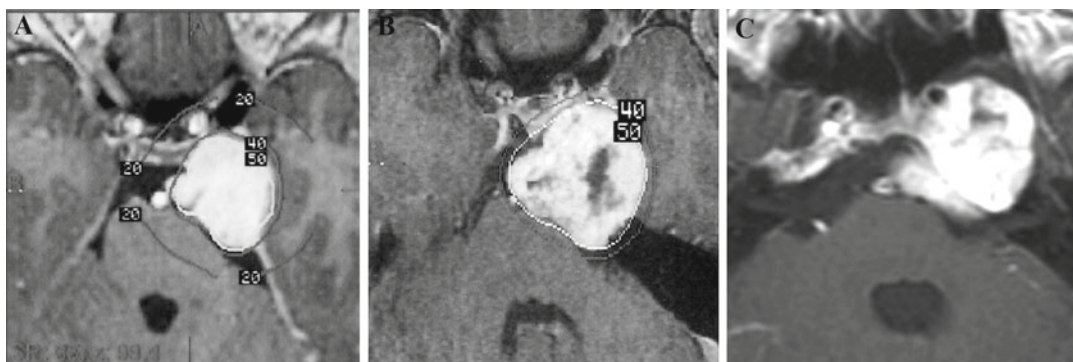


Fig. 30.3 T1-weighted contrast-enhanced axial MR images of a 15-year-old male at the time of SRS show a dumbbell type trigeminal schwannoma (a). Axial T1-weighted contrast-enhanced MR images obtained 26 months after initial SRS showed enlargement of the

enhancing tumor with central contrast loss, which prompted repeat SRS (b). Axial T1-weighted contrast-enhanced MR images obtained 8 years after initial SRS showed slight decrease in volume (c)

Discussion

Tumor Control

It was reported by Pollock et al. (2002) that the outcomes of Gamma Knife radiosurgery in 24 patients with nonvestibular schwannoma; ten patients had trigeminal schwannomas. They reported tumor control in nine of ten patients with trigeminal schwannomas. Three developed

new or worsening trigeminal dysfunction after SRS. It was reported by Pan et al. (2005) that they treated 56 patients with trigeminal schwannomas using Gamma Knife radiosurgery and reported a 93% tumor control rate at a mean follow-up of 68 months. It was reported by Sheehan et al. (2007) that their experience with trigeminal nerve sheath tumors. Twelve (48%) of 26 trigeminal schwannoma patients had tumor regression, ten (40%) had stable lesion, and three (12%) had tumor progression at a

mean follow-up of 48.5 months after SRS. It was reported by Hasegawa et al. (2007) that they treated 37 patients with trigeminal schwannomas and reported actuarial 5- and 10-year tumor control rates of both 84% at a mean follow-up duration of 54 months. They noted that larger tumors (≥ 15 cc) were more likely to progress after SRS. In the present series the 3-year progression free survival of tumor volume of < 8.0 and ≥ 8.0 cc were 94.4% and 64.3%, respectively. A target volume of < 8.0 cc was significantly associated with better progression free survival ($p = 0.0007$). It has reported by Hasegawa et al. (2007) that the importance of tumor location. In their experience all but one tumor located predominantly in the middle fossa was successfully treated. Tumor control was achieved significantly less frequently in patients whose tumors compressed the fourth ventricle ($p = 0.01$).

Functional Preservation

It was reported by Pan et al. (2005) that 25% of patients experienced complete relief of symptoms, 44% had improved symptoms, 14% had no change in symptoms, and 9% had slight worsening of symptoms. It was reported by Sheehan et al. (2007) that 18 (72%) of 26 trigeminal schwannoma patients had clinical improvement, four (16%) had stable symptoms, and three (12%) had worsening of the disease after SRS. It was reported by Hasegawa et al. (2007) that 40% of their patients noted improvement in their symptoms. One patient experienced new symptoms despite good tumor control. One-third of our patients exhibited improvement in neurological symptoms and 58% had no change in their symptoms. Three patients (9%) had delayed worsening of neurological function. Two of these patients developed facial anesthesia and facial pain due to tumor progression. Ten of 22 patients (46%) who underwent SRS as initial management had improvement in neurological symptoms. Ten patients who underwent initial surgical removal had no improvement in neurological symptoms

after SRS. Patients with no prior resection were more likely to improvement ($p = 0.0367$). We noted that patients with associated trochlear or abducens nerve dysfunction improved in approximately half of affected patients.

Summary

SRS is an effective and minimally invasive management option for patients with residual or newly diagnosed trigeminal schwannomas. Patients who have smaller tumor volume and root or ganglion type trigeminal schwannomas are ideal candidates for primary SRS as an alternative to surgical resection. SRS is also valuable for larger dumbbell shaped tumors as a combined approach using both surgical resection and SRS. Most trigeminal schwannoma patients can be expected to maintain their neurological symptoms and signs after SRS.

References

- Al-Mefty O, Ayoubi S, Gaber E (2002) Trigeminal schwannomas: removal of dumbbell-shaped tumors through the expanded Meckel cave and outcomes of cranial nerve function. *J Neurosurg* 96:453–463
- Andrews DW, Suarez O, Goldman HW, Downes MB, Bednarz G, Corn BW, Werner-Wasik M, Rosenstock J, Curran WJ Jr (2001) Stereotactic radiosurgery and fractionated stereotactic radiotherapy for the treatment of acoustic schwannomas: comparative observations of 125 patients treated at one institution. *Int J Radiat Oncol Biol Phys* 50:1265–1278
- Day JD, Fukushima T (1998) The surgical management of trigeminal neuromas. *Neurosurgery* 42:233–240; discussion 240–231
- de Benedittis G, Bernasconi V, Ettorre G (1977) Tumours of the fifth cranial nerve. *Acta Neurochir (Wien)* 38:37–64
- Eisenberg MB, Al-Mefty O, DeMonte F, Burson GT (1999) Benign nonmeningeal tumors of the cavernous sinus. *Neurosurgery* 44:949–954; discussion 954–945
- Flickinger JC, Kondziolka D, Pollock BE, Lunsford LD (1996) Evolution in technique for vestibular schwannoma radiosurgery and effect on outcome. *Int J Radiat Oncol Biol Phys* 36:275–280
- Flickinger JC, Kondziolka D, Niranjan A, Lunsford LD (2001) Results of acoustic neuroma radiosurgery: an analysis of 5 years' experience using current methods [see comment]. *J Neurosurg* 94:1–6

- Goel A, Muzumdar D, Raman C (2003) Trigeminal neuroma: analysis of surgical experience with 73 cases. *Neurosurgery* 52:783–790; discussion 790
- Hasegawa T, Kida Y, Yoshimoto M, Koike J (2007) Trigeminal schwannomas: results of gamma knife surgery in 37 cases. *J Neurosurg* 106:18–23
- Kano H, Niranjana A, Kondziolka D, Flickinger JC, Dade Lunsford L (2009) Stereotactic radiosurgery for trigeminal schwannoma: tumor control and functional preservation Clinical article. *J Neurosurg* 110:553–558
- Konovalov AN, Spallone A, Mukhamedjanov DJ, Tcherekajev VA, Makhmudov UB (1996) Trigeminal neurinomas. A series of 111 surgical cases from a single institution. *Acta Neurochir (Wien)* 138:1027–1035
- McCormick PC, Bello JA, Post KD (1988) Trigeminal schwannoma. Surgical series of 14 cases with review of the literature. *J Neurosurg* 69:850–860
- Niranjana A, Lunsford LD, Flickinger JC, Maitz A, Kondziolka D (1999) Dose reduction improves hearing preservation rates after intracanalicular acoustic tumor radiosurgery. *Neurosurgery* 45:753–762; discussion 762–755
- Pan L, Wang EM, Zhang N, Zhou LF, Wang BJ, Dong YF, Dai JZ, Cai PW (2005) Long-term results of Leksell gamma knife surgery for trigeminal schwannomas. *J Neurosurg* 102(Suppl):220–224
- Pollack IF, Sekhar LN, Jannetta PJ, Janecka IP (1989) Neurilemmomas of the trigeminal nerve. *J Neurosurg* 70:737–745
- Pollock BE, Foote RL, Stafford SL (2002) Stereotactic radiosurgery: the preferred management for patients with nonvestibular schwannomas? *Int J Radiat Oncol Biol Phys* 52:1002–1007
- Regis J, Pellet W, Delsanti C, Dufour H, Roche PH, Thomassin JM, Zanaret M, Peragut JC (2002) Functional outcome after gamma knife surgery or microsurgery for vestibular schwannomas. *J Neurosurg* 97:1091–1100
- Samii M, Babu RP, Tatagiba M, Sepehrnia A (1995a) Surgical treatment of jugular foramen schwannomas. *J Neurosurg* 82:924–932
- Samii M, Migliori MM, Tatagiba M, Babu R (1995b) Surgical treatment of trigeminal schwannomas. *J Neurosurg* 82:711–718
- Sarma S, Sekhar LN, Schessel DA (2002) Nonvestibular schwannomas of the brain: a 7-year experience. *Neurosurgery* 50:437–448; discussion 438–439
- Sheehan J, Yen CP, Arkha Y, Schlesinger D, Steiner L (2007) Gamma knife surgery for trigeminal schwannoma. *J Neurosurg* 106:839–845
- Suh JH, Barnett GH, Sohn JW, Kupelian PA, Cohen BH (2000) Results of linear accelerator-based stereotactic radiosurgery for recurrent and newly diagnosed acoustic neuromas. *Int J Cancer* 90:145–151
- Taha JM, Tew JM Jr, van Loveren HR, Keller JT, el-Kalliny M (1995) Comparison of conventional and skull base surgical approaches for the excision of trigeminal neurinomas. *J Neurosurg* 82:719–725
- Yasui T, Hakuba A, Kim SH, Nishimura S (1989) Trigeminal neurinomas: operative approach in eight cases. *J Neurosurg* 71:506–511
- Yoshida K, Kawase T (1999) Trigeminal neurinomas extending into multiple fossae: surgical methods and review of the literature. *J Neurosurg* 91:202–211

Vestibular Schwannoma: Gamma Knife Radiosurgery

31

Dong Gyu Kim

Abstract

Vestibular schwannoma is the most common tumor of the cerebellopontine angle. Gamma knife radiosurgery was originally recognized as an alternative treatment modality to microsurgery for small vestibular schwannoma in non-surgical candidates. However, the characteristics of minimal invasiveness and low complication rate have made it an attractive treatment option. Advances in neuroimaging and stereotactic techniques reflect the evolution of gamma knife radiosurgery. The high preservation of neurologic functions and tumor control after radiosurgery for vestibular schwannoma have established it as an important alternative to microsurgery.

Introduction

Vestibular schwannoma is a benign neoplasm that occurs predominantly on the vestibular branch of the eighth cranial nerve. It is the most common tumor of the cerebellopontine angle and constitutes approximately 8% of all intracranial tumors. Tinnitus is typically the initial symptom, and dizziness and hearing disturbances follow. There are several therapeutic options for patients with vestibular schwannoma, including clinical observation, surgical resection, fractionated radiotherapy and stereotactic radiosurgery. In the past, most patients suffered from severe symptoms due to

large tumor sizes, and microsurgery was the primary treatment modality. Advances in microsurgical techniques, neurophysiological monitoring and sophisticated anesthetic care have greatly improved the outcome of surgical removal by skull-based approaches. However, for many neurosurgeons, it is not always easy to obtain a favorable surgical outcome. In addition, despite acceptable tumor control rates, complications following microsurgery still occur, and this may place a burden on patients (Samii et al. 1997).

In recent years, vestibular schwannoma has been diagnosed more often in the earlier stage after introduction of magnetic resonance imaging (MRI), leading to some very small and asymptomatic tumors. The various sized vestibular schwannomas were showed in Fig. 31.1. Due to the complications of microsurgery, observation has typically been the safest option for these mild or asymptomatic patients. Although there are

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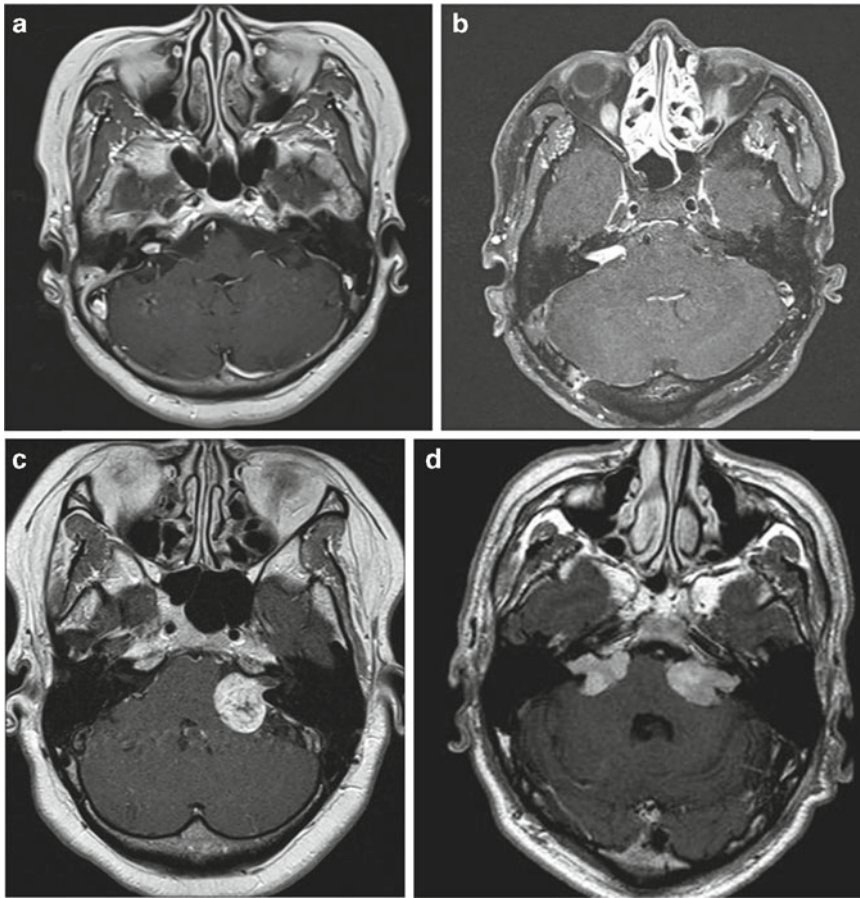


Fig. 31.1 Magnetic resonance images demonstrate the typical features of various sizes of vestibular schwannomas (a–c) and a bilateral vestibular schwannoma in neurofibromatosis type 2 (d). As the tumor grows, the widening of the

internal acoustic canal is more prominent, and the mass in the cistern is enlarged. Tumors are roughly spherical in their intracranial location and are located in the cerebello-pontine angle, extending into the internal acoustic canal

several conflicting results, vestibular schwannomas are believed to progress gradually. Some tumors exhibit no growth within certain period, whereas others grow rapidly. In fact, tumor growth has been reported to vary from 15% to 85%, and the preservation of auditory function ranges between 11% and 100% during the natural course of the disease. Both patients and surgeons agree that long-term tumor control, preservation of cranial nerve function, and maintenance of a high quality of life represent desirable outcome goals. However, a consensus on the optimal management paradigm has not been established because the natural history of the disease remains to be elucidated.

When an optimal management paradigm is not available, the tumor control rate and associated complication rate of each of the treatment policies are the determining factors for clinicians. A meta-analysis by Yamakami et al. (2003) revealed that half of the reported vestibular schwannomas grew in size, a third of patients lost useful hearing, and 20% of tumors ultimately required surgical intervention after employing a wait-and-see policy. In contrast, gamma knife radiosurgery controlled tumor growth in 92% of patients, and only 4.6% of patients underwent microsurgery at the last follow-up (Pollock et al. 2006). The tumor recurrence, mortality and major disability rates following gamma knife radiosurgery

were 1.8%, 0.63%, and 2.9%, respectively. Experienced hands with modern microsurgical technique accomplished 98% complete removal rate of vestibular schwannomas, however, the mortality and major disability rates were 1.1% and 6.5%, respectively. The quality of life was also better for patients who underwent stereotactic radiosurgery compared to surgical resection or clinical observation group. With this background, many patients prefer radiosurgery to surgical resection because of the lower morbidity of the procedure and the similar rates of long-term tumor control. The advantages and disadvantages of using gamma knife radiosurgery for treatment of vestibular schwannomas will be assessed in this chapter.

Radiation Effects of Radiosurgery for Vestibular Schwannomas

The concept of gamma knife radiosurgery was developed by Lars Leksell for the precise destruction of a chosen target using ionizing radiation delivered in a single session without significant destruction of adjacent tissues. Originally, radiosurgery resulted in a total destruction of tissue within the target volume, and gamma knife radiosurgery was used as an alternative to electrode-based lesioning in functional disorders.

Generally, the megavoltage radiation delivered during radiosurgery causes free radical-mediated physical damage to the DNA of the target. A single-strand break has a high likelihood of repair, whereas two or more breaks on the same DNA strand are more likely to cause permanent damage to the cell. The double-strand breaks in the DNA lead to the physical disruption of DNA structures. Consequently, the damage to the DNA in the cell leads to the desired therapeutic effect (Levivier 2008).

However, it is not fully understood how ionizing radiation controls vestibular schwannoma. The activation or triggering of programmed cell death or irreversible growth arrest of tumor cells with injured DNA has been suggested as the radiobiologic mechanism for radiosurgery

in the case of vestibular schwannoma. In addition, the radiation may impact the stroma of vestibular schwannoma and result in endothelial destruction and damage to the vessel wall, inducing ischemia of the tumor cell. This study also identified a high incidence of hemosiderin deposits and vascular mural hyalinization in the tissue from the vestibular schwannoma xenografts in mice that underwent stereotactic radiosurgery (Linskey et al. 1993). In the surgical specimens obtained by microsurgery after radiosurgery, the most prominent feature was a necrotic core surrounded by a transitional zone that was covered by an outer capsule. The central necrosis comprised shrunken apoptotic tumor cells with narrow cytoplasmic rims and basophilic nuclei, which is compatible with coagulation necrosis. The surrounding transitional tissue contained the loose structure of shrunken neoplastic cells. However, a densely packed Schwann cell nest between scar tissue was commonly found in the outer capsule (Yeung et al. 2009). The scar tissue consisted of thick collagen bundles and hyaline degeneration. These features suggest that the primary effect of radiosurgery was the radiation-induced tumor cell death via coagulation necrosis, although vestibular schwannoma may sustain viability over a long period. In addition, endothelial destruction or wall damage in vascular channels were commonly found in the specimens, and results of immunohistochemical studies have suggested that the endothelial cell layer of vessels is very sensitive and reacts before any changes in the tumor tissue (Szeifert et al. 2004).

Data from the experiments and surgical specimens indicate that high dose radiation derived from gamma knife radiosurgery controls vestibular schwannoma via direct radiation-induced cell death and indirect ischemia in the context of endothelial damage. However, these histopathological features may not fully represent the effect of radiosurgery on vestibular schwannomas because the documented findings were from surgical specimens in patients who failed to control the tumor following radiosurgery.

Indication of Gamma Knife Radiosurgery for Vestibular Schwannomas

The management choices for vestibular schwannoma include conservative observation, fractionated radiotherapy, stereotactic radiosurgery, microsurgery, and a combination of these therapies. The treatment for vestibular schwannoma depends on the patient's symptoms, life expectancy, and therapeutic goals together with the tumor's size, content, and growth rate. The current therapeutic goals for many patients and physicians emphasize tumor control and the importance of patient-oriented outcomes, including preservation of hearing and facial function, post-procedural pain, alleviation of present symptoms, and cosmetics.

The natural history of vestibular schwannoma plays a key role in the choice of management strategy. The analysis of 903 patients by Yamakami et al. revealed that 51% of tumors grew, 20% eventually required some form of intervention, and 37% of patients ultimately lost serviceable hearing over a mean follow-up of 3.1 years (Yamakami et al. 2003). A meta-analysis of 1,345 patients reported similar results, showing that 43% of tumors grew, 20% required treatment, and 51% of patients experienced deterioration of hearing that ultimately lost a serviceable hearing over a mean follow-up of 3.2 years (Smouha et al. 2005). Interestingly, there was little correlation between symptoms and tumor size, and quality of life may diminish without tumor growth. These findings should be the cornerstone for making decisions regarding treatment plans and deciding what information to share with patients.

Gamma knife radiosurgery was originally recognized as an alternative treatment modality to microsurgery for small vestibular schwannoma in non-surgical candidates, offering similar tumor control rates with a lower economic burden compared to microsurgery. With reports with gamma knife radiosurgery for vestibular schwannoma treatment, it became clear that gamma knife radiosurgery was superior to microsurgery with respect to preservation of facial and auditory

function. Moreover, gamma knife radiosurgery showed lower treatment-associated morbidity and a better post-operative functional outcome, good patient satisfaction, and a swifter return to independent functioning (Pollock et al. 2006). Thus, gamma knife radiosurgery has become one of the primary treatment options for vestibular schwannoma.

Indications of gamma knife radiosurgery in the treatment of vestibular schwannoma are not very different from those for other brain tumors. The size of tumor is critical. Large tumors make the fall-off line of radiosurgical planning less steep, and radiation to the surrounding tissues increases. Thus, operators usually reduce the prescribed marginal dose to avoid complications, causing a lower tumor control rate than in the case of small vestibular schwannoma (Hasegawa et al. 2005; Yang et al. 2008). The intentional subtotal resection of large vestibular schwannoma following gamma knife radiosurgery could be an answer for both successful and functional preservation and tumor control (Park et al. 2006).

Patients with cystic vestibular schwannoma belong to a well-defined subgroup of patients with poor outcomes following gamma knife radiosurgery. After the report by Pendle et al., these patients were considered as poor candidates for gamma knife radiosurgery, although the control failure rates varied significantly (Pendle et al. 1996). So, primary treatment option of a cystic vestibular schwannoma is microsurgery and residual tumors can be managed after microsurgery. Tumor control of residual vestibular schwannomas after gamma knife radiosurgery was not affected by the pre-operative cystic component (Yang et al. 2008).

Recurrent vestibular schwannomas following radiosurgery may be managed by repeated gamma knife radiosurgery, with a favorable tumor control response. However, adverse radiation effects on the brainstem and deterioration of facial nerve function has led to the prescription of relatively low marginal doses (range: 12–13 Gy).

In summary, the small to medium sized vestibular schwannomas less than 2–3 cm in diameter are good indication of gamma knife radiosurgery.

Patients with poor medical condition for surgery or old age can be treated with radiosurgery, too. Residual tumors after craniotomy and recurred vestibular schwannomas are also well controlled by gamma knife surgery.

Treatment Planning

It is essential to choose appropriate imaging modalities and to acquire high-resolution images because gamma knife radiosurgery is an exclusively image-guided surgical procedure. The thin slice MRI is sufficient to accurately localize the vestibular schwannoma, although some physicians recommend using fusion images of MRI and computer tomography images to avoid the magnetic susceptibility artifact and chemical shift artifact. For vestibular schwannoma radiosurgery, a three dimensional volume acquisition MRI using a gradient pulse sequence is used to cover the entire lesion and surrounding critical structures. However, the operator should keep in mind that 3DT1 introduces inaccuracies from magnetic susceptibility distortions and partial volume effects. A T2-weighted 3D volume sequence is strongly recommended to visualize the critical cranial nerves and inner ear structures with better resolution and less distortion (Niranjan and Flickinger 2008). Under post-surgical conditions, a saturation sequence may be helpful.

Dose planning is the most critical step in the radiosurgical treatment procedure. It is important to determine the clinical results in terms of efficacy and safety. Conformal coverage of the tumor and preservation of the function of surrounding critical nerves such as the facial, cochlear and trigeminal nerve are high-priority for dose planning. The control of vestibular schwannoma after radiosurgery depends on high conformity to the margin, and radiation-related complications may be avoided by the precise delineation of the tumor. The facial and cochlear nerves usually course along the anterior and anterior-inferior margin of the tumor; thus, high conformal planning is essential for this component (Niranjan and Flickinger 2008).

In the extracanalicular portion of vestibular schwannoma, small-diameter (4- and 8-mm) beams have been combined to achieve the ideal target volume and for dose planning purposes. However, a series of 4-mm isocenters may be needed to create the ideal sharp fall-off of radiation in the intracanalicular component of the vestibular schwannoma. This is because the average diameter of the normal internal auditory canal is only 4 mm, and all nerves (i.e., the cochlear, vestibular, and facial nerves) pass through this small space. These rationales and techniques may enhance the clinical results in terms of hearing preservation following gamma knife radiosurgery.

The prescribed doses for gamma knife radiosurgery for the treatment of vestibular schwannoma have been significantly reduced. In the early era of gamma knife radiosurgery, a high marginal dose was used to control the tumor. However, the post-radiosurgical neuropathy of facial and trigeminal nerves may force physicians to reduce the irradiated dose, and trials of gradual reductions in dose have improved both hearing preservation and prevention of neuropathy. Currently, 12–13 Gy is recommended as the tumor margin dose based on the clinical results of large-scale studies, which have confirmed the effective control of the tumor and a low incidence of radiation-related complications (Lunsford et al. 2005). Moreover, a lower dose (10–11 Gy) may be used in cases of large vestibular schwannomas or for preservation of serviceable auditory function (Regis et al. 2008). During treatment planning, the fall-off of radiation should be checked on the cochlea, semicircular canal, and the brainstem, including the cochlear nucleus. One should keep the radiation below the tolerance level to avoid complications. In a study of diverse brainstem pathologies after radiosurgery, it was found that exposure of the brainstem more than 12 Gy at volumes as low as 0.1 cm³ produced an adverse radiation effect and new neurological deficits (Sharma et al. 2008). In vestibular schwannoma, the cochlea is known to receive significantly high radiation doses in patients with a worsening of hearing following gamma knife radiosurgery. A significant association

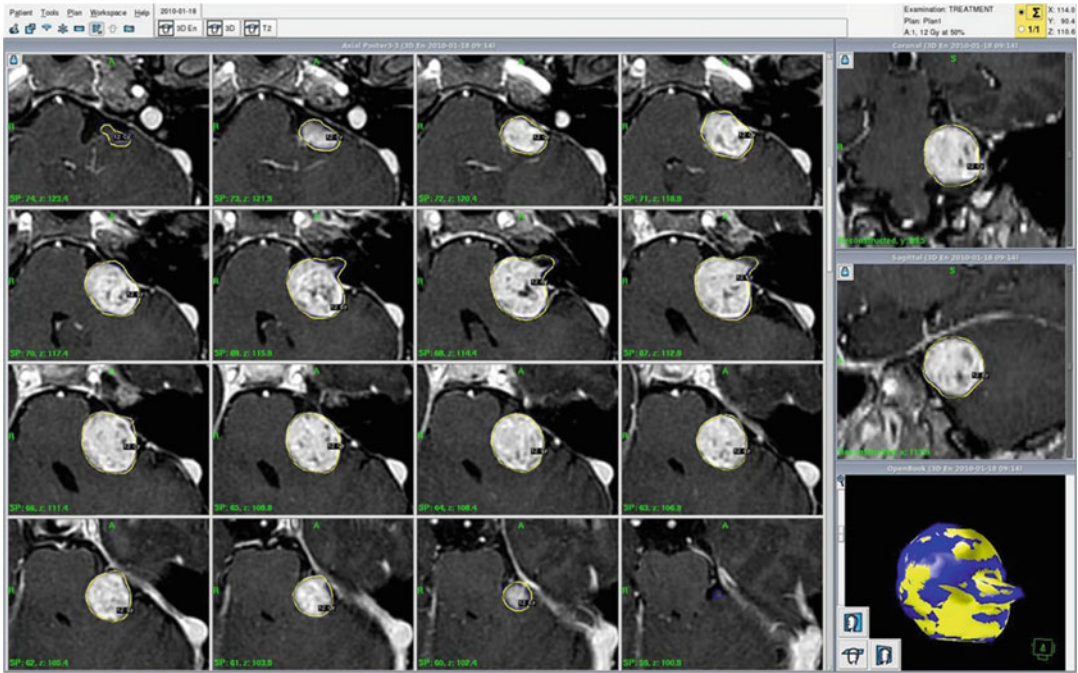


Fig. 31.2 Planning of gamma knife radiosurgery with the Gamma plan®. The prescribed dose ranged from 12 to 13 Gy helps avoid complications and improves control rates. The over-irradiated dose to the cochlear nucleus,

cochlea and brainstem should be avoided to preserve serviceable hearing with the help of the plugging and beam weighting technique

between the cochlear and the intracanalicular dose of radiation delivered during gamma knife radiosurgery has been reported. Another study concluded that patients who received a radiation dose of less than 4.2 Gy to the central cochlea had significantly better hearing preservation that was of the same Gardner-Robertson class (Kano et al. 2009). Similar results have been reported elsewhere, showing a better hearing preservation when the dose to the cochlea was lower than 4 Gy (Regis et al. 2008). Paek et al. measured the radiation doses delivered to the cochlea, cochlear nerve, and cochlear nucleus in patients treated with gamma knife radiosurgery for vestibular schwannoma. They concluded that the maximum dose delivered to the cochlear nucleus (<10 Gy) was the most significant prognostic factor for hearing preservation (Paek et al. 2005). Thus, beam weighting and plugging techniques are frequently used to restore the margin of the brainstem and structures of the inner ear system as shown in Fig. 31.2.

Evaluation of Patients Before and After Radiosurgery

Patients with vestibular schwannoma should be evaluated from both the clinical and radiological perspectives. Radiological diagnosis is based on high-resolution MRI and computed tomography and may be helpful to differentiate vestibular schwannoma from other tumors in the cerebello-pontine angle before radiosurgery. Clinical evaluations include a meticulous patient history, neurological examination, and functional tests for facial and audiologic status. The pure-tone audiogram, speech discrimination score and electroneuronography are frequently included in the pre- and post-radiosurgical work-up. Audiologic function is typically graded with the Gardner-Robertson modification of the Silverstein and Norell classification, and facial nerve function is classified according to the House-Brackmann grading system. Serial MRIs with gadolinium

should be requested at 6-month intervals for 2 years and then annually for other several years. The audiologic evaluations, namely the pure-tone audiogram and speech discrimination, are recommended close to the time of MRI follow-up in patients with serviceable auditory function before radiosurgery.

Tumor Control After Gamma Knife Radiosurgery

The goal of radiosurgery is to control tumor growth, which means growth arrest or the regression of the tumor. Thus, the outcome of radiosurgery is determined by follow-up MRIs after radiosurgery, also known as radiologic control. However, there is no consensus as to the definition of tumor control after radiosurgery. A widely cited definition of tumor growth is that suggested by Flickinger et al. (2004). They defined tumor growth as a 1-mm increase in tumor diameter in any two directions or a 2-mm growth in one direction. Another definition of growth is a greater than 10% increase in tumor volume (Wowra et al. 2005).

In most cases, there is some expansion in tumor size and a distinct loss of the central tumor contrast enhancement at 6 months following gamma knife radiosurgery. And then, the tumor volume gradually decreases and most tumors return to the original size at the time of radiosurgery. Afterwards, tumors continue to decrease gradually over time as shown in Fig. 31.3. Temporary tumor expansion and loss of contrast enhancement create confusion as to whether or not further treatment should be performed, although this is the typical postradiosurgical course. The tumor size should be mapped with serial MRI in order to differentiate true tumor growth from tumor expansion and avoid unnecessary microneurosurgery or repeated radiosurgery. Sometimes, the transient expansion is correlated with facial palsy and facial dysesthesia, which usually improves with time due to tumor shrinkage. Thus, it is important to inform patients thoroughly of these potential sequelae. Noren (1996) demonstrated that all patients

whose vestibular schwannomas were considered to be under control at 5 years after gamma knife radiosurgery remained so after 20 years. Similarly, Kondziolka et al. (1998) reported long-term outcomes in 162 patients with 5–10 years of follow-up. They identified no treatment failures in any patients more than 3 years after undergoing gamma knife radiosurgery. Thus, radiologic follow-up appears to be necessary for at least 3 years following surgery.

In an early series of Swedish experiences of gamma knife radiosurgery in vestibular schwannoma, 14 patients were managed based on images from CT, cisternography and/or pneumoencephalography. The irradiated marginal dose of the tumor ranged from 7 to 45 Gy. The follow-up images revealed the control of the tumor in 10 patients, tumor progression in 3, and a good or excellent general condition in 13 patients. These were encouraging results considering the crude imaging modalities and relatively inaccurate calculations without a computerized registration system, although a very high dose was prescribed in comparison with modern radiosurgical procedures.

In recent reports, the tumor control rate in the context of gamma knife radiosurgery for vestibular schwannoma ranged from 87% to 98% (Flickinger et al. 2004; Hasegawa et al. 2006; Lunsford et al. 2005). Certain factors impacted the results of gamma knife radiosurgery for vestibular schwannoma. First, reliable and accurate imaging and registration systems are basic factors. The quality of images and spatial distortion of anatomical structures result in inaccurate planning for radiosurgery which directly affects the radiation that is administered to the tumor. The technical improvement of modern imaging and dosimetry has dramatically improved clinical outcomes.

Second, the size of tumor is critical to efficacy and safety. In a large series, a poor control rate was identified in large tumors ($\geq 15 \text{ cm}^3$) compared to small tumors (57% and 96% 10-year progression-free survival, respectively) (Hasegawa et al. 2005). In general, gamma knife radiosurgery is not a suitable treatment modality for large brain tumors (mean diameter greater than 3 cm)

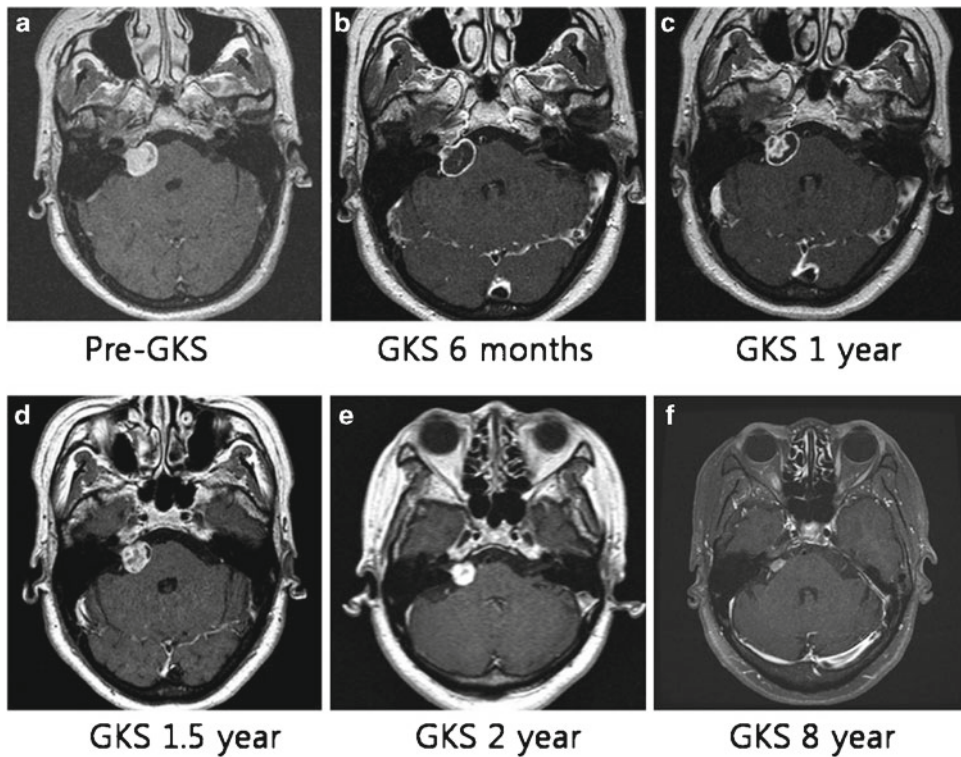


Fig. 31.3 An example of consecutive radiologic changes in vestibular schwannomas after gamma knife radiosurgery. A 53-year-old woman underwent gamma knife radiosurgery with 13 Gy of a marginal 50% isodose (a). The follow-up magnetic resonance image 6 months after gamma knife radiosurgery indicated a loss of the central tumor contrast enhancement with an expansion in tumor

size and without clinical deterioration (b). As the enhancement pattern of the tumor content gradually recovered to its previous status, the tumor shrunk progressively on serial follow-up magnetic resonance images (c–e). The 8-year follow-up magnetic resonance image shows marked regression of the tumor (f)

because a large tumor size increases the risk of complications compared to small tumors. In vestibular schwannoma, significant peri-tumoral edema also invariably occurs after gamma knife radiosurgery for vestibular schwannomas larger than 3 cm in diameter (Yang et al. 2008). The peri-tumoral edema potentially causes headaches, vomiting, dizziness and aggravation of brainstem compression, which may lead to hydrocephalus and neurological deterioration. To avoid the complications, a low prescribed dose is recommended in gamma knife radiosurgery for large vestibular schwannoma. However, marginal doses less than 12 Gy have been associated with increased tumor recurrence.

Third, the content of a given vestibular schwannoma may affect the results of gamma knife radiosurgery. A vestibular schwannoma

with an intratumoral cyst can be problematic because it has been associated with a high failure rate due to the enlargement of the cyst after gamma knife radiosurgery. The Marseille group found that the risk of failure increased in 54 patients who had cysts at the time of gamma knife radiosurgery, compared with 946 patients who did not have cysts (control rate: 93.6% vs. 98%) (Delsanti and Regis 2004). The high failure rate of gamma knife radiosurgery was found in cases of cystic vestibular schwannoma or in different tumors with a cystic component, such as cranio-pharyngiomas or hemangioblastomas (Yang et al. 2008). The reaction of pre-existing intratumoral cysts to radiosurgery is not yet fully understood. One theory is that the transudation from the vessel in the wall of the tumor enlarges the cyst (Pendl et al. 1996). The enforced coagulation

necrosis of tumor tissue may enhance the osmotic gradient. In another study, however, no correlation was found between pretreatment of intratumoral cysts and the risk of post-treatment tumor expansion. Cyst formation after gamma knife radiosurgery, however, was significantly correlated with tumor expansion associated with neurologic changes.

Finally, the genetic background of the tumor also affects the outcome. Bilateral vestibular schwannomas are the hallmark of neurofibromatosis type 2 (NF2), which impose a significant risk of deafness at a young age. Gamma knife radiosurgery has emerged as one of the first-line treatment options based on promising gamma knife radiosurgery results for the treatment of sporadic vestibular schwannoma, compared with the high probability of hearing loss and cranial nerve damage after microsurgical resection (Samii et al. 1997). In a recent study of NF2, the actuarial tumor control rate, measured by serial MRI volumetry, was 74% in the second year and 66% in the fifth year following gamma knife radiosurgery (Phi et al. 2009). These values are clearly lower than those for sporadic vestibular schwannomas. The reasons why vestibular schwannomas in NF2 patients do not respond as well as sporadic tumors are not fully understood. Genetic background, multilobulated tumor stereology, and younger patient age are assumed to play a role in the radiobiology of NF2-associated vestibular schwannomas. Despite the differences in results compared to sporadic tumors, gamma knife radiosurgery does seem to have a role in treating vestibular schwannomas in NF2 patients.

Hearing Preservation After Gamma Knife Radiosurgery

Hearing preservation in cases involving vestibular schwannomas is the most important functional outcome after treatment. Preservation of overall serviceable hearing (Gardner-Robertson class 1 or 2) has been reported in 27–47% of patients who underwent microsurgical resections conducted by experienced surgeons (Fahlbusch

et al. 1998; Samii et al. 1997). In a series of patients who underwent gamma knife radiosurgery, the rate of preserved hearing ranged from 33% to 79% with varying follow-up periods (Litvack et al. 2003).

The mechanism for hearing deterioration after gamma knife radiosurgery is not exactly known. It may be caused by ischemic damage to the cochlear nucleus or the auditory nerve, direct radiation damage to the auditory system, and tumor enlargement in the internal auditory canal (Paek et al. 2005; Yang et al. 2009).

The relationship between the volume on the day of gamma knife radiosurgery and hearing preservation has been under debate. In some reports, hearing preservation rates worsened as the tumor size increased (Litvack et al. 2003). However, the transient expansion of the tumor was not exactly correlated with hearing loss. Moreover, no statistical difference was found in a meta-analysis, which suggests the tumor volume may not be a critical clinical prognostic factor for hearing preservation in all patients who undergo radiosurgery for vestibular schwannoma (Yang et al. 2009). In this analysis, patients with small tumors received a higher average radiation dosage of 17.8 Gy (± 6.6 Gy) and exhibited worse hearing preservation rates. Conversely, a lower average radiation of 15.0 Gy (± 4.9 Gy) was prescribed for patients with large tumors, and their hearing preservation rate improved.

The radiation dose plays an important role in hearing preservation. Patients who received a low marginal dose (≤ 13 Gy) also exhibited favorable hearing preservation rates with excellent tumor control rates (Yang et al. 2009). With a marginal dose, the low radiation (≤ 4 Gy) administered to the cochlear structure was also found to be critical. The cochlea received significantly higher radiation doses in patients who reported worse hearing. Moreover, a significant relationship between cochlear damage and the intracanalicular dose of radiation delivered has been reported.

The auditory function on the day of gamma knife radiosurgery has a predictive value in the long-term preservation of serviceable hearing. A longitudinal study demonstrated that the

hearing of patients with vestibular schwannoma declined more rapidly if the initial hearing was worse (Massick et al. 2000). Furthermore, the earlier treatment of small vestibular schwannoma using modalities such as radiosurgery, radiotherapy, and microsurgery generally led to better hearing outcomes (Hasegawa et al. 2005; Samii et al. 1997). These findings may explain why good pre-gamma knife radiosurgery hearing function is a positive prognostic factor.

The location of the tumor also affects the auditory outcome. In a series using a marginal dose of 13 Gy, unchanged hearing preservation was possible in 50–77% of patients. In the same series, the rate increased to 90% in a subgroup of patients with purely intracanalicular tumors (Lunsford et al. 2005). It has also been shown that 58 (78.4%) of 74 patients with Gardner-Robertson Class I hearing maintained their functional hearing and all patients in the Koos Stage I subgroup reported functional hearing at last follow-up.

Complications

There have been several reports on the adverse effects of gamma knife radiosurgery for vestibular schwannoma. However, the incidence was very rare, and most of the adverse effects were transient.

Approximately 2–4% of patients developed hemifacial spasm after gamma knife radiosurgery, and this typically developed between 1 and 2 years following gamma knife radiosurgery. Irritations of the facial nerve or a delayed vascular insult with tumor expansion were postulated as the cause of hemifacial spasms (Pollock et al. 2006). The symptoms improved with time or responded to carbamazepine. With the low marginal dose of 12–13 Gy, permanent facial palsy has been shown to be a rare event, ranging from 0% to 0.5%, and transient facial palsy has been reported at a little bit higher rate (Hasegawa et al. 2005).

Following radiosurgery, trigeminal neuropathy was found to be the most common cranial neuropathy, except for cranial nerves VII or VIII. Rates for trigeminal symptoms ranged from 0.2% to 4% (Flickinger et al. 2004; Wowra et al. 2005).

Patients receiving more than 13 Gy were significantly more likely to develop trigeminal nerve neuropathy than those receiving less than 13 Gy.

Hydrocephalus in the absence of progressive tumor growth has been reported to occur in 0–7% of patients. Hydrocephalus is believed to be the result of tumor necrosis, with protein debris blocking cerebrospinal fluid flow. The development of hydrocephalus is more common following the treatment of large (>25 mm diameter) tumors. The ventriculoperitoneal shunt is necessary in most cases, although hydrocephalus may resolve spontaneously.

Conclusion

Gamma knife radiosurgery for vestibular schwannoma exhibits tumor control rates of above 90% and leads to lower complication rates compared to microsurgery. Thus, gamma knife radiosurgery is the primary treatment option for small- to medium-sized tumors. Recent advances in neuroimaging and stereotactic techniques contribute to improving the results of gamma knife radiosurgery. The worldwide availability and documented studies have made gamma knife radiosurgery a more attractive treatment modality for vestibular schwannoma. Today, in addition to tumor control, the preservation of cranial nerve functions such as hearing is also a primary concern when employing gamma knife radiosurgery for the treatment of vestibular schwannoma. Trials that define the optimal radiation dose, protect normal tissue, and overcome radio-resistant vestibular schwannoma should improve clinical results. Moreover, basic research that enhances our understanding of radiobiologic effects should be pursued.

References

- Delsant C, Regis J (2004) Cystic vestibular schwannomas. *Neurochirurgie* 50:401–406
- Fahlbusch R, Neu M, Strauss C (1998) Preservation of hearing in large acoustic neurinomas following removal via suboccipito-lateral approach. *Acta Neurochir (Wien)* 140:771–777

- Flickinger JC, Kondziolka D, Niranjan A, Maitz A, Voynov G, Lunsford LD (2004) Acoustic neuroma radiosurgery with marginal tumor doses of 12 to 13 Gy. *Int J Radiat Oncol Biol Phys* 60:225–230
- Hasegawa T, Fujitani S, Katsumata S, Kida Y, Yoshimoto M, Koike J (2005) Stereotactic radiosurgery for vestibular schwannomas: analysis of 317 patients followed more than 5 years. *Neurosurgery* 57:257–265
- Hasegawa T, Kida Y, Yoshimoto M, Koike J, Goto K (2006) Evaluation of tumor expansion after stereotactic radiosurgery in patients harboring vestibular schwannomas. *Neurosurgery* 58:1119–1128
- Kano H, Kondziolka D, Khan A, Flickinger JC, Lunsford LD (2009) Predictors of hearing preservation after stereotactic radiosurgery for acoustic neuroma. *J Neurosurg* 111:863–873
- Kondziolka D, Lunsford LD, McLaughlin MR, Flickinger JC (1998) Long-term outcomes after radiosurgery for acoustic neuromas. *N Engl J Med* 339:1426–1433
- Levivier M (2008) Tissue changes after radiosurgery for vestibular schwannomas. *Prog Neurol Surg* 21:98–102
- Linskey ME, Martinez AJ, Kondziolka D, Flickinger JC, Maitz AH, Whiteside T, Lunsford LD (1993) The radiobiology of human acoustic schwannoma xenografts after stereotactic radiosurgery evaluated in the subrenal capsule of athymic mice. *J Neurosurg* 78:645–653
- Litvack ZN, Noren G, Chougule PB, Zheng Z (2003) Preservation of functional hearing after gamma knife surgery for vestibular schwannoma. *Neurosurg Focus* 14:E3
- Lunsford LD, Niranjan A, Flickinger JC, Maitz A, Kondziolka D (2005) Radiosurgery of vestibular schwannomas: summary of experience in 829 cases. *J Neurosurg* 102:195–199
- Massick DD, Welling DB, Dodson EE, Scholfield M, Nagaraja HN, Schmalbrock P, Chakeres DW (2000) Tumor growth and audiometric change in vestibular schwannomas managed conservatively. *Laryngoscope* 110:1843–1849
- Niranjan A, Flickinger JC (2008) Radiobiology, principle and technique of radiosurgery. *Prog Neurol Surg* 21:32–42
- Noren G (1996) Gamma knife radiosurgery for acoustic neurinomas. In: Gildenberg PL, Tasker RR (eds) *Textbook of stereotactic and functional neurosurgery*. McGraw-Hill, New York, pp 835–844
- Paek SH, Chung HT, Jeong SS, Park CK, Kim CY, Kim JE, Kim DG, Jung HW (2005) Hearing preservation after gamma knife stereotactic radiosurgery of vestibular schwannoma. *Cancer* 104:580–590
- Park CK, Jung HW, Kim JE, Son YJ, Paek SH, Kim DG (2006) Therapeutic strategy for large vestibular schwannomas. *J Neurooncol* 77:167–171
- Pendl G, Ganz JC, Kitz K, Eustacchio S (1996) Acoustic neurinomas with macrocysts treated with gamma knife radiosurgery. *Stereotact Funct Neurosurg* 66(Suppl):103–111
- Phi JH, Kim DG, Chung HT, Lee J, Paek SH, Jung HW (2009) Radiosurgical treatment of vestibular schwannomas in patients with neurofibromatosis type 2: tumor control and hearing preservation. *Cancer* 115:390–398
- Pollock BE, Driscoll CL, Foote RL, Link MJ, Gorman DA, Bauch CD, Mandrekar JN, Krecke KN, Johnson CH (2006) Patient outcomes after vestibular schwannoma management: a prospective comparison of microsurgical resection and stereotactic radiosurgery. *Neurosurgery* 59:77–85
- Regis J, Tamura M, Wikler D, Porcheron D, Levrier O (2008) Radiosurgery: operative technique, pitfalls and tips. *Prog Neurol Surg* 21:54–64
- Samii M, Matthies C, Tatagiba M (1997) Management of vestibular schwannomas (acoustic neuromas): auditory and facial nerve function after resection of 120 vestibular schwannomas in patients with neurofibromatosis 2. *Neurosurgery* 40:696–705
- Sharma MS, Kondziolka D, Khan A, Kano H, Niranjan A, Flickinger JC, Lunsford LD (2008) Radiation tolerance limits of the brainstem. *Neurosurgery* 63:728–732
- Smouha EE, Yoo M, Mohr K, Davis RP (2005) Conservative management of acoustic neuroma: a meta-analysis and proposed treatment algorithm. *Laryngoscope* 115:450–454
- Szeifert GT, Figarella-Branger D, Roche PH, Regis J (2004) Histopathological observations on vestibular schwannomas after gamma knife radiosurgery: the Marseille experience. *Neurochirurgie* 50:327–337
- Wowra B, Muacevic A, Jess-Hempfen A, Hempel JM, Muller-Schunk S, Tonn JC (2005) Outpatient gamma knife surgery for vestibular schwannoma: definition of the therapeutic profile based on a 10-year experience. *J Neurosurg* 102(Suppl):114–118
- Yamakami I, Uchino Y, Kobayashi E, Yamaura A (2003) Conservative management, gamma-knife radiosurgery, and microsurgery for acoustic neurinomas: a systematic review of outcome and risk of three therapeutic options. *Neurol Res* 25:682–690
- Yang SY, Kim DG, Chung HT, Park SH, Paek SH, Jung HW (2008) Evaluation of tumour response after gamma knife radiosurgery for residual vestibular schwannomas based on MRI morphological features. *J Neurol Neurosurg Psychiatry* 79:431–436
- Yang I, Aranda D, Han SJ, Chennupati S, Sughrue ME, Cheung SW, Pitts LH, Parsa AT (2009) Hearing preservation after stereotactic radiosurgery for vestibular schwannoma: a systematic review. *J Clin Neurosci* 16:742–747
- Yeung AH, Sughrue ME, Kane AJ, Tihan T, Cheung SW, Parsa AT (2009) Radiobiology of vestibular schwannomas: mechanisms of radioresistance and potential targets for therapeutic sensitization. *Neurosurg Focus* 27:E2

Vestibular Schwannoma: Gamma Knife Radiosurgery (Method)

32

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Abstract

Vestibular schwannomas are benign intracranial tumors that may be managed using watchful waiting, surgical resection, or radiation. Gamma Knife radiosurgery has been a continually evolving technique for managing such tumors. Over the past two decades, improvements in imaging technology, treatment planning software, and lower dose protocols have helped radiosurgery become a highly efficacious and safe alternative to microsurgical resection. We review the indications that may help patients make an informed decision to choose Gamma Knife for their vestibular schwannoma. The current methods for treatment using the Gamma Knife are discussed, including a description of selecting an isodose line and marginal dose. Follow-up strategies are outlined, for tracking treatment success as well as for patient-centered outcomes such as hearing preservation and monitoring for facial neuropathy. Finally, we describe the best available evidence regarding the efficacy of Gamma Knife radiosurgery for vestibular schwannomas, including the rates of hearing preservation and facial neuropathy being encountered using current treatment protocols.

Introduction

Vestibular schwannomas (VSs) are benign intracranial tumors arising from those cells that myelinate the vestibular nerve. Most occur within the internal acoustic canal (intracanalicular)

and expand to include a significant extracanalicular portion. These extracanalicular masses lie within the cerebellopontine angle, and the resulting mass effect may prove injurious to adjacent cranial nerves, brainstem nuclei, or the cerebellum. Vestibular schwannomas therefore manifest by a spectrum of ipsilateral signs and symptoms including hearing loss, tinnitus, disequilibrium, vertigo, dizziness, facial numbness, and facial paresis. The most common presentation includes gradual sensorineural hearing loss, occurring in 95% of cases according to Matthies and Samii (1997).

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Patients diagnosed with a VS may be candidates for several management options. Watchful waiting involves serial magnetic resonance imaging to track tumor growth and regular follow-up to evaluate symptoms and signs of tumor progression. Treatment options include microsurgery, stereotactic radiosurgery, and fractionated radiotherapy. Unfortunately, evidence based guidelines for choosing a management strategy are relatively weak as reviewed by Pollock (2008). Most data on treatment outcomes arose from retrospective studies, as there have been few prospective comparative studies and no randomized controlled trials. At this point, treatment selection is commonly based on the size of the tumor, associated symptoms and signs, and characteristics of the patient, including age, health, and personal preferences.

Stereotactic radiosurgery using the Gamma Knife (GK) is a promising treatment modality for many patients with VSs. The precise administration of a single dose of radiation, limited to the tumor margin, seeks to arrest growth. Absence of intracranial manipulation may reduce post-operative morbidity and mortality, compared to invasive microsurgery. Therefore, GK radiosurgery may be particularly appealing to patients with smaller VSs and higher functionality prior to treatment. This may be particularly true now, as radiosurgical techniques have evolved considerably over the past decade and the use of lower doses is proving efficacious and safe for treating VSs. In this chapter we will outline the current evidence on the indications, methods, and outcomes associated with the managing VSs using the GK.

Treatment Indications and Patient Counseling

The indications for selecting a particular management strategy for a VS are multifactorial. The lack of robust comparative studies means evidence based guidelines are not available to assist such decisions. Pogodzinski et al. (2004) identified a trend suggesting the strongest predictor of

treatment choice was the specialty of the attending physician. The authors of this chapter advocate a patient-centered approach to selecting treatment. This should be based on the tumor size, tumor growth rate, and the patient's age, health, and personal preferences. Patients should be fully informed of the efficacy and risks associated with each option, but in presenting this evidence the physician should be conscious of their own biases and the limited availability of comparative studies in the literature on VS management. Ultimately, each patient should be made aware of all available options, which may be facilitated by discussions with a radiation oncologist, neurosurgeon, and neuro-otologist prior to treatment.

Watchful waiting offers the least invasive approach to VS management. It is appropriate due to the slow-growing nature of these tumors and the risks associated with the available interventions. On average, most VSs will grow at a rate of 1.9 mm per year in their greatest dimension according to papers by Smouha et al. (2005) and Yamakami et al. (2003). According to their meta-analyses, on average 20% of those managed conservatively will require treatment. This is typically indicated upon additional tumor growth or progression of symptoms.

The greatest challenge in selecting patients for watchful waiting is the unpredictable behavior of a given tumor. Some may grow rapidly, at rates ten-fold greater than average according to Fucci et al. (1999), while others may spontaneously regress. In a large meta-analysis by Smouha et al. (2005), there were few predictors associated with growing VSs. Those identified include previous growth, extracanalicular projections, and younger patients. Separate studies by Battaglia et al. (2006) and Hajioff et al. (2008) identified statistically significant differences in growth rates depending on tumor magnitude, with larger and extracanalicular tumors having greater average growth rates. Furthermore, symptoms do not necessarily correlate with tumor size or growth, and quality of life may diminish even in the case of a static tumor. Smouha et al. (2005) recommend considering watchful waiting for patients over 45 years old and with small tumors, less than 25 mm in largest dimension. With tumor growth,

greater than 2 mm per year in any dimension, or progression of symptoms, patients should consider treatment.

Microsurgery is a highly efficacious treatment for VSs. For patients with large tumors, 30 mm or greater in greatest dimension, surgical removal is strongly indicated and Nikolopoulos and O'Donoghue (2002) have suggested that alternative management "would be unethical." In general, fewer than 2% of patients who undergo resection of their tumor will require additional treatment, according to reviews by Kaylie et al. (2000) and Yamakami et al. (2003). Due to the invasive nature of intracranial surgery, resection is also associated with significant morbidity and a very small risk of mortality (<1%). Hearing preservation was achieved in 49% of patients with tumors smaller than 30 mm, according to a meta-analysis by Yamakami et al. (2003). Postoperative facial neuropathy is a common complication of microsurgery, due to the proximity of the facial nerve to the site of resection. In two comparative studies of microsurgery and radiosurgery, one by Pollock et al. (2006) and another by Myrseth et al. (2005), facial neuropathy was experienced in 17% and 20% of surgical patients, respectively. The only clear indication for microsurgery, as opposed to radiation treatments, is large tumor size. Otherwise, patients should be counseled regarding microsurgery's strong efficacy and associated risks, and an informed decision should be made taking into account their preferences.

Gamma Knife radiosurgery offers a strong alternative to microsurgery for patients with smaller tumors (<30 mm). The goal of radiosurgical treatment is to arrest tumor growth, rather than diminish the tumor volume. Since it can be accomplished without dissecting into the cranium, it may provide less risk of morbidity than surgery. This is especially true with the adoption of improved imaging and planning techniques, as well as lower dose protocols. We will outline the evidence for the efficacy and safety of GK radiosurgery for VSs below in section "[Treatment Outcomes](#)". In general, the paucity of comparative studies between radiosurgery and alternative treatments make it difficult to identify strong indications for preferentially using the

GK. Its use may be well suited for patients with smaller tumors (<30 mm) and those with good hearing function prior to treatment. It is a good option for patients who are not considering surgical resection, either because of health problems or personal preference. As the body of comparative evidence grows, the ability of radiosurgery to provide better patient-oriented outcomes, including functional preservation, will become clear and stronger indications may be identified.

Treatment Planning and Methods

Gamma Knife radiosurgery is a well-established method for treating intracranial lesions using a single dose of radiation concentrated at a precise intracranial location. It is accomplished by utilizing a hemispherical array of 201 fixed Cobalt-60 sources, whose concentric radiation beams sum to a considerable dose at the focus point. The patient's head is immobilized within a frame, defining a precise coordinate system. Gadolinium enhanced magnetic resonance imaging of the head within the coordinate frame allows the tumor volume to be defined in three dimensions. Treatment planning software allows the neurosurgeon or radiation oncologist to map the tumor margin and define the treatment fields. The radiation dose is prescribed such that the 50% isodose line corresponds to the tumor margin. During planning, care should be taken to minimize the dose to adjacent structures of the inner ear or cerebellopontine angle.

The primary goal when using the GK to treat a VS is to permanently arrest tumor growth while preserving cranial nerve function and limiting treatment associated morbidity. Achieving this balance of efficacy and safety requires the administration of a significant dose of radiation to the tumor volume while minimizing the exposure to surrounding tissues. Technological advances in the past decades have been instrumental in accomplishing this goal. These come in the form of high-resolution imaging and more powerful treatment planning software, allowing more precise mapping of the 50% isodose line to the tumor

margin. Additionally, there has been a trend towards using lower marginal doses to treat VSs, therefore decreasing dose dependent toxicity to surrounding structures. Marginal doses below 13 Gy have been utilized with similar efficacy as higher doses, as reported by Flickinger et al. (2004), thereby justifying the trend towards using low dose protocols. The use of improved technologies and lower marginal doses has been instrumental in the improving rates of hearing preservation and reducing post-treatment morbidities, such as facial neuropathy.

In planning a GK treatment for a VS, care should be taken to avoid doses to certain intracranial structures. Irradiating structures of the inner ear and brainstem have been implicated in post-operative hearing loss in a dose-dependent fashion. This association has been identified by Linskey (2008), Paek et al. (2005), and Timmer et al. (2009). Specifically, Linskey (2008) explained that exposure to the cochlea has been postulated to cause high-frequency hearing deficits, while exposure to cochlear nuclei may cause low-frequency deficits. Therefore, avoiding doses to the cochlea and brainstem should be a priority in order to facilitate a good functional outcome. It may be appropriate to demarcate these volumes using the planning software in order to record the dose to these structures. Avoiding additional structures within the cerebellopontine angle can be accomplished by planning a highly conformal dose to the tumor margin.

Selecting a dose is another important aspect to treatment planning. Early on, VSs were treated with marginal doses greater than 15 Gy. Current treatment protocols commonly use 12–13 Gy, which has been shown to offer tumor control as well as improved functional outcomes, particularly hearing preservation. In a recent comparison of low and high doses, published by Yang et al. (2009a, b), there was a statistically significant improvement in hearing preservation and post-treatment facial neuropathy in patients treated with 13 Gy or less. Therefore, the best available evidence suggests that to maximize patient-oriented outcomes, GK radiosurgery should be implemented using a marginal dose of 12–13 Gy, corresponding to the 50% isodose line.

Follow-Up

Patients treated with GK radiosurgery for VS should be subject to close follow-up. This is important to verify the efficacy of the treatment, for early referral for additional interventions (microsurgery or additional radiation) if treatment fails, and to manage treatment associated morbidities. Follow-up records may also be useful for performing retrospective studies and contributing to the body of knowledge on VSs.

The efficacy of radiosurgery is best followed by periodic gadolinium enhanced magnetic resonance imaging. Tumors are typically characterized by linear dimensions, or by volume, which is more typical in the radiosurgical literature. Treatment success is defined by the cessation of growth after radiosurgery. Growth is typically defined as a change in any dimension by more than 1 or 2 mm, providing some margin of error for differences in contrast uptake or image resolution. Transient growth during the first year following GK, followed by cessation of enlargement or tumor regression, has been observed by Nagano et al. (2008). Therefore, minor growth within the first year may not indicate failed radiosurgery, and should prompt close monitoring rather than secondary treatments.

One of the main advantages of GK radiosurgery is the ability to preserve hearing function. Many patients will have been subject to audiometric evaluations prior to radiosurgery, and these should continue to be part of their regular follow-up. For those with vestibular symptoms, formal vestibular testing may also be indicated. Hearing function can be tracked using a variety of statistics. Speech recognition threshold (SRT), which can be approximated by averaging the pure tone hearing function at 500, 1,000, and 2,000 Hz, and speech discrimination scores (SDS) are commonly tracked. In the VS literature, these two statistics are combined to yield a Gardner-Robertson score, first described in Gardner and Robertson (1988), allowing patients to be grouped into five categories of hearing function. Serviceable hearing is defined as a pre-treatment score of 1 or 2, corresponding to an $SRT \leq 50$ dB

and $\text{SDS} \geq 50\%$, and hearing preservation is defined as maintaining such a score following treatment. Tracking pure tone hearing loss may be useful for the purpose of research, especially when combined with pre-treatment data regarding the radiation dose to the cochlea and brainstem nuclei. This may be useful in clarifying the magnitude of dose-dependent toxicity associated with GK radiosurgery.

Similar to hearing function, facial nerve function may be compromised following radiosurgery and should be followed closely. Signs of facial neuropathy are commonly followed using the House-Brackmann grading system, based on the work of House and Brackmann (1985), a somewhat subjective categorization scheme based on a combination of symptoms and signs. Grades range from I to VI, with good facial nerve function commonly defined by having a grade of I or II. A small fraction of patients will experience a permanent decline in facial nerve function from grades I or II to grade III or higher (see section “[Treatment Outcomes](#)”). Some other patients may experience transient symptoms of facial paresis in the acute period following radiosurgical treatment, which has been described in case reports, such as those by Tago et al. (2000) and Pollack et al. (2005).

During follow-up patients may also experience a variety of other treatment-related morbidities, which should be evaluated for during follow-up. Other commonly reported side effects of GK treatment include trigeminal neuropathy and hydrocephalus. Patients may also experience minor morbidities that significantly impact quality of life. These include tinnitus, dizziness, and post-operative headache. Historically, the literature has rarely reported such quality of life measures, and it would be useful to record such data to provide patients a more informed perspective on the experience associated with undergoing GK treatment for their VS. This void in the literature is discussed in depth in the paper by Myrseth et al. (2007). Finally, as with any radiation exposure, there is a small risk of secondary neoplasm after treatment with the GK. Patients and physicians should be aware of this risk, both before treatment and during long-term follow-up.

Treatment Outcomes

There are a number of retrospective studies and a few prospective studies on the use of GK radiosurgery for treating VSs. There have been several large meta-analyses of this data to provide a strong summary of the functional outcomes attained using the GK. These include Kaylie et al. (2000), Shin et al. (2003), Weil et al. (2006), Yamakami et al. (2003), Battaglia et al. (2006), Yang et al. (2009a, b). As described previously, despite the abundance of data, there are only a handful of comparative trials between radiosurgery and alternative treatments, and there have been no randomized trials. Therefore, definitive guidelines have not been established and the reader should be careful not to over-interpret the results below. However, the authors feel that it is appropriate to describe some of the outcomes currently being realized using the GK, with respect to both tumor control and functional preservation.

Radiosurgery for VSs has proven highly efficacious. Tumor control, defined as the cessation of growth, is accomplished in between 91% and 95% of cases based on available meta-analyses. In two studies reporting 10-year actuarial rates, Chopra et al. (2007) and Hasegawa et al. (2005), control was achieved in 91% and 92% of cases. In a retrospective study of 70 patients treated at our center with a mean dose of 12.7 Gy, control was achieved in 94% after an average of 26 months follow-up, as published in Arthurs et al. (2010).

While tumor control, as defined above, remains the traditional measure of success in the radiosurgical literature, greater clinical value may be placed in knowing the fraction of patients who go on to require secondary treatment. This statistic is commonly cited as the outcome in the literature on watchful waiting and microsurgery. For radiosurgery, between 2% and 4% will require additional treatment, based on Yamakami et al. (2003) and Battaglia et al. (2006). This is slightly higher than the 1–2% reported in the microsurgical literature, but the significance of this comparison is limited without additional studies that directly compare the two modalities.

Hearing preservation rates with GK radiosurgery have improved along with improvements in treatment planning and the movement towards lower marginal doses. In the meta-analysis data, between 44% and 63% of patients with serviceable hearing prior to treatment retained function with follow-up. Individual studies published during the past 5 years have achieved rates above 50%. These papers include Arthurs et al. (2010), Lasak et al. (2008), Chopra et al. (2007), Pollock et al. (2006), and Hasegawa et al. (2005).

The GK has also provided good preservation of facial nerve function. In all meta-analyses published after 2000, the average rate of post-operative neuropathy was less than 10%. The cited rates have been from 0% to 5% in a selection of recently published studies by Arthurs et al. (2010), Lasak et al. (2008), Chopra et al. (2007), Pollock et al. (2006), Hasegawa et al. (2005), and Myrseth et al. (2005).

Finally, we will discuss the results of two recently published comparative studies of GK radiosurgery and microsurgical resection. It should be noted that these studies represent small numbers of patients and the results were subject to some confounding based on differences between the treatment cohorts. In a prospective comparison, Pollock et al. (2006) reported that 4% of patients undergoing radiosurgery required additional treatment, as opposed to 0% in the surgical cohort. On the other hand, hearing preservation was achieved in 63% versus 5% and post-operative facial neuropathy in 0% versus 17% in the radiosurgery and microsurgery cohorts, respectively. Myrseth et al. (2005) observed, retrospectively, nearly identical rates of additional treatment (5% versus 6%) and a significant difference in facial neuropathy (5% versus 20%) between radiosurgery and microsurgery, respectively. Both trials favored GK radiosurgery over microsurgery due to the high rate of efficacy and improvements in patient-oriented outcomes. This data alone is insufficient to suggest that radiosurgery is superior to microsurgery, but it does represent an early indication that there may be some patients for whom radiosurgery might be ideal. We hope this will encourage stronger comparative studies in the future

and that better evidence will come available to help patients and their physicians make informed decisions regarding the treatment of their VSs.

References

- Arthurs BJ, Lamoreaux WT, Mackay AR, Demakas JJ, Giddings NA, Fairbanks RK, Cooke BS, Elaimy AL, Peressini B, Lee CM (2010) Gamma knife radiosurgery for vestibular schwannomas: tumor control and functional preservation in 70 patients. *Am J Clin Oncol* 34(3):265–269
- Battaglia A, Mastrodimos B, Cueva R (2006) Comparison of growth patterns of acoustic neuromas with and without radiosurgery. *Otol Neurotol* 27:705–712
- Chopra R, Kondziolka D, Niranjan A, Lunsford LD, Flickinger JC (2007) Long-term follow-up of acoustic schwannoma radiosurgery with marginal tumor doses of 12 to 13 Gy. *Int J Radiat Oncol Biol Phys* 68:845–851
- Flickinger JC, Kondziolka D, Niranjan A, Maitz A, Voynov G, Lunsford LD (2004) Acoustic neuroma radiosurgery with marginal tumor doses of 12 to 13 Gy. *Int J Radiat Oncol Biol Phys* 60:225–230
- Fucci MJ, Buchman CA, Brackmann DE, Berliner KI (1999) Acoustic tumor growth: implications for treatment choices. *Am J Otol* 20:495–499
- Gardner G, Robertson JH (1988) Hearing preservation in unilateral acoustic neuroma surgery. *Ann Otol Rhinol Laryngol* 97:55–66
- Hajioff D, Raut VV, Walsh RM, Bath AP, Bance ML, Guha A, Tator CH, Rutka JA (2008) Conservative management of vestibular schwannomas: third review of a 10-year prospective study. *Clin Otolaryngol* 33:255–259
- Hasegawa T, Fujitani S, Katsumata S, Kida Y, Yoshimoto M, Koike J (2005) Stereotactic radiosurgery for vestibular schwannomas: analysis of 317 patients followed more than 5 years. *Neurosurgery* 57:257–265; discussion 257–265
- House JW, Brackmann DE (1985) Facial nerve grading system. *Otolaryngol Head Neck Surg* 93:146–147
- Kaylie DM, Horgan MJ, Delashaw JB, McMenomey SO (2000) A meta-analysis comparing outcomes of microsurgery and gamma knife radiosurgery. *Laryngoscope* 110:1850–1856
- Lasak JM, Klish D, Kryzer TC, Hearn C, Gorecki JP, Rine GP (2008) Gamma knife radiosurgery for vestibular schwannoma: early hearing outcomes and evaluation of the cochlear dose. *Otol Neurotol* 29:1179–1186
- Linskey ME (2008) Hearing preservation in vestibular schwannoma stereotactic radiosurgery: what really matters? *J Neurosurg* 109 Suppl:129–136
- Matthies C, Samii M (1997) Management of 1000 vestibular schwannomas (acoustic neuromas): clinical presentation. *Neurosurgery* 40:1–9; discussion 9–10

- Myrseth E, Moller P, Pedersen PH, Vassbotn FS, Wentzel-Larsen T, Lund-Johansen M (2005) Vestibular schwannomas: clinical results and quality of life after microsurgery or gamma knife radiosurgery. *Neurosurgery* 56:927–935; discussion 927–935
- Myrseth E, Pedersen PH, Moller P, Lund-Johansen M (2007) Treatment of vestibular schwannomas. Why, when and how? *Acta Neurochir (Wien)* 149:647–660; discussion 660
- Nagano O, Higuchi Y, Serizawa T, Ono J, Matsuda S, Yamakami I, Saeki N (2008) Transient expansion of vestibular schwannoma following stereotactic radiosurgery. *J Neurosurg* 109:811–816
- Nikolopoulos TP, O'Donoghue GM (2002) Acoustic neuroma management: an evidence-based medicine approach. *Otol Neurotol* 23:534–541
- Paek SH, Chung HT, Jeong SS, Park CK, Kim CY, Kim JE, Kim DG, Jung HW (2005) Hearing preservation after gamma knife stereotactic radiosurgery of vestibular schwannoma. *Cancer* 104:580–590
- Pogodzinski MS, Harner SG, Link MJ (2004) Patient choice in treatment of vestibular schwannoma. *Otolaryngol Head Neck Surg* 130:611–616
- Pollack AG, Marymont MH, Kalapurakal JA, Kepka A, Sathiseelan V, Chandler JP (2005) Acute neurological complications following gamma knife surgery for vestibular schwannoma. Case report. *J Neurosurg* 103:546–551
- Pollock BE (2008) Vestibular schwannoma management: an evidence-based comparison of stereotactic radiosurgery and microsurgical resection. *Prog Neurol Surg* 21:222–227
- Pollock BE, Driscoll CL, Foote RL, Link MJ, Gorman DA, Bauch CD, Mandrekar JN, Krecke KN, Johnson CH (2006) Patient outcomes after vestibular schwannoma management: a prospective comparison of microsurgical resection and stereotactic radiosurgery. *Neurosurgery* 59:77–85; discussion 77–85
- Shin YJ, Lapeyre-Mestre M, Gafsi I, Cognard C, Deguine O, Tremoulet M, Fraysse B (2003) Neurotological complications after radiosurgery versus conservative management in acoustic neuromas: a systematic review-based study. *Acta Otolaryngol* 123:59–64
- Smouha EE, Yoo M, Mohr K, Davis RP (2005) Conservative management of acoustic neuroma: a meta-analysis and proposed treatment algorithm. *Laryngoscope* 115:450–454
- Tago M, Terahara A, Nakagawa K, Aoki Y, Ohtomo K, Shin M, Kurita H (2000) Immediate neurological deterioration after gamma knife radiosurgery for acoustic neuroma. Case report. *J Neurosurg* 93(Suppl 3):78–81
- Timmer FC, Hanssens PE, van Haren AE, Mulder JJ, Cremers CW, Beynon AJ, van Overbeeke JJ, Graamans K (2009) Gamma knife radiosurgery for vestibular schwannomas: results of hearing preservation in relation to the cochlear radiation dose. *Laryngoscope* 119:1076–1081
- Weil RS, Cohen JM, Portarena I, Brada M (2006) Optimal dose of stereotactic radiosurgery for acoustic neuromas: a systematic review. *Br J Neurosurg* 20:195–202
- Yamakami I, Uchino Y, Kobayashi E, Yamaura A (2003) Conservative management, gamma-knife radiosurgery, and microsurgery for acoustic neurinomas: a systematic review of outcome and risk of three therapeutic options. *Neurol Res* 25:682–690
- Yang I, Aranda D, Han SJ, Chennupati S, Sughrue ME, Cheung SW, Pitts LH, Parsa AT (2009a) Hearing preservation after stereotactic radiosurgery for vestibular schwannoma: a systematic review. *J Clin Neurosci* 16:742–747
- Yang I, Sughrue ME, Han SJ, Fang S, Aranda D, Cheung SW, Pitts LH, Parsa AT (2009b) Facial nerve preservation after vestibular schwannoma gamma knife radiosurgery. *J Neurooncol* 93:41–48

Vestibular Schwannoma Surgery: Use of Fat Implant to Prevent Cerebrospinal Fluid Fistula

33

Wolf Olaf Lüdemann

Abstract

For the prevention of cerebrospinal fluid fistula (CSF) after vestibular schwannoma surgery a meticulous sealing of opened air cells in the skull base bone is necessary. To find out whether muscle or fat tissue is superior for this purpose, we performed a retrospective analysis. Four hundred and twenty patients between January 2001 and December 2006 had a retrosigmoidal microsurgical removal in a standardized procedure. Opened air cells at the inner auditory canal and the mastoid bone were sealed with muscle in 283 patients and in 137 patients with fat tissue. Analysis was performed regarding the incidence of postoperative CSF fistula and correlation with the patient's tumor grade and sex. The incidence of postoperative CSF leak after application of fat tissue was lower (2.2%) than after usage of muscle (5.7%). Women suffered less from postoperative CSF leaks (3.4%) than men (5.6%). With the tumor grade we found an inverse correlation. Patients with smaller tumors seemed to have a higher rate of CSF leak than patients with large tumors without hydrocephalus. Only large tumors with severe dislocation of the brainstem and hydrocephalus showed a higher rate of CSF-leaks. Fat implant is superior to muscle implant for the prevention of CSF leak after vestibular schwannoma surgery, and should be used for the sealing of opened air cells in skull base surgery.

Introduction

Vestibular schwannoma surgery has been developed to high standards over the last decades. Today, the main treatment goals are total tumor removal and at the same time preservation of cranial nerve functions. Hearing preservation is

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reported to be possible in 39.5–67% and facial nerve function is preserved in 92–100% depending on the tumor extension as reported by Samii and Matthies (1997b) and Samii et al. (2006). There are three highly standardized surgical procedures with different advantages and indications:

1. the middle fossa approach
2. the translabyrinthine approach
3. the lateral suboccipital retrosigmoidal approach.

One of the most frequent complications of each approach in the early postoperative period is the development of CSF leakage. It is reported to happen in 2–30% after the surgical procedure (Fishman et al. 2004; Hoffman 1994; Leonetti et al. 2001). For the lateral suboccipital approach the incidence differs between 2% and 18.4% (Samii et al. 2006; Sanna et al. 2004). The main origin of CSF fistula is air cells located in the petrous bone around the inner auditory canal (IAC) and mastoid air cells, which are opened during the surgical procedure. Therefore, leakproof sealing of the air cells is fundamental. Besides several artificial materials, this can be achieved by applying autologous muscle or fat tissue, which is taken from the patient's neck during the dissection of the bone. Even after careful closure of IAC and mastoid cells, CSF leakage does happen in rare cases. To find out whether muscle or fat tissue is superior for this purpose, a retrospective analysis was reported by Lüdemann et al. (2008).

Patients and Methods

At the International Neuroscience Institute in the period from January 2001 through December 2006 there were 420 patients operated on vestibular schwannomas, 207 female and 213 male. Two hundred and twenty-one of the tumors were located on the left side and 199 on the right side. Neurofibromatosis Type 2 was found in 26 of the patients with bilateral acoustic neurinomas, and 31 patients suffered from recurrence of acoustic neurinomas after surgical removal. Six patients had been treated with Gamma-knife previously. All of the patients were operated by the same

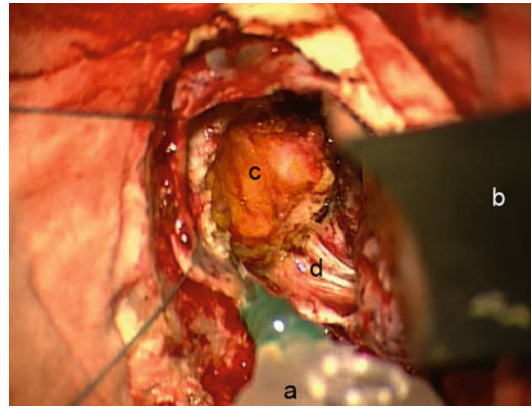


Fig. 33.1 The intraoperative image shows the sealing of the opened inner auditory canal with fat (c) tissue and fibrin (a) glue. The cerebellum is carefully shifted aside using a retractor (b) and the dura mater of the petrous bone (d) is exposed

surgeon with the patient in a semi-seated position and via a lateral suboccipital retrosigmoidal approach by craniectomy.

Meticulous sealing of opened air cells was performed in every surgery after removal of the tumor and coagulation of bleeding vessels. Thus, the bony removal of the petrous bone necessary to reach the IAC and the mastoid bone was always sealed using different materials. From January 2001 through December 2004 we used autologous muscle tissue, which was taken from the patient's neck during the surgical approach. Starting in January 2005 fat tissue was used which was taken from the subcutaneous fat harvested at the site of the surgical approach. Of the 420 patients there were 283 sealed with muscle and 137 sealed with fat tissue. The tissue was fixed in place using fibrin glue in both groups (Fig. 33.1). For the prevention of CSF fistula, the patients were informed preoperatively not to cough, blow the nose or press against the closed glottis (Valsalva maneuver) after surgery. All the patients were closely observed for signs of CSF fistula during the postoperative course. The patient's realization of CSF-leakage is different: some report peculiar impressions as metallic, salty or sweet taste. Some complain about a blocked nose as if they had a cold. The classical CSF dripping out of

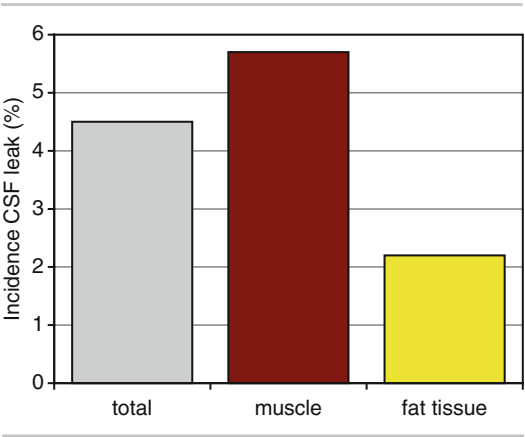
the nose is a rather infrequent first sign. Provoking CSF fistula is easy by asking the patient to bend over for up to 5 min in a sitting position, which leads to elevated intracranial pressure and increased leakage out of the nose as a diagnostic test. In case a CSF fistula is identified, our standard treatment is placing a lumbar drain with continuous flow accompanied by antibiotic therapy, and if that fails, surgical revision is done. Of the 19 patients with postoperative CSF fistula, who were treated with a lumbar drain for 7 days between January 2001 and December 2006 only one patient had to be surgically revised. We analyzed the influence of the material used for sealing the opened air cells, tumor grade, and patient sex on the incidence of CSF fistula. Tumor grade was classified with the magnetic resonance imaging according to Samii and Matthies (1997b) as follows: Class 1 is localized intrameatal only, Class 2 has an extrameatal part, Class 3a fills the cerebellopontine angle, Class 3b reaches the brain stem, Class 4a shows compression of the brain stem, and Class 4b shows severe dislocation of the brain stem and compression of the fourth ventricle leading to hydrocephalus. Statistical evaluation included Chi square test. P values less than 0.05 were considered significant.

Results

Comparison Between Fat and Muscle (Table 33.1)

The patients treated after January 2005 were sealed with fat tissue instead of muscle and showed less CSF leakage than the patients treated with muscle. Sixteen out of 283 patients whose opened air cells were treated with muscle tissue had postoperative CSF leakage. Of the 137 patients from the fat tissue group there were only 3 patients with this complication. The incidence of CSF leakage in the muscle group was 5.7% while the patients in the fat group had CSF leakage only in 2.2%. Still the difference is not significant ($p=0.089$).

Table 33.1 Histogram showing the different incidences of CSF fistulae after sealing of the opened inner auditory canal (IAC) and air cells from the mastoid bone with muscle and fat tissue. Patients from the fat group had CSF leak only in 2.2% compared with 5.7% in the muscle group ($p=0.089$)



Comparison Between Female and Male Regardless of Muscle or Fat Implant (Table 33.2)

Female patients had a lower incidence of postoperative CSF fistula than men. A leak occurred in 3.4% compared with 5.6% in men. This difference is not significant, but shows a striking tendency ($p=0.263$). The single male patient who needed a microsurgical revision was from the muscle group.

Comparison Between Tumor Size Regardless of Muscle or Fat Implant (Table 33.3)

The incidence of CSF fistula had a tendency to be inversely proportional to the tumor size. The highest incidence was 9.7% in patients with intrameatal schwannomas. The lowest incidence was 2.7% in patients with tumor grade 4a. The difference between these two groups was somewhat significant ($p=0.103$). Patients with tumor grade 2 had CSF leak in 6.3% patients with tumor grade 3a 3.6% and those with tumor grade 3b 4.3%. Patients with very large tumors graded 4b, which caused severe compression of the fourth

Table 33.2 Difference in the rate of CSF leak dependent on the patient's sex. Women have less CSF fistula (3.4%) than men (5.6%). This difference exists as well for patients

whose opened air cells were sealed with muscle as for those from the fat tissue group. The difference does not reach significance level ($p=0.27$)

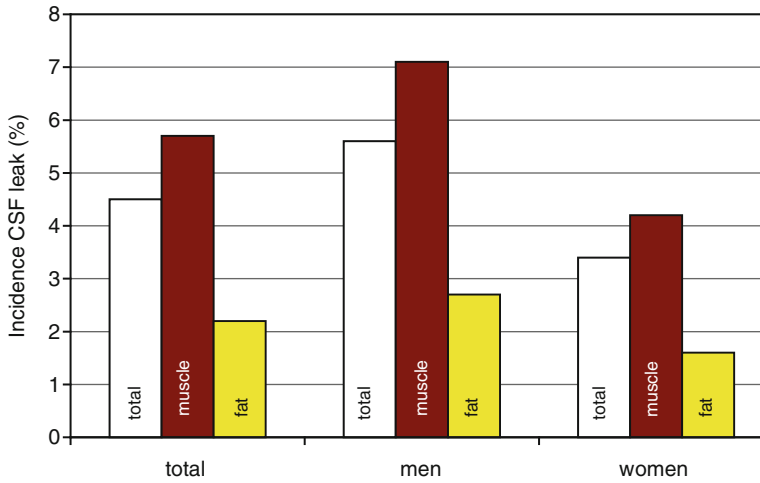
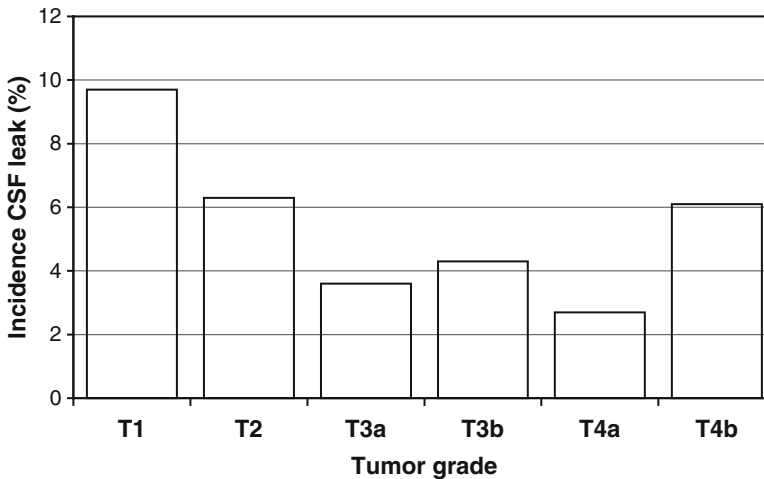


Table 33.3 Correlation between incidence of CSF fistula and tumor grade. Patients with smaller tumors had a higher incidence of postoperative CSF leak. For the intrameatal tumors the incidence was 9.7% while it was only 2.7% for

tumor grade 4a. The difference is not significant ($p=0.103$). Tumors with severe dislocation of the brain stem and compression of the fourth ventricle (T4b) had the highest incidence out of the larger tumors graded 3a–4b (6.1%)



ventricle and hydrocephalus, had a higher incidence of CSF fistula compared with the patients without hydrocephalus and large tumors (2.7% in grade 4a). The incidence for these tumors, graded 4b, was 6.1%. Gender distribution was equal in each group of fat or muscle implant. Sixty-four female patients had a fat implant, while 73 male. Muscle was implanted in 143 female and 140 male patients.

Discussion

Vestibular schwannoma surgery has been developed to high standards since the first case of successful surgery in the cerebellopontine angle by Charles Balance in 1894 described by Koerbel et al. (2005). While this tumor meant lifelong disability or even death in the beginning of the last century a total removal can be achieved in up

to 98.5% (Hori et al. 2006; Samii et al. 2006). The major goals changed towards the preservation of cranial nerve functions. Preservation of the facial nerve function is possible in 94–100% of the patients (Hori et al. 2006; Kaylie et al. 2001; Samii et al. 2006) and of a functional hearing in up to 67% (Samii et al. 2006). With the improvement of the surgical procedures and the preservation of facial and cochlear nerve function in a high percentage of the patients, other complications of the surgical procedure come into focus; for example postoperative CSF fistula, nausea, and dizziness (Stieglitz et al. 2005), ataxia and headache (Ryzenman et al. 2005). Amongst those with an incidence of 12% (range 2–16%) the CSF fistula is a rather frequent complication after tumor removal via a lateral suboccipital approach (Becker et al. 2003; Brennan et al. 2001; Hoffman 1994; Leonetti et al. 2001; Samii et al. 2006; Samii and Matthies 1997a; Sanna et al. 2004; Tonn et al. 2000). As an open connection to the paranasal sinuses which as part of the body surface house a variety of microorganisms a fistula can be the origin of meningitis and represent a constant threat for the patient's health. Reported by Bernal-Sprekelsen et al. (2005) and Santamarta et al. (2003) bacterial meningitis is reported to happen in 8–35% of untreated CSF leaks and as reported by Leonetti et al. (2001) in ~1% of the overall cases. Prevention and sufficient treatment are therefore essential for the surgical result.

The prevention of CSF fistula is based on four pillars:

1. Meticulous sealing of opened air cells as one of the final steps of the surgery.
2. Information of the patient to avoid raising the intracranial pressure unwillingly during the early postoperative course.
3. Immediate discovery of CSF fistula and
4. Sufficient treatment.

Patients should be informed regarding the actions they should avoid postoperatively as to bending over, pressing against the closed glottis or blowing the nose, they should take place before surgery as well as immediately afterwards. This way development of CSF fistula out of an initially watertight situation can be prevented. Early discovery of CSF fistula requires some experience.

It is possible to be diagnosed within 14 days after surgery. Water-like drops which drip out of the nose the first days after surgery can also be water from the irrigation which is collected inside the mastoid air cells. Typically, it becomes apparent after mobilization of the patient. The patient's realization of CSF leakage is different. It can become apparent as a peculiar impression as metallic, salty or sweet taste due to CSF running down the throat. Also, a feeling of a blocked nose and CSF dripping out of the nose can be the first sign. In any suspicious case a provocation test should be performed, but not earlier than 7 days postoperatively to prevent causing a new CSF fistula where there was none before. The patient is asked to sit on the side of the bed and to bend over as if to tie his/her shoelace. In this position he should remain for 5 min under observation whether water-like fluid drips out of the nose or not.

In a CSF-fistula, our standard treatment is immediate placement of a lumbar drain with continuous flow accompanied by antibiotic therapy for 7 days. After removal of the continuous drain the provocation test is repeated. In the case of continued CSF leak, the patient undergoes surgical revision. There was only one male patient in our study who required repeated surgery. For all the prevention and therapy the sufficient closure of opened air cells around the IAC and inside the mastoid bone is basic requirement. We use autologous material, which is taken from the patient's neck during the dissection of the bone a for the surgical approach to the posterior fossa. Until January 2005 we used muscle tissue, which was placed over the opened IAC and over opened mastoid cells in the area of the craniectomy and fixed in place with fibrin glue. From this date on we used fat, which was taken from the subcutaneous tissue around the skin incision, because muscle has the disadvantage to shrink when it is reorganized to a scar and is not as flexible as fat. The use of abdominal fat for this purpose was described earlier with good results for the translabyrinthine approach (Fishman et al. 2004; Sanna et al. 2004). The material was cut into small pieces and meticulously adapted to the opened IAC and opened mastoidal air cells. Visible air cells were not only covered but plugged with fat. All the material

was fixed in position with fibrin glue, which itself to reduces the incidence of CSF fistula (Lebowitz et al. 1995). After surgery the radiologists who evaluate the postoperative CT and MRI scans need to be informed about the fact that fat is applied to the IAC because the space occupying structure can easily be misinterpreted.

To analyze the advantages of either material we performed a retrospective study (Lüdemann et al. 2008). For all the patients disregarding the material used for air cell closure the incidence of CSF leak was an average of 4.5%, which contributes to the incidence published previously which was found to be 2–18% for the lateral suboccipital approach (Samii et al. 2006; Sanna et al. 2004). Considering the two different materials for CSF-fistula prevention this study shows a clear advantage of the fat tissue for the first time. The patients developed CSF fistula in only 2.2% compared with 5.7% in the muscle tissue group (Table 33.1). As reported by Stieglitz et al. (2009) this result becomes significant ($p=0.04$), if recurrent tumors are excluded. There are most unlikely other effects responsible for this improvement. Indeed, the surgeries using fat tissue took place after those using muscle, and the effect could at least partially be due to a learning curve of the surgeon, but on the other hand, the study only includes the last 420 of a series of more than 4,000 successful vestibular schwannoma surgeries. Apart from the change from muscle to fat tissue the surgical procedure was performed as a highly standardized procedure.

Another interesting finding is the higher incidence of CSF fistula in men than in women (Table 33.2). It occurred in 5.6% of the men and only in 3.4% of the women. While there are no differences in the surgical procedure for men and women, an increased pneumatization of the mastoid bone in men as reported by Karakas and Kavakli (2005) might be responsible for this effect. Opening of the IAC would lead to a higher number of opened air cells if the pneumatization of the petrosal bone is increased. Regarding the correlation between tumor size and cerebrospinal fluid leak rate, there is no common opinion in the literature. While Becker et al. (2003) and Bani and Gilsbach (2002) found no correlation,

Brennan et al. (2001) and Santamarta et al. (2003) found significant effects of the tumor size on the leak rate. Our own findings are, that there is an inverse proportionality between tumor grade and CSF fistula incidence (Table 33.3). Among the patients with purely intrameatal schwannomas there was an incidence of 9.7% of CSF fistula, while among patients with large tumors graded 4a, there were only 2.7%. Our theory is that bigger tumors already led to a widening of the IAC as reported by Tsunoda et al. (2000, 2001). Surgical exposure of the intrameatal tumor parts this way may lead to less opening of air cells.

Very large tumors with compression of the fourth ventricle and associated hydrocephalus play an additional role. While the incidence of CSF leak seems to decrease with the tumor size the biggest tumors graded 4b have again a higher rate of 6.1%. This phenomenon may be related to the elevated intracranial pressure that can be found in patients with 4b graded tumors in the early postoperative period. Due to pneumocephalus and blood in CSF, we know that CSF-disturbance can be prolonged even after successful removal of a vestibular schwannoma. Our findings suggest that fat tissue is superior to muscle for the closure of opened air cells at the IAC and around the craniectomy in vestibular schwannoma surgery. As the technique is relatively easy and the material is available in a sufficient amount without a further skin incision in nearly all patients, we suggest the use of fat tissue for the closure of opened air cells.

References

- Bani A, Gilsbach JM (2002) Incidence of cerebrospinal fluid leak after microsurgical removal of vestibular schwannomas. *Acta Neurochir (Wien)* 144:979–982
- Becker SS, Jackler RK, Pitts LH (2003) Cerebrospinal fluid leak after acoustic neuroma surgery: a comparison of the translabyrinthine, middle fossa, and retrosigmoid approaches. *Otol Neurotol* 24:107–112
- Bernal-Sprekelsen M, Alobid I, Mullol J, Trobat F, Tomas-Barberan M (2005) Closure of cerebrospinal fluid leaks prevents ascending bacterial meningitis. *Rhinology* 43:277–281
- Brennan JW, Rowed DW, Nedzelski JM, Chen JM (2001) Cerebrospinal fluid leak after acoustic neuroma surgery: influence of tumor size and surgical approach

- on incidence and response to treatment. *J Neurosurg* 94:217–223
- Fishman AJ, Marrinan MS, Golfinos JG, Cohen NL, Roland JT Jr (2004) Prevention and management of cerebrospinal fluid leak following vestibular schwannoma surgery. *Laryngoscope* 114:501–505
- Hoffman RA (1994) Cerebrospinal fluid leak following acoustic neuroma removal. *Laryngoscope* 104:40–58
- Hori T, Okada Y, Maruyama T, Chernov M, Attia W (2006) Endoscope-controlled removal of intrameatal vestibular schwannomas. *Minim Invasive Neurosurg* 49:25–29
- Karakas S, Kavakli A (2005) Morphometric examination of the paranasal sinuses and mastoid air cells using computed tomography. *Ann Saudi Med* 25:41–45
- Kaylie DM, Gilbert E, Horgan MA, Delashaw JB, McMenomey SO (2001) Acoustic neuroma surgery outcomes. *Otol Neurotol* 22:686–689
- Koerbel A, Gharabaghi A, Safavi-Abbasi S, Tatagiba M, Sami M (2005) Evolution of vestibular schwannoma surgery: the long journey to current success. *Neurosurg Focus* 18:10
- Lebowitz RA, Hoffman RA, Roland JT Jr, Cohen NL (1995) Autologous fibrin glue in the prevention of cerebrospinal fluid leak following acoustic neuroma surgery. *Am J Otol* 16:172–174
- Leonetti JP, Anderson D, Marzo S, Moynihan G (2001) Prevention and management of cerebrospinal fluid fistula after transtemporal skull base surgery. *Skull Base* 11:87–92
- Lüdemann WO, Stieglitz LH, Gerganov V, Sami A, Samii M (2008) Fat implant is superior to muscle implant in vestibular schwannoma surgery for the prevention of cerebrospinal fluid fistulae. *Neurosurgery* 63:38–42
- Ryzenman JM, Pensak ML, Tew JM Jr (2005) Headache: a quality of life analysis in a cohort of 1,657 patients undergoing acoustic neuroma surgery, results from the acoustic neuroma association. *Laryngoscope* 115:703–711
- Samii M, Matthies C (1997a) Management of 1000 vestibular schwannomas (acoustic neuromas): surgical management and results with an emphasis on complications and how to avoid them. *Neurosurgery* 40:11–21
- Samii M, Matthies C (1997b) Management of 1000 vestibular schwannomas (acoustic neuromas): the facial nerve-preservation and restitution of function. *Neurosurgery* 40:684–694
- Samii M, Gerganov V, Samii A (2006) Improved preservation of hearing and facial nerve function in vestibular schwannoma surgery via the retrosigmoid approach in a series of 200 patients. *J Neurosurg* 105:527–535
- Sanna M, Taibah A, Russo A, Falcioni M, Agarwal M (2004) Perioperative complications in acoustic neuroma (vestibular schwannoma) surgery. *Otol Neurotol* 25:379–386
- Santamarta D, Blazquez JA, Maillo A, Munoz A, Caballero M, Morales F (2003) Analysis of cerebrospinal fluid related complications (hydrocephalus, fistula, pseudomeningocele and infection) following surgery for posterior fossa tumors. *Neurocirugia (Astur)* 14:117–126
- Stieglitz LH, Samii A, Kaminsky J, Gharabaghi A, Samii M, Lüdemann WO (2005) Nausea and dizziness after vestibular schwannoma surgery: a multivariate analysis of preoperative symptoms. *Neurosurgery* 57:887–890
- Stieglitz LH, Wrede KH, Gharabaghi A, Gerganov VM, Samii A, Samii M, Lüdemann WO (2009) Factors affecting postoperative cerebrospinal fluid leaks after retrosigmoidal craniotomy for vestibular schwannomas. *J Neurosurg* 111:874–883
- Tonn JC, Schlake HP, Goldbrunner R, Milewski C, Helms J, Roosen K (2000) Acoustic neuroma surgery as an interdisciplinary approach: a neurosurgical series of 508 patients. *J Neurol Neurosurg Psychiatry* 69:161–166
- Tsunoda A, Komatsuzaki A, Suzuki Y, Muraoka H (2000) Three-dimensional imaging of the internal auditory canal in patients with acoustic neuroma. *Acta Otolaryngol Suppl* 542:6–8
- Tsunoda A, Terasaki O, Muraoka H, Komatsuzaki A, Kimura Y (2001) Cross-sectional shapes of the internal auditory canal in patients with acoustic neuromas. *Acta Otolaryngol* 121:627–631

Retrosigmoidal Craniotomy for Vestibular Schwannoma Patients: Postoperative Cerebrospinal Fluid Leak

34

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Abstract

Postoperative Cerebrospinal fluid (CSF) fistulas are one of the most important complications in the early period after vestibular schwannoma surgery. The aim of this study was to identify patients likely to develop CSF leaks using a retrospective analysis. Between January 2001 and December 2006, 420 patients underwent retrosigmoidal microsurgical tumor removal in a standardized procedure. Of these 420 patients, 363 underwent treatment for the first time, and 27 suffered from recurrent tumors. Twenty-six patients had bilateral tumors due to neurofibromatosis type 2, and four patients had previously undergone radiosurgical treatment. An analysis was performed to examine the incidence of postoperative CSF fistulas in all four groups. The incidence of CSF leakage was higher in the tumor recurrence group (11.1%) than in patients undergoing surgery for the first time (4.4%). There were no CSF fistulas in the neurofibromatosis type 2 group or in patients with preoperative radiosurgical treatment. Tumor size and male sex were identified as possible risk factors in a previous study. Low tumor grade, male sex, and surgery for recurrent tumors are significant risk factors for the development of CSF leaks.

Introduction

Vestibular schwannoma surgery today is possible with excellent surgical results, preservation of the facial nerve in the majority of cases and even preservation of a functional hearing in an increasing number of surgical procedures. Despite great advances in surgical technique and experience, CSF leakage remains one of the most frequent complications after vestibular schwannoma surgery. It is reported to occur in 2–18% of cases after retrosigmoidal approach

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by Samii et al. (2006) and Sanna et al. (2004). All three highly standardized procedures for vestibular schwannoma surgery, the middle fossa, translabyrinthine, and retrosigmoidal approaches, do have this complication in common. It can impair the patient's recovery through the development of bacterial meningitis. Prevention of CSF fistulas is very important, because even in case of early identification and proper treatment of this complication it may result in prolonged hospital stay, increased treatment costs, and in some cases even repeated surgery. To identify possible factors that indicate a higher risk of developing CSF fistulas after vestibular schwannoma surgery, we performed a retrospective analysis.

Patients and Clinical Methods

Patient Population

Between January 2001 and December 2006, 420 patients underwent vestibular schwannoma removal at the International Neuroscience Institute in Hannover, Germany. Two hundred seven patients were female and 213 were male. Two hundred and twenty-one of the tumors were located on the left side and 199 on the right. Twenty-six patients had bilateral tumors due to neurofibromatosis type 2. Thirty-one of the patients had recurrent vestibular schwannomas after surgical removal and 6 had undergone radiosurgical treatment previously.

Retrosigmoidal Craniotomy for Vestibular Schwannomas

The same surgeon performed all surgical procedures with the patient in a semiseated position. The patient's head was fixed in a Mayfield headholder frame and his legs were elevated to the level of the head for the prevention of air embolism. We exclusively used the retrosigmoidal approach. Facial nerve function and hearing were monitored constantly throughout the entire procedure. For exposure of the intrameatal tumor part, the posterior wall of the internal auditory canal was

drilled in all cases. The tumor was debulked and the capsule was carefully separated from the cranial nerves. After meticulous hemostasis the internal auditory canal was sealed with muscle or fat tissue, which was harvested from the neck muscles and subcutaneous tissue during the surgical approach. Additionally the tissue was fixed with fibrin glue. In a study published by Lüdemann et al. (2008) we were able to show a reduction of CSF fistula incidence by changing from muscle to fat tissue for this purpose. The dura mater was neatly closed and sealed with fibrin glue as well. Opened mastoid air cells were cleared from mucosa and also sealed using fat tissue and fibrin glue. Finally, the bone defect was closed with polymethylmethacrylate (Palacos, Haraeus Medical GmbH), followed by closure of the deep muscular layer, superficial muscular layer with fascia, subcutaneous tissue, and skin.

Prevention of Postoperative CSF Fistulas

For the prevention of postoperative CSF fistula out of an initially watertight situation, each patient was informed about this complication and ways to avoid preoperatively and in the early postoperative situation. This included avoiding to blow the nose or press against the closed glottis (Valsalva manoeuvre). Postoperatively all patients were closely observed for signs of CSF fistula. Typical symptoms are waterlike fluid dropping out of the nose, metallic, salty or sweet taste or a blocked nose. When a patient was suspected to have a CSF fistula, that patient underwent a provocation test. The patient was asked to sit on the side of the bed and bend over as if to tie his or her shoelace. The patient was observed for 5 min in this position. If there was no clear fluid coming from the nose, the test was judged to have a negative result. If fluid could be collected, it was tested for glucose. A positive result meant, that the fluid was most likely CSF. In cases of unclear results, the fluid was tested for β -trace protein to prove, that it contained CSF. One week after surgery each patient underwent such a provocation test. If the result was negative, the patient was discharged.

CSF Fistula Treatment

If the presence of CSF fistula was proven, the patient received continuous lumbar CSF drainage for 7 days in combination with prophylactic antibiotic treatment. The drainage was adjusted to produce between 150 and 200 ml in 24 h, which was expected to be ~30–40% of the daily CSF production according to Edsbacke et al. (2004). After 7 days and closure of the drainage system, the provocation test was repeated as described above. If the CSF leak continued, the patient underwent surgical revision. During the time of continuous drainage, a CSF sample was sent for microbiological testing and cellular counts every second day. The workflow of patient information, CSF fistula identification and treatment is shown in Fig. 34.1.

Literature Review

To compare our own results with the literature we reviewed 27 studies published between 1991 and 2007. The studies were acquired through a Medline search using key words “CSF”, “vestibular schwannoma” and “fistula” and referred to surgeries that occurred between 1974 and 2007. Some publications referred to the same pool of patients, so only the most recent studies were chosen for evaluation. Altogether 8,589 surgeries were reported in 23 publications (Table 34.1).

Tumor Size

The tumor sizes were measured according to intra- and extrameatal tumor extension as described by Samii and Matthies (1997). Purely intrameatal tumors were described as Class T1. Class T2 tumors had an extrameatal extension. Class T3a were tumors that filled the cerebello-pontine cistern. Class T3b tumors reached the brainstem. Tumors classified as T4a compressed the brainstem, and Class T4b tumors severely dislocated and compressed the brainstem and the

fourth ventricle, causing occlusive hydrocephalus (Fig. 34.2).

Statistical Evaluation

We analyzed the influence of bilateral tumors, previous surgery, and previous radiosurgery on the incidence rate of CSF fistulas and compared our results with those from the literature. Statistical evaluation included using the chi-square test. Probability values <0.05 were considered statistically significant.

Results

Surgical Results

Four hundred twenty patients were operated between January 2001 and December 2006. Three hundred and sixty-three patients underwent surgery for vestibular schwannoma for the first time. Twenty-seven patients were treated for recurrent tumors, 26 showed bilateral tumor growth due to neurofibromatosis type 2 and 4 patients were treated radiosurgically earlier. All patients received drilling of the IAC to allow total tumor removal, which was macroscopically achieved in 411 patients (97.8%). In nine patients total removal was not performed in order to preserve the anatomical integrity of the facial nerve. Three of these patients suffered from neurofibromatosis type 2, one had a malignant schwannoma with infiltration of the brainstem and cranial nerves. In one case an intracanalicular schwannoma showed broad infiltration of the facial and cochlear nerve, three were T4a (one of these was cystic) and one was T4b, also showing infiltration of the facial nerve. All drilled IACs were sealed after tumor removal. Until December 2004 we used fibrin glue and muscle tissue, which was taken from the patient's neck during the surgical approach. Since January 2005 we used fat tissue and fibrin glue instead. Lüdemann et al. (2008) published the reduction of the incidence of postoperative CSF fistula from 4.5% to 2.2% that

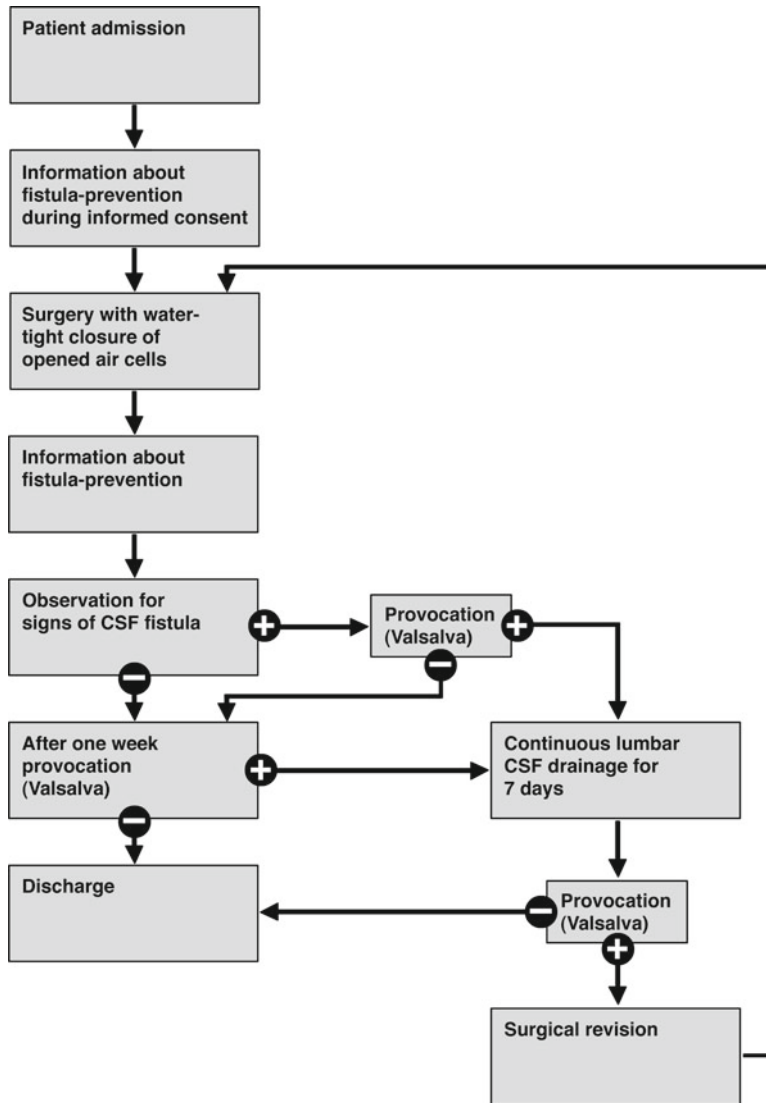


Fig. 34.1 Flow chart for detection and management of postoperative CSF fistula. The patient is informed about the risk of postoperative CSF fistula and about recommendations to avoid elevation of the intracranial pressure postoperatively during the informed consent. This information is given again directly after surgery. During the surgical procedure all opened air cells in the posterior wall of the inner auditory canal and around the craniotomy

are sealed with muscle or fat tissue and fibrin glue. Suspect patients undergo a provocation test, which leads to increased intracranial pressure (Valsalva maneuver). Fistulas are treated with continuous lumbar CSF drainage for 7 days. Success of the conservative treatment is proved by another provocation test after closure of the drain. In case of continued CSF fistula the patients undergo surgical revision

way. Of the total of 420 patients 19 developed a postoperative CSF fistula (4.52%), 7 female and 12 male. All 19 patients showed rhinorrhea. Incisional CSF fistula or otorrhea did not occur. All but one of the patients responded to conservative

treatment with continuous lumbar CSF drainage as described above. Only one male patient finally underwent surgical revision. No patient developed meningitis. Of the 283 patients who were operated before January 2005 and who had their

Table 34.1 Studies reporting cases of postoperative CSF fistula

Author	Year of publication	No. of cases	No. of VS	RS	TL	MF	TL+MF	other
Baird et al.	2007	280	130	130	130			
Bani et Gilsbach	2002	224	224	224				
Becker et al.	2003	300	300	100	100	100		
Brennan et al.	2001	303	303	75	228			
Briggs et al.	1993	43	43		43			
Bryce et al.	1991	319	319	76	203		42	
Cohen et al.	1993	476	146	146				
Cueva et Mastrodimos	2005	343	244	160	149	12		22
Falcioni et al.	1999	200	200		200			
Fayad et al.	2007	389	389		389			
Fishman et al.	2004	215	196	101	71	24		
Gormley et al.	1998	179	179	168	11			
Hoffman	1994	381	381	235	146			
Khrais et al.	2004	710	710		710			
Leonetti et al.	2001	489	455	191	209	3	63	123
Lüdemann et al.	2008	420	420	420				
Meyer et al.	2006	162	162			162		
Miyazaki et al.	2005	1,177	700	1,177				
Nutik et Korol	1995	156	156	156				
Samii et Matthies	1997	1,000	1,000	1,000				
Symon <i>et Pell</i>	1991	38	38	38				
Tonn et al.	2000	508	508	508				
Wu et al.	1999	277	277		277			
Σ		8,589	7,480	4,905	2,866	301	105	145

There were 27 studies chosen for evaluation, which were published between 1991 and 2007. 6 of them refer to the same pool of patients, so only recent ones were taken into consideration. There are 8,589 surgeries reported in 23 publications. Of these 7,480 were removal of vestibular schwannomas. 101 were CPA meningiomas and 930 were various lesions. 4,905 of them were retrosigmoidal (RS), 2,866 translabyrinthine (TL) and 301 were middle fossa (MF) approaches. Another 250 were various or combined approaches (MF+TL=combined middle fossa and translabyrinthine approach)

opened IAC sealed with muscle tissue and fibrin glue, 16 developed a CSF fistula (5.65%), compared with only 3 of the 137 patients who were operated using fat tissue and fibrin glue in the following 2 years (2.19%). This difference does not reach significance level ($p=0.11$). While the difference in the CSF fistula rate between patients of the muscle and the fat tissue group is only a tendency, it reaches significance level, if patients with recurrent tumors are excluded (1.74%, $p=0.04$). Of the 363 patients who underwent surgical tumor removal for the first time 16 (4.41%) and of the 27 patients of the tumor recurrence group 3 patients (11.11%) developed postoperative CSF fistula. There were no CSF fistulas in

patients with previous gamma-knife treatment or bilateral tumors. The difference between patients with first time surgery and those with recurrent tumors reaches significance level ($p=0.03$).

Treatment Outcome

All patients with postoperative CSF fistula received a conservative treatment initially. Of the total of 19 patients with postoperative CSF fistula 18 were successfully treated that way with continuous lumbar CSF drainage for 7 days (94.7%). One male patient, who was from the muscle tissue group, underwent successful surgical revision

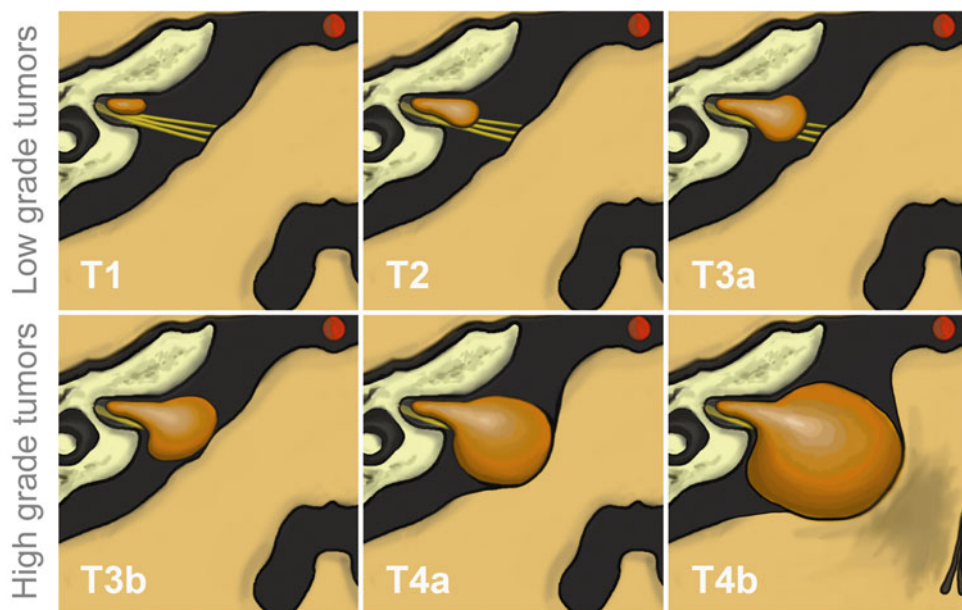


Fig. 34.2 Schematic drawing representing the used tumor classification. The tumor size was measured considering intra- and extrameatal tumor extension. *T1–T3a* Low grade tumors, *T3b–T4b* High grade tumors, *Class T1* Intrameatal tumor, *Class T2* Intra- and extrameatal tumor,

Class T3a Filling the cerebellopontine cistern, *Class T3b* The tumor reaches the brain stem, *Class T4a* Tumor compresses the brain stem, *T4b* Dislocation of the brain stem and compression of the fourth ventricle, leading to occlusive hydrocephalus

with repeated sealing of the IAC with muscle tissue and fibrin glue. There was no case of meningitis due to CSF fistula and no case of meningitis due to continuous lumbar CSF drainage under prophylactic antibiotic treatment. No patient needed middle ear or Eustachian tube obliteration.

Discussion

Advances in Vestibular Schwannoma Surgery

There has been an impressive improvement in patient outcome after vestibular schwannoma surgery over the last decades. In the beginning of the last century, diagnosis of vestibular schwannoma meant death or at least lifelong disability to the patients. While Harvey Cushing (1917) reported a mortality of 20% in 30 patients he operated for vestibular schwannoma, already

24 years later Dandy (1941) reported a reduction of the mortality to 10% in 46 patients. Improvements in microsurgical technique and experience made it possible that Samii and Matthies (1997) reported a mortality rate of 1.1% in 1,000 patients who were operated for vestibular schwannoma. Only a few years later Samii et al. (2006) reported 0% mortality in a series of 200 patients. Today the main goals of vestibular schwannoma surgery have changed, and have moved to complete tumor removal and preservation of function at the same time. In a recent review of literature Stieglitz et al. (2009) found, that in experienced hands complete tumor removal is achieved in 80–99% of cases. At the same time preservation of the facial nerve is achieved in 93–100% and preservation of a functional hearing in as many as 53.3%.

With improvement in surgical outcome, other parameters that influence patient recovery come into focus. Of these, postoperative CSF fistulas are of great interest.

Cerebrospinal Fluid Fistulas and Meningitis

CSF fistulas represent open connections between the subarachnoid space and the possibly contaminated paranasal cavities or body surface. Therefore they mean a risk for the development of meningitis through ascending bacteria. The incidence of CSF leakage in recent publications ranges from 2.2% to 28.6%. Stieglitz et al. (2009) reported an incidence of more than 10% in publications, which were published before 2000. In later publications the incidence is far lower between 2.2% and 5.7%. A study by Symon and Pell (1991) presents an exception with only 5% CSF fistulas, reported after 38 retrosigmoidal approaches for vestibular schwannoma.

The rate of postoperative meningitis is only reported by a small number of publications. Tonn et al. (2000) reported a CSF fistula rate of 12.3% (9.2% subgaleal and 3.1% rhinoliquorrhea) and a 2.9% rate of meningitis in a series of 508 vestibular schwannomas that were operated through retrosigmoidal approaches. Bryce et al. (1991) found a CSF fistula rate of 16% and a 3% rate of meningitis after 235 retrosigmoidal approaches. Samii and Matthies (1997) reported a CSF fistula rate of 9.5% and a 3% rate of meningitis after 1,000 retrosigmoidal approaches to vestibular schwannomas. Altogether, Bernal-Sprekelsen et al. (2005) and Santamarta et al. (2003) reported, that postoperative meningitis occurs in 8–35% of untreated CSF leaks, while Leonetti et al. (2001) reported 1% and Sanna et al. (2004) 0.14% incidence of meningitis in the overall cases. In our own series of 420 patients, who underwent operations between January 2001 and December 2006 at our institute, 19 (4.52%) developed a postoperative CSF fistula. All but one of the patients were successfully treated conservatively. One patient required surgical revision. There were no cases of meningitis. Although the mentioned studies in the literature and our own findings demonstrate continuous improvement of the surgical results, the still high incidence of CSF leaks indicates the need for their prevention, identification, and immediate treatment.

Causes for the Development of CSF Fistulas

For the prevention and early treatment of postoperative CSF fistulas, the identification of possible risk factors for this complication is necessary. The causes can be summarized as originating from: (1) the surgical technique and clinical management, (2) the patient's anatomy, (3) the tumor.

Surgical Approach

There are three standard approaches for the removal of vestibular schwannomas, the retrosigmoidal, translabyrinthine, and middle-fossa approaches. With the here presented results, all vestibular schwannomas are operated on via the retrosigmoidal approach. After careful osteoclastic exposure of the margins of the sigmoid sinus, the dura is opened, CSF is released from the cerebellomedullar cistern and the cerebellum is retracted carefully. The posterior wall of the IAC is drilled for revealing of the intrameatal tumor part, which then is dissected from the seventh and eighth cranial nerve. The tumor mass in the cerebellopontine angle is debulked and the cranial nerves are neatly separated from the tumor together with the arachnoid membrane. This step is followed by sealing of the opened IAC with muscle or fat tissue and fibrin glue, closure of the dura, and cranioplasty using polymethylmethacrylate. Samii and Matthies (1997) reported on the results from 1,000 vestibular schwannoma surgeries that were performed between 1978 and 1993. The CSF fistula leakage rate in this study was 9.2% after retrosigmoidal approaches, which were used exclusively. In a recent study concerning our results between 2001 and 2006, Lüdemann et al. (2008) reported a CSF leakage rate of 4.5%, and only 2.2% since 2004. At other institutes the middle fossa and translabyrinthine approaches are used with good results as well. Table 34.2 shows the CSF fistula rates and surgical approaches of important study groups in comparison. Bryce et al. (1991) reported CSF leaks in 28.6% of patients after combined translabyrinthine and middle fossa approaches before 1990. Pure translabyrinthine

Table 34.2 Incidence of postoperative CSF fistulas in the literature

Institute	Author	Year of publication (years study)	Approach/ No. of cases	Materials for air cell closure	Percentage leak
University of Toronto (Toronto, Canada)	Bryce et al.	1991 (1975–1990)	RS/76	bw	10.5%
			TL/203		11.3%
			TL+MF/42		28.6%
The National Hospital (London, England)	Symon et Pell	1991 (1987–1990)	RS/38	bd, fg	5%
Hudson valley otolaryngology (NY, USA)	Hoffman	1994 (1979–1991)	RS/235	mus, bw	16%
			TL/146		21%
Hudson valley otolaryngology (NY, USA)	Fishman et al.	1996 (1992–1995)	RS/91	fat, fg	21%
			TL/83		13%
Nordstadt Hospital, (Hannover, Germany)	Samii et al.	1997 (1978–1993)	RS/1000	mus, fg	9.2%
George Washington University Medical Center (Washington, USA)	Gormley et al.	1998 (1985–1996)	RS/175	bw, fas, mus,	13.7%
			TL/8	fg	12.5%
			Other/4		25%
Beaujon Hospital (Paris, France)	Wu et al.	1999 (1987–1998)	TL/39	fas, fat	28.2%
			TL/184	fas, fat-flap	21.7%
			TL/54	fat	7.4%
University of Würzburg (Würzburg, Germany)	Tonn et al.	2000 (1991–1997)	RS/508	mus, fg	12.3%
University of Toronto (Toronto, Canada)	Brennan et al.	2001 (1990–1999)	RS/75	bw	5.3%
			TL/228		8.7%
Klinikum Duisburg-Wedau (Duisburg, Germany)	Bani et Gilsbach	2002 (1989–2000)	RS/224	mus	7.5%
University of California (SF, USA)	Becker et al.	2003 (1979–2000)	RS/100	bw	10%
			TL/100	bw	13%
			MF/100	fat	10%
Hudson valley otolaryngology (NY, USA)	Fishman et al.	2004 (1995–2000)	RS/71	bw, fat, stf	8.5%
			TL/101		5%
			MF/24		8.3%
Gruppo Otologico (Piacenza, Italy)	Sanna et al.	2004 (1987–2002)	TL/600	fat	1.8%
			MF/54		3.8%
			RS/38		18.4%
			Other/15		
Medical University South Carolina (Charleston, USA)	Meyer et al.	2006 (1993–2004)	MF/162	mus, fg	5.5%
Johns Hopkins University (Baltimore, USA)	Baird et al.	2007 (2002–2005)	RS/130	bc	2.3%
			RS/150	bw, fat, mus, fg	12%
House Ear Institute (LA, USA)	Fayad et al.	2007 (2003–2005)	TL/389	fat	3.3%
INI (Hannover, Germany)	Lüdemann et al.	2007 (2001–2006)	RS/283	fat, fg	2.2%
			RS/137	mus, fg	5.7%

The table shows the results of different workgroups and gives information about the period, in which the surgical procedures were performed, the surgical technique and the number of patients treated

bw bone wax, *fat* fat tissue, *mus* muscle tissue, *stf* soft tissue flap, *fas* fascia, *bc* hydroxyapatite bone cement, *bd* bone dust, *fg* fibrin glue, *ot* other techniques, *RS* retrosigmoidal approach, *MF* middle cranial fossa approach, *TL* translabry-rinithine approach

and retrosigmoidal approaches involved a risk of CSF leaks in 11.3% and 10.5%, respectively. Brennan et al. (2001) from the same institute reported on surgeries since 1990 and found far lower rates of CSF leaks: 5.3% after retrosigmoidal approaches and 8.7% after translabyrinthine approaches. Hoffman (1994) reported a CSF leak incidence rate of 16% after retrosigmoidal approaches and 8.7% after translabyrinthine approaches until 1991. Fishman et al. (1996) reported on the surgical results between 1992 and 1995 from the same institute. These investigators found a 21% rate of CSF leaks after retrosigmoidal approaches and 13% after translabyrinthine approaches. For surgeries between 1995 and 2000, Fishman et al. (2004) reported a CSF fistula incidence of 8.5% after retrosigmoidal approaches, 5% after translabyrinthine approaches, and 8.3% after middle fossa approaches. A very interesting study by Becker et al. (2003) compared the three standard approaches and found the CSF leak rate after retrosigmoidal and middle fossa approaches was 10%, and the rate after translabyrinthine approaches was 13%. It is clear that all three approaches have shown improvements in the rate of postoperative complications over the years, but it cannot be determined from the literature which of the three approaches has the lowest risk for postoperative CSF leaks.

Drilling of the IAC Posterior Wall

Hoffman (1994) reported on a study of 381 vestibular schwannomas that were treated using either a translabyrinthine (146 cases) or retrosigmoidal (235 cases) approach. He found, that when the IAC was not drilled in retrosigmoidal surgeries, there was a significantly lower incidence of CSF leaks (16% after drilling of IAC, 0% if the IAC was not drilled; $p=0.014$). These findings indicate, that opened air cells in the posterior wall of the IAC are the most important origin of postoperative CSF fistulas. Lang and Kerr (1989) demonstrated, that 22% of the posterior walls of the IAC are pneumatized. Although the drilling of the IAC seems to be the main reason for the development of CSF fistulas, surgeons who do not drill it will not be able to remove the

majority of vestibular schwannomas completely. If the IAC is drilled in vestibular schwannoma surgeries, a meticulous sealing of opened air cells is necessary to prevent CSF leaks to the paranasal sinuses.

Materials Used for Sealing of Opened Air Cells at the Drilled IAC

There are various materials used for the closure of opened air cells, including artificial and autologous materials. Of the artificial materials, bone wax is the most frequently used. Bryce et al. (1991) reported on 76 retrosigmoidal surgeries performed between 1975 and 1990 using only bone wax. The rate of CSF fistulas in this study was 10.5%. Brennan et al. (2001) reported on 75 retrosigmoidal vestibular schwannoma surgeries from the same institute between 1990 and 1999, showing an improved CSF fistula rate of 5.3%. Becker et al. (2003) reported the results from 100 retrosigmoidal surgeries between 1979 and 2000 with use of bone wax for IAC air cell closure, in which the CSF fistula rate was 10%. One disadvantage of bone wax is the possibility of adverse reactions against heterologous material. One report concerning bone wax as a cause of foreign body granuloma in the cerebellopontine angle by Patel et al. (2000) recommended the removal of all bone wax fragments found in soft tissue. One very interesting study by Baird et al. (2007) reported on the use of hydroxyapatite bone cement for the reconstruction of the drilled IAC in 130 cases. It was possible to reduce the CSF fistula rate from 12% to 2.3% using this method in this study. Drawbacks were the 3.8% rate of meningitis in the hydroxyapatite group and the barely acceptable 7.3% rate of meningitis in the control group. Furthermore there were additional incisional CSF leaks in 7.7% of the patients in the hydroxyapatite group compared with 6.7% in the control group. There were no signs of visible complications on the postoperative CT scans, but the median follow-up in the hydroxyapatite group was only 3.8 months. There is no proof that possible resorption of hydroxyapatite cannot lead to late-onset rhinorrhea and meningitis. Also, the use of heterologous material adjacent to possibly nonsterile tissue harbors the danger of bone

cement infection. The still relatively high rate of meningitis (3.8%) makes it difficult to recommend such a technique. On the other hand, there are a high number of autologous materials used for sealing of the drilled IAC. Most of these materials are used in combination with fibrin glue, which Lebowitz et al. (1995) reported to reduce the incidence of CSF fistulas significantly. The most common materials used are muscle and fat tissue, but bone dust, fascia, and soft tissue flaps are used as well. Our own experience published by Lüdemann et al. (2008) has been that muscle tissue used in combination with fibrin glue allows a good sealing of the drilled IAC, but the use of fat tissue in combination with fibrin glue reduced the rate of CSF leaks to 2.2% in 283 vestibular schwannoma surgeries between 2004 and 2006. Fat tissue can be taken from the subcutaneous tissue at both sides of the skin incision for the retrosigmoidal approach in sufficient amounts. Cut into small pieces, it can easily be attached to the drilled IAC and covered with fibrin glue. The rate of postoperative meningitis in this study was 0%. The only drawback of this technique is, that radiologists not familiar with this technique can misinterpret fat tissue at the IAC as remaining or recurrent tumor.

Tumor Size

Lüdemann et al. (2008) reported a tendency, that patients with small tumors have a higher risk of developing postoperative CSF fistulas than patients with large tumors. The highest CSF fistula incidence rate was 9.7% among patients with small intrameatal schwannomas (Class T1) compares with 2.7% among patients with large brainstem compressing tumors (Class T4a). The difference was nearly significant ($p=0.103$). This finding is supported by Sanna et al. (2004), who also found a higher risk of developing a CSF fistula in patients with small tumors. We discussed several possible reasons for this phenomenon. One was, that more extensive drilling of the posterior wall of the IAC was necessary for the exposure of intracanalicular tumors. Large tumors also lead to changes of the IAC and the petrous bone through constant pressure over a long period. This way, large tumors are frequently

found in enlarged IACs. Tumor growth seems to lead to obliteration of air cells in the posterior wall of the IAC and to a reduced risk to develop CSF fistula.

Patient Sex

Again in the study by Lüdemann et al. (2008) we found a tendency, that men have a higher risk to develop CSF fistula after vestibular schwannoma surgery than women. Even though the difference does not reach statistical significance ($p=0.27$) there seemed to be a relevant difference in the anatomy of men and women which led to a higher CSF fistula incidence in men. We discussed, that the higher degree of petrous bone pneumatization in men could be the reason.

Previous Radiosurgical Treatment and Bilateral Vestibular Schwannomas

We compared the incidence of postoperative CSF fistulas of patients who underwent tumor removal for the first time and who showed no obvious additional pathologies with patients suffering from bilateral vestibular schwannomas and patients who underwent previous radiosurgical treatment. The first group, patients undergoing first-time surgery, consisted of 363 patients. Sixteen of these patients (4.4%) subsequently developed CSF fistulas. There were 26 patients with bilateral tumors due to neurofibromatosis type 2. Between 2001 and 2006, there was not a single case of postoperative CSF fistula in patients with neurofibromatosis type 2. Four patients underwent radiosurgery previously, and in this group as well, there were no cases of CSF fistulas. Duong et al. (2000) found a high risk of CSF fistulas after cerebellopontine angle tumor surgery in patients who underwent radiotherapy previously (30% incidence compared with 17% overall risk). These authors stated, that arachnoid scarring might be responsible for this dramatic increase. Unfortunately this study includes only a relatively small number of vestibular schwannomas (29 of 257 tumors; 140 meningiomas, 17 chordomas) and gives no information about the surgical approach used. Furthermore, a CSF leak incidence of 17% is relatively high compared with other studies conducted approximately at

the same time. Despite the fact that previous radiosurgical treatment can make surgical tumor removal far more difficult, we could not find any proof that the rate of CSF fistulas was higher in this group. To clearly delineate previous radiotherapy as a risk factor for CSF leaks, the number of patients in this group is too small.

Repeated Surgery for Vestibular Schwannomas

Similar to previous radiotherapy, previous surgery in the cerebellopontine angle can cause scarring of the tumor and surrounding tissue. We found indications, that previous surgery is an important risk factor for CSF leaks. Of the 27 patients who underwent repeated surgery at the International Neuroscience Institute between 2001 and 2006, 11.1% suffered from CSF fistulas. This is a significantly higher number than in the control group of 363 patients who underwent surgery for the first time with a CSF fistula incidence rate of 4.4% ($p=0.03$). This finding is also supported by Duong et al. (2000) who found a CSF fistula incidence of 50% after previous surgery compared with a 17% overall risk. Because no patient who underwent surgery needed a second surgery, we are not sure about the site of the CSF fistula, that is, whether it was in the IAC or mastoid. Possible CSF fistula sites are: (1) opened air cells at the IAC that were opened in a previous surgery and obstructed only partly by postoperative scarring; and (2) dural leaks at the mastoid air cells resulting from a scarred and hardened dura without physiological microcirculation of the dura due to a loss of capillaries and a decreased healing tendency. Thus, prevention of CSF fistulas in recurrent cases has to focus on a broader covering of the IAC with autologous fat tissue on the one hand. On the other hand, autologous fat sealed with fibrin glue should be placed routinely the second time the dura is opened after proper closure with continuous watertight stitches to reduce the risk of CSF fistulas in this group of patients with a higher rate of developing CSF fistulas. Artificial materials and autologous materials are available for the sealing of the IAC and the dura as well, although autologous materials should be preferred for a lower number of possible com-

plications and to reduce costs. If CSF fistulas are identified, they require immediate treatment for the prevention of possibly life-threatening meningitis. In most cases continuous lumbar CSF drainage is sufficient. If conservative treatment fails, surgical revision with repeated careful sealing of the drilled IAC is necessary. All of the fistulas occurring after vestibular schwannoma surgery were successfully treated using this strategy. More aggressive surgical treatment such as obliteration of the Eustachian tube therefore appears unnecessary. In the future it might be possible to predict which patients are likely to develop postoperative CSF fistulas and to observe them more closely. This might enable the surgeon to include additional precautions in the surgical and pre- and postoperative procedures.

References

- Baird CJ, Hdeib A, Suk I, Francis HW, Holliday MJ, Tamargo RJ, Brem H, Long DM (2007) Reduction of cerebrospinal fluid rhinorrhea after vestibular schwannoma surgery by reconstruction of the drilled porus acusticus with hydroxyapatite bone cement. *J Neurosurg* 107:347–351
- Bani A, Gilsbach JM (2002) Incidence of cerebrospinal fluid leak after microsurgical removal of vestibular schwannomas. *Acta Neurochir (Wien)* 144:979–982; discussion 982
- Becker SS, Jackler RK, Pitts LH (2003) Cerebrospinal fluid leak after acoustic neuroma surgery: a comparison of the translabyrinthine, middle fossa, and retrosigmoid approaches. *Otol Neurotol* 24:107–112
- Bernal-Sprekelsen M, Alobid I, Mullo J, Trobat F, Tomas-Barberan M (2005) Closure of cerebrospinal fluid leaks prevents ascending bacterial meningitis. *Rhinology* 43:277–281
- Brennan JW, Rowed DW, Nedzelski JM, Chen JM (2001) Cerebrospinal fluid leak after acoustic neuroma surgery: influence of tumor size and surgical approach on incidence and response to treatment. *J Neurosurg* 94:217–223
- Briggs RJ, Shelton C, Kwartler JA, Hitselberger W (1993) Management of hydrocephalus resulting from acoustic neuromas. *Otolaryngol Head Neck Surg* 109:1020–1024
- Bryce GE, Nedzelski JM, Rowed DW, Rappaport JM (1991) Cerebrospinal fluid leaks and meningitis in acoustic neuroma surgery. *Otolaryngol Head Neck Surg* 104:81–87
- Cohen NL, Lewis WS, Ransohoff J (1993) Hearing preservation in cerebellopontine angle tumor surgery: the NYU experience 1974–1991. *Am J Otol* 14:423–433

- Cueva RA, Mastrodimos B (2005) Approach design and closure techniques to minimize cerebrospinal fluid leak after cerebellopontine angle tumor surgery. *Otol Neurotol* 26:1176–1181
- Cushing H (1917) Tumors of the nervus acousticus and the syndrome of the cerebellopontine angle. W.B. Saunders Co., Philadelphia, p 247
- Dandy WE (1941) Results of removal of acoustic tumors by the unilateral approach. *Arch Surg* 29:1026–1033
- Duong DH, O'Malley S, Sekhar LN, Wright DG (2000) Postoperative hydrocephalus in cranial base surgery. *Skull Base Surg* 10:197–200
- Edsbacke M, Tisell M, Jacobsson L, Wikkelso C (2004) Spinal CSF absorption in healthy individuals. *Am J Physiol Regul Integr Comp Physiol* 287:R1450–R1455
- Falcioni M, Mulder JJ, Taibah A, De Donato G, Sanna M (1999) No cerebrospinal fluid leaks in translabyrinthine vestibular schwannoma removal: reappraisal of 200 consecutive patients. *Am J Otol* 20:660–666
- Fayad JN, Schwartz MS, Slatery WH, Brackmann DE (2007) Prevention and treatment of cerebrospinal fluid leak after translabyrinthine acoustic tumor removal. *Otol Neurotol* 28:387–390
- Fishman AJ, Hoffman RA, Roland JT Jr, Lebowitz RA, Cohen NL (1996) Cerebrospinal fluid drainage in the management of CSF leak following acoustic neuroma surgery. *Laryngoscope* 106:1002–1004
- Fishman AJ, Marrinan MS, Golfinos JG, Cohen NL, Roland JT Jr (2004) Prevention and management of cerebrospinal fluid leak following vestibular schwannoma surgery. *Laryngoscope* 114:501–505
- Gormley WB, Sekhar LN, Wright DC, Kameron D, Schessel D (1997) Acoustic neuromas: results of current surgical management. *Neurosurgery* 41:50–58; discussion 58–60
- Hoffman RA (1994) Cerebrospinal fluid leak following acoustic neuroma removal. *Laryngoscope* 104:40–58
- Khrais TH, Falcioni M, Taibah A, Agarwal M, Sanna M (2004) Cerebrospinal fluid leak prevention after translabyrinthine removal of vestibular schwannoma. *Laryngoscope* 114:1015–1020
- Lang J, Kerr AG (1989) Pneumatization of the posteromedial air-cell tract. *Clin Otolaryngol Allied Sci* 14:425–427
- Lebowitz RA, Hoffman RA, Roland JT Jr, Cohen NL (1995) Autologous fibrin glue in the prevention of cerebrospinal fluid leak following acoustic neuroma surgery. *Am J Otol* 16:172–174
- Leonetti J, Anderson D, Marzo S, Moynihan G (2001) Cerebrospinal fluid fistula after transtemporal skull base surgery. *Otolaryngol Head Neck Surg* 124:511–514
- Lüdemann WO, Stieglitz LH, Gerganov V, Samii A, Samii M (2008) Fat implant is superior to muscle implant in vestibular schwannoma surgery for the prevention of cerebrospinal fluid fistulae. *Neurosurgery* 63:ONS38–42; discussion 42–33
- Meyer TA, Canty PA, Wilkinson EP, Hansen MR, Rubinstein JT, Gantz BJ (2006) Small acoustic neuromas: surgical outcomes versus observation or radiation. *Otol Neurotol* 27:380–392
- Miyazaki H, Deveze A, Magnan J (2005) Neuro-otologic surgery through minimally invasive retrosigmoid approach: endoscope assisted microvascular decompression, vestibular neurectomy, and tumor removal. *Laryngoscope* 115:1612–1617
- Nutik SL, Korol HW (1995) Cerebrospinal fluid leak after acoustic neuroma surgery. *Surg Neurol* 43:553–556; discussion 556–557
- Patel RB, Kwartler JA, Hodosh RM (2000) Bone wax as a cause of foreign body granuloma in the cerebellopontine angle. Case illustration. *J Neurosurg* 92:362
- Samii M, Matthies C (1997) Management of 1000 vestibular schwannomas (acoustic neuromas): surgical management and results with an emphasis on complications and how to avoid them. *Neurosurgery* 40:11–21; discussion 21–13
- Samii M, Gerganov V, Samii A (2006) Improved preservation of hearing and facial nerve function in vestibular schwannoma surgery via the retrosigmoid approach in a series of 200 patients. *J Neurosurg* 105:527–535
- Sanna M, Taibah A, Russo A, Falcioni M, Agarwal M (2004) Perioperative complications in acoustic neuroma (vestibular schwannoma) surgery. *Otol Neurotol* 25:379–386
- Santamarta D, Blazquez JA, Maillo A, Munoz A, Caballero M, Morales F (2003) Analysis of cerebrospinal fluid related complications (hydrocephalus, fistula, pseudomeningocele and infection) following surgery for posterior fossa tumors. *Neurocirugia (Astur)* 14:117–126
- Stieglitz LH, Wrede KH, Gharabaghi A, Gerganov VM, Samii A, Samii M, Luedemann WO (2009) Factors affecting postoperative cerebrospinal fluid leaks after retrosigmoidal craniotomy for vestibular schwannomas. *J Neurosurg* 111:874–883
- Symon L, Pell MF (1991) Cerebrospinal fluid rhinorrhea following acoustic neurinoma surgery. Technical note. *J Neurosurg* 74:152–153
- Tonn JC, Schlake HP, Goldbrunner R, Milewski C, Helms J, Roosen K (2000) Acoustic neuroma surgery as an interdisciplinary approach: a neurosurgical series of 508 patients. *J Neurol Neurosurg Psychiatry* 69:161–166
- Wu H, Kalamirides M, Garem HE, Rey A, Sterkers O (1999) Comparison of different wound closure techniques in translabyrinthine acoustic neuroma surgery. *Skull Base Surg* 9:239–242

Vestibular Schwannoma (VS) Surgery: Histological Considerations and Operative Results

35

Tomio Sasaki, Koji Yoshimoto, and Suzuki O. Satoshi

Abstract

In vestibular schwannoma (VS) surgery, understanding of the tumor capsules and nerve-tumor interface is very important to achieve extensive tumor removal while preserving facial and cochlear nerve function. We here demonstrate that tumor capsules of VS consist of thin perineurium of the vestibular nerve and remaining vestibular nerve fibers based on our histological examinations. Also, we show our operative concept of “subcapsular” or “subperineurial” dissection to preserve cranial nerve functions. In this procedure, we place the cleavage plane between the tumor capsules and the tumor parenchyma, which enables us to perform nearly total removal of the tumor by tracing this cleavage plane. Finally, we present our surgical outcomes based on this histological consideration, and conclude that we could increase the possibilities of preserving facial and cochlear nerve functions by performing “subcapsular” or “subperineurial” dissection.

Introduction

The ultimate goal of vestibular schwannoma (VS) surgery is to perform extensive tumor removal while preserving facial and cochlear nerve function. Though there have been many reports about surgical techniques to preserve

the function of these nerves in VS surgery, none have discussed operative strategy based on histological consideration of the tumor capsule and facial/cochlear nerve-tumor interface. We have recently published the paper about the surgical cleavage plane for preservation of facial and cochlear nerve functions in VS surgery (Sasaki et al. 2009). In this section, we present histological features of tumor capsules and nerve-tumor interfaces of VS and show our concept of operative technique for VS dissection for preserving facial and cochlear nerve function, as well as our surgical outcomes based on this histological consideration.

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Methods

Patient Population

Between 1980 and 2007, the first author operated on 274 patients with unilateral acoustic tumors through a retrosigmoid, lateral suboccipital approach. There were 115 men and 159 women, and the mean age of the patients was 50.3 years (range from 14 to 76 years). The mean follow-up periods were 100.1 months (range from 2 to 332 months). Patients with bilateral acoustic tumors (neurofibromatosis type 2) were excluded from this study. Measurements of pure tone average (PTA) and the speech discrimination score (SDS) were performed for all patients. Out of these patients, 112 patients retained useful hearing ($PTA \geq 50$ dB and $SDS \leq 50\%$, Gardner-Robertson (GR) Classification 1 and 2) on the tumor side preoperatively. There was no pure intracanalicular tumor in our series. The state of tinnitus is also evaluated for 242 patients with enough information about pre and post operative tinnitus in medical chart.

Patients were classified into four groups depending on the extent of tumor removal (total, near-total, subtotal and partial removal). Total removal means complete removal of both tumor parenchyma and capsule. The operation was defined as having achieved near-total removal when we left a small piece of tumor capsule adhered to the facial or cochlear nerve, which was so tiny tissue that postoperative MR images didn't show any residual mass. Subtotal removal was defined when we left a tumor capsule with a small amount of tumor tissue, which could be detected in the postoperative MR images. When there was a significant amount of residual tumor tissue, the operation was defined as having achieved partial removal (estimated less than 90% removal).

Postoperative facial nerve functions were evaluated routinely at the time of discharge, 6 months and 1 year after the operation, according to the House-Brackmann (HB) grading system

(House and Brackmann 1985). And in our first 108 cases, we examined the taste dysfunction to evaluate the function of the sensory component of the facial nerve. When hearing functions were preserved, measurements of PTA and SDS were performed by audiologists again after surgery.

Histological Examination of the Tumor Capsule

Histological structures of the peripheral nerves are shown in Fig. 35.1. Nerve fibers of the peripheral nerve with surrounding endoneurium hold together to form nerve bundles, which are enveloped by perineurium. And these nerve bundles are further enveloped by outer layer of epineurium. In contrast, cranial nerves in the intradural space lack this epineurium. We hypothesized vestibular schwannoma is enveloped by perineurium and remaining vestibular nerve fibers. To verify this histological consideration, we performed detailed histological examinations of the free surfaces of the tumors in the cerebellopontine cistern in the last 21 consecutive cases with unilateral vestibular schwannomas, which were operated on between 2006 and 2007. In four patients whose useful hearing had been lost preoperatively, the tumors were removed en bloc with the cochlear nerves and the nerve-tumor interfaces were also examined. These surgical specimens were fixed in formaldehyde, cut in a plane perpendicular to the tumor surface, embedded in paraffin and sectioned at 4 μ m. The sections were routinely stained with hematoxylin and eosin, or by Masson's trichrome staining. Immunohistochemical staining was performed using an enhanced indirect immunoperoxidase technique, and the sections were lightly counterstained with hematoxylin as described previously. The primary antibodies used were: rabbit polyclonal anti-human myelin basic protein antibody (DAKO, Denmark, diluted 1:100) and anti-mouse neurofilament antibody (DAKO, Denmark, diluted 1:200).

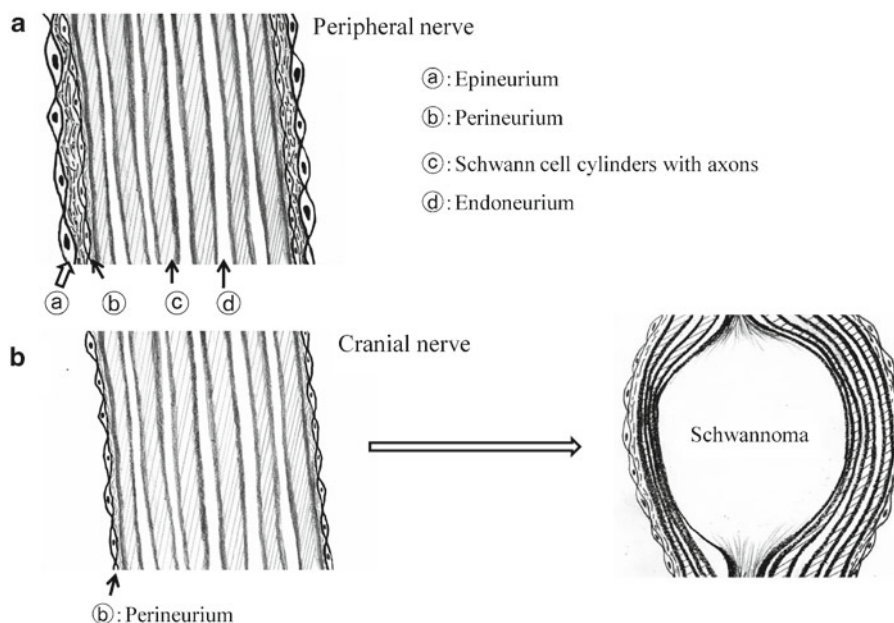


Fig. 35.1 Schematic illustration of the histology of the peripheral (a) and cranial (b) nerve. Although peripheral nerve fibers are enveloped by the thick epineurium[Ⓐ], cranial nerves in the intradural space are enveloped by the

thin perineurium[Ⓑ] without epineurium[Ⓐ]. Therefore, cranial nerves are fragile to the mechanical stress. In the case of vestibular schwannoma, the tumor is enveloped by the remaining vestibular nerve fibers and perineurium

Results

Histological Features of the Tumor Capsules and Nerve-Tumor Interfaces

As shown in Fig. 35.2, free surface of a VS reveal the presence of the thin connective tissue layer which stained blue with Masson trichrome stain (Fig. 35.2b). This connective tissue layer was very thin, being less than 10 μ m thick in most analyzed areas. There are some degenerated vestibular nerves that stained positive with MBP stain beneath this outer layer (Fig. 35.2c, f). The outer thin connective tissue layer was thought to be the perineurium of the vestibular nerve. Therefore, our results show tumor capsules of VS consists of perineurium and vestibular nerves fibers. Beneath the tumor capsule, some nerve fibers were entrapped in the tumor tissue (mixed layer) in some areas (Figs. 35.2c and 35.3a), while the mixed layer beneath the

tumor capsule is not evident in other cases (Figs. 35.2e and 35.4).

In specimens from tumors removed en bloc with the cochlear nerve, we found cochlear nerve fibers attached to the surface of the tumor tissue. Masson trichrome shows both the perineurium of cochlear nerve and the perineurium of the vestibular nerve from which the tumor had originated (Fig. 35.5a).

Operative Techniques for VS Surgery Based on Histological Consideration

Removal of Medium to Large Size Tumors Without Hearing Preservation

In this situation, surgical strategy is to accomplish complete tumor removal while preserving facial nerve function. First step is to expose tumor in the cerebello-pontine cistern by peeling off the arachnoid membrane. Next step after enough debulking

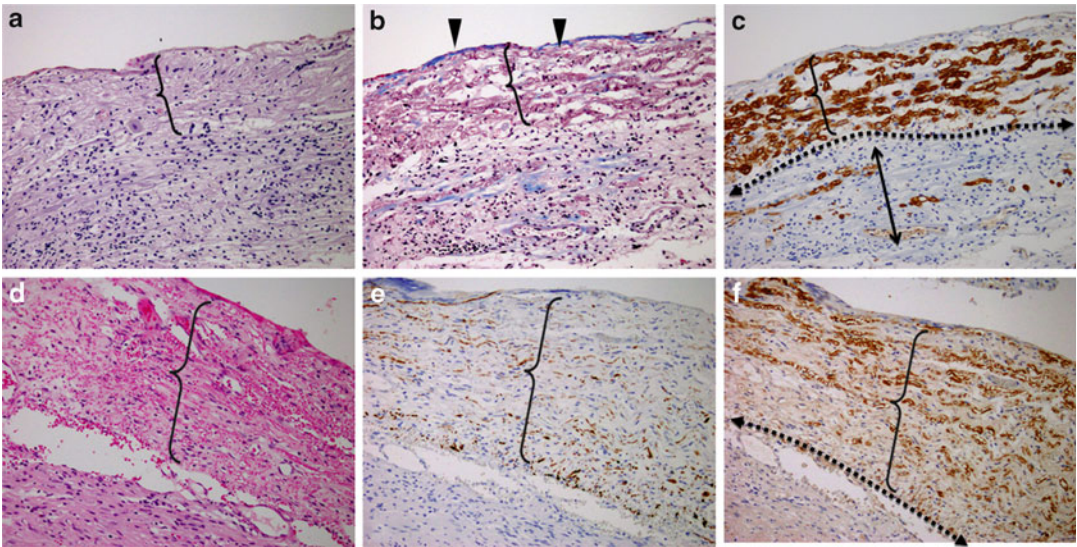


Fig. 35.2 Histological structure of the free surface of the VS. (a) H&E staining. (b) Masson trichrome staining demonstrating thin connective tissue at the surface of the tumor (arrowheads). (c) Myelin basic protein (MBP) staining showing some nerve fibers are trapped in the tumor tissue (mixed layer, *Solid arrow*). (d) H&E staining. (e) Neurofilament staining. (f) MBP staining shows no mixed layer in this specimen. *Solid brace* indicates vestibular nerve fibers, and *dotted arrows* represent an ideal cleavage plane for dissection. Original magnification $\times 20$

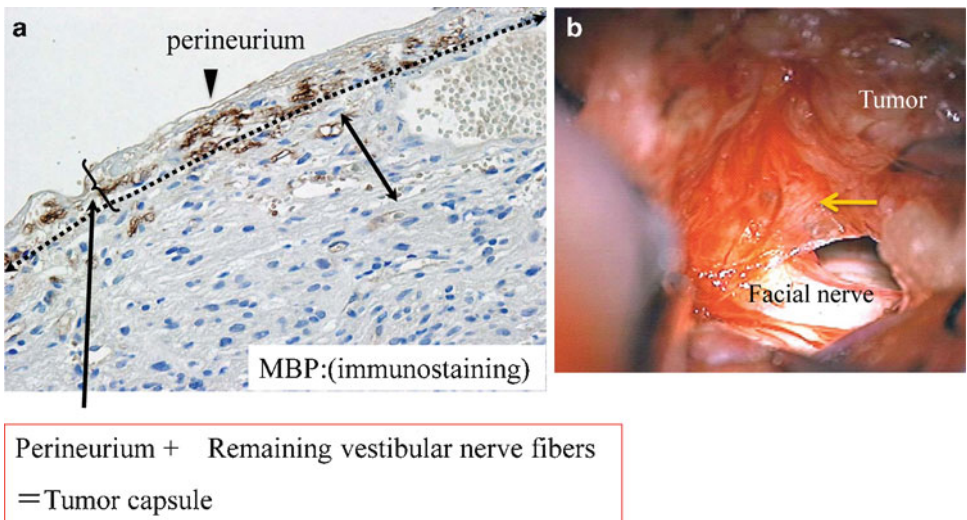


Fig. 35.3 Histological structure of tumor capsule. (a) Tumor capsule consists of perineurium and remaining vestibular nerve fibers. *Arrowhead* indicates perineurium and the *solid arrow* indicates mixed layers of nerve fibers and tumor cells. *Dotted arrow* indicates an ideal cleavage plane for dissection (subperineurial dissection). (b) Intraoperative photograph demonstrating subperineurial dissection. *Arrow* indicates thin tumor capsule

of the tumor by internal decompression is to visualize and identify the root exit (entry) zone of the seventh and eighth nerve in the brain stem. The eighth nerve fibers entering the tumor are divided, and dissection of the tumor capsule along the facial nerve is performed toward internal auditory meatus. It is the most important to understand the histological structure of the tumor capsule and

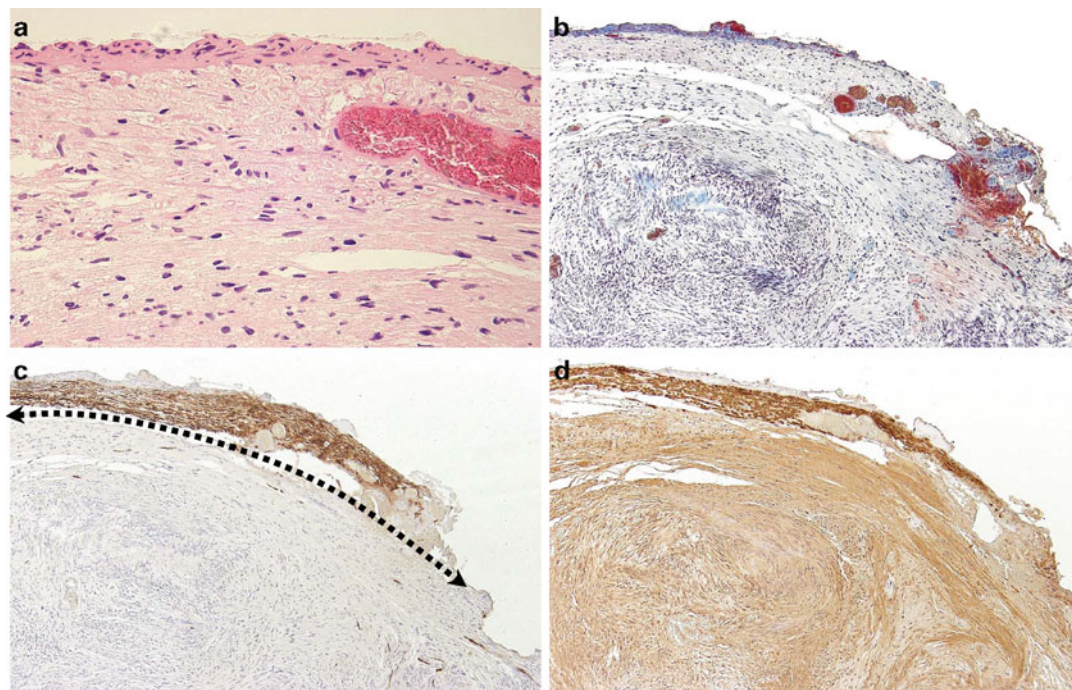


Fig. 35.4 An ideal cleavage plane is shown in *dotted arrow* in this case. (a) H&E staining. (b) Masson trichrome staining. (c) Neurofilament staining. (d) MBP staining

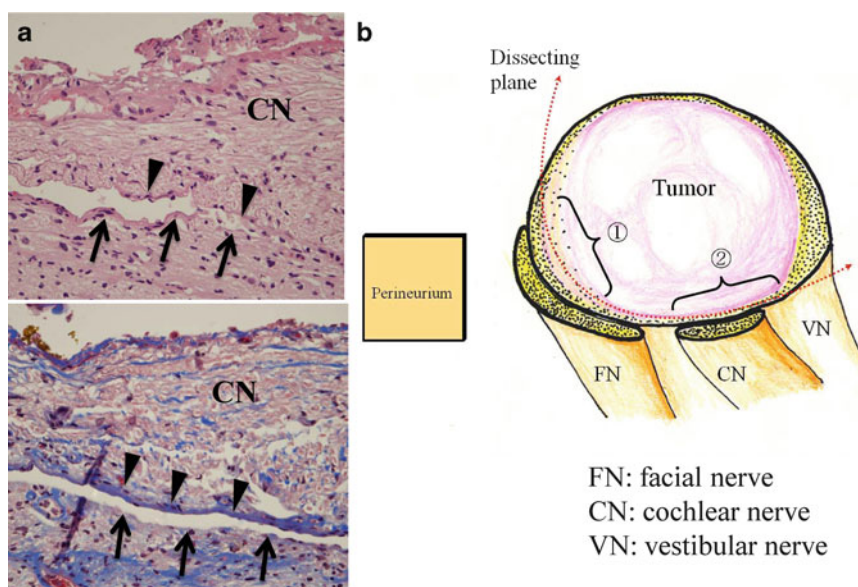


Fig. 35.5 Photomicrograph showing the cochlear nerve-tumor interface of the VS. (a) H&E Stain (*upper*). Masson trichrome staining (*lower*). *Arrowheads* indicate the perineurium of the cochlear nerve and *arrows* indicate the tumor capsule composed of the perineurium of the

vestibular nerve and connective tissue. (b) Schematic illustration of dissection plane based on histological consideration. ① Cleavage plane in the mixed layer of tumor cell and vestibular nerve fiber (mixed layer dissection). ② Subcapsular or subperineurial dissection

tumor-nerve interface during dissection step. Around the root exit zone of the facial nerve, tumor is not tightly adhered to the facial nerve. Therefore, “epicapsular” dissection of the tumor from the facial nerve is possible. Approaching to the internal auditory meatus, however, adhesion of the tumor capsule and facial nerve becomes tighter. In the region around the meatus, the facial nerve are often expanded and flattened by the tumor. In this situation, we apply “subcapsular” or “subperineurial” dissection to preserve facial nerve function (Fig. 35.3b). In this procedure, we place the cleavage plane between the tumor capsule and the tumor parenchyma, which enables us to perform nearly total removal of the tumor by tracing this cleavage plane. However, when the border between the facial nerve and tumor tissue is unclear, we need to develop the cleavage plane intentionally somewhere in the mixed layer of vestibular nerve fibers and tumor cells (mixed layer dissection) as shown in Fig. 35.5b to preserve facial nerve function, though there may be the increasing risk of tumor regrowth.

As the final step, dissection is directed to the tumor in the internal auditory canal. The vestibular and cochlear nerves are divided, and tumor is dissected away from the facial nerve toward the internal auditory meatus. Though the facial nerve can be easily separated from the tumor capsule at fundus, adhesion between the facial nerve and tumor capsule is increasing as closer to the internal auditory meatus. Damage to the facial nerve is preventable by introducing the “subcapsular” or “subperineurial” dissection and mixed layer dissection. We show the schematic description of our concept of dissection plane based on our histological consideration (Fig. 35.5b).

Removal of the Tumor with Hearing Preservation

Hearing preservation is attempted for small to medium-sized tumor with useful hearing. Any stretching to the cochlear nerve is avoided, and it should be stressed that dissection of the tumor from the cochlear nerve is performed in subcapsular or subperineurial fashion where tumor capsule is adherent to the cochlear nerve. The specimen from the tumor removed en block with

the cochlear nerve as shown in Fig. 35.5a, the perineurium can be identified both in cochlear and vestibular nerve, which imply that tumor can be successfully removed without damaging cochlear nerve on condition that dissection plane is beneath the perineurium of the vestibular nerve. With this dissection technique, it is possible to identify the cochlear nerve underneath the perineurium of the vestibular nerve or the thin tumor capsule without touching the cochlear nerve directly. When this cleavage plane is maintained during the whole procedure of tumor dissection, the cochlear nerve function is more likely to be preserved. After removing the tumor in the cerebello-pontine cistern, dissection of the tumor in the internal auditory canal is the final step. At the level of fundus, tumor capsule is not usually adhered to cranial nerves. Therefore, “epicapsular” dissection at fundus can be feasible while “subcapsular” or “subperineurial” dissection should be introduced around internal auditory meatus.

Operative Results

Tumor Removal

In our original series, total or near-total resection was achieved in 94.9% (75/79) of tumors smaller than 2 cm, 86.0% (129/150) of tumors between 2 and 4 cm, and 77.8% (35/45) of tumors larger than 4 cm in size (Table 35.1).

Facial Nerve Function

Preservation of facial nerve function is a primary goal of our cases and anatomical preservation of the facial nerve was attempted in all cases except three cases in which facial nerve was intentionally sacrificed because nerve function was already lost preoperatively due to the previous treatment. In total, anatomical preservation of the facial nerve was achieved in 97.0% (263/271) of the patients (Table 35.2). After 1 year of operation, facial nerve function was scored as good (House-Brackman Grades I and II) in 95.5% cases of anatomically preserved cases. When we confine to the small size of the tumor (<2 cm), anatomical preservation was achieved in all cases (79/79)

Table 35.1 Extent of tumor removal in 274 patients with unilateral vestibular schwannomas

Tumor diameter (cm) (no. of patients)	Total (%)	Extent of tumor removal (no. of patients)		
		Near-total (%)	Subtotal (%)	Partial (%)
Tumor < 1 (14)	14 (100)	0	0	0
1 ≤ tumor < 2 (65)	45 (69.2)	16 (24.6)	4 (6.2)	0
2 ≤ tumor < 4 (150)	73 (48.7)	56 (37.3)	14 (9.3)	7 (4.7)
Tumor ≥ 4 (45)	14 (31.1)	21 (46.7)	6 (13.3)	4 (8.9)
All patients (274)	146 (53.3)	93 (33.9)	24 (8.8)	11 (4.0)

Table 35.2 Postoperative facial nerve functions in 271 patients with unilateral VSs

Tumor diameter (cm)	No. w/anatomical FN preservation	No. w/good FN function		
		At discharge	6 Months FU	1 year FU
<1	100% (14/14)	92.9% (13/14)	92.9% (13/14)	100% (12/12)
1 ≤ tumor < 2	100% (65/65)	81.5% (53/65)	96.9% (62/64)	98.4% (61/62)
2 ≤ tumor < 4	96.6% (144/149)	67.4% (97/144)	86.2% (119/138)	96.2% (125/130)
≥4	93.0% (40/43)	70.0% (28/40)	82.5% (33/40)	87.2% (34/39)
All patients	97.0% (263/271)	72.6% (191/263)	88.7% (227/256)	95.5% (232/243)

Table 35.3 Useful hearing preservation after vestibular schwannoma resection

Tumor diameter (cm)	Percentage of useful hearing preservation (case number)
Tumor < 1	90.9% (10/11)
1 ≤ tumor < 2	57.1% (24/42)
2 ≤ tumor < 4	41.1% (23/56)
Tumor ≥ 4	66.7% (2/3)
All sizes	52.7% (59/112)

and 98.6% (73/74) of patients had good facial nerve functions at 1 year after the surgery. Regarding the function of the intermediate nerve, which is the sensory component of the facial nerve, the taste disturbance based on subjective assessment was detected in 31 patients (28.7%) before surgery and 37 patients (34.3%) after surgery in our first 108 patients (Watanabe et al. 2003). Though preoperative taste disturbance returned to normal function in 51.6% cases (16 out of 31) after surgery, 28.6% of the patients (22 out of 77) without preoperative taste disturbance suffered worsening of the taste function postoperatively. The presence of taste disturbance had no relationship with the preoperative tumor size and the extent of the resection.

Cochlear Nerve Function

Regarding cochlear nerve functions, 112 patients had useful hearing (GR class 1 and 2) on the tumor side preoperatively. In all cases, we attempted preservation of cochlear nerve during the operations, and useful hearing was retained in 52.7% (59/112) of patients postoperatively (Table 35.3). The rate of preserving useful hearing was increased up to 64.2% (34/53) in patients with tumors less than 2 cm. Of the 242 patients evaluated about tinnitus, 171 (70.7%) complained of tinnitus before surgery, and the symptom disappeared in 25.2%, improved in 33.3%, remained unchanged in 31.6%, and worsened in 9.9% of these cases after tumor removal (Kameda et al. 2010). However, postoperative tinnitus didn't show any statistically significant association with cochlear nerve resection nor useful hearing (Tables 35.4 and 35.5).

Tumor Recurrence

Out of total 274 cases, 5 (1.8%) patients required another tumor resection for tumor recurrence. Near-total removal was achieved in three patients, and subtotal or partial removal was performed at the second surgery.

Table 35.4 Effect of cochlear nerve resection on tinnitus in patients with VS

Postop tinnitus status	Preoperative tinnitus (+)		Preoperative tinnitus (–)	
	Cochlear nerve resection (85 patients)	No cochlear nerve resection (86 patients)	Cochlear nerve resection (45 patients)	No cochlear nerve resection (26 patients)
Disappeared (%)	23 (27.1)	20 (23.3)	–	–
Improved (%)	25 (29.4)	32 (37.2)	–	–
Unchanged (%)	31(36.5)	23 (26.7)	42 (93.3)	23 (88.5)
Worsened (%)	6 (7.0)	11(12.8)	3 (6.7)	3 (11.5)

Table 35.5 Postoperative useful hearing and tinnitus

Postop tinnitus status	Preoperative tinnitus present		Preoperative tinnitus absent	
	Postop useful hearing present (40 patients)	Postop useful hearing absent (37 patients)	Postop useful hearing present (12 patients)	Postop useful hearing absent (14 patients)
Disappeared (%)	10 (25.0)	10 (27.1)	–	–
Improved (%)	17 (42.5)	9 (24.3)	–	–
Unchanged (%)	6 (15.0)	13 (35.1)	10 (83.3)	12 (85.7)
Worsened (%)	7 (17.5)	5 (13.5)	2 (16.7)	2 (14.3)

Discussion

Though the importance of gentle and meticulous dissection along the facial and cochlear nerve to preserve these nerve functions has been emphasized in VS surgery, operative concept of dissection plane based on precise histological consideration has not been advocated (Ojemann 2001; Ciric et al. 2005; Gharabaghi et al. 2007). Our histological examinations demonstrate that thin connective tissue exists in the outer layer of VS. This observation is consistent with the previous report by Kuo et al. (1997) who showed that VS is surrounded by a continuous layer of thin connective tissue, which is so tiny that can't be identified during the surgical procedure. Though they speculate that this connective tissue membrane is generated by the tumor or arachnoid layer, we consider that the origin of this thin connective tissue layer is the perineurium of the vestibular nerve. Therefore, we conclude that tumor capsule consists of perineurium and residual fibers of the vestibular nerve.

Yasargil (1996) and Yasargil et al. (1977) demonstrated that VS originate from the intracanalicular segment of the vestibular nerve outside of the arachnoid space, and advocate the theory of the duplication of arachnoid layers during medial growth of VS. According to their theory,

dissection plane is placed between two layers of arachnoid membrane for atraumatic dissection. However, it is well recognized that the facial nerve is sometimes closely adherent to the tumor without any arachnoid layer between the nerve and the tumor. Lescanne et al. (2002) reported a microanatomical study of the internal acoustic meatus and its meningeal layers, and speculated that VS located inside the arachnoid apace, which contradict the theory of the duplication of arachnoid layers.

Concerning nerve-tumor interface, Jaaskelainen et al. (1994) reported immunohistochemical study of the cleavage plane between tumor and facial nerve. In this report, they described a clear-cut histological cleavage plane doesn't exist in some part of the nerve-tumor contact area where facial nerve fibers are embedded in the tumor cells. We speculate that trapped nerve fibers in tumor tissue are derived from vestibular nerve fibers not from facial nerve because perineurium of the vestibular nerve has been expanded and become unclear because of the compression of the tumor cells in case of severe adhesion between the tumor and nerve fibers. As to the cochlear nerve-tumor interface, several authors have reported on tumors without clear-cut border between the cochlear nerve and the tumor tissue (Neely 1981; Forton et al. 1989; Marquet et al. 1990), while Perre et al. demonstrated

that schwannoma cells did not infiltrate the cochlear nerve (Perre et al. 1990). In our surgical specimens obtained from the tumor removed en bloc with cochlear nerve, we found some connective tissue layers bordering the cochlear nerve and tumors. This connective tissue is considered to be the perineurium of vestibular and cochlear nerve. Therefore, there is probably better chance of preserving the nerve functions by dissecting the tumor underneath the perineurium of the vestibular nerve. This subperineurial or subcapsular dissection technique increases the possibility of preservation of functions of these cranial nerves. Though Ojemann (2006) and Ciric I et al. (2005) recommend that the arachnoid layer over the facial and cochlear nerve should be preserved to maximize the possibility of preserving the function of these nerves, our results demonstrate arachnoid layer is not present between the tumor capsule and cochlear nerve. It is possible that they might consider the thin perineurium and tumor capsule as arachnoid layer. We believe these perineurium and tumor capsule are protective layers during surgical manipulation.

The usefulness of our surgical procedure for preservation of facial or cochlear nerve functions is supported by our surgical results (Tables 35.2 and 35.3). In our original series, anatomical preservation of the facial nerve was achieved in 97.0% of the patients and 95.5% of them retained good facial nerve functions (HB grades I and II) at 1 year after surgery. When the tumor diameter was smaller than 2 cm, anatomical preservation was achieved in 100% of the patients and 98.6% of them had good facial nerve functions at 1 year after surgery. These results appeared to be one of the best surgical results of postoperative facial nerve functions among reported series (Samii et al. 2006).

Regarding cochlear nerve functions, six large studies of hearing preservation surgery containing more than 100 surgeries via a retrosigmoid approach have been reported (Ebersold et al. 1992; Glasscock et al. 1993; Samii and Matthies 1997; Koos et al. 1998; Samii et al. 2006; Wanibuchi et al. 2009). However, it is impossible to compare our surgical outcomes directly with other large series of hearing preservation surgery because of the difference of the tumor classification and inclusion criteria. Confining to the small

sized tumor, Koos et al. (1998) reported the best surgical results with overall hearing preservation rate of 78%, ranging from 23.5% to 51% in other reports (Ebersold et al. 1992; Glasscock et al. 1993; Samii and Matthies 1997; Samii et al. 2006; Wanibuchi et al. 2009). Tumor size included in their study was less than 1.8 cm, and the rate of useful hearing preservation in patients with tumors less than 1 cm (designated Grade IIA in their report) was 89%. In the present study, we achieved a high rate of useful hearing preservation (10/11; 90.9%) in the patients with tumors smaller than 1 cm. Recent reports by Samii et al. (2006) appear to describe the best surgical results of hearing preservation in large-sized tumors, although they used their own classification of tumor extension named the New Hannover Classification (Class T1, intrameatal tumor; Class T2, intra- and extrameatal tumor; Class T3a, lesion filling the cerebellopontine cistern; Class T3b, tumor reaching the brainstem; Class T4a, lesion compressing the brainstem; and Class T4b, tumor severely dislocating the brainstem and compressing the fourth ventricle) and the precise sizes of tumors were not described. According to their reports, the overall rate of functional hearing preservation (they defined functional hearing as a PTA of less than 60 dB and an SDS of more than 40%) was 51%, and the rates of functional preservation for each tumor size were 34%, 29%, and 27% in T3b, T4a and T4b tumors, respectively. In the present study, the overall rate of useful hearing preservation (we defined useful hearing as a PTA of less than 50 dB and an SDS of more than 50%) was 52.7% (59 out of 112 patients). Our series included 59 patients with tumors larger than 2 cm, all of which reached the brainstem and corresponded to the New Hannover Classification T3b or higher, and useful hearing was preserved in 25 cases (42.4%) postoperatively. Recently Wanibuchi et al. (2009) reported the good hearing preservation rate, 53.7% for large VS more than 2 cm in diameter, though their definition of useful hearing is different (PTA of less than 80 dB and SDS of more than 40 dB). Our operative results indicated that we could increase the possibilities of preserving hearing functions by performing “subcapsular” or “subperineurial” dissection.

In the literature, the tumor recurrence rate after the removal of vestibular schwannomas ranged from 0.5% to 18%. The completeness of tumor removal is a significant determinant of tumor recurrence. In our series, we experienced five recurrent cases (1.8%), which required additional tumor resection for tumor regrowth. We experienced three tumor recurrences from a small piece of residual tumor capsule after near-total resection. We speculate these tumor recurrences occur from residual tumor cells remaining after dissection of the mixed layer of vestibular nerve fibers and tumor cells. However, all of the recurrent tumors were successfully controlled by the second surgery, and all patients still retain good facial nerve functions at present. Therefore, we consider that the rate of tumor recurrence after “subcapsular” or “subperineurial” dissection is still acceptable in order to give patients a better quality of life.

In conclusion, we should keep in mind that anatomical preservation of facial and cochlear nerves does not always bring functional preservation of these nerves in VS surgery. Intracranial facial and cochlear nerve is very fragile and is susceptible to subtle damage during surgical manipulation. Therefore, our concept of cleavage plane in VS should be integrated to increase the rate of functional preservation of these nerves.

References

- Ciric I, Zhao JC, Rosenblatt S, Wiet R, O'Shaughnessy B (2005) Suboccipital retrosigmoid approach for removal of vestibular schwannomas: facial nerve function and hearing preservation. *Neurosurgery* 56:560–570
- Ebersold MJ, Harner SG, Beatty CW, Harper CM Jr, Quast LM (1992) Current results of the retrosigmoid approach to acoustic neurinoma. *J Neurosurg* 76:901–909
- Forton G, Moeneclaey L, Declau F, Marquet J (1989) The involvement of the cochlear nerve in neurinomas of the eighth cranial nerve. *Arch Otorhinolaryngol* 246:156–160
- Gharabaghi A, Samii A, Koerbel A, Rosahl SK, Tatagiba M, Samii M (2007) Preservation of function in vestibular schwannoma surgery. *Neurosurgery* 60:ONS124–ONS127
- Glasscock ME 3rd, Hays JW, Minor LB, Haynes DS, Carrasco VN (1993) Preservation of hearing in surgery for acoustic neuromas. *J Neurosurg* 78:864–870
- House JW, Brackmann DE (1985) Facial nerve grading system. *Otolaryngol Head Neck Surg* 93:146–147
- Jaaskelainen J, Paetau A, Pyykko I, Blomstedt G, Palva T, Troupp H (1994) Interface between the facial nerve and large acoustic neurinomas. Immunohistochemical study of the cleavage plane in NF2 and non-NF2 cases. *J Neurosurg* 80:541–547
- Kameda K, Shono T, Hashiguchi K, Yoshida F, Sasaki T (2010) Effect of tumor removal on tinnitus in patients with vestibular schwannoma. *J Neurosurg* 112:152–157
- Koos WT, Day JD, Matula C, Levy DI (1998) Neurotopographic considerations in the microsurgical treatment of small acoustic neurinomas. *J Neurosurg* 88:506–512
- Kuo TC, Jackler RK, Wong K, Blevins NH, Pitts LH (1997) Are acoustic neuromas encapsulated tumors? *Otolaryngol Head Neck Surg* 117:606–609
- Lescanne E, Velut S, Lefrancq T, Destrieux C (2002) The internal acoustic meatus and its meningeal layers: a microanatomical study. *J Neurosurg* 97:1191–1197
- Marquet JF, Forton GE, Offeciers FE, Moeneclaey LL (1990) The solitary schwannoma of the eighth cranial nerve. An immunohistochemical study of the cochlear nerve-tumor interface. *Arch Otolaryngol Head Neck Surg* 116:1023–1025
- Neely JG (1981) Gross and microscopic anatomy of the eighth cranial nerve in relationship to the solitary schwannoma. *Laryngoscope* 91:1512–1531
- Ojemann RG (2001) Retrosigmoid approach to acoustic neuroma (vestibular schwannoma). *Neurosurgery* 48:553–558
- Ojemann RG (2006) Suboccipital transmeatal approach to vestibular schwannoma. In: Schmidek HH, Roberts DW (eds) *Operative neurosurgical techniques*, vol 1. Elsevier, Philadelphia, pp 920–931
- Perre J, Viala P, Foncin JF (1990) Involvement of cochlear nerve in acoustic tumours. *Acta Otolaryngol* 110:245–252
- Samii M, Matthies C (1997) Management of 1000 vestibular schwannomas (acoustic neuromas): hearing function in 1000 tumor resections. *Neurosurgery* 40:248–260
- Samii M, Gerganov V, Samii A (2006) Improved preservation of hearing and facial nerve function in vestibular schwannoma surgery via the retrosigmoid approach in a series of 200 patients. *J Neurosurg* 105:527–535
- Sasaki T, Shono T, Hashiguchi K, Yoshida F, Suzuki SO (2009) Histological considerations of the cleavage plane for preservation of facial and cochlear nerve functions in vestibular schwannoma surgery. *J Neurosurg* 110:648–655
- Wanibuchi M, Fukushima T, McElveen JT Jr, Friedman AH (2009) Hearing preservation in surgery for large vestibular schwannomas. *J Neurosurg* 111:845–854
- Watanabe K, Saito N, Taniguchi M, Kirino T, Sasaki T (2003) Analysis of taste disturbance before and after surgery in patients with vestibular schwannoma. *J Neurosurg* 99:999–1003
- Yasargil MG (1996) *Microneurosurgery*, vol IVB. Thieme, New York, pp 100–123
- Yasargil MG, Smith RD, Gasser JC (1977) Microsurgical approach to acoustic neurinomas. *Adv Tech Stand Neurosurg* 4:93–129

Vestibular Schwannomas: Treatment with Bevacizumab

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Abstract

Currently there are no established medical treatments for vestibular schwannoma. Surgery may result in vestibular nerve damage and hearing loss, while radiotherapy can lead to cranial nerve neuropathy and increase the risk of secondary malignancy. Due to its effect on the VEGF pathway, which has been shown to be dysregulated in vestibular schwannomas, bevacizumab has been investigated as a potentially efficacious medical therapy for schwannomas that are unsuitable for surgical resection. Though the place of this treatment in the paradigm of vestibular schwannoma therapy remains to be fully understood, it represents a promising advance for a complex disease for which few satisfactory options currently exist.

Introduction

Vestibular schwannomas are benign neoplasms that arise from the nerve sheath of the eighth cranial nerve in its intracranial portion as it courses from the cerebellopontine angle to the internal auditory meatus. The natural history of these lesions is variable, with some lesions growing continuously, some reaching a stable plateau, and some demonstrating tumor shrinkage. Continuing growth of a cerebellopontine angle tumor results in progressive hearing loss and eventually brainstem or cerebellar compression and fourth ventricle

effacement. Watchful waiting with serial imaging is a widely accepted approach for small, asymptomatic tumors. However, in order to prevent these life-threatening consequences, microsurgical resection and/or radiotherapy is required for symptomatic or rapidly growing tumors. Surgical resection is the only established treatment option for these lesions. However, surgery puts the vestibular nerve at high risk of injury and risks worsening of hearing or even deafness.

Vestibular schwannomas are associated with both neurofibromatosis type 1 (von Recklinhausen disease) and neurofibromatosis type 2, which commonly demonstrates bilateral vestibular schwannomas as a presenting feature. Due to growth of vestibular schwannomas, the majority of patients with neurofibromatosis type 2 (NF-2) develop progressive hearing loss with extinction

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of all functional hearing during early adulthood or middle age. Radiation therapy or microsurgical removal are standard treatment options in patients with growing unilateral acoustic schwannomas, but may achieve tumor control at the expense of hearing loss (Flickinger et al. 2001; Samii et al. 2006). Stereotactic irradiation can result in facial nerve weakness, trigeminal neuropathy, vestibular dysfunction, and increased risk of secondary malignancy (Mautner et al. 2010). Radiotherapy is thus considered unsuitable for bilateral disease. Interventions in patients with NF-2 are associated with higher rates of iatrogenic hearing loss and reduced tumor control (Brackmann et al. 2001; Combs et al. 2005; Samii et al. 1997). Despite the morbidity associated with treatment, brain stem compression associated with progressive tumor growth can result in serious morbidity and mortality. However, there are no medical treatments for patients with vestibular schwannomas, with trials of traditional anticancer agents demonstrating limited clinical effects with considerable toxicity (Babovic-Vuksanovic et al. 2006, 2007; Plotkin et al. 2009).

Rationale for Use of Bevacizumab

Growth of any solid tumor greater than 2–3 mm³ in volume requires angiogenesis (Cayé-Thomasen et al. 2003). Vascular endothelial growth factor-A (VEGF) is one of a large number of factors implicated in this process. VEGF induces vasodilatation, vascular permeability, and endothelial cell proliferation and migration. While it plays a limited role in normal human cells, it is upregulated in a variety of solid tumors, and VEGF is found at higher levels in samples of peripheral blood taken from patients with these tumors. It has most often been identified in association with malignant neoplasms; the detection of increased VEGF and its receptor VEGF-1 in schwannomas is the first implication of this factor in benign neoplastic disease (Cayé-Thomasen et al. 2003; Brieger et al. 2003; Uesaka et al. 2007). Increased levels of VEGF have been correlated with higher rates of tumor growth in patients with vestibular schwannomas (Cayé-Thomasen et al. 2003).

Schwannomas associated with neurofibromatosis type 2 have been shown to have a higher vascular density and vessel diameter than those found in normal nerves, and a surface receptor expression consistent with an angiogenic phenotype (Plotkin et al. 2009).

Nevertheless, schwannomas are not particularly vascular, and the pathophysiology of schwannoma enlargement has not been previously thought to be related to angiogenesis for the most part. The proportion of schwannomas that express VEGF is unclear. Whereas it was seen in all cases in Cayé-Thomasen's series, it was present in only 1 of 34 schwannomas in another previous series (Maurer et al. 2001). In their retrospective analysis of growing vestibular schwannomas, Plotkin et al. (2009), demonstrated the presence of VEGF in 100% of schwannomas, but with a reduced expression of the VEGF receptor-2 (VEGFR-2). This suggests a different mechanism than VEGF-induced stimulation of VEGFR-2, which is a postulated mechanism of angiogenesis in a variety of other tumor types. In contrast, in schwannomas, VEGF-associated growth may be due to an imbalance in pro- and anti-angiogenic stimuli. Within the peripheral nervous system, semaphorins act as negative regulators of VEGF, and the levels of these factors have been shown to be down-regulated in vestibular schwannomas.

Erlotinib is an endothelial growth factor receptor (EGFR) tyrosine kinase inhibitor that has been reported to induce regression in a patients with NF2, and was the first indication that a targeted therapy may be useful in slow-growing, benign tumors as well as malignant neoplasms (Plotkin et al. 2008). Despite its potential, erlotinib was also given to five patients in Plotkin's later series investigating bevacizumab, with toxic effects in one and lack of efficacy in four requiring cessation of therapy (Plotkin et al. 2009).

Bevacizumab (Avastin) is a humanized monoclonal antibody directed against VEGF which has been approved for use in the treatment of colorectal, lung and breast cancers as well as glioblastoma. The treatment is generally used in these solid tumors in combination with established chemotherapeutic regimens, and has shown

efficacy in prolonging overall survival for colorectal and lung cancer patients and progression-free survival for breast cancer patients. Among patients with previously treated glioblastoma, bevacizumab efficacy was demonstrated using a response rate assessed by radiographic criteria and corticosteroid requirements. There was no effect on overall survival for breast cancer of glioblastoma patients. In accordance with its action to prevent activation of the VEGF receptor, bevacizumab has thought to be useful in tumors that express high levels of VEGF. Targeting the VEGF pathway in a mouse model of schwannoma has been shown to lead to decreased permeability, increased tumor cell death, and increased tumor control (Wong et al. 2010). On the basis of these results, the VEGF pathway inhibitor bevacizumab is believed to have great potential in the treatment of vestibular schwannomas.

Bevacizumab is used at a dose of 5 mg/kg body weight by infusion every 2 weeks. The infusion time may vary between 30 and 90 min, and treatment in studies to date has been limited only by adverse effects or tumor progression.

Treatment Outcomes Using Bevacizumab

Treatment of vestibular schwannomas associated with neurofibromatosis Type 2 has been investigated in a recent report by Plotkin et al. (2009). Ten patients who were poor surgical candidates were treated with bevacizumab at a dose of 5 mg per kilogram body weight every 2 weeks for a median duration of treatment of 12 months. Nine out of the ten tumors demonstrated shrinkage, with 6 achieving an imaging response of $\geq 20\%$ reduction in volume. Four of these maintained an imaging response at last follow up of up to 16 months. The median best response was 26% shrinkage, with a range of 44% shrinkage to 32% growth. There was an imaging response in 60% of patients, which was maintained in 4 patients during 11–16 months follow-up. These responses correlated with a decrease in gadolinium

enhancement, though without a change in the pattern of enhancement, during treatment.

Plotkin et al. (2009), also assessed hearing response in seven patients in their cohort. Four out of seven patients eligible for a hearing assessment had a hearing response, two had stable hearing, and there was one case of progressive hearing loss. Among those who improved, two patients gained useful hearing and were able to resume school and work, and these results were durable over the course of follow-up. However, the relationship between radiographic response and hearing response was not straightforward, with one patients reporting decline in word recognition and one patient reporting stable hearing despite tumor shrinkage in both cases.

In another recent report, Mautner et al. (2010), demonstrated that bevacizumab can induce regression of schwannomas in patients with neurofibromatosis type 2. In two cases, bevacizumab induced regression of progressive vestibular schwannomas by more than 40%. Hearing was substantially improved in one case by both subjective and objective audiometric measures, in which the drug was used for a period of 6 months.

The mechanism of hearing improvement with bevacizumab may be related to tumor shrinkage, as well as reduced intra-neural edema (Plotkin et al. 2009). Alternatively, it has been suggested that bevacizumab may promote vascular normalization and improved delivery of other chemotherapeutic agents to the tumor, resulting in more efficient elimination of tumour cells (Diao et al. 2009). This theory has been strengthened by recent work in a mouse model of schwannoma in which bevacizumab was shown to normalize the vasculature of schwannoma xenografts (Wong et al. 2010). In this context, bevacizumab may help to re-establish a natural balance between pro- and anti-angiogenic factors, VEGF and semaphorin 3.

Adverse Effects and Limitations

The most common adverse effects in patients with schwannomas undergoing bevacizumab therapy are raised aminotransferases, proteinuria,

hypertension, delayed wound healing, hyperkalemia, and hyperbilirubinemia. Bevacizumab recently received an FDA black box warning due to an increased risk of gastrointestinal perforation, hemorrhage and an increased risk of wound dehiscence and other surgical complications. Among patients with vestibular schwannomas who have been treated with bevacizumab in the literature, there have been no grade 3 or 4 adverse events, and no patient has discontinued treatment due to adverse effects. It should be noted, however, that only a very small number of patients have so far received treatment with bevacizumab. Further use of the drug in vestibular schwannoma patients should include careful monitoring for these known adverse effects.

Furthermore, bevacizumab has only been investigated in patients who are considered to be poor surgical candidates, and its use remains as a treatment option for those in whom surgery or radiotherapy is contraindicated or in whom the risk of post-procedural hearing loss would be unacceptably high. Despite its promise, bevacizumab remains an investigational agent in the treatment of vestibular schwannomas. Plotkin et al.'s initial report has been criticized for a lack of control arm (Guntinas-Lichius 2009), though the likelihood of shrinkage of untreated schwannomas is low and would have been unlikely to change the observed effect. The efficacy of bevacizumab on its own, as opposed to as an adjuvant therapy to maximise the effect of cytotoxic chemotherapy, remains open (Diao et al. 2009) and will only be definitively elucidated in larger, blinded and randomized trials.

In conclusion, benign nervous system tumors are notoriously resistant to standard cytotoxic chemotherapy. Though the potential of bevacizumab to induce tumor shrinkage and even improve clinical outcomes in some patients, the long term risk-benefit profile and efficacy in randomised, blinded studies remains to be seen. Surgical resection of symptomatic vestibular schwannomas is still the first-line treatment option for the majority of patients. Bevacizumab has accrued some promising early data, with a strong rationale for use in vestibular schwannomas

and early results demonstrating radiographic and clinical response in small series. But the position of bevacizumab in the array of treatment options currently available will require controlled studies in specific patient populations. Whether bevacizumab is ultimately used as a salvage therapy in patients unsuitable for surgery, or as a neo-adjuvant agent to shrink tumors prior to surgical resection, the drug will likely play a role in the future management of this complex and challenging disease.

References

- Babovic-Vuksanovic D, Ballman K, Michels V, McGrann P, Lindor N, King B, Camp J, Micic V, Babovic N, Carrero X, Spinner R, O'Neill B (2006) Phase II trial of pirfenidone in adults with neurofibromatosis type 1. *Neurology* 67:1860–1862
- Babovic-Vuksanovic D, Widemann BC, Dombi E, Gillespie A, Wolters PL, Toledo-Tamula MA, O'Neill BP, Fox E, MacDonald T, Beck H, Packer RJ (2007) Phase I trial of pirfenidone in children with neurofibromatosis 1 and plexiform neurofibromas. *Pediatr Neurol* 36:293–300
- Brackmann DE, Fayad JN, Slattery WH 3rd, Friedman RA, Day JD, Hitselberger WE, Owens RM (2001) Early proactive management of vestibular schwannomas in neurofibromatosis type 2. *Neurosurgery* 49:274–280
- Brieger J, Bedavanija A, Lehr HA, Maurer J, Mann WJ (2003) Expression of angiogenic growth factors in acoustic neurinoma. *Acta Otolaryngol* 123:1040–1045
- Cayé-Thomasen P, Baandrup L, Jacobsen GK, Thomsen J, Stangerup SE (2003) Immunohistochemical demonstration of vascular endothelial growth factor in vestibular schwannomas correlates to tumor growth rate. *Laryngoscope* 113:2129–2134
- Combs SE, Volk S, Schulz-Ertner D, Huber PE, Thilmann C, Debus J (2005) Management of acoustic neuromas with fractionated stereotactic radiotherapy (FSRT): long-term results in 106 patients treated in a single institution. *Int J Radiat Oncol Biol Phys* 63:75–81
- Diao JS, Xia WS, Guo SZ (2009) Hearing improvement after bevacizumab for neurofibromatosis type 2 (letter). *N Engl J Med* 361:1810
- Flickinger JC, Kondziolka D, Niranjan A, Lunsford LD (2001) Results of acoustic neuroma radiosurgery: an analysis of 5 years' experience using current methods. *J Neurosurg* 94:1–6
- Guntinas-Lichius OH (2009) Hearing improvement after bevacizumab for neurofibromatosis type 2 (letter). *N Engl J Med* 361:1809–1810

- Maurer J, Betawanja A, Brieger G, Mann W (2001) Growth factors of vestibular schwannoma [abstract]. *Skull Base* 2:37–38
- Mautner VF, Nguyen R, Kutta H, Fuensterer C, Bokenmeyer C, Hagel C, Friedrich RE, Panse J (2010) Bevacizumab induces regression of vestibular schwannomas in patients with neurofibromatosis type 2. *Neuro Oncol* 12:14–19
- Plotkin SR, Singh MA, O'Donnell CC, Harris GJ, McClatchey AI, Halpin C (2008) Audiologic and radiographic response of NF2-related vestibular schwannoma to erlotinib therapy. *Nat Clin Pract Oncol* 5:487–491
- Plotkin SR, Stemmer-Rachmaninov AO, Barker FG II, Halpin C, Padera TP, Tyrrell A, Sorensen AG, Jain RK, di Tomaso E (2009) Hearing improvement after bevacizumab in patients with neurofibromatosis type 2. *N Engl J Med* 361:358–367
- Samii M, Matthies C, Tatagiba M (1997) Management of vestibular schwannomas (acoustic neuromas): auditory and facial nerve function after resection of 120 vestibular schwannomas in patients with neurofibromatosis 2. *Neurosurgery* 40:696–705
- Samii M, Gerganov V, Samii A (2006) Improved preservation of hearing and facial nerve function in vestibular schwannoma surgery via the retrosigmoid approach in a series of 200 patients. *J Neurosurg* 105:527–535
- Uesaka T, Shono T, Suzuki SO, Nakamizo A, Niino H, Mizoguchi M, Iwaki T, Sasaki T (2007) Expression of VEGF and its receptor genes in intracranial schwannomas. *J Neurooncol* 83:259–266
- Wong HK, Lahdenranta J, Kamoun WS, Chan AW, McClatchey AI, Plotkin SR, Jain RK, di Tomaso E (2010) Anti-vascular endothelial growth factor therapies as a novel therapeutic approach to treating neurofibromatosis-related tumors. *Cancer Res* 70:3483–3493

Management of Vestibular Schwannoma Patients: Quality of Life Outcomes After Treatment

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Abstract

The optimum management of vestibular schwannoma patients remains controversial. With improvement in patient outcomes and differing goals of contemporary treatment modalities, patient evaluated ‘quality of life (QOL)’ as an outcome measure has become important. The literature regarding comparison of QOL outcomes between treatment modalities for vestibular schwannoma patients are reviewed. Three prospective Level II studies were identified. Two studies compared gamma knife radiosurgery with surgery, and one study compared observation with surgery as well as LINAC-based stereotactic radiation treatments (fractionated radiotherapy or radiosurgery). All three studies included measures of QOL using the SF-36 inventory. All three studies demonstrated comparable outcomes with regards to tumor growth control. Grade I-II facial nerve preservation was more frequent for radiation in two studies and equivalent with surgery in the third. Hearing preservation was less frequent in the surgical cohorts. SF-36 scores at last follow-up were equivalent between surgery and radiation therapy in two studies; in the third study, there were only small differences. Despite differences in “physician-evaluated” outcome measures between surgery and radiation therapy, and early retrospective patient surveys showing less favorable surgical outcomes, contemporary patients with vestibular schwannomas appear to enjoy good QOL regardless of treatment modality.

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Introduction

Vestibular schwannoma are benign tumors of the vestibular portion of the eighth cranial nerve with an approximate annual incidence of 1:100,000 (Frohlich and Sutherland 1993). Earlier detection

of these tumors means many are diagnosed at small to medium sizes (i.e., ≤ 3 cm extrameatal diameter), and the vast majority of patients are highly functioning and minimally to mildly symptomatic. Their natural history has been defined in several recent studies: as few as 28.9% of tumors involving the cerebellopontine cistern and 17% of purely intrameatal tumors show demonstrable growth over time (Stangerup et al. 2006). Growth, when documented, is generally slow and varies from 1 to 3 mm/year (Stangerup et al. 2006; Sughrue et al. 2010; Godefroy et al. 2009). A recent review of the literature revealed an overall hearing preservation rate of 54% with observation (Sughrue et al. 2010). QOL measures with observation have also found to be stable over time in many patients (Godefroy et al. 2009; Di Maio and Akagami 2009). Clearly, this favourable natural history requires clinicians to critically address selection criteria for surgery and radiation treatments based on appropriate and comprehensive surrogate markers of treatment efficacy. This chapter discusses the importance of patient-evaluated outcomes with particular focus on QOL and summarizes the currently available literature with regards to comparative studies and assessment of management paradigms for vestibular schwannoma patients.

Evolution in Outcomes of Vestibular Schwannoma Management

Historical outcomes in the treatment of vestibular schwannomas can be divided into eras beginning with the early surgical reports from the early nineteenth century, with nearly unanimous lack of survival, to advances made by Cushing (1917) who published his surgical series, with a substantial decrease in mortality to 10–15%, and Dandy (1925), who reported even better survival. The introduction of the microscope by otolaryngologists by the 1950s, the collaboration between neurotologists and neurosurgeons, and the use of intra-operative monitoring have all subsequently led to excellent facial motor preservation rates and near-zero mortality. Hearing preservation is also reported with varying success, however

generally speaking not to the extent of radiation. In a similar manner, gamma knife radiosurgery and LINAC stereotactic radiotherapy/radiosurgery have evolved substantially with optimization of therapeutic doses and more accurate delivery systems. These modalities offer tumor control rates of 90% or more, and combined rates of persistent facial or trigeminal neuropathy of $<10\%$; hearing preservation for gamma knife varies from 33–79% to 63–100% for stereotactic radiotherapy in recent literature reviews (Sweeney et al. 2009; Sakamoto et al. 2009; Battista 2009). In addition, with better natural history studies and the realization that significant proportion of tumors seem to exhibit a more benign natural history, simple observation has become an accepted management option. Due to these very effective but differing management options and outcomes, traditional outcome measures for comparison of management options are not optimal.

Importance of QOL

Traditional methods to assess the value of an intervention have used “physician-evaluated” outcomes that can be easily objectified. For vestibular schwannoma patients, such measures include survival, tumor growth control or extent of resection, facial nerve function, and serviceable hearing preservation. Despite their advantages, physician-evaluated outcomes may not reflect in a comprehensive manner the relative benefit of a given intervention. “Patient-evaluated” outcomes refer to subjective endpoints derived from patient reports, and include items such as self-reported symptoms (tinnitus, dizziness) and QOL (Akagami et al. 2002; Velanovich 2007).

QOL has been defined as an individual’s perception of their position in life in the context of the culture and value systems in which they live and in relation to their goals, expectations, standards, and concerns (Urbach 2005). Health related QOL has gained popularity in outcomes research including surgical trials. While a patient’s QOL is implicitly taken into consideration with clinical decision-making, its measurement in a robust method remains challenging. QOL inventories can be

generic or disease-specific. Generic measures capture the overall health of the patient at the expense of responsiveness to changes in disease-specific clinical changes (e.g. facial weakness following vestibular schwannoma surgery). Furthermore, they allow comparison of health related QOL across different diseases and different treatments. Disease-specific inventories are in comparison more sensitive to clinically-relevant changes, but may amplify treatment effects and give false impressions about the relative effectiveness of one treatment modality over another. Given their relative benefits, the use of both types of inventories together may have added benefit in certain scenarios.

The Medical Outcomes Trust Short Form 36 (SF-36) is a generic health-related QOL questionnaire that consists of 36 items and assesses eight different health dimensions (Ware 1991). A physical and mental component composite score as well as total score are ultimately derived. The SF-36 has been demonstrated to be both valid and reliable, and is applied extensively in the medical literature across disciplines and disease-states, including studies of vestibular schwannoma management. Whether or not the SF-36 is an appropriate inventory for vestibular schwannoma patients remains a matter of debate. In light of the generally good QOL in vestibular schwannoma patients who are observed (Godefroy et al. 2009), a reasonable hypothesis when assessing either surgery or radiation therapy for vestibular schwannomas would be that a “successful” treatment preserves QOL, in which case the SF-36 would seem to be an appropriate measurement tool. The effect of individual symptoms (facial weakness, hearing, tinnitus, dizziness, etc.) is likely more appropriately addressed with disease/symptom-specific inventories.

Comparative Studies of QOL in Vestibular Schwannoma Patients

Comparative studies of vestibular schwannoma management are inherently prone to a number of limitations. Few patients consent to randomization (the study by Myrseth et al. 2009 provides a

good example of this). While prospective comparative studies offer interesting results for interpretation, their results cannot be extrapolated to outside institutions or lead to generalizing statements regarding superiority of one treatment modality over another. Differences in recruitment, selection, surgical/radiation treatment expertise, and clinical outcomes between centers limit individual studies to appraisals of local management protocols, which are often affected by strong institutional biases regarding who to observe and preference of treatment modality. While the majority of outcomes for surgery can be assessed in the short term (hearing preservation, immediate post-operative facial function, extent of tumor resection), longer follow-up periods are required to assess radiation therapy. Studies with long follow-up periods may demonstrate a “catch-up” effect where the upfront adverse surgical outcome can improve over time while delayed/late effects of radiation can potentially become more significant over time (Di Maio and Akagami 2009). A number of retrospective comparison studies have been reported (Henzel et al. 2009; Pollock et al. 1995; van Roijen et al. 1997; Karpinos et al. 2002; Regis et al. 2002; Myrseth et al. 2005; Martin et al. 2008; Banerjee et al. 2008; Lin et al. 2005), most of which have been reviewed by Pollock (2008). Of these Level III (Sackett and Richardson 2000) studies, only one included QOL measures (Henzel et al. 2009). Henzel et al. (2009) compared 3d volumetric changes and QOL outcomes between 35 LINAC radiosurgery and 39 fractionated radiation therapy patients in a retrospective comparison from Marburg, Germany. SF-36 data did not differ between or within groups with a median follow-up of 50 months.

In the current literature, three prospective non-randomized studies provide the highest available evidence (Level II based on Sackett and Richardson (2000) classification) comparing treatment modalities in vestibular schwannoma patients (Table 37.1 – Di Maio and Akagami 2009; Myrseth et al. 2009; Pollock et al. 2006). In addition, all three included formal QOL assessments using the SF-36. Pollock et al. (2006) published the first prospective study comparing

Table 37.1 Summary of prospective comparative studies of QOL outcomes in vestibular schwannoma patients

Reference; Location	Treatments	N	Mean tumor diameter	Tumor control rates	CN preservation rates		QOL	Other	Follow-up
Pollock et al. (2006); Rochester, U.S.A.	GK SRS	46	12.3 mm	96% control	98%	9/18 ^a	SF-36: higher scores in favour of GK for mental health, bodily pain and mental composite scores	No difference in proportion/intensity of headache or tinnitus at last follow-up	42 months
	Surgery RS = 25 TL = 9	36	14.1 mm	33/36 total resection	83%	0/11 ^a		Better dizziness handicap inventory scores in SRS group at last follow-up	
Myrseth et al. (2009); Bergen, Norway	GK SRS	60	16 mm	NR	59/60	17/25	SF-36: no within- or between-group differences GBI: better 2-year outcomes for total, general and physical scores	No difference in proportion of patients with tinnitus, imbalance or vertigo	24 months
	Surgery (RS)	28	18 mm	23/28 total resection	23/28	0/13			
Di Maio and Akagami (2009); Vancouver, Canada	Observation	47	13 mm	78.9% control	100%	NR	SF-36: no between- and within-group differences for the observation, radiation therapy and surgery ≤3 cm groups from 6 weeks through last follow-up	No difference in self-reported tinnitus, dizziness and useful hearing between observation, radiation therapy and surgery (≤3 cm)	31.8 months
	LINAC XRT	48	19 mm	95.7% control	95.8%	13/21			
	23fXRT								
	25 SRS	134	18 mm (≤3 cm) 37 mm (>3 cm)	91.5% total resection	87.3% ^b	8/31			

VII facial nerve preservation rate, defined as preservation of House-Brackmann grade 1–2 function, *VIII* cochlear nerve, defined as preservation of Gardner-Robertson grade A or B hearing, *CN* cranial nerve, *fXRT* fractionated radiation therapy, *GBI* Glasgow Benefit Index, *GK* gamma knife, *LINAC* linear accelerator, *N* number of patients, *NR* not reported, *QOL* quality of life, *RS* retrosigmoid, *SF-36* short-form 36 survey, *SRS* stereotactic radiosurgery, *TL* translabyrinthine approach, *XRT* radiation therapy

^aRates indicate preservation of Gardner-Robertson grade A hearing only

^bReflects immediate (post-operative day 1) facial function

36 patients undergoing microsurgery with 46 gamma knife radiosurgery patients. Treatment indications (e.g. tumor growth) were not specified, and 21 patients were offered observation but were not included in the study. Patients undergoing surgery were significantly younger. Surgery was either retrosigmoid or translabyrinthine, and the use of intra-operative monitoring was not specified. Mean radiosurgery marginal and maximal doses were 12.2 and 26.4 Gy, respectively. At last follow-up, House and Brackmann (1985) grade I or II facial function (blinded assessment) was more frequent in the radiosurgery group (98% versus 83%). At 1 year, 50% versus 0% of patients had serviceable hearing in the radiosurgery and surgery groups, respectively. There were two cross-overs in the radiosurgery group (4%) who underwent subsequent surgery for progressive tumor growth. Both headache and tinnitus occurred with equal severity between groups, although radiosurgery patients scored better on the dizziness handicap inventory at last follow-up. With regards to QOL, SF-36 physical functioning, role-physical, bodily pain, energy/fatigue and physical composite scores were lower at 3 months in the surgery group. However, at last follow-up, SF-36 data was comparable with small mean between-group differences in mental health (6.8/100 points), bodily pain (5.6/100 points) and mental composite scores (3.3/100 points) only.

Myrseth et al. (2009) prospectively compared 88 vestibular schwannoma patients following either gamma knife radiosurgery or microsurgery with a minimum follow-up period of 2 years. Inclusion criteria included documented tumor growth or maximum extrameatal diameter >20 mm (but ≤25 mm). Patients selected gamma knife radiosurgery more frequently than surgery (60 versus 28 patients); in fact an initial attempt to randomize patients failed because of this. Patients in the gamma knife group were significantly older (57.5 versus 52.5 years). In addition to facial nerve and hearing outcomes, QOL was compared using the SF-36 inventory and Glasgow Benefit Index (GBI). In virtually all surgical patients a retrosigmoid approach was used with facial nerve EMG but no eighth nerve monitoring. Total resection was achieved in 23/28 patients.

Radiosurgery dosing was 12 Gy with a 30–40% isodose curve. Tumor growth control rates were not reported for the radiosurgery group; however within the 2 year follow-up period there was one cross-over of a radiosurgery patient who subsequently underwent surgery for tumor growth. Interestingly, the study “was closed...when the final patient had completed 2 years of follow-up.” It is unclear what the endpoint was which prompted closure of the study with only 28 patients having undergone surgery compared to 60 gamma knife radiosurgery patients. Facial nerve preservation (House and Brackmann grade I or II function) at 2 years (unblinded surgeon assessments) was 23/28 with surgery, and no patients had preserved serviceable hearing, compared to 17/25 patients with hearing preservation in the radiosurgery group. Despite this, SF-36 data remained unchanged from baseline for surgery and radiosurgery, nor were there any inter-group differences at 2 years.

Di Maio and Akagami (2009) reported 264 patients with newly-diagnosed vestibular schwannomas and is the largest study and the only prospective comparison of observation, radiation therapy and surgery. 47 observation patients, 48 LINAC radiation treatment (23 fractionated and 25 LINAC radiosurgery), and 134 surgery patients (97 with tumors ≤3 cm and 37 with tumors >3 cm) were studied. Mean follow-up was 32 months. Surgical patients underwent retrosigmoid, translabyrinthine, or middle fossa craniotomies with brainstem auditory evoked potentials, facial nerve free-running EMG and transcranial motor evoked potential monitoring (Akagami et al. 2005). Minimal peripheral dose for LINAC radiosurgery and fractionated stereotactic radiotherapy was 12 and 45 Gy, respectively. Observation patients had smaller tumors and radiation therapy patients were significantly older. 30/38 observation patients (78.9%) with serial imaging had no tumor growth. Of eight growing tumors, three were treated using surgery, and one was scheduled to be treated using radiation. There were 2/134 patients who crossed over from surgery to radiation treatment for a residual growing tumor. Three patients in the radiation group (n=48) had tumor growth that required surgery. Immediate

post-surgical facial nerve function and hearing preservation was published previously; House and Brackmann grade I-II function was 87.3% day 1 post-operatively for all tumor sizes (Akagami et al. 2005). Hearing preservation (Gardner-Robinson grade A or B) with surgery was 26% again for all tumor sizes (Yong et al. 2008). Previously reported facial nerve preservation for our radiation therapy series was 95.8%, and Gardner-Robertson grade A or B hearing was preserved in 57% of patients (Chung et al. 2004; Thomas et al. 2007). There were no differences in self-reported tinnitus, dizziness, useful hearing and facial weakness between observation, radiation therapy and surgery patients, with the exception of a lower self-reported rate of facial weakness in the observation group. With respect to QOL, there were no between- and within-group differences in SF-36 data for the observation, radiation therapy and surgery ≤ 3 cm groups. Surgery patients with tumors >3 cm had lower baseline SF-36 scores relative to the cohort, but had significant improvements through last follow-up in vitality, social functioning, role emotional, mental and physical components and total score. Patients with large tumors who had poorer QOL compared to the other groups, improved post-operatively to the same QOL status as the other groups on last follow-up. More patients were very satisfied with surgery (79% and 72% for tumors >3 and ≤ 3 cm versus 46% and 45% for observation and radiation respectively). And fewer patients felt the tumor management affected their QOL in the surgery groups. When asked about “the single most important factor affecting QOL right now” and whether it was related to their vestibular schwannoma, roughly one-third of the patients in all groups reported hearing to be the most important factor.

Recommendations for Vestibular Schwannoma Management

Based on the available literature, a number of recommendations regarding vestibular schwannoma management can be made. We demonstrated that observed patients up to 1.3 cm mean

diameter had similar QOL as treated patients (Di Maio and Akagami 2009), suggesting that observation may be underused. For example, mean tumor sizes for radiosurgery and surgery was 1.23 cm in the study by Pollock et al. (2006). In contrast, 36% of patients in our study with tumors <1.5 cm were observed. In our center, many patients with tumors <1.5 cm in diameter who are minimally symptomatic and without tumor growth are observed primarily with serial imaging. Growing tumors, larger diameters or progressively symptomatic patients are the usual indications for treatment. Although larger tumors are more time consuming and stressful for the surgical team, with the use of careful dissection and monitoring techniques, there is no size ‘cut-off’ in our series where facial preservation rates change significantly. This gives us confidence to monitor larger tumors. For tumors <3 cm in size, stereotactic radiation treatments and surgery are presented to patients as having clinical equipoise, although this may not be the case in other institutions. Our bias is towards surgery for younger patients and radiotherapy for older patients or those with significant medical co-morbidity. QOL in our study was decreased preoperatively in patients with tumors >3 cm, suggesting that patients experience deterioration in QOL when their tumor size reaches ~ 3 cm. This finding may have implications in anticipating when patients who have tumors <3 cm are likely to deteriorate in QOL from their VS and supports a policy of early intervention.

Discussion

A growing body of literature showing favourable outcomes for observation, stereotactic radiotherapy and radiosurgery, as well as microsurgery has led to much debate regarding the superiority of one modality over another, and challenged clinicians in providing full disclosure to patients in order to facilitate truly informed consent. The majority of modern surgical series have produced comparable physician-evaluated outcomes to stereotactic radiation treatments, although hearing preservation is more variable with surgery. If one

focuses on health related QOL, two of three prospective studies described above show no difference in SF-36 scores between stereotactic radiation and surgery, and in the third study the QOL differences were small with longer follow-up. This may represent a failure of the SF-36 to detect disease/symptom specific changes in QOL; alternatively, one may draw an impression that many patients enjoy a similar QOL between stereotactic radiation and surgery, and disease-specific inventories may be of less practical clinical value. Unfortunately, many treatment comparison studies for vestibular schwannoma patients lack generalizability for reasons described above, and the onus rests on individual institutions to critically assess and maximize treatment outcomes based on local strengths and weaknesses.

References

- Akagami R, Napolitano M, Sekhar LN (2002) Patient evaluated outcome after surgery for basal meningioma. *Neurosurgery* 50:941–949
- Akagami R, Dong C, Westerberg B (2005) Localized transcranial electrical motor evoked potentials for monitoring cranial nerves in cranial base surgery. *Neurosurgery* 57(1 Suppl):78–85
- Banerjee R, Moriarty JP, Foote RL, Pollock BE (2008) Comparison of the surgical and follow-up costs associated with microsurgical resection and stereotactic radiosurgery for vestibular schwannoma. *J Neurosurg* 108(6):1220–1224
- Battista RA (2009) Gamma knife radiosurgery for vestibular schwannoma. *Otolaryngol Clin North Am* 42(4):635–654
- Chung HT, Ma R, Toyota B, Clark B, Robar J, McKenzie M (2004) Audiologic and treatment outcomes after linear accelerator-based stereotactic irradiation for acoustic neuroma. *Int J Radiat Oncol Biol Phys* 59(4):1116–1121
- Cushing H (1917) Tumors of the nervus acusticus and the syndromes of the cerebellopontine angle. Saunders, Philadelphia
- Dandy WE (1925) An operation for the total removal of cerebellopontine (acoustic) tumors. *Surg Gynecol Obstet* 41:129–148
- Di Maio S, Akagami R (2009) Prospective comparison of QOL before and after observation, radiation, or surgery for vestibular schwannomas. *J Neurosurg* 111(4):855–862
- Frohlich AM, Sutherland GR (1993) Epidemiology and clinical features of vestibular schwannoma in Manitoba, Canada. *Can J Neurol Sci* 20(2):126–130
- Godefroy WP, Kaptein AA, Vogel JJ, van der Mey AG (2009) Conservative treatment of vestibular schwannoma: a follow-up study on clinical and quality-of-life outcome. *Otol Neurotol* 30(7):968–974
- Henzel M, Hamm K, Sitter H, Gross MW, Surber G, Kleinert G, Engenhart-Cabillie R (2009) Comparison of stereotactic radiosurgery and fractionated stereotactic radiotherapy of acoustic neurinomas according to 3-D tumor volume shrinkage and QOL. *Strahlenther Onkol* 185(9):567–573
- House JW, Brackmann DE (1985) Facial nerve grading system. *Otolaryngol Head Neck Surg* 93:146–147
- Karpinos M, Teh BS, Zeck O, Carpenter LS, Phan C, Mai WY, Lu HH, Chiu JK, Butler EB, Gormley WB, Woo SY (2002) Treatment of acoustic neuroma: stereotactic radiosurgery vs. microsurgery. *Int J Radiat Oncol Biol Phys* 54(5):1410–1421
- Lin VY, Stewart C, Grebenyuk J, Tsao M, Rowed D, Chen J, Nedzelski J (2005) Unilateral acoustic neuromas: long-term hearing results in patients managed with fractionated stereotactic radiotherapy, hearing preservation surgery, and expectantly. *Laryngoscope* 115(2):292–296
- Martin TPC, Tzifa K, Kowalski C, Holder RL, Walsh R, Irving RM (2008) Conservative versus primary surgical treatment of acoustic neuromas: a comparison of rates of facial nerve and hearing preservation. *Clin Otolaryngol* 33(3):228–235
- Myrseth E, Moller P, Pedersen PH, Vassbotn FS, Wentzel-Larsen T, Lund-Johansen M (2005) Vestibular schwannomas: clinical results and QOL after microsurgery or gamma knife radiosurgery. *Neurosurgery* 56(5):927–935; discussion 927–935
- Myrseth E, Moller P, Pedersen PH, Lund-Johansen M (2009) Vestibular schwannoma: surgery or gamma knife radiosurgery? A prospective, nonrandomized study. *Neurosurgery* 64(4):654–661; discussion 661–653
- Pollock BE (2008) Vestibular schwannoma management: an evidence-based comparison of stereotactic radiosurgery and microsurgical resection. *Prog Neurol Surg* 21:222–227
- Pollock BE, Lunsford LD, Kondziolka D, Flickinger JC, Bissonette DJ, Kelsey SF, Jannetta PJ (1995) Outcome analysis of acoustic neuroma management: a comparison of microsurgery and stereotactic radiosurgery. *Neurosurgery* 36(1):215–224; discussion 224–219
- Pollock BE, Driscoll CLW, Foote RL, Link MJ, Gorman DA, Bauch CD, Mandrekar JN, Krecke KN, Johnson CH (2006) Patient outcomes after vestibular schwannoma management: a prospective comparison of microsurgical resection and stereotactic radiosurgery. *Neurosurgery* 59(1):77–85; discussion 77–85
- Regis J, Pellet W, Delsanti C, Dufour H, Roche PH, Thomassin JM, Zanarat M, Peragut JC (2002) Functional outcome after gamma knife surgery or microsurgery for vestibular schwannomas. *J Neurosurg* 97(5):1091–1100

- Sackett DS, Richardson WS (2000) Evidence-based medicine: how to practice and teach EBM. Churchill-Livingstone, Philadelphia
- Sakamoto GT, Blevins N, Gibbs IC (2009) Cyberknife radiotherapy for vestibular schwannoma. *Otolaryngol Clin North Am* 42(4):665–675
- Stangerup SE, Caye-Thomasen P, Tos M, Thomsen J (2006) The natural history of vestibular schwannoma. *Otol Neurotol* 27(4):547–552
- Sughrue ME, Yang I, Aranda D, Lobo K, Pitts LH, Cheung SW, Parsa AT (2010) The natural history of untreated sporadic vestibular schwannomas: a comprehensive review of hearing outcomes. *J Neurosurg* 112(1):163–167
- Sweeney P, Yajnik S, Hartsell W, Bovis G, Venkatesan J (2009) Stereotactic radiotherapy for vestibular schwannoma. *Otolaryngol Clin North Am* 42(4):655–663
- Thomas C, Di Maio S, Ma R, Vollans E, Chu C, Clark B, Lee R, McKenzie M, Martin M, Toyota B (2007) Hearing preservation following fractionated stereotactic radiotherapy for vestibular schwannomas: prognostic implications of cochlear dose. *J Neurosurg* 107(5):917–926
- Urbach DR (2005) Measuring QOL after surgery. *Surg Innov* 12(2):161–165
- van Roijen L, Nijs HG, Avezaat CJ, Karlsson G, Linquist C, Pauw KH, Rutten FF (1997) Costs and effects of microsurgery versus radiosurgery in treating acoustic neuroma. *Acta Neurochir (Wien)* 139(10):942–948
- Velanovich V (2007) Using quality-of-life measurements in clinical practice. *Surgery* 141(2):127–133
- Ware JS (1991) The MOS 36-item short-form health survey (SF-36). I. Conceptual framework and item selection. *Med Care* 30:473–483
- Yong RL, Westerberg BD, Dong C, Akagami R (2008) Length of tumor-cochlear nerve contact and hearing outcome after surgery for vestibular schwannoma. *J Neurosurg* 108(1):105–110

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